



SAP Cloud Integration

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SAP Cloud Integration | Cloud

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What Is SAP Cloud Integration?

Support end-to-end process integration through the exchange of messages.

SAP Cloud Integration helps you to connect cloud and on-premise applications with other SAP and non-SAP cloud and on-premise applications. This service can process messages in real-time scenarios spanning different companies, organizations, or departments within one organization.

i Note

SAP Integration Suite combines Cloud Integration, API Management, Integration Advisor, Open Connectors, and other integration capabilities into a cohesive and simplified toolkit for enterprise integration. To provide a comprehensive integration experience, these services are not available separately, but only as part of the Integration Suite service plan. To learn more on different service plans, see the [Integration Suite](#) service catalog.

→ Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development teams for SAP Cloud Integration to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:

- [Feedback](#) > [Create issue](#): Provide feedback about a documentation page. This option opens an issue on GitHub.
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You need a GitHub account to use these options.

More information:

- [Contribution Guidelines](#)
- [Introduction Video: Open Documentation Initiative](#)
- [Blog Post: Introducing the Open Documentation Initiative](#)

Environment

This service is available in the following environments:

- Neo environment
- Cloud Foundry environment

Integration content artifacts designed in the Neo environment are also compatible in Cloud Foundry environment with certain limitations as mentioned in the following.

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#).

- Kyma environment

Features

Implement diverse scenarios

Integrate processes and data in application-to-application (A2A) and business-to-business (B2B) scenarios.

Connect to multiple endpoints

Integrate various applications and data sources from SAP and non-SAP, on premise, as well as the cloud. SAP Cloud Integration comes with a set of prebuilt adapters.

Customize SAP integration scenarios

Benefit from prepackaged integration content to jump-start integration projects and to set up productive scenarios with only minimum effort. You can extend predefined integration flows according to your requirements.

Develop custom adapters

Use the adapter SDK to build your own custom adapters for additional connectivity needs.

Access public APIs

Customize the access to SAP Cloud Integration with our public OData APIs.

Set up secure and reliable communication

Use our core integration and security capabilities for the safe and reliable processing of messages. Configure how messages are exchanged within an integration scenario so that the data involved is protected according to the newest security standards.

Implement various communication modes

Orchestrate business processes and integrate data in synchronous as well as in asynchronous scenarios. SAP Cloud Integration also supports reliable messaging processes based on asynchronous decoupling implemented by using queuing mechanisms.

Integrate with SAP Process Orchestration

Use SAP Cloud Integration and SAP's on-premise integration Platform, SAP Process Orchestration, seamlessly integrated.

Benefit from multitenancy support

Use this service in tenant-aware (multitenant) applications. Run them on a shared compute unit that can be used by multiple consumers (tenants).

Tools

Tools	Description
SAP BTP cockpit	The cockpit is the central point for managing all activities associated with your subaccount and for accessing key information about your applications.
Adapter Development Kit	The Adapter Development Kit allows integration developers to define new adapter types and to integrate them into the Cloud Integration tool environment.
Cloud Connector	It serves as the link between on-demand applications in SAP BTP and existing on-premise systems. You can control the resources available for the cloud applications in those systems.
Integration Suite Dashboard Overview	The Cloud Integration reporting dashboard is part of the Integration Suite content package developed on SAP Analytics Cloud. It is a simple and intuitive widget-based analytics dashboard that provides at-a-glance view of relevant key performance indicators of a Cloud Integration tenant. The widgets in the

Tools	Description
	dashboard display data in a simplified way that helps you visualize the context information with slicing and dicing capabilities. If you're curious to explore, then read the blog on Interactive Reporting Dashboard for SAP Cloud Integration using SAP Analytics Cloud 

System Scope

More information:

- [System Scope in the Cloud Foundry Environment](#)
- [System Scope in the Neo Environment](#)

Pricing and Resources

Get access to SAP Cloud Integration community, pricing models and all the technical resources, tutorials, blogs, and support you need. Get Started with [SAP Cloud Integration](#) .

Key Features

SAP Cloud Integration supports end-to-end process integration across cloud-based and on-premise applications (cloud-cloud and cloud-on-premise integration).

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

Feature Overview

SAP Cloud Integration comprises the following key features:

- Core runtime for processing, transformation, and routing of messages to be exchanged between the involved customer systems
It is ensured that data related to different customers connected to Cloud Integration is isolated. This is important, for example, when using Cloud Integration for business-to-business scenarios.
- Out-of-the-box connectivity support (for example, IDoc, SFTP, SOAP/HTTPS, SuccessFactors, OData, HTTPS)
- Security features such as content encryption and certificate-based communication

Upon purchase, predefined, ready-to-use prepackaged integration content can be made available by SAP without the immediate need for additional hardware or integration skills on the customer's side. This drastically reduces integration project lead times and lowers resource consumption significantly.

Cloud Integration offers full flexibility in how messages can be exchanged between customer systems by the following:

- Leveraging preconfigured integration patterns. These integration patterns provide different options for configuring the data flow between participants, for example, by using routing rules.

- Using various connectivity options. This covers a set of adapters (or endpoint types) that allow participants to connect with different communication protocols to SAP Cloud Integration.

You – as an Cloud Integration customer – can use the integration capabilities without the need to install an integration middleware on your own – as it would be the case with an on premise integration solution.

Compatibility with SAP Process Orchestration

You can also run integration content on the on-premise runtime of SAP Process Integration. For the Cloud Integration Web UI you can select the version of the SAP Process Integration so that the feature set of the integration content designer is adapted to the capabilities of the target runtime.

More information: [Runtime Profiles](#)

Cloud Integration Runtime Features

There is a wide range of supported ways how Cloud Integration can process messages and exchanged them with sender and receiver systems (see: [Integration Capabilities](#) and [Connectivity \(Adapters\)](#)).

For the delivery of messages received from a sender system Cloud Integration supports quality of service *at least once*. This means that the platform guarantees to process an inbound message at least once on the tenant. When you use one of the following adapters, you can configure additional quality of service settings:

Quality of Service	Description
At least once	Inbound message is processed at least once by Cloud Integration. If the same message is received multiple times from a sender, all of them are processed. i Note This quality of service is supported by all sender adapter types.
Best effort	Inbound message is sent synchronously and an immediate response is given back to the sender system.
Exactly once	Inbound message is processed exactly once by Cloud Integration. If a message with identical message ID (for example, XI message ID) is received multiple times from a sender, only the first one is processed by the sender adapter. The subsequent messages can be identified as duplicates and are not processed. i Note In the AS2 sender adapter, duplicate message handling needs to be configured explicitly.

i Note

Quality of service *at least once* is supported by all sender adapter types.

Quality of service *best effort* and *exactly once* are supported by certain adapter types such like, for example, the following ones:

- AS2 sender adapter (see: [Configure the AS2 Sender Adapter](#))
- AS4 sender adapter with ebMS3 receipt (see: [Configure the Sender Channel with ebMS3 Receipt](#))
- XI sender adapter (see: [Configure the XI Sender Adapter](#))

- XI receiver adapter (see: [Configure the XI Receiver Adapter](#))

i Note

The XI receiver adapter also supports the quality of service options *best effort* and *exactly once*.

- Best effort:

The message is sent synchronously; this means that Cloud Integration waits for a response before it continues processing.

- Exactly once:

The message is sent asynchronously. This means that Cloud Integration does not wait for a response before continuing processing. It is expected that the receiver guarantees that the message is processed exactly once.

The XI receiver adapter ensures that the same message is sent with the same XI message ID. That way, the receiver of this message is able to identify this is a duplicate and can handle the message accordingly.

More information: [Configure the XI Receiver Adapter](#)

Security

Various features guarantee that data processed by Cloud Integration during the execution of an integration scenario is protected at a maximum level.

More information:

[Security, Neo Environment](#)

[Security, Cloud Foundry Environment](#)

Cloud Integration OData API

An application programming interface (API) allows you to access Cloud Integration entities such like integration content artifacts, monitoring data, or security content, for example.

More information: [OData API](#)

Partner Directory

The Partner Directory allows you to store information about communication partners and to parameterize integration flows using this information. You can manage the content of the Partner Directory using the OData API.

A use case for the Partner Directory is the design and operation of business-to-business scenarios.

More information: [Partner Directory](#).

Related Information

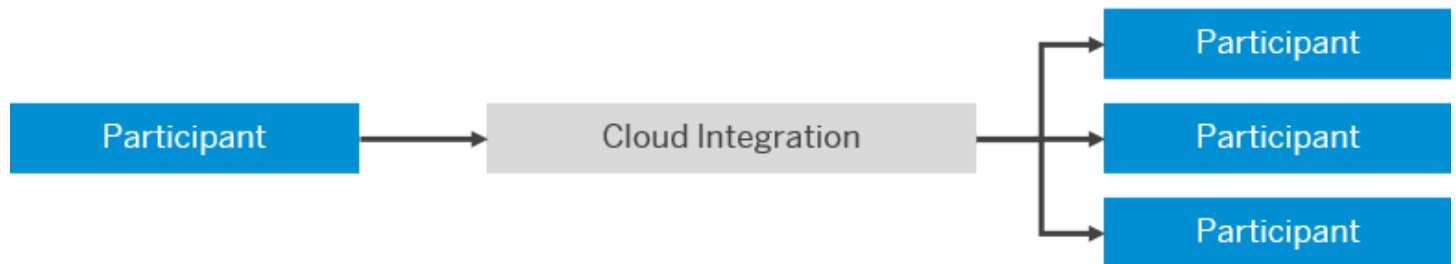
[Tool Access](#)

Integration Capabilities

There is a wide range of integration capabilities that define different ways how messages can be processed on the integration platform and exchanged between sender and receiver systems.

Cloud Integration supports various integration patterns, or ways how applications can be integrated with each other.

The following figure illustrates, as one example, the routing pattern, that allows you to forward a message from one participant to multiple receivers.



When using Cloud Integration, you specify the desired integration pattern by adding a dedicated **integration flow step** or a combination of various integration flow steps to an integration flow.

The following table lists the available integration capabilities, arranged by the related integration flow step types.

Message Transformation

Feature	Description
Mapping	Transforms the data structure and format used by the sender into a structure and format that the receiver can process. Supports the following kinds of mappings: • Message mappings designed with a graphical editor as part of the Cloud Integration toolset (supports XSD and EDMX structures) • Custom-mapping functions defined in scripts • XSLT mappings (defined in an XSLT resource)
ID Mapping	Maps the source message ID to a target message ID. You can use this feature to implement scenarios with exactly once processing of messages.
Content Modifier	Modifies the content of an inbound message by changing the header or body of the message. A message is composed of a message body and message headers. Furthermore, when being processed on a Cloud Integration tenant, additional data associated with the message can be passed along in an additional container (referred to as <i>message exchange</i>) to make it available at a later point in time during message processing. The Content Modifier can read data from and write data to the message body, the message header, and the properties area of the message exchange. That way, the content of a message can flexibly be modified and prepared for a receiver or subsequent processing steps. Certain constraints apply with regard to the supported data formats (as described in the product documentation).
XML Modifier	Modifies the content of an inbound message by removing external DTDs and/or removing XML declarations.
Converter	Transforms an input message into another format. The following converters are available: • XML to JSON: Transforms messages in XML format to JSON format.

Feature	Description
	You can specify streaming (with either the whole XML document or only specified XML elements presented by JSON arrays). • JSON to XML: Transforms messages in JSON format to XML format. • XML to CSV: Transforms messages in XML format to CSV format. • CSV to XML: Transforms messages in CSV format to XML format. • XML to EDI: Transforms a message in XML format to Electronic Data Interchange (EDI) format. • EDI to XML: Transforms a message in EDI format (EDIFACT or ASC-X12 format) to XML format. Certain constraints apply with regard to the supported data formats (as described in the product documentation).
Decoder	Decodes the incoming message to retrieve the original data (for example, if a base64-encoded message has been received). • Base64 Decode: Decodes base64-encoded message content. • GZIP Decompress: Decompresses the message content using GNU zip (GZIP). • ZIP Decompress: Decompresses the message content using zip (only zip archives with a single entry supported). • MIME Multipart Decode: Transforms a MIME multipart message into a message with attachments.
Encoder	Encodes the message using an encoding scheme to secure any sensitive message content during transfer over the network.
Filter	Filters information by extracting a specific node from the incoming message by using an XPath expression.
Message Digest	Calculates a digest of the payload or parts of it and stores the result in a message header.
Script	Executes custom Java script or Groovy script for message processing.

Calling External Systems or Subprocesses

Feature	Description
Request-Reply	Calls an external receiver system in a synchronous step and gets back a response.
Send	Calls an external receiver system for use cases where no reply is expected.
Content Enricher	Calls an external system, accesses resources of this system, and merges the returned content with the original message.
Poll Enrich Step	Polls content from an external component, and enriches the original message with it.
Process Call	Calls a local integration process. A local integration process defines a container for a separate subprocess to be called from the main process. Using local integration processes, a complex message processing sequence can be fragmented and decomposed into smaller parts.
Looping Process Call	Calls a local integration process in a loop.
Idempotent Process Call	Detects if a message ID has already been successfully processed and stores the status of the successful process in the idempotent repository. If there's duplicate execution with the same message ID (for

Feature	Description
	example if there's a retry by the sender system), the called subprocess can either be skipped or the message is marked as a duplicate. You can then decide how to handle the duplicate in the subprocess.

Routing

Feature	Description
Router	Routes a message to one or more receivers. SAP Cloud Integration also supports routing that depends on the content of the message (content-based routing). For example, the tenant detects that a message has a particular field value, and forwards it to the specific receiver participant that handles requests from the sender participant.
Multicast	Sends the same message to more than one receiver. - Parallel multicast: Initiates message transfer to all the receiver nodes in parallel - Sequential multicast: defines the sequence in which the message transfer to the receivers is initiated.
Splitter	Decomposes a composite message into a series of individual messages and sends them to a receiver. Supported splitters: - General splitter: Breaks down a composite message containing 'n' messages into 'n' individual messages. Each individual message is enveloped by the same elements that enveloped the composite message. - Iterating splitter: Splits a composite message into a series of smaller messages without copying the enveloping elements of the composite message - PKCS#7/CMS splitter: Splits a PKCS7 Signed Data message that contains a signature and content (and breaks down the signature and content into separate files) - IDoc splitter: Splits a composite IDoc messages into a series of individual IDoc messages with the enveloping elements of the composite IDoc message - EDI splitter: Splits a bulk EDI message into a series of individual messages and validates and acknowledges the inbound message. A bulk EDI message can contain one or more EDI formats, such as EDIFACT, EANCOM, or ASC-X12. The EDI splitter can process different EDI formats depending on the business requirements of the trading partners. - Zip splitter: Splits an inbound archive file (.zip) into individual files. - Tar splitter: Splits an archive (.tar) file into individual files. Certain constraints apply with regard to the supported data formats (as described in the product documentation).
Join	Merges messages from different routes and combines them into a single message. This feature is used in combination with the Gather feature. Join simply brings together the messages from different routes; it doesn't affect the content of the messages. Certain constraints apply with regard to the usage of this feature (as described in the product documentation).
Gather	Merges messages from different routes (into a single message) with the option to define certain strategies how to combine the initial messages.

Storing Data During Processing

Feature	Description
Persist Message	Stores a message payload so that you can access the stored message and analyze it at a later point in time.
Data Store Operations	Stores messages temporarily for later processing. The following operations are supported: • SELECT • GET • WRITE • DELETE
Write Variables	Specifies values for variables required during message processing.

Protecting Messages

Feature	Description
Encryptor	Encrypts the content of a message. Supported standards: • PGP • PKCS#7/CMS Enveloped Data and Signed Data
Decryptor	Decrypts the content of a message. Supported standards: • PGP • PKCS#7/CMS Enveloped Data and Signed Data
Signer	Signs a message. Supported standards: • PKCS#7/CMS Enveloped Data and Signed Data • XML Digital Signature
Verifier	Verifies a message. Supported standards: • PKCS#7/CMS Enveloped Data and Signed Data • XML Digital Signature

i Note

For mappings, XSLT (Extensible Stylesheet Language Transformations) 2.0 is supported.

i Note

Automatic stream caching mechanism is enabled to support streaming of large data and to avoid out-of-memory problems. This caching mechanism adds an interceptor between two processors, and caches streams either in memory or, if the stream is larger than 64 KB, in the file system. Hence enabling the streams to be read several times from the cache with reduced memory consumption .

Mapping

Mapping transforms (maps) sender into receiver data structures.

In scenarios spanning different application systems or different organizations and enterprises, it is very likely that the structure of the data exchanged between two participants will differ on both sides of a connection due to business-related reasons. To enable a seamless exchange of data, the data structures on both sides of a connection have to be transformed (or: mapped) into each other. There is the option to apply structural mapping of XML documents.

You can re-use existing on-premise content (service interfaces / message mappings / operation mappings / XSLT based mappings) from an SAP Enterprise Services Repository.

Value mappings allow you to map different representations of an object to each other.

Value mappings are useful when performing a dynamic value lookup of an object that has different representations in different contexts. In value mappings, you map these different representations of an object to each other by setting mapping rules in a value mapping table.

i Note

For example: You can use a value mapping to map a Merchant ID to a Customer ID, where Merchant ID is an external application representation of a customer, while Customer ID is an internal SAP representation.

Using Predefined Integration Content

SAP Cloud Integration allows the participating organizations to develop, deploy, and consume services in a standardized manner. SAP provides a predefined set of integration content that covers most of the integration needs for a particular scenario. Customers can use the predefined integration content to implement their integration scenarios with less time and effort. To accomplish this, however, customers need to register with SAP Cloud Integration and complete the onboarding process as recommended by SAP.

Customers can re-use existing on-premise content (message mappings / operation mappings / XSLT based mappings) from an SAP Enterprise Services Repository (EHP 1 for SAP NetWeaver 7.3).

What Is Integration Content

Operating business processes using SAP Cloud Integration implies the exchange of data (messages) between the participants. How messages are exchanged is specified by integration content that is designed based on the requirements of the business process.

As one key part of integration content, integration flows describe how a message sent from one participant is processed by SAP Cloud Integration.

In other words, using integration flows, specific integration pattern like mapping or routing can be specified.

For example, a set of integration flows specifies that a message sent from participant A is forwarded by SAP Cloud Integration to three different receivers B, C, and D, dependent on the business content contained in message. Integration flows also specify mappings of the data structure between sender and receiver or the endpoints of sender and receiver participants.

Connectivity (Adapters)

You have the option to specify which technical protocols should be used to connect a sender or a receiver to the tenant.

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

The following **adapters** are available.

Adapters Provided by SAP

When designing an integration flow, you can choose among the following adapters predefined by SAP.

Adapter	
Feature	Description
AmazonWebServices Sender adapter	Connects SAP Cloud Integration to Amazon Web Services. The adapter supports the following protocols: • S3: Simple Cloud Storage • SQS: Simple Queue Service See: AmazonWebServices Sender Adapter
AmazonWebServices Receiver adapter	Connects SAP Cloud Integration to Amazon Web Services. The adapter supports the following protocols: • S3: Simple Cloud Storage • SQS: Simple Queue Service • SNS: Simple Notification Service • SWF: Simple Workflow Service See: AmazonWebServices Receiver Adapter
AMQP Sender adapter	Enables SAP Cloud Integration to consume messages from queues or topic subscriptions in an external messaging system. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocols: TCP, WebSocket See: Configure the AMQP Sender Adapter
AMQP Receiver adapter	Enables SAP Cloud Integration to send messages to queues or topics in an external messaging system. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocols: TCP, WebSocket

Feature	Description
	See: Configure the AMQP Receiver Adapter
AMQP for SAP Event Mesh Sender adapter	Enables SAP Cloud Integration to consume messages from SAP Event Mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: WebSocket See: AMQP Sender for SAP Event Mesh
AMQP for SAP Event Mesh Receiver adapter	Enables SAP Cloud Integration to send messages to SAP Event Mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: WebSocket See: AMQP Receiver for SAP Event Mesh
AMQP for SAP Integration Suite, advanced event mesh Sender adapter	Enables SAP Cloud Integration to consume messages from SAP Integration Suite, advanced event mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: AMQ Sender for SAP Integration Suite, advanced event mesh
AMQP for SAP Integration Suite, advanced event mesh Receiver adapter	Enables SAP Cloud Integration to send messages to SAP Integration Suite, advanced event mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: AMQP Receiver for SAP Integration Suite, advanced event mesh
AMQP for Microsoft Azure Service Bus Sender adapter	Enables SAP Cloud Integration to consume messages from Microsoft Azure Service Bus. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Sender Adapter
AMQP for Microsoft Azure Service Bus Receiver adapter	Enables SAP Cloud Integration to send messages to Microsoft Azure Service Bus. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter
AMQP for Solace PubSub+ Sender adapter	Enables SAP Cloud Integration to consume messages from Solace PubSub+. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Sender Adapter
AMQP for Solace PubSub+ Receiver adapter	Enables SAP Cloud Integration to send messages to Solace PubSub+. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter

Feature	Description
AMQP for Apache Qpid Broker-J Sender adapter	Enables SAP Cloud Integration to consume messages from Apache Qpid Broker-J. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP, WebSocket See: Configure the AMQP Sender Adapter
AMQP for Apache Qpid Broker-J Receiver adapter	Enables SAP Cloud Integration to send messages to Apache Qpid Broker-J. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP, WebSocket See: Configure the AMQP Receiver Adapter
AMQP for Apache ActiveMQ 5 / Apache ActiveMQ Artemis Sender adapter	Enables SAP Cloud Integration to consume messages from Apache ActiveMQ 5 / Apache ActiveMQ Artemis. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Sender Adapter
AMQP for Apache ActiveMQ 5 / Apache ActiveMQ Artemis Receiver adapter	Enables SAP Cloud Integration to send messages to Apache ActiveMQ 5 / Apache ActiveMQ Artemis. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter
AMQP for IBM MQ Sender adapter	Enables SAP Cloud Integration to consume messages from IBM MQ. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Sender Adapter
AMQP for IBM MQ Receiver adapter	Enables SAP Cloud Integration to send messages to IBM MQ. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter
Ariba Sender adapter	Connects SAP Cloud Integration to the Ariba Network. Using this adapter, SAP and non-SAP cloud applications can receive business-specific documents in commerce eXtensible Markup Language (cXML) format from the Ariba network. The sender adapter allows you to define a schedule for polling data from Ariba. See: Configure the Ariba Sender Adapter
Ariba Receiver adapter	Connects SAP Cloud Integration to the Ariba network. Using this adapter, SAP and non-SAP cloud applications can send business-specific documents in commerce eXtensible Markup Language (cXML) format to the Ariba network. Receiver adapter See: Configure the Ariba Receiver Adapter
AS2 Sender adapter	Enables SAP Cloud Integration to exchange business-specific documents with a partner through the Applicability Statement 2 (AS2) protocol.

Feature	Description
	Sender adapter: Can return an electronic receipt to the sender of the AS2 message (in the form of a Message Disposition Notification (MDN)) See: Configure the AS2 Sender Adapter
AS2 Receiver adapter	Enables SAP Cloud Integration to exchange business-specific documents with a partner through the Applicability Statement 2 (AS2) protocol. See: Configure the AS2 Receiver Adapter
AS4 Sender adapter	Enables SAP Cloud Integration to securely process incoming AS4 messages using Web Services. The AS4 sender adapter is based on the ebMS 3.0 specification that supports the ebMS handler conformance profile. • Supports one-way/ebMS3 push message exchange pattern (MEP). • Support on-way/ebMS3 pull that allows the message party to pick the corresponding message from the partner. • Supports signature verification and decryption of the message. • Generates receipts after processing the incoming AS4 message. • Allows you to set a size limit for the body and attachment of an incoming message. See: AS4 Sender Adapter
AS4 Receiver adapter	Enables SAP Cloud Integration to establish a connection between any two message service handlers (MSHs) for exchanging business documents. The AS4 receiver adapter uses the Light Client conformance policy and supports only message pushing for the sending MSH and selective message pulling for the receiving MSH. Receiver adapter: • Supports one-way/push message exchange pattern (MEP) that involves the transfer of business documents from a sending MSH to a receiving MSH. • Supports one-way/selective-pull message exchange pattern (MEP) that involves the receiving MSH initiating a selective pull request to the sending MSH. The sending MSH responds by sending the specific user message. • Supports storing and verification of receipts. See: AS4 Receiver Adapter
BigCommerce GraphQL Receiver adapter	Connects SAP Integration Suite tenant to a BigCommerce GraphQL Admin APIs using the GraphQL message protocol. The GraphQL message protocol is used to connect to the GraphQL-based Admin APIs of a BigCommerce store. See
BigCommerce REST Receiver adapter	Connects SAP Integration Suite tenant to a BigCommerce REST Admin APIs, using the REST message protocol. The REST message protocol is used to connect to the REST-based Admin APIs of a BigCommerce store. See
Data Store Sender adapter	Enables SAP Cloud Integration to consume messages from a data store. See: Data Store Sender Adapter
ELSTER Receiver adapter	Enables SAP Cloud Integration to send a tax document to the ELSTER server. ELSTER (acronym for the German term <i>Elektronische Steuererklärung</i>) is used in German fiscal management to process tax declarations exchanged over the Internet. The adapter supports the following operations: Getting the version of the ERiC (ELSTER Rich Client) library, validating a tax document, and sending a tax document.

Feature	Description
	See: ELSTER Receiver Adapter
Facebook Receiver adapter	Enables SAP Cloud Integration to access and extract information from Facebook based on certain criteria such as keywords or user data. Using OAuth, the SAP BTP tenant can access resources on Facebook on behalf of a Facebook user. See: Facebook Receiver Adapter
FTP Sender adapter	Enables SAP Cloud Integration to connect to a remote system using TCP (Transmission Control Protocol) to receive files from the system. FTP stands for File Transfer Protocol. The sender adapter allows you to define a schedule for polling data from the connected system. See: Configure the FTP Sender Adapter
FTP Receiver adapter	Enables SAP Cloud Integration to connect to a remote system using TCP (Transmission Control Protocol) to write files to the system. FTP stands for File Transfer Protocol. See: Configure the FTP Receiver Adapter
HTTPS Sender adapter	Establishes an HTTPS connection between SAP Cloud Integration and a sender system. See: HTTPS Sender Adapter
HTTP Receiver adapter	Establishes an HTTP connection between SAP Cloud Integration and a receiver system. Receiver adapter: • Supports HTTP 1.1 only (target system must support chunked transfer encoding and may not rely on the existence of the HTTP Content-Length header) • Supports the following methods: DELETE, GET, HEAD, POST, PUT, TRACE Method can also be determined dynamically by reading a value from a message header or property during runtime. See: HTTP Receiver Adapter
IDoc Sender adapter	Allows SAP Cloud Integration to exchange Intermediate Document (IDoc) messages with a sender system that supports communication via SOAP Web services. A size limit for the inbound message can be configured for the sender adapter. See: Configure the IDoc Sender Adapter
IDoc Receiver adapter	Allows SAP Cloud Integration to exchange Intermediate Document (IDoc) messages with a receiver system that supports communication via SOAP Web services. See: Configure the IDoc Receiver Adapter
JDBC Receiver adapter	Allows SAP Cloud Integration to connect to a JDBC (Java Database Connectivity) database and to execute SQL commands on the database. See: JDBC Receiver Adapter
JDBC for DB2 (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to DB2 (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for DB2 (On-Premise)

Feature	Description
JDBC for Microsoft SQL Server (Cloud) Receiver adapter	Allows SAP Cloud Integration to connect to Microsoft SQL Server (Cloud) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Microsoft SQL Server (Cloud).
JDBC for Microsoft SQL Server (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to Microsoft SQL Server (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Microsoft SQL Server (On-Premise).
JDBC for Oracle (Cloud) Receiver adapter	Allows SAP Cloud Integration to connect to Oracle (Cloud) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Oracle (Cloud).
JDBC for Oracle (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to Oracle (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Oracle (On-Premise).
JDBC for PostgreSQL (Cloud) Receiver adapter	Allows SAP Cloud Integration to connect to PostgreSQL (Cloud) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for PostgreSQL (Cloud).
JDBC for SAP ASE Service (Neo) Receiver adapter	Allows SAP Cloud Integration to connect to SAP ASE Service (Neo) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP ASE Service (Neo).
JDBC for SAP HANA Cloud Receiver adapter	Allows SAP Cloud Integration to connect to SAP HANA Cloud using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP HANA (Cloud).
JDBC for SAP HANA Platform (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to SAP HANA Platform (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP HANA Platform (On-Premise).
JDBC for SAP HANA Service (Neo) Receiver adapter	Allows SAP Cloud Integration to connect to SAP HANA Service (Neo) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP HANA Service (Neo).
JMS Sender adapter	Enables asynchronous messaging by using message queues. The sender adapter consumes messages from a queue. The messages are processed concurrently. To prevent situations where the JMS adapter tries again and again to process a failed (large) message, you can store messages (where the processing stopped unexpectedly) in a dead-letter queue after two retries. Certain constraints apply with regard to the number and capacity of involved queues, as well as for the headers and exchange properties defined in the integration flow before the message is saved to the queue (as described in the product documentation). See: Configure the JMS Sender Adapter
JMS Receiver adapter	Enables asynchronous messaging by using message queues.

Feature	Description
	The receiver adapter stores messages and schedules them for processing in a queue. The messages are processed concurrently. See: Configure the JMS Receiver Adapter
Kafka Sender adapter	Allows SAP Cloud Integration to connect to an external Kafka broker via Kafka protocol and to fetch Kafka records (messages). See: Configure the Kafka Sender Adapter
Kafka Receiver adapter	Allows SAP Cloud Integration to connect to an external Kafka broker via Kafka protocol and to send Kafka records (messages). See: Configure the Kafka Receiver Adapter
Mail Sender for IMAP Sender adapter	Enables SAP Cloud Integration to read e-mails from an e-mail server using the Internet Message Access Protocol (IMAP) protocol. To authenticate against the e-mail server, you can send the user name and password in plain text or encrypted (the latter only if the e-mail server supports this option). You can protect inbound e-mails at the transport layer with IMAPS and STARTTLS. The sender adapter allows you to define a schedule for polling data from the connected system. For more information on possible threats when processing e-mail content with the Mail adapter, see the product documentation. See: Mail Sender for IMAP
Mail Sender for POP3 Sender adapter	Enables SAP Cloud Integration to read e-mails from an e-mail server using the Post Office Protocol (POP3) protocol. To authenticate against the e-mail server, you can send the user name and password in plain text or encrypted (the latter only if the e-mail server supports this option). You can protect inbound e-mails at the transport layer with POP3S and STARTTLS. The sender adapter allows you to define a schedule for polling data from the connected system. For more information on possible threats when processing e-mail content with the Mail adapter, see the product documentation. See: Mail Sender for POP3
Mail Receiver adapter	Enables SAP Cloud Integration to send e-mails to an e-mail server. To authenticate against the e-mail server, you can send the user name and password in plain text or encrypted (the latter only if the e-mail server supports this option). - You can protect outbound e-mails at the transport layer with STARTTLS or SMTPS. - You can encrypt outbound e-mails using S/MIME (supported content encryption algorithms: AES/CBC/PKCS5Padding, DESede/CBC/PKCS5Padding). See: Configure the Mail Receiver Adapter
Microsoft Dynamics CRM Receiver adapter	Connects SAP Cloud Integration to Microsoft Dynamics Customer Relationship Management (CRM). See: Microsoft Dynamics CRM Receiver Adapter
OData Sender adapter	Connects SAP Cloud Integration to systems using the Open Data (OData) protocol in either ATOM or JSON format (only synchronous communication is supported).

Feature	Description
	Supported versions: OData version 2.0 - The adapter receives incoming requests in either ATOM or JSON format. - Supported operations: Create (POST), Delete (DELETE), Query (GET), Read (GET), Update (PUT) Using the GET or POST method, the sender adapter can also invoke operations that are not covered by the standard CRUD (Create, Retrieve, Update, and Delete) methods (function import). See: Configure the OData Sender Adapter
OData Receiver adapter	Connects SAP Cloud Integration to systems using the Open Data (OData) protocol. Supported versions: - OData version 2.0 Supported operations: Create (POST), Delete (DELETE), Merge (MERGE), Query (GET), Read (GET), Update (PUT), Patch (PATCH) - OData version 4.0 Supported operations: Create (POST), Query (GET), Delete (DELETE), Update (PUT), and Patch (PATCH) - The outgoing request payload must be in XML format. See: Configure the OData V2 Receiver Adapter Configure the OData V4 Receiver Adapter
ODC Receiver adapter	Connects SAP Cloud Integration to SAP Gateway OData Channel (through transport protocol HTTPS). Supported operations: Create (POST), Delete (DELETE), Merge (MERGE), Query (GET), Read (GET), Update (PUT) See: ODC Receiver Adapter
OpenConnectors Receiver adapter	Connects SAP Cloud Integration to more than 150 non-SAP Cloud applications that are supported by SAP Open Connectors. - Uses APIs to fetch data from specific third-party applications. - Is designed to handle large volumes of incoming data. - Supports messages in both JSON and XML format, for request and response calls. - Allows you to define specific values for variables. See: OpenConnectors Receiver Adapter
ProcessDirect Sender adapter	Connects an integration flow with another integration flow deployed on the same tenant. An integration flow with a ProcessDirect sender adapter (as consumer) consumes data from another integration flow. N:1 cardinality of producer and consumer integration flows is supported. See: Configure the ProcessDirect Sender Adapter
ProcessDirect Receiver adapter	Connects an integration flow with another integration flow deployed on the same tenant. An integration flow with a ProcessDirect receiver adapter (as producer) sends data to another integration flow. N:1 cardinality of producer and consumer integration flows is supported.

Feature	Description
	See: Configure the ProcessDirect Receiver Adapter
RFC Receiver adapter	Connects SAP Cloud Integration to a remote receiver system using Remote Function Call (RFC). RFC is the standard interface used for integrating on-premise ABAP systems to the systems hosted on the cloud using SAP Cloud Connector. The adapter supports SAP NetWeaver, version 7.31 or higher. See: RFC Receiver Adapter
Salesforce Sender adapter	Connects SAP Cloud Integration to Salesforce. See: Salesforce Sender Adapter
Salesforce Receiver adapter	Connects SAP Cloud Integration to Salesforce. See: Salesforce Receiver Adapter
SFTP Sender adapter	Connects SAP Cloud Integration to a remote system using the SSH File Transfer protocol to read files from the system. SSH File Transfer protocol is also referred to as Secure File Transfer protocol (or SFTP). Supported versions: SSH version 2 (as specified at The Secure Shell (SSH) Protocol Architecture ), SSH File Transfer Protocol (SFTP) version 3 or higher. The sender adapter allows you to define a schedule for polling data from the connected system. See: Configure the SFTP Sender Adapter
SFTP Receiver adapter	Connects SAP Cloud Integration to a remote system using the SSH File Transfer protocol to write files to the system. SSH File Transfer protocol is also referred to as Secure File Transfer protocol (or SFTP). Supported versions: SSH version 2 (as specified at The Secure Shell (SSH) Protocol Architecture ), SSH File Transfer Protocol (SFTP) version 3 or higher. See: Configure the SFTP Receiver Adapter
SOAP SOAP 1.x Sender adapter	Exchanges messages with a sender system that supports Simple Object Access Protocol (SOAP) 1.1 or SOAP 1.2. The message exchange patterns supported by the sender adapter are one-way messaging or request-reply. The adapter supports Web services Security (WS-Security). A size limit for the inbound message can be configured for the sender adapter. See: Configure the SOAP (SOAP 1.x) Sender Adapter
SOAP SOAP 1.x Receiver adapter	Exchanges messages with a receiver system that supports Simple Object Access Protocol (SOAP) 1.1 or SOAP 1.2. The adapter supports Web services Security (WS-Security). See: Configure the SOAP (SOAP 1.x) Receiver Adapter
SOAP SAP RM Sender adapter	Exchanges messages with a sender system based on the SOAP communication protocol and SAP Reliable Messaging (SAP RM) as the message protocol. SAP RM is a simplified communication protocol for asynchronous Web service communication that does not require the use of Web Service Reliable Messaging standards. A size limit for the inbound message can be configured for the sender adapter. See: Configure the SOAP (SAP RM) Sender Adapter

Feature	Description
SOAP SAP RM Receiver adapter	Exchanges messages with a receiver system based on the SOAP communication protocol and SAP Reliable Messaging (SAP RM) as the message protocol. SAP RM is a simplified communication protocol for asynchronous Web service communication that does not require the use of Web Service Reliable Messaging standards. See: Configure the SOAP (SAP RM) Receiver Adapter
SuccessFactors REST Sender adapter	Connects SAP Cloud Integration to a SuccessFactors sender system using the REST message protocol. The adapter supports the following operations: GET See: Configure the SuccessFactors REST Sender Adapter
SuccessFactors REST Receiver adapter	Connects SAP Cloud Integration to a SuccessFactors receiver system using the REST message protocol. The adapter supports the following operations: GET, POST See: Configure the SuccessFactors REST Receiver Adapter
SuccessFactors SOAP Sender adapter	Connects SAP Cloud Integration to SOAP-based Web services of a SuccessFactors sender system (synchronous or asynchronous communication). The adapter supports the following operations: Query See: Configure the SuccessFactors (SOAP) Sender Adapter
SuccessFactors SOAP Receiver adapter	Connects SAP Cloud Integration to SOAP-based Web services of a SuccessFactors receiver system (synchronous or asynchronous communication). The adapter supports the following operations: Insert, Query, Update, Upsert See: Configure the SuccessFactors SOAP Receiver Adapter
SuccessFactors OData V2 Receiver adapter	Connects SAP Cloud Integration to a SuccessFactors system using OData V2. Features of OData version 2.0 supported by the adapter: • Operations: GET (get single entity as an entry document), PUT (update existing entry with an entry document), POST (create new entry from an entry document), DELETE (Delete an entry from an entry document), UPSERT (combination of Update OR Insert) • Query options: \$expand, \$skip, and \$top • Server-side pagination • Client-side pagination • Pagination enhancement: Data retrieved in chunks and sent to Cloud Integration • Deep insert: Creates a structure of related entities in one request • Authentication options: Basic authentication • Reference links: Link two entities using the <link> tag See: Configure the SuccessFactors OData V2 Receiver Adapter
SuccessFactors OData V4 Receiver adapter	Connects SAP Cloud Integration to a SuccessFactors system using OData V4. Features of OData version 4.0 supported by the adapter: • Operations: GET, POST, PUT, DELETE • Navigation • Primitive types supported according to OData V4 specification

Feature	Description
	Structural types supported for create/update operations: Edm.ComplexType, Edm.EnumType, Collection(Edm.PrimitiveType) and Collection(Edm.ComplexType) See: SuccessFactors OData V4 Receiver Adapter
SugarCRM Receiver adapter	Connects SAP Cloud Integration to SugarCRM. See: SugarCRM Receiver Adapter
Twitter Receiver adapter	Enables SAP Cloud Integration to access Twitter and read or post tweets. Using OAuth, SAP Cloud Integration can access resources on Twitter on behalf of a Twitter user. See: Twitter Receiver Adapter
XI Sender adapter	Connects SAP Cloud Integration to a remote sender system that can process the XI message protocol. See: Configure the XI Sender Adapter
XI Receiver adapter	Connects SAP Cloud Integration to a remote receiver system that can process the XI message protocol. See: Configure the XI Receiver Adapter

Connectivity Options and Communication Security

Various adapters allow you to connect the integration platform to remote systems using different kinds of technical communication protocols.

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

During a scenario, the connected remote systems exchange data with each other based on the configured transport protocol. These protocols support different options to protect the exchanged data against unauthorized access. In addition to security at the transport level, the content of the exchanged messages can also be protected by means of digital encryption and signature.

i Note

Mutual TLS (mTLS) is equivalent to client certificate authentication. While setting up the TLS connection, client and server exchange certificates. With mTLS, not only server certificates, but also client certificates are validated based on the signatures provided by certification authorities. For more information, see [Client Certificate Authentication \(Outbound\)](#) and [Keystore](#).

Transport-Level Security

Each adapter allows you to set up a specific security level based on the underlying transport protocol.

Transport-Level Security Options

Transport Protocol	Transport-Level Security
HTTP(S) (Hypertext Transfer Protocol Secure)	Hyper Text Transfer Protocol (HTTP) over Transport Layer Security (TLS) (HTTPS) TLS is a protocol for secure communication over a computer network that is widely used on the Internet. As TLS is the enhancement of SSL, these terms are often used synonymously.

Transport Protocol	Transport-Level Security
	You can protect communication using Transport Layer Security (TLS). In this case, a symmetric key length of at least 128 bits is used (which is technically enforced). Default length of asymmetric keys provided by SAP is 2048 bits. i Note The following versions are supported: • For inbound communication: TLS 1.2 and 1.3 (in the Cloud Foundry environment) • For inbound communication: TLS 1.2 (in the Neo environment) • For outbound communication: TLS 1.2, and 1.3 i Note The HTTP receiver adapter also allows you to use HTTP URLs. However, we do not recommend using this option when transferring confidential data (including the password for basic authentication). Also, if the network is not entirely trusted, there is no way to verify whether the result of an HTTP request originates from a trustworthy source. Therefore, we do not recommend using this option for productive scenarios over the Internet. Receiver adapters also support principal propagation via SAP Cloud Connector. Various authentication options (basic authentication using user credentials, client certificates, or OAuth) are supported depending on the selected sender or receiver adapter. ⚠ Caution Consider that we do not recommend to use basic authentication in productive scenarios because of the following security aspects: Basic authentication has the risk that authentication credentials, for example, passwords, are sent in clear text. Using TLS (transport-layer security, also referred to as Secure Sockets Layer) as transport-level encryption method (when using HTTPS as protocol) makes sure that this information is nevertheless encrypted on the transport path. However, the authentication credentials might become visible to SAP-internal administrators at points in the network where the TLS connection is terminated, for example, load balancers. If logging is not done properly at such devices, the authentication credentials might become part of log files. Also network monitoring tools used at such devices might expose the authentication information to administrators. Furthermore, the person to whom the authentication credentials belong (in the previous example, the password owner) needs to maintain the password in a secure place. This protocol is supported by all adapter types that allow communication over HTTPS, for example: Ariba, AS2, AS4, ELSTER, Facebook, HTTP, SOAP, IDoc, OData, ODC, OpenConnectors, SAP SuccessFactors, Twitter, XI.
SFTP (Secure Shell File Transfer Protocol)	Secure Shell (SSH) is used to securely transfer files in an open network. SSH is a network protocol that allows you to set up a secure connection to a remote computer, namely an SFTP server. SSH uses a symmetric key length with at least 128 bits to protect FTP communication. Default length of asymmetric keys provided by SAP is 2048 bits. Supported authentication methods: • User name/password authentication (where the SFTP server authenticates the calling component based on the user name and password)

Transport Protocol	Transport-Level Security
	- Public key authentication (where the SFTP server authenticates the calling component based on a public key) Secure data transfer with SFTP is based on a combination of symmetric and asymmetric keys. Symmetric (session) keys are used to encrypt and decrypt data within a session. Asymmetric key pairs are used to encrypt and decrypt the session keys. When asymmetric key pairs are used, SFTP also ensures that only authorized public keys are used by the involved participants. Supported versions: - SSH version 2 (as specified at The Secure Shell (SSH) Protocol Architecture ➦) - SSH File Transfer Protocol (SFTP) version 3 or higher This protocol is supported by the SFTP sender and receiver adapter.
FTPS (Secure File Transfer Protocol)	This protocol is used to transfer files from and to file servers. This is an extension of the File Transfer Protocol (FTP). This protocol is supported by the FTP sender and receiver adapter.
JMS (Java Message Service)	JMS is an API that supports reliable asynchronous communication based on a JMS Message Broker. This protocol is supported by the JMS sender and receiver adapter.
JDBC (Java Database Connectivity)	Java Database Connectivity (JDBC) is a protocol or API that defines how a client accesses a database. This protocol is supported by the JDBC receiver adapter.
RFC (Remote Function Call)	Remote Function Call (RFC) is an SAP-proprietary protocol that is used as standard for the communication of SAP systems. This protocol is supported by the RFC receiver adapter.
TCP (Transmission Control Protocol)	TCP is one of the main communication protocols used on the Internet. TLS, previously mentioned, often (but not always) runs on top of TCP. This protocol is supported by the AMQP (TCP) and the LDAP adapter.
WebSocket	WebSocket is a protocol that supports simultaneous communication in both directions (full-duplex). This protocol is used for setting up connections with a messaging systems. This protocol is supported by the AMQP (WebSocket) adapter.
SMTPS (Simple Mail Transfer Protocol Secure)	This protocol enables a computer to exchange emails with a mail server. With SMTPS, SMTP connections can be secured by SSL or TLS. Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied.

Transport Protocol	Transport-Level Security
	This protocol is used by the Mail receiver adapter.
POP3S (Post Office Protocol version 3 over TLS/SSL)	This option enables email clients to retrieve emails from an email server using the Internet Protocol (IP). Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol can be used by the Mail sender adapter.
IMAPS (Internet Message Access Protocol over TLS/SSL)	Internet Message Access Protocol over TLS/SSL (IMAPS). This option enables email clients to retrieve emails from an email server using a TCP/IP connection. Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol can be used by the Mail sender adapter.

Message-Level Security

On top of the *transport-level security* options, you can also secure the communication at *message level*, where the content of the exchanged messages can also be protected by means of digital encryption and signatures. Various security standards are available to do this, as summarized in the following table.

To configure message-level security options, you use dedicated integration flow steps (for example, the **Encryptor** and **Signer** step types).

The following standards and algorithms are supported:

Message-Level Security Standards and Algorithms

Standard	Security Feature
PKCS#7/CMS Enveloped Data and Signed Data	Encryption/decryption of message content
	Signing/verification of payload

Standard	Security Feature
PKCS#7/CMS Enveloped and Signed Data	Encryption/decryption and signing/verification of payload
Open Pretty Good Privacy (PGP)	Encryption/decryption of message content
	Encryption/decryption and signing/verification of message
XML Signature	Signing/verification of payload
WS-Security	Signing/verification of SOAP body

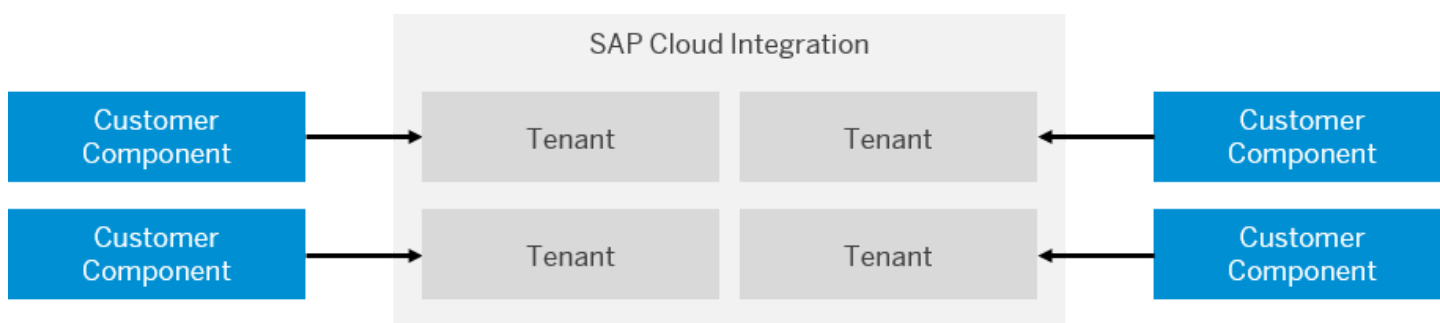
Related Information

[Message-Level Security](#)

Elements of a Cloud-Based Integration Scenario

In a cloud-based integration scenario, different customer components exchange messages with each other, and SAP Cloud Integration acts as an integration hub. Each customer component is securely connected to the Cloud Integration runtime.

In a nutshell, an integration scenario relies on a general landscape and component setup as illustrated in the following figure.



- Secure connections between Cloud Integration and the involved remote components

The Cloud Integration platform is fragmented into different tenants. A **tenant** represents the resources of the platform allocated to a customer and must be securely connected to the associated component in the customer landscape. The chosen transport protocol allows for specific transport-level security options (for example, HTTPS). On top of this, Cloud Integration supports various message-level security options, which allow you to digitally sign and encrypt the transferred data. The security setup relies on digital keys, which are stored in keystores; the creation and management of keystores is part of the security configuration of each component. The type of keystore and digital key used depends on the chosen security option. Therefore, we refer to these elements generally as **security artifacts**.

In order to enable a tenant to securely communicate with a customer component, you have to configure the required security artifacts and deploy them on the tenant. On the other side of the communication, the customer component has to be configured accordingly by the responsible system administrator.

- Integration knowledge specified and accessible to the runtime components

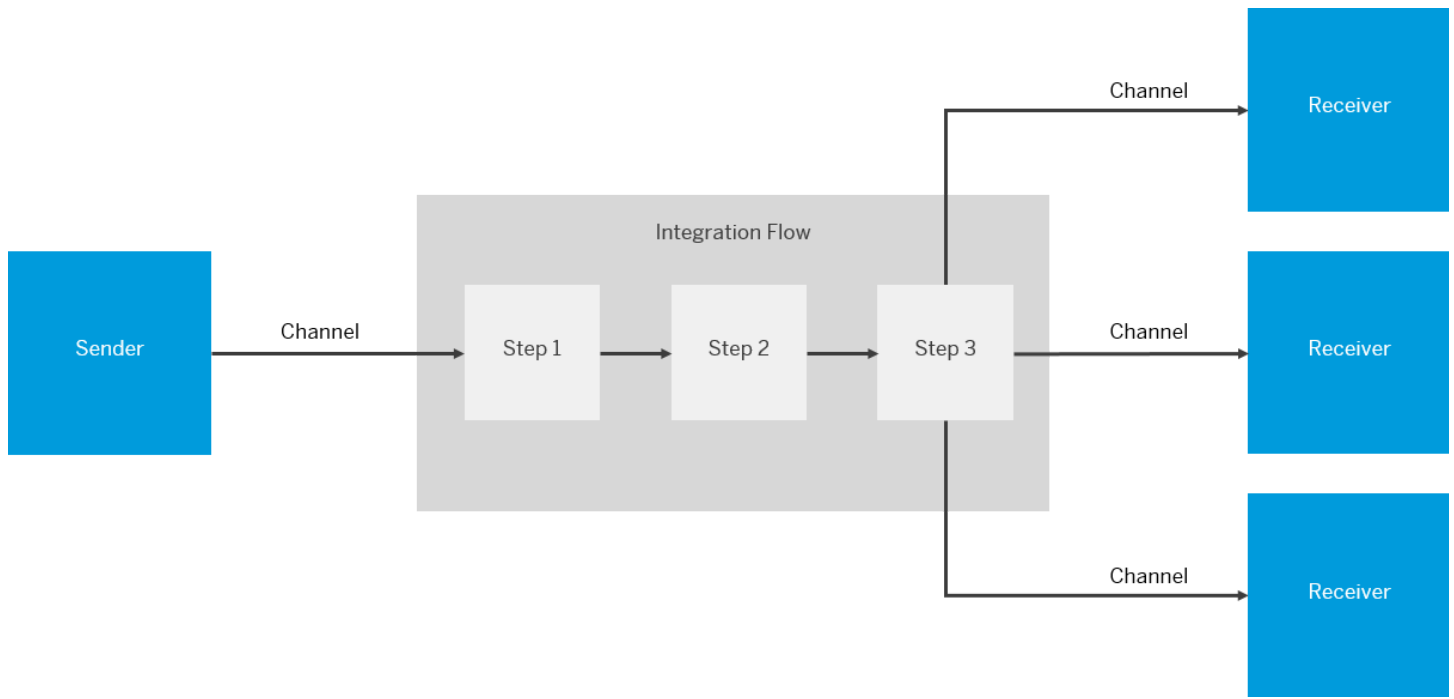
During the operation of an integration scenario, Cloud Integration acts as an integration hub for the message exchange. To ensure a seamless process and data flow during the operation of the scenario, the Cloud Integration runtime needs to access the information on how messages are to be processed. This information is also referred to as *integration knowledge* and is contained in the **integration content** for each tenant. A key part of the integration content is the **integration flow**, which specifies step-by-step how a message is to be processed on a tenant. For example, a mapping step transforms the data contained in a message so that it can be processed by a receiver system, whereas a routing step defines one or more receivers of a message.

During the design time of an integration scenario, you define the required integration flows. To *activate* an integration flow, you have to deploy it on the associated tenant.

Integration Flows

An integration flow allows you to specify how a message is processed on a tenant.

The following figure provides a simplified and generalized representation of an integration flow.



Related Information

[Runtime in Detail](#)

[Connectivity Options and Communication Security](#)

Tool Access

You can access and manage integration content and operate and monitor integration artifacts and messages at runtime.

The software that implements the process integration capabilities is updated on a regular basis.

In addition to the runtime components, you can use a Web-based application to:

- Access predefined integration content provided by SAP at SAP Business Accelerator Hub.
- Design integration flows and other integration artifacts.

An integration flow allows you to specify how a message is processed by Cloud Integration (see [Elements of a Cloud-Based Integration Scenario](#) and [Elements of an Integration Flow](#)). You can design integration flows with a graphical editor.

SAP also provides guidelines for integration flow design (see: [Integration Flow Design Guidelines](#)).

More information: [Developing Integration Content with SAP Cloud Integration](#)

- Monitor the processing of messages and check the status of deployed integration artifacts.

You also manage-related artifacts such as digital keys and certificates.

More information:

[Regions](#)

[Regions in the Neo Environment](#)

Partner Directory

The Partner Directory allows you to store information about communication partners and to parameterize integration flows using this information.

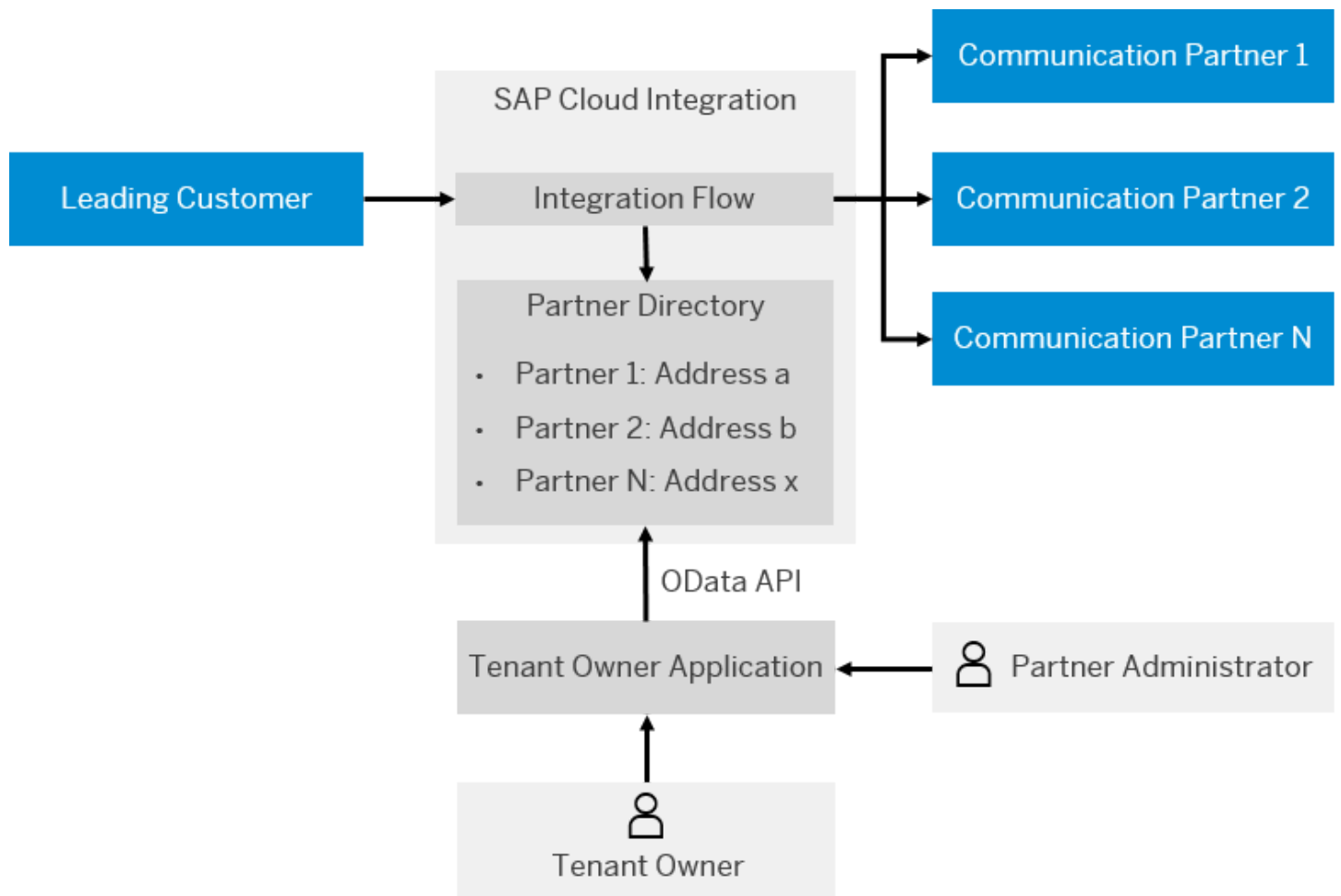
The Partner Directory helps you to set up a communication network between many communication partners efficiently. You use the Partner Directory to store partner-specific information. Those components that are parameterized read this information during runtime from the Partner Directory.

In the context of a business-to-business (B2B) scenario involving a partner network, the person or organization that is responsible for the B2B scenario as a whole is also the owner of the SAP Cloud Integration tenant.

Administrators at the side of each business partner use a dedicated application (referred to as tenant owner application) to maintain entities in the Partner Directory (through an OData API). The tenant owner application is provided to the partners by the tenant owner.

Note that such a tenant owner application is not part of the feature set of SAP Cloud Integration. However, SAP Cloud Integration comes with a set of OData APIs that allow access to the Partner Directory (and can be used to implement such a tenant owner application).

As illustrated in the figure, the Partner Directory is embedded in the system landscape in the following way.



Partner Directory Entities

The Partner Directory contains the following entities (which can be accessed using an OData API):

- Partner ID (PID)

A partner has an ID (PID) that is unique within the Partner Directory. The uniqueness of the PID is ensured by the tenant owner application.

- Alternative Partner

A partner can have several alternative identifiers (Alternative Partner). The same concepts are applied to the Alternative Partner as to party identifiers in SAP Process Integration: Each Alternative Partner has three string fields: Agency, Scheme, and ID.

- Agency

Name of the organization that defines the identification scheme (or schema) and issues names for the objects to be identified.

- Scheme

The reference framework within which objects are uniquely identified by names.

- ID

An ID or name that identifies the object within the given scheme.

For more information on the alternative partner ID and how it is related to the partner ID, check out the following SAP Community blog: [Cloud Integration – Partner Directory – Partner Dependent XML Structures and IDs](#).

For more information on the usage of the fields Agency, Scheme, and ID, see the documentation of SAP Process Integration at <http://help.sap.com>.

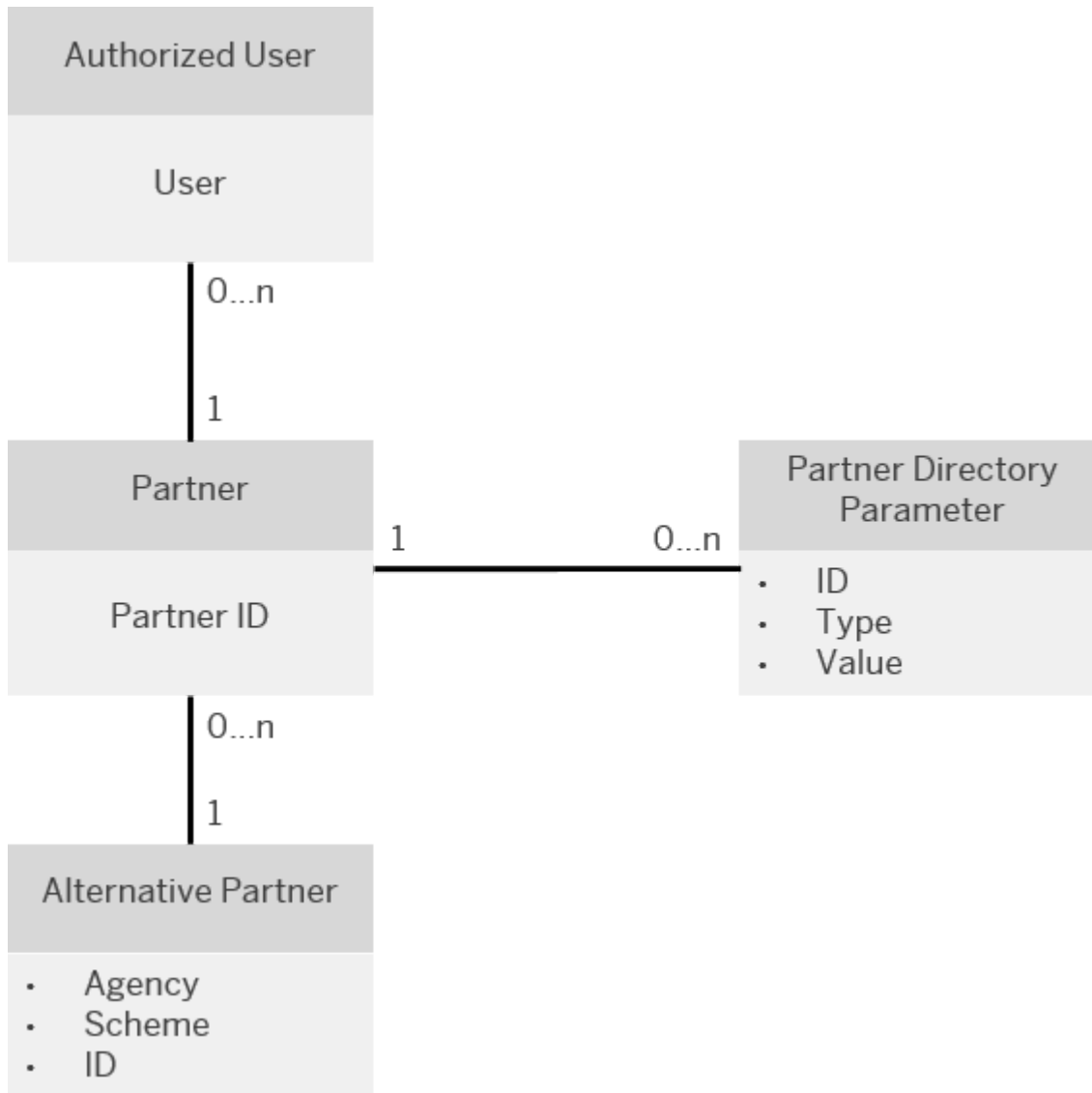
- Authorized User

This user authorizes a sending partner system to log in to SAP Cloud Integration (inbound communication).

If the partner uses HTTPS with client certificate authentication to connect to SAP Cloud Integration, certificate-to-user mappings are applied. One or more Authorized Users can be defined for each partner.

- Partner Directory Parameters for the parameterization of integration flows

The following figure provides an overview of the Partner Directory entities.



The Partner Directory entities related to a partner are maintained by an administrator at the partner organization using the tenant owner application. The Partner Directory entities are accessed using an OData application interface.

The following integration flow components are parameterizable so that partner-specific information (such as partner endpoint address, specific mapping, client certificates for inbound calls) can be used at runtime.

- XSLT: XSLT files are specified by each partner
- XML Schema Validator: XSD files are specified by each partner
- HTTP receiver adapter: receiver address (however, one common user or client certificate is used to call the partner system)

- HTTP receiver adapter: user credential
- HTTP sender adapter: sender partner-specific user or client certificate
- SOAP receiver adapter: user credential
- Script: for accessing partner-specific information and performing partner-specific operations

For Partner Directory parameters of AS2 Sender Adapter, see [Configure the AS2 Sender Adapter](#).

For Partner Directory parameters of AS2 MDN Sender Adapter, see [Configure the AS2 MDN Sender Adapter](#)

i Note

PD parameters are shown in the MPL log as MPL properties.

For a step-by-step example of how to use the Partner Directory, see <https://blogs.sap.com/2017/07/25/cloud-integration-partner-directory-step-by-step-example/> .

⚠ Caution

Limitations

Be aware of the following limitations when working with the Partner Directory:

Size restrictions for the different entity types:

- Maximum number of StringParameters overall: 3,000,000 (corresponds to 10,000 partners each using 300 StringParameters)
- Maximum number of BinaryParameters overall: 400,000 (corresponds to 10,000 partners each using 40 BinaryParameters)
- Maximum number of AlternativePartners overall: 1,000,000 (corresponds to 10,000 partners each using 100 AlternativePartners)
- Maximum number of AuthorizedUsers overall: 500,000 (corresponds to 10,000 partners each using 50 AuthorizedUsers)

The maximum size of a keystore is 2 MB (when using the Neo environment).

- The 2 MB limit corresponds to around 2000 X.509 certificates.
- A key pair with a chain of three X.509 certificates consumes about 3 KB, so if the keystore only contains key pairs of this type, then you can store around 600 key pairs in the keystore.

Limit for certificate-to-user mapping (when using the Neo environment): 2 MB (corresponds to about 2000 X.509 certificates)

The maximum size of a keystore is 6 MB (when using the Cloud Foundry environment).

- The 6 MB limit corresponds to around 6000 X.509 certificates.
- A key pair with a chain of three X.509 certificates consumes about 3 KB, so if the keystore only contains key pairs of this type, then you can store around 1800 key pairs in the keystore.

If you upload a whole keystore (.jks file) to the tenant, the maximum keystore size is limited to 2 MB.

For more information on the entities of the Partner Directory and how to work with them, check out the OData API section of this documentation.

For detailed step-by-step descriptions how to use the Partner Directory, see the following blogs:

- [Cloud Integration – Partner Directory – Step-by-Step Example](#) 
- [Cloud Integration – Partner Directory – Partner Dependent XML Structures and IDs](#) 
- [Cloud Integration – Partner Directory – Sender Partner Connecting with Client Certificate Authentication](#) 
- [Cloud Integration – Partner Directory – Partner Dependent User Credential Selection](#) 
- [Cloud Integration – Partner Directory – Mass Configuration](#) 

Related Information

[Partner Authorization \(Inbound\)](#)

[OData API](#)

[Read and Modify Partner Directory Content](#)

[Dynamically Reading XSLT Mappings from the Partner Directory](#)

[Dynamically Reading XSD Files from the Partner Directory](#)

[Partner Directory Cache](#)

Licensed Capabilities and Features

Licensed Capabilities for Cloud Foundry

For more information about different service plans and their supported feature set, see SAP Note [2903776](#) .

For information related to the service plans available for SAP Cloud Integration, also refer to [SAP Discovery Center](#) .

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

Licensed Capabilities for Neo

The SAP Cloud Integration Enterprise license supports the following additional capabilities of an integration flow.

- **AS2 Adapter** – You use this capability to configure a sender and receiver channel of an integration flow with the AS2 adapter. You can use this adapter and exchange business-specific documents with your partner through AS2 protocol. You can use this adapter to encrypt/decrypt, compress/decompress, and sign/verify the documents.
- **JMS Adapter** – You use this capability (Java Message Service) to connect messaging systems to the Integration Engine.
- **EDI to XML Converter** – You use this capability to transform messages in EDI format to XML format. You can convert EDIFACT and ASC-X12 format into XML format.
- **XML to EDI Converter** – You use this capability to transform a message in XML format to EDI format. You can convert EDIFACT and ASC-X12 format into XML format.

- EDI Splitter – You use this capability to split inbound bulk EDI messages, and during processing you can configure the splitter to validate and acknowledge the inbound messages.
- Enterprise Message Broker – You (tenant admin) can provision message broker to use JMS adapter scenarios.
- Integration Advisor (IA) – You use IA capabilities to define, maintain, share, and deploy integration content for exchanging business document in B2B scenarios. Based on the designed message implementation guidelines and mapping guidelines, the IA automatically generates the required runtime artifacts that can be used in an integration flow. For more information, see [SAP Integration Advisor](#).

System Scope in the Cloud Foundry Environment

This section describes the system scope for Cloud Integration tenants that are deployed in the Cloud Foundry environment. Read the recommendations to know how to optimize the resources when exceeding the scope.

Resource	Scope
Integration content	2 GB Refer to the blog on Content Size Limits  learn how to reduce your integration content size.
JMS queues	9 GB, 150 transactions (default configuration with 30 queues) Can be scaled up to 30 GB, 500 transactions (with 100 queues) See the blog on Cloud Integration – JMS Resource and Size Limits  for further guidance on how to set the queue size to restrict the limit and on how to delete unused queues.
Message processing log persistence	35 GB See: Cloud Integration – Setting the Log Level for Message Processing 
Runtime database	35 GB See: Optimize Performance
Disk space	10 GB Refer to SAP Note 2648415  to learn how to optimize the integration flow development to prevent the integration flow from running into the “No More Space left on Disk” error.

For more information on the available data storage features, refer to [Data Storages](#).

System Scope in the Neo Environment

This section describes the system scope for Cloud Integration tenants that are deployed in the Neo environment. Read the recommendation to know how to optimize the resources when exceeding the scope.

Resource	Scope
Integration content	500 MB

Resource	Scope
	Refer to the blog on Content Size Limits  learn how to reduce your integration content size.
JMS queues	9 GB 9 GB, 150 transactions (default configuration with 30 queues) Can be scaled up to 30 GB, 500 transactions (with 100 queues) See the blog on Cloud Integration – JMS Resource and Size Limits  for further guidance on how to set the queue size to restrict the limit and on how to delete unused queues.
Tenant database	32 GB See the following blogs for information on how to: • Avoid Storing Payloads in the Message Processing Log  • Avoid Excessive Storage Load caused by Using MPL Attachments  • Set the Log Level for Message Processing 
Disk space	2 GB Refer to SAP Note 2648415  to learn how to optimize the integration flow development to prevent the integration flow from running into the “No More Space left on Disk” error.

For more information on the available data storage features, refer to [Data Storages](#).

Software Update

Cloud Integration software is updated monthly by SAP.

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

The following applies for software updates.

- Software updates are performed by SAP. Customers do not have to take any action here.
- Software updates do not require any downtime of productive scenarios running on the integration platform.
- SAP ensures that deployed integration flows continue running after the update.
- In case a tenant is not accessible or shows component failures after an update, SAP is notified by appropriate alert mechanisms and triggers the necessary recovery processes.

The *SAP Cloud Integration Terms and Conditions* specify details of service level agreements and system availability. You can find the SAP Cloud Integration Terms and Conditions at: <http://global.sap.com/corporate-en/our-company/agreements/index.epx> .

- If a specific integration flow stops working unnoticed after update, customers are asked to open a ticket. Choose the corresponding sub component under LOD-HCI-PI-OPS (priority high).

- For SAP to ensure proper service delivery, the customer must collaborate with SAP by providing the necessary support and any required professional services, for example, by adapting custom adapters to the latest runtime version, and testing custom integration content deployed on the tenant, if undocumented features are used. This collaboration is a prerequisite for proper service through SAP.

Operating Model

An operation model clearly defines the separation of tasks between SAP and the customer during all phases of an integration project.

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

SAP BTP and SAP Cloud Integration (also known as Cloud Integration) have been developed on the assumption that specific processes and tasks will be the responsibility of the customer. The following table contains all processes and tasks involved in operating the aforementioned services and specifies how the responsibilities are divided between SAP and the customer for each individual task. It does not include the operation of systems and devices residing at operational facilities owned by the customer or any other third party, as these are the customer's responsibility.

Changes to the operating model defined for the services in scope are published using the *What's New* (release notes) section of the respective product documentation on SAP Help Portal. Customers and other interested parties must review the product documentation on a regular basis. If critical changes are made to the operating model, which require action on the customer side, an explicit notification is sent by e-mail to the affected customers. If customers want to receive such notifications, they can subscribe to the relevant communication channels offered by SAP (for example, by opening a customer incident).

It is not the intent of this document to supplement or modify the contractual agreement between SAP and the customer for the purchase of any of the services in scope. In the event of a conflict, the contractual agreement between SAP and the customer as set out in the Order Form, the General Terms and Conditions of SAP Cloud Services, the supplemental terms and conditions, and any resources referenced by those documents always takes precedence over this document.

Responsibilities for operating the following services are listed in the table below:

- SAP BTP (referred to as *Platform*)
- SAP Cloud Integration

Responsibilities for Operating SAP BTP

Activity	Task	SAP Cloud Service		Responsibility	
		Platform	Cloud Integration	SAP	Customer
Communication Management	Appoint an English-speaking contact person and communicate the name to SAP. This is required to ensure timely processing of configuration change requests affecting the customer system, interacting with SAP for efficient incident processing, and other interaction between SAP and the customer.	✓	✓		✓
	Subscribe to the communication channels offered by SAP for receiving prompt information about any service disruptions,	✓	✓		✓

Activity	Task	SAP Cloud Service		Responsibility	
		Platform	Cloud Integration	SAP	Customer
	critical maintenance activities affecting the customer system, and change requests requiring action on the customer side.				
	Inform the customer about any service disruptions, critical maintenance activities affecting the customer system, and change requests requiring action on the customer side.	✓	✓	✓	
Asset Management	Management of the hardware and infrastructure resources, from acquisition through disposal. This includes the request and approval process, procurement management, life-cycle management, and disposal management.	✓	✓	✓	
	Protect IT assets such as systems, network, and data from threats that arise from unauthorized physical access or physical influence on those assets.	✓	✓	✓	
Provisioning	Provisioning of resources and systems to customers in accordance with ordered package and requirements. This includes the allocation and provisioning of technical (physical and virtual) resources, such as storage, network, compute units, systems, and database hosts, the deployment of the application software and the proper initial configuration of quotas, service subscriptions, permissions, and trust configuration. i Note Provisioning of tenants linked with global accounts is in the responsibility of the customer.	✓	✓	✓	
Integration Content Development	Design, build, deploy, and operate the integration content hosted in the application. This includes proper testing of the integration content under realistic conditions before its productive usage. Integration content may comprise integration flows, adapters, scripts, and so on.		✓		✓
Security Material Management	Create, configure, deploy, and operate (renew) security material hosted in the application. This includes proper testing of the security material under realistic conditions before its productive usage. Security material may comprise user credentials, PGP key rings, certificates, known hosts files, and so on.		✓		✓

Activity	Task	SAP Cloud Service		Responsibility	
		Platform	Cloud Integration	SAP	Customer
	Create, configure, deploy, and operate the keystore artifact hosted in the application. This includes the import of public and private keys used for certificate-based authentication when sending a message from the application.		✓		✓
Message Transmission	Correct transmission of the messages within the according constraints offered by the application. SAP makes no warranty and shall have no liability for the contents of any message transmitted via the application, including the accuracy or completeness of any information contained in a message.		✓	✓	
Change Management	Apply regular product increments, as well as corrections to the application to avoid incidents with minimal possible disruption of normal operations. Ensure that all changes (such as changes in scheduling of administrative jobs, enabling the product capabilities, and so on) are evaluated, authorized, prioritized, planned, tested, implemented, documented, and reviewed prior to implementation.		✓	✓	
	Perform upgrades of the application in a monthly cycle. Emergency changes, for example, triggered by Incident Management processes, have accelerated testing, approval, and implementation.		✓	✓	
	Apply regular product increments, as well as corrections to the infrastructure, systems, and services to avoid incidents with minimal possible disruption of normal operations. Ensure that all changes (such as updates of the Java runtime, operating system patches, and so on) are evaluated, authorized, prioritized, planned, tested, implemented, documented, and reviewed prior to implementation.	✓		✓	
	Perform upgrades of the infrastructure, systems, and services in a bi-weekly cycle. Emergency changes, for example, triggered by Incident Management processes, have accelerated testing, approval, and implementation.	✓		✓	
	Consume latest version of provisioned infrastructure, systems, and services (for example, Java runtime, operating system) to run the application in the customer account.	✓	✓	✓	

Activity	Task	SAP Cloud Service		Responsibility	
		Platform	Cloud Integration	SAP	Customer
	Collaborate with SAP to ensure timely processing of change requests affecting the resources in the customer account.	✓	✓		✓
	Prompt delivery of patches for security vulnerabilities in the operating system and database hosted by the application. This includes reviewing the priority of the relevant patches, assessing the risk, and finally implementing the patch via the Change Management process.	✓	✓	✓	
Incident Management	Process incidents reported by the customer according to the Service Level Agreement. The incident is recorded and prioritized in the incident tracking system (BCP). Monitor the status and progress of the incident throughout its whole lifecycle and give regular status updates to the customer.	✓	✓	✓	
	In the event of incidents, make reasonable effort to support end users and manage their incidents, to explore self-help tools to find already documented solutions, and to liaise with SAP support in the event of new problems to ensure timely processing of incidents affecting the resources in the customer account.	✓	✓		✓
	Confirm incident resolution in the incident tracking system (BCP).	✓	✓		✓
Service Requests	Process service requests reported by the customer according to the Service Level Agreement. The service request is recorded and prioritized in the service request tracking system (BCP). Monitor the status and progress of the service request throughout its whole lifecycle and give regular status updates to the customer.	✓	✓	✓	
	Confirm service request completion in the service request tracking system (BCP).	✓	✓		✓
Backup & Restore	Perform a backup of the database systems hosted in the customer account. A database log backup is done every 10 minutes and stored on the primary storage. Every 2 hours the logs are transferred from primary to secondary storage. Full data backup is done every day.	✓	✓	✓	
	Restore previously backed-up data to recover to a consistent state. Verify the completeness of the restored data based on log files created	✓	✓	✓	

Activity	Task	SAP Cloud Service		Responsibility	
		Platform	Cloud Integration	SAP	Customer
	during the recovery and smoke tests to verify the system's consistency.				
	Give regular status updates to the customer throughout the entire restore procedure.	✓	✓	✓	
	Collaborate with SAP to ensure timely processing of data restores if required.	✓	✓		✓
	Validate logical integrity and consistency of the restored data.	✓	✓		✓
User Access Management	Provide a proper user to SAP, to which the Account Administrator role for the customer account is to be granted by SAP as part of the provisioning process.		✓		✓
	Grant the Administrator role for the customer account to the user nominated by the customer.		✓	✓	
	Manage users, permissions, and security configurations within the customer account.	✓	✓		✓
System Monitoring	Ensure availability of the customer system according to the Service Level Agreements as agreed in the contractual agreement between SAP and the customer, by active monitoring, prompt issue detection, and incident prevention.	✓	✓	✓	
	Monitor the resource consumption (memory, CPU, storage) to detect issues in technical operations.	✓	✓	✓	
Malware Management	Ensure that the infrastructure and platform services are free of viruses, spam, spyware, and other malicious software. If malware is detected, an auto-notification is generated, which is assessed and resolved by the operator.	✓	✓	✓	
Application Management	Design, develop, deploy, configure, maintain, and operate the application within the customer account. This includes maintaining a staged environment for application delivery (if required), application resource management, and managing application availability and performance.	✓			✓
	Provide infrastructure, tools, and application programming interfaces for the lifecycle management and operations of the application in the customer account.	✓			✓

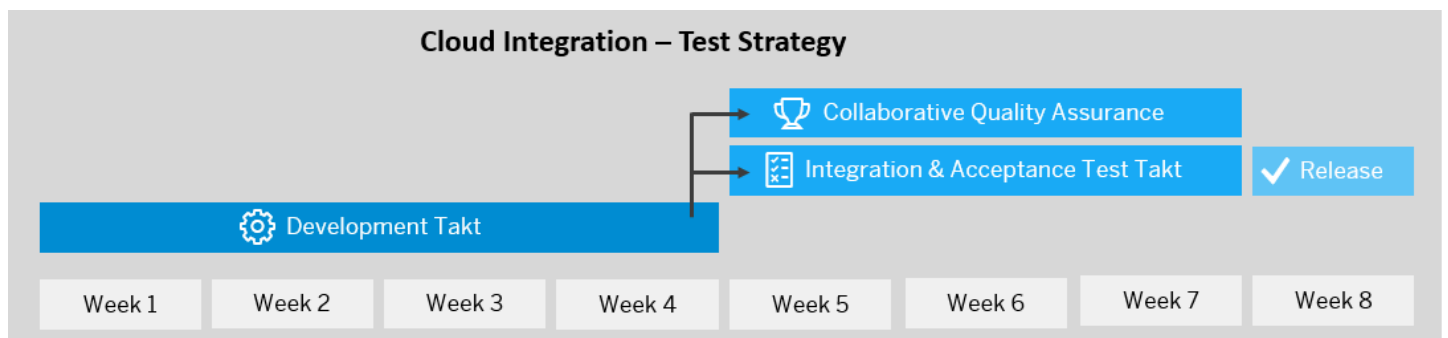
Activity	Task	SAP Cloud Service		Responsibility	
		Platform	Cloud Integration	SAP	Customer
	Regularly adopt the latest versions of the tools for lifecycle management and operations offered at the SAP Development Tools site .	✓			✓
Network Management	Manage the network isolation of the accounts provisioned to the customer.	✓	✓	✓	
	Operate the network infrastructure transparently for customers, ensuring elasticity, high availability, and security.	✓	✓	✓	
	Create and manage own Web domain for the application in the customer account to ensure data isolation.	✓			✓
Penetration Testing	Inform SAP about any penetration testing that shall be performed for the customer system and ask for their approval. Testing is not allowed on any systems or resources shared with other customers. The results, if any, from the test are to be treated strictly as the confidential information of SAP and the customer and are not to be shared with any person or entity without explicit written authorization from SAP. Customers are required to share the results with SAP and work together with SAP operations to mitigate or remedy any security issues.	✓	✓		✓
Decommissioning	Ensure the secure deletion of data and/or hardware disposal. This includes the disassembling of systems along with peripherals and their removal. Before dismantling and handover for further use or return to the vendor, the data is wiped securely from the system. i Note Decommissioning of tenants can be triggered by the customer for the tenants linked with global accounts.	✓	✓	✓	
	Perform a final export of the customer data from the service using the provided data export self services within the time stated in the Terms & Conditions document. The following services are provided: • Export of Message content stored through persist flow-step in an integration flow by means of an API • Export of Customer data in JMS queues		✓		✓

Activity	Task	SAP Cloud Service		Responsibility	
		Platform	Cloud Integration	SAP	Customer
	- Export of Integration Content - Export of Security Key Material only for public keys				

Quality Assurance

Quality Assurance is central to the SAP Cloud Integration development process. SAP invests significantly into holistic product testing, covering for both functional and non-functional qualities to deliver regression free and flawless new feature increments. The Test Strategy is designed and pivoted on the DevOps principle of Continuous Integration & Continuous Delivery (CI/CD) Test pipeline.

SAP Cloud Integration delivers increments after a four-week development cycle and a four-week testing cycle. Each of the two cycles is governed by strong assessment and test criteria (quality KPIs) which form the basis for acceptance or rejection of the increment.



SAP Cloud Integration – Test Strategy

To deliver quality, a release build version is produced which is assessed by all development teams involved. All of the automated tests from both from Development and Integration and Acceptance test teams are executed daily on a dedicated central landscape and make it into the central CI/CD Test pipeline. The following aspects are part of the test pipeline:

- SAP BTP environments: Neo, Cloud Foundry
- Test types: unit, component, system and scenario level
- Test scope: regression and new features covering for both functional and non-functional aspects (performance, software installation, updates).

Part of the functional tests is also: semantic versioning of adapters and flow-steps (those new features for integration flows which require configuration are delivered in new versions only; new component versions are used in new integration flow model creations only; existing integration flows remain unchanged and continue to run without interaction; integration flow compatibility, i.e. seamless migration from Neo to Cloud Foundry.)

i Note

See, the release notes for SAP Cloud Integration for functional increments: [SAP Cloud Integration](#); see also the patch release notes for SAP Cloud Integration [Patch Releases for Cloud Integration](#).

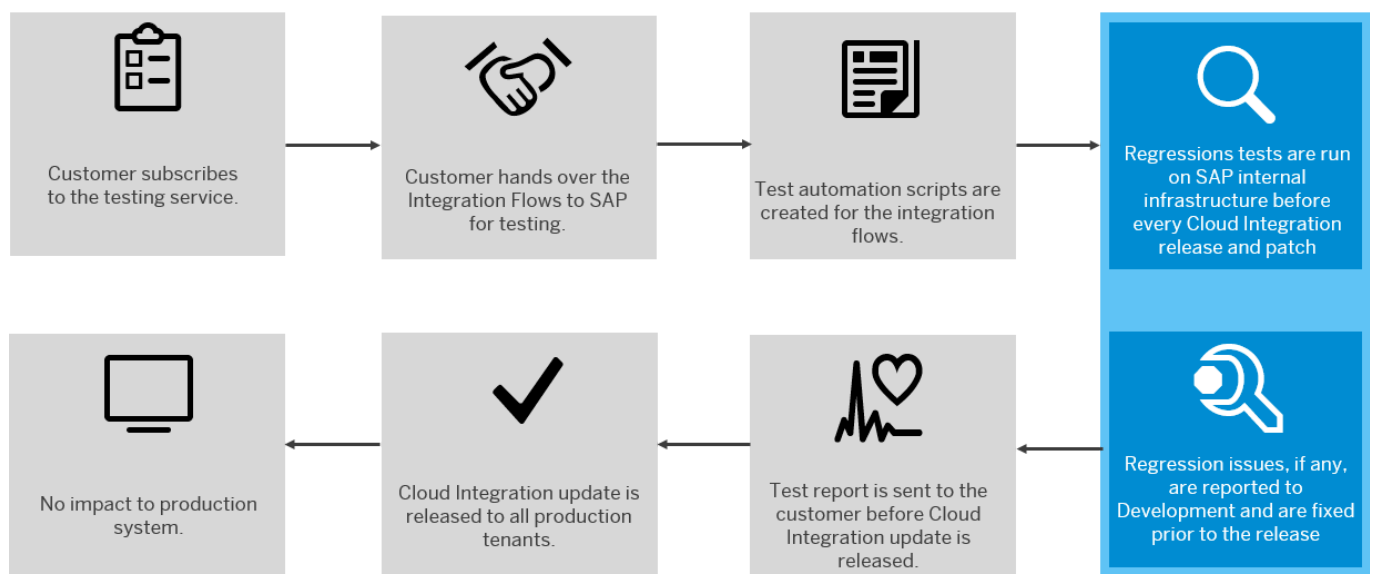
A successful Development Close results in release build version, which is assessed as “ready for productive use” by our first internal customer during the Integration & Acceptance Test takt (IAT takt). The scope during the IAT takt is to simulate and

validate real-time End-to-End (E2E) customer facing scenarios, along with active test engagement with our SAP Application teams and OEM Partners as part of our "Collaborative Quality Assurance".

The acceptance test team challenges the development close assessment by executing the Product Acceptance Tests on a dedicated and well-governed test landscape, covering the following quality aspects like:

- Automated regression tests (functional and performance) for integration packs, adapters, manual exploratory tests on new features, software update, cloud qualities and so on, that are executed by Cloud Integration experts from development organization.
- Manual or automated regression tests (functional and performance) of integration packs and adapters as part of "Collaborative Quality Assurance".
- Automated regression test suites as part of Customer Test Service (CTS):

SAP offers a customer-specific regression test service focused specifically on a customer's integration flows. The service helps businesses test non-standard integration content that will ensure business continuity. It includes test automation and regular execution of a customer's scenarios in the context of one acceptance takt, executed by SAP on SAP internal systems. The SAP Cloud Integration development teams add the test success of a customer's scenarios to the mandatory release criteria (without extra investment on the test landscape for the customer) and won't update any system until the new features have passed the tests. Customers receive a corresponding test report before the system update including resolutions for solving arising issues. During this process, a customer's data is safe and protected.



Customer Test Service

i Note

If you are interested in more details and/or a commercial offer for this optional and complementary service, contact SAP_CPI_Test_Automation@sap.com.

Related Information

[Blog: SAP Cloud Platform Integration - how we do software updates](#) 📄

SAP Cloud Integration

Core Components, 2026

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.47.* • Increment: 2608	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.83.* • Increment: 2608	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.56.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2608	Info only	General Availability	Changed
Edge Integration Cell	Other	Number Range Object Support for Edge Integration Cell	You can now manage number range objects on Edge Integration Cell runtime. You can deploy number range objects to one or more runtimes, copy them across Cloud Integration and Edge Integration Cell runtimes, and monitor the deployed objects per runtime node. See Managing Number Ranges	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Dynamic support for source encoding property in EDI convertors	Now the EDI to XML, XML to EDI converters, and EDI Splitter supports dynamic support for source encoding property. See Define EDI to XML Converter Define XML to EDI Converter Define EDI Splitter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Enabling Prepared Statement Support in JDBC Receiver Adapter	You can now enable prepared statement execution on the JDBC Receiver Adapter. When enabled, the adapter compiles the SQL query once and runs it repeatedly with bound parameters, improving throughput for high-volume database operations. See JDBC Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	RNIF Adapter	Use the RNIF (RosettaNet Implementation Framework) adapter to exchange RosettaNet business messages with trading partners. See: RNIF Adapter	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	W3C Trace Message Header	SAP Cloud Integration adopts the W3C Trace Context (transparent) standard as the universal trace identifier for cross-system tracing. See: Message Processing Log - Adapter Tracing	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Where-Used View for Security Material Available Now	You can now identify all the integration flows that refer to a specific security material. See Where Used	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Fixed Incorrect HTTP Status Code in OData V2 Sender	Fixed an issue where the OData V2 sender adapter returned 401 Unauthorized instead of 403 Forbidden when an authenticated user did not have the required role assigned. The adapter now returns the correct HTTP status code.	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Importing Multiple Value Mapping from Integration Directory Available Now	You can now import multiple value mappings from the Integration Directory and view in Cloud Integration. See Import Value Mapping from Integration Directory	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Deploying an OAuth2 Password Credentials Available Now	You can now deploy an OAuth2 Password Credentials artifact. See Deploying an OAuth2 Password Credentials SAP Cloud Integration customers, see Deploying an OAuth2 Password Credentials	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Deletable Element Available Now	You can now choose the element can be deleted at runtime while editing the data type. See Editing Data Type	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.46.* • Increment: 2607	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.82.* • Increment: 2607	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.55.*	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			- Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2607			
Cloud Integration	Cloud Foundry	Grouping of artifacts	In the Artifacts tab of your integration package, you can now group the artifacts by type. See: Editing Integration Package	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Enhancements in Monitor Message Processing	The Monitor Message Processing UI has been updated and new filters are available. See: Monitor Message Processing Using Additional Filter Criteria	Info only	General Availability	New
Cloud Integration	Cloud Foundry	New Payload Viewer for Data Stores and Message Queues	You can now inspect message payloads using the View Payload action directly from the Manage Data Stores and Manage Message Queues UIs. This action opens a dialogue that shows you the body, headers, properties and attachments from your payload file. You can also download the original file or copy its body content. See: Managing Data Stores Managing Messages Stored in Queues	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Keep SOAP Envelope option Available Now	You can now retain an existing SOAP envelope in the payload when sending messages for XI Adapters. See Configure the XI Sender Adapter Configure the XI Receiver Adapter SAP Cloud Integration customers, see Configure the XI Sender Adapter Configure the XI Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Mass Migration of Integration Configuration Objects Available Now	You can now migrate multiple integration scenarios from SAP Process Integration/SAP Process Orchestration to your SAP Integration Suite tenant. See Modernize	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Enhancements in SuccessFactors OData (V2 or V4) and REST receiver adapters Available Now	For SuccessFactors OData (V2 or V4) and REST receiver adapters, the X-SF-Correlation-Id header sent back as HTTP request header is now deprecated and replaced with X-Correlation-Id. See • SuccessFactors OData Adapter • Configure the SuccessFactors REST Receiver Adapter SAP Cloud Integration customers, see • Configure the SuccessFactors REST Receiver Adapter • SuccessFactors OData V4 Receiver Adapter • SuccessFactors OData V2 Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	AI-based Memory Profiling Available Now	You can now use AI-based integration flow memory profiling feature to enhance the existing integration flow simulation. See AI-based Memory Profiling	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Accurate Execution of Multiple Basic Timer Schedules Available Now	You can now configure an integration with more than one basic timer schedule to execute all configured schedules accurately. See Define a Timer Start Event	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Handling Message Mapping Resources Available Now	In migration tooling, you can now choose how to handle the dependent message mapping resources when you are importing message mapping as local resource. See • Standard Approach • Pipeline Approach	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Integration Content Migration Agent Available Now	You can now use the integration content migration agent to convert java mappings into groovy scripts while using migration tooling. See Integration Content Migration Agent	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.45.* • Increment: 2606	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.81.* • Increment: 2606	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.54.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2606	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Custom Status for Message Queues	You can now define and view a custom status for messages stored in message queues to monitor and diagnose message processing outcomes. See Managing Messages Stored in Queues. SAP Cloud Integration customers, see Managing Messages Stored in Queues.	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Fault Message Type Artifact Available Now	You can now create and configure Fault Message Type Artifacts. See • Creating Fault Message Type • Configuring Fault Message Type • Consuming Fault Message Type in Service Interface	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	New Integration flow design guidelines available now	New design guideline under ensure upgrade readiness available now See Use import statements supported by the script runtime Restrict usage to supported frameworks and libraries SAP Cloud Integration customers, see Use import statements supported by the script runtime Restrict usage to supported frameworks and libraries	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.44.* - Increment: 2605	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.80.* - Increment: 2605	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.53.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2605	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	PIPO XSD support in EDI Splitter now	For EDI Splitter version 2.10 and above, you can use the same XSDs in your tenant as in your SAP Process Integration and SAP Process Orchestration system for the VDA message standard. See Define EDI Splitter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Update in endpoint for content transport	- Supported endpoint for content transport in SAP Cloud Integration via SAP Cloud Transport Management Service is updated as part of SAP Cloud Deployment Service enhancements. - The legacy endpoint <code>https://deploy-</code>	Recommended	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			<code>service.cfapps.</code> <code><domain></code> is now deprecated. Action: Switch to the new <code>https://deploy-service.cf.<domain></code> endpoint before Q4 to ensure future compatibility. See Creating HTTP Destinations and Transport Route SAP Cloud Integration customers, see Creating HTTP Destinations and Transport Route			
Cloud Integration	Cloud Foundry	GitHub support in SAP Integration Suite Available Now	You can now integrate GitHub repositories and perform Git operations in your integration flows. See Integrating with GitHub	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Updated Rate Limit for Design Time Artifacts Public APIs	The tenant rate limit for all public Design Time Artifacts APIs has been changed from 20 requests per second to 15 requests per second. See Rate Limits on SAP Cloud Integration SAP Cloud Integration customers, see Rate Limits on SAP Cloud Integration	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Configuring Multiple Runtimes Available Now	You can now perform shared configurations across multiple runtimes for external parameters of an integration flow. See Configure Externalized Parameters of an Integration Flow	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Message Mapping Import with Function Library Support Available Now	You can now import Message Mappings along with their Function Libraries when importing a Message Mapping from within an Integration Flow using the References tab. See Manage References SAP Cloud Integration customers, see Manage References	Info only	General Availability	New
Cloud Integration	Cloud Foundry	HTTP Receiver Adapter Retry Configuration Enhancements Available Now	You can now access and manage enhanced retry configurations for the HTTP Receiver Adapter through the Retry tab. • Retry on Exception and its related options are now available in the Retry tab.	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			• Retry Failed Request field has been renamed to Retry on Error Response. • Throw Exception on Failure option and its dependent options have been moved to the Retry tab. See HTTP Receiver Adapter SAP Cloud Integration customers, see HTTP Receiver Adapter			
Cloud Integration	Cloud Foundry	OData V2 Adapter Retry Configuration Support Available Now	You can now configure retry behavior for the OData V2 adapter using the dedicated Retry tab. See Configure the OData V2 Receiver Adapter SAP Cloud Integration customers, see Configure the OData V2 Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Memory Exhaustion Chart	There's now a separate memory exhaustion chart on the Inspect Memory overview page. See: Inspect System Memory Usage	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.43.* • Increment: 2604	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.79.* • Increment: 2604	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.52.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2604	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	VDA document support in EDI converters	Now the EDI to XML, XML to EDI converters, EDI Splitter and Extractor supports VDA standard. See Define EDI to XML Converter Define XML to EDI Converter Define EDI Extractor Define EDI Splitter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	New Attribute on Aggregator Step	You can now choose when to start measuring the timeout period on the aggregator step. See Define Aggregator	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	OData API for JMS Queues and Message	New OData APIs are provided to support JMS queue monitoring and message operations such as delete, move and retry. This enhancement allows users to monitor and manage queues and messages. It also supports advanced filtering to improve message management efficiency. Both exclusive and non-exclusive queues are supported.	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Principal Propagation Support for SOAP (SAP RM and SOAP 1.x) Receiver Adapter in Cloud Foundry	You can now use principal propagation as the authentication method in the SOAP (SAP RM and SOAP 1.x) receiver adapters when connecting to on-premise systems with the on-premise proxy type in the Cloud Foundry environment. See Configure the SOAP (SAP RM) Receiver Adapter https://help.sap.com/docs/integration-suite/sap-integration-suite/configure-soap-soap-1-x-receiver-adapter?version=CLOUD	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Review Changes Before Deployment in Known Hosts Editor	When editing an existing Known Hosts (SSH) artifact, you can now review and compare your changes before deploying them. Enable the Review Changes toggle to view a side-by-side comparison of the currently deployed configuration and the modified configuration. Differences such as changed host keys, updated algorithms, newly added entries, and removed entries are visually highlighted.	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			See Maintaining SSH Known Hosts for SFTP Connectivity			
Cloud Integration	Cloud Foundry	OData V4 Receiver CSRF Token Handling Improvements	You can now correctly process CSRF token handling and retry scenarios in the OData V4 Receiver adapter when HTTP session reuse is disabled for the integration flow.	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Usage of the Duplicate Subtree	For OpenAPI Spec JSON, you can now use the Duplicate Subtree action for a node that is of the type array. See Creating Message Mapping as a Flow Step SAP Cloud Integration customers, see Creating Message Mapping as a Flow Step	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Ensuring valid syntax available now	You can now detect Java Script compilation and syntax errors through new Upgrade Readiness design guideline checks. See Ensure Valid Syntax SAP Cloud Integration customers, see Ensure Valid Syntax	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Default Traceparent Header Support for OData Receiver Adapter Available Now	You can now use the OData V2 and V4 Receiver adapters with the <code>traceparent</code> header included by default in Request and Response headers. See • Configure the OData V2 Receiver Adapter • Configure the OData V4 Receiver Adapter SAP Cloud Integration customers, see • Configure the OData V2 Receiver Adapter • Configure the OData V4 Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Default Traceparent Header Support for HTTP Receiver Adapter Available Now	You can now use the HTTP Receiver adapter with the <code>traceparent</code> header included by default in Request headers. See HTTP Receiver Adapter	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			SAP Cloud Integration customers, see HTTP Receiver Adapter			
Cloud Integration	Cloud Foundry Neo	Upgrading Java Script available now	You can now upgrade javascript to version 2.0. See Upgrading Script SAP Cloud Integration customers, see Upgrading Script	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Enhancements on Inspect Feature	- You can now use the Previous and Next buttons on the time filter to switch the time interval. - When you navigate to Inspect Top Consumption, the selected time period is automatically highlighted in the heatmap. - You can see the count of memory exhaustion events on Inspect Memory Usage. See: Inspect	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.42.* - Increment: 2603	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.78.* - Increment: 2603	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.51.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2603	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Empty String Handling in JDBC Receiver Adapter	Now you can select how empty string values are interpreted in the payload when generating SQL statements. See JDBC Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	New External Logging Destinations	JSON and OTLP are now available as external logging destinations. See: External Logging	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Enhanced connection handling for OData	Improved management of idle and closed connections in high-volume and long-running scenarios ensures more	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
		V2 Receiver is Available Now	reliable outbound OData communication. Upstream idle connections are closed after a fixed timeout.			
Cloud Integration	Cloud Foundry	Restoring Subaccount is Available Now	You can now restore your subaccount, when deleted from the BTP cockpit. See Restore Subaccount	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Non-XML Payloads Supported in XI Adapter	The XI adapter now supports non-XML payloads such as JSON and plain text in XI messages. This enhancement applies to both inbound and outbound messages on the XI sender and receiver adapters, and it allows any message format to be exchanged via the XI protocol. For more information on the XI Adapter, see XI Adapter .	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	XI to XI Application Fault Handling	You can now see a new option, XI to XI Application Fault Handling , in the XI sender adapter when using Quality of Service Best Effort . When you enable this option, the XI sender will return the receiver's error response with an HTTP status code of 200 whenever the XI receiver adapter encounters an application error. This feature is available for the XI sender adapter version 1.20 onwards. For more information, see Configure the XI Sender Adapter .	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Best practices for Timer and Schedule based Integration Flows Available Now	You can now use the best practices to design and operate schedule-based integration flows. See Timer and Schedule based Integration Flows SAP Cloud Integration customers, see Timer and Schedule based Integration Flows	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Script API for version 2 Available Now.	You can now use Script API for version 2. See SDK API SAP Cloud Integration customers, see SDK API	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Comprehensive deployment status view available now from design time	You can now view the Runtime Profiles where an integration flow is deployed in the Deployment Status view. See Deployment Status View SAP Cloud Integration customers, see Deployment Status View	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.41.* • Increment: 2602	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.77.* • Increment: 2602	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.50.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2602	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.40.* • Increment: 2601	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.76.* • Increment: 2601	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.49.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2601	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Enhancements to the custom Time filter in Monitor Message Processing	You can now select Last Updated to display messages based on when their status last changed within a specific time period. See Monitor Message Processing	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Message Counting during High Load Scenarios	During high load scenarios in Monitor Message Processing , the system displays three dots instead of the message count while counting is in progress. This way, you can see all	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			retrieved messages even if counting fails. See Monitor Message Processing			
Cloud Integration	Cloud Foundry	Recommendations Button in Inspect	When resource consumption becomes critical, a recommendations button becomes available. See Inspect	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Inspect Integration Flows By System Memory Usage	You can now see the integration flows with the largest message sizes. See Inspect Integration Flows By System Memory Usage	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Ensure Upgrade readiness guideline now applicable to java script step	The Upgrade readiness design guidelines now apply to Script step including Java scripts and Groovy scripts. See Scripts SAP Cloud Integration customers, see Scripts	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	JDBC Data Source Connection Pool Management	Enhanced JDBC data source configuration with connection pooling capabilities to optimize database connectivity performance and resource utilization. See Managing JDBC Data Sources SAP Cloud Integration customers, see Managing JDBC Data Sources	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.39.* • Increment: 2513	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.75.* • Increment: 2513	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.48.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2513	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Uploading groovy script version 1.x	You can now upload groovy script 1.x and 2.0 in the script collection artifact and integration flow.	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
		and 2.x available now	See • Upload a Script Collection • Adding Local Resources SAP Cloud Integration customers, see • Upload a Script Collection • Adding Local Resources			
Cloud Integration	Cloud Foundry	Design guidelines applicable to the respective script collection available now	You can now view and execute the design guidelines applicable to a particular script collection. See Design Guidelines View for Script Collection SAP Cloud Integration customers, see Design Guidelines View for Script Collection	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Support of PI/PO Message Mapping with Adapter Type Parameters	In Migration tooling and standalone message mapping import, you can now import PI/PO message mapping objects with Adapter Type parameter category of type Import. See Creating Message Mapping as an Artifact SAP Cloud Integration customers, see Creating Message Mapping as an Artifact	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.38.* • Increment: 2512	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.74.* • Increment: 2512	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.47.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2512	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry Neo	Support for 202 response code for Create operation in ODATA V4 Receiver adapter.	The OData V4 receiver adapter now supports HTTP 202 responses with an empty body for Create operations.” See Configure the OData V4 Receiver Adapter SAP Cloud Integration customers, see Configure the OData V4 Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Configure Reuse Connection time	You can now configure the maximum live connection time for OData V2 Receiver Adapter. See Configure the OData V2 Receiver Adapter SAP Cloud Integration customers, see Configure the OData V2 Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Known Hosts Management Enhancements	Maintaining SSH Known Hosts is now easier with a dialog-based experience. The Create Known Hosts (SSH) dialog lets you create, edit, delete, and import known host entries in one place, streamlining the SFTP setup process. See Maintaining the Known Hosts File for SFTP Connectivity SAP Cloud Integration customers, see Maintaining the Known Hosts File for SFTP Connectivity	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Updating and Validating of OData API configuration parameters available now	You can now Update and Validate OData API configuration parameters to explicitly trigger validation after completing all updates. See Update and Validate Configuration Parameters SAP Cloud Integration customers, see Update and Validate Configuration Parameters	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Generating a test XML file from the source message structure available now	You can now generate a test XML file from the source message structure. See Test Message Mapping	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			SAP Cloud Integration customers, see Test Message Mapping			
Cloud Integration	Cloud Foundry	Importing unsupported function libraries available now	Migration tooling now supports import of unsupported Function Libraries. See: • Standard Approach • Pipeline Approach	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Enhanced XML-to-EDI Converter Schema Flexibility	You can now seamlessly convert XML payloads with Process Integration /Process Orchestration structure using Integration Advisor (IA) XSDs in the XML-to-EDI converter. This enhancement allows you to maintain existing partner XML formats while leveraging the latest Integration Advisor schema definitions for both EDIFACT and X12 conversions. See Define XML to EDI Converter SAP Cloud Integration customers, see Define XML to EDI Converter	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Changes in AI Review Report for Groovy Scripts	When you optimize Groovy scripts with AI, you can now also download the original script. See: Optimize Groovy Scripts	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	New design guidelines for upgrade readiness is now available	The new guideline under Ensure Upgrade Readiness provides improved clarity on resource handling within Groovy scripts. This supports smoother behavior during runtime upgrades and enhances overall script reliability. See Access resources appropriately. SAP Cloud Integration customers, see Access resources appropriately.	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Dynamic selection for operation details is now available	You can now select Dynamic option while configuring OData V4 Adapter. See Configure the OData V4 Receiver Adapter SAP Cloud Integration customers, see Configure the OData V4 Receiver Adapter	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry Neo	Query options for all operations now available	You can now add query options for all operations while configuring OData V4 Adapter. See Configure the OData V4 Receiver Adapter SAP Cloud Integration customers, see Configure the OData V4 Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Additional Attributes Available for Filtering Messages Stored in Queues	You can now choose to display additional message attributes in Manage Message Queues through the Messages table settings. These attributes can be used as filter criteria, enabling you to narrow down messages more efficiently. The following attributes are now supported for filtering: Sender, Receiver, Application Message Type, Application Message ID, and Correlation ID. For a description of all message attributes displayed in the message details, see Managing Messages Stored in Queues.	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Deprecated Grant Type client_x509	The <code>client_x509</code> grant type will reach its end of life in January 2026, and it will be replaced by <code>client_credentials</code>. If you operate scenarios using inbound authentication with client certificates and service instances with <code>client_x509</code> grant type, please update them to the grant type <code>client_credentials</code>. For information on how to update affected service instances, check Update Service Instance that Contains Deprecated client_x509 Grant Type.	Recommended	Deprecated	Changed

Patch Releases for Cloud Integration

This topic provides information on patch releases for hotfixes, bugfixes, and code enhancements.

This topic covers patches for SAP Cloud Integration and Integration Advisor.

The following patch release information covers the most recent changes made to the latest version of the software.

→ Tip

Each software patch always contains the current bug fix as well as the bug fixes provided to the previous patches.

The following example illustrates this rule:

Let's assume that SAP has recently provided the following patches:

Software Version	Description
6.20.12	Bugfix C
6.20.11	Bugfix B
6.20.10	Bugfix A

Let's assume your previous software update was applying patch 6.20.10.

If you now update to software version 6.20.12, this patch provides you with bugfix B **and** bugfix C.

Patches for different components are associated with different major software version numbers. For example, patches for Cloud Integration in the Cloud Foundry environment have major software version 6 (for example, 6 . 53 . 23). For more information, see [Understanding Software Version Numbers](#).

September 2026

Software Increment: 2607

Technical Component	Software Version	Description
Cloud Integration	8.46.27	This patch includes the following updates: - Prevents unnecessary DNS resolution of SAP Cloud Connector virtual hostnames in the JDBC adapter. - Enhances the Configuration API UPDATE and GET operations as a stop-gap solution. When multiple runtime profiles exist, updating a value will now reflect across all runtime profiles.
Cloud Integration	7.55.19	This patch prevents unnecessary DNS resolution of SAP Cloud Connector virtual hostnames in the JDBC adapter.
Cloud Integration	6.82.110	This patch includes the following updates: - Optimizes the business logic for delete operations, effectively reducing database CPU load. - Resolves a database full table scan issue and optimizes query performance during data extraction and evaluation. - Prevents unnecessary DNS resolution of SAP Cloud Connector virtual hostnames in the JDBC adapter. - Enhances the Configuration API UPDATE and GET operations as a stop-gap solution. When multiple runtime profiles exist, updating a value will now reflect across all runtime profiles.

Technical Component	Software Version	Description
Cloud Integration	8.46.25	This patch includes the following updates: - Fixed trace enhancement that missed retaining the incoming trace parent in the context. - Corrects custom separator handling, avoids corrupted data in agreements and agreement templates, and guarantees proper GS Controller number usage within agreements.
Cloud Integration	7.55.18	This patch includes the following updates: - Fixed missing imports that prevented the artifacts and filter search in the Message Processing Logs page. - Certificate chains using the connectivity test can be viewed and downloaded. - Fixed trace enhancement that missed retaining the incoming trace parent in the context.
Cloud Integration	6.82.105	This patch includes the following updates: - User experience improvement on mass migration jobs. - Fixed missing imports that prevented the artifacts and filter search in the Message Processing Logs page. - Numbering of custom rules in Partner Directory now correctly start with 1. - Certificate chains using the connectivity test can be viewed and downloaded. - XPath Injection in Alert Rule Merge Processor fix (Single Quote Issue). - Deployment issue for Integration Adapter where adapters get stuck in the starting state and never go to the STARTED state is fixed. - Fix for the concurrent requests causing MaaS push failures. - Resolved the issue with graph messages not being metered for tenants that have Graph activated but do not have Cloud Integration activated. - Fixed trace enhancement that missed retaining the incoming trace parent in the context. - Corrects custom separator handling, avoids corrupted data in agreements and agreement templates, and guarantees proper GS Controller number usage within agreements.
Cloud Integration	8.46.24	This patch resolved an issue due to missing Spring XML bean definitions in the IDoc sender adapter Remove and Substitute options for Invalid XML Character Handling.
Cloud Integration	6.82.96	This patch includes the following updates: - Unified Customer Landscape integration has been restored by migrating the Unified Customer Landscape registration to the MAP v1 REST API (mTLS). This resolves a broken integration caused by the previously used GraphQL API being deprecated, and ensures the ISUITE_SUBSCRIPTION_LIST is correctly populated during provisioning. - Retry mechanism was added to resolve Message Processing Log clean up jobs rare limit issue.

Technical Component	Software Version	Description
		Resolved an issue due to missing Spring XML bean definitions in the IDoc sender adapter Remove and Substitute options for Invalid XML Character Handling.

August 2026

Software Increment: 2607

Technical Component	Software Version	Description
Cloud Integration	8.46.22	This patch fixes the issue where a custom query would yield an invalid URL when the OData V4 adapter invokes a call to the service document to fetch the CSRF token.
Cloud Integration	7.55.13	This patch fixes the issue where a custom query would yield an invalid URL when the OData V4 adapter invokes a call to the service document to fetch the CSRF token.
Cloud Integration	6.82.91	This patch includes the following updates: - Subsidiary Communication channel is now able to add an Agreement Template by taking the accurate ID value. - It fixes an inconsistent switch of artifact status on the Web UI for the Integration Cell runtime. - Recent entities API is optimized to improve performance, showing the correct deploy status for MCP server and few API related artifacts in APIM GA tenant. - Fixed the issue where a custom query would yield an invalid URL when the OData V4 adapter invokes a call to the service document to fetch the CSRF token.

August 2026

Software Increment: 2606

Technical Component	Software Version	Description
Cloud Integration	8.45.18	This patch fixes the following: - Fixed an issue where the B2B archiving POST endpoint required authentication with specific scopes, which could prevent task execution. - Fixed an issue preventing the automatic removal of content in EIC due to an incorrect status reported by the runtime.
Cloud Integration	8.45.17	This patch includes the following updates: - Resolved an issue where the AS2 filename configured in AS2 Receiver CCT could not be pushed to Process Direct. - Resolved a Process Direct exception "Failed to determine address from expression" triggered after a TPM upgrade. - Resolved a TPM application crash. - Resolved an out-of-memory issue in the migration tool summary report.

Technical Component	Software Version	Description
		- Resolved an issue where updating configurations via the Configuration OData API reset values to defaults for edge locations. - Resolved an issue preventing users from starting and stopping message queues. - Resolved an issue preventing users from assigning and unassigning PGP keyrings when an inactive runtime is assigned. Inactive runtimes can now be unassigned.
Cloud Integration	7.54.12	This patch includes the following updates: - Existing Graph customers can now onboard API Management onto Integration Cell. The activation button for Integration Cell is now enabled in the user interface, allowing customers to click and activate this capability. - Enhanced state transition handling to reduce delays in heavily loaded systems. - Resolved an issue preventing users from uploading new public and secret keyrings in the PGP view. - Enhanced logging to provide insights into how Kafka headers are used. This change does not affect runtime behavior. - Resolved an issue where a failure in SAP-Passport header parsing logic prevented Message Processing Log (MPL) creation. - Resolved an issue preventing users from applying table settings in the keystore view
Cloud Integration	6.81.49	This patch includes the following updates: - Resolved an issue where the <code>it-op-multicloud-rest</code> component was not released properly during the IAT phase. - Resolved an issue where the message body and headers were missing in the exception subprocess when an XI receiver adapter failed during the send process before any HTTP response was received (for example, due to an <code>UnknownHostException</code>). Failures where a response was received (such as HTTP 500) were not affected and continued to work correctly. - A function library containing a UDF (User-Defined Function) with dynamic configurations was failing to deploy. The severity has been reduced from error to warning, allowing the function library to deploy successfully. - Addressed inconsistencies related to managing tenant resources. - Fixed an issue that prevented the use of proxy host/port settings for the XI adapter on Edge Integration Cell. Previously, the generator did not write the corresponding parameters into <code>beans.xml</code>. The XI adapter generator now includes these parameters, so proxy host/port settings work as expected. - Enhanced state transition handling to reduce delays in heavily loaded systems. - Existing Graph customers can now onboard API Management onto Integration Cell. The activation button for Integration Cell is now enabled in the user interface, allowing customers to click and activate this capability.
Cloud Integration	8.45.11	This patch includes the following updates: Resolved an issue where the message body and headers were missing in the exception subprocess when an XI receiver adapter failed during the send process before any HTTP

Technical Component	Software Version	Description
		response was received (for example, due to an <code>UnknownHostException</code>). Failures where a response was received (such as HTTP 500) were not affected and continued to work correctly. Enhanced state transition handling to reduce delays in heavily loaded systems.

Value
Resolved an out-of-memory issue in the migration tool summary report.
Resolved an issue where updating configurations via the Configuration OData API reset values to defaults for edge locations.

July 2026

Software Increment: 2606

Technical Component	Software Version	Description
Cloud Integration	7.54.8	This patch includes the following updates: - Fixed an issue where the it-op-multicloud-rest component was not released correctly during the IAT phase, causing the backend to not recognize MCP Server and Connector instances when loading the Message Status Overview page. - Fixed an issue where the message body and headers were missing in the exception subprocess when an XI receiver adapter failed during the send process before receiving an HTTP response (for example, due to an <code>UnknownHostException</code>). The original message content is now preserved and available in the exception subprocess.

July 2026

Software Increment: 2605

Technical Component	Software Version	Description
Cloud Integration	6.80.91	This patch fixes a persistence error during data extraction.

Technical Component	Software Version	Description
Cloud Integration	8.44.24	This patch includes the following updates: - Fixes failures with generating integrations with AI by updating the UCL Discovery host URL and adding a Null Pointer Check. - Resolves an issue causing <code>SAP_AS2_SND_*</code> parameters to go missing in the PD environment after deploying a second bind-mode agreement.
Cloud Integration	6.80.90	This patch includes the following updates: - Upgrades the HANA driver version to enable use of the latest HANA features. - Fixes agreement activation failures caused by undefined signature encoding. - Corrects bind-mode agreement activation being incorrectly blocked by incongruent ruleset IDs. - Restores the ability to update or reactivate CCT-used active security through the TPM backend. - Adds the <code>javax.sql</code> and <code>javax.sql.rowset</code> packages to the message mapping artifact manifest to enable JDBC API use cases. - Fixes failures with generating integrations with AI by updating the UCL Discovery host URL and adding a Null Pointer Check. - Resolves an issue causing <code>SAP_AS2_SND_*</code> parameters to go missing in the PD environment after deploying a second bind-mode agreement.
Cloud Integration	7.53.23	This patch includes the following updates: - Fixes a security issue in CXF and Knative HTTP endpoints by adding inbound header filtering to CXF filter strategy. - Upgrades the HANA driver version to enable use of the latest HANA features.

Technical Component	Software Version	Description
Cloud Integration	8.44.23	This patch includes the following updates: - Disables the CSV-to-XML schema cache by default to prevent collisions between same-named schemas and eliminate conversion errors. - Introduces enhanced security hardening for OData sender endpoints. - Fixes RNIF adapter conflicts with the CiG custom adapter by renaming the standard adapter and removing redundant bundling to prevent runtime issues. - Fixes a security issue in CXF and Knative HTTP endpoints by adding inbound header filtering to CXF filter strategy.
Cloud Integration	6.80.84	This patch includes the following updates: - Fixes RNIF adapter conflicts with the CiG custom adapter by renaming the standard adapter and removing redundant bundling to prevent runtime issues. - Enables SPC communication for the BIC Trial plan. - Introduces DSI 2.0, with dsi-package instance metering now executed for the standard, premium, enhanced, and enterprise agreement editions, supported by multiple connectors per instance and metering configuration updates. - Fixes a security issue in CXF and Knative HTTP endpoints by adding inbound header filtering to CXF filter strategy.
Cloud Integration	7.53.21	This patch includes the following updates: - Adds the missing User ID parameter and fixes download errors for the Secret and Secret, Public entry types on the PGP Entry page. - Resolves filtering issues on the PGP Management page.

Technical Component	Software Version	Description
		- Disables the CSV-to-XML schema cache by default to prevent collisions between same-named schemas and eliminate conversion errors. - Introduces enhanced security hardening for OData sender endpoints.
Cloud Integration	6.80.80	This patch includes the following updates: - Fixes the issue where Aggregator V1.1 remains active indefinitely. A race condition caused timeout-based completion to fail because of optimistic locking. - Fixes agreement activation failures caused by custom mapping and adapter type/direction incompatibilities. - Disables the CSV-to-XML schema cache by default to prevent collisions between same-named schemas and eliminate conversion errors. - Introduces enhanced security hardening for OData sender endpoints. - Restores missing OAuth2 configuration settings for the HANA database administration service to ensure reliable service-to-service authentication in multi-tenant environments. - Fixes reference-mode agreement activation and export defects by correcting identical sender/receiver IDs and ensures successful import of trading-partner parameters from templates.

June 2026

Software Increment: 2604

Technical Component	Software Version	Description

Technical Component	Software Version	Description
Cloud Integration	6.79.80	This patch fixes the issue where Aggregator V1.1 remains active indefinitely. A race condition caused timeout-based completion to fail because of optimistic locking.
Cloud Integration	8.43.23	This patch fixes the issue where Aggregator V1.1 remains active indefinitely. A race condition caused timeout-based completion to fail because of optimistic locking.
Cloud Integration	8.43.22	This patch includes the following updates: - Updated the Neethi library to version 3.2.2 and Apache CXF to 3.4.8-sap-10 to resolve issues caused by circular references during WS-Policy normalization. - Implemented JMX-based monitoring for the RFC Receiver adapter to automatically detect long-running integration flows and raise Dynatrace notifications. - Fixed a backward compatibility issue that prevented integration flows from opening or deploying after a system rollback.
Cloud Integration	6.79.79	This patch includes the following updates: - Updated the Neethi library to version 3.2.2 and Apache CXF to 3.4.8-sap-10 to resolve issues caused by circular references during WS-Policy normalization. - Implemented JMX-based monitoring for the RFC Receiver adapter to automatically detect long-running integration flows and raise Dynatrace notifications. - Fixed a backward compatibility issue that prevented integration flows from opening or deploying after a system rollback. - Improved the MPL cleanup job by optimizing a long-running query to handle high data volumes more efficiently. - Fixed an issue that prevented trading partner parameters from being saved in agreement templates and caused errors during agreement export.
Cloud Integration	8.43.20	This patch includes the following updates: - This fix prevents the manifest name in the outgoing XI message from being modified when you have an attachment name with a + character. It avoids replacing + with a space ' '. - Removed a database migration script that caused datasource initialization failures when you use agreement versioning. - Optimized the data source housekeeping threads to avoid out-of-memory errors in the application. - Removed incorrectly stored PIDs inside the agreement that caused activation failures.
Cloud Integration	6.79.73	This patch includes the following updates: - This fix prevents the manifest name in the outgoing XI message from being modified when you have an attachment name with a + character. It avoids replacing + with a space ' '. - A patch release for <code>it-b2b-calm</code> and <code>it-b2b-archiving</code> was performed without including the corresponding assembly. - Removed a database migration script that caused datasource initialization failures when you use agreement versioning.

Technical Component	Software Version	Description
		- Optimized the data source housekeeping threads to avoid out-of-memory errors in the application. - Removed incorrectly stored PIDs inside the agreement that caused activation failures.
Cloud Integration	6.79.69	This patch includes the following updates: - Fixed a race condition issue that occurred in the <i>it-rc-web</i> application. - Resolved an issue with time zone parsing in extracted data. - Adjusted API rate limits to accommodate higher traffic loads, reducing the occurrence of throttling errors (HTTP 429) during peak usage. - Enhanced the data cleanup job to handle larger volumes more efficiently by supporting configurable time ranges, improving overall system reliability. - Significantly improved loading performance when navigating to the Agreements page, eliminating long wait times especially for customers with large agreement portfolios. - Fixed an XML parsing issue. - Resolved read time-out errors that occurred during Kafka topic setup as part of the deployment process. - Removed the lower bound restriction for the JDBC Netty Worker Thread Count setting. It can now be configured below the default value of 256, providing greater flexibility for resource-constrained environments.
Cloud Integration	8.43.19	This patch includes the following updates: - Fixed a race condition issue that occurred in the <i>it-rc-web</i> application. - Significantly improved loading performance when navigating to the Agreements page, eliminating long wait times especially for customers with large agreement portfolios. - Removed the lower bound restriction for the JDBC Netty Worker Thread Count setting. It can now be configured below the default value of 256, providing greater flexibility for resource-constrained environments.
Cloud Integration	6.79.60	This patch includes the following updates: - Fixed an issue where rollback did not update the correct version in the UI. - Security-relevant fixes have been back-ported to the Camel 3.14.7 branch, along with a custom Aggregator fix. - The B2B Document cast error has been resolved through a new implementation and native query. - Added the missing dependency <code>liquibase-extension-lockservice</code> to the class path. - Resolved a security compliance vulnerability.
Cloud Integration	7.52.11	This patch includes the following updates: Introduction of a certificate validation method to support the new <code>sap_cloudintegrationcertificate</code> issued by the SAP Cloud Platform Client CA.

Technical Component	Software Version	Description
		Security-relevant fixes have been back-ported to the Camel 3.14.7 branch, along with a custom Aggregator fix.
Cloud Integration	8.43.17	This patch includes the following updates: - Security-relevant fixes have been back-ported to the Camel 3.14.7 branch, along with a custom Aggregator fix. - Resolved a security compliance vulnerability.

May 2026

Software Increment: 2603

Technical Component	Software Version	Description
Cloud Integration	6.78.63	This patch includes the following updates: - Fixed issues in Service Interface artifact creation and proxy generation when using local XSD references. The patch also resolves agreement import failures by ensuring imported data is properly rolled back during failed import operations, preventing duplicate resource entries during re-import. - Fixed an issue where OFTP Sender Channel matching did not work correctly for different VFNs.
Cloud Integration	8.42.14	This patch includes the following updates: Fixed an issue where OFTP Sender Channel matching did not work correctly for different VFNs.
Cloud Integration	6.78.61	This patch includes the following updates: - Enhanced certificate handling by migrating Ruby PEM certificate processing to use the native Ruby OpenSSL library. This update improves performance, removes external CLI dependencies, and supports FIPS 140-2 compliance. - Improved readiness feature handling to safeguard against 404 errors until the configured timeout period is reached. The update also improves worker startup reliability during software updates. - Applied a fix for aggregator processing in Camel. - Fixed an issue with parsing deeply nested XML data.
Cloud Integration	7.51.12	This patch includes the following update: Applied a fix for aggregator processing in Camel.
Cloud Integration	6.78.56	This patch includes the following updates: - Removed the commercial logo from community content search results. - Filtered out Groovy Script Version 2 artifacts from Message Mappings, as this version is currently not supported in Message Mapping scenarios.

Technical Component	Software Version	Description
		- Updated Spring Boot and Spring Security patch versions to address identified security vulnerabilities. - Added explicit handling for large CLOB data when using Oracle drivers. - Fixed an issue where TPM could fail to load due to stale cache handling during background cleanup operations. This enhancement improves datasource cache cleanup stability and prevents recurring TPM loading issues. - Fixed a regression issue where custom mappings stopped functioning after editing and reactivating agreements. Required Partner Directory parameters are now correctly updated during reactivation. - Resolved a Message Mapping validation issue introduced after migrating backend validation services to Java 25. The fix addresses schema parsing restrictions affecting deeply nested XML structures.
Cloud Integration	7.51.11	This patch includes the following updates: - Filtered out Groovy Script Version 2 artifacts from Message Mappings, as this version is currently not supported in Message Mapping scenarios. - Added explicit handling for large CLOB data when using Oracle drivers.
Cloud Integration	8.42.11	This patch includes the following updates: - Fixed a regression issue where custom mappings stopped functioning after editing and reactivating agreements. Required Partner Directory parameters are now correctly updated during reactivation. - Resolved a Message Mapping validation issue introduced after migrating backend validation services to Java 25. The fix addresses schema parsing restrictions affecting deeply nested XML structures.
Cloud Integration	6.78.49	This patch includes the following updates: - Enables you to add user-centric comments during the PUT call when updating an integration flow. - Implemented code enhancements in the design-service to address out-of-memory issues caused by delays in report summary generation. - Improved deployment stability in the IT-CO application by optimizing the handling of large data volumes during deployment-related operations, resulting in more efficient execution under load. - Fixed an issue where TPM could fail to load after updating to increment 2603. - Updated Apache Commons subsystem library versions to the latest supported release. - Fixed a subsystem deployment order issue that could cause integration flows to intermittently enter Starting or Error states before returning to Started status. - Corrected keystore navigation and update handling to improve processing of section-based names and ensure reliable validation of popup options.
Cloud Integration	7.51.9	This patch includes the following updates: Implemented code enhancements in the design-service to address out-of-memory issues caused by delays in report summary generation.

Technical Component	Software Version	Description
		- Updated Apache Commons subsystem library versions to the latest supported release. - Corrected keystore navigation and update handling to improve processing of section-based names and ensure reliable validation of popup options.
Cloud Integration	8.42.10	This patch includes the following updates: Corrected keystore navigation and update handling to improve processing of section-based names and ensure reliable validation of popup options.

April 2026

Software Increment: 2602

Technical Component	Software Version	Description
Cloud Integration	6.77.47	This patch includes the following items: - Fixes intermittent failures in Script Optimization requests caused by false positives from Azure prompt filtering by improving prompt handling. - Ensures compatibility by filtering out unsupported version 2 Groovy scripts in Message Mappings. - Removes unnecessary logging of database details to improve logging efficiency. - Implements a security fix by removing unnecessary logging of database details.
Cloud Integration	7.50.10	This patch ensures compatibility by filtering out unsupported version 2 Groovy scripts in Message Mappings.
Cloud Integration	6.77.42	This patch includes the following items: - Upgrades Netty from version 4.2.6.Final to 4.2.11.Final to address security vulnerabilities, including protection against potential DDoS and HTTP request smuggling attacks. - Fixes security vulnerabilities by upgrading Spring Boot and Spring Security to newer versions, ensuring improved application security.

Technical Component	Software Version	Description
		- Fixes an issue where message processing failed when trace log level was enabled for the mail receiver using AES-GCM encryption. - Improves performance by reducing excessive database calls during each flow deployment, resulting in better overall system performance. - Fixes a data inconsistency issue in NS2 where deleted bindings were still visible in the UI, preventing users from performing further delete operations.
Cloud Integration	8.41.16	This patch fixes an issue where message processing failed when trace log level was enabled for the mail receiver using AES-GCM encryption.
Cloud Integration	6.77.38	This patch includes the following items: - Removes unnecessary explicit transactions (BEGIN/COMMIT) for read-only (SELECT) queries in the StatementExecutor, preventing database connection retention and avoiding connection pool exhaustion under load. - Prevents out-of-memory conditions that caused the IT-TPM application to crash intermittently, improving overall application stability. - Adds additional logging and retry mechanisms in the JDBC Adapter to address intermittent service reference null issues and improve stability. - Fixes an issue where re-adding alternative partner agencies after recreating a Partner Directory (PID) entry could result in duplicate entry errors. All agencies can now be added successfully without errors in EIC.
Cloud Integration	7.50.7	This patch adds additional logging and retry mechanisms in the JDBC Adapter to address intermittent service reference null issues and improve stability.
Cloud Integration	8.41.15	This patch adds additional logging and retry mechanisms in the JDBC Adapter to

Technical Component	Software Version	Description
		address intermittent service reference null issues and improve stability.
Cloud Integration	6.77.32	This patch includes the following items: • Configures the thread pool size of the RFM poller to enhance resiliency by increasing the pool size from 5 to 10, reducing the risk of thread unavailability for RFM poller tasks. • Fixes an issue where CMS encryption and signing were always disabled on the OFTP Receiver channel. • Resolves a metering processing failure caused by missing mandatory MPL IDs after enabling the new metering database structure, especially when the log level was set to None. This ensures proper handling of records in the metering database. • Removes an additional logger that was unnecessarily logging database details, improving logging efficiency. • Fixes failures in Script Optimization requests observed in Canary due to rejection by the Azure content filter.
Cloud Integration	8.41.14	This patch includes the following items: • Configures the thread pool size of the RFM poller to enhance resiliency by increasing the pool size from 5 to 10, reducing the risk of thread unavailability for RFM poller tasks. • Fixes an issue where CMS encryption and signing were always disabled on the OFTP Receiver channel. • Removes an additional logger that was unnecessarily logging database details, improving logging efficiency. • Fixes failures in Script Optimization requests observed in Canary due to rejection by the Azure content filter.

March 2026

Technical Component	Software Version	Description
Cloud Integration	8.40.27	This patch fixes the issue of network errors in the OData V2 receiver, caused by stale connections remaining in the pool after a timeout, by implementing a background job that periodically clears these stale connections when the system variable <code>SAP_ODATA2_CONNECTION_POOL_CLEANUP_ENABLED</code> is set to <code>True</code> .
Cloud Integration	6.76.46	This patch fixes the issue of network errors in the OData V2 receiver, caused by stale connections remaining in the pool after a timeout, by implementing a background job that periodically clears these stale connections when the system variable <code>SAP_ODATA2_CONNECTION_POOL_CLEANUP_ENABLED</code> is set to <code>True</code> .
Cloud Integration	8.40.26	This patch fixed an issue that prevented the deletion of service keys for Process Integration Runtime service instances.
Cloud Integration	6.76.45	This patch fixed an issue that prevented the deletion of service keys for Process Integration Runtime service instances.
Cloud Integration	8.40.25	This patch includes the following: • It enhances partner DeLet e for PD in Edge Integration Cell (EIC). • Ensures that the exception subprocess functionality for API artifacts in EIC works correctly, preventing deployment failures in environments running older worker versions. • Improves the reliability of the HTTP Receiver Adapter by introducing configurable retry support for I/O-related connection errors, reducing message failures caused by intermittent network connection resets. (Retry is disabled by default and can be enabled by SAP operations).
Cloud Integration	7.49.13	This patch includes the following items: • Introduces Retry mechanisms to the JDBC adapter to mitigate short-term failures with connectivity services or database interruptions. Retry operations are configured via tenant level ENV variables to enhance stability. • Resolves errors with the OData V4 adapter failing to parse 202 Accepted responses when the <code>Prefer: respond-async</code> header is included. This patch ensures successful processing of async requests. • Enhances the JDBC adapter with additional loggers and Retry mechanisms to resolve Service Reference Null issues.
Cloud Integration	6.76.44	This patch includes the following items: • It enhances the JDBC adapter with additional loggers and Retry mechanisms to resolve Service Reference Null issues. • Stale artifact metadata in the microservice database is now automatically cleaned up, fixing issues where artifacts were incorrectly displayed as Starting during rapid deploy/undeploy or redeploy actions. This ensures accurate monitoring status. • The cleanup expiry configuration was reduced from 7 to 3 days to prevent the buildup of outdated artifacts. • Calling integration flow endpoints with an expired OAuth token now returns the correct HTTP 401 response code instead of HTTP 500, ensuring accurate authentication

Technical Component	Software Version	Description
		error reporting. • Partner Delete enhancement for PD in EIC.
Cloud Integration	6.76.36	This patch includes the following items: • Resolves issues with software updates getting stuck by fixing the deployment automation script. • Fixes an issue with the generation of the evaluation report. • It enhances the JDBC adapter with additional loggers and Retry mechanisms to resolve Service Reference Null issues.

March 2026

Software Increment: 2513

Technical Component	Software Version	Description
Cloud Integration	6.75.51	This patch includes the following items: • It fixes an issue with the generation of the evaluation report. • Resolves errors with the OData V4 adapter failing to parse 202 Accepted responses when the Prefer: respond-async header is included. This patch ensures successful processing of async requests.
Cloud Integration	6.75.48	This patch fixes the following issues: • Ensures that Exception Subprocess functionality for API artifacts in Edge Integration Cell (EIC) works correctly, preventing deployment failures in environments running older worker versions. • Fixes a data conversion issue during data extraction that could cause the extraction process to fail. • Updates third-party dependencies identified by the OSS security scanner as vulnerable to denial-of-service attacks. • Introduces a retry mechanism in the JDBC adapter to improve resilience during temporary connectivity service interruptions or database outages. Retry behavior can be configured using tenant-level environment variables.

February 2026

Software Increment: 2513

Technical Component	Software Version	Description
Cloud Integration	8.39.12	This patch fixes the following issues:

Technical Component	Software Version	Description
		Ensures headers configured under Metadata Details are sent only for metadata and service document calls across all message processing logs, preventing unintended header propagation.
Cloud Integration	7.48.13	This patch fixes the following issues: - Ensures headers configured under Metadata Details are sent only for metadata and service document calls across all message processing logs, preventing unintended header propagation. - Improves the reliability of the HTTP Receiver Adapter by introducing configurable retry support for I/O-related connection errors, reducing message failures caused by intermittent network connection resets. (Retry is disabled by default and can be enabled by SAP operations.)
Cloud Integration	6.75.44	This patch fixes the following issues: - Ensures headers configured under Metadata Details are sent only for metadata and service document calls across all message processing logs, preventing unintended header propagation. - Improves the reliability of the HTTP Receiver Adapter by introducing configurable retry support for I/O-related connection errors, reducing message failures caused by intermittent network connection resets. (Retry is disabled by default and can be enabled by SAP operations.) - Enhances HTTP runtime resilience and Java compatibility by enabling configurable retries for I/O exceptions, supporting both Kafka client 3 and Kafka client 4, and extending compatibility with JDK 17 and higher. - Enables visibility of the API Provider option for the Edge Integration Cell (EIC) profile, improving configuration consistency for EIC scenarios.
Cloud Integration	6.75.38	This patch fixes the following issues: - Ensures recently created or updated artifacts are displayed correctly on the home page. - Fixes health check failures that caused incorrect system status across multiple landscapes. - Resolves issues loading  Company Profile  Security Configurations  in the UI and during Agreement exports.
Cloud Integration	6.75.36	This patch does the following: - Corrects the release version for Migration Assessment and Integration Assessment to 1.50. - Deactivates the script to create custom domains.

February 2026

Software Increment: 2512

Technical Component	Software Version	Description
Cloud Integration	6.74.44	This patch fixes the following issues: - Fixed tenant migration failures caused by a duplicate Liquibase changeset ID. - Resolved UCL formation creation, deletion, and synchronization failures with Integration Suite caused by Provisioning App callbacks not working. Updated configuration has restored successful callback. - Fixed an OSS vulnerability in the transitive dependency qs by updating the affected component. - Optimized database query handling by reducing the number of update operations, improving efficiency and overall interaction stability.
Cloud Integration	8.38.13	This patch optimized database query handling by reducing the number of update operations, improving efficiency and overall interaction stability.

January 2026

Software Increment: 2512

Technical Component	Software Version	Description
Cloud Integration	6.74.39	This patch fixes the following issues: - Resolved an MTMS scaling configuration issue where the instance count of <code>it-ai-service-java</code> was left at 1 instead of the expected 3. - Fixed an issue in API creation where menu options were incorrectly displayed for Edge Integration Cell integrations due to profile matching logic. Strict detection now ensures only relevant options are shown. - Resolved frequent not-found errors when loading individual TPM resources. - Improved performance of Config Service APIs to handle higher request volumes efficiently.
Cloud Integration	8.38.12	This patch improved the performance of Config Service APIs to handle higher request volumes efficiently.

December 2025

Software Increment: 2510

Technical Component	Software Version	Description
Cloud Integration	8.36.29	This patch added the <code>MALLOC_ARENA_MAX</code> environment variable to optimize memory allocation. This improves garbage collection behavior in the old generation heap.
Cloud Integration	6.72.69	This patch added the <code>MALLOC_ARENA_MAX</code> environment variable to optimize memory allocation. This improves garbage collection behavior in the old generation heap.

Technical Component	Software Version	Description
Cloud Integration	6.72.67	This patch enhances the Metabase information and also improves the efficiency of the ObjectStoreCleanupTask by limiting the cleanup scan to the /tasks folder instead of scanning the entire object store.
Cloud Integration	7.45.31	This patch enhances the Metabase information and fixes the Decommission task.
Cloud Integration	6.72.65	This patch fixes the following issues: • Improved Azure Shared File System lock handling to prevent frequent 404 errors • Resolved intermittent issues when saving API artifacts and importing OpenAPI definitions, improving overall reliability and user experience. • Enhanced the CXF library with a scheduled cleanup mechanism to ensure temporary files created during message processing are properly removed after a configurable delay. • Fixed a NullPointerException in the Xi Sender Adapter that could occur when payload attachments were missing from the Xi manifest, preventing message processing failures in this scenario.
Cloud Integration	7.45.29	This patch fixes the following issues: • Enhanced the CXF library with a scheduled cleanup mechanism to ensure temporary files created during message processing are properly removed after a configurable delay. • Fixed a NullPointerException in the Xi Sender Adapter that could occur when payload attachments were missing from the Xi manifest, preventing message processing failures in this scenario.
Cloud Integration	8.36.29	This patch fixes the following issues: • Improved Azure Shared File System lock handling to prevent frequent 404 errors • Enhanced the CXF library with a scheduled cleanup mechanism to ensure temporary files created during message processing are properly removed after a configurable delay. • Fixed a NullPointerException in the Xi Sender Adapter that could occur when payload attachments were missing from the Xi manifest, preventing message processing failures in this scenario. • Improved the efficiency of the ObjectStoreCleanupTask by limiting the cleanup scan to the /tasks folder instead of scanning the entire object store.

Technical Component	Software Version	Description
Cloud Integration	6.72.61	This patch fixes the following issues: - Added enhanced logging to help investigate an edge case where the jMS retry interval behaved differently from the configured value, even without a worker crash - Updated the list of supported database drivers to include the IBM DB2 z/OS driver with the required license, enabling customers to use the approved driver provided by IBM. - Fixed an issue with identifying Camel temporary files after a file-naming change introduced in newer Camel versions, preventing duplicate reporting of the same files. - Fixed an issue in EDI flow steps where X12 payloads ending with the TA1 keyword were not processed correctly, ensuring such payloads are now processed as expected.
Cloud Integration	7.45.27	This patch fixes the following issues: - Fixed ConcurrentModificationException issue in message processing flow. - Updated the list of supported database drivers to include the IBM DB2 z/OS driver with the required license, enabling customers to use the approved driver provided by IBM. - Fixed an issue in EDI flow steps where X12 payloads ending with the TA1 keyword were not processed correctly, ensuring such payloads are now processed as expected.
Cloud Integration	8.36.26	This patch fixes the following issues: - Updated the list of supported database drivers to include the IBM DB2 z/OS driver with the required license, enabling customers to use the approved driver provided by IBM. - Fixed an issue in EDI flow steps where X12 payloads ending with the TA1 keyword were not processed correctly, ensuring such payloads are now processed as expected.
Cloud Integration	6.72.59	This patch fixes the following issues: - Closed a gap in the PGP Decryptor related to checksum handling. - Fixed an issue where the Virtual Host was shown incorrectly as the default. - Upgraded the Eric library to versions 43.2.6 and 43.3.2. - Added malloc-arena-max in the worker configuration to reduce crashes. - Fixed an issue in the IDoc Receiver Adapter where MessageId determination using the Map option did not work as expected. - Fixed a cache-related issue. - Fixed permission management errors for TPM dedicated roles. - Removed the time restriction on archived data to ensure all data is archived when the job is triggered. - Fixed ConcurrentModificationException issue in message processing flow. - Made the ConnectorInstance artifact type available in the itco service. - Applied memory optimization parameters to reduce micro service crashes. - Fixed some errors for the UI and DBSL context fix.
Cloud Integration	7.45.24	This patch fixes the following issues:

Technical Component	Software Version	Description
		- Upgraded the Eric library to versions 43.2.6 and 43.3.2. - Fixed an issue in the IDoc Receiver Adapter where MessageId determination using the Map option did not work as expected.
Cloud Integration	8.36.23	This patch fixes the following issues: - Upgraded the Eric library to version 43.2.6. - Fixed ConcurrentModificationException issue in message processing flow.

December 2025

Software Increment: 2509

Technical Component	Software Version	Description
Cloud Integration	6.71.49	Fixed a regex issue in Inspect to correctly detect Camel temporary files after the naming convention update in Camel.

November 2025

Software Increment: 2509

Technical Component	Software Version	Description
Cloud Integration	6.71.48	This patch fixes the following issues: - Fixed TPM issues and connection problems between TPM and TRM. - Fixed an issue by adopting the correct IA versions for NS2 to ensure proper version alignment. - Fixed an issue with spreadsheet report generation where excessive attributes caused problems with cell style limits. - Added a health check for it-is-app-prov-route. - Fixed a checksum handling issue in the PGP Decryptor.
Cloud Integration	8.35.15	Fixed a checksum handling issue in the PGP Decryptor.
Cloud Integration	6.71.42	Improved performance for Message Mapping validation by optimizing memory consumption, preventing design service crashes due to high memory usage.
Cloud Integration	6.71.40	This patch fixes the following issues: - Fixed an issue with the deploy dialog during cross-deployment. - Fixed an issue with Custom Mapping Payloads when Archiving is enabled. - Fixed an issue causing a continuous increase in JVM threads in the worker application.

Technical Component	Software Version	Description
		Fixed an issue in the OAuth token server response missing the expires_in field, ensuring correct handling of such responses and proper formation of subsequent token requests.
Cloud Integration	7.44.14	This patch fixes the following issues: - A message mapping deployment issue that occurs in certain Process Integration/Process Orchestration migrated message mappings which contain parameters. - The calculation of offsets and tokens in the generation of the next link of OData results. This link is required for client-side pagination of result sets greater than 1000 rows. - Fixed an issue with the deploy dialog during cross-deployment. - Fixed an issue with Custom Mapping Payloads when Archiving is enabled. - Fixed an issue in the OAuth token server response missing the expires_in field, ensuring correct handling of such responses and proper formation of subsequent token requests. - Improved performance for Message Mapping validation by optimizing memory consumption, preventing design service crashes due to high memory usage.
Cloud Integration	8.35.14	This patch fixes the following issues: - Fixed an issue causing a continuous increase in JVM threads in the worker application. - Fixed an issue in the OAuth token server response missing the expires_in field, ensuring correct handling of such responses and proper formation of subsequent token requests. - Improved performance for Message Mapping validation by optimizing memory consumption, preventing design service crashes due to high memory usage.
Cloud Integration	6.71.36	This patch fixes the following issues: - The calculation of offsets and tokens in the generation of the next link of OData results. This link is required for client-side pagination of result sets greater than 1000 rows. - Switched back to SAP JBP v1 to resolve the OutOfMemory issues. - The namespace location to resolve dependencies when calculating dependent schemas in the primary artifact. This allows users to import artifacts with dependents even when the namespace format includes a pipe symbol.
Cloud Integration	6.71.33	This patch fixes the following issues: - A message mapping deployment issue that occurs in certain Process Integration/Process Orchestration migrated message mappings which contain parameters. - A validation issue in message mapping that occurs when multiple XSDs with different namespaces are used as source or target schemas. - This patch improves the catching mechanism.

October 2025

Technical Component	Software Version	Description
Cloud Integration	6.71.31	This patch fixes the following issues: • It fixes the order of the ArtifactInstanceType enum to correctly mark artifacts with the correct instance type. • It brings the latest code to Edge Integration Cell and maintains version parity. • It fixes default configuration for service deployment. • It introduces a flag that enables or disables the MCP feature. • It restores the rendering of the Analytics user interfaces. • It fixes KPI data rendering for classic runtime in tenants with multi-runtime enabled. • It fixes the standalone message mapping simulation issue for certain payloads. • It enhances connection handling in key rotation scheduler. • This patch resolves the issue where you can't select any defined number ranges other than the first number range in the Receiver Interchange. You can now also define an identifier with a custom scheme code.
Cloud Integration	8.35.12	This patch brings the latest code to Edge Integration Cell and maintains version parity.

October 2025

Software Increment: 2507

Technical Component	Software Version	Description
Cloud Integration	8.33.25	This patch fixes deployment issues for integration flows using XI adapters, ensuring successful execution for version 1.15 and above. It supports Quality of Service settings: At Least Once (ALO), Exactly Once (EO), and Handled by Integration Flow (HBI).
Cloud Integration	6.69.64	This patch fixes deployment issues for integration flows using XI adapters, ensuring successful execution for version 1.15 and above. It supports Quality of Service settings: At Least Once (ALO), Exactly Once (EO), and Handled by Integration Flow (HBI).
Cloud Integration	8.33.24	Any receiver adapter connecting to an on-premise system may fail with an HTTP 429 error from XSUAA during token fetch from connectivity service. This patch introduces a robust retry mechanism, as recommended in the XSUAA help documentation, to prevent such iFlow failures.
Cloud Integration	6.69.63	This patch fixes the following issues: • This patch makes the timeout for physical database creation configurable. The issue is fixed by handling it from the application. • Any receiver adapter connecting to an on-premise system may fail with an HTTP 429 error from XSUAA during token fetch from connectivity service. This patch introduces a robust retry mechanism, as recommended in the XSUAA help documentation, to prevent such iFlow failures.

Technical Component	Software Version	Description
Cloud Integration	6.69.61	This patch adds mitigation mechanism to toggle gzip compression for message mapping save.
Cloud Integration	6.69.60	This patch fixes the following issues: - Renewing the refresh token for Authorization Code Grant credentials was not possible due to a misconfigured audit log. This could lead to integration flow startup failures or message processing errors. - This patch updates the build pack from sap_java_buildpack to sap_java_buildpack_to_be_removed for the Java 8 and Java 17 aliases. The original name of SAP JBP v1 remains available only until October 26. After that date, it will be removed from the landscapes.
Cloud Integration	8.33.23	This patch fixes the following issues: - It enables automatic generation of the thread if any artifacts are stuck in an intermediate state. - Issues with generating the thread dump, as the stack trace depth is currently limited to eight. This limitation hides important details that aid in detailed analysis. Additionally, there's a plan to implement a named thread pool for deploy threads. This feature is helpful for debugging issues. - The renewal of the Refresh Token of an Authorization Code Grant credentials was not possible due to a misconfigured auditlog. This caused integration flows startup to fail or messages processing failures.

September 2025

Software Increment: 2507

Technical Component	Software Version	Description
Cloud Integration	8.33.22	This patch fixes the following issues: - A validation issue in message mappings in which the recursive node of a duplicated node is expanded and mapped while the recursive node in the original node of the duplicated node is not expanded. - Some nodes in message mappings created using recursive and duplicate operations don't resolve in the UI. This patch improves the logic for rendering such nodes and makes them accessible.
Cloud Integration	6.69.58	This patch fixes the following issues: - A validation issue in message mappings in which the recursive node of a duplicated node is expanded and mapped while the recursive node in the original node of the duplicated node is not expanded. - Some nodes in message mappings created using recursive and duplicate operations don't resolve in the UI. This patch improves the logic for rendering such nodes and makes them accessible. - An issue with length at DB side.

Technical Component	Software Version	Description
		- An issue with evaluation. - An issue with the application in integration policy.
Integration Advisor	2.3.2	This patch fixes incorrect Venetian Blind XSD file name generation for EDIFACT MIGs with Syntax Version 2.
Cloud Integration	7.42.15	This patch includes fixes for the following items: - A validation issue in message mappings in which the recursive node of a duplicated node is expanded and mapped while the recursive node in the original node of the duplicated node is not expanded. - Some nodes in message mappings created using recursive and duplicate operations don't resolve in the UI. This patch improves the logic for rendering such nodes and makes them accessible. - The issue where not all message queues are shown.
Cloud Integration	8.33.21	This patch fixes the following issues: - The issue where not all message queues are shown. - The issue with the reduced snapshot size not being logged. - This patch adds fallback support for changing log level via custom or default JMX MBean. - This patch addresses the issue observed in Ariba's RNIF servlet-based adapter, where the servlet fails to register after an Out of Memory (OOM) error. As a result, customers were receiving a 404 error for inbound calls to that node.
Cloud Integration	7.42.14	This patch addresses the issue observed in Ariba's RNIF servlet-based adapter, where the servlet fails to register after an Out of Memory (OOM) error. As a result, customers were receiving a 404 error for inbound calls to that node.
Cloud Integration	7.42.13	This patch fixes the issue with opening of Groovy scripts.
Cloud Integration	6.69.57	This patch fixes the following issues: - The syntax version always shows Release 2.8.0-SNAPSHOT in the display mode, even when it should display a different value. - Reduced snapshot size not being logged. where
Cloud Integration	6.69.56	This patch addresses the following issues: - An issue with synchronous interfaces showing "0" in message throughput. - The intermittent issue that occurs when opening integration flows, particularly those that are complex. - This patch adds fallback support for changing log level via custom or default JMX MBean.
Cloud Integration	6.69.54	This patch includes the following items: LLM model tuning for improved AI service efficiency and accuracy.

Technical Component	Software Version	Description
		• Fix for synchronous interface throughput issue, addressing incorrect display of "0" messages. • Intermittent integration flow opening issue when complex integration flows are opened. • Fixes the rate limit 429 issue.
Cloud Integration	6.69.49	This patch includes the following items: • JDBC adapter moved to a stable version. • Enhancements related to load distribution. • Updates related to Message Processing Log Attachment view on Monitor Message Processing screen.
Cloud Integration	6.69.45	This patch fixes the following issues: • When the syntax version is null, the UI incorrectly displays "Release undefined." The UI now shows this field as empty. • Fixes an internal dependency alignment with Jakarta-compliant libraries identified during DTS consumption. • Basic Auth was failing for API Management capability + Integration Cell use cases due to credential store separation for Integration Cell specific scenarios. The patch fixes this by creating the key in the generic secure store. • The patch addresses issues related to the Design Page during API Management capability and Integration Cell reactivation.
Cloud Integration	7.42.11	This patch includes the following items: • JDBC adapter moved to a stable version. • Updates related to Message Processing Log Attachment view on Monitor Message Processing screen.
Cloud Integration	8.33.18	This patch includes the following items: • JDBC adapter moved to a stable version. • Enhancements related to load distribution.

September 2025

Software Increment: 2506

Technical Component	Software Version	Description
Cloud Integration	6.68.26	This patch fixes issues with Unicode characters parsing in extracted XML files.

August 2025

Technical Component	Software Version	Description
Cloud Integration	8.32.11	This patch fixes the issue with artifact deployments timeouts and incorrect status displayed on the monitoring page, due to a race condition caused by frequent deployments of Number Range Objects. Artifacts now show the correct status, and deployments complete successfully in the latest software version.
Cloud Integration	6.68.24	This patch fixes the issue with artifact deployments timeouts and incorrect status displayed on the monitoring page, due to a race condition caused by frequent deployments of Number Range Objects. Artifacts now show the correct status, and deployments complete successfully in the latest software version.
Cloud Integration	8.32.10	This patch fixes the following issues: - During concurrent message processing, following the startup of workers—often triggered by a system restart or software update—a surge of simultaneous token requests is sent to the XSUAA server. This sudden burst can exceed the server's rate limit of 60 requests per second. As a result, the message flow may fail. - When splitter is used as part of exception subprocess.
Cloud Integration	6.68.22	This patch fixes the following issues: - This patch makes the job scheduler initialization asynchronous. This change reduces the startup time of the it-design-service and minimizes potential crashes. - During concurrent message processing, following the startup of workers—often triggered by a system restart or software update—a surge of simultaneous token requests is sent to the XSUAA server. This sudden burst can exceed the server's rate limit of 60 requests per second. As a result, the message flow may fail. - When splitter is used as part of exception subprocess.
Integration Advisor	2.2.4	This patch resolves the issue about MIG Migration deleting segments which follow qualified nodes.
Cloud Integration	6.68.16	This patch enhances the check for Trading Partner Management enablement to avoid issues during dedicated database creation.
Cloud Integration	7.41.11	This patch fixes the issue when splitter is used as part of exception subprocess.
Cloud Integration	6.68.15	This patch includes the following items: - Adds improvements to the Zip Splitter, ensuring better stability and efficiency across multiple tenants and data centers. - Enables dynamic log level adjustment through a new MBean (LogLevelManager). If unavailable, the system now falls back to the default Logback JMXConfigurator MBean.
Cloud Integration	8.32.7	This patch includes the following items: - Adds improvements to the Zip Splitter, ensuring better stability and efficiency across multiple tenants and data centers. - Enables dynamic log level adjustment through a new MBean (LogLevelManager). If unavailable, the system now falls back to the default Logback JMXConfigurator MBean.

Technical Component	Software Version	Description
Cloud Integration	7.41.10	This patch includes the Improvements to the Zip Splitter, ensuring better stability and efficiency across multiple tenants and data centers.

August 2025

Software Increment: 2505

Technical Component	Software Version	Description
Cloud Integration	6.67.36	This patch includes the following items: - Improvements to the Zip Splitter, ensuring better stability and efficiency across multiple tenants and data centers. - Optimizes memory usage during design time with the latest version of the Saxon library (12x).
Cloud Integration	8.31.18	This patch ensures optimized memory usage during design time with the latest version of the Saxon library (12x).
Cloud Integration	7.40.18	This patch ensures optimized memory usage during design time with the latest version of the Saxon library (12x).

July 2025

Software Increment: 2505

Technical Component	Software Version	Description
Cloud Integration	6.67.34	This patch includes the following items: - JDBC adapter has been updated to a stable version with enhanced logging to help identify problematic JDBC URLs. - Supports the message mappings that contain recursive nodes within duplicate nodes, ensuring they are correctly generated in the mapping output..
Cloud Integration	8.31.17	With this patch, the JDBC component has been updated to a stable version with enhanced logging to help identify problematic JDBC URLs.
Cloud Integration	7.40.17	This patch includes the following items: - JDBC adapter has been updated to a stable version with enhanced logging to help identify problematic JDBC URLs. - Supports the message mappings that contain recursive nodes within duplicate nodes, ensuring they are correctly generated in the mapping output..
Cloud Integration	7.40.15	This patch includes the following items:

Technical Component	Software Version	Description
		- Improved performance for integration scenarios that use XSLT Mapping with dynamic resolution such as Partner Directory, which have higher memory consumption. - Improved logic for rendering nodes in message mapping that were created using a combination of recursive and duplicate operations. This improved logic makes the nodes accessible ensuring resolution. - Support for upcoming HTTP feature updates. - Resolved <code>StackOverflowError</code> by integrating Camel patch (CAMEL-22147) for transactions split that could cause deep stack frames.
Cloud Integration	6.67.33	This patch improves logic for rendering nodes in message mapping that were created using a combination of recursive and duplicate operations. This improved logic makes the nodes accessible ensuring resolution.
Cloud Integration	8.31.16	This patch includes the following items: - Fixes to the liquibase script execution avoiding app crashes during startup. - Enabled stream download for Poll Enrich to support large file processing with SFTP sender. - Support for upcoming HTTP feature updates. - Enhanced support for on-premise connections enabled within the starter edition runtime profile. - Resolved <code>StackOverflowError</code> by integrating Camel patch (CAMEL-22147) for transactions split that could cause deep stack frames.
Cloud Integration	6.67.32	This patch includes the following items: - Fixes to the liquibase script execution avoiding app crashes during startup. - Enhanced logging capabilities to track when credentials are retrieved after a recent software update. - Enabling of the G1 Garbage Collector (G1GC). - Support for upcoming HTTP feature updates. - Enhanced support for on-premise connections enabled within the starter edition runtime profile. - Resolved <code>StackOverflowError</code> by integrating Camel patch (CAMEL-22147) for transactions split that could cause deep stack frames.

July 2025

Software Increment: 2504

Technical Component	Software Version	Description
Cloud Integration	8.30.17	This patch fixes a bug in Camel that receives the file size as integer instead of long. Files with sizes greater than 2.3 GB aren't processed properly, and the application throws an out of memory exception.

Technical Component	Software Version	Description
Cloud Integration	6.66.47	This patch fixes the following issues: • Customers of a specific tenant might see or modify Security Artifacts or view Data Store/JMS entries belonging to another customer's tenant. This issue can occur when there is a high load on the Key Management microservice or the Runtime API. • A bug in Camel that receives the file size as integer instead of long. Files with sizes greater than 2.3 GB aren't processed properly, and the application throws an out of memory exception..

June 2025

Software Increment: 2504

Technical Component	Software Version	Description
Cloud Integration	7.39.18	This patch fixes issues with Role Check modifications in the Script Collection Deploy scenario.
Cloud Integration	6.66.44	This patch fixes issues with Role Check modifications in the Script Collection Deploy scenario.
Cloud Integration	8.30.16	This patch adds an additional logs to facilitate detailed analysis of customer incidents.
Cloud Integration	7.39.17	This patch adds an additional logs to facilitate detailed analysis of customer incidents.
Cloud Integration	6.66.41	This patch fixes the following items: • Compatibility issues with main XSDs that reference types from external XSDs within mappings. • Incorporates an additional logs to facilitate detailed analysis of customer incidents. • Mitigates CVE-2025-49146 by deploying an updated version of the PostgreSQL library • Regeneration failures during artifact deployment by resolving a database insert error. • Migration mapping that have names exceeding 60 characters. • Security enhancements. • Validation for the undeployment activity in the OData DELETE API.
Cloud Integration	7.39.16	This patch adds a check for the old standard sap hciertificate in security artifact transport.
Cloud Integration	6.66.37	This patch fixes the following items: • Memory usage issue in <code>it-co</code>.

Technical Component	Software Version	Description
		• Regeneration failures during artifact deployment by resolving a database insert error. • Validation for the undeployment activity in the OData DELETE API.

May 2025

Software Increment: 2503

Technical Component	Software Version	Description
Cloud Integration	8.29.18	This patch fixes the issue with the Kafka Sender Adapter, restoring full functionality to the "Retry Failed Messages" feature.
Cloud Integration	6.65.40	This patch fixes the following items: • Memory usage issue in <code>it-co</code>. • Management of suspended tenants, ensuring effective xsuaa key rotation within the configuration service. • Kafka Sender Adapter where the "Retry Failed Messages" functionality is not working. • Enabling tenant-specific configuration service for xsuaa key rotation. • Security enhancements. • Retention of archival settings in integration flow monitoring during a Camel upgrade.
Cloud Integration	8.29.17	This patch includes security enhancements.
Cloud Integration	7.38.21	This patch fixes the following items: • Kafka Sender Adapter where the "Retry Failed Messages" functionality is not working. • Queue monitor failing issues to load messages due to multiple JMS broker connection requests. • Security Enhancements.
Cloud Integration	6.65.36	This patch fixes the following items: • Queue monitor failing issues to load messages due to multiple JMS broker connection requests. • Security Enhancements.
Cloud Integration	8.29.16	This patch fixes the following items: • Gather issue with <code>ZipAggregationStrategy</code>. • In camel-3.x runtime, when multiple calls are made to MDI using the MDI receiver adapter in a single message processing, the local ID sent to the MDI service was incorrect. • Kafka sender adapter where the "Retry Failed Messages" functionality is not working.

Technical Component	Software Version	Description
Cloud Integration	7.38.18	This patch fixes the following items: • The issue with the Kafka sender adapter, restoring full functionality to the "Retry Failed Messages" feature. • Security Enhancements. • The issue where security artifact migration from the NEO tenant to the CF tenant fails if the CF tenant is on an extension landscape. For example, an extension landscape URL looks like this: https://tenantxy.integrationsuite.cfapps.eu10-004.hana.ondemand.com/shell/home. The "004" is the extension number. • Gather issue with ZipAggregationStrategy.
Cloud Integration	6.65.34	This patch fixes the following items: • Unable to set the MPLL log level due to a redirection issue from the config service to TRM. The redirection URL formation has been updated to resolve this issue. • Gather issue with ZipAggregationStrategy. • In camel-3.x runtime, when multiple calls are made to MDI using the MDI receiver adapter in a single message processing, the local ID sent to the MDI service was incorrect. • An update for the Apache HTTP Client is required to fix CVE-2025-27820. • Kafka sender adapter where the "Retry Failed Messages" functionality is not working • Input validation for old service instances improved.
Integration Advisor	1.99.6	This patch fixes the issue where MAG simulation fails if one of the MIGs uses a custom codelist.
Integration Advisor	1.99.5	This patch fixes the following items: • Pretransformation fail due to the use of wrong ArtifactValue. • Incorrect SAP telemetry destination name in the code.

April 2025

Software Increment: 2502

Technical Component	Software Version	Description
Cloud Integration	6.64.31	This patch fixes the following items: • Hanging issues and facilitates a smooth Camel 3x migration with transaction-enabled XSLT Mapping Components. • Intermittent access issue affecting over 10 customers in Integration Suite, following the "@sap/approuter" update to version 19.0.0 to address a vulnerability identified by Blackduck. • Failure of MTMS Software Update on i t - r c - web in larger systems.

Technical Component	Software Version	Description
		Non-functional learn more banners and learning journey links.
Cloud Integration	7.37.18	This patch fixes the hanging issues and facilitates a smooth Camel 3x migration with transaction-enabled XSLT Mapping Components.
Cloud Integration	8.28.10	This patch fixes the hanging issues and facilitates a smooth Camel 3x migration with transaction-enabled XSLT Mapping Components.
Cloud Integration	6.64.28	This patch fixes the following items: - Improved handling of migrated message mappings, enabling the opening of expanded recursive nodes with duplicate subtrees. - Splitter component by setting a default value to the SapGroup header when it is empty.
Cloud Integration	7.37.17	This patch fixes the following items: - Improved handling of migrated message mappings, enabling the opening of expanded recursive nodes with duplicate subtrees. - Splitter component by setting a default value to the SapGroup header when it is empty.
Cloud Integration	8.28.9	This patch fixes the issue with the Kafka Sender Adapter, restoring full functionality to the "Retry Failed Messages" feature.
Cloud Integration	6.64.26	This patch fixes the following items: - Deployment failure for <code>cp_it_resource_cockpit</code> due to revoked admin privileges from the Postgres user. - Kafka Sender Adapter where the "Retry Failed Messages" functionality is not working. - Customers to deploy Message Mappings containing JSON schemas. - Queue monitor not loading messages due to multiple JMS broker connection requests.
Integration Advisor	1.98.4	This patch fixes the issue where Push to Partner Directory or Export of runtime artifacts of a MAG was taking considerably longer time.
Cloud Integration	7.37.15	This patch fixes the following items: - Performance enhancements related to content deployment. - Kafka Sender Adapter where the "Retry Failed Messages" functionality is not working. - Customers to deploy Message Mappings containing JSON schemas. - Queue monitor not loading messages due to multiple JMS broker connection requests.
Cloud Integration	6.64.21	This patch fixes the following items: Temp store accumulation by cleaning up ZIP files.

Technical Component	Software Version	Description
		Critical CPU usage in Inspect's dedicated PostgreSQL, suspected to be caused by Azure hyperscaler, observed exclusively in eu20-001.
Cloud Integration	7.37.11	This patch includes performance enhancements related to content deployment.

April 2025

Software Increment: 2501

Technical Component	Software Version	Description
Cloud Integration	8.27.23	This patch fixes the following items: - Splitter issue in Camel by enabling rollback capability. - The 'Handled By Integration Flow' option in an XI sender channel causes errors due to the absence of a payload in synchronous responses when sending XI messages with Quality of Service BestEffort.
Cloud Integration	6.63.47	This patch fixes the following items: - Splitter issue in Camel by enabling rollback capability. - The 'Handled By Integration Flow' option in an XI sender channel causes errors due to the absence of a payload in synchronous responses when sending XI messages with Quality of Service BestEffort.
Cloud Integration	7.36.30	This patch fixes the following items: - Performance enhancements related to content deployment. - Splitter issue in Camel by enabling rollback capability. - The 'Handled By Integration Flow' option in an XI sender channel causes errors due to the absence of a payload in synchronous responses when sending XI messages with Quality of Service BestEffort.
Cloud Integration	8.27.21	This patch fixes the following items: - Truncated message after GZIP flow step. - Groovy metadata cache clearance. - Integration flows failing during worker instance restart due to network issues impacting Audit Logger activation. The dependency causing errors has been removed to ensure smoother operations. - Feature to cache the KeyManager in OData V2 Receiver is reverted.
Cloud Integration	6.63.45	This patch fixes the following items: - Truncated message after GZIP flow step. - MPL Status for asynchronous messages.

Technical Component	Software Version	Description
		- Exclusive queue-related fixes: - Reordering on reload. - Duplicate entries in the move messages dialog. - Issue with route generation when Quality Of Service type Handled By Integration Flow is selected. - An error message was added for long loading times. - Deployment and staging of application <code>it-op-eic-admincockpit</code> on Ali Cloud tenants. - Groovy metadata cache clearance. - Integration flows failing during worker instance restart due to network issues impacting Audit Logger activation. The dependency causing errors has been removed to ensure smoother operations. - Feature to cache the <code>KeyManager</code> in OData V2 Receiver is reverted. - Wrong claims in LLM-generated Script Optimization Report. - Redacted strings showing in script optimization proposals.
Cloud Integration	7.36.28	This patch fixes the following items: - Groovy metadata cache clearance. - Truncated message after GZIP flow step. - MPL Status for asynchronous messages. - Exclusive queue-related fixes: - Reordering on reload. - Duplicate entries in the move messages dialog. - Issue with route generation when Quality Of Service type Handled By Integration Flow is selected. - An error message was added for long loading times.

March 2025

Software Increment: 2501

Technical Component	Software Version	Description
Cloud Integration	8.27.16	This patch fixes the following issues: - Security enhancements. - Exchange headers are retained during exceptions, aligning functionality with Camel 2.x.

Technical Component	Software Version	Description
		Attachment content is consistently of type String in Camel 3, maintaining compatibility with Camel 2.
Cloud Integration	6.63.39	This patch fixes the following issues: - Retrieval of MPL Attachments through MPL OData APIs to 1000 items per request, ensuring more efficient data handling with the endpoint <code>/MessageProcessingLogs(<MPL ID>)/Attachments</code>. - Security enhancements. - Exchange headers are retained during exceptions, aligning functionality with Camel 2.x. - Attachment content is consistently of type String in Camel 3, maintaining compatibility with Camel 2.
Cloud Integration	7.36.24	This patch fixes the following issues: - Security enhancements. - Exchange headers are retained during exceptions, aligning functionality with Camel 2.x. - Attachment content is consistently of type String in Camel 3, maintaining compatibility with Camel 2.
Cloud Integration	7.36.26	This patch fixes the issue with the feature to cache the KeyManager in OData V2 Receiver.

March 2025

Software Increment: 2413

Technical Component	Software Version	Description
Cloud Integration	8.26.19	This patch fixes the following issues: - PD usage in the mail receiver adapter for encryption. - Message Mapping, Value Mapping, Function Library, and Imported Archive stuck in the starting state by providing exception handling and ensuring accurate artifact status updates in the UI.
Cloud Integration	6.62.57	This patch fixes the following issues: - PD usage in the mail receiver adapter for encryption. - Frequent crashing issue of the IT-CO application caused by low disk space. - Message Mapping, Value Mapping, Function Library, and Imported Archive stuck in the starting state by providing exception handling and ensuring accurate artifact status updates in the UI.

Technical Component	Software Version	Description
Cloud Integration	7.35.29	This patch fixes the following issues: • PD usage in the mail receiver adapter for encryption. • Download of the Certificate Signing Request in NEO Camelupdate Version with Java 17 for Elliptic Curve keys.

February 2025

Software Increment: 2413

Technical Component	Software Version	Description
Cloud Integration	8.26.17	This patch fixes the aggregator issue on Camel 3.x.
Cloud Integration	6.62.51	This patch fixes the aggregator issue on Camel 3.x.
Cloud Integration	7.35.27	This patch fixes the aggregator issue on Camel 3.x.
Cloud Integration	8.26.16	This patch fixes the following issues: • Metaspace consumption during script execution. • Runtime artifact generation by making stage timeout configurable. • Message processing fails when using the Mail receiver adapter with signature/encryption and a dynamic alias set via property. • Errors while using SOAP Sender adapter with OneWay, WS Standard, and WS Addressing configurations. • Errors during message processing on Camel 3.x with larger attribute size by increasing the XML max attribute to 120.000.
Cloud Integration	6.62.50	This patch fixes the following issues: • Metaspace consumption during script execution • Design time loading issue by disabling Telemetry Kafka topics. • Runtime artifact generation by making stage timeout configurable. • Function Library cannot access certain packages included in the Java core library due to missing import-package entries in those java core libraries. • Message Mapping, Value Mapping, Function Library, and Imported Archive remain in the starting state for an extended period. • Message processing fails when using the Mail receiver adapter with signature/encryption and a dynamic alias set via property. • Switchable legacy behavior for:

Technical Component	Software Version	Description
		- String functions with line separator in XSLT mapping. - Message History, which is by default switched off in Camel 3.x. - Errors while using SOAP Sender adapter with OneWay, WS Standard, and WS Addressing configurations. - Thread blocking in the design service that causes crashes. - Errors during message processing on Camel 3.x with larger attribute size by increasing the XML max attribute to 120.000.
Cloud Integration	7.35.26	This patch fixes the following issues: - Function Library cannot access certain packages included in the Java core library due to missing import-package entries in those java core libraries. - Message Mapping, Value Mapping, Function Library, and Imported Archive remain in the starting state for an extended period. - Message processing fails when using the Mail receiver adapter with signature/encryption and a dynamic alias set via property.
Cloud Integration	7.35.23	This patch fixes the following issues: - Switchable legacy behavior for: - String functions with line separator in XSLT mapping. - Message History, which is by default switched off in Camel 3.x. - Errors while using SOAP Sender adapter with OneWay, WS Standard, and WS Addressing configurations. - Artifacts were undeployed by the sync task even when those were not marked for undeployment. - Thread blocking in the design service that causes crashes. - Errors during message processing on Camel 3.x with larger attribute size by increasing the XML max attribute to 120.000.

January 2025

Software Increment: 2412

Technical Component	Software Version	Description
Cloud Integration	6.61.41	This patch fixes the following issues: - Skip regeneration for artifacts that are deployed in Cloud Integration runtime location will. - Credential rotation is optimized for unused keys. - Data conversion affecting the interpretation of search results in the PI/PO system.

Technical Component	Software Version	Description
Cloud Integration	5.69.15	This patch fixes: • The issue that occurs while editing a datasource in the JDBC adapter. • The issue with process direct adapter that occurs while identifying addresses with spaces along with their respective integration flow names.
	7.34.20	
	6.61.37	
Cloud Integration	6.61.36	This patch fixes the issue of excessive file descriptors usage detected in camel stream cache.
Cloud Integration	5.69.15	This patch fixes the issue of message mapping script functions.
	7.34.20	
	6.61.35	
Cloud Integration	5.69.14	This patch fixes the issue of content discovery for neo tenants.
	7.34.19	
Cloud Integration	5.69.13	This patch fixes the blocking issue caused by the codenarc scans for groovy scripts during deployment.
	7.34.18	
	6.61.34	
Cloud Integration	6.61.32	This patch fixes login related issue.
Cloud Integration	6.61.30	This patch fixes the loading issue of the Capabilities section on the Home Page.
Cloud Integration	6.61.29	This patch fixes the billing related issue.
Integration Advisor	1.95.4	MAG Simulation and export of runtime artifacts suddenly started failing while looking-up for some codelist after the release of 1.95.3. This patch fixes this issue.
Cloud Integration	6.61.28	This patch fixes the issue with creating and rendering design time artifacts.
Cloud Integration	7.34.16	With this patch, for FTP Sender adapter, Streaming is enabled only if Change Directory Stepwise is disabled.
	6.61.27	
Cloud Integration	7.34.16	With this patch, for FTP Sender adapter, Streaming is enabled only if Change Directory Stepwise is disabled.
	6.61.27	
Cloud Integration	7.34.15	This patch fixes the issue with integration flows when using Process Direct Adapter.
	6.61.25	
Cloud Integration	7.34.14	This patch fixes an issue with class loading during tenant restarts that occur during software updates and adds logs for further verification.
Cloud Integration	5.69.11	This patch restores the download option for PGP Keys.
	7.34.13	
	6.61.24	

Technical Component	Software Version	Description
Cloud Integration	7.34.12	The 7.34.10 release introduced a change in request handling that caused errors in the SOAP Sender channel when using the Asynchronous One-Way Message Exchange pattern. This patch resolves the issue by reverting to the original behavior.

January 2025

Software Increment: 2410

Technical Component	Software Version	Description
Cloud Integration	7.32.21	JDBC exceptions and file descriptor overflow were reported due to too many open files.
	5.67.21	This patch resolves the issue by advising the user to keep the Datasource limit at 49, as recommended.
	6.59.53	

December 2024

Software Increment: 2410

Technical Component	Software Version	Description
Cloud Integration	7.32.15	In the Camel 3.x migration, internal properties in Exchange weren't being propagated when using the Process Direct Adapter.
	6.59.39	This patch enables the selective migration of certain properties based on an environment variable. If the environment variable is turned ON, only specific properties migrate. If the environment variable is absent or turned OFF, the default behavior allows the migration of all properties, both internal and external.
Integration Advisor	1.93.4	This patch resolves issues with the Pre, Post, and Mag XSLTs generated during the MAG runtime export, when the MIG contains a default namespace with unqualified nodes. Before, the default namespace was incorrectly assigned to certain nodes.

December 2024

Software Increment: 2409

Technical Component	Software Version	Description
Cloud Integration	7.31.21	Internal properties in Exchange were not being propagated when using ProcessDirect adapter in Camel 3x migration. This patch resolves the issue.
	8.22.17	
Cloud Integration	6.58.40	On Cloud Foundry, the Cloud Integration worker application failed to connect to the Credential Store Service due to network issues, causing the keystore to be unavailable. This patch fixes the issue.
	8.22.15	

Software Increment: 2409

Technical Component	Software Version	Description
Cloud Integration	6.58.31	The JDBC Adapter version 1.5 is currently not available in Camel 3.x tenants due to a version difference in the command (cmd) between the Camel 2.x and 3.x branches. This patch resolves the issue.
Cloud Integration	7.31.12	This patch fixes the usage of \${property} expressions in the ID Mapper flow step for tenants on Camel 3.x.

October 2024

Software Increment: 2409

Technical Component	Software Version	Description
Cloud Integration	7.31.10 8.22.9	The Process Direct Adapter with Camel 3.x software was stuck in the processing state when called for Dynamic Address Resolution for the process direct consumer. This patch resolves the issue.
Cloud Integration	5.66.6 7.31.8 6.58.24	The JDBC Adapter version 1.5 was unavailable in Camel 3x tenants due to the command version difference between the Camel 2x and 3x branches. This patch fixes the issue.

September 2024

Software Increment: 2408

Technical Component	Software Version	Description
Cloud Integration	6.57.31 7.30.18	This patch fixes the unexpected long processing time of the integration flows.
Cloud Integration	6.57.28	Fix for content deployment failures caused by authentication issue between the underlying microservices.
Integration Advisor	1.91.6	The following issues have been fixed with this patch: • Display problems with mapping lines in MAG when an invalid mapping element is present. • MIG Simulation not working correctly with EDI Flat files. • EDI Flat files not being accepted for the customization of MIG during its creation.

August 2024

Software Increment: 2407

Technical Component	Software Version	Description
Integration Advisor	1.90.2	Issues were reported when activating a MAG with target MIG based on a custom message. This patch fixes the issue.
Cloud Integration	5.64.14 7.29.18	Integration flows were undeployed automatically because database queries were executed in non-transactional manner along with lack of synchronization between related tasks. This patch fixes the issue.
Cloud Integration	5.64.14 7.29.18 6.56.33	In the tenants using Apache Camel 3.14 runtime the integration flows utilizing AS2 and AS4 adapters experience frequent retries during message processing failures. For more details and solution, see 3516264 🗑️ This patch fixes the issue.

August 2024

Software Increment: 2406

Technical Component	Software Version	Description
Integration Advisor	1.89.3	Issues were reported when activating a MAG with target MIG based on a custom message. This patch fixes the issue.
Cloud Integration	5.63.17 7.28.20 6.55.21 8.19.14	A recent security fix for GHAS/CodeQL message mapping runtime, which blocked XML payloads containing Document Type Declarations (DTDs), has been reverted due to customer impact. This patch withdraws the security fix.
Cloud Integration	7.28.17	There was the issue that the payload got lost after being fetched by the Mail sender adapter if mail was signed/encrypted. This patch fixes the issue.
Cloud Integration	7.28.14 6.55.13	The following issues have been fixed with this patch: - The conversion from property to exchangeProperty was not properly implemented for the generation step, thus preventing the initiation of integration flows. - The default to use breadcrumb has been changed with Camel 3.x from <code>true</code> to <code>false</code>. As some customers rely on that header, the value has been reverted back to <code>true</code>
Cloud Integration	6.55.12 5.63.11	OEM adapters will be upgraded to Apache Camel version 3.x during the migration to Camel 3.x. This patch provides a change that makes this step transparent on the user interface.

July 2024

Software Increment: 2405

Technical Component	Software Version	Description
Cloud Integration	5.62.17	Message failures were reported in IDoc packages when the field DOCNUM was not set for some messages. This patch fixes the issue.
Cloud Integration	5.62.16 7.27.19	Timer failures were reported when worker IJS was restarted(OOM) after camel 3x migration. This patch fixes the issue.

June 2024

Software Increment: 2404

Technical Component	Software Version	Description
Cloud Integration	7.26.10	Message processing failures with the message Too many open files have been observed after migrating to Camel 3x. This patch fixes the issue.
Cloud Integration	6.53.31 7.26.8	A null pointer exception was thrown when sending an XI message. This patch fixes the issue.
Cloud Integration	6.53.29 7.26.7 5.61.10	During software update, a Timer step configured with a start date triggered the writing of a message processing log at an unscheduled time. This patch fixes the issue.
Cloud Integration	6.53.27 5.61.9	The JDBC Kafka consumer took too much time in case of an error. The patch fixes the issue so that processing happens in asynchronous mode now.
Cloud Integration	6.53.25	Content transport was impacted by an outdated service key stored in the credential store. The patch fixes this issue and ensures a consistent update of the affected entry in the credential store. and ensures a consistent update of the affected entry in the credential store.
Cloud Integration	5.61.8 6.53.23 7.26.6	stream. Special characters got converted to invalid characters. This patch fixes the issue.

May 2024

Software Increment: 2403

Technical Component	Software Version	Description
Cloud Integration	6.52.20	There have been DNS cache issues with JCo library resulting in the RFC receiver adapter to throw the error Opening socket to partner failed: error 111 - Connection refused. This patch fixes the issue.
Cloud Integration	5.60.9 7.25.10	The missing DOCNUM field in the request payload was causing an error "Mandatory IDoc SOAP header missing". This patch fixes the issue.

Technical Component	Software Version	Description
	6.52.18	
Cloud Integration	6.52.17	There was an issue with number range redeployment that lead to artifacts getting stuck. This patch fixes the issue.
Cloud Integration	5.60.8 7.25.9	There have been issues with integration flow deployment. This patch fixes the issues.

April 2024

Software Increment: 2402

Technical Component		Description
Cloud Integration	5.59.13 7.24.10	After a software update, the following issue has been observed: Integration flows with Timer events that were already in Deployed or Started state went to Failed state during software update (if Timer was scheduled to run with the past date). This patch fixes the issue.
Cloud Integration	6.51.28	There have been problems to load Monitor page. This patch fixes the issue.
Cloud Integration	6.51.27	There were some issues with the generation of key-pairs with X.509 certificates using elliptic curve algorithms (for example, secp192k1, secp256k1, secp192r1, and others). This patch fixes the issue.
Cloud Integration		There has been a performance issue with the Inspect feature. This patch fixes the issue.
Cloud Integration	6.51.25	There was an issue with a potential negative impact on the deployment and undeployment of integration artifacts. This patch fixes the issue.
Cloud Integration	5.59.12 7.24.9	Integration flows that use the Timer event have been discarded after the upgrade to Camel 3x. This patch fixes the issue and also improves the upgrade process to Camel 3x.
Cloud Integration	5.59.11 6.51.23 7.24.8	System performance has been improved and truncation of value mapping has been fixed by changing the way XML files are read.

March 2024

Software Increment: 2401

Technical Component	Software Version	Description
Cloud Integration	5.58.11 6.50.23 7.23.15	There have been outages where scenarios stopped working. This patch fixes the issue.

Technical Component	Software Version	Description
Cloud Integration	5.58.10 7.23.14 6.50.22	For some customers, scenarios using the SFTP adapter stopped working. Even after already having patched this issue, few customers observed continuation in following error message being raised: SSH_MSG_DISCONNECT: 2 Failed to read binary. This patch fixes the issue.
Cloud Integration	5.58.9 7.23.13	There were multiple timer executions during camel 3x migration. This patch fixes the issue. There were multiple timer executions, during camel 3x migration. This patch fixes the issue.
Cloud Integration	5.58.9 7.23.12	Some customers experienced issues with SFTP scenarios which stopped working and receiving connection errors from servers with message SSH_MSG_DISCONNECT: 2 Failed to read binary packet data! this patch fixes the issues.
Cloud Integration	7.23.11 8.14.8	As the behavior of Camel 3x has changed, some customers face an execution error in an edge case, when setting and reading the body within the same Groovy script. This patch fixes the issue.

March 2024

Software Increment: 2313

Technical Component	Software Version	Description
Cloud Integration	5.57.10 7.22.9	Due to an OData metadata version reported as "1.0", modern OData client applications are refusing to parse service definitions correctly. This patch changes the version to "2.0" and resolves the issue.

February 2024

Software Increment: 2313

Technical Component	Software Version	Description
Cloud Integration	5.57.10 7.22.9	Due to an OData metadata version reported as "1.0", modern OData client applications are refusing to parse service definitions correctly. This patch changes the version to "2.0" and resolves the issue.
Cloud Integration	6.49.27	This patch ensures that the API returns the details of the artifact version which is available on worker even if the multiple versions are available in CO db after executing regeneration.
Cloud Integration	5.57.9 7.22.7 6.49.25	There have been partly broken integration flow processes, due to Database Timeouts happening upon degraded encryption/decryption performance. This patch fixes this issue.
Cloud Integration	5.57.7 6.49.4	EDI splitter threw an exception in integration flow for the Partners sending the EDIFACT payload without line feed and '?' (escape character + segment terminator) in one segment. This patch fixes the issue.
Cloud Integration	5.57.7	There was a performance degradation for message processing caused by repetitively calling NM functionality to retrieve the tenant name. This patch fixes the issue by caching this single value to avoid

Technical Component	Software Version	Description
	7.22.5	redundant calls.
	6.49.21	

January 2024

Software Increment: 2312

Technical Component	Software Version	Description
Cloud Integration	6.48.22	The patch addresses performance degradation issues.
	8.12.2	
Cloud Integration	6.48.20	There has been an issue with upsert operations when using the JDBC adapter. This patch fixes the issue.
	7.21.3	
	8.12.1	
Cloud Integration	5.54.18	With this patch, the AMQP adapters have been updated to consume the ActiveMQ 5.16.7 version.
	7.19.16	

Understanding Software Version Numbers

Understand the software version numbers indicated for each patch.

A software version designation is composed of three digits separated by a period, for example, 6 . 53 . 23.

- The major version is 6 in this example.
- The second digit denotes the monthly software increment (which corresponds to increment 2404).
- The third number is incremented with each software patch.

Component	Major Version	Software Version Example
Cloud Integration (Cloud Foundry environment; runtime based on Apache Camel version 3.*)	8	8.24.13
Cloud Integration (Cloud Foundry environment; runtime based on Apache Camel version 2.*)	6	6.53.23
Cloud Integration (Neo environment; runtime based on Apache Camel version 3.*)	7	7.26.6
Cloud Integration (Neo environment; runtime based on Apache Camel 2.*)	5	5.61.8

Component	Major Version	Software Version Example
Integration Advisor	1	1.74.3

Archive - Patch Release Notes for Cloud Integration

This page contains a historical archive of all patch release notes for Cloud Integration.

December 2023

Software Increment: 2310

Technical Component	Software Version	Description
Cloud Integration	6.46.40	Tenant provisioning was failing in some cases. This patch fixes the issue.
Cloud Integration	6.46.38	The following issues have been fixed with this patch: Some security material artifacts were deactivated after being associated with one or more runtimes (and the system returned an error message). Furthermore, the consumption logic for Kafka acknowledgment messages has been improved.
Cloud Integration	5.54.17 6.46.36	Issues have been observed when two EDI to XML converter steps were executed with different values for the Exclude Interchange and Group envelopes null parameter. This patch fixes the issues.
Cloud Integration	5.54.15 6.46.27	There is an issue occurring intermittently, affecting customers using the HTTP sender adapter under heavy load, resulting in MPLs in state COMPLETED and an HTTP 500 status response/ HTTP 200 with empty response. This patch fixes the issue.

December 2023

Software Increment: 2309

Technical Component	Software Version	Description

November 2023

Software Increment: 2309

Technical Component	Software Version	Description
Cloud Integration	5.53.11 5.54.12	There is an issue with integration flows disappearing during a standard content update scenario. This patch fixes the issue

Technical Component	Software Version	Description
Cloud Integration	6.45.22	The following issues have been observed when designing message implementation guidelines (MIGs): • The MIG message type of the imported agreement has been wrong. • Import stopped when an agreement failed before agreement was executed. • Customers were unable to define custom separators. This patch fixes the issues.
Cloud Integration	7.18.7 6.45.18	There was an issue with integration flows containing SFTP sender adapters that stopped processing and could not be redeployed. This patch fixes the issue.

November 2023

Software Increment: 2309

Technical Component	Software Version	Description
Cloud Integration	5.53.11 5.54.12	There is an issue with integration flows disappearing during a standard content update scenario. This patch fixes the issue
Cloud Integration	6.45.22	The following issues have been observed when designing message implementation guidelines (MIGs): • The MIG message type of the imported agreement has been wrong. • Import stopped when an agreement failed before agreement was executed. • Customers were unable to define custom separators. This patch fixes the issues.
Cloud Integration	7.18.7 6.45.18	There was an issue with integration flows containing SFTP sender adapters that stopped processing and could not be redeployed. This patch fixes the issue.

October 2023

Software Increment: 2308

Technical Component	Software Version	Description
Cloud Integration	5.52.11	There is an issue with customer database connection, as existing drivers are incompatible with the customer database. To allowlist the new SQL Server Driver 11.2.3 with this patch fixes this issue.
Cloud Integration	6.44.12	Back-up keystore entries were not shown due to an issue in the UI. This patch fixes the issue.
Integration Advisor	1.78.5	Customers are unable to activate their MAGs due to a code issue as the UI generates duplicate mappings for the target nodes. This patch fixes this issue.

Technical Component	Software Version	Description
Cloud Integration	6.44.13	As the external logging activation is not synchronized with Kafka activation, depending on the timing, the external logging activation can terminate during start-up and activation changes are no longer processed. External logging activation must be manually restarted. This patch fixes the issue.

September 2023

Software Increment: 2305

Technical Component	Software Version	Description
Cloud Integration	2.64.1	On some tenants, it was impossible to create or update integration flows using the JMS enterprise capability, due to a missing configuration on the on the enterprise licensing information. This patch fixes the issue.

August 2023

Software Increment: 2305

Technical Component	Software Version	Description
Cloud Integration	5.49.14	The cluster migration to different brokers using the system property <code>com.sap.core.messaging.admin.client.group</code> was causing a mismatch between CMN and TMN leading to a break in the communication. This patch fixes the issue by clearing out the system property in the TMN.
Cloud Integration	6.41.32	mTLS based XSUAA keys might get expired. This patch fixes the issue by extending the validity of mTLS based keys to 150 days.
Cloud Integration	6.41.30	The integration flow execution was getting impacted during Neo to Cloud Foundry migration. This patch fixes the issue by improving the integration flow execution time.

July 2023

Software Increment: 2305

Technical Component	Software Version	Description
Cloud Integration	6.41.26	Transporting integration content from Transport Management Service or CTS+ to Cloud Integration was failing for some customers in the Cloud Foundry environment. This patch fixes the issue by allowing to fetch the required parameter before constructing the URL attribute.
Cloud Integration	6.41.26	When <code>itspace</code> was included in the URL path of the Monitor > Integrations (message processing log monitoring) application of SAP Integration Suite, the navigation between Monitor > Integrations (message processing log monitoring) and Monitor > B2B Scenarios was not possible. To fix the issue, with this patch <code>itspace</code> was removed from the SAP Integration Suite address.

Technical Component	Software Version	Description
Cloud Integration	6.41.25	Customers were facing sporadic issues during script collection deployment. This patch fixes the issues.
Cloud Integration	6.41.24	1. Due to object store issues, customers were not able to perform deployments or undeployments. This patch has enhanced logs to help identify root cause of the issue. 2. There were CO crashes observed due to high CPU usage. This patch has a fix which improves the efficiency.
Cloud Integration	6.41.23	When trying to deploy a message mapping artifact, the deployment was failing. This patch fixes the issue.

July 2023

Software Increment: 2304

Technical Component	Software Version	Description
Integration Advisor	1.74.3	There was an issue with activating agreements. This patch fixes the issue.
Cloud Integration	6.40.20	There was the issue that when designing APIs (using the API Management capability) deployed integration flows could not be fetched. This patch fixes the issue.
Cloud Integration	6.40.18 5.48.14	There was an issue with long-running transactions that resulted in exhaustion of database resources. This patch fixes the issue.

June 2023

Software Increment: 2303

Technical Component	Software Version	Description

May 2023

Software Increment: 2303

Technical Component	Software Version	Description
Cloud Integration	6.39.18	There was an issue with the de-commissioning of tenants. This patch fixes the issue.
Integration Advisor	1.73.2	For pre-transformation of message implementation guidelines, only a limited amount of source nodes was selectable. This patch fixes the issue.

Technical Component	Software Version	Description
Cloud Integration	6.39.17 5.47.9	There have been issues with the usage of JMS adapters. This patch fixes the issues.

April 2023

Software Increment: 2302

Technical Component	Software Version	Description
Integration Advisor	1.72.3	You were not able to finish the expected pre-transformation step, as some nodes from the source message implementation weren't shown in the drop down box and couldn't be selected. Therefore it was not possible to save the mapping guidelines. This patch fixes the issue.
Cloud Integration	6.38.22	The deployment of artifacts from packages with names containing non ASCII characters was impossible in the Cloud Foundry environment with latest software version. This patch fixes the issue.
Cloud Integration	6.38.21 5.46.13	The upcoming Netty version upgrade requests additional loggers to collect request and response headers for the AS2 adapter, to avoid incompatibility. This patch is for runtime data collection.
Cloud Integration	6.38.20	There have been issues in productive tenants with too much logging, when the database connection is used in an uncategorized context. This makes the log analysis difficult. The default log level should be "Info" and only changed on demand. This patch fixes the issues
Cloud Integration	6.38.16	There has been an issue with the script collection not being available after undeployment and redeployment. This causes the message processing to fail for the integration flows referring to the script collection. This patch fixes the issue.

March 2023

Software Increment: 2301

Technical Component	Software Version	Description
Cloud Integration	6.37.26	There has been an issue with the Trace log level that resulted in message processing failure when multiple integration flows have been traced. This patch fixes the issue.
Cloud Integration	6.37.25	There has been an issue with the activation of endpoints with self-signed certificates (for example, involved when using external logging with Splunk trial instances). This patch fixes the issue.
Cloud Integration	5.45.13	There has been an issue with the aggregator integration flow step resulting in aborted message processing and error messages. This patch fixes the issue.
Cloud Integration	5.45.11	There was an issue with the failover feature. This patch fixes the issue.
Integration Advisor	1.71.3	There has been an issue with the export of MIGs based on old custom messages. This patch fixes the issue.

February 2023

Software Increment: 2213

Technical Component	Software Version	Description
Cloud Integration	6.36.20	There was an issue with integration flows that processed many calls to adapters writing attributes (for example, RFC adapter). The time required to persist message processing logs increased quadratically with the number of adapter attributes. This resulted in failures and infinite retries. This patch fixes the issue.

February 2023

Software Increment: 2212

Technical Component	Software Version	Description
Cloud Integration	6.35.20	There was an issue with the data archiving feature as customers were unable to activate data archiving with client credentials. This patch fixes the issue.
Cloud Integration	6.35.19	There was an issue in the DB when a large amount of data is being fetched, setting a transaction into "read only" mode and blocking the message transfer. The patch now allows the explicit closing of the transaction, setting the "read only" mode to "false" and restores the dirty connection.
Cloud Integration	6.35.18	There were issues in some micro-services: if the Operations applications are restarted under load, a data race condition can cause the startup to fail and require a manual restart. No database connections can be made and all requests are rejected until the application is manually restarted. This patch fixes the issue.
Cloud Integration	6.35.14	There has been an issue with the OData API resource <code>IntegrationRuntimeArtifacts</code> (of the <code>Integration Content</code> API) . This patch fixes the issue, and you can now use the following paths to fetch artifact information in the CCloud Foundry environment: <code>/IntegrationRuntimeArtifacts(Id='{id}')/{property}/\$value</code> <code>/IntegrationRuntimeArtifacts(Id='{id}')</code>

January 2023

Software Increment: 2212

Technical Component	Software Version	Description
Cloud Integration	5.43.7 6.35.15	There was an issue with the XI sender adapter: Attachments with content types <code>text/*</code> and without <code>Content-Transfer-Encoding</code> header have been lost when parameter Quality Of Service was set to Exactly Once . This patch fixes the issue.
Cloud Integration	6.35.13	There was a coding error that had an impact on the data consistency on the customer tenant. This patch fixes the issue.
Cloud Integration	5.43.6	There was the issue that older versions of the SuccessFactors receiver adapter didn't show all available data centers in the Address field. This patch fixes the issue.

January 2023

Software Increment: 2210

Technical Component	Software Version	Description
Cloud Integration	6.33.32	In high volume situations, few customers observed intermittent SQL exception errors such as Cannot execute INSERT/DELETE/UPDATE/SELECT FOR UPDATE in read-only mode during runtime processing of messages. The patch fixes this issue.

December 2022

Software Increment: 2210

Technical Component	Software Version	Description
Cloud Integration	6.33.26	With this patch, the following change has been implemented: A check has been introduced for the process of copying an integration adapter package from the Discover to the Design section. This new check validates if the integration adapter contained in the package is supposed to be available for your service plan.
Integration Advisor	1.67.5	There have been issues with the mapping documentation and the import and export feature. This patch fixes the issue.
Cloud Integration	6.33.25	With this patch, the following change has been implemented: The pretransformation feature was available for Integration Advisor as part of standalone SAP Cloud Integration (but not in SAP Integration Suite). With this patch, the feature is also available for SAP Integration Suite.

November 2022

Software Increment: 2209

Technical Component	Software Version	Description
Integration Advisor	1.66.9	A coding error has resulted in customizing data being wrongly displayed. This patch fixes the issue.
Cloud Integration	5.40.9	There was the issue that messages aren't processed by the following sender adapters: SOAP SOAP 1.x, SOAP SAP RM, XI, IDoc. This patch fixes the issue.
Migration Tooling (Beta)	6.32.24	There have been issues with certain Migration Tooling (Beta) templates. This patch fixes these issues.
Cloud Integration	6.32.23	There was an issue with the integration flow simulation feature (mapping simulations run into a timeout). This patch fixes the issue.
Cloud Integration	6.32.19	There had been the following issue with the XI receiver adapter. Under certain conditions, the adapter doesn't send messages as expected: The message contains an attachment that already contains a content

Technical Component	Software Version	Description
		ID. The content ID is used to create the request without checking if the content ID is in accordance with the specification. The patch provides a bugfix that enables the XI receiver adapter to generate a new content ID in the described case and, that way, to make sure that message processing doesn't fail.
Cloud Integration	6.32.18	There had been the following issue: An integration flow disappears from the Web UI after the first deployment, when the runtime location status transitions from NEW to ACTIVE. This patch fixes the issue.

October 2022

Software Increment: 2208

Technical Component	Software Version	Description
Cloud Integration	6.31.31	Customers are unable to access/navigate to the Integration Suite capabilities via home page, as the XUSAA token is giving an incorrect URL and impacting the token exchange flow. This patch fixes the issue.
Cloud Integration	6.31.30	Integration flow simulations time out if kafka consumer rebalancing occurs during the simulation process, impairing the use of the simulation feature. This patch fixes the issue.
Cloud Integration	6.31.26	If a stuck artifact is identified, deployment of new artifacts is disabled. This fix allows deployments/undeployment of artifacts and improves exception handling during the same.
Cloud Integration	6.31.25	Due to a missing path in the Integration Studio router, customers could not use the CPI Discovery feature for APIM in the Integration Studio context. This patch fixes the issue.
Cloud Integration	6.31.24	Due an existing bug, message processing can occasionally continue even if an artifact was undeployed. This patch fixes the issue.
Cloud Integration	6.31.23	There were issues when using the environment variable <code>IT_TENANT_UX_DOMAIN</code> for URL calculation on Integration Studio tenants, as the environment variable value was changed. The value of <code>IT_TENANT_UX_DOMAIN</code> has been reset, and we introduced a new environment variable <code>IT_TENANT_ISTUDIO_UX_DOMAIN</code> to be used on Integration Studio tenants for URL calculation pointing to the Web UI. For non-Integration-Studio tenants you still have to use <code>IT_TENANT_UX_DOMAIN</code> for all URL calculations.
Cloud Integration	6.31.20	There is an issue in the JMS queue, as large messages are blocking the queue and stopping all messages deliveries. This patch fixes the issue in the API.
Cloud Integration	5.39.13	The integration flow deployment is impossible if a test artifact stays in deploying state. This patch fixes this issue.

September 2022

Software Increment: 2207

Technical Component	Software Version	Description
Cloud Integration	6.30.16 5.38.13	There was an issue with the XMLFactory due to missing libraries in OData V2 receiver. This patch fixes the issue and contains log enhancements for trouble shooting.

Technical Component	Software Version	Description
Cloud Integration	6.30.13	Due to an inconsistency in the unlocking/unlocking processes, timer based integration flow executions are overlapping. This patch fixes the issue
Cloud Integration	6.30.11	There is an issue in the simulation request execution for integration flows, message mappings and other artifacts. This patch fixes the issue.
Cloud Integration	5.38.12 6.30.10	In some cases, DB connections are not closed in case of errors. This patch fixes the issue.
Cloud Integration	6.30.9	The Integration flow deployment was failing due to worker configurations. The default worker was updated configuration to resolve the issue.
Cloud Integration	5.38.11	Due to a stricter URL-parsing introduced by a recent SAP JVM patch, LDAP operations are failing. This patch fixes the issue.
Cloud Integration	5.38.10	There is an issue in reading the metadata of the Cluster Lock table, resulting in locks not being acquired, and message processing failure for timer-based messages, as they are moved to "discarded" state. This patch fixes the issue.

August 2022

Software Increment: 2206

Technical Component	Software Version	Description
Cloud Integration	6.29.17	Improved the application parameters for optimized database communication during "Determination of Artifact status" and "On-demand deployment of Artifacts"
Integration Advisor	1.63.6	Due to missing name documentation for some nodes, the MIGs' migration based on a custom message and containing qualified nodes is not possible. This patch fixes the issue.
Integration Advisor	1.63.6	Since the validation check for messages requiring prefixes is failing, the use of MIG simulation for MIGs based on Edifact, Eancom and X12 TypeSystems is impossible. This patch fixes the issue.
Cloud Integration	6.29.16	The TRM deployment experienced issues due to circular dependencies. This patch fixes the issue.
Cloud Integration	6.29.15 5.37.7	In some cases, prepackaged content from "Order Management Foundation" (OMF) is adding debug information via scripts to MPLs, even if the log level is set to "info" due to a change in CPI behavior. This patch makes sure that the restriction on log level "debug" is considered again by the prepackaged content.
Cloud Integration	6.29.14 5.37.6	The IDoc adapter generates and sets a new DOCNUM field in the payload when starting, but usually, the receiver system rejects this field. This patch deactivates this functionality for the customer.
Cloud Integration	6.29.14	Optimize Database access to improve performance.
Integration Assessment	6.29.13	Activation of SAP Integration Suite Integration Assessment capability was failing. This patch fixes this issue.
Cloud Integration	6.29.12 5.37.5	Concurrent connections to the same FTP server could run into communication errors. This patch fixes this issue.

July 2022

Software Increment: 2204

Technical Component	Software Version	Description
Cloud Integration	6.27.24	Since object store calls are taking more time than expected, artifacts are stuck in intermediate state, impacting subsequent deployment and undeployment. To considerably improve the situation and allow further analysis, we removed redundant object store calls and added a log statement for the time taken for each call .
Cloud Integration	6.27.22	Exceptions are not caught up properly on the worker instance when number ranges artifacts are not able to create DB sequences. This leads to artifacts not being deployed, and blocks further deployments as well. This patch fixes the issues.

June 2022

Software Increment: 2204

Technical Component	Software Version	Description
Integration Advisor	1.61.3	Customers were not able to use the export/import feature if source and target tenant were in the same region. This patch fixes the issue.
Cloud Integration	5.35.9 6.27.16	In some high load situations and using JMS, the following issues have been observed: - Cloud Integration raises the following error message: 400: Too Many Producers. - A JMS thread blocking Issue This patch fixes the issue.
Cloud Integration	5.35.8 6.27.16	If using the AS2 adapter requesting a signed MDN with SHA1 algorithm and configured to verify MIC, verification of the MDN fails because value of Received-content-MIC has been changed from sha1 to sha - 1. This patch fixes the issue.
Cloud Integration	6.27.13	The SAP Master Data Integration adapter didn't process the parameters Address , ODM Entity Type , and ODM Entity Version when configured dynamically using a header or an exchange property. This patch fixes the issue.
Cloud Integration	6.27.11	If there's a runtime error during the start of an integration flow, an error message is shown. This error can also be accessed using the Cloud Integration OData API (Error Information of Runtime Artifact resource of Integration Content API ). However, when displaying the error message using the OData API, an HTTP 204 empty body was retrieved. This patch fixes this issue.
Cloud Integration	6.27.10	Integration flows containing script collections often started before the dependent script collection. This inconsistency of order resulted in messages not being processed. The patch addresses the issue by correcting the deployment order.
Cloud Integration	5.35.6	Cloud Integration displayed the deployment and runtime status as Not Deployed even for deployed artifacts. This patch fixes this issue.

June 2022

Software Increment: 2203

Technical Component	Software Version	Description
Cloud Integration	6.26.21	SOAP-based sender adapters didn't work on Alibaba Cloud. This patch fixes this issue.
Cloud Integration	6.26.20	Integration flows with LDAP connections failed. This patch fixes this issue.
Cloud Integration	5.34.13	There have been issues with performing software update on certain tenants. This patch fixes these issues.

May 2022**Software Increment: 2203**

Technical Component	Software Version	Description
Cloud Integration	6.26.19	Customers have been billed for their test tenant. This patch fixes this issue.
Cloud Integration	6.26.18	On undeployment of erroneous artifact, the artifact state is stuck in stopping state. The patch fixes this issue.
Cloud Integration	5.34.12	If there are multiple worker nodes, the <code>IntegrationRuntimeArtifacts</code> resource of the <code>Integration Content</code> API doesn't return all artifacts' runtime errors. Only those errors are returned that have occurred on the last active node. This patch fixes the issue.
Cloud Integration	5.34.11	This patch fixes the following issue: OData batch processing response parsing was failing because of a bug in the Apache Olingo library. While processing the response, Apache Olingo threw an exception if there was a mismatch of the numbers of requests and responses. With this patch, this situation is now handled smartly.
Cloud Integration	5.34.10	This patch fixes the following issue: On demand deployment and un-deployment has been disabled in presence of stuck artifacts. This patch fixes the issue so that on demand deployments and un-deployments are accepted even if some artifacts are stuck on some workers. Furthermore, the patch provides an improvement of the related error messages.
Integration Advisor	1.60.2	For preprocessing of message implementation guidelines, the qualification of simple content was missing. As a result, wrong runtime artifacts have been created for Cloud Integration. This caused errors in processing of the affected integration flows. This patch fixes the issue.

May 2022**Software Increment: 2202**

Technical Component	Software Version	Description
Cloud Integration	6.25.22	There was the issue that customers have been billed for their test tenant. This patch fixes this issue.
Cloud Integration	6.25.20	There have been issues with processing UPDATE requests to the ValueMapping resource from the Integration Content API of the Cloud Integration OData API . This patch fixes these issues.
Cloud Integration	6.25.19	There have been issues with processing GET requests to the ValueMapping resource from the Integration Content API of the Cloud Integration OData API . This patch fixes these issues.
Cloud Integration	5.33.14	There were issues with connecting to the Partner Directory (HTTP 503 error code was raised). This patch fixes these issues
Cloud Integration	6.25.18	There was an issue with processing timestamps that led to cases where system log entries showed dates in the future. This patch fixes the issue.

April 2022

Software Increment: 2202

Technical Component	Software Version	Description
Cloud Integration	6.25.17	Cloud integration deployments fail after software update, and artifacts can't be viewed in the package view, because of the rate limiting applied to the service manager connection. With this patch, the service manager binding, which is defined as optional, is removed.
Integration Advisor	6.25.17	There is an issue with the Integration Advisor showing a busy status when qualifying a node using a MIG Local Codelist. This patch fixes the issue. Customers are unable to migrate their MIGs if multiple values are selected in a business context. This patch fixes the issue.
Cloud Integration	5.33.13	Before this patch has been applied, artifact redeployment worked in the following way: The existing artifact is deleted, and the new artifact is inserted into the database. These two operations are part of separate transactions. This resulted in issues with the update of certificate-to-user mappings: If a database insertion error occurs during the update of a certificate-to-user mapping, all existing certificate-to-user mappings are deleted. With this patch, both operations are now part of a single transaction. This results in the following behaviour: If inserting a new artifact into the database fails, the existing artifact remains in the database as well. This patch, therefore, fixes the issue with the certificate-to-user mapping update.
Cloud Integration	6.25.15 5.33.12	In some cases, there have been issues with multiple concurrent SuccessFactors OData requests resulting in network errors. This patch fixes the issues.
Cloud Integration	6.25.14	If the creation of a service instance fails, the service instance will be in state Creation Failed and the instance can't be used or deleted. This patch allows the deletion of the service instance.

March 2022

Software Increment: 2201

Technical Component	Software Version	Description
Cloud Integration	6.24.29 5.32.14	In some cases, there have been issues with multiple concurrent SF ODATA requests, as our current connection reuse implementation has shortcomings, and multiple connections are left idle. This is leading to network errors. This patch fixes the issues.
Cloud Integration	5.32.13	As of now, the receiver information of the MPL wasn't available in the Cloud Reporting. It has been added and is now visible there.
Cloud Integration	5.32.12	There have been issues with telemetry results when using script collection or message mapping. This patch fixes the issue.
Cloud Integration	5.32.11	There was an issue during data extraction, preventing the data from being displayed in the Cloud Reporting. This patch fixes the issue.
Cloud Integration	1.58.4	There was an issue in activating multiple versions of a Custom message. This patch fixes the issue. Creation of a draft version of a MIG or a MAG was prohibited, if an active version doesn't exist. This limitation has been removed.
Cloud Integration	6.24.26	A security vulnerability was found that can lead to denial-of-service (DoS) attacks. This patch fixes the issue.
Cloud Integration	5.32.10	There have been issues updating or adding certificates to the keystore, because of a gap in the keystore profile discovery. This patch fixes the issue.
Cloud Integration	6.24.25	There have been issues with execution of timer-based integration flows because of an incorrect datasource reference. This was only seen with subset of tenants who uses JDBC driver for their JDBC adapter scenarios. This patch fixes the issue.
Cloud Integration	6.24.20 6.24.22	There has been an issue that the Cloud Integration database reached its limit that resulted in deployment failures. This patch fixes this issue.
Cloud Integration	6.24.21 5.32.9	There was an issue in the connection management logic of the OData Adapter during retry causing an exception and leading to message failures. This patch fixes the issue.

February 2022

Software Increment: 2113

Technical Component	Software Version	Description
Cloud Integration	6.23.13	A security vulnerability was found that can lead to denial-of-service (DoS) attacks. This patch fixes the issue.
Cloud Integration	6.23.12	Activation of SAP Integration Suite capabilities was failing. This patch fixes this issue.
Cloud Integration	5.31.9	There have been issues with the synchronization of integration content. These issues resulted in situations where integration flows and security material metadata have been removed from the tenant without any notice. This patch fixes these issues.
Cloud Integration	6.23.11	There have been the following issues: Only on trial tenants: Issues using Cloud Integration and deploying artifacts.

Technical Component	Software Version	Description
		Issues with the deployment of Cloud Integration artifacts in the correct sequence. This resulted in situations such like the following one: An integration flow using a value mapping is started before the value mapping. As consequence, the integration flow can't find and process the value mapping at runtime. This patch fixes these issues.
Cloud Integration	5.31.7	There have been issues with the deployment of Cloud Integration artifacts in the correct sequence. This resulted in situations such like the following one: An integration flow using a value mapping is started before the value mapping. As consequence, the integration flow can't find and process the value mapping at runtime. This patch fixes this issue.
Cloud Integration	6.23.10	There have been issues with the Cloud Integration user interface: It was either not opened, or a <i>session expired</i> error message being shown. This patch fixes this issue.
Cloud Integration	6.23.9	In some cases, under high load, customers get JMS transaction-related errors like: <code>javax.jms.JMSEException: Error rollback - internal error (Operation ROLLBACK disallowed in state COMMITTING.)</code> or <code>com.solacesystems.jcsmp.InvalidOperationException: Operation CREATEFLOW disallowed in state COMMITTING.</code> The error messages can vary but they all refer to transactional operations. This patch prevents the occurrence of these errors.

January/February 2022

Software Increment: 2112

Technical Component	Software Version	Description
Cloud Integration	6.22.14	This bugfix prevents Cloud Integration from consuming too many platform resources.
Cloud Integration	6.22.13	The initialization of the repository destination XXX failed because of a library update, causing an RFC principal propagation issue. This patch fixes the issue.
Cloud Integration	6.22.12	In some cases, during integration flow deployment, “the request reply generation” is skipped. This behaviour stops the execution of the integration flow. As the current log size is insufficient to identify the root cause, we increase the log size with this patch.
Cloud Integration	6.22.11 5.30.11	Sometimes, the aggregator component does not release the lock set on the aggregate. This leads to a continuous aggregation until the lock is released manually. This patch fixes the issue.
Cloud Integration	5.30.10	Redundant data logging from CI applications caused the system to report a log volume size issue and increased the load on the platform infrastructure. This patch fixes the issue.
Cloud Integration	6.22.8 5.30.9	There has been an issue when storing the headers of a message processing log in a customer hosted CMS system. This patch fixes the issue.
Cloud Integration	6.22.8	There have been concurrency issues with the process that updates the artifact instances in the database. Because of these issues, it took time for the system to get integration flows from status Starting to status Started . This patch fixes the issue.

Technical Component	Software Version	Description
Cloud Integration	6.22.6	There was the following issue with scenarios using the HTTP receiver adapter: If the set-cookie header (from the response message) isn't stored in a cookie, in some cases the HTTP Session Reuse feature doesn't work. The reason is that the Cloud Integration runtime code doesn't check if the set-cookie header is case insensitive. This patch fixes the issue.

December/January 2021

Software Increment: 2110

Technical Component	Software Version	Description
Cloud Integration	5.28.17	Cloud Integration raised an error when transferring ELSTER messages (for LStA, LStB and ELStAM) using the ELSTER adapter. The error was caused by a wrong version of the ERiC library included in Cloud Integration. This patch fixes the issue. Now, the correct ERiC library version is included. For more information, see SAP Note 3137796 🙌.
Cloud Integration	6.20.22	There have been issues with the deployment of integration content. This patch fixes these issues.
Cloud Integration	6.20.20 5.28.16	A concurrency conflict caused an error in JMS processing and resulted in a retry in message processing. This patch resolves this issue.
Cloud Integration	6.20.19	There have been issues with the deployment of Number Range Object artifacts. This patch fixes these issues.
Cloud Integration	6.20.17 5.28.15	Improvement in Content Security Policy (CSP).

October/November 2021

Software Increment: 2109

Technical Component	Software Version	Description
Cloud Integration	6.19.25	There have been issues with successfully saving changes for existing integration flows, as well as with integration flows not getting unlocked. This was corrected.
Cloud Integration	6.19.24 6.19.23	There have been issues with scenarios using the HTTP receiver adapter with Authentication parameter set to OAuth2 Client Credentials or OAuth2 SAML Bearer Assertion . The default Timeout setting wasn't used as expected. This patch fixes this issue.
Cloud Integration	6.19.22 5.27.14	When using an FTP sender adapter with the Post Processing parameter set to Move File , Cloud Integration waits for a timeout during command completion. The patch fixes the issue so that, first, the command is completed and, secondly, the file is moved. That way, the file processing order is kept.
Cloud Integration	6.19.21	With this patch, we have optimized CPU usage for the Kafka broker.

Technical Component	Software Version	Description
Cloud Integration	6.19.20 5.27.13	With this patch, you get the option to relax the name check in the JSON-to-XML converter. The relaxed check does allow that JSON member names can contain letters, digits, hyphens ('-'), underscores ('_'), periods ('.'), hash characters ('#'), spaces, and at-signs ('@'), but the original name check does only allow JSON names that are compliant with the name specification of XML names (see: https://www.w3.org/TR/2008/REC-xml-20081126/#NT-NameChar). By default, the relaxed check isn't active. In order to activate the relaxed check, open a ticket on component LOD-HCI-PI-CON-SOAP and request that the Java System Property <code>com.sap.it.xmljson.name.checker.simplified.active=true</code> is to be configured for the worker nodes. See also: 3112970 - JSON-to-XML Converter Exception Caused by Invalid JSON Member Name (Knowledge Base Article)
Cloud Integration	6.19.19	Improvement in telemetry (payload size).
Cloud Integration	6.19.18	Improvement in telemetry (message size).
Cloud Integration	6.19.17 5.27.12	This patch fixes the following issues: • Several JMS problems such as blocking undeployment of integration flows have been fixed. • Logging support for scenarios that involve the OpenConnectors adapter has been improved.
Cloud Integration	6.19.15	There have been issues on tenants using the XI adapter with a JMS license but without provisioned broker. This patch fixes these issues.
Cloud Integration	5.27.11	This patch fixes the following issues: • Issues on tenants using the XI adapter with a JMS license but without provisioned broker. • Several JMS problems such as blocking undeployment of integration flows.
Cloud Integration	5.27.10 6.19.14	Messages have no grace period to finish processing, when integration flows are getting undeployed. This patch fixes this issue.
Cloud Integration	5.27.9 6.19.12	The Ariba adapter failed to read the content from the attachment after fetching it from pending queue. This was because of a bug in the adapter, and the content was lost. This patch fixes this issue with the Ariba adapter.
Cloud Integration	5.27.8 6.19.11	The Ariba adapter failed to fetch messages with attachment from pending queue. This patch fixes this issue with the adapter.
Cloud Integration	5.27.7	An unexpected response was obtained while reading keystore secret. This patch fixes the issue with <code>getTokenCredential</code> API to provide expected outcome when reading the <code>TokenCredential</code> from keystore, and when no keystore with specific name is provided.

September 2021

Software Increment: 2108

Technical Component	Software Version	Description
Cloud Integration	6.18.17	This patch fixes the mapping simulation failures because the high number of Kafka producer threads on CO.
Cloud Integration	5.26.15	There was an error during internal database schema optimization. This patch fixes the issue.
Cloud Integration	6.18.16	There have been the following issues: • Users weren't able to open message monitoring and to deploy integration flows. • Integration adapter redeployment failed after deployment failure. This patch fixes these issues.
Cloud Integration	5.26.14 6.18.14	There has been an issue with archiving of message processing logs if a message contained multiple attachments with the same name. This patch fixes the issue.
Cloud Integration	5.26.13 6.18.13	This patch fixes the following issues: • An issue with PGP keyring caching: In some cases, the system didn't cache PGP keyrings correctly on the worker nodes. • An issue with failed messages when using the SuccessFactors OData receiver adapter with the OAuth2 SAML Bearer Assertion authentication option. • An issue with errors during database maintenance
Cloud Integration	6.18.12 5.26.12	There was an issue with the tenant database when Number Ranges artifacts were involved in the scenario. This patch fixes the issue.

August 2021

Software Increment: 2107

Technical Component	Software Version	Description
Cloud Integration	5.25.11 6.17.21	The errors in the interceptors caused by undeployment can lead to a successful HTTP response code for inflight messages although the messages failed. This patch fixes this issue.
Cloud Integration	5.25.10	The integration flow deployment failure issue caused by insufficient column length to store the artifacts description is resolved with this patch.
Cloud Integration	5.25.8 6.17.18	In scenarios with an outbound connection to an on-premise system (with Proxy Type set to On-Premise and the Location ID parameter specified in the HTTP receiver channel), the following issue was observed: Custom headers with a wildcard character (*) specified by the Request Headers parameter have not been sent to the receiver system. As a result, Cloud Integration didn't send any X-CSRF token to the receiver system, causing in an HTTP 403 error. This patch fixes the issue.

16 August 2021

Software Version

SAP Cloud Integration	6.16.23
	5.24.21

Bugfix

When using the OpenConnectors adapter, there have been issues handing responses from the connected system. The system raised error messages that contained the following string: `org.apache.http.TruncatedChunkException: Truncated chunk`.

The issue has been fixed with this patch.

10 August 2021

Software Version

SAP Cloud Integration	5.24.20
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Bugfix

AMQP connections to a AWS hosted ActiveMQ broker fail with a TLS handshake fatal alert .

A new system property is introduced with this patch, that allows to limit the allowed TLS versions for the AMQP adapter to 1.2, since the issue only happens with TLS 1.3. If the system property is not set, the change has not impact at all.

The reported issue is fixed with this patch.

05 August 2021

Software Version

SAP Cloud Integration	5.24.19
	6.16.21

Bugfix

Kafka sender adapters leak file descriptors in case of poll exceptions.

If a Kafka sender runs into an exception during `consumer.poll()`, the consumer is not properly closed. Still, a new one is instantiated and the old one leaks. The old consumer holds up to 4 file descriptors, which is a limited resource. As this resource is much more limited on CF, customers running on CF have a higher risk to run into a subsequent “too many files” exception. Once the node reaches this state, it needs to be restarted.

This issue is fixed with this patch.

28 July 2021

Software Version

SAP Cloud Integration	5.24.17
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Logging Improvement

There is an issue after a node restart: in some cases the authentication filter for the CXF servlet does not get registered on some worker instances. As a consequence all incoming SOAP calls to the affected workers requiring any type of role based authentication fail with a 401 response. (This issue can be resolved through a restart of the affected worker node.)

The patch improves logging for the servlet filter registration to help further analysis of the actual primary issue.

28 July 2021

Software Version

SAP Cloud Integration	5.24.17
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Feature Revert

HTTP OAuth Client was changed with 2016 release for better and robust error handling. The older OAuth client used to continue the message processing despite OAuth errors, eventually to fail in the actual HTTP call. This approach coincidentally worked well and the iflow executions were successful till our recent software update. The changes were proactively reverted to avoid further issues.

21 July 2021

Software Version

SAP Cloud Integration	5.23.16
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Performance Issue

Cleanup Monitoring Data Job is not able to cope with the load on that tenant and DB Space tends to get exhausted. This issue is fixed with this patch.

15 July 2021

Software Version

SAP Cloud Integration	6.15.17
	5.23.15

Bugfix

This patch fixes the issue of socket factory reset during disconnection of the FTP adapter in between subsequent requests.

15 July 2021

Software Version

SAP Cloud Integration	6.15.16
	5.23.14

Bugfix

This patch fixes the transport failure at target tenant during import phase, which was caused by the missing ACL role for technical user in transport service configuration.

15 July 2021

Software Version

SAP Cloud Integration	6.15.15
	5.23.13

Bugfix

If an exception occurred during the closing of an aggregate, in certain cases locks from the in-progress repository haven't been removed, leading to aggregates that have been kept open for long time. This patch ensures that in such exceptional cases the locks are removed from the in-progress repository.

08 July 2021

Software Version

SAP Cloud Integration	6.15.14
	5.23.12

Bugfix

When you use uppercase letters to configure key aliases in SFTP adapter, the integration flow failed to deploy because the alias was not found in the keystore anymore. This casing-related problem with the keystore entries is now fixed.

08 July 2021

Software Version

SAP Cloud Integration	6.15.13
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Bugfix

Software update was failing for tenants deployed on Cloud Foundry environment. With this patch, a code fix was provided to resolve the issue.

07 July 2021

Software Version

SAP Cloud Integration	6.15.12
	5.23.11

Bugfix

Intermittent failure with OData v4 adapter was being improperly logged. This patch fixes this issue with the OData v4 adapter for better logging in case of runtime errors.

02 July 2021

Software Version

SAP Cloud Integration	6.15.12
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Bugfix

The tenant provisioning is unsuccessful, and the tenant URL is exposed to the customer. When you click the URL the tenant does not work as expected.

This issue has been fixed with this patch.

01 July 2021

Software Version

SAP Cloud Integration	6.15.11
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Tenant Provisioning Issue

The timeout value for tenant provisioning has been made configurable.

This has been done with this patch.

30 June 2021

Software Version

SAP Cloud Integration	6.15.10
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Bugfix

There were thread problems occurring under high load.

This issue has been fixed with this patch.

22 June 2021

Software Version

SAP Cloud Integration	5.22.15
	6.14.14

Bugfix

There has been an issue with the FTP adapter when using it with the Cloud Connector (**Proxy Type** set to **On-Premise**). Due to problems establishing the connection through the Cloud Connector, certain files haven't been stored on the FTP server.

This issue has been fixed with this patch.

16 June 2021

Software Version

SAP Cloud Integration	6.14.13
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Bugfix

Deployment of integration flows took an unexpected long time or even failed in certain special situations. This issue has been fixed with this patch.

14 June 2021

Software Version

SAP Cloud Integration	5.22.14
	6.14.12

Bugfix

The following issues have been fixed with this patch:

- Processing of integration flows using JMS queues has been stopped, undeployment failed, and the affected runtime node had to be restarted.
- When using integration flows with a JMS sender adapter, messages went into blocked state if multiple large messages were processed in parallel.

14 June 2021

Software Version

SAP Cloud Integration	5.22.13
	6.14.11

Bugfix

There have been issues when deploying integration flows that contained message mappings using WSDL/XSD with external references.

Further-on, issues have been reported related to the deployment of OAuth2 Client Credentials Artifacts. Both issues have been fixed with this patch.

09 June 2021**Software Version**

SAP Cloud Integration	5.22.12
	6.14.10

Bugfix

Integration scenarios configured with XI adapter against a PO adapter engine stopped working because of a null pointer exception. With this patch, we've removed the null pointer exception.

31 May 2021**Software Version**

SAP Cloud Integration	5.22.11
	6.14.9

Bugfix

Size of package increased after export. This issue has been fixed with this patch.

23 May 2021**Software Version**

SAP Integration Advisor	1.47.3
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Bugfix

While generating runtime artifacts in Cloud Integration, you would have encountered an error in IDoc preprocessing. This was because the preprocessing XSLT was not generated in a way the messages with nested qualifiers could be processed correctly. This issue has been fixed with this patch.

16 May 2021**Software Version**

SAP Cloud Integration	6.13.16
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Bugfix and Performance Improvement

With this patch, we have provided the following:

- Added additional processing steps for better analysis, when you face a problem while storing MPLs.
- A video on SAP Integration Suite, appearing on the Suite dashboard, was locked as private as part of content clean-up. Now, we have changed the video's privacy settings and is made public.

Software Version

SAP Cloud Integration	5.21.8
	6.13.16

Enhancement

After the recent software update, you've encountered an error while transporting integration packages between Cloud Integration tenants. The error here was displayed when you did a configuration check.

Unable to fetch OAuth Token, Token value is null

A bug in the code was identified and is fixed with this patch.

16 May 2021

Software Version

SAP Cloud Integration	6.13.15
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Enhancement

With this patch, we have now enabled the Database (DB) connection pool for the tenant deployed on Cloud Foundry environment and the connections to the DB are regulated.

9 May 2021

Software Version

SAP Cloud Integration	6.13.14
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Bugfix

With this patch, we have resolved an issue with the wrong user view of designer workspace generated in message monitoring. This issue was fixed by using the ID to calculate the URL instead of name.

9 May 2021

Software Version

SAP Integration Advisor	1.47.2
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Bugfix

There was an issue with breadcrumbs navigation from MAG Details screen. The screen was retaining the information related to the previously opened MAG's model and was giving incorrect results or proposals during runtime. With this patch, the issue has been fixed.

1 May 2021

Software Version

SAP Cloud Integration	5.21.06
	6.13.12

Bugfix

With this patch, we have:

- Fixed the synchronization issue caused by duplicate sequence numbers of NRO that got generated because of the race condition. This issue was fixed by introducing Postgres Advisory locks.
- Improved the outbound error handling. Now, you can view the details of the outbound errors, which were displayed directly in MPLs.
- Fixed an issue that occurred while deploying security materials. You couldn't earlier deploy security materials of the type **OAuth2 SAML bearer assertion** for target systems of type SAP BTP, Neo and SAP BTP, Cloud Foundry.

28 April 2021**Software Version**

SAP Cloud Integration	3.36.8
	5.20.10

There have been issues with correct charging of test connections with the purchased SAP Cloud Integration tenant.

These issues have been fixed with this patch.

28 April 2021**Software Version**

SAP Integration Advisor	1.46.4
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There was an issue with integration content development using SAP Integration Advisor.

The mapping list table wasn't loaded with mapping information. Instead of this, the message **No Data** was shown. This patch contains the fix that gracefully handles a key customer scenario affected by this issue.

21 April 2021**Software Version**

SAP Integration Advisor	1.46.3
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There was an issue when SAP Integration Advisor loaded a message implementation guideline (MIG) or a mapping guideline (MAG) with a property of type **direction** or **status**. In certain cases, a health check was caused and an alert in Service Provider Cockpit was initiated. This issue has been fixed with this patch.

16 April 2021

Software Version

SAP Cloud Integration	3.36.7
	5.20.9

Deployment of OData APIs failed.

This issue is fixed with this patch.

09 April 2021

Software Version

SAP Cloud Integration	6.12.12
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Performance issue

There was an XSUAA service broker issue.

This issue is fixed with this patch.

31 March 2021

Software Version

SAP Cloud Integration	3.35.9
	5.19.9
	6.11.13

Bugfix

Update XSUAA to version 1.14.16.

The issue is fixed with this patch.

31 March 2021

Software Version

SAP Cloud Integration	6.11.12
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Bugfix

The OData v4 adapter couldn't serialize decimal values less than "0.1". Any payload containing valid decimal values less than "0.1" was being invalidated by olingo and as such the customer was blocked.

This issue is fixed with this patch.

25 March 2021

Software Version

SAP Cloud Integration	1.45.4
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Bugfix

Customer reported an issue in integration as well as unavailability of Integration Advisor.

This issue is fixed with this patch.

25 March 2021

Software Version

SAP Cloud Integration	6.11.11
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Bugfix

When using AS2 sender channels with Quality of Service *Best effort*, negative MDN has been received. This was also the case when the integration flows have been processed successfully and messages have been reaching the target system.

This issue is fixed with this patch.

17 March 2021

Software Version

SAP Cloud Integration	6.11.10
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Performance Issue

Customer has reported performance problems using ProcessDirect call.

These issues are fixed with this patch.

16 March 2021

Software Version

SAP Cloud Integration	5.19.6
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Bugfix

Customers have reported performance problems processing files via the SFTP adapter.

These problems are fixed with this patch.

15 March 2021

Software Version

SAP Cloud Integration	6.11.9
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Bugfix

Problems with the access policies update (for JMS queues and data store content) are fixed with this patch.

15 March 2021

Software Version

SAP Cloud Integration	6.11.8
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Bugfix

Issues with the Web user interface (problems with filter and search in [Discover](#) section) are fixed with this patch.

10 March 2021

Software Version

SAP Cloud Integration	3.35.5
	5.19.5
	6.11.8

Bugfix

Usage of the OData V4 adapter `$batch` feature was affected due to an issue with the Olingo libraries (hard-coded timeout configured for requests that can't be overridden today).

03 March 2021

Software Version

SAP Cloud Integration	3.34.11
	5.18.13

Bugfix

This patch was provided to enable the JMS retry behavior.

02 March 2021

Software Version

SAP Cloud Integration	6.10.15
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Bugfix

The patch was released to mitigate an issue with the Kafka Root Certification.

26 February 2021

Software Version

SAP Cloud Integration	3.34.10
	5.18.12

Security fix

A security vulnerability was found with XMLBeans (2.6.0 version) and it didn't protect the user from malicious XML input. To prevent such attacks, the XMLBeans was upgraded to 4.0.0.

26 February 2021

Software Version

SAP Cloud Integration	6.10.14
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Bugfix

High usage of CPU and thread exhaustion was leading to downtime of those microservices that were consuming configuration services. This issue has been fixed with this patch.

23 February 2021

Software Version

SAP Cloud Integration	6.10.13
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Bugfix

A bug was discovered with the Solace message broker. With this patch, a fix is applied.

22 February 2021

Software Version

SAP Cloud Integration	3.34.8
	5.18.10
	6.10.12

Bugfix

The patch fixes the following issues:

10/6/26, 11:37 AM

- Users with zonified account type were unable to make RFC connections. This was because the RFC connection was using the Tenant ID instead of Subaccount ID.
- Escape character “_” is not handled properly during migration from Process Integration system to a Cloud Integration tenant.

19 February 2021

Software Version

SAP Cloud Integration	6.10.11
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Bugfix

The URL in instances for OData sender response contained wrong HTTP scheme and port.

This issue has been fixed with this patch.

10 February 2021

Software Version

SAP Cloud Integration	3.33.13
	5.17.16

Bugfix

There was an issue with repeated deployments of artifacts (for example, integration flows) on the worker nodes resulting in system downtimes.

This issue has been fixed with this patch.

9 February 2021

Software Version

SAP Cloud Integration	6.10.10
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Bugfix

There was an issue with the creation of service instances for your tenant.

This issue has been fixed with this patch.

9 February 2021

Software Version

SAP Cloud Integration	6.10.8
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Bugfix

There was an issue with getting updates for integration packages (copied via the OData API) after migration to Cloud Foundry.

This issue has been fixed with this patch.

3 February 2021

Software Version

SAP Cloud Integration	3.33.12
	5.17.15
	6.9.22

Bugfix

The following issues have been fixed with this patch:

- The system by default appended a charset parameter in the Content - Type header when the content-type was `text/*`.
This caused problems for endpoints that do not expect charset parameter.
- In certain cases, scenarios using the Mail sender adapter run into concurrency problems that delayed message processing. It could happen, that unrelated integration flows with a Mail sender adapter shared the same lock. Note that the Mail sender adapter requires a lock in order to poll messages (**Lock Timeout** parameter).

1 February 2021

Software Version

SAP Cloud Integration	6.9.20
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Bugfix

The issue with system deployment getting stuck in the step for the App Router is solved.

28 January 2021

Software Version

SAP Cloud Integration	6.9.19
	3.33.11
	5.17.14

Bugfix

The issue with the "parser expanding external entities by default. An attacker can nest external entities in what is known as a "Billion Laughs Attack" that causes excessive memory consumption and potentially crash the Jersey instance" has been fixed.

Furthermore, a Job Scheduler issue has been fixed.

27 January 2021

Software Version

SAP Cloud Integration	6.9.18
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Design issue fix

We provide the patch for : "the lack of loading of the keys of resources in the root web app in the corresponding framework".

20 January 2021

Software Version

SAP Cloud Integration	3.31.29
	5.15.30

Bugfix

When a GET request is triggered for OData Sender, integration flow for different operation is getting triggered. Logs have been added to analyze the issue.

20 January 2021

Software Version

SAP Cloud Integration	3.31.28
	5.15.29
	6.7.44

Bugfix and Performance improvement

We improved the performance and the fix solves the issue with integration flow deployment stuck in "starting" state.

18 January 2021

Software Version

SAP Cloud Integration	5.15.28
	6.7.43

Bugfix

The fix ensures that no Empty Cookie header is populated. Cookie header is added only when there are valid cookies stored for the endpoint.

14 January 2021

Software Version

SAP Cloud Integration	3.31.27
	5.15.26

Bugfix

A new system property is set on the worker nodes of the customer tenant, to be able to update the customer to the newer CPI release.

09 January 2021**Software Version**

SAP Cloud Integration	5.15.25
	6.7.40

Bugfix

An integration flow deployed with the HTTP Receiver adapter (version: 5.x) encountered an error stating “Too many open files”. This error occurred when the file descriptors upper limit is reached. With this patch, the issue has been fixed by sharing the resources across all adapters in the tenant.

08 January 2021**Software Version**

SAP Cloud Integration	3.31.25
	5.15.24
	6.7.39

Bugfix

When you encounter the [UniquenessViolationException](#) while importing an integration package from TMS/CTS+, the package and artifacts gets locked. As a result, the subsequent import fails and throws an “Could not acquire lock” error. You can release the lock by logging in to your Cloud Integration tenant and unlocking the package and the artifacts.

21 December 2020**Software Version**

SAP Cloud Integration	3.31.23
	5.15.22
	6.7.37

Bugfix

In version 1.0 of the ProcessDirect adapter, a regex constraint check was provided for the **Address** field that didn't allow the address to end with a special character. This check has been removed in version 1.1 of the adapter.

Assume that before the bugfix the address `MY_ADDRESS_{{My_ID}}` has been specified. In that case, the value `My_ID` couldn't be found by the system, which resulted in an effective address `MY_ADDRESS_`. As a consequence, an infinite regex check loop crashed the design service.

18 December 2020

Software Version

SAP Cloud Integration	3.31.22
	5.15.21
	6.7.36

Bugfix

TLS connection error occurred when you deployed an integration flow that had OData or AS2 receiver adapter. This error was caused when the tenant keystore contained multiple key pairs. We have resolved this error by changing the keystore (from the JCEKS keystore to the IAIIKeyStore).

18 December 2020

Software Version

SAP Cloud Integration	6.7.35
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Bugfix

Fixed: Ongoing issues with outbound message failures. (AS2 & OData receiver adapter)

15 December 2020

Software Version

SAP Cloud Integration	6.7.35
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Bugfix

This patch fixes the issue with the software update (DB call).

09 December 2020

Software Version

SAP Cloud Integration	6.7.31
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Bugfix

There were issues with *xstream* version 1.4.11. Upgrade to version 1.4.14 fixes this issue.

09 December 2020

Software Version

SAP Cloud Integration	6.7.31
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Bugfix

This patch fixes an issue in message mapping. (An error message came up if the number of nodes in message mapping was greater than 10).

26 November 2020

Software Version

SAP Cloud Integration	5.14.18
	6.6.20

Code Change

Content-Length entity header belong to an HTTP request didn't pass through the HTTP servers leading to a failure of an integration scenario. With this patch, a code fix was provided to the library used by the HTTP Receiver adapter to rectify the failure.

24 November 2020

Software Version

SAP Cloud Integration	3.30.22
	5.14.18

Bugfix

When using RFC adapter, you would have encountered "Maximum number of RFC connections reached" error. This patch enables JCo connections log for monitoring the connections. The information in the log helps you to troubleshoot the reason during communication.

20 November 2020

Software Version

SAP Cloud Integration	3.30.21
	5.14.17

Code Change

An error occurred while accessing the data source configuration due to absence of null check. With this patch fix, null check is implemented along with the necessary actions to be performed upon a null value detection. Even if data source configuration is not redeployed after software update, the system will work as expected without any error.

12 November 2020

Software Version

SAP Cloud Integration	3.30.20
	5.14.16

Bugfix

This patch resolves issues with delayed message processing in case JDBC data sources are involved.

11 November 2020

Software Version

SAP Cloud Integration	3.30.18
	5.14.14
	6.6.18

Bugfix

This patch resolves an issue with the number range service (returned duplicate number in concurrent scenarios).

3 November 2020

Software Version

SAP Cloud Integration	3.30.17
	5.14.13
	6.6.17

Code Change

This patch fixes an issue found when you use JDBC Receiver adapter to execute stored procedure on SAP ASE database.

29 October 2020

Software Version

SAP Cloud Integration	3.30.16
	5.14.12

Code Change

Japanese characters were lost from the payload when UPSERT requests and responses are made to SuccessFactors system from an integration flow. The SuccessFactors OData Adapter didn't correctly handle the UTF-8 encoding of Japanese characters in the payload. The issue is resolved with this patch.

28 October 2020

Software Version

SAP Cloud Integration	5.14.11
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Bugfix

This patch resolves an incompatibility of Java Cryptography Extension (JCE) policy with newer version of Apache Karaf runtime 2.56.0.

27 October 2020

Software Version

SAP Cloud Integration	3.30.15
	5.14.10
	6.6.16

Design Change

This patch resolves the issue related with JDBC Receiver adapter. Now a mechanism is introduced to handle the situation wherein the Kafka event is not received while creating a data source. This makes the adapter more robust at runtime.

21 October 2020

Software Version

SAP Cloud Integration	3.30.14
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Bugfix

An error occurred while deploying credentials from Data Store at runtime. When you edit and save the credentials in the Data Store, the credentials were supposed to remain in the password storage. But during deployment these credentials were deleted from the password storage. This issue is resolved with this patch.

20 October 2020

Software Version

SAP Cloud Integration	3.30.12
	5.14.10
	6.6.15

Code Change

This patch provides an update for the following adapters:

- OData V2
- SuccessFactors OData V2
- OData V4

The OData query had generated a faulty XSD schema and the schema was unusable in the mapping step modeled in an integration flow. With this patch, the fault in XSD has been fixed.

10 October 2020

Software Version

SAP Cloud Integration	3.29.24
	5.13.13

Design Change

With this patch update, we have now optimized the credential deployment to reduce the delay in refreshing the credentials during runtime.

07 October 2020

Software Version

SAP Cloud Integration	4.16.26
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Bugfix

This patch solves the CPI Number Range Service Duplicate issue.

07 October 2020

Software Version

SAP Cloud Integration	4.16.25
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Design Change

The TRM now aborts the software update task for a tenant that has been stuck in a particular state for a certain amount of time, in order not to block other tenants. This issue is resolved with this patch.

01 October 2020

Software Version

SAP Cloud Integration	3.29.23
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Bugfix

This patch fixes the issue with the mailbox locking mechanism: the username will now be considered for the lock, so that there will be no concurrent polling of the mailbox.

01 October 2020

Software Version

SAP Cloud Integration	3.29.23
	5.13.12

Bugfix

This patch fixes the issue with the SFTP server: temporary file name will now be set correctly on SFTP server.

29 September 2020

Software Version

SAP Cloud Integration	3.29.22
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Feature Gap

This patch

- allows you to handle endpoints in exception subprocesses.
- improves parent-child relationship in case of multiple nesting levels between the main integration process and the local integration process.
- allows to treat all as single connection based on property if property names are same, but with lowercase/uppercase letters in between.
- allows the handling of SAP endpoints originating from the same tenant host.
- fixes the issues related to the parsing host `https://host:port/${property.path}`
- allows the adding of integration flow details along with sender adapter type/receiver adapter type info, if multiple connections exist in the same integration flow.

25 September 2020

Software Version

SAP Cloud Integration	3.29.21
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Feature Gap

Ensured UTF-8 encoding was being honored while providing argument to XML Parser. The default encoding of the library ISO-8859-1 was being followed before.

25 September 2020

Software Version

SAP Cloud Integration	3.29.20
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Feature Gap

The new alert for critical Solace queue capacity does now also yield when APIs throw an exception.

25 September 2020

Software Version

SAP Cloud Integration	3.29.19
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Extended Feature

JMS move feature was extended by adding short sleep statements and optimized connection handling . The patch enables the extended move feature.

23 September 2020

Software Version

SAP Cloud Integration	4.16.21
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Bugfix

This fix removes the validation on parameter length that was introduced as part of a security feedback for the Content Transport implementation in CF, and allows now SAP shipped standard content being transported in customer's QA tenant if the package ID is larger than 190 characters.

18 September 2020

Software Version

SAP Cloud Integration	4.16.20
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Bugfix

This patch allows the outbound communication to work on the tenants using sap_cloudintegration certification (with an SAP provided keypair), by identifying and migrating all certificates without complete chain to have the complete chain.

16 September 2020

Software Version

SAP Cloud Integration	3.17.4
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Bugfix

This patch removes the incompatible change introduced by platform with CIS 2.0.

16 September 2020

Software Version

SAP Cloud Integration	3.17.4
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Bugfix

This patch updates the `sap_cloudintegrationcertificate` with missing certificate chain.

08 September 2020

Software Version

SAP Cloud Integration	3.29.17
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Bugfix

This patch release fixed an issue related to the cluster lock mechanism (logging has been improved).

02 September 2020

Software Version

SAP Cloud Integration	3.29.15
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Bugfix

This patch release provides the following bug fixes:

- Platform resiliency related to temporary network unavailability has been improved. In particular, temporary network unavailability for the password store caused the caller applications to fail. With this fix, there won't be any downtime any more caused by such issues. Before the fix, calls to the password store used to fail with this exception: `[CONTENT] [CONTENT_DEPLOY][ErrorRetrievePassword]:Error retrieving password for alias: <alias_name>, An error occurred while trying to get password with alias.`
- A two-minutes spike has been observed in every call that is made to the platform in a high fidelity usage (where accuracy on the retrieval of password fetch in a high load situation is expected). This issue reduced the turnaround time and induced delays in message processing. With the patch, the performance has been improved. As a result, you do not face any delay in a continuous message processing load any more.

- A change has been made in the data store coding that might work around an issue with the JDBC driver for the tenant database (error message: Cursor 'jconnect_implicit_17' was declared with a FOR UPDATE clause. This cursor was found to be read only.).

31 August 2020

Software Version

SAP Cloud Integration	3.29.14
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Bugfix

The fix is related to the ELSTER adapter. The version of the ERiC libraries has been updated to 31.7.8.0 according to a requirement by the German Tax authorities.

06 August 2020

Software Version

SAP Cloud Integration	4.15.14
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Bugfix

1. Life time reduction of refresh token issue

With the fix on expiry of the refresh token, a new token will be requested for all CF Trial and Prod. customers.

2. Software update got stuck due to unreceived notifications of deleted CPI tenants.

The fix avoids calling CIS and update task can be performed.

06 August 2020

Software Version

SAP Cloud Integration	3.28.13
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Bugfix

When high number of rfc calls are executed in parallel(more than 50),it can be that response for two different requests are getting mixed up

Code is now more threadsafe using threadlocal variable.

04 August 2020

Software Version

SAP Cloud Integration	4.15.13
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Bugfixes

The fix is to construct always CMDVariantUri instead of using already existing one

Regeneration of Custom Adapter & fetch All Capabilities APIs has to be thread Safe in Regeneration Tool. The fix makes those APIs Thread Safe.

In Message Flow Check there is a Validation to check whether Nested Externalized Values are present in Prop file or not. Parameterized Values are coming as empty to Message Flow Check because the values are not set. The fix is we set the values so that the Validation Error doesn't occur.

Application logs have been strengthened.

IFLW file doesn't have BPMNElementId for BPMNPlane, this value is collaborationId. Due to this, there is a Null Pointer Exception. The fix adds a Null check. If collaborationId is null, then the will be from Collaboration Model.

28 July 2020

Software Version

SAP Cloud Integration	4.14.17
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Bugfixes

With this patch, the "Retry Exhausted" issue on the SAP Integration Suite is fixed.

15 July 2020

Software Version

SAP Cloud Integration	3.27.13
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Bugfix

With this patch, the JDBC adapter has been enabled to support batch processing using PreparedStatement objects for sending SQL statements to the database, provided the system property for alias has been set.

9 July 2020

Software Version

SAP Cloud Integration	3.27.12
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Bugfix

When you use a Parallel Splitter step in an integration flow configured with an OData v4 receiver adapter, the message splitting fails due to the sharing of tenant resources by these multiple split messages. The issue with parallel processing is fixed with this patch.

7 July 2020

Software Version

SAP Cloud Integration	3.27.11
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Bugfix

Uploading key-pairs into your tenant keystore using the signature algorithm [SHA256withRSAandMGF1](#) in the X.509 certificate was not possible. But now with this patch you can upload them to the keystore.

7 July 2020

Software Version

SAP Cloud Integration	3.27.10
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Bugfix

An exception occurred while processing message in JDBC adapter. This was caused due to high memory consumption from the destination database. To resolve this *PreparedStatement* was used and *dynamic_prepare* property was enabled. This patch contains these changes made to the JDBC adapter.

30 June 2020

Software Version

SAP Cloud Integration	4.14.12
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Bugfix

- Software update/rollback was failing for couple of tenants deployed on Cloud Foundry environment. With this patch this issue is resolved.
- A bug was found while instantiating the configuration service. This issue occurred due to a bug in the code and with this patch the issue has been fixed.

30 June 2020

Software Version

SAP Cloud Integration	3.27.6
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Bugfix

Umlaut or special characters, found in the request and response payloads, are not supported by OData v4 receiver adapter. Earlier these characters were replaced by some unknown values. With this patch OData v4 receiver adapter supports umlaut characters.

16 June 2020

Software Version

SAP Cloud Integration	3.26.9
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Bugfix

The enhancements for session csrf reuse which cause intermittent failure with session reuse are reverted with this patch version.

16 June 2020

Software Version

SAP Cloud Integration	4.13.17
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Downport of prescript

Format has been adapted: additional lines were removed.

15 June 2020

Software Version

SAP Cloud Integration	3.26.8
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Bugfix

The following issue has been solved:

Issue in the mail adapter. A configuration change solved the issue.

07 May 2020

Software Version

SAP Cloud Integration	3.24.14
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Bugfix

The following issue has been solved:

In case of an error with a connection using the OData Sender adapter, incorrect JSON content has been returned.

04 May 2020

Software Version

SAP Cloud Integration	3.24.12
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Bugfix

API used to perform *DELETE* operation for removing integration packages (configure-only content) from your workspace failed to respond. This issue has been fixed with this patch.

03 April 2020

Software Version

SAP Cloud Integration	3.22.17
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Bugfix

When inputs for the ASE database service (connected with the JDBC adapter) occurred at a high rate, in certain cases the database pool reached its limit and caused an insufficient procedure cache error. With this patch the issue has been fixed.

25 March 2020

Software Version

SAP Cloud Integration	3.22.16
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Bugfix

The following issues have been solved:

- You encountered an error while configuring the [Key Info Content](#) parameter in XML Digital Signer (version 1.2). Now this error is fixed.
- A bug was noticed while configuring relative XPath expression in General Splitter. As per the configuration the splitter processed only the first entries of the payload and the rest of the entries in the payload were ignored. The bug has been fixed now.

20 March 2020

Software Version

SAP Cloud Integration	3.22.15
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Bugfix

Database query timeout value was changed from seconds to milliseconds causing the DB index update job to fail. This was also the cause of incompatibility issue with the EclipseLink update. With this patch the issue has been fixed by setting the unit of time to seconds.

17 March 2020

Software Version

SAP Cloud Integration	3.22.14
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Bugfix

Refer resolution provided for OData Sender adapter on 6 March, 2020 (version 3.21.23).

6 March 2020

Software Version

SAP Cloud Integration	3.21.23
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Bugfix

The following issues have been solved with this patch:

- When using an OData Sender adapter, the conversion of the original request payload to XML was not working as expected. This was because the EDMX schema of the OData Sender had same navigation property name for different navigation entities.
- During high load message processing scenarios, BAT worker nodes used to get into out of memory state. Unfortunately, proper logs weren't generated to analysis the issue. With this patch we put a mechanism in place to capture the runtime behavior logs for better analysis.

28 February 2020

Software Version

SAP Cloud Integration	3.21.21
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Bugfix

When sending an acknowledgment, the AS4 adapter was failing while parsing the document. This was due to the fact that the incoming document did not contain a namespace prefix. With this patch, this condition is now handled in the right way.

27 February 2020

Software Version

SAP Cloud Integration	3.21.20
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Bugfix

Processing of integration flows that contained HTTPS sender adapter version 1.0.0 failed.

With this patch, this issue has been fixed

21 February 2020

Software Version

SAP Cloud Integration	3.21.18
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Bugfix

Messages (containing mail attachments) were not processed by the receiver system due to wrong transfer encoding on mail attachments.

With this patch, this issue has been fixed

11 February 2020

Software Version

SAP Cloud Integration	3.20.14
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Bugfix

Scenarios using AS4 in the PEPPOL network were failing after partner AS4 endpoints have been updated with the eDelivery profile.

With this patch, this issue has been fixed.

09 February 2020

Software Version

SAP Cloud Integration	3.20.11
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Bugfix

As per the security configuration or requirement you must allowlist XML namespaces used in an integration flow. The new version of XML to CSV converter (1.1) introduced validation to support only the allowlisted namespace. But if you have the older version of the converter (1.0), then the validation caused an issue and the payload returned empty from the converter. With this patch, we have enhanced the versioning of the feature to support your existing integration scenario.

26 January 2020

Software Version

SAP Cloud Integration	4.8.13
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Bugfix

Access was denied to Cloud Integration service broker instance while performing authorization using User Account and Authentication (UAA)-API. This issue is fixed with this patch.

24 January 2020

Software Version

SAP Cloud Integration	3.20.8
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Bugfix

This patch contains a correction of a connectivity problem with the XI receiver adapter that may have occurred under specific circumstances.

20 November 2019

Software Version

SAP Cloud Integration	3.18.10 or 5.3.9
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Bugfix

Runtime node (worker node) crashed when integration flow using ELSTER receiver adapter was deployed. With this patch update, the issue has been fixed and now you can send tax documents to the ELSTER server.

06 November 2019

Software Version

SAP Cloud Integration	3.17.11 or 5.2.9
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Bugfix

The following issues have been solved with this patch:

- Integration flow CRUD actions have been blocked, and customers were unable to modify their integration flows caused by an issue with the handling of the related OSGi bundles of the SAP Cloud Integration framework.
- A memory shortage issue has been solved which was caused by a high number of SAP Cloud Integration OData API requests and a memory leak in the OData API framework.

21 August 2019

Software Version

SAP Cloud Integration	4.2.0
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Bugfix

This patch improves the stability of the tenant onboarding process.

24 July 2019

Software Version

SAP Cloud Integration	3.13.9
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Bugfix

There have been inconsistencies in the infrastructure caused additional nodes being launched after restart of multiple runtime nodes. This issue has been fixed with this patch.

18 July 2019

Software Version

SAP Cloud Integration	3.13.8
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Bugfix

An error occurred in the design workspace when you imported an earlier version of an integration package, and the actual integration package (to be overwritten by the import) had some unsaved changes. This error was due to an issue with the auto-save functionality and has been fixed with this patch.

09 July 2019

Software Version

SAP Cloud Integration	3.13.7
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Bugfix

Tenant update failed because latest tenant cluster model was unavailable in the tenant management node. This issue has been resolved with this patch and now the latest tenant cluster is available on the tenant management node.

30 June 2019

Software Version

SAP Cloud Integration	3.11.26
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Bugfix

You would have encountered 403 Forbidden error while connecting with SAP's Europe data center. The cause of the error was due to a problem occurred while establishing TSL/SSL communication. With this patch the issue has been resolved.

14 June 2019

Software Version

SAP Cloud Integration	3.11.24
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Bugfix

This patch fixes the issue that the rendering of the mapping editor depended on the order in which mapping steps are performed. A possible implication was that when the customer modified its mapping, the editor stopped opening.

Software Version

SAP Cloud Integration	3.11.22
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Bugfix

After updating the tenant with the latest Cloud Integration software version, the following improvements are available:

- Integration flows are deployed faster on the runtime node (worker node) during software update and unplanned restarts or crashes.
- The length of the key material supported has been increased.
- Memory-related crashes with the tenant management node due to incorrectly deployed content has been fixed.

4 June 2019

Software Version

SAP Cloud Integration	3.11.21
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Bugfix

Deployment of integration content was failing due to an issue with persisting a certain artifact in the runtime.

This issue has been fixed with this patch.

31 May 2019

Software Version

SAP Cloud Integration	2.53.19 or 3.11.20
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Bugfix

You have integrated a Cloud Integration tenant with Ariba system and have experienced a missing multipart payload during inbound AS2 communication. This issue was caused while verifying the signature of the multipart files and it was found that a mandatory **AS2 header: Content-Description** was missing. With this patch, the header is made optional and the error is resolved.

23 May 2019

Software Version

SAP Cloud Integration	2.53.18 or 3.11.19
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Bugfix

With this patch, the size limitation of the keystore and certificate-to-user mapping (originally, 1 MB) has been increased to 2 MB.

21 May 2019

Software Version

SAP Cloud Integration	2.53.17 or 3.11.18
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Bugfix

The optional [Scope](#) parameter has been added to the [OAuth2 Credentials](#) artifact (when as [Grant Type](#) the option [OAuth2SAMLBearerAssertion](#) is selected).

17 May 2019

Software Version

SAP Cloud Integration	2.53.16 or 3.11.17
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Bugfix

If you have experienced integration flows in failed state after deployment with an error `class not found exception javax.sql.DataSource`, this issue occurred due to an error in the backend. With this patch, the issue is fixed.

16 May 2019

Software Version

SAP Cloud Integration	2.53.15 or 3.11.16
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Bugfix

If you use XSLT mapping version 1.2 for processing a payload that has an attachment exceeding 100 KB, then the message processing goes to failed state. This issue has been fixed with this patch.

15 May 2019

Software Version

SAP Cloud Integration	2.53.14 or 3.11.14
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Bugfix

This patch fixes all performance related issues experienced during monitoring phase. Due to this issue, the integration flow deployment took longer than usual time.

14 May 2019

Software Version

SAP Cloud Integration	2.53.13 or 3.11.13
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Bugfix

Due to a recent change in software, if there are multiple components with same name then the integration flows corresponding to those components stay in STARTING state. This patch version fixes this issue.

4 May 2019

Software Version

SAP Cloud Integration	2.52.12 or 3.10.12
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Bugfix

If you store JSON attachment in a message using Content Modifier, the content of the attachment was not being displayed on the monitoring page. This issue has been resolved with this patch. Now the content of the attachment is being displayed in the monitoring page.

3 April 2019

Software Version

SAP Cloud Integration	2.51.13 or 3.9.13
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Bugfix

Recent optimizations to the message mapping feature missed to account for an edge case with variables. Due to this, the XPath expressions used in the variables got corrupted on edit and save of mappings. The patch fixes this problem.

2 April 2019

Software Version

SAP Cloud Integration	2.51.12 or 3.9.12
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Bugfix

You would have encountered a failure while deploying an integration flow. This issue is due to a limitation in the field size for artifact metadata (to be stored during deployment). The patch fixes this problem.

7 March 2019

Software Version

SAP Cloud Integration	2.50.16 or 3.8.15
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Bugfix

You would have encountered a failure while deploying an integration flow. This issue is due to the incompatibility of the manifest file (version 1) with Karaf runtime. Now the issue is fixed with this patch update and you need to redeploy the integration flow.

27 February 2019

Software Version

SAP Cloud Integration	2.50.15 or 3.8.14
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Bugfix

During integration flow deployment the file upload scanner was rejecting the integration flow bundle. The issue occurred because the file upload scanner identified the bundle as corrupted and rejected the bundle. The issue has been resolved with this patch release.

24 February 2019

Software Version

SAP Cloud Integration	2.50.14 or 3.8.13
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Bugfix

IntegrationRuntimeArtifacts API was designed to deploy the integration flow bundle sent through the API at runtime. It was found during deployment it was not considering the configured values. This bug has been fixed in this patch.

It is recommended to deploy any design time integration flow artifact by using DeployIntegrationDesignTimeArtifact entity found in IntegrationDesignTimeArtifact API.

19 February 2019

Software Version

SAP Cloud Integration	2.50.13 or 3.8.12
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Bugfix

The design time page was not responding after the content package update. This issue affected all SAP Cloud Integration tenants. It occurred due to an unexpected code error in the back end. A patch has been released and the issue is resolved.

16 February 2019

Software Version

SAP Cloud Integration	2.50.12 or 3.8.11
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Bugfix

User was not able to set the scope for [OAuth2](#) client credentials due to a bug. This issue has been fixed now.

31 January 2019

Software Version

SAP Cloud Integration	2.49.14
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Bugfix

You were unable to open Message Mapping and an error was displayed. This issue was caused because the schema contained a definition that had a very huge value, such as "maxOccurs=9999999". This issue has been fixed.

30 January 2019

Software Version

SAP Cloud Integration	2.49.13
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Bugfix

Integration flows were not getting deployed if the HTTPS Sender endpoint contained "*", because the wildcard was not recognized. This issue has been fixed now.

22 January 2019

Software Version

SAP Cloud Integration	2.49.12
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Bugfix

An issue was found in integration flow scenarios connected to an OData API using OData Receiver adapter. During Update or Delete operations the adapter encountered issues when the Entity set had a composite key. The message processing log, returns an error with the message "The request URI contains an invalid key predicate". This issue has been fixed with this patch.

7 January 2019

Software Version

SAP Cloud Integration	2.48.8
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Bugfix

In HTTP receiver adapter, when you use Client Certificate Authentication and provide a private key alias as a dynamic expression, for example \${header.abcd}. The timeout provided by the customer was not working. The default timeout of 60 seconds was getting automatically applied. This issue has been fixed and timeout provided by the customer is being applied.

17 December 2018

Software Version

SAP Cloud Integration	2.47.20
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Bugfix

The patch fixes an issue where integration flow endpoints were not accessible for a certain time period due to redeployment by the system.

05 December 2018

Software Version

SAP Cloud Integration	2.47.16
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Bugfix

The patch fixes the WSDL download for SOAP adapter endpoints. The error was that the downloaded WSDL did not contain the generated policies anymore.

05 December 2018

Software Version

SAP Cloud Integration	2.47.15
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Bugfix

This patch fixes the following issue with integration content transport:

During the export of a package that contains an artifact that was auto-saved, the auto-saved one is also exported along with the package. This should not be case.

With the patch we have fixed the export of the content package. Also in case the package was previously exported, the system will not allow that the package is imported.

04 December 2018

Software Version

SAP Cloud Integration	2.47.14
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Bugfix

This patch fixes an issue with the WebService interoperability with the tax authority of the Canary Islands.

Furthermore, the following issue has been fixed:

When a tenant has more than one runtime nodes and when an AS2 adapter is involved, it can happen that updates to message processing logs get lost. This is due to the fact that in such a scenario messages can be written to and read from a JMS queue during a short time period where, parallel to this, the processing of the integration flow continues. As such steps are logged in different message processing log (MPL) runs, the involvement of multiple runtime nodes could imply that different MPLs are written nearly at the same time (leading to a Duplicate Key exception).

03 December 2018

Software Version

SAP Cloud Integration	2.47.13
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Bugfix

This patch fixes the problem that the import of an integration package fails in case the package contains auto-saved artifacts.

30 November 2018

Software Version

SAP Cloud Integration	2.47.12
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Bugfix

This patch fixes the following problem: The alias for data source was on class level, which was not working on further calls to an endpoint. This was changed to local variable.

29 November 2018

Software Version

SAP Cloud Integration	2.47.11
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Bugfix

This patch fixes a problem that occurs when you use the SOAP receiver adapter in conjunction with the trace feature (that enables the tracing of the processed payload). Certain combinations of elements in an integration flow can cause a type conversion error during message processing if the message processing log level **Trace** has been activated. The error occurs in the SOAP receiver channel. An example for such a combination is an HTTP call via HTTP adapter before the SOAP call. This can block the integration flow development process.

26 November 2018

Software Version

SAP Cloud Integration	2.47.10
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Bugfix

The new version of the OData V2 adapter (adapter version 1.12) overwrote the existing version (1.11). Therefore, existing integration flows that contained the adapter version 1.11 generated an error during design time. This issue has been solved.

17 November 2018

Software Version

SAP Cloud Integration	2.46.16
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Bugfix

There was a bug in the Apache Olingo library which implied the following behavior: batch responses with exactly 8192 objects resulted in a `BufferOverflowException` which was then followed by a failure of message processing. This issue is fixed now.

Furthermore, this patch provides a resolution for an issue reported on loading of artifact lists in the Design tab of the Web UI.

10 November 2018

Software Version

SAP Cloud Integration	2.46.15
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Bugfix

The test and production tenant configurations for ATO are different. While fetching the SAML token from Vanguard the AS4 adapter uses the destination URL. During this process, the destination URL was assigned to the `AppliesTo` field and this resulted in message failure. This issue is solved by specifying the header `SAP_AS4_Outbound_ATO_SAML_AppliesTo` with a value provided by ATO.

30 October 2018

Software Version

SAP Cloud Integration	2.46.13
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Bugfix

Failed Artifacts Monitor was introduced to report failed artifacts. When this monitor was applied to all clusters, there was an issue occurred to content in failed state. The alert level for failed content was raised to 'Aggregated tenant availability'. This issue is now fixed and actual alerts detected.

29 October 2018

Software Version

SAP Cloud Integration	2.46.12
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Bugfix

While implementing OData APIOData API, an exception was thrown when Deep Insert functionality was used. The error was caused due a bug in the Apache Olingo library. This error has been fixed.

16 October 2018

Software Version

SAP Cloud Integration	2.44.21
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Bugfix

There was no mechanism to detect failed integration flows on worker nodes. Now build a managed component monitor called **ContentStateMonitor** whose display name is "Failed Artifacts Monitor" to check for failed integration flows and it reports if a failed integration flow is found.

10 October 2018

Software Version

SAP Cloud Integration	2.44.20
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Bugfix

Liquibase change logs were not getting applied to some clusters due to already held locks and issues were encountered while launching the clusters. This issue has been fixed by clearing all change logs older than 10 minutes.

30 September 2018

Software Version

SAP Cloud Integration	2.44.19
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Bugfix

An exception was thrown when database could not save custom header attribute values exceeding 200 bytes. It was found that the database reserved only 200 bytes for specific data types. This issue has been fixed and now when characters exceed 200 bytes it is rendered to UTF-8 standards and truncated.

29 September 2018

Software Version

SAP Cloud Integration	2.44.18
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Bugfix

StreamClosedException error occurred while running the **EnrichArtifactManifestTask**. During runtime the task adds `javax.sql` in the manifest file of the OData APIs and integration flow packages. This error has been fixed.

21 September 2018

Software Version

SAP Cloud Integration	2.44.17
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Bugfix

XML escape characters such as `&`, `<`, and `so on`, appeared as it is during runtime and this caused deployment issues. This issue is now fixed.

16 September 2018

Software Version

SAP Cloud Integration	2.44.16
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Bugfix

During design when an external parameter in [Write Variable](#) is selected a check error was thrown. The workaround is if the integration flow is editable and not a standard content, then you must change the value in [Type](#) field to a constant instead of external parameter.

Bugfix

During runtime the XSLT Mapping created empty output files. This issue is now fixed.

15 September 2018

Software Version

SAP Cloud Integration	2.44.15
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Bugfix

The integration flow fails when you add '&' character while externalizing the Endpoint field in a SOAP receiver adapter. This issue is now fixed.

12 September 2018

Software Version

SAP Cloud Integration	2.44.14
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Bugfix

During runtime, integration flow sometimes do not record complete logging information in the MPL. This issue is now fixed.

11 September 2018

Software Version

SAP Cloud Integration	2.44.13
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Bugfix

The route to send asynchronous messages in an XI receiver adapter is not generated during runtime. This issue is now fixed.

11 August 2018

Software Version

SAP Cloud Integration	2.43.12
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Bugfix

Before the values in the [Maximum Characters Retrieved from Tweet](#) field could not be externalized, but now you can externalize the values.

09 August 2018

Software Version

SAP Cloud Integration	2.43.11
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Bugfix

Content Modifier component was not displaying headers or exchange properties for pre-externalized parameters of a Scheduler. This issue has been fixed.

27 July 2018

Software Version

SAP Cloud Integration	2.42.14
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Bugfix

This issue occurs when you have not requested for an acknowledgment and [Process Invalid Messages](#) option is selected during EDI Splitter runtime. If an error occurs at the interchange level of an EDIFACT message type. It was not possible for an integration developer to resolve this error because no exception was thrown. This issue has been fixed and now an exception will be thrown for every error occurring at the interchange level.

21 July 2018

Software Version

SAP Cloud Integration	2.42.13
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Bugfix

Intermittent calls to a Hybris OData endpoint createlatform a new session on Hybris. This causes the Hybris service to return HTTP Status **403** or **502** or **Target Server Failed to Respond** errors with high load. This issue has been fixed.

18 July 2018

Software Version

SAP Cloud Integration	2.42.12
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Bugfix

Integration scenarios using OData V2 adapter returned with HTTP status codes and this impacted the business logic during runtime. This issue has been fixed for OData V2 adapter from version 1.7 and above.

30 June 2018

Software Version

SAP Cloud Integration	2.41.10
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Bugfix

It was observed during OData v2 adapter runtime, **401 Unauthorized** error caused credential cache update to fail. This issue has been fixed.

23 June 2018

Software Version

SAP Cloud Integration	2.41.9
SAP Integration Advisor	1.9.5

Bugfix

After management node restart, the content would get stuck in **Starting** state. This issue is now fixed.

Bugfix

Under heavy load, it was observed that content synchronization was taking a long time and you could not deploy new content. This issue is now fixed.

Bugfix

For SAP Integration Advisor, when a qualifier value contained invalid XML QName character, the generated mapping XSLT was invalid as it contained the invalid character. This is fixed now.

18 June 2018

Software Version

SAP Cloud Integration	2.41.8
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Bugfix

Verification of incoming message signature has been reverted to the old way. It does not involve any changes to channel properties.

2 June 2018

Software Version

SAP Cloud Integration	2.40.14
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Bugfix

A few Ariba transactions fail with exception `com.sap.it.rt.edi.exception.EDIHandlerException: Invalid Payload: EDI EXTRACTION cannot process the payload`. Extra logs have been added to AS2 adapter for checking incoming payload/headers.

29 May 2018

Software Version

SAP Cloud Integration	2.40.13
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Bugfix

This fix is applicable only for integration flows with mail sender adapter. An issue in the mail sender adapter that unpacked the performance has been fixed. Redeploy the integration flows with mail sender adapter to activate the changes.

26 May 2018

Software Version

SAP Integration Advisor	1.18.4
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Bugfix

Upload of new type system revision was failing due to timeout. This has been fixed now.

19 May 2018

Software Version

SAP Cloud Integration	2.40.12
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Bugfix

When you externalize the authentication parameters of SOAP and IDoc adapters version 1.0, it was not being displayed in the integration flow quick configuration. This is now fixed.

Bugfix

In HTTP sender adapter, if you have enabled adapter tracing and send a message with empty body, the message processing would be in **Error** state. This is now fixed.

Bugfix

In case of an exception triggered by [XML Validator](#), message processing log (MPL) attachment, which is an XML Validator error document, was not being created. This is now fixed.

5 May 2018

Software Version

SAP Integration Advisor	1.7.4
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Bugfix

EDI to XML converter would deliver an XML output with namespace that is incorrectly qualified. With this fix, the EDI preprocessing XSLT script corrects the document namespace.

7 February 2018

Software Version

SAP Integration Advisor	1.4.7
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Bugfix

While using SAP Integration Advisor, the mapping functionality would be unavailable until you removed documentation from the message guidelines. This is fixed now.

In SAP Integration Advisor, exported mappings using the UN-EDIFACT Type System failed at runtime because the generated namespace name is incorrect.

27 January 2018

Software Version

Node Assembly	2.36.11
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Bugfix

In scenarios with Ariba receiver adapter, the CamelHttpResponseCode in the exchange was wrongly set as a string instead of an integer. This resulted in you being unable to create RFP and sourcing project. This issue is fixed now.

23 January 2018

Software Version

Node Assembly	2.36.10
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Bugfix

When you try to configure integration flows in standard content like [eDocument: Electronic Invoicing for Spain](#) where base version of SOAP, OData or SuccessFactors adapters are used, you see an empty error and the configuration will not be possible. This is fixed now.

Bugfix

In [XML Signature](#) steps that use XADES-BES with [Data Object Format](#) element, the attribute `ObjectReference` of `Data Object Format` element was being generated without the '#' character at the beginning. This is fixed now.

23 December 2017

Software Version

Node Assembly (Cluster 2.x)	2.35.8
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Bugfix

When the SuccessFactors OData API returned a server error to SuccessFactors OData adapter, the response XML was invalid due to erroneous XML encoding. This issue is fixed now.

30 November 2017

Software Version

Node Assembly (Cluster 2.x)	2.34.9
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Bugfix

Transactions were failing for some partners due to PD cache entries. Fix is provided by invalidating PD cache in such scenarios.

Patched Component

Node Assembly (Cluster 2.x)	2.34.8
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Bugfix

Configuration changes are made to the database to fix excessive resource consumption issue. This fix will not require any additional downtime to reflect the changes.

27 October 2017

Software Version

Patched Component

Node Assembly (Cluster 1.x)	1.55.5
Node Assembly (Cluster 2.x)	2.33.5

Bugfix**Message processing error:**

The integration flow processing fails and throws stack overflow error, if the package contains more number of messages. The issue is fixed by correcting the returned metadata.

20 September 2017

Software Version

Patched Component

Node Assembly (Cluster 1.x)	1.54.7
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Bugfix

Quick configure on the prepackaged integration flows:

- If you try to configure and enter a value for the empty field, the prepackaged integration flow, all empty values for the keys of configurable parameters are updated with the new value and it leads to the wrong configuration. This causes failure during message processing.

i Note

This issue has no impact on the already deployed integration flows.

- When a custom integration flow is built with content modifier in eclipse and is configured in Web UI, the empty values are set for the externalized keys, SAP Cloud Integration throws validation error. The issue has been resolved and setting the empty values for the keys is allowed irrespective of externalizing the parameters in Eclipse or Web UI environment.

20 September 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.32.10
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Bugfix

The following issues have been solved:

Quick configure on the prepackaged integration flows:

- If you try to configure and enter a value for the empty field, the prepackaged integration flow, all empty values for the keys of configurable parameters are updated with the new value and it leads to the wrong configuration. This causes failure during message processing.

i Note

This issue has no impact on the already deployed integration flows.

- When a custom integration flow is built with content modifier in eclipse and is configured in Web UI, the empty values are set for the externalized keys, SAP Cloud Integration throws validation error. The issue has been resolved and setting the empty values for the keys is allowed irrespective of externalizing the parameters in Eclipse or Web UI environment.

AS2 and JMS Sender adapter with dead letter handling: The error occurs during processing of integration flows that may have a AS2 sender adapter and JMS sender adapter with dead letter handling feature. Due to this error, the messages remain infinitely in the processing state. The workaround in such scenario is to disable the dead letter handling feature and retry again.

12 September 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.31.10
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Bugfix

The following issue have been solved:

- In certain cases, the following error message is displayed: *Error during polling for JMS messagesjavax.jms.JMSEException: Error creating consumer - internal error (503: Max Client Queue and Topic Endpoint Flow Exceeded).*

This error only comes up in case large messages are processed in conjunction with external problems (for example, network issues). It is caused by a bug in the code that is in charge of handling large messages. This bug has been removed with the patch.

- Unprocessed messages may remain in the JMS queue.

This situation may occur under heavy load and with several active consumers. To avoid such problems, the settings for the interaction of SAP Cloud Integration software and 3rd-party components have been optimized.

i Note

In order to benefit from this correction, you need to redeploy affected integration flows.

02 September 2017

Software Version

Patched Component

Node Assembly (Cluster 1.x)	1.53.6
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Bugfix

The following issue has been solved:

Task logs have been cleaned up to prevent database bloating which can cause outage.

25 August 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.31.9
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Bugfix

The following issue has been solved:

Web IDE: Due to some unknown issues pop up appeared several times and hindered the usage of the product. Pop up is disabled now.

12 August 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.30.17
Node Assembly (Cluster 1.x)	1.52.12

Bugfix

The following issue has been solved:

Web IDE: Due to some unknown issues pop up appeared several times and hindered the usage of the product. Pop up is disabled now.

12 August 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.30.16
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Bugfix

The following issue has been solved:

OData Query: In case of multilevel response, some attributes were missing when the data was received from the server via SAP Cloud Integration. The properties are now generated correctly.

29 July 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.29.16
Node Assembly (Cluster 1.x)	1.51.13

Bugfix

The following issues have been fixed:

- **WebUI Design Time Issue:** After editing a mapping with target groupings, saving that mapping was not possible. This is fixed now and you can save the mapping.
- **JMS Adapter Message Handling:** You would see message status as **COMPLETED** instead of **FAILED** in case of handled errors.

13 July 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.29.16
Node Assembly (Cluster 1.x)	1.51.13

Bugfix

The following issue has been solved:

- Selecting JMS messages from a queue, in certain situations leads to a minimal or none message throughput and/or errors. This affects JMS adapters as well as monitoring like Lock- and Queue Monitor.

13 July 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.29.15
Node Assembly (Cluster 1.x)	1.51.12

Bugfix

The following issue has been solved:

- OData query calls are not following the metadata constraints.
This scenario occurs when the metadata defines some property to be nullable false, but the property contains null values.

07 July 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.29.13
Node Assembly (Cluster 1.x)	1.51.10

Bugfix

The following issue has been solved:

- A cleanup job removes on a daily basis log configurations for integration flows which do no longer exist. Due to a wrong query, also for existing integration flows with different integration flow ID, the configuration gets removed.

03 July 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.29.12
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Bugfix

The following issue has been solved:

- Due to buffering of JMS consumers, the number of consumers exceeded a limit in the Solace messaging service.

30 June 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.29.10
Node Assembly (Cluster 1.x)	1.51.8

Bugfix

The following issue has been solved:

- When the Data Store is locked due to long running transactions, no monitoring is possible.

28 June 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.29.8
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Bugfix

The following issues have been solved:

- In some cases, you would see a runtime error in integration flows containing **Subprocess** with looping enabled, and invoking a **Local Integration Process** with **Multicast**.

05 June 2017

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.28.10
Node Assembly (Cluster 1.x)	1.50.9

Bugfix

The following issues have been solved:

- Unable to import integration package in WebUI workspace (**Design** tab)

Due to a bug while exporting the integration package, configuration values of **Value Mapping** artifact was getting exported. Import function does not recognize the content and the action to import this package in the workspace (**Design** tab) fails.

01 June 2017 - SAP Cloud Integration

Software Version

Patched Component

Node Assembly (Cluster 2.x)	2.28.9
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Bugfix

The following issues have been solved:

- Problem with state transitions for nodes

There was no transition from ERROR state to LIVE state for a node. Therefore, a node that had moved to ERROR state always remained in state ERROR even after it was restarted and working without an error (since there was no transition to move it back to LIVE). Due to this, component monitors generated alerts.

- Problems when using Multicast in a looping process

18 May 2017 - SAP Cloud Integration**Software Version**

Patched Component

Node Assembly (Cluster 2.x)	2.27.12
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Bugfix

The following issue has been solved:

Decryption of large PGP messages that are encrypted on the file system during streaming is not possible.

12 May 2017 - SAP Cloud Integration**Software Version**

Patched Component

Node Assembly (Cluster 1.x)	1.49.8
Node Assembly (Cluster 2.x)	2.27.11

Bugfix

The following issues have been solved:

- Deployment of scenario fails due to checks error.
- PGP Secret Keyring Artifact synchronization issue.

The following issues have been fixed in the OData adapter:

- Adding a tag `xsi:nil="true"` to explicitly set a field as null on the server is now allowed.

Example: `<DiscontinuedDate xsi:nil="true" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"/>`

- Support of format HH:mm:ss(12 Hours)

Software Version

Patched Component

Node Assembly (Cluster 1.x)	1.49.6
Node Assembly (Cluster 2.x)	2.27.9

Bugfix

The following bug has been fixed: If one integration flow contains the following elements (in conjunction), deploying the integration flow was not possible:

- Multicast
- Exception Subprocess
- Send

Archive - Release Notes for SAP Cloud Integration

Archive of SAP Cloud Integration release notes.

Related Information

[2018 SAP Cloud Integration \(Archive\)](#)

[2017 SAP Cloud Integration \(Archive\)](#)

[2016 SAP Cloud Integration \(Archive\)](#)

[2015 SAP Cloud Integration \(Archive\)](#)

[2014 SAP Cloud Integration \(Archive\)](#)

[2013 SAP Cloud Integration \(Archive\)](#)

2025 What's New for SAP Cloud Integration (Archive)

Core Components, 2025

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
Cloud Integration	Neo Cloud Foundry	ELSTER Adapter Supports New ERiC Version	A new version 43.3.2.0 of the Elster Rich Client (ERiC) library is available for the ELSTER adapter.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.36.* • Increment: 2510	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.72.* - Increment: 2510	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.45.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2510	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Unified Destinations UI in BTP Cockpit	The BTP Cockpit now provides a unified Destinations interface where you can now access comprehensive destination configuration and management tools through the simplified interface. See Creating and Modifying RFC Destination SAP Cloud Integration customers, see Creating and Modifying RFC Destination	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	New Memory Allocation Filter	There is a new filter on the Inspect Memory Usage heatmap that allows you to display the top integration flows that consume the most memory. See Inspect Integration Flows By System Memory Usage	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Importing unsupported function library now available	You can now import unsupported function libraries for a created function library artifact. See Import Function Library from ES Repository	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Enhancements in the Migration Tooling- Standard Approach	For Standard Approach, you can now migrate scenarios with multiple sender channels. Each sender	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			channel will be mapped to an integration flow. See Known Limitations of Migration Tooling					
Cloud Integration	Cloud Foundry	Create and edit service interface artifacts now available	You can create and edit service interface artifacts in your Cloud Integration tenant. See • Creating Service Interface Artifact • Configuring Service Interface	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Dynamic selection for operation details now available	You can now select Dynamic option while configuring OData V2 Adapter. See Configure the OData V2 Receiver Adapter SAP Cloud Integration customers, see Configure the OData V2 Receiver Adapter	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry Neo	Manifest Handling for XI Sender and Receiver Adapters	The XI sender and receiver adapters now handle the content of the manifest of the XI message to the Camel message, which is transformed into properties of the Camel Exchange available in the integration flow. For more details, see: XI Adapter .	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry Neo	New Groovy runtime with upgrade of Groovy script step to version 2.0 is now available	You can now upgrade your Groovy script step to version 2.0, which uses the Groovy runtime 4.0.26. See Upgrading Groovy Script Step SAP Cloud Integration customers, see Upgrading Groovy Script Step	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
Cloud Integration	Cloud Foundry Neo	New guideline to ensure upgrade readiness is now available	Utilizing the new design guideline under ensure upgrade readiness, you can now automatically update your scripts with recommended libraries and classes. See Use recommended libraries and classes SAP Cloud Integration customers, see Use recommended libraries and classes	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry Neo	Dynamic Headers Support for XI Sender and Receiver Adapters	The XI sender and receiver adapters now handle dynamic headers within XI messages, which are transformed into system properties available in the integration flow. With this enhancement, you can access the <code>DynamicConfiguration</code> within the integration flow and read or modify its properties using a content modifier or script. For more details, see: XI Adapter .	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Predecessor ID in Monitor Message Processing	You can use the Predecessor ID to navigate to the previous message processing log in a chain of interconnected integration flows. See Using IDs to Filter Messages	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.35.* - Increment: 2509	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.71.* - Increment: 2509	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Mod Busir Proce
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.44.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2509	Info only	General Availability	Changed	Technology	Not applic
Cloud Integration	Cloud Foundry	TA1 acknowledgement support for ASC-X12 messages is now available	EDI extractor version 2.2 and above supports TA1 segment in acknowledgment messages (997). See Define EDI Extractor SAP Cloud Integration customers, see Define EDI Extractor	Info only	General Availability	New	Technology	Not applic
Cloud Integration	Cloud Foundry	Error log enhancements for groovy script failure in Message Mapping	You can now view detailed message-processing logs when the execution of a Groovy script used in message mapping fails.	Info only	General Availability	New	Technology	Not applic
Cloud Integration	Cloud Foundry	Deploy on Edge Integration Cell runtime	You can now deploy an integration artifact on the with 'View-only' roles. See Tasks and Permissions for Cloud Integration	Info only	General Availability	New	Technology	Not applic
Cloud Integration	Cloud Foundry	Enhancements in the Migration Tooling	The migration tool now automatically maps integration flow steps while migrating integration scenarios of SAP Process Integration/Process Orchestration that include: - File based adapters with decryption and encryption parameters - SFSF adapter channel - For pipeline approach, while migrating scenarios based on the multiple receiver interface connected	Info only	General Availability	New	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			to same channel, each migrated integration flow will now have a receiver channel created. See Known Limitations of Migration Tooling					
Cloud Integration	Cloud Foundry	Empty segment processing is now available	You can now select how to handle empty segments in a EDI to XML and XML to EDI converter. See • Define EDI to XML Converter • Define XML to EDI Converter SAP Cloud Integration customers, see: • Define EDI to XML Converter • Define XML to EDI Converter	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	New grant type configuration option for OAuth2 client credentials is now available	You can now select how to use Grant Type while deploying OAuth2 client credentials. See Deploying an OAuth2 Client Credentials Artifact SAP Cloud Integration customers, see Deploying an OAuth2 Client Credentials Artifact	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Importing unsupported function libraries is now available	You can now import message mapping with unsupported function libraries. See Creating Message Mapping as an Artifact SAP Cloud Integration customers, see Creating Message Mapping as an Artifact	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
Cloud Integration	Cloud Foundry	Top Integration Flows Filter	You can now use the Top Integration Flows filter to adjust the number of integration flows shown on the Top Integration Flows chart.	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Download Groovy Script	You can now download the AI report generated with the Optimize feature to your local file system. See: Optimize Groovy Scripts	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry Neo	Min-max cardinality support in message mapping	Message mapping now supports min-max cardinality for group & choice elements.	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	New enhancements for Message Usage Dashboard	You can now view more detailed information about your integration flows. See Analyze Message Usage	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	New FTP Sender Poll Enrich adapter now available	You can now use the FTP Sender Poll Enrich adapter in your integrations. See Configure the FTP Sender Adapter Used with the Poll Enrich Step SAP Cloud Integration customers, see Configure the FTP Sender Adapter Used with the Poll Enrich Step	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	SFTP Poll Enrich Metadata retrieval now available	You can now retrieve list of files from an SFTP server without reading its content. See Configure the SFTP Sender Adapter Used with the Poll Enrich Step SAP Cloud Integration customers, see Configure the SFTP Sender Adapter Used with the Poll Enrich Step	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Rate Limiting	The rate limiting feature helps prevent denial-of-	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			service attacks on the entire product or its individual applications. See: Rate Limits on Cloud Integration					
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.33.* - Increment: 2507	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.69.* - Increment: 2507	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.77.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2507	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.42.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2507 The software version 7.42.* is being rolled out in a phased approach. All tenants running on software version 5.76.* would be automatically moved to 7.42.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Service Interface Artifact now available	You can now import service interface artifacts from your PI/PO system. See Working with Service Interface	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
Cloud Integration	Cloud Foundry	New design guidelines for upgrade readiness now available	New design guidelines ensuring upgrade readiness are now available for groovy script of your integration flow. See: • Ensure Upgrade Readiness • Groovy Script	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry Neo	Enabling ADK persistency with adapter API Available	You can now utilize an adapter API to store and retrieve critical information ensuring data persistence even after node restarts or crashes. See Enabling ADK Persistency	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	GenAI Based Integration Flow Generation with Mediation Steps Available	You can now generate a more comprehensive integration flow with more mediation steps. See Generate Integrations with AI Assistance	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Offline Audit log	The offline audit log feature stores audit log entries locally on when the connection to SAP Cloud Integration is unavailable. Once the connection is restored, the system automatically replicates these entries to the cloud.	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Filter Message Processing Logs with Same Correlation ID.	You can now use the Correlation ID link in Monitor Message Processing to filter all message processing logs with the same correlation ID. Using IDs to Filter Messages	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	Streaming option for HTTP Receiver Adapter Available	You can now split a large composite message for HTTP Receiver Adapter by streaming.	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			See HTTP Receiver Adapter SAP Cloud Integration customers, see HTTP Receiver Adapter					
Cloud Integration	Neo Cloud Foundry	Configure Target Root Element in EDI to XML Converter for X12 standard	You can now choose to use the same target root element as the incoming payload in the outgoing payload. See Define EDI to XML Converter SAP Cloud Integration customers, see Define EDI to XML Converter	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.32.* - Increment: 2506	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.68.* - Increment: 2506	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Monitor Message Status Overview Time Filter	For technical reasons, the collected data in Monitor Message Status Overview can be slightly incomplete and might not cover the entire period specified by the user. See: Monitor Message Status Overview	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.76.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2506	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.41.* - Adapter Development Kit for	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			SAP Cloud Integration: 2.53.* Increment: 2506 The software version 7.41.* is being rolled out in a phased approach. All tenants running on software version 5.76.* would be automatically moved to 7.41.* or a higher version available at that point of time, in the near future.					
Cloud Integration	Cloud Foundry Neo	Support for propagation of technical user for OData V2 Receiver Adapter	You will now be able to use technical users for principal propagation authentication for OData V2 Receiver adapter with on-premise proxy type in Cloud Foundry environment. See: Configure the OData V2 Receiver Adapter SAP Cloud Integration customers see: Configure the OData V2 Receiver Adapter	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Temporary Storage Entitlement	The new entitlement for temporary storage usage is 6 GB.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	ELSTER Adapter Supports New ERiC Version	A new version 42.2.2.0 of the Elster Rich Client (ERiC) library is available for the ELSTER adapter. This update includes enhanced data type versions in the ElsterLager2 procedure to improve data processing capabilities and revised interfaces as part of ongoing improvements in data exchange protocols.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.31.* - Increment: 2505	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.67.* - Increment: 2505	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.75.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2505	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.40.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2505 The software version 7.40.* is being rolled out in a phased approach. All tenants running on software version 5.75.* would be automatically moved to 7.40.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	Regex Filtering in File Name for SFTP Sender Adapter used with the Poll Enrich step	You can now use regex filtering to evaluate the entered filename as a real expression or a simple expression in Source for SFTP Sender Adapter used with the Poll Enrich step version 1.8 and above. See SFTP Adapter and SAP Cloud Integration customers, see SFTP Adapter.	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Neo	Advanced Timer in Scheduler for SFTP	You can now use advanced timer in Scheduler for SFTP	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
	Cloud Foundry	and FTP Adapters	and FTP adapters version 1.20 and 1.4 and above respectively. See SFTP Adapter and FTP Adapter. SAP Cloud Integration customers, see SFTP Adapter and FTP Adapter.					
Cloud Integration	Neo Cloud Foundry	Read Lock Check Interval for SFTP Sender Adapter	You can now set read lock interval for SFTP adapters to pause between attempts to acquire a read lock. See SFTP Adapter SAP Cloud Integration customers, see SFTP Adapter.	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	Update Version of Integration Flow Components	- The button previously labeled Migrate is now renamed to Update Version. - You can now also update the version of integration adapters and flow steps. See Update Version of Integration Flow Component SAP Cloud Integration customers, see Update Version of Integration Flow Component.	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.30.* - Increment: 2504	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	SAP Cloud Integration: 6.66.*	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			Increment: 2504					
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.74.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2504	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.39.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2504 The software version 7.39.* is being rolled out in a phased approach. All tenants running on software version 5.74.* would be automatically moved to 7.39.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	Enhancements in XSD Handling for EDI Splitter version 2.9, EDI to XML Converter version 2.6, and XML to EDI Converter version 2.6	You can now use the same XSDs in your tenant as in your Process Integration and Process Orchestration system. See Define EDI Splitter Define EDI to XML Converter Define XML to EDI Converter	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	Configure Target Root Element in EDI to XML Converter	You can now choose to use the same target root element as the incoming payload in the outgoing payload.	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			See Define EDI to XML Converter SAP Cloud Integration customers, see Define EDI to XML Converter					
Cloud Integration	Neo Cloud Foundry	Empty File Handling in SFTP Adapter	You can now choose how to process empty files. See SFTP Sender Adapter SAP Cloud Integration customers, see SFTP Sender Adapter	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.29.* - Increment: 2503	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry Neo	Changes to the General and Iterating Splitter XPath Expression and Grouping	Now you can enter a reference to a header or property instead of a fixed value for XPath Expression and Grouping in the configuration of General and Iterating Splitter.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.65.* - Increment: 2503	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.73.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2503	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.38.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2503	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			The software version 7.38.* is being rolled out in a phased approach. All tenants running on software version 5.73.* would be automatically moved to 7.38.* or a higher version available at that point of time, in the near future.					
Cloud Integration	Neo Cloud Foundry	Configuring Namespace and Prefix for Externally Referenced Schemas in Message Mapping	You can now configure the namespace and prefixes for externally referenced schema in the External Namespace Handling. See Creating Message Mapping as a Flow Step SAP Cloud Integration customers, see Creating Message Mapping as a Flow Step	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Enhancement in Message Mapping Import	You can now import a message mapping from Enterprise Repository, which has imported archives as dependent object. See Importing Message Mapping as an Artifact SAP Cloud Integration customers, see Importing Message Mapping as an Artifact	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	AS2/AS2 MDN Sender Adapter HeaderSuppression	The AS2/AS2 MDN Sender Adapter adapter response includes the header X-content-type-options, which is not compliant with the AS2 protocol. It will be suppressed from all versions of the AS2/AS2 MDN sender adapter.	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 8.28.* - Increment: 2502	Info only	General Availability	Changed	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.64.* - Increment: 2502	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.72.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2502	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.37.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2502 The software version 7.37.* is being rolled out in a phased approach. All tenants running on software version 5.72.* would be automatically moved to 7.37.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	Not applicable
Cloud Integration	Cloud Foundry	Scaling Disk Space and Integration Content Size	All existing disk sizes scale to 10 GB, and integration content scales to 2 GB. See: System Scope in the Cloud Foundry Environment	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	Inspect Total Values of Data Store Usage	You can now view the exact values of data store usage in the Data Store bar chart. See: Inspect Data Store Usage	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	New Show Flow Step Function in Memory Heatmap	You can use the Show Flow Step function to identify the integration flow step that consumes the most memory.	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busir Proce
			See: Inspect Integration Flows By System Memory Usage					
Cloud Integration	Neo Cloud Foundry	Modifying Externalized Parameter descriptions in Integration Flow	You can now modify the information about the parameter from the Externalized Parameters view. For more information, see Externalized Parameters View SAP Cloud Integration customers, see Externalized Parameters View	Info only	General Availability	New	Technology	Not applic
Cloud Integration	Neo Cloud Foundry	Append custom parameters to service document call for OData V4 receiver adapter	For OData V4 receiver adapter version 1.17 and above, the custom parameters configured for \$metadata call will also be appended to the service document call.	Info only	General Availability	New	Technology	Not applic
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 8.27.* • Increment: 2501	Info only	General Availability	Changed	Technology	Not applic
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.63.* • Increment: 2501	Info only	General Availability	Changed	Technology	Not applic
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.71.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2501	Info only	General Availability	Changed	Technology	Not applic
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.36.* • Adapter Development Kit for SAP Cloud Integration: 2.53.*	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			Increment: 2501 The software version 7.36.* is being rolled out in a phased approach. All tenants running on software version 5.71.* would be automatically moved to 7.36.* or a higher version available at that point of time, in the near future.					
Cloud Integration	Cloud Foundry	New Statuses in Temporary Storage Chart	Statuses are now included in the Temporary Storage chart in the Inspect feature. See: Inspect Temporary Storage	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Cloud Foundry	New Message Size Chart in Data Store	- A chart for message sizes is added to Data Store in the Inspect feature. - The Top Data Stores heatmaps now display top data stores by usage and message size. See: Inspect Data Store Usage	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	CSV Header Improvements	While using XML to CSV converter, you can now include the field names as headers in the converted CSV file. For more information, see Configure XML to CSV Converter SAP Cloud Integration customers, see Configure XML to CSV Converter	Info only	General Availability	New	Technology	Not applicable
Cloud Integration	Neo Cloud Foundry	New Access Type Option for the JMS Adapter	You can now configure your queues to either exclusive or non-exclusive access types. With the non-exclusive access type, messages are processed concurrently by multiple	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module Business Process
			worker nodes, offering higher throughput. While with exclusive access type messages are processed sequentially by a single consumer, maintaining the order in which messages are received, which is essential for scenarios requiring strict sequencing. For more information on these configuration options, see: Configure the JMS Sender Adapter , and Configure the JMS Receiver Adapter .					
Cloud Integration	Neo Cloud Foundry	Enhanced AS4 Response Handling and Streamlined Push Operations	You can now - Choose to add an attachment to the payload of AS4 message. For more information, see Configure Receiver Channel with ebMS3 Push SAP Cloud Integration customers, see Configure Receiver Channel with ebMS3 Push - Decide how to handle the response type, as user message or receipt message. For more information, see Configure Receiver Channel with ebMS3 Push SAP Cloud Integration customers, see Configure Receiver Channel with ebMS3 Push	Info only	General Availability	New	Technology	Not applicable

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Module
			Choose the two way sync communication pattern as the response type. For more information, see Configure the Sender Channel with ebMS3 Push SAP Cloud Integration customers, see Configure the Sender Channel with ebMS3 Push					
Cloud Integration	Neo Cloud Foundry	Customizable Proxy Settings Enabled for Edge Integration Cell Runtime	You can now set the proxy type settings if you have Edge Integration Cell runtime activated. For more information, see Configure the AS2 Sender Adapter Configure the AS2 Receiver Adapter SAP Cloud Integration customers, see Configure the AS2 Sender Adapter Configure the AS2 Receiver Adapter	Info only	General Availability	New	Technology	Not applicable

2024 What's New for SAP Cloud Integration (Archive)

2024 What's New for SAP Cloud Integration (Archive)

Core Components, 2024

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.62.* - Increment: 2413	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.70.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2413	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.35.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2413 The software version 7.35.* is being rolled out in a phased approach. All tenants running on software version 5.70.* would be automatically moved to 7.35.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Addition of error response code and body to exchange for OData V4 receiver adapter	OData V4 Receiver adapter version 1.16 onwards sets the exchange header <code>CamelHttpResponseCode</code> with the HTTP response code and sets the exchange body with the error response received from the OData V4 service.	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.61.* - Increment: 2412	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.69.* - Adapter Development Kit for SAP Cloud Integration: 2.53.*	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			Increment: 2412					
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.34.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2412 The software version 7.34.* is being rolled out in a phased approach. All tenants running on software version 5.69.* would be automatically moved to 7.34.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Expiring Refresh Token for OAuth2 Authorization Code with Generic Provider	Expiring refresh tokens are now supported on the Cloud Foundry platform for OAuth2 Authorization Code with Generic Provider. For detailed setup and configuration, see: Deploying an OAuth2 Client Authorization Code with Generic Provider.	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Stop on Exception in SFTP Sender Adapter	You can stop the processing of the current file batch if any exception or error is encountered. See: Configure SFTP Sender Adapter SAP Cloud Integration customers, see: Configure SFTP Sender Adapter	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	New quality of service option for XI sender and receiver adapter	A new quality of service option, Handled by Integration Flow, is now available for the XI sender and receiver adapters. This option allows quality of service to be managed within the Integration Flow. For more information, please check: Configure the XI Sender Adapter.	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	SAP Cloud Integration: 6.59.*	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			Increment: 2410					
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.67.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2410	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.32.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2410 The software version 7.32.* is being rolled out in a phased approach. All tenants running on software version 5.67.* would be automatically moved to 7.32.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Changes to Timer Flow Step	A new version of the timer flow step is introduced. You can: - Create multiple fixed schedules with one timer flow step to repeatedly trigger the message processing of the same integration flow. Earlier, you had to create multiple timer flow steps within an integration flow to configure such a scheduling. - Experience a revamped scheduler design giving you much more flexible options to configure your recurring patterns. Define a Timer Start Event.	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Neo Cloud Foundry	Fix Incompatibilities of Local Groovy Scripts	As a first step towards keeping your custom codes clean and future-ready, you can now easily remove unused and duplicate import statements from your local Groovy scripts. Fix Script Incompatibilities.	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	Enable Regex Filtering in SFTP adapter	You can now apply Regex filtering on file name in SFTP sender adapter . See: SFTP Sender Adapter SAP Cloud Integration customers choose: SFTP Sender Adapter	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.58.* - Increment: 2409	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.66.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2409	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.31.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2409 The software version 7.31.* is being rolled out in a phased approach. All tenants running on software version 5.66.* would be automatically moved to 7.31.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Managing Partner Directory Entries	You can now manage Partner Directory entries with the SAP Integration Suite user interface (under  Monitor )	Info only	General Availability	New	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			Integrations and APIs >). You can use Partner Directory entries to parameterize integration flows. See: Managing Partner Directory Entries SAP Cloud Integration customers choose: Managing Partner Directory Entries					
Cloud Integration	Neo Cloud Foundry	Support for Group Number and Transaction Set Number in EDI Splitter	You can now modify group number and transaction set number segments in the functional acknowledgment message of EDI Splitter . See: EDI Splitter SAP Cloud Integration customers choose: EDI Splitter	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	Support for Dynamic Properties in CSV to XML Converter Using Headers, Properties, and Partner Directory	You can now use dynamic configurations for various fields. See: Configure CSV to XML Converter SAP Cloud Integration customers choose: Configure CSV to XML Converter	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	Support for Adding and Deleting Custom Key-Value Pair for the OAuth2 Client Credentials Artifact	You can now add or delete additional parameters in the custom parameter table. See: Deploying an OAuth2 Client Credentials Artifact SAP Cloud Integration customers choose: Deploying an OAuth2 Client Credentials Artifact	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.57.* - Increment: 2408	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.65.* - Adapter Development Kit for SAP Cloud Integration: 2.53.*	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			Increment: 2408					
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.30.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2408 The software version 7.30.* is being rolled out in a phased approach. All tenants running on software version 5.65.* would be automatically moved to 7.30.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	New Message Status Overview	Message Status Overview allows you to quickly get an overview of some of the collected statistics of your messages according to your custom filters. See: Monitor Message Status Overview SAP Cloud Integration customers choose: Monitor Message Status Overview	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	New Content Encryption Algorithm for Mail Adapter	A new content encryption algorithm is supported by the Mail adapter: AES/GCM/NoPadding. See: Configure the Mail Receiver Adapter SAP Cloud Integration customers choose: Configure the Mail Receiver Adapter	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	New Parameter for ELSTER Adapter	The ELSTER receiver adapter allows you to download objects from an object store (OTTER) provided by the financial authorities. The adapter supports ERiC Version 40.2.10.0. See: ELSTER Receiver Adapter SAP Cloud Integration customers choose: ELSTER	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			Receiver Adapter					
Cloud Integration	Neo Cloud Foundry	SFTP Sender Adapter	The SFTP Sender adapter now allows you to specify a maximum file size for processing. See: Configure the SFTP Sender Adapter	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	New Inspect Integration Flows Feature	You can inspect the top five integration flows that consume the most resources. See: Inspect Top Consuming Integration Flows	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Support of PI/PO Message Mapping with Simple Type Parameters	You can now import PI/PO message mapping objects with Simple Type parameter category. See: Creating Message Mapping as an Artifact SAP Cloud Integration customers choose: Creating Message Mapping as an Artifact	Info only	General Availability	New	New	
Cloud Integration	Cloud Foundry	New Inspect Temporary Storage Feature	You can inspect the total storage usage that temporary files allocate in the database for a given time period. See: Inspect Temporary Storage	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	Customize acknowledgement message envelope	You can now modify the envelope of the acknowledgment message (for EDIFACT and X12 EDI formats) generated by EDI Splitter. See: EDI Splitter	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Inspect Feature Offers Embedded Recommendations for High Usage Levels	Users can now quickly access tailored suggestions and assistance when usage levels reach warning or critical stages. When resource consumption becomes critical, a new Recommendations button will appear, providing timely and relevant guidance to help resolve potential issues.	Info only	General Availability	New	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.56.* - Increment: 2407	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.64.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2407	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.29.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2407 The software version 7.29.* is being rolled out in a phased approach. All tenants running on software version 5.64.* would be automatically moved to 7.29.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry Neo	Connectivity to MariaDB	Now, JDBC receiver adapter supports connectivity to Maria DB. See: and JDBC for MariaDB (On-Premise) . JDBC for MariaDB (Cloud) .	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry Neo	Edit namespace for message type	You can now edit the namespace for message type from the overview tab. See: Configuring Message Type	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry Neo	New AMQP Sender Adapter Parameter	A new parameter in the AMQP sender adapter allows you to disable reply-to for a message. See: Configure the AMQP Sender Adapter SAP Cloud Integration customers choose Configure the AMQP Sender Adapter	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Cloud Foundry Neo	New Special Character Allowed for Partner Directory	The ID of the Partner, StringParameter, BinaryParameter, and UserCredentialParameter can now contain the character '@'. See: Partner Directory SAP Cloud Integration customers choose Partner Directory	Info only		Changed	Technology	
Cloud Integration	Cloud Foundry Neo	New Parameter in SFTP Sender Adapter	A new parameter in the SFTP sender adapter, Poll on one worker only, allows you to specify that the polling process is performed on a single worker node only. See: Configure the SFTP Sender Adapter SAP Cloud Integration customers choose Configure the SFTP Sender Adapter	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.55.* - Increment: 2406	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.63.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2406	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.28.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2406 The software version 7.28.* is being rolled out in a phased approach. All tenants running on software version 5.63.* would be automatically moved to 7.28.* or a higher version	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			available at that point of time, in the near future.					
Cloud Integration	Cloud Foundry Neo	New Content Encryption Algorithm Options in PKCS7/CMS Encryptor	The PKCS7/CMS Encryptor offers the following options for Content Encryption Algorithm (when for Signatures the option Enveloped Data Only has been chosen): AES/CCM/NoPadding, AES/GCM/NoPadding. See: Encrypt and Sign the Message Content with PKCS#7/CMS Encryptor SAP Cloud Integration customers choose Encrypt and Sign the Message Content with PKCS#7/CMS Encryptor	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Automatic porting of consumed custom integration adapters when software is updated with Apache Camel runtime 3.14.	The latest versions of your already consumed AmazonWebServices, MicrosoftDynamicsCRM, Salesforce, and SugarCRM adapters will automatically be ported to your tenants with Apache Camel 3.14 runtime. Upload Custom Adapters	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Principal Propagation of Technical Users in HTTP Receiver Adapter	You can now propagate the principal of the inbound technical user with the Principal Propagation authentication method in the HTTP receiver adapter. Earlier, this was supported only in the Neo environment. HTTP Receiver Adapter.	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	New Inspect Content Size Feature	You can inspect the number of integration artifacts deployed on your tenant and the file size they allocate. See: Inspect Content Size	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry Neo	PATCH Operation in OData V4 Receiver Adapter	You can now use PATCH operation in the OData V4 receiver adapter to update existing entities. See: Configure the OData V4 Receiver Adapter.	Info only	General Availability	New	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.54.* - Increment: 2405	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.62.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2405	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.27.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2405 The software version 7.27.* is being rolled out in a phased approach. All tenants running on software version 5.62.* would be automatically moved to 7.27.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry Neo	Size Column In References Tab of Property Sheet	A new Size column is introduced in the References tab of the property sheet to display the file size of local resources that you upload to an integration flow. See: Manage References .	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry Neo	Skip Design Guidelines With Consent	You can now skip the non-compliant design guideline by providing a reason and consent. See: Design Guidelines View .	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.53.* - Increment: 2404	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.61.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2404	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.26.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2404 The software version 7.26.* is being rolled out in a phased approach. All tenants running on software version 5.61.* would be automatically moved to 7.26.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry Neo	Message Digest Supports New Canonicalization Methods	The Message Digest step supports now the following additional canonicalization methods: XML Canonicalization Version 1.1 and XML Canonicalization with Comments Version 1.1. See: Define Message Digest (SAP Cloud Integration customers choose Define Message Digest)	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry Neo	New Design Guidelines Added	Additional design guidelines are added to help you design integration flows in a robust fashion. The new design guidelines include checks to efficiently use scripts, handle security, optimize memory usage, and a few more. For more information on how to enable design guidelines, see: Design Guidelines.	Info only	General Availability	New	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Cloud Foundry Neo	Search Support For Message Flow (Adapter)	Search is enabled for Message Flows (Adapters) in the integration flow editor. See: Overview of Integration Flow Editor .	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Generate unique interchange number from EDI Splitter	You can now generate a unique interchange number while generating an acknowledgement number from EDI Splitter. See EDI Splitter . Cloud Integration customers choose: EDI Splitter	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Decimal Character for outgoing payload	For XML to EDI convertor, you can now choose the decimal character of the converted message. See XML to EDI Converter . Cloud Integration customers choose: XML to EDI Converter	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.52.* • Increment: 2403	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.60.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2403	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.25.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2403 The software version 7.25.* is being rolled out in a phased approach. All tenants running on software version 5.60.* would be automatically moved	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			to 7.25.* or a higher version available at that point of time, in the near future.					
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.51.* - Increment: 2402	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.59.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2402	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.24.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2402 The software version 7.24.* is being rolled out in a phased approach. All tenants running on software version 5.59.* would be automatically moved to 7.24.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry Neo	Cloud Integration - Integration Flow Design Guidelines Changes	The following new integration flow design guideline (including integration flow) has been newly added: Handle IDoc Bulk Messages.	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry Neo	Improvements for Advanced Scheduler	You can now define start and end dates for recurring schedules using the advanced scheduler. Define a Timer Start Event.	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry Neo	Support For Sub-levels in OData V4 Receiver Adapter	You can now use the query modeller in the OData V4 receiver adapter to construct a query containing a deep structure for the GET operation.	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			Configure the OData V4 Receiver Adapter.					
Cloud Integration	Cloud Foundry Neo	Introducing Design Guidelines	Design guidelines enable you to design integration flows in a robust fashion to safeguard your business processes. - As the integration lead, learn about all available design guidelines and enable the applicable ones. See: Design Guidelines. - As an integration developer, abide by the enabled design guidelines. See: Design Guidelines View.	Info only	General Availability	New	Technology	
Cloud Integration	Cloud Foundry Neo	ELSTER Adapter Supports New ERiC Version	On February 9, 2024, the German tax authority communicated that on April 29, 2024 the Elster Rich Client (ERiC) library version will be increased to 39.4. The ELSTER adapter supports the new ERiC version as of SAP Integration Suite increment 2402.	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry Neo	Target encoding field in XML to EDI converter	You can now select the encoding format for the outgoing payload in XML to EDI converter. See: Define XML to EDI Converter	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.50.* - Increment: 2401	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.58.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2401	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.23.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2401 The software version 7.23.* is being rolled out in a phased approach. All tenants running on software version 5.58.* would be automatically moved to 7.23.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Dynamic support for address and credential in SuccessFactors SOAP Receiver Adapter	In the SuccessFactors SOAP Receiver Adapter, you can now dynamically define the Address and Credential Name fields in the message exchange. See: Configure the SuccessFactors SOAP Receiver Adapter	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Defining Access Policies at Integration Package Level	You can now use access policies to protect all artifacts of an integration package. This enhancement simplifies administration efforts for tenant administrator considerably. If an access policy has been defined for an integration package, access policies for individual artifacts (within this package) will still be taken into account. Managing Access Policies	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	New Property Available to Set Time Zone for XAdES SigningTime	A new property allows you to set the time zone for the XAdES SigningTime element of the signature created by the XML Signer step. Signing the Message Content with XAdES-BES (1)	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
Cloud Integration	Neo Cloud Foundry	Support for Custom Headers and Queries in OData V4 Receiver Adapter's Metadat Call	In the OData V4 receiver adapter, you can now send custom request headers and queries in the metadata call, which is internally triggered during the message processing. Configure the OData V4 Receiver Adapter	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Target encoding field in EDI to XML converter	You can now select the encoding format for the outgoing payload in EDI to XML converter. See: Define EDI to XML Converter	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.49.* - Increment: 2313	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.57.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2313	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.22.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2313 The software version 7.22.* is being rolled out in a phased approach. All tenants running on software version 5.57.* would be automatically moved to 7.22.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	JMS Receiver Adapter Stores Integration Flow ID in Property	A new property (SAP_IntegrationFlowID) is available that contains the ID of the integration flow that	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			sent a message through the JMS receiver adapter. The JMS consumer (reading the message through the JMS sender adapter) can use this property to define further steps that depend on the integration flow ID. See: Headers and Exchange Properties Provided by the Integration Framework (SAP Cloud Integration customers choose Headers and Exchange Properties Provided by the Integration Framework)					
Cloud Integration	Cloud Foundry	AdvancedEventMesh Adapter Available	The AdvancedEventMesh adapter enables to connect to SAP Integration Suite, advanced event mesh. See: Advanced Event Mesh Adapter	Info only	General Availability	Changed	Technology	
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.48.* • Increment: 2312	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.56.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2312	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 7.21.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2312 The software version 7.21.* is being rolled out in a phased approach. All tenants running on software version 5.56.* would be automatically moved to 7.21.* or a higher version	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			available at that point of time, in the near future.					
Cloud Integration	Neo Cloud Foundry	IDoc Receiver Adapter Has New Parameter to Set Message ID	The adapter provides a parameter to configure the message ID. You can use this feature to implement scenarios that guarantee end-to-end Exactly Once delivery. See: Configure the IDoc Receiver Adapter (SAP Cloud Integration customers choose Configure the IDoc Receiver Adapter)	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Kafka Sender and Receiver Adapter Support Additional SASL Mechanism	You can now also use the following SASL mechanism for the authentication of Kafka servers when using the Kafka sender and receiver adapter: SCRAM-SHA-512. See: See: • Configure the Kafka Sender Adapter (SAP Cloud Integration customers choose Configure the Kafka Sender Adapter) • Configure the Kafka Receiver Adapter (SAP Cloud Integration customers choose Configure the Kafka Receiver Adapter)	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Message Monitoring Performance Improved	The search logic for messages has been improved in such a way that it also performs efficiently under high load conditions. The retrieval of the number of messages has been decoupled from the message search. As a result, the list of messages is displayed even while the system is still determining the number of messages. See: Monitor Message Processing (SAP Cloud	Info only	General Availability	Changed	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	
			Integration customers choose Monitor Message Processing)					
Cloud Integration	Neo Cloud Foundry	Integration Artifact Filter Improved	When you navigate from the message monitor to the integration content monitor, instead of a pre-filtered list, the currently deployed active artifact will be opened. See: Monitor Message Processing (SAP Cloud Integration customers choose Monitor Message Processing)	Info only	General Availability	Changed	Technology	
Cloud Integration	Neo Cloud Foundry	Usability Improvement After Creating An Integration Flow	You can now directly open an integration flow in the integration flow editor from the creation screen. Earlier, you had to create the integration flow, go the the Artifacts list view of the package, and then navigate to the editor. See: - Integration Suite: Creating an Integration Flow Creating an Integration Flow	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	Search Existing Flow Steps in an Integration Flow	You can now search for an existing flow step in an integration flow by different parameters. See: Overview of Integration Flow Editor	Info only	General Availability	New	Technology	
Cloud Integration	Neo Cloud Foundry	Adapter Proxy Settings in Edge	You can now define proxy settings for the following adapters: ELSTER Receiver Adapter Configure the SFTP Sender Adapter Configure the SFTP Receiver Adapter	Info only	General Availability	New	Technology	

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			Configure the SFTP Sender Adapter Used with the Poll Enrich Step				
Cloud Integration	Neo Cloud Foundry	Newer Version of Exception Subprocess Available	A newer version of Exception Subprocess pool – version 1.1 – is now available. You can easily move to the newer version in a single click. See: Update Version of an Integration Flow Component .	Info only	General Availability	New	Technology

2023 What's New for SAP Cloud Integration (Archive)

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Core Components, 2023

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.46.* - SAP Integration Advisor: 1.80.* - Increment: 2310	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.54.* - SAP Integration Advisor: 1.80.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2310	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.19.* - SAP Integration Advisor: 1.80.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2310 The software version 7.19.* is being rolled out in a phased approach. All tenants running on software version 5.54.* would be automatically moved to 7.19.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Inspect System Memory Usage	You can inspect system memory usage by integration flows that are processed on the tenant. See: Inspect System Memory Usage	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	New Feature to Analyze Message Usage	You can now analyze the number of messages processed by integration flows for a selected time range. See: Analyze Message Usage	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Support for the EDI (Electronic Data Interchange) Standard TRADACOMS	You can now design integration scenarios that use the EDI (Electronic Data Interchange) standard TRADACOMS. The related integration flow steps have been enhanced accordingly. See: • Define EDI Extractor • Define EDI Splitter • Define EDI to XML Converter • Define XML to EDI Converter	Info only	General Availability	Changed
Cloud Integration	Neo Cloud Foundry	ELSTER Adapter Supports New Version of ERiC Library	The ELSTER receiver adapter supports now version 39 of the Eric library. The new version provides the annual versions for the employment tax notification (ETNotif.) and advance return tax on sales/purchases (UVA). See: SAP note 3381451 📄 SAP note 3399546 📄	Info only	General Availability	New
Cloud Integration	Neo Cloud Foundry	Message Mapping – New Standard Functions Added	Two new standard functions – getHeader and getProperty – are added so that you can fetch the headers and properties from the message exchange without having to write custom scripts. See: • Creating Message Mapping as a Flow Step	Info only	General Availability	New
Integration Advisor	Cloud Foundry Neo	Custom Type System	You can now create a custom type system and upload WSDL files for SOA messages. See: Custom Type Systems	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Integration Advisor	Cloud Foundry Neo	DateTime Mapping	You can now provide Fixed Values for the missing parts of a DateTime format of a target node. See: Date Time Conversion	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.45.* - SAP Integration Advisor: 1.79.* - Increment: 2309	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.53.* - SAP Integration Advisor: 1.79.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2309	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 7.18.* - SAP Integration Advisor: 1.79.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2309 The software version 7.18.* is being rolled out in a phased approach. All tenants running on software version 5.53.* would be automatically moved to 7.18.* or a higher version available at that point of time, in the near future.	Info only	General Availability	Changed
Cloud Integration	Neo	Support for Inbuilt Retry in HTTP Receiver Adapter	You can now configure connection retry to the target system in case of failed HTTP requests. The feature is available only with the tenants running on the Neo environment with software version 7.18.xx and above. See: HTTP Receiver Adapter .	Info only	General Availability	New
Integration Advisor	Cloud Foundry Neo	Custom Type System	You can now create a custom type system to upload and use custom IDoc messages. See: Custom Type Systems	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.44.* - SAP Integration Advisor: 1.78.*	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			Increment: 2308			
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.52.* - SAP Integration Advisor: 1.78.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2308	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Adjust the Validity Time of Access Token during Instance Creation	You can now manually adjust the validity time of your access token during the creation of your service instance. See: Creating Service Instance and Service Key for Inbound Authentication	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Deactivate Pin Certificates Option during Service Instance Creation	You can now manually deactivate the pin certificate functionality to be able to use renewed certificates without additional configuration. See: Creating Service Instance and Service Key for Inbound Authentication	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Multi Mapping for JSON Schemas	In message mapping, you can now design a multi-mapping definition by using JSON schemas.	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Move Properties and Headers In A Content Modifier Step	You can now move the properties and headers in a content modifier step up and down to decide the order of processing them. See: Define Content Modifier .	Info only	General Availability	New
Integration Advisor	Cloud Foundry Neo	Mapping Guidelines	- You can now search and filter the mapping entities in a MAG based on a specific function pattern. See: Working with Mapping Guidelines (MAGs) - You can now delete multiple mapping entities together in a Mapping Guideline. See: Mapping the Source and Target Nodes	Info only	General Availability	Changed
Integration Advisor	Cloud Foundry Neo	Export a single MIG/MAG version	You can now export a single version of MIG or MAG into a .zip file. See: Exporting MIG/MAG	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Integration Advisor	Cloud Foundry Neo	Code Value Mapping	Code value mapping in a MAG now supports N:1 mapping cardinality. See: Code Value Mapping	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.43.* • SAP Integration Advisor: 1.77.* • Increment: 2307	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.51.* • SAP Integration Advisor: 1.77.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2307	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Inspect Usage of Integration Resources Extended by Monitoring Storage Capacity	You can now inspect usage of the monitoring storage database for a given time period. See: Inspect Monitoring Storage Usage	Info only	General Availability	Changed
Integration Advisor	Cloud Foundry Neo	Migrating a Message Implementation Guideline	You can now migrate a Message Implementation Guideline to a different Message Type with different root node. See: Migrating a Message Implementation Guideline	Info only	General Availability	Changed
Integration Advisor	Cloud Foundry Neo	DateTime Mapping in a Mapping Guideline	The Date Time Mapping now supports N:1 mapping between the source and target mapping structure. See: Date Time Conversion	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.41.* • SAP Integration Advisor: 1.75.* • Increment: 2305	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.49.* • SAP Integration Advisor: 1.75.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2305	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	SFTP JSchi Library Update	The new release supports several more recent algorithms and stops using deprecated ones. For backward compability with older SFTP servers, you can still enable using several deprecated algorithms for SFTP polling. See: • Configure the SFTP Sender Adapter • Configure the SFTP Sender Adapter Used with the Poll Enrich Step • Configure the SFTP Receiver Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Binary Parameter for Partner Directory	The maximum size of the binary parameter for the Partner Directory has been increased to 1.5 MB. See: Partner Directory.	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	System Settings Tab	The existing ES Repository tab under Settings is renamed as System. The setting is improved so that you can: • connect to more than one SAP Process Orchestration systems • test the connection to all configured SAP Process Orchestration systems from one place • avoid creating destinations in SAP BTP subaccount which now SAP Integration Suite takes care See: Importing Content from SAP Process Orchestration System.	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.40.* • SAP Integration Advisor: 1.74.* • Increment: 2304	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.48.* • SAP Integration Advisor: 1.74.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2304	Info only	General Availability	Changed
Cloud Integration	Neo	Enhanced Cloud Integration OData API	The Security Content API contains the new resource Secure Parameter. You can use this resource to create, update, read, and delete Secure Parameter artifacts. See: Security Content	Info only	General Availability	Changed
Integration Advisor	Cloud Foundry Neo	Extended Namespace Support in MIGs	There is an update on how Namespace declarations are handled for message root nodes in Message Implementation Guidelines. See: Exporting Runtime Artifacts	Info only	General Availability	New
Integration Advisor	Cloud Foundry Neo	Conditional Mapping in MAGs	You can now set conditional mapping for nodes with Code Value Mapping in a Mapping Guideline. See: Mapping the Source and Target Nodes	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	JMS Queue Name – Maximum Length Increased	Using the JMS sender and receiver adapter, you can now define queue names with more than 20 characters length (maximum 80 characters). See: • Configure the JMS Sender Adapter • Configure the JMS Receiver Adapter	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Disable Attachments for Message Processing Failures	In the latest versions of the HTTP, OData V2, and OData V4 receiver adapters, you have an option to disable the creation of attachments when the message processing fails.	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			See: • HTTP Receiver Adapter • Configure the OData V2 Receiver Adapter • Configure the OData V4 Receiver Adapter			
Cloud Integration	Cloud Foundry	Inspect Usage of Integration Resources Extended and Enabled for SAP Integration Suite	You can now inspect the usage of integration resources also when using SAP Integration Suite. Additionally, with this increment, the usage of database transactions is covered. For data store usage, a new feature has been added: the data volume is displayed that is used by global data stores. Therefore, this feature now covers the following integration resources: • Database connections • Data stores • Database transactions See: Inspect	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.39.* • SAP Integration Advisor: 1.73.* • Increment: 2303	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.47.* • SAP Integration Advisor: 1.73.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2303	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Update Required for Dedicated Adapters	If you're using an adapter (provided in partnership with Advantco) that supports connectivity with one of the following components, make sure to switch to a newer adapter version: A new feature has been added that allows you to inspect usage of Salesforce (ADVANTCO-SFO adapter), Amazon Web	Required	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			Service (ADVANTCO-AWS adapter), Microsoft Dynamics CRM (ADVANTCO-MSD adapter), and Sugar CRM (ADVANTCO-SGR adapter). adapter), Microsoft Dynamics CRM (ADVANTCO-MSD adapter), and Sugar CRM The older adapter versions are in maintenance and don't get further updates. For more information, see SAP note 3001980 . Furthermore, these older adapter versions aren't supported in the Cloud Foundry environment (and, therefore, they also aren't supported for SAP Integration Suite). For more information on the new adapter versions, see: • AmazonWebServices Sender Adapter • AmazonWebServices Receiver Adapter • Microsoft Dynamics CRM Receiver Adapter • Salesforce Sender Adapter • Salesforce Receiver Adapter • SugarCRM Receiver Adapter Certain integration packages use adapters that are affected by this change. If you're using one of these integration packages, we recommend you to switch to the latest version of the integration package (see: SAP note 3001980 . Perform this step as soon as possible, but no later than 1 March 2024. This will give you enough time to test your integration scenarios. For more information on the required steps, see SAP note 3001980 .			
Cloud Integration	Cloud Foundry	New Feature Allows You to Inspect Usage of Integration Resources	updates. For more information, see SAP note You can, for example, identify those integration flows that contribute Advantco) will no longer run as of 1 January 2024. guidelines how to change	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			your integration design in order to improve the situation. The feature covers the following integration resources: - Database connections (used by integration flows) - Data stores (used by integration flows) This feature is only available for SAP Cloud Integration, Cloud Foundry environment. See: Inspect			
Integration Advisor	Cloud Foundry Neo	GS1 XML Type System	A new type system GS1 XML has now been added to the Type System library. See: Overview Of B2B Standards	Info only	General Availability	Changed
Integration Advisor	Cloud Foundry Neo	Namespace Prefix in Message Implementation Guideline	You can now view the Namespace Prefix of a XML qualified node in a Message Implementation Guideline. See: Working with a Node	Info only	General Availability	Changed
Integration Advisor	Cloud Foundry Neo	Conditional Mapping in Mapping Guidelines	Conditional Mapping is now extended to support leaf-to-leaf mapping. See: Mapping the Source and Target Nodes	Info only	General Availability	Changed
Integration Advisor	Cloud Foundry Neo	Global Code Value Mapping	You can now create a Global Code Value Mapping directly from the Mapping Guidelines tab. See: Global Code Value Mappings	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Auto Layout Feature in Integration Flow Editor	You can now prettify your complex Integration Process pools in one click using the auto layout feature in the Integration Flow Editor. See: Overview of Integration Flow Editor.	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.38.* - SAP Integration Advisor: 1.72.* - Increment: 2302	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.46.* - SAP Integration Advisor: 1.72.*	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			- Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2302			
Cloud Integration	Cloud Foundry Neo	PGP Keys Monitor Provides Access to All PGP Key- Related Features	The features for the management of PGP keys has been disabled for the Security Material tile (under Manage Security). All features to manage PGP keys are now accessible from the PGP Keys tile (under Manage Security). See: Managing PGP Keys	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	New Parameter Compress Stored Message Introduced for XI Sender and Receiver Adapter	A new parameter has been introduced for the XI sender and receiver adapter that allows you to compress the stored message if JMS queue is used as temporary storage. See: Configure the XI Sender Adapter Configure the XI Receiver Adapter	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Renaming Product Profiles	Product profiles are being renamed as "Runtime Profiles". See: Runtime Profiles .	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Support for EDIFACT Syntax Version 2	EDI to XML Converter, XML to EDI Converter, EDI Splitter, EDI Validator now support EDIFACT Syntax Version 2. See: Define EDI to XML Converter , Define XML to EDI Converter , Define Splitter , Define EDI Validator	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.37.* - SAP Integration Advisor: 1.71.* - Increment: 2301	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.45.* - SAP Integration Advisor: 1.71.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2301	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	Handling of Duplicate	Handling of duplicate attachment names was changed for the SOAP SOAP 1.x	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
	Neo	Attachment Names in SOAP SOAP 1.x Sender Adapter	sender adapter. If an attachment name occurs several times, now also a Guid is added to the first of these attachment names (before this increment, no Guid was added to the first one). See: Configure the SOAP (SOAP 1.x) Sender Adapter			
Cloud Integration	Cloud Foundry Neo	Support to Define Source of Partner ID in AS2 Sender Adapter	You can now specify the source of Partner ID in AS2 Sender adapter. See: Configure the AS2 Sender Adapter	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Support for On-Premise and Cloud SAP ASE Service Databases in JDBC Receiver Adapter	JDBC Receiver adapter now supports SAP ASE Service database on both On-Premise and Cloud infrastructures. See: JDBC for SAP ASE Platform (On-Premise) and JDBC for SAP ASE Service (Cloud) .	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Advanced Scheduler Configuration in the Timer Flow Step	You can now configure complex and granular schedules using combination of various units of time measurement. See: Define a Timer Start Event .	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Transport Owner Propagation	You can now propagate the logged in user as the owner of the particular transport action while transporting artifacts using Cloud Transport Management. See: Content Transport Using Cloud Transport Management .	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	New Configuration Option for OData Receiver Adapter Variants Allow Reuse of Connections Across HTTP Requests	In the latest versions of the OData receiver adapter variants, you've an option to reuse connection objects from the internal connection pool that improves the network turnaround time. See: • Configure the OData V2 Receiver Adapter • Configure the OData V4 Receiver Adapter • Configure the SuccessFactors OData V2 Receiver Adapter • SuccessFactors OData V4 Receiver Adapter	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry Neo	New Filter Option for Query in the Model Operation Wizard of OData V2 Receiver Adapter	You can now filter the Fields list when you're trying to choose the right field name while modeling a query using the Model Operation wizard. See: Configure the OData V2 Receiver Adapter .	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Keyboard Shortcut to Create Artifacts	In the integration flow creation dialog, you can now press the Enter Return key to create an integration flow. See: Creating an Integration Flow . The feature is available for creation of Message Mapping, Value Mapping, and Script Collection too.	Info only	General Availability	New
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.36.* • SAP Integration Advisor: 1.70.* • Increment: 2213	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.44.* • SAP Integration Advisor: 1.70.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2213	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry	RabbitMQ Adapters Available	The RabbitMQ sender adapter allows you to consume messages from queues on the RabbitMQ server. In addition, you use the adapter to send acknowledgements to the RabbitMQ server. The RabbitMQ receiver adapter allows you to send messages from exchanges or queues on the RabbitMQ server.	Info only	General Availability	New
Integration Advisor	Cloud Foundry Neo	Update Mapping Guidelines	The new feature Update MAGS now allows you to select and update a group of mapping guidelines together. See: Updating Mapping Guidelines	Info only	General Availability	New
Cloud Integration	Cloud Foundry	External Logging feature	The External Logging feature enables customers to send message processing logs to an external system, independently of available database storage.	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			See: External Logging Cloud Foundry Environment			
Cloud Integration	Cloud Foundry Neo	New version of Integration Process Pool Available	1.2 version of Integration Process pool is available with a default Transaction Handling value that improves the processing performance of your tenant. See: Define Transaction Handling .	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	Persisting the Cardinality in Message Mapping	For multi-mapping schema, the cardinality is persisted if the source or target schema is replaced. Earlier, the cardinality used to revert to the default value if the schema was replaced.	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	New Option to Update an RSA Key	You can now update an RSA key in the keystore. See: Updating an RSA Key .	Info only	General Availability	Announcement
Cloud Integration	Cloud Foundry	New AzureStorage Adapter Available	The new adapter type allows you to exchange data between Azure Storage and .	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Update Required for Dedicated Adapters	If you're using an adapter that supports connectivity with one of the following components, make sure to switch to a newer adapter version: Salesforce, Amazon Web Service, Microsoft Dynamics CRM, and Sugar CRM. The older adapter versions are in maintenance and don't get further updates. For more information, see SAP note 3001980  In particular, these older adapter versions aren't supported in the Cloud Foundry environment (and, therefore, they also aren't supported for SAP Integration Suite). For more information on the new adapter versions, see: • AmazonWebServices Sender Adapter • AmazonWebServices Receiver Adapter	Recommended	General Availability	Announcement

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
			• Microsoft Dynamics CRM Receiver Adapter • Salesforce Sender Adapter • Salesforce Receiver Adapter • SugarCRM Receiver Adapter Certain integration packages use adapters that are affected by this change. If you're using one of these integration packages, we recommend you to switch to the latest version of the integration package (see: SAP note 3001980 ).			
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.35.* • SAP Integration Advisor: 1.69.* • Increment: 2212	Info only	General Availability	Changed
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.43.* • SAP Integration Advisor: 1.69.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2212	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Support to Define Source of Partner ID in AS2 MDN Sender Adapter	You can now specify the source of Partner ID in AS2 MDN sender adapter. See: Configure the AS2 MDN Sender Adapter	Info only	General Availability	Announcement
Cloud Integration	Cloud Foundry Neo	New Option to Upload an RSA Key to Keystore	You can now upload an RSA key to the keystore. See: Uploading an RSA Key	Info only	General Availability	New
Cloud Integration	Cloud Foundry Neo	New Operations Added to Script Collections Resource of Integration Content OData API	You can now navigate to a script collection from an integration package and save a script collection as version. See: Integration Content	Info only	General Availability	Changed

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry	Role-Based Protection of Connectivity Test Has Been Changed	The new role template <code>CredentialsRead</code> has been introduced that is required in addition to role template <code>CredentialsEdit</code> in order to perform connectivity tests. The role templates for the integration developer and tenant administrator persona that contain permission to perform connectivity tests have been adapted accordingly. See: Tasks and Permissions	Info only	General Availability	Changed
Cloud Integration	Neo	Role-Based Protection of Connectivity Test Has Been Changed	Role <code>NodeManager.readcredentials</code> is required to perform this task (instead of role <code>NodeManager.deploycredentials</code>). The role authorization groups for the integration developer and tenant administrator persona that contain permission to perform connectivity tests have been adapted accordingly. See: Tasks and Permissions	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	AMQP Adapters Support Client Certificate Authentication	The AMQP sender and receiver adapters now support client certificate authentication for TCP transport protocol. See: • Configure the AMQP Sender Adapter • Configure the AMQP Receiver Adapter	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Integration Flow Design Guidelines Changes	The integration flow design guideline EOIO via Aggregator (with integration flow Pattern Quality Of Service - Scenario 08b) has been deleted.	Info only	Deleted	Changed
Cloud Integration	Cloud Foundry Neo	Integration Flow Design Guidelines Changes	The following new integration flow design guidelines (including integration flows) have been newly added: • Create Attachments • Read Multiple Attachments Based on Filter Criteria	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Cloud Integration	Cloud Foundry Neo	Integration Flow Design Guidelines Changes	The following integration flow design guidelines have been changed: • Decouple Sender and Flows Using Data Store now uses a data store based on Data Store sender adapter instead of a Timer event. • Decouple Sender and Flows Using Data Store and Polling Consumer now uses a data store and polling consumer based on new data store sender adapter instead of timer event. • The integration flow design guideline (and integration flow Pattern Quality Of Service - Scenario 08) has been replaced by the new guideline Resequencer (see Resequencer).	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	Handling of Duplicate Attachment Names by SOAP SOAP 1.x Sender Adapter	Handling duplicate attachment names has been improved for the SOAP SOAP 1.x sender adapter. See: Configure the SOAP (SOAP 1.x) Sender Adapter	Info only	General Availability	Changed
Cloud Integration	Cloud Foundry Neo	XI Sender Adapter Sets New Header for Quality of Service	The new header SapQualityOfService is set by the adapter. It contains the quality of service of the sender system. See: • Configure the XI Sender Adapter • Headers and Exchange Properties Provided by the Integration Framework	Info only	General Availability	
Integration Advisor	Cloud Foundry Neo	Display of XML Tag Name	You can now view the XML Tag Name of a node in the Details section of a Message Implementation Guideline. You can use this value while maintaining the XSD assertions. See: Working with a Node	Info only	General Availability	New
Integration Advisor	Cloud Foundry Neo	Custom Type System	You can now delete the active version of a custom message. See: Deleting a Custom Message	Info only	General Availability	New

Technical Component	Environment	Title	Description	Action	Lifecycle	Type
Integration Advisor	Cloud Foundry Neo	Version History of a MIG and MAG	The version history of a MIG and MAG now also displays the import details if applicable. See: Message Implementation Guidelines (MIGs) Mapping Guidelines (MAGs)	Info only	General Availability	New

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Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.33.* - SAP Integration Advisor: 1.67.* - Increment: 2210	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.41.* - SAP Integration Advisor: 1.67.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2210	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	Import/Export of a Mapping Guideline	Import and export of a mapping guideline now also includes its pre-transformation in the zip file. See: Import and Export	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	Filtering MIGs and MAGs	You can now filter message implementation guidelines and mapping guidelines using two new filters: Last Imported By	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			- Last Imported Between See: Message Implementation Guidelines (MIGs) Mapping Guidelines (MAGs)				
Integration Advisor	Cloud Foundry Neo	Download and Upload of standard codelists	- You can now download the standard codelist from any type system in csv format. - You can now upload and modify the standard codelists in a message implementation guideline. See: Creating MIG Codelists	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	Pre-Transformation of a Message Implementation Guideline	You can now use the Pre-Transformation feature in a mapping guideline to transform the structure of your message implementation guideline before mapping. See: Pretransforming a Message Implementation Guideline	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Neo to Cloud Foundry Migration Enhanced	Migration from the Neo environment to the multi-cloud foundation supports the following new steps: Migrating data stores See: Migrating Cloud Integration from the Neo Environment to the Multi-Cloud Foundation	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry	New Slack Adapter Available	The new adapter type allows you to exchange data between Slack storage and .	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Splunk Adapter Available	The new adapter type allows you to exchange data between Splunk storage and .	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Cloud Foundry Neo	Handling of Duplicate Attachment Names by Mail Sender Adapter	Handling duplicate attachment names has been improved for the Mail sender adapters (POP3 and IMAP4) by the introduction of GUIDs. See: Configure the Mail Sender Adapter	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Dynamic support for MDN properties in AS2 Sender adapter	You can now update the MDN properties of AS2 sender adapter dynamically. See: AS2 Sender Adapter: MDN	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Support for incoming message decryption in AS2 Sender adapter	AS2 adapter has been extended to configure decryption option from incoming payload. See: AS2 Sender Adapter: Security	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Support for On-Premise Postgres database in JDBC Receiver adapter	JDBC adapter now supports On-Premise Postgres database for both Neo and Cloud Foundry tenants. See: JDBC for Postgres (On-Premise)	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Feature for Timer Start Event to Avoid Validation Exception	A new checkbox is introduced to make sure that the runtime status of the integration artifact doesn't go into Error status. See: Define a Timer Start Event	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.32.* - SAP Integration Advisor: 1.66.* - Increment: 2209	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.40.* - SAP Integration Advisor: 1.66.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2209	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	Exporting a Mapping Guideline	When you export a mapping guideline (MAG), the global code value Mapping of the MAG also gets exported now. See: Import and Export	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	New Codelists collection	GS1 global codelists is now available as a new content in the Type System Library. See: Overview Of B2B Standards	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Content	New content has been added to the Integration Flow Design Guidelines - Learn the Basics integration package. Both documentation and a sample integration flow are available. See: Attachment Handling	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New entry in the Message Processing Log	A new entry ExecutedMapping is added to the message processing log that indicates which message mapping artifact was executed. See: Message Processing Log - Text View.	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Improvements for OAuth2 Client Credentials Artifact	New fields are introduced to leverage shared secret between services. See: Deploying an OAuth2 Client Credentials Artifact.	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Fixing issues with Content Enricher	There were issues with Content Enricher where it was enriching the content of an original message with unexpected content that wasn't defined as part of the content enrich strategy. The issue is fixed with 1.2 version of Content Enricher.	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Cloud Foundry Neo	New Operations Added to Script Collections Resource of Integration Content OData API	You can now delete and update a script collection. See: Integration Content	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Handle Invalid XML Characters in XML Modifier	You can now configure how to handle invalid xml characters in the XML Modifier step. See: Define XML Modifier	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.31.* - SAP Integration Advisor: 1.65.* - Increment: 2208	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.39.* - SAP Integration Advisor: 1.65.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2208	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Operations Added to Script Collections Resource of Integration Content OData API	You can now read and download a script collection and its resources. See: Integration Content	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	Creating a Message Implementation Guideline	You can now directly create a Message Implementation Guideline from a specific message type of a type system/custom type system. See: Creating Message Implementation Guidelines	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Keep-Alive for SOAP (SOAP 1.x) Receiver	You can now activate the keep-alive functionality to signal to the server that the connection should remain open.	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			See: Configure the SOAP (SOAP 1.x) Receiver Adapter				
Cloud Integration	Cloud Foundry Neo	TSL Connectivity Tests: Option to Add Root Certificate to Keystore	During a TSL connectivity test, you can now add a root certificate directly to the keystore. See: TLS Connectivity Tests	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry	Changed Length for Secure Parameter	You can now use up to 4096 characters for your secure parameter in Cloud Foundry. See: Deploying a Secure Parameter Artifact	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry	New Tutorial for Inbound OAuth Client Credentials Grant Authentication	There's a new tutorial available on developers.sap.com describing how to set up inbound authentication for API clients calling the Cloud Integration OData API. The Cloud Integration OData API provides access to various Cloud Integration resources such as message processing logs, integration content, log files, etc. See: Tutorial: Set Up Inbound OAuth Client Credentials Grant Authentication for API Clients with SAP-Generated Certificate 	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Authenticate SOAP (SOAP 1.x) Receiver with OAuth 2.0 SAML Bearer Assertion Grant when connecting to target system type SuccessFactors	You can now select OAuth 2.0 SAML Bearer Assertion Grant as authentication option for SOAP (SOAP 1.x) Receiver Adapter when connecting to target system type SuccessFactors. See: Configure the SOAP (SOAP 1.x) Receiver Adapter	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Support for Client Certificate authentication in OData V4 receiver adapter	Client Certificate is now available as an authentication type from version 1.10 and onwards of the OData V4 receiver adapter. See: Configure the OData V4 Receiver Adapter	Info only	General Availability	New	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Cloud Foundry Neo	Dynamically Assign Message Mapping Artifacts to a Message Mapping	From version 1.3 and onwards of the message mapping flow step, you can dynamically assign message mapping artifacts using a header or property, or via partner directory. This way, you can execute different message mappings from a single integration flow. See: Creating Message Mapping as a Flow Step .	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	AS2 Receiver Adapter Enhancement	On configuring AS2 receiver channel for the Request-Reply integration flow element, the AS2 receiver adapter version 1.8 and above sets the exchange header Sap_AS2MessageID with originalMessageID. See: Configure the AS2 Receiver Adapter	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	API to Publish Status of Connection to Integration Flow Monitoring	Support for a public API to publish the status of the connection to integration flow monitoring for ADK sender integration adapters. See: Enabling Connection Status for Integration Flow	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.30.* - SAP Integration Advisor: 1.64.* - Increment: 2207	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.38.* - SAP Integration Advisor: 1.64.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2207	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Deployment of Integration	You can now deploy and undeploy an integration adapter.	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
		Adapter	See: Manage Integration Content				
Cloud Integration	Cloud Foundry Neo	AS2 Receiver adapter enhancement	AS2 receiver adapter has been enhanced for Dynamic support in Proxy Type, Authentication Type, Content Transfer Encoding, and MDN Type. See Configure the AS2 Receiver Adapter	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	Import and Export MIG/MAG	- You can now import and overwrite existing status of MIG and MAG in a tenant. - You can now import new versions or overwrite existing versions of a MIG/MAG in the same tenant where it was exported from. See: Import and Export	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content has been added to the Integration Flow Design Guidelines - Enterprise Integration Patterns integration package. Both documentation and a sample integration flow are available. See: ID Mapping (with Multicast)	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Signature Algorithms Supported for PKCS7 Signer Step	The PKCS7 Signer integration flow step now supports the following additional signature algorithms: SHA3-224/RSA, SHA3-256/RSA, SHA3-384/RSA, SHA3-512/RSA, SHA3-512/DSA, SHA3-384/DSA, SHA3-256/DSA, SHA3-224/DSA, SHA512/DSA, SHA384/DSA, SHA3-224/ECDSA, SHA3-256/ECDSA, SHA3-384/ECDSA, SHA3-512/ECDSA, SHA512/ECDSA, SHA384/ECDSA, SHA256/ECDSA, SHA224/ECDSA, SHA1/ECDSA. See: Message-Level Security	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			Sign the Message Content with PKCS#7/CMS Signer				
Cloud Integration	Cloud Foundry Neo	New Signature Algorithms Supported for Simple Signer Step	The Simple Signer integration flow step now supports the following additional signature algorithms: SHA3-224/RSA, SHA3-256/RSA, SHA3-384/RSA, SHA3-512/RSA, SHA3-512/DSA, SHA3-384/DSA, SHA3-256/DSA, SHA3-224/DSA, SHA512/DSA, SHA384/DSA, SHA3-224/ECDSA, SHA3-256/ECDSA, SHA3-384/ECDSA, SHA3-512/ECDSA, SHA512/ECDSA, SHA384/ECDSA, SHA256/ECDSA, SHA224/ECDSA, SHA1/ECDSA. See: Message-Level Security Sign the Message Content with Simple Signer	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Option to Configure JMS Receiver Adapter with Dynamic Queue Name	You can now dynamically configure queue names with the JMS Receiver Adapter. See: Configure the JMS Receiver Adapter	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Support for OAuth2 SAML Bearer in SuccessFactors OData V4 Adapter	OAuth2 SAML Bearer Assertion is now available as an authentication type from version 1.7 and onwards of the SuccessFactors OData V4 receiver adapter. See: SuccessFactors OData V4 Receiver Adapter.	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.29.* - SAP Integration Advisor: 1.63.* - Increment: 2206	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.37.* - SAP Integration Advisor: 1.63.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2206	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Scheduler Support for ADK Sender Adapter	You can now use scheduler component in your custom adapter. See: Enabling Scheduler Support for ADK Sender Adapter	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Accessing On-Premise Application using Cloud Connector via TCP	Now, Adapter API supports on-premise connectivity using Transmission Control Protocol also. See: Accessing On-Premise Application using Cloud Connector	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content has been added to the Integration Flow Design Guidelines - Enterprise Integration Patterns integration package. Both documentation and a sample integration flow are available. See: Decoupling via Data Store	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Compress Stored Messages Option in the JMS Receiver Adapter	You can now compress messages in the JMS queue. See: Configure the JMS Receiver Adapter	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Parameter for PKCS#7/CMS Signer	A new parameter has been added to the PKCS#7/CMS Signer step that allows you to specify the object identifier for the content type. See: Sign the Message Content with PKCS#7/CMS Signer	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry	New Operations Added to	New POST and DELETE operations have been added to	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
	Neo	Message Mappings Resource of Integration Content OData API	the Message Mappings resource of the Integration Content API. These new operations support the creation and deletion message mappings through the API. See: Integration Content				
Cloud Integration	Cloud Foundry Neo	New Operation for Value Mappings Resource of Integration Content OData API	A new operation has been added to the Value Mappings resource of the Integration Content OData API that allows you to delete entries from value mappings. See: Integration Content	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Passthrough Option for Message Header Name	Using the AMQP receiver adapter, you can now use the Passthrough option to forward the message header name without transformation. See: Configure the AMQP Receiver Adapter	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Script Collection Resource Added to Integration Content OData API	The new resource Script Collection has been added to the Integration Content OData API. You can create and upload a script collection, add resources to a script collection, and deploy a script collection. See: Integration Content	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	PATCH operation in HTTP Receiver Adapter	The HTTP receiver adapter now supports PATCH operation to partially update resources. See: HTTP Receiver Adapter	Info only	General Availability	New	Technology
Integration Advisor	Cloud Foundry Neo	Export and Import of MIGs and MAGs	- The Export dialog now displays the number of MIGs and MAGs available in the tenant. See: Exporting MIG/MAG - You can now view the detailed information of the import result in the Import dialog. See: Importing MIG/MAG	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			You can now overwrite draft and active MIGs and MAGs. See: Import and Export				
Cloud Integration	Cloud Foundry Neo	New Dropbox Adapter Available	The Dropbox adapter allows you to connect to a user's Dropbox account from and to perform different operations as supported by the Dropbox APIs.	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Integration Adapters available in SAP Business Accelerator Hub	New integration adapters are now available in SAP Business Accelerator Hub with easy-to-consume experience. See: Consuming Integration Adapters from SAP Business Accelerator Hub	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.27.* - SAP Integration Advisor: 1.61.* - Increment: 2204	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.35.* - SAP Integration Advisor: 1.61.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2204	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Self-Service to Delay Software Update	Tenant administrators can now configure their tenants to delay the monthly updates from SAP. See: .	Recommended	General Availability	New	Technology
Cloud Integration	Cloud Foundry	New Adapter for SAP Master Data Integration	Synchronize your master data from SAP and other third-party applications with SAP Master Data Integration service.	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Importing Value Mappings from ES Repository	You can now import value mappings from ES Repository in your PI landscape to Cloud	Info only	General Availability	New	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			Integration. See: Creating Value Mapping .				
Cloud Integration	Cloud Foundry Neo	Settings for JSON Target Schema in Message Mapping	You can now configure the settings of a message mapping resource to handle basic data types for JSON target schemas. See: Creating Message Mapping as a Flow Step .	Info only	General Availability	New	Technology
Integration Advisor	Cloud Foundry Neo	Import and Export Message Implementation Guidelines and Mapping Guidelines	You can now import message implementation guidelines and mapping guidelines. See: Import and Export Importing MIG/MAG Exporting MIG/MAG	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Enhancing Batch Payload for JDBC Adapters	For JDBC receiver adapters (version 1.5 and above), you can now use multiple access tags in INSERT Mode of Batch Payload functionality. See: Batch Payload and Operation	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Message Mappings Resource Added to Integration Content OData API	The new resource Message Mappings has been added to the Integration Content API. A number of GET operations and query options allow you to access message mappings through the API. See: Integration Content	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.26.* - SAP Integration Advisor: 1.60.* - Increment: 2203	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.34.* - SAP Integration Advisor: 1.60.* - Adapter Development Kit for SAP Cloud Integration: 2.53.*	Info only	General Availability	New	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			Increment: 2203				
Cloud Integration	Cloud Foundry Neo	OAuth2 SAML Bearer Credential in HTTP Receiver Adapter for SuccessFactors System	You can now use a OAuth2 SAML Bearer Assertion credential, that has a Key Pair Common Name and SuccessFactors target system, in HTTP Receiver adapter. This helps you remove dependency on using basic authentication to connect to a SuccessFactors OData V2 system. See: Deploying an OAuth2 SAML Bearer Assertion .	Recommended	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Server-Side Pagination in OData V4 Receiver Adapter	OData V4 Receiver adapter now supports pagination. See: Configure the OData V4 Receiver Adapter .	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Signature Algorithms and Canonicalization Methods Supported for XML Digital Signer	The XML Digital Signer step now supports new signature algorithms and canonicalization methods. See: Sign the Message Content with XML Digital Signature	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Enhancing AS2 Adapters to set Message Status to Failed on Negative MDN	You can now set the message status to Failed on negative MDN for both Asynchronous and Synchronous MDN type. See AS2 Adapter	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Extending Batch Support to Native SQL Queries for JDBC Receiver Adapters	Batch support is now enabled for native SQL queries also. See, JDBC Receiver Adapter	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.25.* - SAP Integration Advisor: 1.59.* - Increment: 2202	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	SAP Cloud Integration: 5.33.*	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			- SAP Integration Advisor: 1.59.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2202				
Cloud Integration	Cloud Foundry Neo	Consuming Adapters from SAP API Business Hub	You can easily consume adapters that are published in SAP API Business Hub while designing your integration. See: Import Integration Adapters .	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content has been added to the Integration Flow Design Guidelines - Enterprise Integration Patterns integration package. Both documentation and a sample integration flow are available.	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Query Option Available for Message Store OData API	The Cloud Integration Message Stores API supports a new query option to get stopped JMS queues. See: JMS Resources Example Requests JMS Resources 	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Data Store Sender Adapter Available	The new adapter enables Cloud Integration to consume messages from a data store. This feature helps you to enable asynchronous decoupling of inbound and outbound processing by using the data store as temporary storage. See: Data Store Sender Adapter	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.24.* - SAP Integration Advisor: 1.58.* - Increment: 2201	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	SAP Cloud Integration: 5.32.*	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			- SAP Integration Advisor: 1.58.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2201				
Cloud Integration	Cloud Foundry Neo	Support for Retry in SuccessFactors OData V2 Receiver Adapter	The new version of SuccessFactors OData V2 receiver adapter supports retry for HTTP response code 429 for all operations now. See: Configure the SuccessFactors OData V2 Receiver Adapter .	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Log Level Error Available	A new log level Error has been introduced. Choose this log level to records basic information for failed message executions only. See: Setting Log Levels	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Support for Null Values in OData V4 and SuccessFactors OData V4 Receiver Adapters	The new versions of OData V4 and SuccessFactors OData V4 receiver adapters now support representing null values in both request and response. See: Configure the OData V4 Receiver Adapter SuccessFactors OData V4 Receiver Adapter	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Connection Metering	Tenant administrators can now generate report on details of the connections associated with the tenant for a particular date that is metered and billed using the Provisioning application.	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	Mapping Guideline	You can now filter and search for code values in the Global code value mapping of a mapping guideline. See: Mapping Guidelines (MAGs) .	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content has been added to the Integration Flow Design Guidelines - Handle Errors Gracefully integration package. Both documentation and a	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			sample integration flow are available. It shows how to handle exceptions raised in a receiver connected through the HTTP receiver adapter. See: Don't Throw Exception on Failure				
Cloud Integration	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.23.* • SAP Integration Advisor: 1.57.* • Increment: 2113	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	• SAP Cloud Integration: 5.31.* • SAP Integration Advisor: 1.57.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2113	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content has been added to the Integration Flow Design Guidelines - Enterprise Integration Patterns integration package. Both documentation and a sample integration flow are available. Additionally, the documentation of the various use cases and guidelines has been improved. New example scenarios show how to use the Idempotent Process Call step in order to guarantee quality of service Exactly Once. See: • Idempotent Process Call Handles Duplicates • Idempotent Process Call Handles Duplicates (with JMS and Data Store Operations)	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			Idempotent Process Call Handles Duplicates (With Alternative Response)				
Cloud Integration	Cloud Foundry	Archiving Destination Supports OAuth Authentication	The archiving function now supports the OAuth authentication for the archiving destination. See: Configuring Destination	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	Kafka Adapter Supports TLS 1.3	The Kafka adapter supports Transport Layer Security (TLS) 1.3 protocol for outbound and inbound communication. See: Configure the Kafka Sender Adapter Configure the Kafka Receiver Adapter	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	Guidelines have been added related to the communication of integration flows deployed on the same tenant. See: Communication between Integration Flows	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content has been added to the Integration Flow Design Guidelines - Handle Errors Gracefully integration package. Both documentation and a sample integration flow are available. Additionally, the documentation of the various use cases and guidelines has been improved. See: Handle Errors Gracefully Handle Exceptions in Dependent Integration Flows (Simple Scenario) Handle Exceptions in Subprocess (Simple Scenario)	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Integration Advisor	Cloud Foundry Neo	Message Implementation Guideline	You can now download the code values from the codelist in a message implementation guideline. See: Creating MIG Codelists	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.22.* - SAP Integration Advisor: 1.56.* - Increment: 2112	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Software Version Update	- SAP Cloud Integration: 5.30.* - SAP Integration Advisor: 1.56.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2112	Info only	General Availability	Changed	Technology
Cloud Integration	Neo	Handling of Artifacts with Status Stopping Has Been Improved	Handling of integration artifacts in Stopping state has been improved in the Monitor section (under Manage Integration Content). During undeployment of an artifact, the artifact is not anymore immediately removed from the artifact list. Instead of this, the artifact is still shown, but its status changes to Stopping. Furthermore, for artifacts with status Stopping the following applies: - The functions Restart, Undeploy, and Download aren't available. - You can still navigate to the artifact (integration flow model) and to the message monitor. See: Runtime Status	Info only	General Availability	Changed	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content has been added to the Integration Flow Design Guidelines - Enterprise Integration Patterns integration package. Both documentation and a sample integration flow are available. One example integration scenario has been added that shows how to use the ID Mapping step together with the AS2 sender adapter. Another example scenario shows how to set up quality of service Exactly Once when the receiver system isn't idempotent. See: • ID Mapping (with AS2 Sender Adapter) • XI Sender Adapter Handles Duplicates	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Enhanced Cloud Integration OData API	With the enhancement of the OData API, you can now get the build and deploy status of an integration artifact. This status indicates the deployment status of an artifact triggered for deployment. See: Integration Content Get Runtime Status of Deployed Integration Flow	Info only	General Availability	Changed	Technology
Cloud Integration	Cloud Foundry	New Optional Parameters for Data Archiving	We introduced new optional parameters: • ClientCompression • ResponseCompression • CookiesForAuthentication • ConnectTimeout • ReadTimeout See: Configuring Destination	Info only	General Availability	New	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
Cloud Integration	Cloud Foundry Neo	Option to Disable Throwing Exceptions in HTTP Receiver Adapter	You now have an option to ignore the HTTP failure responses from remote server and proceed with the message processing. Earlier, by default, when there were HTTP failure responses, the message processing failed too. See: HTTP Receiver Adapter .	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	TLS 1.3 Protocol Version Supported	You can now use Transport Layer Security (TLS) 1.3 protocol for outbound communication. See Connectivity Options and Communication Security	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Process Call Step	You can now execute a process call step to check if an incoming message was already processed, and skip the processing of this message. See Define Idempotent Process Call	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	New Integration Flow Design Guideline Content	New content that shows how to transfer files has been added to the Integration Flow Design Guidelines – Learn the Basics integration package. Both documentation and a sample integration flow are available. See: • Combine XML Files via Poll Enrich • Poll and Merge Folder	Info only	General Availability	New	Technology
Cloud Integration	Cloud Foundry Neo	Allow Header Folding in MIME Encoder	You can now unfold long headers in the MIME encoder to comply to the mail protocol. See Define a MIME Multipart Encoder .	Info only	General Availability	New	Technology
Integration Advisor	Cloud Foundry Neo	Deleting a Qualified Instance in a Message Implementation Guideline	Deleting one or more qualified instances has now been enhanced to retain the values of the node. See: Additional Options For Qualification	Info only	General Availability	Changed	Technology
Integration Advisor	Cloud Foundry Neo	XSD Assertion in Message Implementation Guidelines	You can now extend the MIG validation to validate dependencies between different fields of the MIG structure using XSD assertions and XSD patterns	Info only	General Availability	New	Technology

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business
			of a group node and leaf node respectively. See: Working with a Node				

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Core Components, 2021

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.20.* - Increment: 2110	Info only	Changed	2021-12-04
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 5.28.* - SAP Integration Advisor: 1.64.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2110	Info only	Changed	2021-12-04
Cloud Integration	Integration Suite	Cloud Foundry Neo	Billing of Messages Using SAP-to-SAP Content	Starting with software increment 2110, all messages processed by modified standard SAP-to-SAP integration content (from SAP Business Accelerator Hub) are now charged according to the official metric. For more information, see SAP Note 294234  .	Info only	Changed	2021-12-04
Integration Advisor	Integration Suite	Cloud Foundry Neo	Creating a Mapping Guideline	You can now copy a mapping guideline using two different options. See: Creating Standard Graphical Mapping Guidelines	Info only	Changed	2021-12-04
Integration Advisor	Integration Suite	Cloud Foundry	Working with a Mapping Guideline	You can now search for a	Info only	New	2021-12-04

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
		Neo		particular node in the source and target structure of a mapping guideline. The mapping type is now indicated by an icon above the mapping line. See: Mapping the Source and Target Nodes			
Cloud Integration	Integration Suite	Cloud Foundry Neo	New Parameter for AMQP Sender Adapter	The mapping type is now indicated by an icon above theThe AMQP sender adapter comes with a new parameter that allows you to consume expired messages. See: Configure the AMQP Sender Adapter	Info only	Changed	2021-12-04
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Design Guideline Content	New integration flow design guidelines have been added to the Integration Flow Design Guidelines - Enterprise Integration Patterns integration package. Both documentation and a sample integration flow are available. See: • Variant: Dynamic Routing Using JMS Message Queues • Quality of Service Exactly Once	Info only	New	2021-12-04
Cloud Integration	Integration Suite	Neo Cloud Foundry	Generic Provider for OAuth2 Client Authorization Code	You can now use the Generic Provider for OAuth2 Client Authorization Code. See: Deploying an OAuth2 Client Authorization Code with Generic Provider	Info only	New	2021-12-04
Cloud Integration	Integration Suite	Neo Cloud Foundry	Batch Mode and Operations in JDBC Receiver Adapter	You can now perform batch operations like modifying multiple	Info only	New	2021-12-04

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				documents in one transaction. See: JDBC Receiver Adapter			
Cloud Integration	Integration Suite	Neo Cloud Foundry	Introducing Community Packages	Community Packages are prepackaged, editable, open-source integration content developed by the integration experts in the community. See Working with Prepackaged Integration Content .	Info only	New	2021-12-04
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.19.* - Increment: 2109	Info only	Changed	2021-10-16
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 5.27.* - SAP Integration Advisor: 1.63.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2109	Info only	Changed	2021-10-16
Integration Advisor	Integration Suite	Neo Cloud Foundry	Migrating a Message Implementation Guideline	You can now successfully migrate a message implementation guideline that is based on a IDoc. See: Migrating a Message Implementation Guideline	Info only	New	2021-10-16
Integration Advisor	Integration Suite	Neo Cloud Foundry	Extending a qualification	You can now extend an existing qualification of a message implementation guideline by adding an additional qualifier. See: Creating a Qualified Instance	Info only	Changed	2021-10-16
Cloud Integration	Integration Suite	Neo Cloud Foundry	Validation Step Introduced for JSON-to-XML Converter	The JSON-to-XML converter now checks for each JSON member name if it can be converted into a valid XML element or attribute name. If not, the	Recommended	Changed	2021-10-16

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				system raises an exception. The introduced enhancement can have an impact on existing integration flows containing the previous version of the JSON-to-XML converter. Because of the newly introduced validation check, scenarios that have processed invalid JSON member names before can now result in an error. Action: Check if your existing integration scenarios have processed invalid JSON member names (not convertible into valid XML element or attribute names) before. If that's the case, change your scenarios so that only that kind of JSON content is processed that can be converted into a valid XML. To find more information on which characters are allowed in your JSON content so that Cloud Integration can convert it into valid XML, see: Limitations for JSON to XML Conversion. See also: 3112970 - JSON-to-XML Converter Exception Caused by Invalid JSON Member Name  (Knowledge Base Article)			
Cloud Integration	Integration Suite	Neo Cloud Foundry	Use Fast Exists Check for SFTP Adapter	You can now enable the Fast Exists Check for the SFTP sender and receiver adapter. See: Configure the SFTP Sender	Info only	New	2021-10-16

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				Adapter Configure the SFTP Sender Adapter Used with the Poll Enrich Step Configure the SFTP Receiver Adapter			
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Design Guideline Content	A new integration flow design guideline has been added to the Integration Flow Design Guidelines – Use Scripting Appropriately integration package. Both documentation and a sample integration flow are available. See: Parse JSON	Info only	New	2021-10-16
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.18.* - Increment: 2108		Changed	2021-09-18
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 5.26.* - SAP Integration Advisor: 1.62.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2108		Changed	2021-09-18
Integration Advisor	Integration Suite	Neo Cloud Foundry	Mapping Guidelines	Mapping guideline now supports group to leaf mapping. See: Mapping the Source and Target Nodes		New	2021-09-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improvements to Access Policies	The scope of access policies now includes the protection of: Script collection artifacts		Changed	2021-09-18

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				- Runtime operations like deploy and undeploy, restart, download, log level change, and configure archiving of the artifacts See: Creating Custom Roles for Access Policies, Cloud Foundry Environment Creating Custom Roles for Access Policies, Neo Environment			
Cloud Integration	Integration Suite	Neo Cloud Foundry	Deployment Error Message in the Integration Flow Editor	Deployment error message is now shown in the Deployment Status tab of the integration flow editor. See: Deployment Status View .		New	2021-09-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Enhancements to Copy and Paste of Integration Flow Steps	You can now copy and paste multiple flow steps in one go. See: Overview of Integration Flow Editor .		Changed	2021-09-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Enhancements to HTTP Receiver Adapter	In case of message failures while a HTTP Receiver adapter calls the target system, message processing log attachments are created. HTTP request headers, response headers, and response body are created as unique attachments depending upon the message failure scenario. See: HTTP Receiver Adapter .		Changed	2021-09-18
Cloud Integration	Cloud Integration	Neo Cloud Foundry	Kafka Polling Monitor for the Kafka Sender Adapter	After deployment of the integration flow with the Kafka Sender adapter, if there's an exception thrown during polling for whichever reason, the		New	2021-09-18

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				corresponding error details are shown in the Monitor section. See: Configure the Kafka Sender Adapter			
Cloud Integration	Integration Suite	Neo Cloud Foundry	New ID Mapping Step Available	You can use the ID Mapping step to map a source message ID to a target message ID. You can use this feature to implement scenarios with exactly once processing of messages, for example. See: Define ID Mapping		New	2021-09-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Design Guideline Content	A new integration flow design guideline has been added to the Integration Flow Design Guidelines – Use Scripting Appropriately integration package. Both documentation and a sample integration flow are available. See: • Access URL Get Parameters in Scripts • Access URL Paths in Scripts		New	2021-09-18
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.17.* • Increment: 2107		Changed	2021-08-21
Cloud Integration	Integration Suite	Neo	Software Version Update	• SAP Cloud Integration: 5.25.* • SAP Integration Advisor: 1.61.* • Adapter Development Kit for SAP Cloud Integration: 2.53.* • Increment: 2107		Changed	2021-08-21

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Archiving Data	You can connect Cloud Integration to a remote content management system and use this system to archive data. See: Archiving Data, Cloud Foundry Environment		New	2021-08-21
Integration Advisor	Integration Suite	Cloud Foundry	Migrating a Message Implementation Guideline	Migration of a message implementation guideline will now consider Compound qualified nodes as well. See: Migrating a Message Implementation Guideline		Changed	2021-08-21
Integration Advisor	Integration Suite	Neo Cloud Foundry	Simulating a Message Implementation Guideline	You now have the option to rerun simulation on a message implementation guideline based on example data or payload data. See: Simulating Message Implementation Guidelines		Changed	2021-08-21
Integration Advisor	Integration Suite	Neo Cloud Foundry	Simulating a Mapping Guideline	You can now view all the potential instances of a node after simulating a mapping guideline, independent of whether the instances are created or not. See: Simulating Mapping Guidelines		New	2021-08-21
Cloud Integration	Cloud Integration	Neo Cloud Foundry	New Integration Flow Design Guideline Content	A new integration flow design guideline has been added to the Integration Flow Design Guidelines - Learn the Basics integration package. Both documentation and a sample integration flow are available. See: Decouple Sender and Flows Using JMS Message Queues A new integration flow design guideline has been added to the Integration		Changed	2021-08-21

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				Flow Design Guidelines - Apply Highest Security Standards integration package. Both documentation and a sample integration flow are available. See: Protect Data Integrity			
Cloud Integration	Integration Suite	Neo Cloud Foundry	Enhancements to Copy and Paste of Integration Flow Steps	You can now copy flow steps and paste them across different integration flows. See: Overview of Integration Flow Editor .		Changed	2021-08-21
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improvements to Access Policies in Design Time	Message mappings and value mappings can also be protected using access policies. See • Creating Custom Roles for Access Policies, Cloud Foundry Environment • Creating Custom Roles for Access Policies, Neo Environment		Changed	2021-08-21
Cloud Integration	Integration Suite	Neo Cloud Foundry	Change in Delta Sync Range for SuccessFactors SOAP Adapter	For SuccessFactors SOAP Receiver adapter, the delta sync time range is changed to a 3-month timeframe. Earlier, the time range was from 01-01-1970. See: Configure the SuccessFactors SOAP Receiver Adapter .		Changed	2021-08-21
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.16.* • Increment: 2106		Changed	2021-07-26
Cloud Integration	Integration Suite	Neo	Software Version Update	• SAP Cloud Integration: 5.24.* • SAP Integration Advisor: 1.60.* • Adapter Development Kit		Changed	2021-07-26

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				for SAP Cloud Integration: 2.53.* Increment: 2106			
Integration Advisor	Integration Suite	Neo Cloud Foundry	Conditional Mapping	You can now view the details of the input and outcome of a conditional mapping after simulating a mapping guideline. See: Simulating Mapping Guidelines		Changed	2021-07-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	New On-Premise Database	The JDBC adapter now supports the connection to on-premise HANA databases. See: JDBC Receiver Adapter .		New	2021-07-26
Integration Advisor	Integration Suite	Neo Cloud Foundry	Message Implementation Guideline	- You can now simulate a message implementation guideline using the payload data. - You can now view the distinction between nonexistent and empty leaf nodes after simulating the mapping structure of a message implementation guideline. See: Simulating Message Implementation Guidelines		New	2021-07-26
Integration Advisor	Integration Suite	Neo Cloud Foundry	Code Value Mapping	You can now delete or change the deprecated values found in the code value mapping in a mapping guideline. See: Value Transformations		Changed	2021-07-26

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Design Guideline Content	A new integration flow design guideline has been added to the Integration Flow Design Guidelines - Handle Errors Gracefully integration package. Both documentation and a sample integration flow are available. See: Handle Errors in Successful Responses		Changed	2021-07-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Throw Exception in Poll Enrich	You can now enable the option to throw an exception in the message processing of the Poll Enrich step if no message is found. See: Define Poll Enrich		Changed	2021-07-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Deployment Status view in the integration artifact editor	Deployment Status is now available in the property sheet of the integration artifact editor. See: Deployment Status View.		New	2021-07-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Copy and Paste Integration Flow Steps while Modeling Integration artifacts	You can now copy flow steps and paste within the artifact. See: Overview of Integration Flow Editor.		New	2021-07-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improvements of LDAP Receiver Adapter	- Search operation now returns multi-valued attributes using semicolon or xml tags. - You can now define the input type for a modify operation. - Modify operation now supports add, remove, and replace of the entities. See: LDAP Receiver Adapter.		Changed	2021-07-26

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improvements of Access Policies in Design Time	OData APIs can also be protected using access policies. See • Creating Custom Roles for Access Policies, Cloud Foundry Environment • Creating Custom Roles for Access Policies, Neo Environment		Changed	2021-07-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Support for OAuth2 SAML Bearer Certificate Authentication in SF Adapters	SuccessFactors SOAP Sender Adapter and the Query Modeling wizards of SuccessFactors V2 OData and SuccessFactors SOAP Receiver adapters now support OAuth2 credentials. See: • Configure the SuccessFactors (SOAP) Sender Adapter • Configure the SuccessFactors OData V2 Receiver Adapter • Modifying SuccessFactors SOAP Entity and Operation		New	2021-07-26
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	• SAP Cloud Integration: 6.15.* • Increment: 2105		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo	Software Version Update	• SAP Cloud Integration: 5.23.* • SAP Integration Advisor: 1.59.* • Adapter Development Kit for SAP Cloud Integration: 2.53.*		Changed	2021-06-26

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				Increment: 2105			
Integration Advisor	Integration Suite	Neo Cloud Foundry	Compound Qualification	You can now qualify a node using more than one qualifying nodes in a message implementation guideline. See: Creating a Qualified Instance		New	2021-06-26
Integration Advisor	Integration Suite	Neo Cloud Foundry	Mapping Guideline	You can now view the distinction between nonexisting and empty leaf nodes after simulating the mapping structure of a mapping guideline. See: Working with Mapping Guidelines (MAGs)		New	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Poll Interval for XI Adapter	You can now define the poll interval for both the XI Sender and the XI Receiver adapter, specifying the waiting time before a new attempt is made to consume messages from the data store. See: Configure the XI Sender Adapter Configure the XI Receiver Adapter		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Selective Transport of Integration Artifacts	Now you can transport one or more integration artifacts from your integration package to another integration tenant hosted on the same environment. See: Content Transport		Changed	2021-06-26

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Modifying Java User Defined Functions (UDFs)	You can now view and edit Java UDF mapping content that was imported from ES Repository. See: Importing Mapping Content from ES Repository		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	JDBC Driver	Now, you can upload and deploy type-4 compliant IBM DB2 JDBC drivers in runtime to access DB2 database. See: Configure JDBC Drivers		New	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Design Guideline Content	A new integration flow design guideline has been added to the Integration Flow Design Guidelines - Learn the Basics: Message Mapping integration package. Both documentation and a sample integration flow are available. See: Message Mapping		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Message Mapping as Artifact	Now, you can create message mapping artifact and reuse them by reference across different integration flows within the same integration package. See: Creating Message Mapping as an Artifact		New	2021-06-26
Cloud Integration	Integration Suite	Cloud Foundry	Improvements to Malware Scanning Capability	With this release, script collection artifacts and integration packages are also scanned for malware before upload. See: Malware Scanner .		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Support for OAuth2 SAML Bearer Certificate Authentication in SF Adapter	SuccessFactors SOAP Receiver Adapter now supports OAuth2 credentials. See: Deploying an OAuth2 SAML Bearer Assertion .		New	2021-06-26

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Update related to ELSTER Adapter	This software release contains an update for the ELSTER receiver adapter: The German tax authorities have released a new version (33.4.4.0) of the ERiC (ELSTER Rich Client) library. This update requires version 12 for the ELSTER data collection (Datenabholung). If you use the ELSTER adapter, in particular, together with the data collection method for ETStmt (LStB) and ELStAM, make sure that you implement the following SAP note 3067520  in your HR system. See: ELSTER Receiver Adapter		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Keep File and Process Again option for the SFTP Sender Adapter used with the Poll Enrich Step	You can now select the option Keep File and Process Again for the SFTP Sender Adapter used with the Poll Enrich Step to allow repeated and parallel access to the same file without moving or deleting it. See: Configure the SFTP Sender Adapter Used with the Poll Enrich Step		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Data Store Default Expiry Period	The data store default expiry period was changed from 90 to 30 days. See: Define Data Store Write Operations		Changed	2021-06-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improvement of Access Policies in Design Time	REST and SOAP APIs can also be protected using access policies. See • Creating Custom Roles for Access Policies, Cloud		Changed	2021-06-26

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				Foundry Environment Creating Custom Roles for Access Policies, Neo Environment			
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.14.* - Increment: 2104		Changed	2021-05-29
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 5.22.* - SAP Integration Advisor: 1.58.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2104		Changed	2021-05-29
Integration Advisor	Integration Suite	Cloud Foundry Neo	Message Implementation Guideline	You can now activate a message implementation guideline that is based on a custom message by activating the custom message first. See:		Changed	2021-05-29
Cloud Integration	Integration Suite	Cloud Foundry Neo	PDF Guides in SAP Cloud Integration	To improve access to information, we're discontinuing to support all PDF guides, except for the Feature Scope Description guide. The PDF guides will no longer be updated, and it will be deleted. Going forward we do not recommend using the existing PDFs.		Deleted	2021-05-29
Integration Advisor	Integration Suite	Neo Cloud Foundry	Migrating a Message Implementation Guideline	Migration of user-defined qualifier markers and qualifications based on newly added global codelists are now supported. See: Migrating a Message Implementation Guideline		Changed	2021-05-29

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Introducing Script Collection Artifact	You can create script collection artifacts that can contain supported script resources like Groovy script, JavaScript, and Jar (archive) files in it. Script collection comes with benefits like reusable script resources, reduced memory usage, ease of maintenance. See Working with Script and Script Collection .		New	2021-05-29
Cloud Integration	Integration Suite	Neo	JDBC Receiver Adapter	You can now use JDBC receiver adapter to connect with HANA 2.0 Database (DB) for Cloud Integration tenants hosted on Neo Environment. Existing users can upgrade their HANA DB from 1.0 to 2.0.		Changed	2021-05-29
Cloud Integration	Integration Suite	Neo Cloud Foundry	Consuming Access Policies in Design Time	As a tenant administrator, you can now apply access policies for the integration artifacts at design time. Earlier, you could only protect the business data created during the runtime execution of integration artifacts. See Creating Custom Roles for Access Policies, Cloud Foundry Environment Creating Custom Roles for Access Policies, Neo Environment		Changed	2021-05-29
Cloud Integration	Integration Suite	Neo Cloud Foundry	Enhanced Cloud Integration OData API	The Idempotent Repository resource of the Message Processing Logs API has been deprecated and replaced by a new one. See: Message Processing Logs		Changed	2021-05-29

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Listing Allowed Headers in the HTTP Receiver Adapter	You can now list the request headers that must go from and the response headers that must come to the HTTP Receiver Adapter. Earlier, the adapter exchanged all the headers by default leading to few instances of message processing failure. See HTTP Receiver Adapter .		New	2021-05-29
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Design Guideline Content	New guidelines for integration flow design have been added to the integration packages Integration Flow Design Guidelines - Relax Dependencies to External Components and Integration Flow Design Guidelines - Run an Integration Flow Under Well-Defined Boundary Conditions. See: • Perform OData Batch Requests • Reduce Size of OData Content Enricher Response • Reduce the Memory Consumption for Splitter Scenarios		Changed	2021-05-29
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Features Added to SFTP Adapter for Poll Enrich Step	Additional features are supported when connecting to an external component using the Poll Enrich step: You can now dynamically configure the Proxy Type and Authentication parameters. See: Configure the SFTP Sender Adapter Used with the Poll Enrich Step		Changed	2021-05-29

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.13.* - Increment: 2103		New Changed	2021-05-01
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 5.21.* - SAP Integration Advisor: 1.57.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2103		New Changed	2021-05-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	Parameters of PGP Encryptor/Decryptor Dynamically Configurable	You can now configure the following parameters dynamically based on headers or properties: PGP Encryptor: Encryption key user ID and signer user ID See: Define PGP Encryptor PGP Decryptor: Public key user ID See: Define PGP Decryptor Assume that you like to send an encrypted and signed message to multiple receivers and use specific keys for each receiver. Furthermore, assume that the receivers are determined at runtime from the content of the inbound message. You can use this feature to dynamically get the required encryption and signing key dynamically at runtime.		Changed	2021-05-01

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Poll Enrich Integration Flow Step Type	You can use this step together with the SFTP sender adapter to poll (read) content from an external component and to enrich the original message with this content. See: Define Poll Enrich		New	2021-05-01
Cloud Integration	Integration Suite	Neo	Configure the Kafka Sender or Receiver Adapter	You can now configure the Kafka Sender Adapter or the Kafka Receiver Adapter to connect to an external Kafka broker via Kafka protocol. See: Kafka Adapter .		New	2021-05-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	Perform Kafka Adapter Connectivity Tests	You can now perform connectivity tests for the Kafka adapter.		New	2021-05-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	Define the Number of Concurrent Processes for JMS Queues in XI Adapter	You can now define the number of concurrent processes for JMS queues for the XI sender and the XI receiver. See: • Configure the XI Sender Adapter • Configure the XI Receiver Adapter		New	2021-05-01
Cloud Integration	Integration Suite	Cloud Foundry	Malware Scanning Capability	With this release, OData API projects and keystore files are also scanned for malware before upload. See: Malware Scanner .		Changed	2021-05-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improvements of Design Time Artifact Locks	A tenant administrator can now release a locked design time artifact immediately after a user locked it. Earlier, the tenant admin could unlock the artifact only 24 hours after the user locked it. See: Design Time Artifact Locks .		Changed	2021-05-01

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improvements of Timer Start Event	Improvements are made to the timer start event to avoid several triggers of the same integration flow at the same time.		Changed	2021-05-01
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.12.* - Increment: 2102		Changed	2021-04-09
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 3.36.* / 5.20.* - SAP Integration Advisor: 1.56.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2102		Changed	2021-04-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Adapters in SAP Cloud Integration	SAP has released two adapters to enhance your integration and connectivity options. See:		New	2021-04-09
Cloud Integration	Integration Suite	Neo	Access Policy Update	Access policies now also guard the access to variables. Creating Custom Roles for Access Policies, Neo Environment Creating Custom Roles for Access Policies, Cloud Foundry Environment		Changed	2021-04-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Message Status	The new message status ABANDONED is set in the MPL, when the message processing is interrupted		Changed	2021-04-09

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				because of a re- or undeployment of an integration flow, or a controlled worker node shutdown. This status is not final and the processing might be resumed if retries are configured. Message Status			
Cloud Integration	Integration Suite	Neo Cloud Foundry	Feature Update	The search of Custom Header Properties is available in the Message Monitor of the Web UI. See: Headers and Exchange Properties Provided by the Integration Framework		Changed	2021-04-09
Integration Advisor	Integration Suite	Neo Cloud Foundry	Automotive Edifact Subsets Supported	SAP Integration Advisor now provides the B2B Libraries for the automotive EDIFACT subsets (JAIF EDIFACT, Odette EDIFACT, and VDA EDIFACT) and the associated automotive codelists.		New	2021-04-09
Integration Advisor	Integration Suite	Neo Cloud Foundry	Filter MIGs and MAGs based on user identifier and a date range	You can now filter MIGs and MAGs based on the following filter criteria: Created By Modified By Created Between Modified Between See: Message Implementation Guidelines (MIGs). Mapping Guidelines (MAGs).		New	2021-04-09
Integration Advisor	Integration Suite	Neo Cloud Foundry	Conditional Mapping	You can now specify conditions on leaf nodes to control the creation and cardinality of the target group node instances in a group-to-group mapping. See: Mapping the Source and Target Nodes.		New	2021-04-09

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
Cloud Integration	Integration Suite	Cloud Foundry	Malware Scanning Capability	As tenant administrators, you can now enable malware scan for the design time files that are uploaded by the tenant users. See: Malware Scanner .		New	2021-04-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Data Compression for SFSF OData V2 Receiver Adapter	SuccessFactors OData V2 Receiver Adapter supports data compression that improves the message processing time. See: Configure the SuccessFactors OData V2 Receiver Adapter .		New	2021-04-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Support for OAuth2 SAML Bearer Certificate Authentication in SF OData Adapter	SuccessFactors OData V2 Receiver Adapter now supports OAuth2 for technical user propagation. Earlier, only support for principal propagation was available. See: Deploying an OAuth2 SAML Bearer Assertion .		New	2021-04-09
Cloud Integration	Integration Suite	Neo	Improvements of Message Structure Using JSON in Message Mapping	Some of the limitations with message structure while using Swagger/OpenAPI spec JSON file in message mapping are fixed now. You can now use untyped, mixed-type, array-of-arrays type, and array-at-root type structures. See: Creating Message Mapping as a Flow Step .		Changed	2021-04-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Packaging in XI Sender Adapter	Packaging is now supported by the XI Sender Adapter. See: Configure the XI Sender Adapter .		New	2021-04-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Design Guideline Content	The following new content has been added to the integration flow design guidelines: A set of new integration flows and documentation		Changed	2021-04-09

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				illustrate how to define transaction handling properly. See: Define Proper Transaction Handling - A new topic summarizes the naming conventions relevant for integration developers. See: Naming Conventions - A new topic explains how to optimize the memory footprint of your scenario. See: Optimize Memory Footprint			
Cloud Integration	Integration Suite	Neo Cloud Foundry	Enhanced Cloud Integration OData API	You can now access data stores, data store entries, and variables using the OData API. See: Message Stores		New	2021-04-09
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.11.* - Increment: 2101		Changed	2021-03-06
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 3.35.* - SAP Integration Advisor: 1.55.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2101		Changed	2021-03-06
Integration Advisor	Integration Suite	Neo Cloud Foundry	Migrating a Message Implementation Guideline	You can now migrate your MIG to a different version (newer or older) of the same Type System.		New	2021-02-14

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				See: Migrating a Message Implementation Guideline .			
Cloud Integration	Integration Suite	Neo Cloud Foundry	Enhanced Cloud Integration OData API	The supported system query options for JMS Resources were extended to now allow the query option \$expand. See: Message Stores .		Changed	2021-03-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simple Object Access Protocol (SOAP) 1.2	The SOAP (SAP RM) Adapter and SOAP (SOAP 1.x) Sender Adapter now support Simple Object Access Protocol (SOAP) 1.2. See: • Configure the SOAP (SAP RM) Sender Adapter • Configure the SOAP (SOAP 1.x) Sender Adapter		Changed	2021-03-06
Cloud Integration	Integration Suite	Cloud Foundry	Subscribing to the Service	Now, experience the simplified way to subscribe to Process Integration or Cloud Integration service via the Service Marketplace in your SAP BTP cockpit account. See: Initial Setup .		New	2021-03-06
Cloud Integration	Integration Suite	Cloud Foundry	Configure the Kafka Sender or Receiver Adapter	You can now configure the Kafka Sender Adapter or the Kafka Receiver Adapter to connect to an external Kafka broker via Kafka protocol. See: Kafka Adapter .		New	2021-03-06
Cloud Integration	Integration Suite	Neo	OAuth Inbound Authentication Supports JSON Web Token	SAP Cloud Integration now supports the usage of JSON Web Token (JWT) for inbound authentication with OAuth client credentials grant. Usage of JSON Web Token (JWT) is also supported for authentication. The		Changed	2021-03-06

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
				advantage of using JWT is that at runtime no additional steps are required to have an identity provider validate the token. Therefore, this feature results in a better performance under high load when a token is used for multiple calls within the limit of its validity period. See: OAuth Client Credentials Grant			
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	The integration flow design guidelines have been enhanced in the following way: - One guideline (including integration content) has been added to show how to retrieve only delta data from a source system. See: Delta Synchronization - A new topic contains detailed information on how to modify content during integration flow processing. See: Modify Content		Changed	2021-03-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	Self-Service Capability to Unlock Design Time Artifacts	As tenant administrators, you can now unlock the design time artifacts that are locked by any user of the tenant. Design time Artifact Locks.		New	2021-03-06
Cloud Integration	Integration Suite	Neo	Enhancements for Flow Step Recommendation	You can now see flow step recommendations on the sequence connectors		Changed	2021-03-06

Technical Component	Capability	Environment	Title	Description	Action	Type	Available as of
		Cloud Foundry		between two flow steps. Earlier, the machine learning based recommendation was available only for the flow steps. See Using Flow Step Recommendation .			
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.10.* - Increment: 2013		Changed	2021-02-14
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 3.34.* - SAP Integration Advisor: 1.54.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2013		Changed	2021-02-14
Integration Advisor	Integration Suite	Neo Cloud Foundry	Deleting a Mapping Guideline	You can now choose to delete a MAG including its history or only the specific version of a MAG. See: Deleting Mapping Guidelines		Changed	2021-02-14
Cloud Integration	Integration Suite	Neo Cloud Foundry	AmazonWebServices Sender Adapter	The Amazon Web Services (AWS) sender adapter enables your tenant to transfer data to AWS cloud platform. See: AmazonWebServices Sender Adapter .		New	2021-02-14
Cloud Integration	Integration Suite	Neo Cloud Foundry	API-based Integration Artifacts	You can now create new integration artifacts of type REST, SOAP, and OData APIs. Certain constraints might apply with regard to the usage of this feature based on your license model. See: Develop API-Based Integration Artifacts .		New	2021-02-14

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Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.9.* - Increment: 2012	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 3.33.* - SAP Integration Advisor: 1.53.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2012	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	RFC Receiver Adapter	RFC Receiver Adapter now supports complex parameter - such as <i>table parameter inside structure parameter, structure parameter inside table parameter, and nested table parameters</i> are now supported.	Changed	2021-01-24
Integration Advisor	Integration Suite	Neo Cloud Foundry	Filter MIGs and MAGs on metadata	You can filter MIGs and MAGs based on various filter criteria. See: Message Implementation Guidelines (MIGs) Mapping Guidelines (MAGs)	New	2021-01-24
Integration Advisor	Integration Suite	Neo Cloud Foundry	MIG Payload Import is Optional	You can choose to use payload values as example values in your MIG. See: Creating Message Implementation Guidelines	New	2021-01-24
Integration Advisor	Integration Suite	Neo Cloud Foundry	Using Functions Without a Source Node Mapping	You can create and assign a function to a target leaf node that isn't mapped to any source node. See: Using Functions Without a Source Node Mapping	New	2021-01-24
Integration Advisor	Integration Suite	Neo Cloud Foundry	Deletion of a Mapping Line	You can now delete an individual mapping line from a mapping entity. See: Mapping the Source and Target Nodes	New	2021-01-24
Integration Advisor	Integration Suite	Neo Cloud Foundry	New Versions for Odette, ASC X12 and UN/EDIFACT Message Standard	Integration Advisor and Cloud Integration now support the Odette Message Standard. This comprises the original Odette messages (like ORDERR:2) published in the 1980s and 1990s.	New	2021-01-24

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				See: Integration Advisor – Odette Message Standard 📄 The following new versions are added to the Library of Type Systems: • ASC X12: 007050, 007060, 008010 • UN/EDIFACT: D.18A, D.18B, D.19A, D.19B, D.20A See: Integration Advisor – new versions for B2B Libraries ASC X12 and UN/EDIFACT 📄		
Cloud Integration	Integration Suite	Neo Cloud Foundry	New OEM Adapters in Cloud Integration	SAP has partnered with Rojo Consultancy to release four adapters to enhance your integration and connectivity options. See: • AmazonWebServices Receiver Adapter • Microsoft Dynamics CRM Receiver Adapter • Salesforce Receiver Adapter • SugarCRM Receiver Adapter	New	2021-01-24
Cloud Integration	Integration Suite	Neo	Connection Metering	Tenant Administrators can now download and view additional details of the connections associated with your tenant that are metered and billed using the Provisioning application.	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData Service Project is renamed	OData service project is renamed as OData API Project. See: Developing an OData API Project .	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Predefined values in Custom Tags	You can now add predefined values to a custom tag so that integration developers can choose from the list when they create integration packages. See: Creating Custom Tags .	New	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Generating XML Schema Definition is controlled by integration developers	For OData V2 and V4 adapters, you can now decide whether you want to generate an XSD file for modeling operations. Earlier, XSD file was generated by default every time you edited or updated the operation. See: Configure the OData V2 Receiver Adapter .	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Dynamic Alias in PKCS7 Verifier and Decrypter	You can now set a dynamic alias in the PKCS7 Verifier and Decryptor. See Verify the PKCS#7/CMS Signature and Define PKCS#7/CMS Decryptor.	New	2021-01-24

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Increased Max. Length of Alternative Partner ID	The maximum length of Alternative Partner Ids has been increased to 255 characters.	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo	Start/Stop Message Queue Option in Message Queue Monitor	You can now stop a queue to prevent that it runs full. Likewise, you can manually start a queue. See: Managing Message Queues	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Access Token of OAuth 2.0 Authorization Code Credential accessible in Script Step	You can now fetch access tokens of an OAuth2 Authorization Code Credential in a Script Step in your integration flow. See: Define a Local Script Step .	New	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Flow Step: XML Modifier	Use the XML Modifier flow step to ignore external DTDs during processing. See: Define XML Modifier .	New	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Allow Dynamic Encryption for FTP Receiver Adapter	You can now dynamically configure the encryption parameter for the FTP Receiver Adapter. Configure the FTP Receiver Adapter .	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	SOAP (SOAP 1.x) Sender Adapter: Updated Behaviour of 'Signing Order' checkbox	With the release of version 1.8. of the SOAP (SOAP 1.x) Sender Adapter, the 'Signing Order' checkbox is visible only if the option 'Verify and Decrypt Message and Sign and Encrypt Response' has been selected for 'WS-Security Type'. See: Configure the SOAP (SOAP 1.x) Sender Adapter .	Changed	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Message Monitoring: Value Help Offered for Extended Filter Settings	When monitoring messages, next to properties such like the Time , Status , Artifact (name), and ID , you can use additional filter attributes such like Sender , Receiver , Custom Status , Application Message Type , and Custom Header . For these extended filter attributes, value help is offered now. To filter for a dedicated Custom Header , you need to enter the name of the property and its value. See: Monitor Message Processing	Changed	2021-01-24
Cloud Integration	Integration Suite	Cloud Foundry	Handling of Artifacts with Status Stopping Has Been Improved	Handling of integration artifacts in Stopping state has been improved in the Monitor section (under Manage Integration Content). During undeployment of an artifact, the artifact is not anymore immediately removed from the artifact list. Instead of this, the	Changed	2021-01-24

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				artifact is still shown, but its status changes to Stopping. Furthermore, for artifacts with status Stopping the following applies: - The functions Restart, Undeploy, and Download aren't available. - You can still navigate to the artifact (integration flow model) and to the message monitor. See: Manage Integration Content Runtime Status		
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Role Available to Override Access Policies	The new role <code>AccessPoliciesArtifacts.AccessAll</code> allows you to override access policies and grant access to data such like message queue content or message processing logs. See: Tasks and Permissions	New	2021-01-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	AMQP Sender Adapter: Prefetch value configurable in the AMQP Sender Adapter	You can now configure the max. number of messages that may be prefetched by one worker. See: Configure the AMQP Sender Adapter	New	2021-01-24
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.7.* - Increment: 2010	Changed	2020-12-05
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 3.31.* - SAP Integration Advisor: 1.51.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 2010	Changed	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	The integration flow design guidelines have been enhanced in the following way: - Two guidelines (including integration content) have been added about storing messages on the tenant database. See: Use the Persist Step and Store/Retrieve Messages in/from the Data Store - One guideline (including integration content) has been added in the	Changed	2020-12-05

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				context of adapter configuration. See: Consume a Public HTTP Service with Query Parameters - One guideline (including integration content) has been added to show how to expose an endpoint to a Timer-based integration flow. See: Expose an Endpoint for a Scheduled Integration Flow - One guideline (including integration content) has been added to show how to use the Script step for writing custom header properties. See: Use Custom Header Properties to Search for Message Processing Logs		
Cloud Integration	Integration Suite	Neo Cloud Foundry	Monitoring Message Processing	- You can now display the package information in the Message Processing log. - You can now display the custom header properties in the MPL Monitor. - You can now search for all MPL header attributes. See: Monitor Message Processing	Changed	2020-12-05
Integration Advisor	Integration Suite	Neo Cloud Foundry	Mapping Leaf to Group Node	You can now map a source leaf node to a target group node. The target group node repeats based on the source leaf node occurrence. See: Mapping the Source and Target Nodes	New	2020-12-05
Cloud Integration	Integration Suite	Cloud Foundry	Period Is Now Allowed Character When Creating User Roles	A restriction was lifted regarding allowed characters for user role names. You can now also use a period (.) when defining custom roles. See: Managing User Roles, Cloud Foundry Environment	Changed	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	Invalid XML Character Handling in IDoc Sender Adapter	You can now define how to handle invalid XML characters in the IDoc Sender Adapter. Choose between Throw Error, Remove, and Substitute. See: Configure the IDoc Sender Adapter	Changed	2020-12-05
Cloud Integration	Integration Suite	Neo	JDBC Receiver Adapter	Now JDBC adapter allows connecting your tenant to On-Premise databases and it also supports additional cloud databases.	New	2020-12-05

Technical Component	Capability	Environment	Title	Description	Type	Available as of
		Cloud Foundry		See: JDBC Receiver Adapter		
Cloud Integration	Integration Suite	Neo Cloud Foundry	AS4 Receiver Adapter	AS4 Receiver adapter with Push Message Protocol now supports Type attribute. This attribute helps the receiver participant to identify the payload. See: Configure Receiver Channel with ebMS3 Push	Changed	2020-12-05
Cloud Integration	Integration Suite	Cloud Foundry	RFC Receiver Adapter	RFC adapter is now available for Cloud Integration tenants hosted on Cloud Foundry environment. See: RFC Receiver Adapter	New	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	Overview of an Integration Flow Editor	You can now experience the new version of an integration flow editor that comes with the highly responsive features. See: Overview of Integration Flow Editor .	New	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	Configure Multiple Integration Flows	You can configure multiple integration flows one after the other in the configure view. You can save and deploy all integration flows just by one click. See: Configure Multiple Integration Flows .	Changed	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	Assign Sender and Receiver Systems	You can key in value help for the sender and receiver system at the time of creating an integration flow artifact. See: Creating an Integration Flow .	New	2020-12-05
Cloud Integration	Integration Suite	Cloud Foundry	OData APIs for an integration adapter	You can use OData remote APIs to deploy integration adapters. See: Integration Content .	New	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData Remote API for Updating Sender and Receiver Parameter	You can now update the sender and receiver parameter while updating integration flow using OData API.	Changed	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simulation of an Integration Flow	You can now see the simulation tool embedded in the new version of integration flow editor tool bar.	Changed	2020-12-05
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Metadata for Integration Flow Artifacts	While viewing the metadata of an integration flow, you see a new entry Integration: SAP-to-SAP . This metadata is applicable for standard integration flows from one SAP system to another.	New	2020-12-05
Cloud Integration	Integration Suite	Neo	Deleting an Integration Package	You must provide consent before deleting a package. After deleting the package, you can't	Changed	2020-12-05

Technical Component	Capability	Environment	Title	Description	Type	Available as of
		Cloud Foundry		recover the package and its content. See: Editing an Integration Package .		
Cloud Integration	Integration Suite	Neo Cloud Foundry	Support for .jar file format while uploading a Value Mapping artifact	You can now upload a .jar file when you create a new Value Mapping artifact using the OData API. See: Integration Content .	Changed	2020-12-05
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	- SAP Cloud Integration: 6.6.* Major increment of the version is due to the update of Camel runtime and dependent open source components. - Increment: 2009	Changed	2020-10-24
Cloud Integration	Integration Suite	Neo	Software Version Update	- SAP Cloud Integration: 3.30.* - SAP Integration Advisor: 1.50.* - Adapter Development Kit for SAP Cloud Integration: 2.52.* - Increment: 2009	Changed	2020-10-24
Integration Advisor	Integration Suite	Neo Cloud Foundry	Date Time Format	Integration Advisor now supports additional DateTime formats for Message Implementation and Mapping Guidelines. See: Value Transformations .	Changed	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Connection Metering API	Based on customer feedback and request, we have made further optimizations to our connection metering. For more details, refer 2962718  .	Changed	2020-10-24
Cloud Integration	Integration Suite	Cloud Foundry	JDBC Receiver Adapter	JDBC receiver adapter is now available in Cloud Foundry Environment. See: JDBC Receiver Adapter	New	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V4 Receiver Adapter	Now, you can use Batch Processing (\$batch) operation to request OData V4 service in \$batch mode. See: Configure the OData V4 Receiver Adapter	New	2020-10-24
Cloud Integration	Integration Suite	Neo	Message Status	Additional Message Status DISCARDED added.	Changed	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	The tenant administrator can now create and download custom tag keys in a tenant using the OData API.	New	2020-10-24

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	JDBC Driver	Now, you can upload and deploy type-4 compliant JDBC drivers in CPI runtime. See: Configure JDBC Drivers	New	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	- You can now save a value mapping with version details using the OData API. - You can now fetch value mapping based on the version details using the OData API. See: Integration Content	Changed	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Using Flow Step Recommendation	You get recommendations for the next step when adding a new flow step in the integration flow development. See Using Flow Step Recommendation .	New	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simulation of an integration flow	- Tar and Zip splitters are now supported during the integration flow simulation. - You can now simulate the integration flow in edit mode without saving it. See Simulation of an Integration Flow .	New	2020-10-24
Cloud Integration	Integration Suite	Cloud Foundry	OData APIs for an integration adapter	You can use OData remote APIs to import and delete integration adapters.	New	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	You can now invoke the OData API to check if an update is available for a package. See: Integration Content	New	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Message Mapping	Message Mapping now supports Swagger JSON with OpenAPI Spec version 2.0 and 3.0.	New	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	Attachments for XI Adapter	With the release of the version 1.14. of the XI Adapter, both the XI Sender adapter and the XI Receiver Adapter support attachments. Configure the XI Sender Adapter Configure the XI Receiver Adapter	Changed	2020-10-24
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	The integration flow design guidelines have been enhanced by two new integration packages. Each integration package already contains a first set of integration flows (and corresponding documentation has been provided):	Changed	2020-10-24

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				- A new integration package Integration Flow Design Guidelines - Learn the Basics contains integration flows that show you how to model the first, simple integration scenarios and how to use the elementary integration flow components. With this update, you find integration flows that cover the following learning targets: - Model your first, simple integration flow end-to-end. - Use the Converter step (to convert JSON to XML and CSV to XML, and vice versa). - Access headers and properties. - Encode and decode content (Covers: MIME Multipart Encoder and Decoder steps, Base64 Encoder and Decoder steps) - Set the mapping context. See: Learn the Basics - A new integration package Integration Flow Design Guidelines - Scripting Guidelines provides integration flows that show you how to apply the Script step. With this update, you find integration flows that cover the following learning targets: - Access secure parameters in a Script. - Access headers and properties in a Script. - Access a value mapping in a Script. The related documentation also provides recommendations when using the Script step, in particular, examples of how you don't use the Script step. See: Use Scripting Appropriately		
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: SAP Cloud Integration: 4.16.*	Changed	2020-08-29

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				Increment: 2007		
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: - SAP Cloud Integration: 3.29.* - SAP Integration Advisor: 1.49.* - Adapter Development Kit for SAP Cloud Integration: 2.57.* - Increment: 2007	Changed	2020-08-29
Cloud Integration	Integration Suite	Neo	SAP Cloud Integration Reporting Dashboard	Now you can easily get report on relevant key performance indicators of a CPI tenant using Cloud Integration reporting dashboard. See: The tools section in What Is SAP Cloud Integration?	New	2020-08-29
Integration Advisor	Integration Suite	Neo Cloud Foundry	Message Implementation Guideline	- You can now have nodes of type Date, Time and DateTime in an MIG message structure. - ExampleValues and XSDPatterns present in the MIG won't get published now during the activation. See: Working with a Message Implementation Guideline	New	2020-08-29
Integration Advisor	Integration Suite	Neo Cloud Foundry	Exporting Runtime Artifacts	You can now export runtime artifacts of Message Implementation Guidelines (MIG) and Mapping Guidelines (MAG) in Excel format. See: Exporting Runtime Artifacts	New	2020-08-29
Integration Advisor	Integration Suite	Neo Cloud Foundry	Mapping Guidelines(MAG)	You can now map nodes of type Date, Time and DateTime between the source and target structure in MAG editor. See: Working with Mapping Guidelines (MAGs)	Changed	2020-08-29
Cloud Integration	Integration Suite	Neo Cloud Foundry	Generate Message ID in SAP RM Receiver	You can now configure how the Message ID is generated in the SOAP (SAP RM) Receiver Adapter. See: Configure the SOAP (SAP RM) Receiver Adapter	Changed	2020-08-29
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V2 Receiver Adapter	The Function Import feature now supports more return types. See: Configure the OData V2 Receiver Adapter	Changed	2020-08-29

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V4 Receiver Adapter	OData v4 receiver adapter now supports metadata caching. Supported Receiver Adapter Versions: • OData V4: 1.6 • SuccessFactors OData V4: 1.5	Changed	2020-08-29
Cloud Integration	Integration Suite	Neo Cloud Foundry	OAuth for Mail Sender and Receiver Adapter	You can now configure the Mail Sender adapter and the Mail Receiver adapter with OAuth2 authentication to Microsoft 365 Mail server. • Configure the Mail Sender Adapter • Configure the Mail Receiver Adapter	Changed	2020-08-29
Cloud Integration	Integration Suite	Cloud Foundry	Importing Integration Adapter	You can now import your developed integration adapters in the Cloud Foundry environment. See: Importing Custom Integration Adapter, Cloud Foundry Environment.	New	2020-08-29
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.15.* • Increment: 2006	Changed	2020-08-01
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.28.* • SAP Integration Advisor: 1.48.* • Adapter Development Kit for SAP Cloud Integration: 2.52.* • Increment: 2006	Changed	2020-08-01
Integration Advisor	Integration Suite	Neo Cloud Foundry	Mapping Guidelines	You can now push mapping artifacts from Mapping Guidelines(MAG) editor to your SAP Cloud Platform Integration tenant. See: Inject Mapping Artifacts to SAP Cloud Integration	Changed	2020-08-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	Connection Metering API	We have made some fixes for identified gaps in our connection metering. For more details, refer 2962718 🐞.	New	2020-08-01
Integration Advisor	Integration Suite	Neo Cloud Foundry	Message Implementation Guidelines(MIG)	Integration Advisor now provides limited support for recursive nodes in the MIG Editor. See: Working with a Message Implementation Guideline	Changed	2020-08-01

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Integration Advisor	Integration Suite	Neo Cloud Foundry	Exporting Runtime Artifacts	Message XSDs for an EANCOM MIG will have a new file name complying to the new unified naming convention. See: Exporting Runtime Artifacts	Changed	2020-08-01
Cloud Integration	Integration Suite	Cloud Foundry	Content Transport	You can now enable Content Transport in Cloud Integration Cloud Foundry environment. See: Enabling Content Transport in Cloud Foundry Environment	New	2020-08-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	You can now invoke the Value Mapping configurations with \$filter option using the OData API. See: Value Mapping Configuration Requests	Changed	2020-08-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	The integration flow design guidelines have been enhanced. - A new integration package is available that contains example integration content covering how to work with the Partner Directory. See: Use the Partner Directory Appropriately - A new guideline (including example integration flows) has been added covering the topic of exception handling in dependent integration flows. See: Handle Exceptions in Dependent Integration Flows - A new guideline has been added that covers the topic of streaming. See: Optimize Integration Flow Design for Streaming	New	2020-08-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	XI Sender and Receiver Adapter	You can now configure the number of days after which stored messages are deleted. See: Configure the XI Sender Adapter Configure the XI Receiver Adapter	Changed	2020-08-01
Cloud Integration	Integration Suite	Neo Cloud Foundry	EDI Extractor and EDI Validator	Capabilities of EDI Extractor and EDI Validator are now available in your Cloud Integration tenant. See:	New	

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				• Define EDI Validator • Define EDI Extractor		
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.14.* • Increment: 2005	Changed	2020-07-04
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.27.* • SAP Integration Advisor: 1.47.* • Adapter Development Kit for SAP Cloud Integration: 2.52*: 3.27.* • Increment: 2005	Changed	2020-07-04
Integration Advisor	Integration Suite	Cloud Foundry	Availability	Integration Advisor is now available in Cloud Integration Cloud Foundry environment. See: Initial Setup of %ica-long-name% in Cloud Foundry Environment	New	2020-07-04
Integration Advisor	Integration Suite	Neo Cloud Foundry	Message Implementation Guideline	Values present in the uploaded XML payload will now be considered as example values while creating the MIG. See: Creating a New Message Implementation Guideline	Changed	2020-07-04
Integration Advisor	Integration Suite	Neo Cloud Foundry	Audit Logs	You can now view the audit logs for security-relevant events in the Integration Advisor. See: Audit Logging for %ica-long-name%	New	2020-07-04
Cloud Integration	Integration Suite	Cloud Foundry	Managing Access Policies in CF	The access policies monitor allows you to show and maintain access policies in the Cloud Foundry environment. see: Creating Custom Roles for Access Policies, Cloud Foundry Environment	New	2020-07-04
Cloud Integration	Integration Suite	Neo	FTP Connectivity Tests	You can perform FTP connectivity tests to check the settings required by the FTP adapter. See: FTP Connectivity Tests	New	2020-07-04
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	You can now update the custom tags in a configure-only package using the OData API.	New	2020-07-04
Cloud Integration	Integration Suite	Neo	FTP Sender and Receiver Adapter	The FTP adapter allows you to configure transport protocolThe Send Step now also supports connections to the FTP FTP/FTPS	New	2020-07-04

Technical Component	Capability	Environment	Title	Description	Type	Available as of
		Cloud Foundry		for the connection to the FTP server to send Receiver adapter. messages to the FTP server or to receive messages from theThe FTP adapter allows you to configure transport protocol FTP server. FTP/FTPS for the connection to the FTP server to send For the FTP Sender Adapter, see: Configure the FTP Sender Adapter For the FTP Receiver Adapter, see: Configure the FTP Receiver Adapter		
Cloud Integration	Integration Suite	Neo Cloud Foundry	Send Step	See: Define a Send Step	Changed	2020-07-04
Cloud Integration	Integration Suite	Neo Cloud Foundry	SFTP Sender and Receiver Adapter	In order to improve user guidance, parameters have been rearranged on the configuration user interface of the SFTP sender and receiver adapter. See: Configure the SFTP Sender Adapter Configure the SFTP Receiver Adapter	Changed	2020-07-04
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	A new design guideline has been added that shows you how to apply message signing and encryption. See: Apply Message-Level Security	Changed	2020-07-04
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simulation of an Integration Flow	- The elements aggregator, splitter, and looping process call have been allowed in the simulation. - You can now upload input payload from the file system. See Simulation of an Integration Flow.	New	2020-07-04
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: - SAP Cloud Integration: 4.13.* - Increment: 2004	Changed	2020-06-06

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.26.* • SAP Integration Advisor: 1.46.* • Adapter Development Kit for SAP Cloud Integration: 2.52.* • Increment: 2004	Changed	2020-06-06
Integration Advisor	Integration Suite	Neo Cloud Foundry	Library of Custom Type Systems	You can now delete the uploaded custom messages from the library. Codelists and <i>xsd:enumeration</i> in XSD are now supported while uploading a custom message. See: Library of Custom Type Systems	Changed	2020-06-06
Integration Advisor	Integration Suite	Neo Cloud Foundry	Mapping Guidelines	You can now use the new mapping type String Processing to connect the mapping elements of type String and Token. See: Working with Mapping Guideline (MAG)	New	2020-06-06
Cloud Integration	Integration Suite	Cloud Foundry	Partner Directory Available in Cloud Foundry Environment	The Partner Directory has been made available in the Cloud Foundry environment. Before it was only available in the Neo environment. See: Parameterizing Integration Flows Using the Partner Directory	New	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	You can now create, read and update configurations in value mapping using the OData API. See: Integration Content	Changed	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	You can now use the following parameters along with the GET method for Custom tags. • \$top • \$skip • \$orderby • \$select	Changed	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	Quality Assurance	Familiarize yourself with the Quality Standards for SAP Cloud Platform Integration for holistic product testing, covering for both functional and non-functional qualities. See: Quality Assurance	New	2020-06-06

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Tar Splitter and Gather Step	You can now split and gather archive (.tar) files. See Define Tar Splitter and Define Gather and Join	New	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	Show Subject DN and Issuer DN in the Keystore	You can now see the Subject DN and Issuer DN in the Keystore. See: Managing Keystore Entries	Changed	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	Alias for SSH Keys	You can now assign aliases when creating or adding SSH Keys. See: Uploading an SSH Key	Changed	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	A new integration flow has been added to the integration flow design guidelines. See: Specify Proper Session Handling .	Changed	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	Enhancements for the SFTP Sender and Receiver Adapter	The SFTP sender adapter has been enhanced by an additional authorization option (Dual). The SFTP sender adapter has been enhanced by an additional authorization option (Dual). The SFTP receiver adapter has been enhanced by additional authorization options (Dual and Dynamic). Furthermore, various parameters can now be configured dynamically using headers and properties. For the dynamic configuration of specific parameters, SAP provides predefined properties. See: • Configure the SFTP Sender Adapter • Configure the SFTP Receiver Adapter	Changed	2020-06-06
Cloud Integration	Integration Suite	Cloud Foundry	Role-Based Access to OData API	For API clients, you can now configure secure, role-based access to the OData API. See: OAuth with Client Credentials Grant for API Clients	New	2020-06-06
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simulation of an Integration Flow	The element Gather has been allowed in the simulation. See: Simulation of an Integration Flow .	Changed	2020-06-06
Cloud Integration	Integration Suite	Cloud Foundry	Product Profiles	You can enable or disable the product profiles in the tenant settings. The support packages are disabled by default. See: Runtime Profiles	New	2020-06-06

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	ODATA APIs to update an integration flow	You can now use APIs to update an integration flow name, artifact content and save as specified version.	New	2020-06-06
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: - SAP Cloud Integration: 4.12.* - Increment: 2003	Changed	2020-05-09
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: - SAP Cloud Integration: 3.24.* - SAP Integration Advisor: 1.45.* - Adapter Development Kit for SAP Cloud Integration: 2.52.* - Increment: 2003	Changed	2020-05-09
Integration Advisor	Integration Suite	Neo Cloud Foundry	Library of Custom Type Systems	You can now import XSD with multiple messages in your custom type system. See: Library of Custom Type Systems	Changed	2020-05-09
Integration Advisor	Integration Suite	Neo Cloud Foundry	Message Implementation Guideline	You can now use the new and improvised wizard for creating a Message Implementation Guideline. You can now upload xml file while creating a MIG to design your message structure. See: Creating a New Message Implementation Guideline	Changed	2020-05-09
Integration Advisor	Integration Suite	Neo Cloud Foundry	Message Implementation Guideline	The Local Codelist of MIG has been renamed to MIG Codelist.	Changed	2020-05-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simulation of an Integration Flow	The element Multicast has been allowed in the simulation tool. See: Simulation of an Integration Flow .	Changed	2020-05-09
Cloud Integration	Integration Suite	Cloud Foundry	Creation of Custom Domains	You can now create Custom Domains for the Cloud Integration Platform in the Cloud Foundry environment. See: Setting-Up Custom Domains in CF	New	2020-05-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData APIs for Number Ranges	You can now use APIs to Create, Delete and Update number ranges.	New	2020-05-09
Cloud Integration	Integration Suite	Neo	Product Profiles	You can enable or disable the product profiles in the tenant settings.	New	2020-05-09

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				See: Runtime Profiles		
Cloud Integration	Integration Suite	Neo Cloud Foundry	Configure Multiple Integration Flows	You can mass configure and deploy integration flows in the cloud foundry environment. See: Configure Multiple Integration Flows	Changed	2020-05-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Custom Tags	The tenant administrator can now create and export custom tags in the Settings tab in the tenant. The integration developers can now maintain the values of the custom tags in the packages. See: Creating Custom Tags	New	2020-05-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	You can now read and update custom tags using the OData API.	New	2020-05-09
Cloud Integration	Integration Suite	Neo	Content Transport	The tenant administrator can now check the configuration details of the selected transport mode in the tenant. See: Enabling Content Transport	New	2020-05-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	The following guidelines have been added for integration flow developers: - Guideline Control the Number of Simultaneously Opened Database Connections has been enhanced by an example integration flow and more detailed documentation. See: Control the Number of Simultaneously Opened Database Connections - A new guideline and related integration content has been added for error handling. See: Handle Exceptions When Using the Splitter Pattern - A new guideline and related integration content has been added to the security section. See: Upload WSDLs as Integration Flow Resource	Changed	2020-05-09

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Integration Flow Step Zip Splitter and Zip Aggregation Algorithm	The new Zip Splitter integration flow step decomposes an inbound archive file (.zip). Likewise, a Zip aggregation strategy has been added to the Gather step. See: • Define Zip Splitter • Define Gather and Join	New	2020-05-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Improved Retry Handling for the AMQP Sender adapter	You can now define Max. Number of Retries and Delivery Status After Max. Retries. See: Configure the AMQP Sender Adapter	Changed	2020-05-09
Cloud Integration	Integration Suite	Neo Cloud Foundry	Set message type in AMQP Receiver adapter	The AMQP Receiver adapter offers now the possibility to specify the message type: automatic, binary or text. See:	Changed	2020-05-09
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.11.* • Increment: 2002	Changed	2020-04-11
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.23.* • SAP Integration Advisor: 1.44.* • Adapter Development Kit for SAP Cloud Integration: 2.52.* • Increment: 2002	Changed	2020-04-11
Cloud Integration	Integration Suite	Neo	Managing Access Policies	In SAP Cloud Integration, user permissions are granted based on tasks that can be performed on all artifacts and data. Access Policies provide a way to additionally protect a subset of artifacts and data. See Creating Custom Roles for Access Policies, Neo Environment	Changed	2020-04-11
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simulation of an Integration Flow	You can simulate an integration flow without the need to deploy it on the tenant (activating of tracing supported). See: Simulation of an Integration Flow	New	2020-04-11

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Integration Advisor	Integration Suite	Neo Cloud Foundry	Library of Custom Type Systems	You can now upload custom messages – this enables you to create MIGs and MAGs based on your own message structures. See: Library of Custom Type Systems	New	2020-04-11
Integration Advisor	Integration Suite	Neo Cloud Foundry	Library of Type Systems	Integration advisor has now introduced a new type system called GS1 EANCOM. This is a subset of the UN/EDIFACT standard.	New	2020-04-11
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V4 Receiver Adapter	- The adapter now supports DELETE operation. - You can now build your query using the query modeler. See: Configure the OData V4 Receiver Adapter	Changed	2020-04-11
Cloud Integration	Integration Suite	Neo	OData V2 API	You can now import and download an integration package using the OData V2 API. See: Importing Integration Packages	New	2020-04-11
Cloud Integration	Integration Suite	Neo	OData V2 API	You can now create, read, deploy and download value mapping using the OData V2 API. See: Value Mapping Requests	New	2020-04-11
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Content Added for Integration Flow Design Guidelines	The following guidelines have been added for integration flow developers: - The guideline Use CSRF Protection has been added. Two reference integration flows have been added to the integration package Integration Flow Design Guidelines - Apply Highest Security Standards. See: Use CSRF Protection - The guideline Use Client Certificate Authentication has been added. Two reference integration flows have been added to the integration package Integration Flow Design Guidelines - Apply Highest Security Standards. See: Use Secure Authentication Methods - More details have been added to guideline Anticipate Message Throughput When Choosing a Storage Option.	Changed	2020-04-11

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				See: Anticipate Message Throughput When Choosing a Storage Option		
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Post-Processing Options for Mail Sender Adapter	When you have created a Mail sender adapter and selected as Transport Protocol the option IMAP4 , the following new Post-Processing options are available: Archive and Archive and Mark as Read . See: Configure the Mail Sender Adapter	Changed	2020-04-11
Cloud Integration	Integration Suite	Neo Cloud Foundry	Looping Process Call	You can now define an action when the maximum iterations count for loop processing is reached. See: Define Process Call	Changed	2020-04-11
Cloud Integration	Integration Suite	Neo Cloud Foundry	Timestamps in Data Store Get Operation	From component version 1.5 onwards, the Created At (header: SAP_DataStoreCreatedAt) and Retain Until (header: SAP_DataStoreExpiresAt) timestamps of the data store entry are included in the message. See: Headers and Exchange Properties Provided by the Integration Framework	Changed	2020-04-11
Cloud Integration	Integration Suite	Neo Cloud Foundry	Content Transfer Encoding in Mail Receiver Adapter	You can now choose the content transfer encoding in which you send attachments to the mail server. See: Configure the Mail Receiver Adapter	Changed	2020-04-11
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.10.* • Increment: 2001	Changed	2020-03-14
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.22.* • SAP Integration Advisor: 1.43.* • Adapter Development Kit for SAP Cloud Integration: 2.51.* • Increment: 2001	Changed	2020-03-14
Cloud Integration	Integration Suite	Neo Cloud Foundry	Message Mapping	You can now download MMAP files along with their dependent resources. See: Manage Resources of an Integration Flow	New	2020-03-14
Cloud Integration	Integration Suite	Neo	OData Public API	You can now customize your GET query using \$top and \$skip parameters. See: Integration Package Example Requests	Changed	2020-03-14

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo Cloud Foundry	Include Original Mail in Mail Sender Adapter	You can now include the original email in the SAP_MAIL_ORIGINAL_MESSAGE property for further processing such as verification of the original email. See: Configure the Mail Sender Adapter	Changed	2020-03-14
Cloud Integration	Integration Suite	Neo Cloud Foundry	Lock Timeout in Mail Sender Adapter	You can now specify the amount of time a lock is active during a polling process. These locks also appear in the Manage Locks tile. See: Configure the Mail Sender Adapter	Changed	2020-03-14
Cloud Integration	Integration Suite	Neo Cloud Foundry	Simple Signer Alias	In Simple Signer the alias field can now also be set as an exchange property. See: Sign the Message Content with Simple Signer	Changed	2020-03-14
Cloud Integration	Integration Suite	Neo Cloud Foundry	Timeouts in XI Receiver Adapter	You can now define two new timeouts for the XI Receiver Adapter: • Timeout (in ms) specifies the amount of time that the client waits for a responsive before the http connection is interrupted. • Lock Timeout specifies how long the client should wait before trying to process the message again, for instance in the event of a cluster outage. See: Configure the XI Receiver Adapter	Changed	2020-03-14
Cloud Integration	Integration Suite	Neo Cloud Foundry	AS4 Sender Adapter	You can now use AS4 Sender adapter for receiving data from a trading partner and the following message exchange patterns are supported for an inbound communication: See: • Configure the Sender Channel with ebMS3 Pull • Configure the Sender Channel with ebMS3 Push To know more on how to configure the sender adapter to generate a receipt upon successfully processing the incoming AS4 push message. See: Configure the Sender Channel with ebMS3 Receipt	New	2020-03-14
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.9.*	Changed	2020-02-15

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				Increment: 1913		
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: - SAP Cloud Integration: 3.21.* - SAP Integration Advisor: 1.42.* - Adapter Development Kit for SAP Cloud Integration: 2.50.* - Increment: 1913	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	HTTPS Sender Adapter	Now, you can return an exception to the sender system during an HTTPS call.Adapter Development Kit for SAP Cloud Platform Integration: 2.64.* See: HTTPS Sender Adapter	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Integration Content Entity Types	Integration Content Entity TypesAdapter Development Kit for SAP Cloud Platform supports additional parameters for the entity ServiceEndpoints.	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Write Variables and Content Modifier	The step has been leveraged to use the capabilities of XPath 3.1 Enterprise Edition (EE). See: Define Write Variables Define Content Modifier	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Message Mapping	You can now copy MMAP files to your integration flow from other integration flow from the same or different package. See: Manage Resources of an Integration Flow	New	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Download of artifacts	You can now download an integration artifact without losing the Sender and Receiver information.	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V2 Receiver Adapter	Connecting to OData backend has been improvised with the introduction of connection pool. See: Configure the OData V2 Receiver Adapter	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Polling Information in Manage Integration Content	The Polling Information (in the Web UI Operations view under Manage Integration Content) now provides information on the date and time of the latest polls. By using this feature, you can check on the status of your polls, see whether further polls are scheduled	Changed	2020-02-15

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				or not and get detailed error messages in case of failed polls. See: Manage Integration Content		
Cloud Integration	Integration Suite	Neo Cloud Foundry	Size Limits for Uploading Certificates, Key pairs, and Signing Responses to the Keystore	The size limits for uploading certificates, key pairs, and signing responses to the keystore have been increased. You can now upload certificates up to the size of 10240 bytes, key pairs up to the size of 30720 bytes and signing responses up to the size of 30720 bytes.	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Return HTTP Response Code as Header in IDoc Receiver Adapter	The IDoc receiver adapter contains a feature that, when activated, writes the value of the HTTP response code provided by the connected receiver system into the header . See: Configure the IDoc Receiver Adapter	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Return HTTP Response Code as Header in XI Receiver Adapter	The XI receiver adapter contains a feature that, when activated, writes the value of the HTTP response code provided by the connected receiver system into the header . See: Configure the XI Receiver Adapter	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	Return HTTP Response Code as Header in SOAP (SAP RM) Receiver Adapter	The SOAP (SAP RM) receiver adapter contains a feature that, when activated, writes the value of the HTTP response code provided by the connected receiver system into the header . See: Configure the SOAP (SAP RM) Receiver Adapter	Changed	2020-02-15
Cloud Integration	Integration Suite	Neo Cloud Foundry	SapCmsSignedData in the PKCS#7/CMS Signer	With the release of the version 1.3 of the PKCS#7/CMS Signer, the signed data in the SapCmsSignedData can now be included in the property. See: Sign the Message Content with PKCS#7/CMS Signer	Changed	2020-02-15
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.8.* • Increment: 1912	Changed	2020-01-18

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.20.* • SAP Integration Advisor: 1.41.* • Adapter Development Kit for SAP Cloud Integration: 2.49.* • Increment: 1912	Changed	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	User Interface	• The user announcement is finally here. Be updated with all the latest releases and new features by clicking the  in your web application. • You can now access integration flow design guidelines and the troubleshooting guide through the user menu  in your web application.	New	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Message Mapping	The following features have been introduced for message mapping: • Export as Spreadsheet • Copy Expression • Paste Expression See: Creating Message Mapping	New	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	LDAP Adapter	The LDAP adapter now supports Search and Delete operations. See: LDAP Receiver Adapter	New	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V2 Receiver Adapter	\$batch mode is now supported for GET query operation. See: Configure the OData V2 Receiver Adapter	Changed	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Receiver Party and Receiver Service in XI Sender Channel	The request response of the XI sender adapter is now configurable for the communication party and the communication component. See: Configure the XI Sender Adapter	Changed	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	AMQP Sender and Receiver Adapter	AMQP sender and receiver adapter now support connectivity to on-premise messaging systems using the SAP Cloud Connector See: • Configure the AMQP Sender Adapter	New	2020-01-18

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				Configure the AMQP Receiver Adapter		
Cloud Integration	Integration Suite	Neo Cloud Foundry	Return HTTP Response Code as Header in SOAP 1.x Receiver Adapter	The SOAP 1.x receiver adapter contains a feature that, when activated, writes the value of the HTTP response code provided by the connected receiver system into the header. Use the header <code>CamelHttpResponseCode</code> to get the response from the receiver system. See: Configure the SOAP (SOAP 1.x) Receiver Adapter	Changed	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Write Variables	Write variable definitions supports type expression for creating variable. For example, you can use type <code>\${header.source}</code>. See: Define Write Variables	Changed	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Local Integration Process	Validation checks have been improved for local integration processes. If the integration flow includes some empty elements and sequences, a clear message is shown while displaying problems. See: Define Local Integration Process	Changed	2020-01-18
Cloud Integration	Integration Suite	Neo Cloud Foundry	Externalization	There are major improvements for the externalization feature in the areas of integration flow web editor, configuration view, and download capabilities. The enrichments in these areas show clear separation in the responsibilities. See: Externalize Parameters of an Integration Flow	Changed	2020-01-18
Cloud Integration	Integration Suite	Cloud Foundry	Check Feature Available in Queue Monitor	The tile Managing Message Queues (in the Web UI Operations view under Manage Stores) now provides the Check function when using SAP Cloud Integration in the Cloud Foundry environment. Using this option, you can find unused and missing queues. See: Managing Message Queues	Changed	2020-01-18
Cloud Integration	Integration Suite	Neo	JMS OData API Extensions	You can address additional resources of the used JMS queues.	Changed	2020-01-18

2019 SAP Cloud Integration (Archive)

Cloud Integration 2019

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	Software Version Update (release skipped)	The versions have been updated: - SAP Cloud Integration: 3.19.* - Increment: 1911	Changed	2019-12-21
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update (release skipped)	The versions have been updated: - SAP Cloud Integration: 4.7.* - Increment: 1911	Changed	2019-12-21
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: - SAP Cloud Integration: 4.6.* - Increment: 1910	Changed	2019-12-07
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: - SAP Cloud Integration: 3.18.* - SAP Integration Advisor: 1.39.* - Adapter Development Kit for SAP Cloud Integration: 2.61.* - Increment: 1910	Changed	2019-12-07
Cloud Integration	Integration Suite	Neo Cloud Foundry	User Interface	The latest version of SAP Cloud Integration has introduced an impressive visual experience with the new user interface theme. You can notice the change has been made to appearance of windows, dialogs, and controls.	New	2019-12-07
Cloud Integration	Integration Suite	Neo Cloud Foundry	SuccessFactors SOAP Adapter	SuccessFactors SOAP adapter now internally uses the <code>startRow</code> parameter to fetch the next page in case of session timeout. See: Configure Communication Channel with SuccessFactors(SOAP) Adapter	Changed	2019-12-07
Cloud Integration	Integration Suite	Neo Cloud Foundry	New Adapter for AMQP Messaging Systems	To connect to AMQP messaging systems, the AMQP sender and receiver adapter has been made available. See: Configure the AMQP Sender Adapter Configure the AMQP Receiver Adapter	New	2019-12-07
Cloud Integration	Integration Suite	Neo Cloud Foundry	AS4 Receiver Adapter	Now you can partition AS4 messages between the exchange participants. See: Configure Receiver Channel with ebMS3 Push .	Changed	2019-12-07

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	AS4 Sender Adapter	Partner Directory support is now available for AS4 Sender Adapter. Partner Directory parameters are shown in the MPL log as MPL properties.	Changed	2019-12-07
Cloud Integration	Integration Suite	Neo Cloud Foundry	Queue Status Added to JMS Resource View	The Manage Message Queues editor now provides the queue status in the JMS resource view. See: Managing Message Queues	Changed	2019-12-07
Cloud Integration	Integration Suite	Neo Cloud Foundry	Configuration of individual JMS queues available in Message Queue Monitor	To prevent any blockages in the processing due to overloaded JMS queues, you can now configure the maximum size of individual JMS queues. See: Managing Message Queues	New	2019-12-07
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.5.* • Increment: 1909	Changed	2019-10-26
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.17.* • SAP Integration Advisor: 1.38.* • Adapter Development Kit for SAP Cloud Integration: 2.60.* • Increment: 1909	Changed	2019-10-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	XML to CSV Converter	The namespace information used in the schema will now be considered provided the namespaces are declared at the integration flow level. See: Configure XML to CSV Converter	Changed	2019-10-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V4 Receiver Adapter	You can now connect to OData V4 service using OAuth2 Client Credentials authentication method. See: Configure OData V Receiver Adapter	Changed	2019-10-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	SuccessFactors OData V2 Receiver Adapter	You can now construct the required payload for successfactors OData V2 Upsert operation. See: • Configure SuccessFactors OData V2 Receiver Adapter • Blog: Payload Structure for SuccessFactors Upsert 📄	Changed	2019-10-26

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	Importing Integration Packages	You can now import an integration package (zip import) over an existing package without overwriting its externalized parameters' configured values. See: Importing Integration Packages	Changed	2019-10-26
Cloud Integration	Integration Suite	Neo	OData V2 Remote API	OData V2 API for OAuth2ClientCredentials is now available.	New	2019-10-26
Cloud Integration	Integration Suite	Neo	Accessing On-Premise Application using Cloud Connector	Now you can use APIs to build a ADK project for accessing an op-premise application. See: Accessing On-Premise Application using Cloud Connector .	New	2019-10-26
Cloud Integration	Integration Suite	Neo Cloud Foundry	Validating Message Payload against XML Schema	XML Validator will now show the result of an output in property instead of headers. See: Validating Message Payload against XML Schema .	Changed	2019-10-26
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.4.* • Increment: 1908	Changed	2019-09-28
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.16.* • SAP Integration Advisor: 1.37.* • Adapter Development Kit for SAP Cloud Integration: 2.59.* • Increment: 1908	Changed	2019-09-28
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData API	All the headers available in the integration flow pipeline at the time of message processing will now be returned as response headers when the OData API is invoked. See: Invoking an OData API .	Changed	2019-09-28
Cloud Integration	Integration Suite	Cloud Foundry	Cloud Connector Connectivity Test	Cloud Connector Connectivity Test is now supported. This test checks if the cloud connector is connected to the Cloud Integration tenant. See: Cloud Connector Connectivity Tests	Changed	2019-09-28
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.3.* • Increment: 1907	Changed	2019-08-31

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.15.* • SAP Integration Advisor: 1.36.* • Adapter Development Kit for SAP Cloud Integration: 2.58.* • Increment: 1907	Changed	2019-08-31
Cloud Integration	Integration Suite	Neo Cloud Foundry	OData V4 Receiver Adapter	You can now ensure that your integration flow is protected against malicious attack by enabling the CSRF option while using the OData V4 receiver adapter. See: Configure OData V4 Receiver Adapter	Changed	2019-08-31
Cloud Integration	Integration Suite	Neo Cloud Foundry	Filter	Now Filter component supports Enterprise Edition capabilities of XPath 3.1. See: Define Filter	Changed	2019-08-31
Cloud Integration	Integration Suite	Cloud Foundry	Where-Used Feature Available in Queue Monitor	The tile Managing Message Queues (in the Web UI Operations view under Manage Stores) now provides the Where-Used function when using SAP Cloud Integration in the Cloud Foundry environment. Using this option, you can find out the integration flows in which a queue is used and whether the integration flows write to or consume a queue, or both. See: Managing Message Queues	Changed	2019-08-31
Cloud Integration	Integration Suite	Cloud Foundry	Cloud Connector Support for Receiver Adapters	The following receiver adapter types support now usage of Cloud Connector to connect to an on premise system: SOAP 1.x, SOAP SAP RM, XI, IDoc, SFTP, and Mail. See: Outbound: SAP Cloud Connector	Changed	2019-08-31
Cloud Integration	Integration Suite	Cloud Foundry	Elster Receiver Adapter Available	The Elster receiver adapter is now supported when using SAP Cloud Integration in the Cloud Foundry environment. See: ELSTER Receiver Adapter	Changed	2019-08-31
Cloud Integration	Integration Suite	Neo Cloud Foundry	Integration Flow Design Pattern Document	A new section provides an overview of patterns how to to design enterprise-grade integration flows. See: Integration Flow Design Guidelines	New	2019-08-31
Cloud Integration	Integration Suite	Neo	Lock Timeout for In-Progress Repository Now	The timeout lock for the in-progress repository of the XI sender adapter is now configurable. It	New	2019-08-31

Technical Component	Capability	Environment	Title	Description	Type	Available as of
		Cloud Foundry	Configurable in XI Sender Adapter	is displayed in the Delivery Assurance tab of the XI adapter. See: Configure the XI Sender Adapter		
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.2.* • Increment: 1906	Changed	2019-08-03
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.14.* • SAP Integration Advisor: 1.35.* • Adapter Development Kit for SAP Cloud Integration: 2.57.* • Increment: 1906	Changed	2019-08-03
Cloud Integration	Integration Suite	Cloud Foundry	JMS Resource View Now Available in the Cloud Foundry Environment	In the Manage Message Queue Monitor you can now see the used JMS resources in the JMS instance. See: Managing Message Queues	New	2019-08-03
Cloud Integration	Integration Suite	Neo	Content Transport	You can now see the mode of the transport configured by the tenant administrator while triggering the transport. See: Content Transport Using CTS+ Content Transport Using Transport Management Service Content Transport Using MTAR Download	Changed	2019-08-03
Cloud Integration	Integration Suite	Cloud Foundry	Managing Custom Roles in the Cloud Foundry Environment	Learn how to manage custom roles in the Cloud Foundry environment. A new Web UI is now available in the Monitor section under Manage Security. More information: Managing User Roles, Cloud Foundry Environment	New	2019-08-03
Cloud Integration	Integration Suite	Neo Cloud Foundry	Create/Upload an Integration Flow	You can now edit the ID field while creating or uploading an integration flow. See: Creating an Integration Flow	Changed	2019-08-03
Integration Advisor	Integration Suite	Neo	Library of Type Systems	SAP specific type systems are now available in your type system library, for creating interfaces	Changed	2019-08-03

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				for SAP specific scenarios. See: Library of Type Systems and read the blog on Latest B2B/A2A Libraries  .		
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.13.* • SAP Integration Advisor: 1.34.* • Adapter Development Kit for SAP Cloud Integration: 2.56.* • Increment: 1905	Changed	2019-07-06
Cloud Integration	Integration Suite	Neo	ServiceEndpoints Entity	You can now apply filter based on Protocol while retrieving the service endpoints registered in the tenant. See: ServiceEndpoints Example Requests	Changed	2019-07-06
Cloud Integration	Integration Suite	Neo	OData V2 receiver adapter	You can now enable server-side (__next link) or client-side pagination in the SuccessFactors and generic OData V2 receiver adapter. See: Configure SuccessFactors OData V2 Receiver Adapter Configure OData V2 Receiver Adapter	Changed	2019-07-06
Cloud Integration	Integration Suite	Neo	Increased Size Limit for Keystore Monitor and Certificate to User Mapping	The size limit for the keystore and the User to Certificate Mapping increased from 1MB to 2MB. See: Managing Keystore Entries Managing Certificate-to-User Mappings, Neo Environment	Changed	2019-07-06
Integration Advisor	Integration Suite	Neo	Message Implementation Guidelines (MIGs)	• You can now compare your Message Implementation Guideline (MIG) with other MIGs in your workspace so that you can easily detect changes, deviations between the artifacts. • You can now create local code lists in your Message Implementation Guidelines (MIG), providing you an opportunity to further customize your MIGs to ensure that they suit your specific business needs.	Changed	2019-07-06

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				See: Message Implementation Guidelines (MIGs) and also find this useful blog on Importance of Customized Codelists in MIGs  .		
Cloud Integration	Integration Suite	Cloud Foundry	Software Version Update	The versions have been updated: • SAP Cloud Integration: 4.0.* • Increment: 1904	New	2019-06-10
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 3.*.* • SAP Integration Advisor: 1.33.* • Adapter Development Kit for SAP Cloud Integration: 2.55.* • Increment: 1904	Changed	2019-06-10
Cloud Integration	Integration Suite	Cloud Foundry	SAP Cloud Integration	SAP Cloud Integration is now available in Cloud Foundry environment. → Remember There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note 2752867  . See: Initial Setup of SAP Cloud Integration in the Cloud Foundry Environment	New	2019-06-10
Integration Advisor	Integration Suite	Neo	Message Implementation Guidelines (MIGs)	You can now export a Message Implementation Guideline (MIG) in RTF Format. See: Message Implementation Guidelines (MIGs) .	Changed	2019-06-10
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 2.53.* or 3.11.* • SAP Integration Advisor: 1.32.* • Adapter Development Kit for SAP Cloud Integration: 2.54.* • Increment: 1903	Changed	2019-05-11
Cloud Integration	Integration Suite	Neo	Content Transport	You can now view the error codes along with the error message if there is a configuration or transport failure. See: Content Transport	Changed	2019-05-11
Cloud Integration	Integration Suite	Neo	OData V2 Receiver Adapter	Now during message processing for non-GET operations, the OData V2 receiver adapter	Changed	2019-05-11

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				accepts and processes the HTTP 2xx response code from the server. See: Configure OData V2 Receiver Adapter		
Cloud Integration	Integration Suite	Neo	Mail Receiver Adapter	The mail receiver adapter now supports dynamic configuration of the public key used for encryption. See: Configure the Mail Receiver Adapter	New	2019-05-11
Cloud Integration	Integration Suite	Neo	Add SSH Keys in Keystore Monitor	You can now upload SSH keys or putty keys to the keystore monitor. See: Uploading an SSH Key	New	2019-05-11
Cloud Integration	Integration Suite	Neo	SuccessFactors OData V2 Receiver Adapter	You can now link an entity to a different navigation entity with different key parameters in the Upsert operation. See: Configure SuccessFactors OData V2 Receiver Adapter	Changed	2019-05-11
Cloud Integration	Integration Suite	Neo	Integration Flow Editor	You can now recover the unsaved version of you script or XSLT resource through the Auto-Save feature. See: New and Improvised Integration Flow Editor for SAP Cloud Integration	Changed	2019-05-11
Cloud Integration	Integration Suite	Neo	Integration Flow Editor	You can now view the help information for a specific adapter or flow step directly using the Context Sensitive help. See: New and Improved Integration Flow Editor for SAP Cloud Integration	New	2019-05-11
Integration Advisor	Integration Suite	Neo	Message Implementation Guidelines (MIGs) and Mapping Guidelines (MAG)	- You can now edit the SAP specific content such as Logo, Document Title, Contact and more for these exported artifacts before generating the documentation, offering you better customization capabilities. - You can now get an overview of the Mapping Guidelines (MAG) in which a specific Message Implementation Guideline (MIG) is used. You can also see whether the MIG is used in the source or target of the MAG, providing you a better understanding of the implications in case you change something. See: Message Implementation Guidelines (MIGs)	Changed	2019-05-11

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				Mapping Guidelines (MAGs)		
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: - SAP Cloud Integration: 2.52.* or 3.10.* - SAP Integration Advisor: 1.31.* - Adapter Development Kit for SAP Cloud Integration: 2.53.* - Increment: 1902	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	OData V4 receiver adapter	The OData V4 receiver adapter now supports On-Premise connectivity. See: Configure OData V4 Receiver Adapter	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	Message Mapping	You can now upload an OData V4 metadata file with extensions <i>.edmx</i> and <i>.xml</i> as the source and target messages while creating a message mapping. See: Creating Message Mapping	New	2019-04-13
Cloud Integration	Integration Suite	Neo	Content Transport	The Transport Management Service is now generally available. More information : Content Transport using Transport Management Service	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	Content Modifier	You can now maintain the data type value for the type Expression in the Content Modifier. See: Define Content Modifier	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	Text Area Externalization	Externalization of the text area has been improvised to provide better usability. See: Externalize Parameters of an Integration Flow	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	XSLT Mapping	You can now utilize the XSLT 3.0 specification through XSLT mapping version 1.2. See: Create XSLT Mapping	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	SAP Cloud Integration	Now you can save a complete URL of a tenant that contains specific strings related to the resources. Bookmarking URLs provides you an easy way of direct access to the resource.	New	2019-04-13
Cloud Integration	Integration Suite	Neo	SAP Cloud Integration	The Cloud Integration service tile has been renamed to Process Integration.	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	Creation of SSH keys now possible with EC algorithms	SSH keys now support Elliptic Curve (EC) algorithms to connect to the SFTP server. In parallel, the DSA key creation has been deprecated (only DSA keys with 1024 bit key length are supported).	Changed	2019-04-13

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				See: Creating a Key Pair/SSH Key Pair		
Cloud Integration	Integration Suite	Neo	Mail Receiver Adapter	Mails that are sent out to email recipients can now be signed in the Mail receiver adapter. Before that, it was only possible to encrypt the mails. See: Configure the Mail Receiver Adapter	New	2019-04-13
Cloud Integration	Integration Suite	Neo	SOAP Header Script API	SOAP headers received by a sender channel can now be accessed and further processed using a SOAP Script API. See: Setting and Getting SOAP Headers	New	2019-04-13
Cloud Integration	Integration Suite	Neo	JSON to XML Converter	The JSON to XML Converter now has the option to add XML root elements. Before that, you could only convert JSON documents with one root element.	Changed	2019-04-13
Integration Advisor	Integration Suite	Neo	Mapping Guidelines (MAG)	You can now maintain changes to code list mappings centrally. The application will ensure that these changes are reflected in all the Mapping Guidelines (MAGs) where the code list mappings are used. See: Mapping Guidelines (MAGs) and do read the blog on how to create individual code value mappings in MAGs.	Changed	2019-04-13
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 2.51.* or 3.9.* • SAP Integration Advisor: 1.30.* • Adapter Development Kit for SAP Cloud Integration: 2.52.* • Increment: 1901	Changed	2019-03-16
Cloud Integration	Integration Suite	Neo	Integration Content Entity Types	New OData API called ServiceEndpoints introduced to access the service endpoints exposed by SAP Cloud Integration on a tenant. See: Integration Content	New	2019-03-16
Cloud Integration	Integration Suite	Neo	Open Connectors Receiver Adapter	Use Open Connector adapter to integrate and enable message exchange with over 150 non-SAP cloud applications. See: OpenConnectors Receiver Adapter	New	2019-03-16
Cloud Integration	Integration Suite	Neo	OData V4 Receiver Adapter	You can now allowlist the HTTP request and response headers for OData V4 outbound adapter. See: Configure OData Receiver Adapter V4	New	2019-03-16

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	OData V2 Receiver Adapter	The Modeling Operation wizard can now read the Externalized parameters of the OData V2 connection details. Configure OData Receiver Adapter V2See:	Changed	2019-03-16
Cloud Integration	Integration Suite	Neo	Dynamic Configuration of Username Token in SOAP 1.x Receiver Adapter	You can dynamically configure the Username Token credentials as property by specifying either a header or a property name in one of the following ways: \$ {header.headername} or \$ {property.propertyname}. See: Parameters That Support Dynamic Configuration Configure the SOAP (SOAP 1.x) Receiver Adapter	New	2019-03-16
Cloud Integration	Integration Suite	Neo	Web UI Enhancement of JMS Resources	The numbers of tenant management nodes is shown in the JMS Resources. Before the update, this was not explicitly shown in the Web UI. See: JMS Resource Limits and Optimizing their Usage Managing Message Queues	Changed	2019-03-16
Cloud Integration	Integration Suite	Neo	Downloading Key Pairs from SAP History Tab in Keystore	You can now download a root certificate from the SAP History tab in the Keystore . See: Downloading a Key Pair from the Key History	New	2019-03-16
Cloud Integration	Integration Suite	Neo	Minimum Limit of Max Retry Interval in XI Sender and Receiver Adapter and JMS Sender Adapter	To avoid filling the message processing logs, the minimum limit of the Max Retry Interval is now set to 10 minutes. See: Configure the XI Receiver Adapter Configure the XI Sender Adapter JMS Adapter	Changed	2019-03-16
Cloud Integration	Integration Suite	Neo	Data Store Write Operation	The data store write operation is now able to also store headers. Get operation would be able to read messages including headers already. See: Define Data Store Write Operations	Changed	2019-03-16

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 2.50.* or 3.8.* • SAP Integration Advisor: 1.29.* • Adapter Development Kit for SAP Cloud Integration: 2.51.* • Increment: 1813	Changed	2019-02-16
Cloud Integration	Integration Suite	Neo	SuccessFactors OData V2 Receiver Adapter	You can now filter the fields using the IN operation when editing the query manually. See: Configure SuccessFactors OData V2 Receiver Adapter	Changed	2019-02-16
Cloud Integration	Integration Suite	Neo	AS4 Sender Adapter	You can now configure a sender channel with the AS4 adapter as a receiving MSH to securely process incoming business documents. See: AS4 Sender Adapter	New	2019-02-16
Cloud Integration	Integration Suite	Neo	AS4 Receiver Adapter	Security enhancements have been implemented for SOAP-based messages. You can now save incoming signed receipts and verify the signature. See: Configure Receiver Channel with Push Message Protocol	Changed	2019-02-16
Cloud Integration	Integration Suite	Neo	Creating an Integration Flow	The upload of an integration flow prefills the Name and the ID field. See: Creating an Integration Flow	Changed	2019-02-16
Cloud Integration	Integration Suite	Neo	JMS Resource and Size Limits	The maximum capacity for a single JMS queue has changed from 4 GB to 95% of the total queue capacity. See: Configure Communication Channel with JMS Adapter	Changed	2019-02-16

Technical Component	Capability	Environment	Title	Description	Type	Available as of
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 2.49.* • SAP Integration Advisor: 1.28.* • Adapter Development Kit for SAP Cloud Integration: 2.50.* • Increment: 1812	Changed	2019-01-19
Cloud Integration	Integration Suite	Neo	JMS Adapter	You can now activate JMS resources on Cloud Integration tenants without having the Enterprise Edition. More information about JMS resource and size limits: Configure Communication Channel with JMS Adapter	New	2019-01-19
Cloud Integration	Integration Suite	Neo	OData V2 Receiver Adapter	You can now provide the HTTP request and response header values for the adapter in the Processing section. See: Configure OData Receiver Adapter V2	New	2019-01-19
Cloud Integration	Integration Suite	Neo	JDBC Data Sources	You need not redeploy an integration flow after editing the data source. See: Managing JDBC Data Sources	New	2019-01-19
Cloud Integration	Integration Suite	Neo	SuccessFactors OData V2 Receiver Adapter	You can now set the pagination type for the adapter in the Processing section. See: Configure SuccessFactors OData V2 Receiver Adapter	New	2019-01-19
Cloud Integration	Integration Suite	Neo	Manage Resources of an Integration Flow	You can now upload a valid XML file via the EDMX uploader in the Resources view. See: Manage Resources of an Integration Flow	Changed	2019-01-19
Cloud Integration	Integration Suite	Neo	XSL Documents and XSD Documents Reference in Partner Directory	You are able to reference XSL documents and XSD documents in the Partner Directory via Partner URI (using the <code>xsl:import</code> or "document" feature). See: Dynamically Reading XSLT Mappings from the Partner Directory	Changed	2019-01-19

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				Dynamically Reading XSD Files from the Partner Directory.		
Cloud Integration	Integration Suite	Neo	Downloading New Key Pairs in the Keystore	You have now the option to download a root certificate in the New SAP Keys tab See: Downloading a New Key Pair Provided by SAP	New	2019-01-19
Cloud Integration	Integration Suite	Neo	Message Routing	You can now define how many concurrent processes to use in the General Splitter and Iterating Splitter to process split messages. See: Define General Splitter Define Iterating Splitter	Changed	2019-01-19
Cloud Integration	Integration Suite	Neo	OData API	We have enhanced the Odata API by <code>IdempotentRepositoryEntries</code> in the Message Processing Logs. See: Message Processing Logs	New	2019-01-19
Cloud Integration	Integration Suite	Neo	XI Sender Adapter Enhancements	When configuring the XI sender adapter, you can now select among an updated selection of Quality of Service options. The updated list of options is: Best Effort , At Least Once , and Exactly Once . To learn more, see: Configure XI Sender Adapter	Changed	2019-01-19
Cloud Integration	Integration Suite	Neo	Software Version Update	The versions have been updated: • SAP Cloud Integration: 2.48.* • SAP Integration Advisor: 1.27.* • Adapter Development Kit for SAP Cloud Integration: 2.49.* • Increment: 1811	Changed	2019-01-07
Cloud Integration	Integration Suite	Neo	Provisioning Enterprise Messaging	You can now increase or decrease the enterprise messaging queues for a specific subaccount. See: Activating Enterprise Messaging .	New	2019-01-07
Cloud Integration	Integration Suite	Neo	Message-Mapping Editor	You can now search for a particular node element and also view its occurrences within a structure of the Message-Mapping Editor. See: Creating Message Mapping .	New	2019-01-07
Cloud Integration	Integration Suite	Neo	SuccessFactors OData V2 and SOAP Adapters	HTTP request headers relevant for correlation when the message is processed, are now sent to the SuccessFactors backend . See: SuccessFactors OData V2 Adapter	Changed	2019-01-07

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				Configure Communication Channel with SuccessFactors (SOAP) Adapter.		
Cloud Integration	Integration Suite	Neo	SuccessFactors OData V2 and SOAP Adapters	Message Retry now happens for Upsert and other Query operations when using SuccessFactors OData V2 or SOAP Adapters. See: Configure SuccessFactors OData V2 Receiver Adapter. Configure Communication Channel with SuccessFactors (SOAP) Adapter.	Changed	2019-01-07
Cloud Integration	Integration Suite	Neo	ProcessDirect Adapter	Now it is not mandatory to configure a payload body between a Timer and a ProcessDirect receiver adapter. This functionality is made available from ProcessDirect adapter 1.1 version onwards.	New	2019-01-07
Cloud Integration	Integration Suite	Neo	Twitter Adapter	If your retweet contains 280 characters, then the adapter fetches the entire tweet. See: Twitter Receiver Adapter.	Changed	2019-01-07
Cloud Integration	Integration Suite	Neo	Integration Flow Editor	You can find the Unsaved Changes text appearing under the name of an integration flow, when you do not save the changes made to the integration flow. See: New and Improved Integration Flow Editor for SAP Cloud Integration.	New	2019-01-07
Cloud Integration	Integration Suite	Neo	Integration Flow Editor	You can now copy and paste the adapter configurations in an integration flow. See: New and Improved Integration Flow Editor for SAP Cloud Integration.	New	2019-01-07
Cloud Integration	Integration Suite	Neo	Content Modifier	You can now set the type as Expression or Constant for a payload in a message body in an Integration Flow. See: Define Content Modifier.	Changed	2019-01-07
Cloud Integration	Integration Suite	Neo	OData Receiver Adapter V4	You can now define a new version of the OData receiver adapter in an Integration Flow. See: Configure OData Receiver Adapter V4.	New	2019-01-07
Cloud Integration	Integration Suite	Neo	OData Receiver Adapter V2	You can now enable/disable the CSRF protected option of your OData receiver V2 adapter in an Integration Flow. The adapter also uses Function Import which can now be consumed in \$batch mode. See: Configure OData Receiver Adapter V2.	Changed	2019-01-07
Cloud Integration	Integration Suite	Neo	Invoking an OData API	You can now invoke an active OData API OData API from your calling application.	New	2019-01-07

Technical Component	Capability	Environment	Title	Description	Type	Available as of
				See: Invoking an OData API .		
Cloud Integration	Integration Suite	Neo	Creating Message Mapping	You can now add external reference to WSDL and Schema in Message Mapping. See: Creating Message Mapping .	Changed	2019-01-07
Cloud Integration	Integration Suite	Neo	PKCS#7/CMS Signer, PKCS#7/CMS Encryptor	You can now use headers and exchange properties to dynamically configure the Private Key Alias parameter in the PKCS#7/CMS Signer, and the Receiver Public Key Alias and Private Key Alias parameters in the PKCS#7/CMS Encryptor. See: Sign the Message Content with PKCS#7/CMS Signer , Encrypt and Sign the Message Content with PKCS#7/CMS Encryptor	Changed	2019-01-07
Cloud Integration	Integration Suite	Neo	Data Store Operations	You can now use exchange properties to dynamically define the Data Store Name attribute for the Select, Write, Get, and Delete operations. You can also use exchange properties to define the Entry ID attribute for the Write, Get, and Delete operations. See: Define Data Store Select Operations , Define Data Store Write Operations , Define Data Store Get Operations , Define Data Store Delete Operations	Changed	2019-01-07

2018 SAP Cloud Integration (Archive)

24 November 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration	2.47.*
SAP Integration Advisor	2.5.*
Adapter Development Kit for SAP Cloud Integration	2.48.*
Increment	1810

Enhanced
OData API, Neo Environment
The Odata API was enhanced by IdMapToIds and IdMapFromIds in the Message Processing Logs.
More information: Message Processing Logs

New

JDBC Receiver Adapter, Neo Environment

You can now use adapter tracing in JDBC Receiver Adapter.

Enhanced**Keystore Entries, Neo Environment**

You can now download the public content of the backup keystore to your local disk.

More information: [Downloading Backed-Up Keystore Entries](#)

Enhanced**SFTP Sender Adapter, SFTP Receiver Adapter, Neo Environment**

You can now use the SFTP Sender Adapter to poll messages via the SAP Cloud Connector to the SFTP Server.

You can now use the SFTP Receiver Adapter to send messages via the SAP Cloud Connector to the SFTP Server.

More information: [Configure the SFTP Sender Adapter](#), [Configure the SFTP Receiver Adapter](#)

27 October 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration	2.46.*
SAP Integration Advisor	1.6.*
Adapter Development Kit for SAP Cloud Integration	2.47.*
Increment	1809

New**Activating Enterprise Messaging, Neo Environment**

Use [Enterprise Messaging](#) service to design and deploy integration flows configured with JMS capabilities like JMS and AS2 adapters.

More information: [Enabling and Configuring a Tenant](#).

New**JDBC Receiver Adapter, Neo Environment**

The JDBC Receiver Adapter enables you to connect integration flows with HANA or ASE databases.

More information: [JDBC Receiver Adapter](#).

New**Integration Flow Editor, Neo Environment**

Now you can use Auto-Save functionality to recover the unsaved version of the integration flow.

More information: [Overview of Integration Flow Editor](#).

New

Enabling Authentication for Servlet Based Adapters, Neo Environment

Now you can enable Basic and Certificate based authentication for custom adapters.

More information: [Enabling Authentication for Servlet Based Adapters](#).

New

Query Parameters Supported for IntegrationDesignTimeArtifacts, Neo Environment

Use query parameters for controlling the amount and order of data for IntegrationDesignTimeArtifacts.

More information: [Query Parameters Supported for IntegrationDesignTimeArtifact Requests](#).

New

JDBC Data Sources, Neo Environment

You can now use the JDBC Data Sources to create and manage a cluster of artifact connections to interact with a database.

More information: [Managing JDBC Data Sources](#).

Enhanced

OData Receiver Adapter, Neo Environment

OData adapter now supports patch operation.

More information: [Configure the OData V2 Receiver Adapter](#).

Enhanced

LDAP Adapter, Neo Environment

You can update attribute with multiple values.

More information: [LDAP Receiver Adapter](#).

Enhanced

OData Receiver Adapter, Neo Environment

OData adapter now supports patch operation for both single and multiple (\$Batch) entities.

More information: [Configure the OData V2 Receiver Adapter](#).

Enhanced

OData Receiver Adapter, Neo Environment

All the response headers will be converted to message headers. This may overwrite or interfere with any of headers you have defined in your integration flow, which has to be explicitly handled. For example, in such a scenario, you should take back-up of your defined headers key and value via script step.

More information: [Configure the OData V2 Receiver Adapter](#).

Enhanced

Keystore Entries, Neo Environment

You can now download a root certificate to your keystore.

More information: [Downloading Single Keystore Entries](#).

Enhanced**OAuth2 Credentials Artifact, Neo Environment**

You can now use principal propagation to authenticate users to access applications running on SAP BTP Cloud Foundry version.

More information: [Deploying an OAuth2 Client Credentials Artifact](#).

New**API Content Package, Neo Environment**

You can now use APIs to generate integration flows and add it to a pre-existing integration package.

More information: [Generating Integration Content using APIs](#).

New**Custom Adapters, Eclipse Environment**

If you are deploying multiple adapters, make sure you provide different endpoint scheme.

More information: .

Enhanced**XML Validator, Neo Environment**

You can now use XML Validator 2.0 version to help you to validate XML files against an XML schema.

More information: [Validating Message Payload against XML Schema](#).

Enhanced**Mail Adapter, Neo Environment**

The sender mail adapter can now decrypt encrypted mails and can check the signature of a signed message.

More information: [Configure a Channel with Mail Adapter](#).

New**OData Receiver Adapter, Neo Environment**

You can now use **OAuth2 SAML Bearer Assertion** authentication method to forward sender's credentials to the receiver system.

More information: [Configure the OData V2 Receiver Adapter](#).

Enhanced**Creating an Integration Flow, Neo Environment**

You can now copy an integration flow artifact from the **Artifact** **Actions** view.

More information: [Creating an Integration Flow](#).

Enhanced**OData V2 Adapter, Neo Environment**

Odata V2 adapter supports creation or insertion of data with payload having primary key(s). If the key(s) are auto generated in the service the primary key is optional and need not be provided in the payload.

Enhanced**SuccessFactors OData V2 Receiver Adapter, Neo Environment**

Retry for **Upsert** operation is now enabled for inner error code 412. Retry will be executed once and after 1 minute only.

More information: [Configure the SuccessFactors OData V2 Receiver Adapter](#).

Enhanced**OData Receiver Adapter, Neo Environment**

OData receiver adapter makes a \$metadata call, before the actual endpoint call.

More information: [Configure the OData V2 Receiver Adapter](#).

New**Copying an Integration Package, Neo Environment**

You can call the remote API for copying an integration package from **Discover** section to **Design** section.

More information: .

Enhanced

The list of subprocessors for SAP BTP has been updated and a new version is available on SAP Support Portal:

[SAP Subprocessors](#) 

Please find the direct link to the updated list at: [Subprocessor List](#) 

New subprocessors have been added:

- SAP affiliates:
 - Hipmunk, Inc.
 - SAP ČR, spol. s r.o.
 - SAP National Security Services, Inc.
 - SAP New Zealand Ltd.
 - SAP Österreich GmbH
 - Success Factors, Inc.

- Non-SAP Affiliates:

- Microsoft Corporation

The role description for Tata Consultancy Services Deutschland GmbH has been enhanced by the tasks OS Support as well as Incident/Outages handling (24x7).

A subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws).

Note that services of SAP BTP such as, for example, Neo environment, are covered by this document.

Please note that a new functionality has been made available on the [SAP Subprocessors Support Portal page](#) 📄 :

- For more transparency of the current SAP subprocessors as well as advanced notifications a Subscribe button was added to each of the subprocessor documents:
 - It is possible to subscribe/unsubscribe to the SAP subprocessors list(s) which are of interest.
 - Once subscribed, an automatic email notification will be sent out each time the list has been changed or updated.
- Further details are described in SAP Note [2645947](#) 📄.

7 September 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.47.*
SAP Cloud Integration Node Assembly	2.44.*
SAP Integration Advisor Node Assembly	2.2.*

New

You can now set a maximum size limit for processing inbound messages.

More information: [HTTPS Sender Adapter](#).

New

You can now use **OAuth2 SAML Bearer Assertion** authentication methods to allow the tenant authenticate itself against the receiver using the **Credential Name**.

More information: [HTTP Receiver Adapter](#).

New

You can now:

- Set a maximum size limit for processing inbound messages.
- Store messages for 90 days, after which the messages are deleted.
- Retain the message for two days, by which the messages have to be fetched before an alert is raised.
- Encrypt messages in the data store.
- Use **Trusted Root Certificate** to verify AS2 inbound message.

More information: [Configure the AS2 Sender Adapter](#).

Enhanced

You can not only update but also call the remote API to create an integration flow.

More information:

Enhanced

You must enable **HTTP Session Reuse**, either **On Exchange** level or **On Integration Flow** level for SuccessFactors OData V2 Receiver Adapter and SuccessFactors (SOAP) Adapter.

More information:

[Configure the SuccessFactors OData V2 Receiver Adapter](#)

[SuccessFactors SOAP Adapter](#)

Enhanced

You can browse and select SuccessFactors data center URL by using the **Select** option for SuccessFactors OData V2 Receiver Adapter and SuccessFactors (SOAP) Adapter.

More information:

[Configure the SuccessFactors OData V2 Receiver Adapter](#)

[SuccessFactors SOAP Adapter](#)

Enhanced

OData API Projects (ODP) can now be visualized on SAP Business Accelerator Hub.

Enhanced

Odata V2 adapter supports **Function Import** for functions which return entity or collection of entities. Below return types are not supported:

- Complex types
- Collection of complex types
- Simple types
- Collection of simple types
- Void

Enhanced

You can view and download exchange properties during tracing.

More information: [Message Processing Log](#)

Enhanced

You can see the custom script that you have created with the name you have provided in the **Custom** section of **Mapping Expression**. You can now open and edit the script file.

Enhanced

You can also create and update a resource with content in zip folder. In this case, you cannot use [ReferencedResourceType](#) parameter.

More information:

4 August 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.46.*
SAP Cloud Integration Node Assembly	2.43.*
SAP Integration Advisor Node Assembly	2.1.*

New

You can now use SuccessFactors OData V2 adapter with [OAuth2 SAML Bearer Assertion](#) authentication.

More information: [Configure the SuccessFactors OData V2 Receiver Adapter](#).

New

You can now select the maximum numbers of characters that are fetched from a tweet.

More information: [Twitter Receiver Adapter](#).

Enhanced

You cannot use property propagation across producer and consumer integration flows.

More information: [ProcessDirect Adapter](#).

Enhanced

OData receiver adapter supports sending error response in exception subprocess. The error response body will be part of expression [\\${in.body}](#).

More information: [Configure the OData V2 Receiver Adapter](#).

Enhanced

You can now dynamically configure the following parameters of the XI receiver adapter:

- Address
- Credential Name
- Private Key Alias
- XI-specific identifiers (Communication Party (for sender and receiver), Communication Component (for sender and receiver), Service Interface (for receiver), and Service Interface Namespace (for receiver))

More information: [Configure the XI Receiver Adapter](#).

Enhanced

You can now minimize, maximize and restore model diagram screen while viewing message processing log of an integration flow.

Enhanced

You can now update a keystore entry by uploading a new certificate or key pair.

More information:

[Updating a Key Pair](#) (Web UI)

[Updating a Certificate](#) (Web UI)

(OData API)

(OData API)

New

You can call the remote API to download integration artifact.

More information: .

New

You can call the remote API for reading, creating, updating, deleting and downloading resources of an integration flow. You can also perform multiple operations on resources in a single call using batch request.

More information:

New

You can call the remote API to delete integration artifact.

More information: .

Enhanced

You can pass custom HTTP headers to OData receiver if you have defined the header in content modifier or script element and the element is placed before OData receiver adapter in an integration flow.

More information: [Configure the OData V2 Receiver Adapter](#).

Enhanced

You can now save value mapping entries with version number in value mapping editor, by clicking on **Save as version**.

Enhanced

You can deploy OData API artifact for **SAP Cloud Integration** and **SAP Cloud Integration OEM** runtime profiles only.

More information: [Developing an OData API Project](#).

Enhanced

Even if the integration flow is not in edit mode you can execute simulate and display queue test.

You can refer to input xml file uploaded for simulation, for display queue also, and vice versa.

More information: [Message Mapping](#).

Enhanced

The list of subprocessors for SAP BTP has been updated and a new version is available on SAP Support Portal:

[SAP Subprocessors](#) 

Please find the direct link to the updated list at: [Subprocessor List](#) 

New subprocessors have been added:

- SAP affiliates:
 - CNQR Operations Mexico S. de. R.L. de. C.V.
 - Sybase Software (India) Private Ltd.
 - SAP Norge AS
- Non-SAP Affiliates:
 - Amazon Web Services Inc.
 - Cloud Elements Inc.
 - Computer Sciences Corporation India Pvt. Ltd.

The following SAP Affiliates subprocessors have been removed:

- SAP Australia Pty. Ltd.
- SAP México S.A. de C.V
- Fedem Technology AS

The role description for Tata Consultancy Services Deutschland GmbH has been enhanced by the tasks OS Support as well as Incident/Outages handling (24x7).

A subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws).

Note that services of SAP BTP such as, for example, Neo environment, are covered by this document.

Please note that a new functionality has been made available on the [SAP Subprocessors Support Portal page](#) :

- For more transparency of the current SAP subprocessors as well as advanced notifications a Subscribe button was added to each of the subprocessor documents:
 - It is possible to subscribe/unsubscribe to the SAP subprocessors list(s) which are of interest.
 - Once subscribed, an automatic email notification will be sent out each time the list has been changed or updated.
- Further details are described in SAP Note [2645947](#) .

7 July 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.45.*
SAP Cloud Integration Node Assembly	2.42.*

Enhanced

You can now capture OData receiver adapter's error response body, HTTP response code using a header, and send custom message headers as HTTP headers to OData server.

More information: [Configure the OData V2 Receiver Adapter](#).

New

You can now use **Delete** operation in SuccessFactors OData V2 adapter.

Enhanced

You can now by default, **Exclude Interchange and Group Envelopes** found in an EDI document.

More information: [Define EDI to XML Converter](#).

New

You can now use a new header value to notify the AS4 receiver adapter to perform operations on already compressed payloads.

More information: [Configure Receiver Channel with ebMS3 Push](#).

New

You now use **Number Ranges** to add a unique interchange number to each document during EDI processing and view the artifacts with the corresponding status. This feature is now available for all license types.

More information: [Managing Number Ranges](#).

New

You can now use **Best Effort** to transmit customized MDN acknowledgments to the AS2 sending partner.

More information: [AS2 Adapter](#).

New

You can now use **Number Range** type attribute in the Content Modifier to fetch values from predefined number range.

More information: [Define Content Modifier](#).

New

You can now use the **System ID** to fetch the requests from a specific business system ID belonging to a supplier in the Ariba Network.

More information: [Ariba Adapter](#).

New

You can now view a display queue in which you can see a list of the field values that you are passing to the mapping expression. You can also provide a **Test Input** and visualize how the output of the mapping expression will look like.

More information: [Message Mapping](#).

New

You can now use **Delete** operation in SuccessFactors SOAP adapter.

New

You can call the remote API to create integration flow.

Enhanced

You can now minimize, maximize and restore property sheet screen.

9 June 2018 - SAP Cloud Integration

i Note

Development, maintenance, and support for SAP Cloud Integration in Eclipse has been discontinued from **09 June, 2018**. Going forward development, maintenance, and support would be available only for SAP Cloud Integration in Web and ADK based adapter development using Eclipse. For further information please read the [blog](#) .

Software Version

SAP Cloud Integration Tooling	2.44.*
SAP Cloud Integration Node Assembly	2.41.*
SAP Integration Advisor Node Assembly	1.9.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

This feature/enhancement is available for Web and Eclipse.

You can now connect a tenant with a remote system over the XI protocol using the XI adapter.

More information:

[Configure the XI Sender Adapter](#)

[Configure the XI Receiver Adapter](#)

New

You can call the remote API for reading an integration artifact which will return properties of an integration flow. You can also call the remote API for reading and updating parameters that are externalised in an integration flow.

More information:

New

This feature is available for Web.

Now tenant administrators can disable a tenant in Cloud Integration service.

More information: .

New

This feature is available for Web.

You can now configure a custom adapter with endpoint visualization.

More information: [Enabling Endpoint Visualization for Custom Adapter](#).

New

This feature is available for Web.

You can now build custom adapter with tracing feature.

More information: [Enabling Tracing for Custom Adapter](#).

Enhanced

This enhancement is available for Web.

You can now add dependent resources while adding a particular resource from another integration flow.

More information: [Manage References](#).

Enhanced

This enhancement is available for Web.

You can now browse for predefined xpath for content modifier and write variable.

More information:

[Define Content Modifier](#)

[Define Write Variables](#)

New

This feature/enhancement is available for Web.

You can now copy and work with integration content published by SAP partners on SAP Business Accelerator Hub. This content will also get updates if the SAP partner publishes an update, similar to the integration content published by SAP.

More information on working with prepackaged integration content: [Working with Prepackaged Integration Content](#).

Enhanced

This feature/enhancement is available for Web.

In the OData sender adapter, you can now see a list of available user roles on your tenant by choosing **Select** when you select the **Authorization** as **User Role**. For example, if you want to use **ESBMessaging.send** user role for authorization, you can select that from a list instead of manually typing it. This method is less error prone.

More information: [Configure the OData Sender Adapter](#).

12 May 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.43.*
SAP Cloud Integration Node Assembly	2.40.*
SAP Integration Advisor Node Assembly	1.8.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhanced

This enhancement is available for Web.

You can now select among the roles that are defined for the runtime node to define permissions to process inbound messages. The enhancement is available for the following sender adapters: SOAP (SOAP 1.x), SOAP (SAP RM), IDoc.

In case you have as **Authorization** chosen the option **User Role**, the value help for **User Role** contains the default role **ESBMessaging.send** and all custom roles defined on runtime node level.

More information:

[SOAP \(SAP RM\) Adapter](#)

Enhanced

This enhancement is available for Web.

You can now benefit from a system job that automatically deletes queues that are not used any more in deployed integration flows (in case they don't contain any message).

More information: [JMS Adapter](#)

Enhanced

This enhancement is available for Web.

You can now upload files with the following extension when adding a keystore (under ► **Monitor** ► **Manage Security** ► **Keystore** ►):

- .pfx
- .p12

More information:

[Uploading a Keystore](#)

[Uploading a Key Pair](#)

New

This feature is available for Web.

You can now import and use CSV files in value mapping artifacts.

More information: [Formatting Guidelines for CSV Files used in Value Mapping](#).

New

This new feature is available for Web.

You can now use predefined header values for defining a value for the individual header type in Content Modifier and Write Variables.

More information:

- [Define Content Modifier](#)
- [Define Write Variables](#)

New

This new feature is available for Web.

You can now externalize values for individual cell in a table, checkboxes, and dropdowns.

More information: [Externalize Parameters of an Integration Flow](#).

Enhanced

This enhancement is available for Web.

You can now upload multiple resources and archived dependent resources to an integration flow.

More information: [Manage References](#).

New

This new feature is available for Web.

You can now use AS4 Light Client Conformance Profile for exchanging business messages securely using web services. The AS4 profile is compliant with the ebMS 3.0 standards and supports one-way/push and one-way/pull message exchange patterns.

More information: [AS4 Receiver Adapter](#).

New

This feature is available for Web.

You can now visualize endpoint definitions for integration flows using IDoc, SOAP or HTTP adapters.

More information: [Endpoint Visualization](#).

New

This feature is available for Web.

You can now use multiple messages in the source and target of message mapping definition resource.

More information: [Creating Message Mapping as a Flow Step](#)

New

This feature is available for Web.

You can now import WSDL files from ES repository.

More information: [Importing Mapping Content from ES Repository](#).

New

This feature/enhancement is available for Web.

You can now use Principal Propagation as an Authentication setting in the XI Receiver Adapter.

More information:[Configure the XI Receiver Adapter](#) .

Enhanced

This enhancement is available for Web.

You can now use headers and exchange properties for providing adapter configuration values for OData, SuccessFactors OData V2, and SuccessFactors OData V4 adapters. For example, you can specify the value of **Address** field as `${header.address}` or `${property.addr}` and provide these values during runtime.

More information:

- [SuccessFactors OData V2 Adapter](#)
- [SuccessFactors OData V4 Receiver Adapter](#)
- [OData Adapter](#)

New

This enhancement is available for Web.

You can now use adapter tracing in HTTP receiver and HTTPS sender adapters.

More information:

- [Message Processing Log - Adapter Tracing](#)
- [HTTP Receiver Adapter](#)
- [HTTPS Sender Adapter](#)

New

This feature is available for Web.

You can now see the endpoint information for integration flows containing HTTPS sender, AS2 in the **Manage Integration Content** section of operations view.

More information:

- [HTTPS Sender Adapter](#)
- [AS2 Adapter](#)

Enhanced

The list of subprocessors for SAP BTP has been updated and a new version is available on SAP Support Portal:

[SAP Subprocessors](#) 

Please find the direct link to the updated list at: [Subprocessor List](#) 

The following new subprocessors have been added:

- SAP affiliates:
 - Fedem Technology AS
 - SAP Labs Finland Oy
 - SAP Polska Sp. z o.o.

Furthermore, the subprocessor *BIT Group GmbH & Co. KG* was renamed to *Itelligence Global Managed Services GmbH*.

A subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws).

Note that services of SAP BTP such as, for example, Neo environment, are covered by this document.

14 April 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.42.*
SAP Cloud Integration Node Assembly	2.39.*
SAP Integration Advisor Node Assembly	1.7.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

This feature is available for WebUI.

You can now display the adapter tracing for adapters that transform the message either before sending or upon reception. The log level has to be set to *Trace*.

The option **Enable Debug Trace** has been removed from the **Runtime Configuration** tab in the **Design** section (for product profile SAP Cloud Integration selected). You can instead of this do the related configuration settings in the **Monitoring** application now.

As of now, the option **Enable Debug Trace** is only available in the **Runtime Configuration** tab, when you have selected a product profile for SAP Process Orchestration.

More information: [Message Processing Log - Adapter Tracing](#)

New

This feature is available for WebUI.

You can now configure the RFC adapter to create a new RFC connection in the backend every time a new call is made to the target system. This option is mandatory when you are using principal propagation for RFC adapter.

More information: [RFC Receiver Adapter](#)

New

This feature is available for WebUI.

You can now import **Operation Mapping** content from **ES Repository**. For example, if you have mapping content from your process orchestration system that you want to reuse in your SAP Cloud Integration integration flow, you can directly import this from the **ES**

Repository of your process orchestration system.

More information: [Importing Mapping Content from ES Repository](#)

Enhancement

This feature is available for WebUI.

You can now import single certificates and key pairs into the tenant keystore.

More information:

[Uploading a Certificate](#)

[Uploading a Key Pair](#)

Enhancement

Keystore lifecycle management has been enhanced in the following way:

A system job makes sure that in case a key provided by SAP is due to expire, the new key is automatically is activated one day before the expiration.

Enhancement

This enhancement is available for WebUI.

You can now use the new query modeler for building a query when you are working with SuccessFactors OData V2 adapter and OData adapter. The new query modeler offers improved look and feel, in addition to making the process of building a query easy through a step-by-step approach

More information:

- [SuccessFactors OData V2 Adapter](#)
- [OData Adapter](#)

New

This is a documentation update.

You can now use the Eclipse Developer's Guide as a PDF only.

More information: [Developer's Guide for Eclipse \(PEF\)](#)

17 March 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.41.*
SAP Cloud Integration Node Assembly	2.38.*
SAP Integration Advisor Node Assembly	1.6.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhanced

This feature is available in Web application and in Eclipse.

You can now specify a specific cloud connector instance during message processing using Camel exchange property.

More information:

- [SOAP \(SAP RM\) Adapter](#)
-
- [HTTP Receiver Adapter](#)

Enhanced

This feature is available in Web application and in Eclipse.

You can now use headers and exchange properties to dynamically configure the Private Key Alias property for WS-Security in the SOAP 1.x Sender and Receiver Adapter.

More information: [SOAP \(SOAP 1.x\) Adapter](#).

15 February 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.40.*
SAP Cloud Integration Node Assembly	2.39.*
SAP Integration Advisor Node Assembly	1.5.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

The **ProcessDirect** (Sender and Receiver) adapter is introduced in both Eclipse and Web UI.

For more information see, Web UI: [ProcessDirect Adapter](#)

New

This feature is available in Web application.

SAP BTP now offers Consumption-based model for consuming services in Cloud Integration.

For more information, see .

New

You can now transport integration content from SAP Cloud Integration to **CTS+** system directly without any manual intervention with a single click.

This feature is available only in the SAP Cloud Integration web application.

For more information, see [Content Transport](#).

Enhanced

This feature is available in Web application.

You can change the alias of a keystore entry.

More information: [Changing the Alias of a Keystore Entry](#)

Enhanced

You can now see some changes in the user interface of SuccessFactors SOAP adapter versions 1.0, 1.1 and 1.2 in the **Processing** tab.

New

You can now import integration content (mainly mapping artifacts) from ES Repository to SAP Cloud Integration. This helps you in easily reusing existing integration content from your on premise ES Repository in your SAP Cloud Integration web application.

For more information, see [Importing Content from SAP Process Orchestration System](#).

New

The **Externalize** feature now allows you to externalize a text area of an integration component.

For more information see, [Externalize Parameters of an Integration Flow](#).

Enhanced

In **Problems** view, you can click the Location ID link to identify the issue related to integration components or resources.

For more information see, [Problems View](#).

Enhanced

The list of subprocessors for SAP BTP has been updated and a new version is available on SAP Support Portal:

[SAP Subprocessors](#) 

Please find the direct link to the updated list at: [Subprocessor List](#) 

The following new subprocessors have been added:

- SAP affiliates:
 - SAP Labs France SAS
- Non-SAP affiliates:
 - SELLBYTEL Services Malaysia Sdn Bhd.
 - Westhouse Consulting GmbH
 - Yoh of Canada Corporation

A subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws).

Note that services of SAP BTP such as, for example, Neo environment, are covered by this document.

18 January 2018 - SAP Cloud Integration

Software Version

SAP Cloud Integration Tooling	2.39.*
SAP Cloud Integration Node Assembly (Cluster 2.x)	2.36.*
SAP Integration Advisor Node Assembly	1.4.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New/Enhanced

This feature is available in Web application and in Eclipse.

In XML Signer you can now use headers in private key alias field to dynamically sign the message based on specific conditions.

More information: [Sign the Message Content with XML Digital Signature](#)

New

This new view available in Web application lists all the errors and warnings related to an integration flow.

For more information see, [Problems View](#).

Enhanced

RFC receiver adapter:

- supports property along with headers as regular expression to create dynamic destinations.
- allows you to deploy the integration flows without provisioning the destinations first.

→ Remember

It is recommended you to ensure the destination configured in the RFC adapter does exists and is up and running.

- supports table parameter as a part of import parameters for RFC functions.
- shows MPL logs indicating if the RFC function is invoked with BAPI transaction commit or not.

Enhanced

This feature is available in Web application.

Keystore management (for the tenant administrator) has been enhanced by the following feature.

A system job makes sure that in this case the key is automatically activated (within a day after it has expired).

Enhanced

The OData API was enhanced for the `UserCredentialParameter` of the Partner Directory by the query option `returnHashedPassword=SHA256`.

You can now request that the user credential parameter returns a hashed value for the password instead of a NULL value

More information: [Requests for UserCredentialParameter](#)

2017 SAP Cloud Integration (Archive)

09 December 2017 - SAP Cloud Integration

Software Version

Tooling	2.38.*
Node Assembly (Cluster 2.x)	2.35.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhanced

RFC receiver adapter now supports auto commit feature for BAPI functions that require `BAPI_TRANSACTION_COMMIT` to be invoked implicitly by the RFC.

For more information, see:

- [RFC Receiver Adapter](#)
-

New

The integration flow editor available in Web application has a new improved interface that helps the integration developers to work efficiently.

For more information, see [Configure Integration Flow Components](#).

Enhanced

You can now upload a single or multiple resources either from the file system or integration flow that is within the tenant.

For more information, see [Manage References](#)

Enhanced

This feature is available in Web application.

The keystore management feature facilitates the renewal of SAP keys.

You can now activate a new key pair provided by SAP in order to replace an old key pair which is supposed to expire soon. To access new SAP keys (provided by SAP), a new **New SAP Keys** section has been added to the keystore management feature.

A new **SAP Key History** section shows expired SAP keys which have been replaced by new ones. You can also restore a key pair from the **SAP Key History**.

For more information, see:

[Managing the Lifecycle of Keys](#)

[Activating a New Key Pair Provided by SAP](#)

[Restoring a Key Pair from the Key History](#)

Enhanced

This feature is available in Web application.

You can view all parameters of the component using **For expired keys, the end of validity period is highlighted in All Parameters** option, when you configure externalized parameters of an integration flow.

More information:

[Configure Externalized Parameters of an Integration Flow](#).

New

This feature is available in Web application.

New features are introduced through new versions of the components. To consume this new feature you must migrate to new version.

More information:

[Update Version of an Integration Flow Component](#).

Enhanced

This feature is available in Web application and in Eclipse.

In the JMS Receiver Adapter you can select this option to also transfer the exchange properties to the JMS queue.

For more information, see:

[JMS Adapter](#)

Enhanced

This feature is available in Web application.

With the new status **Blocked** in the Message Queue Monitor you can now see which messages were involved in multiple node crashes and were therefore not processed.

For more information, see:

- [Managing Message Queues](#)

Enhanced

The list of subprocessors for SAP BTP has been updated and a new version is available on SAP Support Portal:

[SAP Subprocessors](#) 

Please find the direct link to the updated list at: [Subprocessor List](#) 

The following changes have been made compared to the previous version:

- All SAP Affiliates subprocessors have been added to the list now.
- Scale Focus AD has been added to the list of non SAP Affiliates subprocessors.

A subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws).

Note that services of SAP BTP such as, for example, Neo, are covered by this document.

11 November 2017 - SAP Cloud Integration

Software Version

Tooling	2.37.*
Node Assembly (Cluster 1.x)	1.56.*
Node Assembly (Cluster 2.x)	2.34.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhanced

This feature is available in Web application.

You can display the properties of a selected keystore entry.

For more information, see:

[Displaying Properties of a Keystore Entry](#)

Enhanced

RFC receiver adapter now supports creation of dynamic destination by using headers.

For more information, see:

- [RFC Receiver Adapter](#)
-

14 October 2017 - SAP Cloud Integration

Software Version

Tooling	2.36.0
Node Assembly (Cluster 1.x)	1.55.0
Node Assembly (Cluster 2.x)	2.33.0

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhanced

This feature is available in Web application and in Eclipse.

The Scheduler of SFTP and Mail sender adapter has been changed so that the option **Run Once** has been removed. Furthermore, default values for the interval under **Schedule on Day** and **Schedule to Recur** have been changed so that the scheduler runs by default every 10 seconds between 00:00 and 24:00 o'clock.

For more information, see:

[Mail Adapter](#)

[Configure the SFTP Sender Adapter](#)

Enhanced

This feature is available in Web application and in Eclipse.

In the WS Security settings, you can now specify a signature algorithm to be applied when signing the response message.

For more information, see:

New

This feature is available in Web application.

You can now import or add different file types to an integration content.

For more information, see [Manage References](#)

New

You now access the User Credential and Key Store in SAP ADK project to authenticate and validate a user.

For more information, see:

- [Accessing Trust and Key Managers](#)
- [Accessing User Credentials](#)

New

This feature is available in Web application.

In the Message Queue Monitor you can now display the integration flows in which a queue is used and show unused and missing queues.

For more information, see [Managing Message Queues](#)

New

EDIFACT syntax version 4 is now supported.

For more information, see [Define EDI Splitter](#).

Enhanced

RFC receiver adapter version 1.2.0 and higher supports:

- function modules that contain "/" in their names
- table in import parameter

For more information, see:

[RFC Receiver Adapter](#)

16 September 2017 - SAP Cloud Integration

Software Version

Tooling	2.35.0
Node Assembly (Cluster 1.x)	1.54.0
Node Assembly (Cluster 2.x)	2.32.0

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhancement

This feature is available in Web application.

You can now view externalised values by selecting relevant details of externalised components in **More** tab.

More information: [Configure Externalized Parameters of an Integration Flow](#)

New

You can now transport integration content using **Change and Transport System (CTS+)** tool.

For more information, see [Content Transport](#)

New

Now you can include additional properties in the URI, for retrieving specific information on a custom adapter (SAP ADK) during runtime.

For more information, see .

19 August 2017 - SAP Cloud Integration

Software Version

Tooling	2.34.0
Node Assembly (Cluster 1.x)	1.53.0

Node Assembly (Cluster 2.x)	2.31.0
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→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhanced The following feature is available in Web UI in Cloud Integration. You can now back up and restore keystore entries which are owned by the tenant administrator. More information: Backing Up Keystore Entries Restoring Backed-Up Keystore Entries
Enhanced You can now use content assist feature for groovy script which means that you can view list of existing methods of message class, once you start typing initial letters of the required method. You can add content assist jar file in integration project to use this feature. For more information, see .
Enhanced The Credential Name attribute can now be configured dynamically in the IDoc, SOAP SOAP 1.x and SOAP SAP RM receiver adapters. This feature is available in the Web UI and in Eclipse. For more information, see (for the Web UI): SOAP (SAP RM) Adapter
New The Partner Directory has been released for SAP Cloud Integration product profile. The Partner Directory contains information on partners that are connected to a cross-partner tenant in the context of a larger network You can use the Partner Directory when setting up a network of many communication partners. Partner-specific information can be parameterized in a few integration flows (which dynamically read the partner-specific information from the Partner Directory). That way, you can easily add new partners to the network without changing or redeploying integration flows.
New The following feature is available in Eclipse or Web UI in Cloud Integration. Cloud Connector Proxy in Mail Sender Adapter is supported. For more information, see • • Mail Adapter
New

As of this tact, SOAP (SAP RM adapter) can also be used in combination with the Send step type.

This feature is available in the Web UI and in Eclipse.

For more information, see

[Define a Send Step](#)

New

The **Dead-letter** queue option has been introduced in AS2 adapter. This option enables you to place those messages that cannot be processed after two retries.

- For more information, see .
- For more information, see .

New

SAP ADK framework has introduced a new adapter project creation wizard with maven plugin support. It allows you to build and deploy custom adapters.

For more information, see .

Enhanced

The authentication option has been enhanced in POP3 Connectivity Test, IMAP Coonectivity Test and SMTP Connectivity Test.

For more information, see

[Performing Connectivity Tests](#)

Enhanced

The usability of the Manage Locks, the Manage Queues and the Manage Keystore areas of the MonitorWeb UI has been improved.

The content of the [Entry](#) property provides a link to the message in the [Managing Message Queues](#) monitor.

A filter and a quick search field has been introduced.

For more information, see

[Message Locks](#)

[Managing Message Queues](#)

[Managing Keystore Entries](#)

New

You can now disable a field in the target structure in a mapping definition resource. This helps you in testing or simulating the mapping without mapping the mandatory fields.

For more information, see .

Enhanced

Operational Aspects: List of subprocessors (non SAP affiliates) for SAP BTP updated

The list of subprocessors (non SAP affiliates) for SAP BTP has been updated. A subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws).

Note that services of SAP BTP such as, for example, SAP Cloud Integration, are covered by this document.

The following subprocessors have been added:

Accenture GmbH

Dynatrace GmbH

Kaavian Systems

Find the updated list at: [Subprocessor \(non SAP Affiliates\) List](#)👉

21 July 2017 - SAP Cloud Integration

Software Version

Tooling	2.33.0
Node Assembly (Cluster 1.x)	1.52.0
Node Assembly (Cluster 2.x)	2.30.0

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

RFC Receiver Adapter is generally available now and can be used from Eclipse or Web UI in Cloud Integration.

RFC is the standard interface used for integrating On-premise ABAP systems to the systems hosted on cloud using SAP Cloud Connector. The adapter supports NetWeaver 7.31 version or higher.

For more information, see

-
- [RFC Receiver Adapter](#)

Enhanced

Logging can be implemented by using Simple Logging Facade for Java (SLF4J).

For more information, see .

Enhanced

Developing an OData API is generally available now. This service was previously in beta version.

New

The following feature is available in Eclipse or Web UI in Cloud Integration.

Cloud Connector Proxy in Mail Receiver Adapter is supported. You can now use the Mail Receiver Adapter to send emails via the SAP Cloud Connector to the receiver.

For more information, see

-
- [Mail Adapter](#)

Enhancement

You can now configure the script in the custom function to return multiple string values.

For more information, see .

New

The following feature is available in Eclipse or Web UI in Cloud Integration.

You can now place messages in the dead-letter queue if it cannot be processed after two retries.

For more information, see

-
- [JMS Adapter](#)

New

The following feature is available in Eclipse or Web UI in Cloud Integration.

You can now configure HTTP Session Handling in the Runtime Configuration.

For more information, see

-
- [Specify the Runtime Configuration](#)

New

The SAP Cloud Integration, enterprise edition supports additional capabilities of an integration flow.

More information: [Licensed Capabilities and Features](#)

Enhanced

Keystore management in the Web UI has been enhanced in the following way:

The tenant administrator can now also download SSH keystore entries for SFTP connections (with alias `id_rsa` or `id_dsa`) in OpenSSH format.

More information: [Downloading Single Keystore Entries](#)

New

Authentication via principal propagation is now available in the OData adapter.

For more information, see .

Enhanced

Message Monitoring/Managing Locks allows you to deal with messages that cannot be processed and are placed in the dead-letter queue.

More information: [Message Locks](#)

24 June 2017 - SAP Cloud Integration

Software Version

Tooling	2.32.0
Node Assembly (Cluster 1.x)	1.51.0
Node Assembly (Cluster 2.x)	2.29.0

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

A new Keystore Monitor in the Web UI allows the tenant administrator to display entries of the tenant keystore and to manage those entries which are owned by the tenant administrator.

The Keystore Monitor provides you with an overview of the entries of the keystore (deployed on the tenant).

Furthermore, the Keystore Monitor provides you with the following options:

- Uploading a keystore
- Downloading the public content of a keystore or a single keystore entry
- Deleting a keystore entry

For more information, see [Managing Keystore Entries](#)

New

Externalize feature is now available in the Web UI of Cloud Integration. It allows you to declare a parameter as a variable and reuse the same variable in more than one integration component.

For more information, see [Externalize Parameters of an Integration Flow](#).

New

The Resources viewer in the Web UI of Cloud Integration helps you to manage different resources associated within an integration content.

For more information, see [Manage References](#).

New

Now you can introduce custom classes using Blueprint metadata during runtime for custom adapters.

For more information, see:

	. Blueprint Metadata
Enhanced	The OData API was enhanced to support keystore management activities by the tenant administrator. For more information, see
Enhanced	The OData API allows you to address certificate-to-user mappings. For more information, see
New	Provisioning message broker allows you (tenant admin) to use JMS adapter scenarios only if you have Enterprise Edition license. For more information, see Activating Enterprise Messaging
New	JMS Adapter and Message Queue Monitor is available for Cloud Integration Customer only if you have Enterprise Edition license. For more information, see
Enhanced	The Web UI now supports also an additional transaction handling configuration option Required for JMS. For more information, see Define Transaction Handling
New	Parameter CRAM-MD5 in Sender/Receiver Mail Adapter was renamed to Encrypted User/Password For more information, see
Enhanced	Operational Aspects: List of subprocessors (non SAP affiliates) updated The list of subprocessors (non SAP affiliates) has been updated. A subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws). Note that services such as, for example, SAP Cloud Integration, are covered by this document. Find the updated list at: List of Subprocessors (non SAP Affiliates) 

27 May 2017 - SAP Cloud Integration

Software Version

Tooling	2.31.0
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Node Assembly (Cluster 1.x)	1.50.0
Node Assembly (Cluster 2.x)	2.28.0

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

You can now create or upload value mapping artifacts to your integration package.

For more information, see [Creating Value Mapping](#).

New

The following feature is available in Eclipse and in the Web UI of Cloud Integration.

For methods GET, DELETE, and HEAD you can now send the body of a message with the request.

More information:

- [HTTP Receiver Adapter](#)

New

The following are the new additions included while developing an new adapter:

- In component metadata, a new child element for FixedValue(s) property has been introduced.
- In develop adapters chapter mentions the format used to construct an application URL for calling a servlet.

More information:

- [Component Metadata](#)
-

New

The following feature is available in Eclipse and in the Web UI of SAP Cloud Integration.

You can now process and route failed EDI messages using EDI splitter.

More information:

- [Define EDI Splitter](#) (Web UI)
- (Eclipse)

Enhanced

The SMTP Outbound Connection Test has been enhanced in the Eclipse tooling of Cloud Integration.

You can now download certificates, check for mail addresses and validate the server certificate.

More information:

Enhanced

The following feature is available in the Eclipse tooling of Cloud Integration.

The **Data Store** viewer now also downloads headers.

More information:

Enhanced

The **Managing Locks** editor in the Web UI of Cloud Integration has been improved and made more user-friendly. You can now filter or search for entries, for example.

More information: [Message Locks](#)

29 April 2017 - SAP Cloud Integration

Software Version

Tooling	2.30.0
Node Assembly (Cluster 1.x)	1.49.0
Node Assembly (Cluster 2.x)	2.27.0

→ **Tip**

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

The feature **Managing Locks** is available in the Web UI of Cloud Integration.

It allows you to display and manage lock entries that are created when more than one runtime nodes try to process a file at the same time

More information: [Message Locks](#)

New

Cloud Integration tools are supported on Eclipse Neon only. You will get update for Cloud Integration software through this Eclipse update site:.

New

The following feature is available in Eclipse and in the Web UI of Cloud Integration.

In the SAP SOAP 1.x Sender Adapter, you can now configure the Message Exchange Pattern manually.

More information:

-
-

New

The following feature is available in Eclipse and in the Web UI of Cloud Integration.

The SOAPAction Header can now be used as a Camel Header for the following receiver adapters:

- SOAP 1.x
- SAP RM
- SOAP IDoc

More information: [Headers and Exchange Properties Provided by the Integration Framework](#)

New

The **documentation** now contains a detailed list showing which **single roles** are required in order to perform the various tasks related to Cloud Integration.

1 April 2017 - SAP Cloud Integration

Software Version

Tooling	2.29.*
Node Assembly (Cluster 1.x)	1.48.*
Node Assembly (Cluster 2.x)	2.26.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

New

The following feature is available in Eclipse and in the Web UI of Cloud Integration.

In the SAP SOAP 1.x Receiver Adapter, you can now clean up the adapter-specific headers after the receiver call.

More information:

New

The following feature is available in Eclipse and in the Web UI of Cloud Integration.

You can use Location ID to connect a cloud connector instance to your account.

You define the Location ID in the destination configuration on the cloud side.

More information:

-
-
-
-

Enhancement

In the SOAP (SAP RM) Adapter the processing settings have been changed to one default setting.

The following feature has been changed in Eclipse and in the Web UI of Cloud Integration.

The default setting is identical with the setting **Robust** in former releases.

The provider invokes service synchronously and the processing errors are returned to the consumer.

More information:

-
- [SOAP \(SAP RM\) Adapter](#)

New

The following feature is available in Eclipse and in the Web UI of Cloud Integration.

For the SOAP (SOAP 1.x) receiver adapter **Principal Propagation** is now available as an **Authentication** setting.

More information:

-
-

New

The following feature is available in the Web UI of Cloud Integration.

The **JSON to XML Conveter** is now available.

More information:[Define JSON to XML Converter](#)

New

Usability enhancements to Mapping Definition Resource Editor

In the mapping definition resource editor of Cloud Integration WebUI, you can map two selected fields and all the fields with identical names in their corresponding sub-tree by choosing . You can delete all the definitions by choosing .

For more information, see .

New

4 March 2017 - SAP Cloud Integration

Software Version

Tooling	2.28.*
Node Assembly (Cluster 1.x)	1.47.*
Node Assembly (Cluster 2.x)	2.25.*

→ Tip

To check if you have this node assembly version in your Eclipse tooling, open the **Integration Operations** perspective in Eclipse, and in the **Node Explorer**, position the cursor on the tenant name. The node assembly version is then displayed in the tooltip.

Enhancement

New Integration Flow Steps Available in WebUI

The following security-related steps have been made available in the Web UI.

- XML Digital Signer

More information: [Sign the Message Content with XML Digital Signature](#)

- PKCS#7 Verifier

More information: [Verify the PKCS#7/CMS Signature](#)

- XML Verifier

More information: [Verify the XML Digital Signature](#)

11 February 2017 - SAP Cloud Integration

These release notes correspond to the following released software versions:

These release notes correspond to the customer shipment on **2017-02-11**.

Software Version

Tooling: **2.27.***

Node Assembly (Cluster 1.x): **1.46.***

Node Assembly (Cluster 2.x): **2.24.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

No new features or releases for **Web UI, Integration Designer (Eclipse), Integration Operations (Eclipse), Service Development, and SAP Cloud Integration API.**

2016 SAP Cloud Integration (Archive)

17 December 2016

These release notes correspond to the customer shipment on **2016-12-17**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.25.***

Node Assembly (Cluster 1.x): **1.44.***

Node Assembly (Cluster 2.x): **2.22.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Enhanced authorization option in sender adapters	Enhanced	For the following adapter types, the authorization option has been enhanced to allow also to enter custom roles. The following adapter types have been enhanced:	
Dynamically provide Data Store Name	Enhanced	You can now dynamically provide the Data Store Name for the transient data store using headers.	Define Data Store Operations
Technical information available for integration flow elements	New	This feature enables you to view the technical information like step id and version of a component (element). For channels, this technical information can be accessed using context menu. For other elements you can click on the information icon in the editor when you move the cursor on the component (element).	
Message Monitoring user interface enhanced by paging	Enhanced	You can now browse through the list of processed messages using a paging option.	Monitor Message Processing

Function	Type of Change	Description	More Information
Update operation support for SuccessFactors OData V4 receiver	New	You can now use the SuccessFactors OData V4 adapter to perform Update operation.	SuccessFactors OData V4 Receiver Adapter
Context switch in source message for mapping	New	You can change the context of a field in the source message of the mapping definition resource.	

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Enhanced authorization option in sender adapters	Enhanced	For the following adapter types, the authorization option has been enhanced to allow also to enter custom roles. The following adapter types have been enhanced:	
Dynamically provide Data Store Name	Enhanced	You can now dynamically provide the Data Store Name for the transient data store using headers.	
Tracing retention period modified	Enhanced	Post message processing, the trace data is deleted after 60 minutes.	
Update operation support for SuccessFactors OData V4 receiver	New	You can now use the SuccessFactors OData V4 adapter to perform Update operation.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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Function	Type of Change	Description	More Information

Service Development

Function	Type of Change	Description	More Information

SAP Cloud Integration API

Function	Type of Change	Description	More Information
Deploy integration artifacts with OData API	Enhanced	You can now deploy an integration artifact (integration flow, value mapping, or OData API) using the OData API.	OData API

Additional Information

Function	Type of Change	Description	More Information

19 November 2016

These release notes correspond to the customer shipment on **2016-11-19**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.24.***

Node Assembly (Cluster 1.x): **1.43.***

Node Assembly (Cluster 2.x): **2.21.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

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More information: <https://tools.hana.ondemand.com/#hci>

Redeploy Integration Flows with Run Once Setting in Timer or Scheduler

If you have deployed integration flows with **Run Once** option selected in **Timer/Scheduler**, you have to manually **Undeploy** the integration flows and **Deploy** them again. This prevents the integration flow from triggering message after software update.

Redeployment of Integration Flows Might be Required in These Cases

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Inbound authorization for SOAP-based adapters changed	Enhanced	Configuration of inbound authorization for SOAP-based adapters (SOAP 1.x, SAP RM and IDoc) is now accomplished per adapter. Accordingly, inbound authorization cannot be performed anymore in the sender participant.	SOAP (SAP RM) Adapter Assign Sender and Receiver Components Changed procedures: Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment Setting Up Inbound HTTP Connections (with Certificate-to-User Mapping), Neo Environment Setting Up Inbound HTTP Connections (with Client Certificate Authentication), Neo Environment
Canceling a message not possible any more	Enhanced	The feature to cancel messages has been removed from the Message Monitor.	
Example for the HTTP receiver query string	Enhanced	You can now find an example how to use the query string in the HTTP Receiver Adapter.	HTTP Receiver Adapter
Timer/Scheduler Run Once Enhancement	Enhanced	This fix ensures that integration flows with Run Once setting in Timer/Scheduler trigger messages only when the	Define a Timer Start Event

Function	Type of Change	Description	More Information
		integration flow bundles are deployed.	
Assign mapping definition resource	Enhanced	You can now assign a mapping definition resource to the message mapping step in addition to creating a new mapping definition resource.	Working with Mapping
Pass filter conditions via header or property for SuccessFactors SOAP asynchronous operations	Enhanced	You can now pass filter conditions via header or properties while performing asynchronous or ad-hoc query using SuccessFactors SOAP adapter.	SuccessFactors SOAP Adapter

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Inbound authorization for SOAP-based adapters changed	Enhanced	Configuration of inbound authorization for SOAP-based adapters (SOAP 1.x, SAP RM and IDoc) is now accomplished per adapter. Accordingly, inbound authorization cannot be performed anymore in the sender participant.	
Canceling a message not possible any more	Enhanced	The feature to cancel messages has been removed from the Message Monitor.	
Technical information available for integration flow elements	New	This feature enables you to view the technical information like step id and version of a component (element). For channels, this technical information can	

Function	Type of Change	Description	More Information
		be accessed using context menu. For other elements you can click on the information icon in the editor when you move the cursor on the component (element).	
Example for the HTTP receiver query string	Enhanced	You can now find an example how to use the query string in the HTTP Receiver Adapter	
Timer/Scheduler Run Once Enhancement	Enhanced	This fix ensures that integration flows with Run Once setting in Timer/Scheduler trigger messages only when the integration flow bundles are deployed.	
Pass filter conditions via header or property for SuccessFactors SOAP asynchronous operations	Enhanced	You can now pass filter conditions via header or properties while performing asynchronous or ad-hoc query using SuccessFactors SOAP adapter.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Server certificate chain enhanced by SAN	Enhanced	The server certificate chain now also contains the SAN (SubjectsAlternativeNames).	

SAP Cloud Integration API

Function	Type of Change	Description	More Information
OData API has been enhanced	Enhanced	The OData API has been enhanced by the following features: A new entity MessageProcessingLogAttachment allows	OData API

Function	Type of Change	Description	More Information
		you to access MPL attachments. - A new entity <code>MessageStoreEntryAttachmentProperties</code> you to access properties on a message store entry attachment. - You can now undeploy an integration artifact by applying the DELETE operation on an <code>IntegrationRuntimeArtifact</code> instance.	

Additional Information

Function	Type of Change	Description	More Information
Configuring OAuth for inbound communication	New	You can now configure OAuth for inbound communication.	Setting Up Inbound HTTP Connections (with OAuth), Neo Environment

22 and 10, October 2016

These release notes correspond to the customer shipment on **2016-10-22**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.23.***

Node Assembly (Cluster 1.x): **1.42.***

Node Assembly (Cluster 2.x): **2.20.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Script feature allows usage of some more methods	Enhanced	You can now add, set, get, remove headers to/from an attachment and add, set attachment objects as a map,using message processing log through some new methods.	Define a Local Script Step
Timer/Scheduler Run Once setting	Enhancement	If you configure the Timer or Scheduler with Run Once setting, message is triggered only when you deploy the integration flow. Restarting the integration flow bundle will not trigger a message.	Define a Timer Start Event

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Script feature allows usage of some more methods	Enhanced	You can now add, set, get, remove headers to/from an attachment and add, set attachment objects as a map,using message processing log through some new methods.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Trace Configuration Editor	New	Integration flow developers and/or tenant admin personas can now enable	

Function	Type of Change	Description	More Information
		tracing for specific integration flows using the self service feature. This feature is available in the Trace Configuration tab in Integration Operations perspective. You do not have to depend on SAAS admin for enabling trace anymore.	
Timer/Scheduler Run Once setting	Enhancement	If you configure the Timer or Scheduler with Run Once setting, message is triggered only when you deploy the integration flow. Restarting the integration flow bundle will not trigger a message.	

Service Development

Function	Type of Change	Description	More Information

SAP Cloud Integration API

Function	Type of Change	Description	More Information
OData API extended by new entities	Enhanced	The OData API has been extended by the following entities: MessageProcessingLogAdapterAttribute Allows to access adapter-specific attributes in line with message processing log data. IntegrationRuntimeArtifact and RuntimeArtifactErrorInformation Addresses deployed integration artifacts (such as integration flows, value mappings or OData APIs) and related error information.	

These release notes correspond to the customer shipment on **2016-09-24**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.22.***

Node Assembly (Cluster 1.x): **1.41.***

Node Assembly (Cluster 2.x): **2.19.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

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For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
ODC Adapter	New	ODC adapter enables you to communicate with systems that expose data through OData channel for SAP Gateway	ODC Receiver Adapter
General and iterating splitter enhancements	Enhanced	Line Break has been introduced as new Expression Type to support handling large inbound messages (non-XML files).	Define General Splitter Define Iterating Splitter
MIME multipart encoder step has been enhanced.	Enhanced	You can dynamically add headers in the MIME multipart encoder step (Include Headers option). This was up to now only possible by using the Eclipse Integration Designer.	MIME Multipart Messages
Message Monitor, message processing log shows real	Enhanced	In the Message Monitor, message processing log (Web	

Function	Type of Change	Description	More Information
endpoint URL/name used by SOAP/HTTP/SFTP adapter.		UI), as of now the endpoint URL / name which is actually been used at runtime is being displayed (property <code>RealDestinationUrl</code> for the SOAP/HTTP adapter, property <code>ProducedFile</code> for the SFTP adapter). You have several options to specify an endpoint URL in a receiver channel. Examples: • HTTP receiver adapter: In the Address field you can manually enter an HTTP address or you can dynamically override a manually entered address using the Camel header <code>CamelHttpUri</code>. • SFTP adapter: In the File Name field you can manually enter a file name (to specify the file to be read from/written to the SFTP server) or you can dynamically override a manually entered file name using the Camel header <code>CamelFileName</code>.	
Integration Content Monitor now shows detailed error information.	Enhanced	In case an error occurs during the lifecycle of an artifact, detailed information on the error is displayed under Status Details. Before that change, if artifact deployment ran into an error, only the Task Log of the Eclipse Operations tool provided more detailed information on the error.	Manage Integration Content
Option to select product profile during creation of integration flow	Enhanced	You can now select product profiles while adding integration flow to integration package.	Creating an Integration Flow

Function	Type of Change	Description	More Information
Configuring Mail sender adapter enabled	New	You can now use the mail sender adapter for the following tasks: • Downloading e-mails from mailboxes using IMAP or POP3 protocol • Accessing the content of the e-mail body • Accessing e-mail attachments	Mail Adapter

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Option to select product profile during creation of integration flow	Enhanced	You can now select product profiles while creating integration project for an integration flow.	
XSLT Mapping version 1.1 enhancement	Enhanced	You can now select mapping source from partner directory and also set header and exchange properties.	
Configuring Mail sender adapter enabled	New	You can now use the mail sender adapter for the following tasks: • Downloading e-mails from mailboxes using IMAP or POP3 protocol • Accessing the content of the e-mail body • Accessing e-mail attachments	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information

Service Development

Function	Type of Change	Description	More Information
ODC as a data source	New	You can now create and deploy an OData API that exposes data from an IW_BEP component on an on-premise SAP Gateway system (ODC).	Importing from ODC Binding to ODC

Operational Aspects

The list of Subprocessors (non SAP Affiliates) for SAP BTP and its services, SAP Financial Services Network, and SAP Cloud Identity Access Governance has been updated. A Subprocessor is any entity or individual, which has or potentially will have access to or process personal data (as defined in applicable data protection laws). Find the updated list in the Support Portal at: [Services Subprocessor \(non SAP Affiliates\) List](#) 

Note that services such as, for example, SAP Cloud Integration service are covered by this document.

27 August 2016

These release notes correspond to the customer shipment on **2016-08-27**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.21.***

Node Assembly (Cluster 1.x): **1.40.***

Node Assembly (Cluster 2.x): **2.18.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

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Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Deleting headers and properties in content modifier	Enhanced	You can now not just create but also delete headers and properties.	Define Content Modifier

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Deleting headers and properties in content modifier	Enhanced	You can now not just create but also delete headers and properties.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Outbound connectivity test tool enhanced	Enhanced	Outbound connectivity test for SSL and SSH connections has been enhanced in the following way: - During the test of SSL outbound connectivity you have the option to get the certificate chain from the server (in order to compare it with the keystore entries). An additional option (Validate Server Certificate) has been added to the SSL outbound connectivity test. - During the test of SSL outbound connectivity you have	

Function	Type of Change	Description	More Information
		the option to verify the host key and adapt the related known_hosts file accordingly. You can also test SSH connection with the different available authentication options.	

Service Development

Function	Type of Change	Description	More Information

30 and 16, July 2016

These release notes correspond to the customer shipment on **2016-07-30**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.20.***

Node Assembly (Cluster 1.x): **1.39.***

Node Assembly (Cluster 2.x): **2.17.***

i Note

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Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

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Features

Web UI

Function	Type of Change	Description	More Information
SOAP (SAP RM) Adapter	Enhanced	You can now select None as authentication method.	SOAP (SAP RM) Adapter
Message processing log property SAP_Receiver can be reset	Enhanced	You can now use the Content Modifier to reset the header SAP_Receiver.	

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
SOAP (SAP RM) Adapter	Enhanced	You can now select None as authentication method.	

Service Development

Function	Type of Change	Description	More Information
OData sender adapter is now read-only.	Enhanced	With this release, OData sender adapter is not available for editing in the integration flow. It is prepopulated with data you have provided when binding OData objects to a data source.	Editing an Integration Flow
Managing unused bindings	New	SAP Cloud Integration now gives you the ability to reconfigure or delete unused bindings.	Managing Unused Bindings
Support for \$expand	New	You can now use \$expand as a system query option when calling an OData API developed in SAP Cloud Integration.	Developing an OData API Project

SAP Cloud Integration API

Function	Type of Change	Description	More Information
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Function	Type of Change	Description	More Information
OData API released for customers.	New	An Open Data Protocol (OData) application programming interface (API) has been released for customers that allows you to access data (for example, monitoring data).	OData API

These release notes correspond to the customer shipment on **2016-07-16**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.19.***

Node Assembly (Cluster 1.x): **1.38.***

Node Assembly (Cluster 2.x): **2.16.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

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Features

Web UI

Function	Type of Change	Description	More Information
Use Temporary File	New	You can use the Use Temporary File function to write the data to a temporary file initially. Once the write procedure is finished, the temp file is renamed to the target file.	Configure the SFTP Receiver Adapter
Monitoring application user interface changed	Enhanced	In the Integration Content Monitor the data is presented slightly different than before.	Manage Integration Content
Done File Expected	New	You can use the new Read Lock Strategy Done File Expected to signal that the file to be processed is ready for consumption.	Configure the SFTP Sender Adapter
Partner content	New	You can now access and use integration content from partners in the public catalog.	
Timeout option for SuccessFactors adapter with SOAP message protocol	New	You can now configure timeout or the maximum time the adapter waits for a response in the SuccessFactors adapter with SOAP message protocol.	SuccessFactors SOAP Adapter
Create(POST) operation available for SuccessFactors adapter with OData V4 message protocol in the receiver channel	New	You can now use Create(POST) operation with SuccessFactors adapter with OData V4 message protocol in the receiver channel.	SuccessFactors OData V4 Receiver Adapter

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Use Temporary File	New	You can use the Use Temporary File function to write the data to a temporary file initially. Once the write procedure is finished, the temp file is renamed to the target file.	
Done File Expected	New	You can use the new Read Lock Strategy Done File Expected to signal that the file to be processed is ready for consumption.	
Create(POST) operation available for SuccessFactors	New	You can now use Create(POST) operation with	

Function	Type of Change	Description	More Information
adapter with OData V4 message protocol in the receiver channel		SuccessFactors adapter with OData V4 message protocol in the receiver channel.	
Timeout option for SuccessFactors adapter with SOAP message protocol	New	You can now configure timeout or the maximum time the adapter waits for a response in the SuccessFactors adapter with SOAP message protocol.	

No new features or releases for **Integration Operations (Eclipse)**, and **Service Development**.

04 June 2016

These release notes correspond to the customer shipment on **2016-06-04**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.18.***

Node Assembly (Cluster 1.x): **1.37.***

Node Assembly (Cluster 2.x): **2.15.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

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You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

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Features

Web UI

Function	Type of Change	Description	More Information
Maximum number of iterations increased for looping process call	Enhanced	The maximum number of iterations for the looping process call has been increased to 9999.	Define Looping Process Call
Web UI Monitoring - certificate-to-user mapping capabilities enhanced	Enhanced	You can map multiple certificates to the same user (n:1 certificate-to-user mappings are now possible).	Authentication and Authorization Options (Inbound) Client Certificate Authentication and Certificate-to-User Mapping (Inbound) , Neo Environment
Web UI Monitoring - Managing Integration Content	Enhanced	You cannot download configure-only content any more.	Manage Integration Content

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Integration Operations - Deployed Artifacts	Enhanced	You cannot download configure-only content any more.	
New externalizable parameter for IDoc (IDoc-SOAP), SOAP (SAP RM) and SOAP (SOAP 1.x) receiver adapter	Enhanced	You can externalize the Allow Chunking parameter.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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Service Development

Function	Type of Change	Description	More Information
Developing OData APIs in Beta	Enhanced	This feature is now in beta version.	Developing an OData API Project

07 May 2016

These release notes correspond to the customer shipment on **2016-05-07**.

These release notes correspond to the following released software versions:

Software VersionTooling: **2.17.***Node Assembly (Cluster 1.x): **1.36.***Node Assembly (Cluster 2.x): **2.14.*****i Note**

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/mars/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

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Features

Web UI

Function	Type of Change	Description	More Information
HTTP receiver adapter: dynamic configuration of Credential Name	Enhanced	You can dynamically configure the Credential Name property (when basic authentication is specified) by entering either a header or a parameter name.	HTTP Receiver Adapter
Web-based Monitoring - new message processing log property added	Enhanced	The CorrelationId was added that allows you to identify correlated messages.	Monitor Message Processing

Function	Type of Change	Description	More Information
SFTP adapter enhanced by new capabilities	Enhanced	The SFTP adapter has been enhanced by the following new capabilities: • SFTP sender and receiver adapter: For the connection to the SFTP server, authentication based on user name and password (defined by a User Credential artifact) has been enabled. Before, only authentication based on a public key was possible. • SFTP sender adapter: New version of Scheduler tab is available. • SFTP sender adapter: As File Name, you can now enter file name patterns (using * and ? character).	Configure the SFTP Sender Adapter Configure the SFTP Receiver Adapter

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Eclipse Mars now supported	Enhanced	The Integration Designer and the Integration Operations feature now run on Eclipse Mars edition.	
HTTP receiver adapter: dynamic configuration of Credential Name	Enhanced	You can dynamically configure the Credential Name property (when basic authentication is specified) by entering either a header or a parameter name.	
SFTP adapter enhanced by new capabilities	Enhanced	The SFTP adapter has been enhanced by the following new capabilities: • SFTP sender and receiver adapter: For the connection to the SFTP server, authentication based on user name and password (defined by a User Credential	

Function	Type of Change	Description	More Information
		artifact) has been enabled. Before, only authentication based on a public key was possible. • SFTP sender adapter: New version of Scheduler tab is available. • SFTP sender adapter: As File Name, you can now enter file name patterns (using * and ? character).	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Eclipse Mars now supported	Enhanced	The Integration Designer and the Integration Operations feature now run on Eclipse Mars edition.	
Message Monitoring - new message processing log property added	Enhanced	The CorrelationId was added that allows you to identify correlated messages.	

09 April 2016

These release notes correspond to the customer shipment on **2016-04-09**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.16.***

Node Assembly (Cluster 1.x): **1.35.***

Node Assembly (Cluster 2.x): **2.13.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Configure OData adapter assigned to sender channel	Enhanced	You can now assign an OData adapter to the sender channel in an integration project and configure it.	
New product profile	Enhanced	A new product profile SAP Process Orchestration 7.5 SP3 can now be activated under Settings (when your tenant is configured accordingly by SAP).	
Support for OData V4 message protocol in SuccessFactors Adapter	New	You can now access SuccessFactors OData V4 service using the SuccessFactors adapter.	SuccessFactors OData V4 Receiver Adapter
Configure B2B Integration	New	The configure B2B integration area provides an overview of number ranges related artifacts.	Managing Number Ranges

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Configure OData adapter assigned to sender channel	Enhanced	You can now assign an OData adapter to the sender channel in an integration project and configure it.	

Function	Type of Change	Description	More Information
Support for OData V4 message protocol in SuccessFactors Adapter	New	You can now access SuccessFactors OData V4 service using the SuccessFactors adapter.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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Service Development

Function	Type of Change	Description	More Information
Developing OData APIs	New	You can now develop and provision OData APIs from existing data sources such as SOAP, REST and OData.	Developing an OData API Project

12 March 2016

These release notes correspond to the customer shipment on **2016-03-12**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.15.***

Node Assembly (Cluster 1.x): **1.34.***

Node Assembly (Cluster 2.x): **2.12.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Web-based Monitoring UI changed	Enhanced	the user interface of the Monitor Message Processing editor has been changed (master-detail view enhanced).	Monitor Message Processing
Dynamic Parameters	Enhanced	You can find a list of parameters that can be set dynamically at runtime.	Dynamic Parameters (Example)

No new features or releases for **SAP Integration Advisor**.

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Dynamic Parameters	Enhanced	You can find a list of parameters that can be set dynamically at runtime.	Dynamic Parameters (Example)
Externalizing Parameters of Integration Flow	Enhanced	You can find a list of parameters that can be externalized.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Properties view for nodes: new property Profile for nodes	New	A new property Profile is displayed for nodes in the Properties view.	

13 February 2016

These release notes correspond to the customer shipment on **2016-02-13**.

These release notes correspond to the following released software versions:

Software Version

Tooling: **2.14.***

i Note

To check if you have this node assembly version, open the **Integration Operations** perspective in Eclipse, and in the Node Explorer, position the cursor on the tenant name. The node assembly version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Web UI Monitoring: changes in Message Monitor	Enhanced	Web UI Monitoring: Integration Flow Name was renamed to Artifact Name (to other artifact types support future). Attribute Artifact Type has been added.	Monitor Message Processing
Web UI Monitoring: changes in Integration Content Monitor	Enhanced	Web UI Monitoring shows under Managing Integration Content (in the attribute details section) the field Deploy State was renamed to State and provides the state of the artefact with	Manage Integration Content

Function	Type of Change	Description	More Information
		regard to configure-only content.	
Auto-update of Integration Packages	Enhanced	Integration packages get auto-updated once the date of manually updating them expires.	Add Integration Packages to the Customer Workspace
Monitoring: Managing certificate-to-user mappings	New	A new combination of authentication and authorization for inbound messages is supported (for HTTPS connections). A new artifact (Certificate-to-User Mapping) has been introduced for that purpose that can be managed in the Monitoring section of the Web application. Using a certificate-to-user mapping, a user can be authenticated based on a certificate. Certificate-to-user mappings make sure that a user is always associated with the certificate as a whole, not only with one attribute of it (for example the common name (CN)).	Managing Certificate-to-User Mappings, Neo Environment Client Certificate Authentication and Certificate-to-User Mapping (Inbound), Neo Environment
Adding additional header field to message using the Content Modifier or Script step	Enhanced	You can add an additional header (SAP_MessageType) to a message using the Content Modifier or Script step. This header will also be displayed in the message processing log.	

2015 SAP Cloud Integration (Archive)

December 2015

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.32.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Monitoring: Managing certificate-to-user mappings	New	A new combination of authentication and authorization for inbound messages is supported (for HTTPS connections). A new artifact (Certificate-to-User Mapping) has been introduced for that purpose that can be managed in the Monitoring section of the Web application. Using a certificate-to-user mapping, a user can be authenticated based on a certificate. Certificate-to-user mappings make sure that a user is always associated with the certificate as a whole, not only with one attribute of it	Managing Certificate-to-User Mappings, Neo Environment Client Certificate Authentication and Certificate-to-User Mapping_(Inbound), Neo Environment

Function	Type of Change	Description	More Information
		(for example the common name (CN)).	
Adding additional header field to message using the Content Modifier or Script step	Enhanced	You can add an additional header (SAP_MessageType) to a message using the Content Modifier or Script step. This header will also be displayed in the message processing log.	

No new features or releases for **SAP Integration Advisor, Integration Designer (Eclipse), and Service Provisioning in SAP Cloud Integration**.

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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20 December 2015

These release notes correspond to the customer shipment on **2015-12-20**.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.31.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Facebook adapter	New	You can use the Facebook receiver adapter to extract information from Facebook based on certain criteria such as keywords, user data, for example.	Facebook Receiver Adapter
Monitoring/Manage Security Artifacts: deploying additional artifact types	Enhanced	You can now deploy the following artifact types: - SSH Known Hosts - Secure Parameter	Deploying an SSH Known Hosts Artifact Deploying a Secure Parameter Artifact
Monitoring/Manage Integration Content: Only one status for integration content artifacts.	Enhanced	There is only one status (per node) for integration content artifacts. Before, two different statuses have been a displayed in the integration content monitor: Deploy status (which indicated the status of the artifact distribution on the tenant cluster) and runtime status (which indicated the actual health of the artifact per node as determined by its monitor).	Runtime Status
Monitoring: adaptation of user interface design	Enhanced	The design of the pages has been adapted: Functions that relate to selected elements in the table (for example, to Edit a selected table entry) are located on top of the table. Other functions (for example, to Add a new element) are located at the bottom of the editor.	Monitor
Monitoring/Monitor processed Messages: attachments displayed as plain text	Enhanced		Monitor Message Processing

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Facebook adapter	New	You can use the Facebook receiver adapter to extract information from Facebook based on certain criteria such as keywords, user data, for example.	
Encoding/Decoding Messages	Enhanced	You can use the encoding/decoding scheme MIME multipart encode/decode to transform the message content.	
XML Digital Signature	Enhanced	You can add an enveloped transform to a detached signature for XML Digital Signature using the header CamelXmlSignatureTransformMethods	

21 November 2015

These release notes correspond to the customer shipment on **2015-11-21**.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.30.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Delete Method in HTTP Adapter	Enhanced	You can use the delete method in HTTP Adapter	HTTP Receiver Adapter
Monitoring	Enhanced	The following new features are available: - Sub sections have been renamed and restructured. The following sub sections are now available: Monitor, Message Processing, Manage Integration Content (covers now integration content such as integration flows), and Manage Security Material (covers now security-related artifacts). - When adding or changing a tile for the Manage Integration Content section, you can now also specify the artifact Type (which allows you to specify if you like to display only integration flows, only value mappings, or both content types). - You can now deploy User Credentials artifacts.	Monitor

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
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Function	Type of Change	Description	More Information
Open PGP: verification of uncompressed data packets	Enhanced	For input messages to be verified using Open PGP, the Compressed Data packet is now optional (it has been mandatory before this release).	How OpenPGP Works
Delete Method in HTTP Adapter	Enhanced	You can use the delete method in HTTP Adapter	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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Additional Information

Function	Type of Change	Description	More Information
Information on how to avoid encoding issues	New	There is a new topic on how to avoid encoding issues.	Avoiding Encoding Issues

24 October 2015

These release notes correspond to the customer shipment on **2015-10-24**.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.29.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including StreamingClick on the version-dependent internal Eclipse Update Site released for the

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Twitter Adapter	New	Click on th and onYou can use the Twitter receiver adapter to extract information from the Twitter platform based on certain criteria such as keywords, user data, for example.	Twitter Receiver Adapter
Message-ID Handling Soap (SAP RM) Adapter	New	You can set the message-id manually	SOAP (SAP RM) Adapter

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Twitter Adapter	New	You can use the Twitter receiver adapter to extract information from the Twitter platform based on certain criteria such as keywords, user data, for example.	
	New	You can set the message-id manually	
Message-ID Handling Soap (SAP RM) Adapter	New	You can set the message-id manually	

26 and 14, September 2015

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Software Version (Runtime)

1.28.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

These release notes correspond to the following released software versions:

These release notes correspond to the customer shipment on **2015-09-26**.

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
For SOAP messages, an error message containing a URL (to access message processing log) is sent back to the sender (when configured accordingly)	Enhanced		Define Error Configuration
Send step	New	You can configure a Send step to specify a service call to a receiver system for scenarios and adapters where no reply is expected.	Define a Send Step
Web-based Monitoring user interface has been refined	Enhanced	Web-based Monitoring user interface has been refined. You can now, for example, display message attachments in separate tabs.	Monitor

Function	Type of Change	Description	More Information
In SOAP Adapter (SOAP 1.x) new parameters can be externalized	Enhanced	You can externalize different parameters in WS-Security configuration in SOAP Adapter (SOAP1.x)	
Add Integration Package or Integration Flow	New	You have to add the integration package or integration flow to your customer workspace. This enables you to access the artifacts in that package, configure, and deploy them.	Add Integration Packages to the Customer Workspace
Configure Product Profile	New	The tenant admin can view and configure the product profile to mark a particular product profile as default, for the tenant. This enables you to make no more changes to the product profile.	Set Default Runtime Profile
Define Switching Product Profile	New	You can switch product profiles if you want to build integration flows for different products on the same customer tenant.	Configure Runtime Profile for an Integration Flow
Editing Scripts of a Mapping	New	You can now modify the script of a mapping.	

SAP Integration Advisor

Function	Type of Change	Description	More Information
SAP Integration Advisor	New	SAP Integration Advisor allows business partners to easily specify and describe business requirements of business-to-business (B2B) interfaces, map and test them.	SAP Integration Advisor

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
For SOAP messages, an error message containing a URL (to access message processing log) is sent back to the sender (when configured accordingly)	Enhanced		
Working with Product Profiles	New	Product profile is a collection of capabilities such as success factor adapter, splitter or datastore elements, available in the	

Function	Type of Change	Description	More Information
		product. You can consume these capabilities at the time of designing integration flows.	
Deleting Adapters	New	You delete an adapter once an adapter is no more required in the system.	
Multiple key-value pairs in SFAPI Parameters	Enhanced	You can now specify multiple key value pairs in SFAPI parameters while configuring SuccessFactors adapter.	

Software Version (Runtime)

1.27.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

These release notes correspond to the following released software versions:

These release notes correspond to the customer shipment on **2015-09-14**.

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Redeployment of Integration Flows Might be Required in Cases Including Streaming

For scenarios that include streaming data is written to the file system when the data volume exceeds a specific threshold value. The files are encrypted. The encryption algorithm has been changed from RC4 to AES/CTR/NoPadding, and the threshold has been increased from 64/128 KB to 1 MB.

The new values are used for new or redeployed integration flows. Already deployed integration flows will still use the old values. Therefore, you need to redeploy the integration flows if the new values are to be used for existing integration flows.

Features

Web UI

Function	Type of Change	Description	More Information
Dynamic attributes for the HTTP adapter	Enhanced	You can dynamically configure the Address and Query field of the HTTP adapter.	HTTP Receiver Adapter
Body MIME type and Body Encoding editable	New	You can set Body MIME type and Body Encoding	Mail Adapter
Add all message attachments	New	You can add all attachments contained in the message exchange to the e-mail	Mail Adapter
Create the JSON message without the root element tag	New	You can create the JSON message without the root element tag	Define XML to JSON Converter
Setting SOAP headers with Groovy script	Enhanced	You can set SOAP headers using Groovy script.	Read and Modify SOAP Headers

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Dynamic attributes for the HTTP adapter	Enhanced	You can dynamically configure the Address and Query field of the HTTP adapter.	
Body MIME type and Body Encoding editable	New	You can set Body MIME type and Body Encoding	
Add all message attachments	New	You can add all attachments contained in the message exchange to the e-mail	
Create the JSON message without the root element tag	New	You can create the JSON message without the root element tag	
Setting SOAP headers with Groovy script	Enhanced	You can set SOAP headers using Groovy script.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Generic WSDL download	Enhanced	For sender channels for which no WSDL has been specified, a generic WSDL file can be downloaded (Integration Operations, Properties view, Services tab).	

01 August 2015

These release notes correspond to the customer shipment on **2015-08-01**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.26.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Features

Web UI

Function	Type of Change	Description	More Information
Modelling PKCS#7 Encryptor and Signer steps	Enhanced	You can now model PKCS#7 Encryptor and Signer steps using the Web-based Integration Designer.	Sign the Message Content with PKCS#7/CMS Signer Encrypt and Sign the Message Content with PKCS#7/CMS Encryptor
Configuration setting Proxy Type available for HTTP adapter in Web-based Integration Designer	Enhanced	SAP Cloud Connector is supported for the HTTP adapter. The corresponding setting (Proxy Type) is now	HTTP Receiver Adapter

Function	Type of Change	Description	More Information
		also available in the Web-based Integration Designer.	
Modelling a channel with the Mail Adapter	Enhanced	You can now model a channel with the Mail Adapter using the Web-based Integration Designer.	Mail Adapter
Dynamically configure the mail adress and the attachment names in Mail adapter on receiver side	New	You can now dynamically configure the mail adress and the attachment names in Mail adapter on receiver side	Mail Adapter
Updating Integration Packages and their Contents	New	You can now choose to update your integration packages and their contents with the latest changes.	
Reworked Monitoring pages	Enhanced	The Monitoring pages (Message Monitor and Integration Content Monitor) have been reworked and allow the user to do the following customize tables by adding or removing columns. For integration flows both the display name and the technical name are shown in the filter settings and in the result tables.	Monitor
XML Validator	Enhanced	You can now validate the incoming message payload against the configured XML schema	

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Modelling a channel with the Mail Adapter	Enhanced	You can now model a channel with the Mail Adapter using the Web-based Integration Designer.	
Dynamically configure the mail adress and the attachment names in Mail adapter on receiver side	New	You can now dynamically configure the mail adress and the attachment names in Mail adapter on receiver side	
Product Profiles	Enhanced	You can now consume the capabilities in product profiles at the time of designing integration flows.	
XML Validator	Enhanced	You can now validate the incoming message payload	

Function	Type of Change	Description	More Information
		against the configured XML schema	
Parameterization of Timer	New	You can externalize the timer parameters. Refer to documentation for handling older integration flows with timer.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Integration flow display names	Enhanced	The integration flow display name is now showed in the Operations UI, for example, in the Message Monitoring editor, the Deployed Artifacts editor, and the Component Status view.	
Integration flow version	Enhanced	The integration flow version is now displayed in the Message Monitoring editor.	

Adapter Developer Kit

Function	Type of Change	Description	More Information
Developing Adapters	New	You can now develop new SAP Cloud Integration adapter types on eclipse platform to extend the connectivity of SAP Cloud Integration with remote systems.	Developing Adapters

04 July 2015

These release notes correspond to the customer shipment on **2015-07-04**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.25.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Features

Web UI

Function	Type of Change	Description	More Information
Modeling data store operations	New	The Web-Based Designer allows you to model and specify data store operations (for the transient data store).	Define Data Store Operations
Escalation event	New	The Web-Based Designer allows you to model and specify an escalation event.	Define an Escalation Event
PKCS#7 Encryptor and PKCS#7 Signer	Enhanced	The Web-Based Designer allows you display PKCS#7 Signer and Encryptor steps for imported integration flows. You cannot yet create new steps of that type.	Sign the Message Content with PKCS#7/CMS Signer Encrypt and Sign the Message Content with PKCS#7/CMS Encryptor
Ariba Adapter for Sender and Receiver Channels	New	You configure sender and receiver channels of an integration flow with the Ariba adapter. These channels enable the SAP and Non-SAP cloud applications to send and receive business specific documents in cXML format to and from Ariba network. Examples for business documents are purchase order, invoice, etc.	Ariba Adapter
Define Script	New	You use this task to execute custom java script or groovy script for message processing	Define a Local Script Step

Function	Type of Change	Description	More Information
Define the XML-to-JSON Converter	New	The XML-to-JSON converter enables you to transform messages in XML format to JSON format	Define XML to JSON Converter
Custom Functions in Message Mapping	New	You can define or modify a mapping expression by using the associated Custom Functions of a message mapping	

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Accessing MPL in the script step	Enhanced	There are the following additional Java interfaces for the message processing log (MPL) which you can address with the script step (either in Groovy Script or JavaScript): MessageLogFactory, MessageLog.	
Ariba Adapter for Sender and Receiver Channels	New	You configure sender and receiver channels of an integration flow with the Ariba adapter. These channels enable the SAP and Non-SAP cloud applications to send and receive business specific documents in cXML format to and from Ariba network. Examples for business documents are purchase order, invoice, etc.	
SAP Cloud Connector support for IDoc adapter (receiver channel)	New	You can use the IDoc adapter to connect to on-premise systems via SAP Cloud Connector.	
Namespace support for Xpath in Gather step	Enhanced	You can specify the namespace in Xpath if the incoming XML contains namespace.	
Header and property variables support for SuccessFactors SOAP adapter parameters	Enhanced	You can specify the key and value using header or property variables in the parameters during channel configuration	
Character encoding for request payload in OData adapter	New	You have the option of specifying UTF-8 as the character encoding format for encoding the request	

Function	Type of Change	Description	More Information
		payload while configuring the OData adapter.	
Using Custom Functions in Message Mapping	New	You can create your own custom functions by using groovy scripts and use them as required.	
SAP Cloud Connector supported for HTTP adapter	Enhanced	You can use the SAP Cloud Connector with HTTP adapter receiver channels to connect to on-premise systems.	

Additional Functions

Function	Type of Change	Description	More Information
User management functions enabled for customers	New	Customers can now specify the members of their account and define their permissions.	Adding Members to an Account

18 June 2015

These release notes correspond to the customer shipment on **2015-06-18**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.24.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/luna/>

More information: <https://tools.hana.ondemand.com/#hci>

Eclipse Luna for SAP Cloud Integration

You have to use the Eclipse Luna release when you like to install the Integration Designer and the Integration Operations user interface.

There is **no support for Eclipse Kepler any more**.

An own BPMN editor comes with Eclipse Luna which provides features like the following ones:

- Automatic connecting of shapes for sequence flows
- Automatic adjustments of bending points of connections
- Resizing shapes

Features

SAP Cloud Integration Web Application

Function	Type of Change	Description	More Information
Display names for integration flows	Enhanced	For integration flows display names (instead of technical names) are now shown in the message monitor. In earlier versions of the software, for integration flows only a technical name (without any spaces) could be specified. As of this version, you have the option to specify an additional display name for integration flows in the Web-based Integration Designer. This display name can be a <i>human readable</i> name with multiple words separated by spaces. In particular, when importing an integration flow into the Web-based Integration Designer, you have to specify the display name mandatory. Note the following implication when you have edited an integration flow in the Web-based Integration Designer and after that you like to import this integration	

Function	Type of Change	Description	More Information
		flow into the Eclipse-based Integration Designer. In that case, the display name of the integration flow is shown, which might lead to confusion (as the integration flow name in Eclipse-based Integration Designer might have <i>changed</i> that way).	

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Error handling strategy for SOAP messages	Enhanced	When defining the error handling strategy for SOAP messages, you can now define if in case of an exception the SOAP fault exception is to be returned to the sender system. If you don't select this option, an error template containing the MPL ID is sent to the sender system instead.	
Custom query options for OData adapter	Enhanced	You can define custom query options other than the options available as a part of operations modeler when you configure the OData adapter receiver channel.	
SAP Cloud connector support for SOAP and OData adapter in receiver channel	New	You can use the SAP Cloud Connector with SOAP and OData adapter receiver channels to connect to on-premise systems.	• • • • SAP Cloud Connector
OData support for content enricher	New	You can use the OData adapter and SuccessFactors OData adapter with content enricher.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Outbound connection test for SMTP connections	Enhanced	You can now test an outbound connection (for a sender mail adapter).	
Monitoring shows now display names for integration flows.	Enhanced	Instead of technical integration flow names, display names are now	

Function	Type of Change	Description	More Information
		shown in the message monitor.	

07 May 2015

These release notes correspond to the customer shipment on 2015-05-07.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.23.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

Features

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
Web-based monitoring enhancements of user interface	Enhanced	In the Message Monitor you can now open a configuration dialog for the externalized parameters of the integration flow. When you click on the status, you open the details (message processing log). When you click on an integration flow name, the	Monitor

Function	Type of Change	Description	More Information
		graphical representation of the integration flow is shown (read-only). In the Integration Content Monitor you can now undeploy and configure artifacts. - Specifying SAP_sender and SAP_Receiver With the Content Modifier you can specify additional header fields that can be used for end-to-end tracing. You can specify now the following additional fields: SAP_Sender and SAP_Receiver.	
Edit support for local integration process and sequential multicast elements	New	You can edit the properties of local integration process and sequential multicast elements in the integration flow editor.	• Define Local Integration Process • Define Multicast

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
New standard can be used to send out encrypted mails/attachments	Enhanced	You can configure the mail adapter on the receiver to send encrypted e-mails/attachments using S/MIME standard.	
New scheduler tab for SFTP sender adapter	Enhanced	You can find the polling parameters under the new Scheduler tab	
Specifying SAP_sender and SAP_receiver as header elements	Enhanced	the Content Modifier allows you to specify SAP_sender and SAP_receiver as header element (that can be used for end-to-end tracing).	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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09 April 2015

These release notes correspond to the customer shipment on 2015-04-09.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

Features

SAP Cloud Integration Spaces (Web UI)			
Function	Type of Change	Description	More Information
Redesign of the Monitoring pages	Enhanced	To improve usability for integration developers, the <i>Monitoring</i> pages of the Web-based SAP Cloud Integration application have been redesigned. The Monitoring entry page is composed now of tiles that show the status of messages or integration content artifacts for specific filter criteria (for example, for a specific status). You can now personalize the Monitoring entry page by adding new tiles or changing existing ones or by re-arranging the tiles on the page. The pages are designed that way that the full screen size of your device can be utilized. To filter which messages or integration content artifacts are to be displayed, you can now select among all available statuses for messages and artifacts. The URLs of the individual pages of the Web-based SAP Cloud Integration application can now be bookmarked.	Monitor

Function	Type of Change	Description	More Information
Receiver mail adapter	New	You can now configure a receiver mail adapter to send out messages by e-mail.	
Streaming for XML-to-JSON converter	Enhanced	The XML-to-JSON converter supports streaming.	
Escalation event has	New	You can use this new step to specify an escalation event. An escalation event stops message processing without triggering further message processing retries. For synchronous messages, an error messages is sent to the sender. A new message status ESCALATED has been introduced for the message monitoring.	
SFAPI Parameters Support for SuccessFactors SOAP adapter	New	You can specify additional SFAPI parameters for SuccessFactors SOAP adapter when you are configuring the adapter.	

No new features or releases for **Integration Operations (Eclipse)**.

12 March 2015

These release notes correspond to the customer shipment on 2015-03-12.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.21.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

Features

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
Progress bar for integration flow opening	New	You can see a progress bar when you open an integration flow. You can also see relevant prompts in case of exceptions.	
Testing message mapping	New	You can validate the correctness of message mapping with the given test input at design time.	Testing Mappings

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
New configuration settings for the usage of XML Advanced Electronic Signature (XAdES).	Enhanced	You can use XML Advanced Electronic Signature (XAdES) to sign messages. The Integration Designer now provides full support of the XAdES-BES and XAdES-EPES forms.	Signing the Message Content with XML Advanced Electronic Signature
New algorithm can be used for message signing.	Enhanced	For the Simple Signer, PKCS#7/CMS Signer and PKCS#7/CMS Signed and Enveloped Data, you can now use the following additional signature algorithm RIPEMD256/RSA.	Message-Level Security
Change in path traversal default in SFTP Receiver.	Enhanced	If the file contains any backward path traversals, this can lead to a potential risk of directory traversal. In such a case message processing is stopped with an error.	

Function	Type of Change	Description	More Information
New parameter of Receiver SOAP (SOAP1.x) Adapter	New	You can now select the required layout type. Options are strictor lax .	
Activating Tenant and Integration Flow Tracing	Enhanced	The documentation of this feature has been improved.	
Usage of Gather after Splitter in integration flow modeling	Enhanced	You can use Gather after Splitter while modeling an integration flow.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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12 February 2015

These release notes correspond to the customer shipment on **2015-02-12**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

2.0.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released SAP Cloud Integration software (runtime components).

You will get the actual Eclipse tool version (according to the actually released SAP Cloud Integration software) through this Eclipse update site:

<https://tools.hana.ondemand.com/kepler>

Features

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
Editing an Integration Package	Enhancement	You can now save versions of artifacts irrespective of their editing status.	Editing an Integration Package

No new features or releases for **Integration Designer (Eclipse)**.

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Outbound connection test tool	New	You can test an outbound connection for a tenant (calling a receiver system). Both protocols SSL and SSH are supported.	

2014 SAP Cloud Integration (Archive)

18 December 2014

These release notes correspond to the customer shipment on **18.12.2014**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)

1.18.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

Features

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
Adapter modeling dialog	New	You can use the adapter modeling dialog to add an adapter to a communication channel and configure it. The dialog appears when you create a communication channel and guides you to the adapter configuration step.	Accessing Integration Content Using SAP Cloud Integration Spaces

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Upsert operation for SuccessFactors adapter with OData message protocol	New	You can use the Upsert operation to perform both Insert and Update operations in one communication cycle.	
Content enricher enhancement	Enhancement	The enhanced message from content enricher contains all the content from the lookup message referred by multiple entries of the key element.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Message monitor allows to display MPL of dependent messages in aggregation scenarios	Enhanced	In case you have configured a message aggregation use case (using the Aggregator step in the integration flow), you have the option to show the status of the source messages (that are to be aggregated) and of the aggregated message.	

22 November 2014

These release notes correspond to the customer shipment on **22.11.2014**.

i Note

Note that these dates refer to planning and can be changed without further notice.

Related Software Versions

Software Version (Runtime)

1.17.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

i Note

Make sure that for the Integration Designer and Integration Operations tools (to be locally installed on your client) you use the same version as for the released software (runtime components).

You will get the actual Eclipse tool version (according to the actually released software) through this Eclipse update site:

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

Features

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
Message Monitoring enhanced by additional filter options	Enhanced	As an administrator who likes to get an overview of the message flow, you can now also search for messages associated with a specific integration flow name.	

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Procedure how to convert messages in XML to JSON format and messages in JSON format to XML format	New	Using the XML-to-JSON Converter you can now transform messages in XML format to JSON format and messages in JSON format to XML format.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
Message monitoring user interface enhanced by additional filter options	Enhanced	As an administrator who likes to get an overview of the message flow, you can now use additional search options. - Filtering for messages associated with a specific integration flow name - Filtering for messages associated with an application ID An application ID can flexibly be configured in the Content Modifier step of an integration flow. At runtime, the application ID is written into the message header and can be detected by message monitoring.	

Additional Functions

Function	Type of Change	Description	More Information
Connecting a Customer System to SAP Cloud Integration has been revised	Enhanced	This section of the documentation has completely been revised. It also contains now the process flow for the customer-managed operating model that has been made available within the SAP Cloud Integration Partner Edition.	

25 October 2014

These release notes correspond to the customer shipment on **25.10.2014**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.16.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

More information: <https://tools.hana.ondemand.com/#hci>

Features

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
Viewing PRO Package Details	New		

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Defining Write Variables	New	You use Write Variables to define variables, which you access across message flows for a specific integration flow or across integration flows(s). You use Content Modifier to read variables in to headers and properties, and Write Variables to create/update variables.	
XML Digital Signer offers additional settings	Enhanced	The XML Digital Signer has been enhanced by the following attributes: - You can now select an encoding scheme for the output XML document. - You can now specify the value of the Id attribute of the Signature element. - You can now also configure the Signer that way that the parent element of the Signature element is to be specified by an XPath expression (in case the Enveloped XML Signature option is selected).	
Signing the Message Content with Simple Signer	New	Simple Signer makes it easy to sign messages to ensure authenticity and data integrity	

Function	Type of Change	Description	More Information
		when sending an XML resource to participants on the cloud.	
Aggregator step	New	The Aggregator step allows you to combine multiple incoming messages into a single message.	

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information

27 September 2014

These release notes correspond to the customer shipment on **27.09.2014**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Software Version (Runtime)

1.15.*

(is provided by SAP)

i Note

You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.

Link to Eclipse Update Site

<https://tools.hana.ondemand.com/kepler>

More information: <https://tools.hana.ondemand.com/#hci>

Features

No new features or releases for **SAP Cloud Integration Spaces (Web UI)**.

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Data Store Operations	New	You can use Data Store to store messages. Data Store supports four types of operations. Data Store supports the following operations:	

Function	Type of Change	Description	More Information
		• Write – You can use this operation to store the messages into the data store. • Delete – You can use this operation to trigger the deletion of messages in the data store. • Select – You can use this operation to fetch messages in bulk from the data store. You can also specify the number of messages you fetch per poll. • Get - You can use this operation to fetch a specific message from the data store.	
Multicast Operation	New	You can use multicast to route a single message to more than one receiver in a single integration process.	
Content Enricher	New	You can enrich the existing message with the contents of a lookup message during the course of an integration process using the content enricher. You can also specify the key for enriching the content.	
Security enhancement	Enhancement	A security enhancement is added to the server side. You need to update the operations UI and integration tooling to the latest version (1.15.0). i Note You might encounter errors if you deploy integration content without updating to 1.15.0 release.	NA

Integration Operations (Eclipse)

Function	Type of Change	Description	More Information
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30 and 16, August 2014

These release notes correspond to the customer shipment on **30.08.2014**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.14.* (is provided by SAP)	1.14.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci
i Note You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.	

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
Working with value mappings	New	You can add and edit value mapping artifacts in an integration package.	Editing Value Mapping
Editing mapping details	Enhanced	You can now edit message mappings in an integration flow.	Editing an Integration Package
Editing integration packages	Enhanced	You can now save versions of an artifact	Editing an Integration Package
Deploying data flows	Enhanced	The old deploy icon has been changed and replaced with the new one.	Deploying Data Flows

Integration Designer (Eclipse)

Function	Type of Change	Description	More Information
Defining an exception subprocess	New	You can use this task if you want to catch any thrown exception in the integration process and perform additional processing on it.	
Using headers to dynamically override HTTP adapter configuration	Enhanced	You can set headers before calling the HTTP adapter in case you want to dynamically	

Function	Type of Change	Description	More Information
		override the configuration of the adapter.	
XML Digital Signature	Enhanced	Detached XML Signatures are now supported.	

These release notes correspond to the customer shipment on **16.08.2014**.

Planned future shipments: 30.08.2014.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.13.* (is provided by SAP) i Note You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.	1.13.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description	More Information
UI Changes	Enhanced	The My Projects label has been replaced by Run . This tab page enables you to access the integration flows you have deployed. The Manage label has been changed to Design . This tab page enables you to access the integration packages and artifacts.	
SuccessFactors adapter: REST protocol support	Enhanced	You can use the REST protocol to communicate with the SuccessFactors system. You can access the LMS module of SuccessFactors system through this.	
SuccessFactors (SOAP)	Enhanced	You can edit system properties and adapter specific attributes for the SFSF (SOAP) adapter.	

Function	Type of Change	Description	More Information
SuccessFactors adapter: Odata Capability in receiver	New	The SuccessFactors adapter supports OData protocol for communication in the receiver channel.	
Quick configure scheduler	Enhanced	Quick configure feature is available for the scheduler element	
Sender system and Receiver system edit	New	You can edit the system properties for sender and receiver systems.	
SOAP 1.x adapter	New	You can add and edit configuration details for SOAP 1.x adapter.	

Integration Designer (Eclipse Feature)

Function	Type of Change	Description	More Information
Web Service Security Username Token Profile 1.1 supported	Enhanced	You can configure username token (password digest, plain text) authentication with WS-Security to connect to the backend.	
HTTP Method PUT	New	Use this method to update or create the enclosed data on the receiver side.	

Integration Operations (Eclipse Feature)

Function	Type of Change	Description	More Information
Basic Authentication artifact renamed to User Credentials	Enhanced	You can configure username token (password digest, plain text) authentication with WS-Security to connect to the backend.	

19 July 2014

These release notes correspond to the customer shipment on **19.07.2014** .

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.12.*	1.12.*

Software Version (Runtime)	Link to Eclipse Update Site
(is provided by SAP) i Note You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.	https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

Integration Designer (Eclipse Feature)

Function	Type of Change	Description	More Information
End of Juno Eclipse support	Enhanced	Support of Juno Eclipse for the Integration Designer and Integration Operations feature has been stopped. These features can now be used with the following Eclipse version only: Kepler release (Eclipse 4.3).	
Configuring a Channel with SOAP (SOAP 1.x) Adapter	Enhanced	The SOAP 1.x Adapter allows the configuration of WS-Security options.	
Properties/Variable concept for Content Modifier	Enhanced	You use content modifier if you want to modify the content of the incoming message by providing additional information in the header, property or body of a message before sending it to the receiver.	
Integration Flow Tracing		Tracing helps to track the message flow of processed messages and view relevant message payload at different points of message flow. It also helps to know if there is any error in the message execution. This feature has now been made available externally.	
Defining Script	New	You use this task to assign javascript or groovy script for message processing.	
Properties/Variable concept for Content Modifier	Enhanced	You use content modifier if you want to modify the content of the incoming message by providing additional information in the header, property or body of a message	

Function	Type of Change	Description	More Information
		before sending it to the receiver.	
Integration Flow Tracing	New	Tracing helps to track the message flow of processed messages and view relevant message payload at different points of message flow. It also helps to know if there is any error in the message execution.	

Integration Operations (Eclipse Feature)

Function	Type of Change	Description	More Information
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SAP Cloud Integration Spaces

Function	Type of Change	Description	More Information
Locking Integration Packages	New	You can restrict the editing of integration package to only one user at a time.	
Changing Source and Target Message Structuring	New	You can now change the source and target message without changing the file extension.	
Copying Integration Packages	New	You can copy existing integration packages to your workspace.	
Editing an Integration Package	Enhanced	You can now copy an integration package from the Discover tab to the Manage tab and edit it.	Editing an Integration Package
Working with an Integration Package	Enhanced	Previously named as Viewing an Integration Package has been renamed to Working with an Integration Package.	Working with an Integration Package
Mass Configuration	New	You can configure multiple integration flows at the same time in a single screen.	Configure Multiple Integration Flows
OData Support: Query editing	New	You can edit the OData query inline.	
Timer: edit	New	You can edit the attributes of the timer element in the integration flows.	
Quick configure: SFSF adapter and Odata Adapter	New	You can perform a quick configuration of SuccessFactors and OData adapters without accessing the integration flow editor.	
PGP Encryptor/Decryptor edit	New	You can edit the system properties for integration flows with PGP Encryptor/Decryptor.	Define PGP Encryptor Define PGP Decryptor
SFTP adapter edit	New	You can edit the system properties for integration flows with SFTP adapter.	
Converter edit	New	You can edit the attributes of the converter element (CSV to XML) in the integration flows.	

These release notes correspond to the customer shipment on **21.06.2014** .

Planned future shipments: 19.07.2014, 02.08.2014, 30.08.2014

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.11.* (is provided by SAP) i Note You can check whether you have this software version as follows: Open the Integration Operations perspective in Eclipse, and in the Node Explorer position the cursor on the tenant name. The software version is then displayed in a tooltip.	1.11.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

Integration Designer (Eclipse Feature)

Function	Type of Change	Description
Encrypting and Signing the Message Content with PKCS#7/CMS	Enhanced	SAP Cloud Integration also supports Signed and Enveloped Data for PKCS#7/CMS. In other words, both encryption and signing can be applied in one step.
Decrypting and Verifying the Message Content with PKCS#7/CMS		
Defining Encoders	Enhanced	The Encoder now also supports zip and gzip compression.
Configuring a Channel with SFTP Adapter	Enhanced	The adapter allows you to configure several actions after message processing, for example, deleting or moving the file.
Configuring Channels with HTTP Adapter	Enhanced	Parameters and values for the receiver HTTP adapter have been enhanced.
Defining Service Calls	New	A service call is used to call an external system. Such calls enable data to be transferred to or from the target system. It can be used for Request-Reply and Content Enrichment operations.
Partial Parameterization	Enhanced	Partial parameterization enables you to change part of a field rather than the entire field. This variable entity of the field is entered within curly braces.

Function	Type of Change	Description
SuccessFactors Adapter: OData message protocol supported	New	You can use the OData message protocol to fetch data from the SuccessFactors system.
SuccessFactors Adapter: REST message protocol supported	New	You can use the REST message protocol to communicate with the Learning Management System (LMS) of the SuccessFactors system.
SuccessFactors Adapter: UI Enhancements	Enhanced	The user interface is modified and tooltips are provided to improve usability.
OData Adapter: UI Enhancements	Enhanced	The user interface is modified and tooltips are provided to improve usability.
OAuth2 credentials deployment	New	You can deploy OAuth2 credentials on your cluster by using the deploy credentials wizard for use with the SuccessFactors REST protocol of the Learning Management System (LMS).
Converter: CSV to XML Converter	New	The CSV to XML converter converts files in .csv format to .xml format.
Encrypting the Message Content with PGP	New	You have the option to protect the message using Open Pretty Good Privacy (PGP).
Decrypting the Message Content with PGP		

10 May 2014

These release notes correspond to the customer shipment on **10.05.2014** .

Planned future shipments: 21.06.2014, 19.07.2014, 02.08.2014, 30.08.2014

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.10.* (is provided by SAP) i Note You can check for this software version in the following way: Open the Integration Operations perspective in Eclipse and in the Node Explorer position the cursor on the tenant name. Then the software version is displayed in a tooltip.	1.10.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

Integration Designer (Eclipse Feature)

Function	Type of Change	Description
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Integration Operations (Eclipse Feature)

Function	Type of Change	Description
SAP Cloud Integration Integration Operations feature available on Eclipse Kepler edition	New	SAP Cloud Integration Integration Operations feature can also be used with Eclipse Kepler release (Eclipse 4.3).
New artifacts to support PGP	New	New following artifact types have been introduced to support message level security based on Open Pretty Good Privacy (PGP): - PGP Public Keyring - PGP Secret Keyring
SAP Solution Manager systems can be registered as alert consumers	New	You can enable SAP Solution Manager to display SAP Cloud Integration alerts. To support this feature, a new parameter has been introduced for the tenant configuration.

12 April 2014

These release notes correspond to the customer shipment on **12.04.2014**.

Planned future shipments: 10.05.2014, 21.06.2014, 19.07.2014, 02.08.2014, 30.08.2014

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.9.* (is provided by SAP) i Note You can check for this software version in the following way: Open the Integration Operations perspective in Eclipse and in the Node Explorer position the cursor on the tenant name. Then the software version is displayed in a tooltip.	1.9.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

Integration Designer (Eclipse Feature)

Function	Type of Change	Description
HTTP outbound adapter	New	The HTTP adapter allows you to configure an outbound HTTP connection from SAP Cloud Integration to a receiver.

Integration Operations (Eclipse Feature)

Function	Type of Change	Description
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15 March 2014

These release notes correspond to the customer shipment on **15.03.2014** .

Planned future shipments: 12.04.2014, 10.05.2014, 21.06.2014, 19.07.2014, 02.08.2014, 30.08.2014

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.8.* (is provided by SAP) i Note You can check for this software version in the following way: Open the Integration Operations perspective in Eclipse and in the Node Explorer position the cursor on the tenant name. Then the software version is displayed in a tooltip.	1.8.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

General

Function	Type of Change	Description
SFTP Polling on Multiple Runtime Nodes	Enhanced	SFTP polling is supported in the following way: the same file can be polled by multiple endpoints configured to use the SFTP channel. This means that you can now deploy an integration flow with a configured SFTP channel on multiple runtime nodes (which might be necessary to meet failover requirements) without the risk of creating duplicates by polling the same file multiple times. Note that to enable the new option, integration flows (configured to use SFTP channels) that have been developed prior to the introduction of this feature have to be re-generated.

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description
Monitoring (SAP Cloud Integration Spaces)	Enhanced	

Integration Designer (Eclipse Feature)

Function	Type of Change	Description
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Integration Operations (Eclipse Feature)

Function	Type of Change	Description
Job Scheduler tab allows you to schedule jobs	New	You have the option to schedule jobs required to operate the cluster efficiently.

15 February 2014

These release notes correspond to the customer shipment on **15.02.2014** .

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.7.* (is provided by SAP) i Note You can check for this software version in the following way: Open the Integration Operations perspective in Eclipse and in the Node Explorer position the cursor on the tenant name. Then the software version is displayed in a tooltip.	1.7.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description
Monitoring (SAP Cloud Integration Spaces)	Enhanced	

Integration Designer (Eclipse Feature)

Function	Type of Change	Description
Signing the Message Content with XML Digital Signature: new/deleted attributes	Enhanced	The following changes have been made: - Added: Key Info Content field. - Deleted: Key Info Id, Signature Id, Object Id will now be generated and cannot be configured any more with the Integration Designer.

Function	Type of Change	Description
Verifying the XML Digital Signature: new/deleted attributes	Enhanced	The following changes have been made: • • Added: Public Key Aliases can now be selected, Check For Key Info option has been enabled • Validate Signature option
Verifying the PKCS7/CMS Signature	Enhanced	Several public key aliases are now allowed.
Decrypting the Message Content (PKCS#7)	Enhanced	The Private Key Alias has been deleted.
Plain SOAP renamed as SAP RM and added additional Parameters for IDoc SOAP and SOAP RM	Enhanced	To harmonize with the existing the SOAP (SOAP 1.x) adapter configurations, these additional parameters are added for IDoc SOAP and SOAP RM
Support of SOAP1.x for content enricher to external resource	Enhanced	Connection between content enricher and external resource can be now configured with SOAP 1.x. also.

Integration Operations (Eclipse Feature)

Function	Type of Change	Description
Implementing context-sensitive help	New	You can now access those parts of the Operations Guide that cover the Integration Operations feature (in particular, the subsection <i>Monitoring (Integration Operations Feature in Eclipse)</i>) directly from the tool by selecting  Help  Help Contents. In addition to that, you can open the documentation covering the following elements of the Integration Operations feature as context-sensitive help by clicking the question mark icon (context-sensitive help): Tenant Configuration wizard and Deploy Artifacts wizards.
Tenant Configuration editor	New	You have the option to specify a set of parameters that determine a tenant cluster – in other words: a target configuration of a tenant cluster. The target configuration includes the state and topology of the tenant cluster in terms of number and type of contained nodes, and other related attributes. You can open the relevant configuration user interface by double-clicking a tenant in the Node Explorer and opening the Tenant Configuration editor for the selected tenant.

18 January 2014

These release notes correspond to the customer shipment on **January 18 2014**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.6.* (is provided by SAP) i Note You can check for this software version in the following way: Open the Integration Operations perspective in Eclipse and in the Node Explorer position the cursor on the tenant name. Then the software version is displayed in a tooltip.	1.6.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description
Monitoring (SAP Cloud Integration Spaces)	Enhanced	

Integration Designer (Eclipse Feature)

Function	Type of Change	Description
Configuring sender and receiver channel	New	You can configure the sender and receiver channel of the SuccessFactors connector to transfer data.
Operations modeler	New	You can create a SFQL (SuccessFactors Query Language) query and use it to transfer data.
Enabling overwrite of Existing Message	Enhanced	This feature enables you to overwrite an existing persisted message with the same ID.
WSDL Storage on tenant management node (TMN)	New	This feature is used for accessing the FSN WSDL file on the TMN so that it can be used for configuring the WS-RM adapter.
Enabling Chunking Option	New	This feature enables chunking of messages from the sending system.
Compress Message option for Plain SOAP Adapter	Enhanced	This feature enables compressing of Plain SOAP adapter messages.

Integration Operations (Eclipse Feature)

Function	Type of Change	Description
Usability improvements of Integration Operations feature	Enhanced	The following improvements have been made to the Integration Operations feature: - Version information pop-ups are displayed when starting Eclipse. - New context menu functions in the views/editors enable you to copy

Function	Type of Change	Description
		object names or to open the properties information, for example. - You can now download the MPL to your hard disk from the Properties view (when message selected in Message Monitoring view). - A new status <i><scheduled></i> was introduced for tasks in the Tasks view.

2013 SAP Cloud Integration (Archive)

The following features are new or have been enhanced in the current version of SAP Cloud Integration (for process integration).

07 December 2013

These release notes correspond to the customer shipment on **07.12.2013**.

Planned future shipments: 01-16-2014

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.5.* (is provided by SAP) i Note You can check for this software version in the following way: Open the Integration Operations perspective in Eclipse and in the Node Explorer position the cursor on the tenant name. Then the software version is displayed in a tooltip.	1.5.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description
Viewing Integration Packages	Enhanced	You can now download all artifacts at one go
Monitoring (SAP Cloud Integration Spaces)	Enhanced	There are the following enhancements in the SAP Cloud Integration Spaces

Function	Type of Change	Description
		Monitoring application: • Autorefresh on dashboard/start page • Displaying notifications in a separate bar at the bottom of the Web UI • Restarting Integration Artifacts • Dropdown listbox in Artifact view that allows to open detail page for another artifact • Tooltips for message errors • Highlighted information on message errors in the message processing log • Cancelling erroneous messages

Integration Designer (Eclipse Feature)

Function	Type of Change	Description
Assigning the Sender and Receiver Participants	Enhanced	You can authenticate a sender system having any SOAP (SOAP 1.x, Plain SOAP, SOAP WS-RM) or IDoc (IDoc SOAP) connector using Basic Authentication apart from the already available feature of authenticating using an authorized client certificate.
Defining Splitter	Enhanced	Two new type of splitters, called the General Splitter and Iterative Splitter, are available in addition to the existing IDoc Splitter and PKCS#7/CMS Signature-Content Splitter..
Defining Content Modifier	Renamed	Content Enricher is now renamed to Content Modifier. The feature functionality has no change. It's corresponding pattern is no more available in the Integration Flow wizard during the integration flow creation.

Integration Operations (Eclipse Feature)

Function	Type of Change	Description
Deploying a Basic Authentication Artifact	Enhanced	The wizard for Basic Authentication (CREDENTIALS) artifacts has been enhanced to support scenarios with basic authentication (receiver side/outbound).
Properties View for Deployed Artifacts	Enhanced	When you select a Basic Authentication artifact in the Deployed Artifacts editor, the Properties view shows additional information on the artifact.

Function	Type of Change	Description
Monitoring External Reachability of Runtime Node	New	You can monitor if runtime nodes (assigned to the tenant management node) can be reached by external calls. For this purpose, an external SSL call of a runtime node is simulated and monitored using a specific component in the Component Status view.

09 November 2013

These release notes correspond to the customer shipment on **09.11.2013**.

i Note

Note that these dates refer to planning and can be changed without further notice.

These release notes correspond to the following released software versions:

Related Software Versions

Software Version (Runtime)	Link to Eclipse Update Site
1.4.* (is provided by SAP) i Note You can check for this software version in the following way: Open the Integration Operations perspective in Eclipse and in the Node Explorer position the cursor on the tenant name. Then the software version is displayed in a tooltip.	1.4.* https://tools.hana.ondemand.com/juno More information: https://tools.hana.ondemand.com/#hci

SAP Cloud Integration Spaces (Web UI)

Function	Type of Change	Description
Deploying Data Flows	New	You can deploy data flows through the integrated Data Services application available on SAP Cloud Integration Spaces.
Monitoring (SAP Cloud Integration Spaces)	New	A new section on SAP Cloud Integration Spaces provides capabilities to monitor SAP Cloud Integration clusters and message processing. In particular, the following information can be accessed: - Information on the runtime status of deployed integration content - Information on the status of messages processed on a tenant cluster

Integration Designer (Eclipse Feature)		
Function	Type of Change	Description
Defining Gateway	Enhanced	You can create conditions for non-XML messages. Such conditions are allowed for header expressions using a set of supported operators and regex.
Creating Multi Mappings	Enhanced	This is an enhancement of the mapping feature. You can now perform mapping using multiple source and target messages of type XSD or WSDL.

Integration Operations (Eclipse Feature)		
Function	Type of Change	Description
Deployed Artifacts Editor	Enhanced	For security artifacts deployed on a tenant of a cluster based on the 1.4.* software version, the node type TENANT_MGMT is displayed in the Deployed Artifacts editor.
Component Status View	Enhanced	The new sub system <i>Persistence</i> has been introduced. It allows you to monitor for each node if the write access to the data base works correctly.

What's New for Integration Advisor

Core Components, 2024

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo			Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Create MIGS by Payload Supports IDocs and Custom IDocs	The recently introduced wizard that uses payloads to create message implementation guidelines now also supports SAP IDocs and custom IDocs. See Creating Message Implementation Guidelines by Payload. As Cloud Integration (standalone) customer, see Creating Message Implementation Guidelines by Payload.	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.17.* - Increment: 2608	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Create MIGs by Payload	Create new message implementation guidelines using an existing payload. Integration Advisor takes a payload you provide as base for a new MIG and suggests the best fitting objects during the creating process. See Creating Message Implementation Guidelines by Payload. As Cloud Integration (standalone) customer, see Creating Message Implementation Guidelines by Payload.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Automatic Reordering of Repeating Composites	You can now enable automatic reordering of repeating composite data elements in UN/EDIFACT segments to match the message implementation guideline sequence. Use this feature to handle business partner payloads where composite instances arrive in varying orders, removing the need for complex mapping logic to accommodate different positioning. See Enabling Arbitrary Composite Ordering. As Cloud Integration (standalone) customer, see Enabling Arbitrary Composite Ordering.	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.16.* - Increment: 2607	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Updated Behavior for Fixed Value in Date Time Conversion	When defining fixed values in date time conversion, you can now set Z as a fixed value for the UTC marker. Additionally, Integration Advisor now supports the new date time format CCYY'-MM'-DD'T'hh':mm':ss['.ff*]Z. See Date Time Conversion. As Cloud Integration (standalone) customer, see Date Time Conversion.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.15.* - Increment: 2606	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Additional Matching Rules for Overlay Mapping Guidelines	Additional matching rules for overlay mapping guidelines have been added: It's now easier to match segments that moved to segment groups and data elements that moved to composites with their counterparts in the base mapping guideline. See Matching Rules for Overlay Mapping Guidelines. As Cloud Integration (standalone) customer, see Matching Rules for Overlay Mapping Guidelines.	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	New Documentation Structure and Links	With this release, you can find the documentation for the SAP Integration Suite capability SAP Integration Advisor in a new location: Integration Advisor . Update your bookmarks to not miss any guide updates.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.14.* - Increment: 2605	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Extended Matching Behavior for EDIFACT Overlay Mapping Guidelines	The mapping behavior for EDIFACT base and overlay mapping guidelines has been extended. In case of mismatches in the segment group, the matching now automatically recognizes trigger segments instead and draws a mapping line. This way, errors are avoided. See Matching Rules for Overlay Mapping Guidelines. As Cloud Integration (standalone) customer, see Matching Rules for Overlay Mapping Guidelines.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	VDA Type System Available	Integration Advisor now includes the VDA message standard. The type system content is available a few days after the upgrade to the new software version. See Overview of B2B Standards. As Cloud Integration (standalone) customer, see Overview of B2B Standards.	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Mass Update	When deleting MIGs or MAGs, you can now use mass actions to delete multiple MIGs or MAGs at once. Note that they can only be deleted if they're not referenced by any other artifacts and that deletion is irreversible. See Deleting Mapping Guidelines and Deleting Message Implementation Guidelines. As Cloud Integration (standalone) customer, see Deleting Mapping Guidelines and Deleting Message Implementation Guidelines.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	• SAP Integration Advisor: 2.12.* • Increment: 2603	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Envelope Support for cXML	You can now combine cXML messages with cXML envelopes in your message implementation guidelines. See Overview of B2B Standards. As Cloud Integration (standalone) customer, see Overview of B2B Standards.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Protecting Custom Codelists	You can now protect custom codelists, allowing you to restrict editing rights to certain custom codelists. See Protecting Artifacts. As Cloud Integration (standalone) customer, see Protecting Artifacts.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry	Software Version Update	• SAP Integration Advisor: 2.11.*	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
	Neo		Increment: 2602					
Integration Advisor	Cloud Foundry Neo	Protecting Global Code Value Mappings	You can now also protect global code value mappings, which allows you to restrict editing rights to certain global code value mappings. See Protecting Artifacts. As Cloud Integration (standalone) customer, see Protecting Artifacts.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Unified Envelope Handling in MIG Preprocessing	The envelope handling in MIG preprocessing has been unified. The MIG preprocessing now automatically keeps or removes envelopes depending on the MIG definition and expectation of following processing steps. If your integration flow in Cloud Integration implements any special handling on these payloads, check if it continues to work as expected. See Exporting Runtime Artifacts. As Cloud Integration (standalone) customer, see Exporting Runtime Artifacts.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	ZIP Files as Source for Custom Messages	In addition to XSD and WDSL files, you can now also use ZIP files to upload custom messages to a custom type system library. See Adding a Custom Message. As Cloud Integration (standalone) customer, see Adding a Custom Message.	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Migration of Existing Duplicates in Overlay MAGs	Since target node duplicates from base MAGs are now visible in the overlay MAG, all overlay MAGs created before this release are automatically updated to the new behavior. In most cases, this migration doesn't affect your work, but in edge cases, existing mapping can be affected. Therefore, it's advised that you check that your mappings still work as intended in special cases. See Limitations. As Cloud Integration (standalone) customer, see Limitations.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Changed Target Node Duplication Behavior	Overlay MAGs now automatically inherit target node duplication rules from their base MAG. While you can't modify or delete these inherited rules, you can define additional, overlay-specific rules. See Working with an Overlay Mapping Guideline. As Cloud Integration (standalone) customer, see Working with an Overlay Mapping Guideline.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.10.* - Increment: 2601	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	No Validation of Code Values When Working with Codelists	While editing the codelist of a message implementation guideline, you can work with code values. If you don't want	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			the extended semantic validation of your MIG to check the value of the field against the code values, you can select the new checkbox No validation of code values. See Working with Codelists. As Cloud Integration (standalone) customer, see Working with Codelists.					
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.9.* - Increment: 2513	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Upload Expected Target Payload for Mapping Guidelines Simulation	When simulating a mapping guideline, you can now upload an expected target payload. After the simulation, the simulation data is then compared against the expected data. Going through the differences, you can make corrections in the payload to reach your desired outcome. See Simulating a Mapping Guideline. As Cloud Integration (standalone) customer, see Simulating a Mapping Guideline.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Extended Handling for UN/EDIFACT Segment Groups During MIG Migration	If a node can't be fully matched during the migration of a MIG, the migration finds an alternative match. In the case of segment groups for UN/EDIFACT, the handling has been changed as follows: If SGnn of the old message has become SGmm in the new message, with the same trigger segment,	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			then segment group SGnn is automatically migrated to segment group SGmm. See Migrating a Message Implementation Guideline. As Cloud Integration (standalone) customer, see Migrating a Message Implementation Guideline.					
Integration Advisor	Cloud Foundry Neo	Automatically Create 1:1 Mappings	You can now automatically create mappings between identical nodes using the Create 1:1 Mappings function. See Working with Mapping Guidelines (MAGs). As Cloud Integration (standalone) customer, see Working with Mapping Guidelines (MAGs).	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Protect MIGs and MAGs	You can now restrict write access to MIGs and MAGs by protecting them. Only users with the relevant rights can edit protected artifacts, as well as grant and revoke protection. See Protecting Artifacts. As Cloud Integration (standalone) customer, see Protecting Artifacts.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.8.* - Increment: 2512	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	XSLT as Source for MAGs	You can now create mapping guidelines that use XSLT files as source. To create an XSLT mapping guideline, go to the Mapping Guidelines tab and choose Create XSLT MAG. After completing the creation wizard, you can upload an	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			XSLT file that defines your desired mapping. Additionally, the original standard and overlay MAGs are now grouped under the term graphical mapping guidelines to emphasize the distinction with XSLT mapping guidelines. See Creating XSLT Mapping Guidelines. As Cloud Integration (standalone) customer, see Creating XSLT Mapping Guidelines.					
Integration Advisor	Cloud Foundry Neo	Import Code Value Mappings Using CSV Files	You can now populate code value mappings through a CSV file by using the Import function. The CSV file should contain predefined code value mappings. Before uploading, you can download a template and fill it with your data to ensure that you're using the correct format. See Global Code Value Mappings. As Cloud Integration (standalone) customer, see Global Code Value Mappings.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.6.* - Increment: 2510	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Additional Edifact Versions Available	Integration Advisor now offers the early Edifact versions 90.2 S2, 91.1 S2, and 91.2 S2, which are all associated with Edifact syntax version 2. See Overview of B2B Standards.	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			As Cloud Integration (standalone) customer, see Overview of B2B Standards .					
Integration Advisor	Cloud Foundry Neo	New Technical IDOC Node for Custom IDoc Messages	Newly created custom IDoc messages now include the additional technical IDOC node. This node allows additional mapping scenarios in which bulk IDocs need to be mapped into multiple line items of one EDI message, or the other way around. See Adding a Custom Message. As Cloud Integration (standalone) customer, see Adding a Custom Message.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Copy Mapping Guidelines with MIGs	You now have increased flexibility when copying an existing mapping guideline. With the option Copy MAG with MIGs..., you can now choose if source MIG, target MIG, or both MIGs should be copied as well. See Creating a New Mapping Guideline. As Cloud Integration (standalone) customer, see Creating a New Mapping Guideline.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.5.* - Increment: 2509	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Changed Behavior When Assigning Codelists to Leaf Nodes	The behavior of the checkbox Use all code values of the codelist during the assignment of codelists to leaf nodes has been changed: If you manually select all code	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			values of a list one by one without selecting the checkbox, the checkbox is no longer checked automatically. This behavior has consequences, for example, if you migrate your MIG to a new message version and the new version provides more code values: If you didn't select the checkbox, each code value is migrated by itself, and any new code values stay unselected. See Assigning Codelists to Leaf Nodes. As Cloud Integration (standalone) customer, see Assigning Codelists to Leaf Nodes.					
Integration Advisor	Cloud Foundry Neo	Global Parameter Suggestions	When creating global parameters inside a MAG, you're now provided with suggestions for frequently used parameters. This makes the creation of global parameters more convenient. See Working with Mapping Guidelines. As Cloud Integration (standalone) customer, see Working with Mapping Guidelines.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Setting the UN/EDIFACT Message Version (Field 0052) for Older UN/EDIFACT Versions	In your MIGs, you can now define the expected UN/EDIFACT message version value. The default value D is applied automatically, but you can edit it. This setting influences the file name of the message XSD used in the EDI flow steps of Cloud Integration.	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			For older UN/EDIFACT releases, you can change the value for field 0052 as described in the documentation. See Exporting Runtime Artifacts. As Cloud Integration (standalone) customer, see Exporting Runtime Artifacts.					
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.3.* - Increment: 2507	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Harmonization of User Interface for Code Value Mappings with 1:1 and n:1 Cardinality	The user interface for code value mappings for 1:1 and n:1 cardinality has been harmonized. See Code Value Mapping. As Cloud Integration (standalone) customer, see Code Value Mapping.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Combining Overlay MAGs with Pretransformation	You can now combine an overlay MAG with pretransformation. The following scenarios are possible: - Base MAG with PTS and overlay MAG without pretransformation - Base MAG with PTS and overlay MAG with modified pretransformation - Base MAG without PTS and overlay MAG with pretransformation - Disable pretransformation of base MAG without pretransformation	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			of overlay MAG (Indirectly) To learn about these different scenarios, see Working with an Overlay Mapping Guideline. As Cloud Integration (standalone) customer, see Working with an Overlay Mapping Guideline.					
Integration Advisor	Cloud Foundry Neo	Download MAG Simulation Results as EDI Flat File	You can now download the result of a MAG simulation as an EDI flat file. You can only use this additional option if the target message is an EDI message from one of the type systems ASC X12, Odette, TRADACOMS, UN/EDIFACT or one of its subsets. After a simulation has run, instead of choosing  <i>Download simulation result</i>, open the download menu. There, choose Download as EDI Flat File. See Simulating a Mapping Guideline. As Cloud Integration (standalone) customer, see Simulating a Mapping Guideline.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	• SAP Integration Advisor: 2.2.* • Increment: 2506	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	• SAP Integration Advisor: 2.1.* • Increment: 2505	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	AI Settings Available	The AI settings for MIGs and MAGs are now also available under  Settings > Artificial Intelligence > . See Artificial Intelligence .	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	MIG Creation with Multiple Payloads	You can now create message implementation guidelines using multiple payload files in a ZIP archive. For more information, see Creating a New Message Implementation Guideline . As Cloud Integration (standalone) customer, see Creating a New Message Implementation Guideline .	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Mass-Update the Base MAG Version in Overlay MAGs	You can now mass-update mapping guidelines by replacing the base MAG version in the overlay MAGs. This option is a second method of updating mapping guidelines besides replacing the MIG for a set of MAGs. To mass-update a group of mapping guidelines, go to the MAG overview page and choose  Mass-Update MAGs > Replace Base MAG Version in Overlay MAGs > , then follow the wizard. See Updating Mapping Guidelines . As Cloud Integration (standalone) customer, see Updating Mapping Guidelines .	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	New Type System Versions	The following new type system versions are available: • ASC X12: 008050, 008060	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			cXML: 1.2.065, 1.2.066 See Overview of B2B Standards. As Cloud Integration (standalone) customer, see Overview of B2B Standards.					
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 2.0.* - Increment: 2504	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Use Full or Reduced Message XSD File	In your message implementation guideline (MIG) for EDI type systems, you can now choose if you want to use the full or a reduced message XSD file. The message XSD file is used in the EDI flow steps. For new MIGs, the default is the full XSD file, whereas existing MIGs continue to use reduced XSD files for compatibility reasons. See Exporting Runtime Artifacts. As Cloud Integration (standalone) customer, see Exporting Runtime Artifacts.	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Mass Validate XSLT Snippets	You can now mass validate all your XSLT snippets inside your mapping guideline (MAG). See Working with a Mapping Guideline (MAG). As Cloud Integration (standalone) customer, see Working with a Mapping Guideline (MAG).	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.99.* - Increment: 2503	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.98.* - Increment: 2502	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Exporting Message Types	You can now directly export EDI Type System Message Types as XSD files, eliminating the need to create a Message Implementation Guideline. See Exporting Runtime Artifacts. As Cloud Integration (standalone) customer, see Exporting Runtime Artifacts	Info only	General Availability	New	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.97.* - Increment: 2501	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Overlay Mapping Guidelines	You can now create overlay Mapping Guidelines allowing you to customize and extend base Mapping Guidelines. See Creating an Overlay Mapping Guideline. As Cloud Integration (standalone) customer, see Creating an Overlay Mapping Guideline	Info only	General Availability	New	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.95.* - Increment: 2412	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Improved Mapping Proposal	To enhance the Proposal feature's capabilities, Draft MIGs and MAGs will now be published to the knowledge base, in addition to active ones. You can opt-in/opt-out of this feature in the settings. See: Settings	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Creating a Custom IDoc	When creating a custom IDoc, you can now define a naming convention in the format Message Type.IDoc Type.Extension Type. See: Adding a Custom Message	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Creating a MIG	When creating a MIG from a payload, you can now select specific nodes for qualification. See: Creating a Message Implementation Guideline	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Validating a XSLT Snippet	You can now validate XSLT code for a function directly in the MAG editor at compile time. This feature displays detailed error messages, including line numbers for quick resolution: See: Using Functions	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Simulating a MAG	Simulation errors in the MAG editor now displays the specific nodes and corresponding links for easy identification. See: Simulating a Mapping Guideline	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	• SAP Integration Advisor: 1.93.* • Increment: 2410	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	xCBL Type System	The Type System library now includes the new xCBL type system. The content of the system will be rolled-out in a phased manner post-release. See: Overview Of B2B Standards	Info only	General Availability	New	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Neo Cloud Foundry	Functions in Mapping Guidelines	Functions in Mapping Guideline editor now supports XSLT 3.0 and XPath 3.1. See: Functions	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	Software Version Update	- SAP Integration Advisor: 1.92.* - Increment: 2409	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	Versioning a Global Code Value Mapping	Global Code Value mappings can now be activated providing consistency and standardized code-value associations. See: Global Code Value Mapping	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.91.* - Increment: 2408	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	New Read-Only Access Role	Support personas can now view, simulate and export artefacts using a new read-only access role. See: - For Neo: Assigning Users for SAP Integration Advisor - For CF: Assigning Users for SAP Integration Advisor	Info only	General Availability	New	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Creating a Mapping Guideline	The Mapping Guideline creation wizard has been enhanced to allow for modifications to key information, including Name, Business Context, and Summary. See: Creating a Mapping Guideline	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.90.* - Increment: 2407	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Envelopes in MIGs	You can now include envelopes when creating a Message Implementation Guideline(MIG). This applies to ASC X12, UN/EDIFACT and Odette. i Note Envelopes will only be visible in the MIG creation wizard after relevant content has been added to your system. The content will be released gradually over the next few days/weeks, starting with UN/EDIFACT and ASC X12, followed by Edifact subsets and Odette. See: Creating a Message Implementation Guideline	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Integration Advisor Migration Guide	We have now published a Migration Guide for Integration Advisor that details the steps needed for customer who are planning to migrate their data from Neo environment to Integration Suite. See: Integration Advisor Migration Guide	Info only	General Availability	New	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.89.* - Increment: 2406	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	Custom Codelists	You can now copy code values from existing codelists and add them to your custom codelists. See: Adding a Custom Codelist	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Support of EDI Flat file payload	You can now upload EDI flat files directly when creating MIGs, as well as during the simulation of MIGs and MAGs. See: • Creating a Message Implementation Guideline • Simulating a Message Implementation Guideline • Simulating a Mapping Guideline	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	• SAP Integration Advisor: 1.88.* • Increment: 2405	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Enable/Disable Proposal Service	You can now control the collection of data for the Proposal service by enabling or disabling the service according to your requirements. See: Settings	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	• SAP Integration Advisor: 1.87.* • Increment: 2404	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Disabling the Proposal Service	To avoid data collection for proposal service, customers can now opt out of proposal service completely. In the next release, an extension to	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			the data collection feature is planned where all Integration Advisor artifacts (MIGs, MAGs, etc.) are automatically collected. If you don't want to use the proposal service and don't want your content to be collected, you need to opt out of the proposal before this new feature gets released before 28th June, 2024. See: Settings					
Integration Advisor	Cloud Foundry Neo	Global Custom Codelists	You can now create custom codelists and use them across all artifacts. See: Custom Type Systems	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.85.* - Increment: 2402	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Disabling Proposal Service	You can now choose to opt out of the Proposal service that is used to collect data and provide proposals when working with a MIG/MAG. See: Settings	Required	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Redesign of the Mapping List	The Mapping List of a MAG now displays more information about the source and target nodes such as constants and global parameters. See: Working with Mapping Guidelines (MAGs)	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
Integration Advisor	Cloud Foundry Neo	XPath Expressions for MIGs	The MIG editor now displays two versions of XPath expressions which can be used before and after the preprocessing of the message. See: Working with a Node	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Filter Code Values in MIG	The code values of a node in MIG editor can now be filtered using multiple options. See: Working with Codelists	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.84.* - Increment: 2401	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	DateTime Conversion	You can now add global parameters to a source node in DateTime conversion of your mapping guideline. See: Date Time Conversion	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.83.* - Increment: 2313	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	Duplicate Target Nodes	You can now duplicate target nodes in the mapping editor of a Mapping Guideline. See: Mapping the Source and Target Nodes	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	Global Code Value Mapping	You can now create a global code value mapping with multiple source code lists and one target code list (N:1 cardinality).	Info only	General Availability	Changed	Technology	Not applic

Technical Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modu Busin Proce
			See: Global Code Value Mappings You can now map multiple code values with a target code value in a global code value mapping in the mapping editor of a MAG. See: Code Value Mapping					
Integration Advisor	Cloud Foundry Neo	Software Version Update	- SAP Integration Advisor: 1.82.* - Increment: 2312	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	Display xsd:nillable value in a MIG	You can now view if an element/node in a MIG allows nil values through the Nillable field. See: Working with a Node	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	Administrative Data in the Documentation Template	You can now select whether you want to display the administrative data of your MIG and MAG in the documentation template. See: Settings	Info only	General Availability	Changed	Technology	Not applic
Integration Advisor	Neo Cloud Foundry	Support of xsd:nillable in Custom Messages	Custom Messages now support xsd:nillable. See: Custom Type Systems	Info only	General Availability	Changed	Technology	Not applic

Initial Setup

Includes links to concepts and activities required to set up and start using Cloud Integration.

Related Information

[Initial Setup of SAP Cloud Integration in the Neo Environment](#)

[Initial Setup of SAP Cloud Integration in the Cloud Foundry Environment](#)

[Environment-Specific Aspects Integration Developers Should Know](#)

Initial Setup of SAP Cloud Integration in the Cloud Foundry Environment

Quickly get started with SAP Cloud Integration in the Cloud Foundry (CF) environment.

i Note

You can't subscribe to Process Integration or Cloud Integration service independently anymore. To provide a comprehensive integration experience, Cloud Integration is only available as a capability of the SAP Integration Suite. For a new subscription of Cloud Integration, subscribe to SAP Integration Suite. See: [Initial Setup of SAP Integration Suite](#).

Trial Account

Trial accounts are intended for personal exploration, and not for production use or team development. The features included in a trial account are limited, compared to an enterprise account. Consider the following before using a trial account:

- Every trial user gets one trial account only.
- Cloud Foundry trial accounts expire after 30 days. You can extend the trial period to a maximum of 90 days, after which your account is automatically deleted.
- Usage of runtime resources are limited only for functional evaluations. Processing of large message payloads is not supported.
- A subaccount in your trial account is created automatically. Each subaccount is associated with exactly one Cloud Foundry organization in which you can create additional spaces.
- You can manage members in your trial account.
- You can activate SAP Event Mesh with limited capabilities.
- You can use production and beta services in trial accounts.
- A trial account includes 4 GB of memory for applications.
- You can use 8 GB of instance memory.
- SAP does not provide support to establish secure connection using private keys and authentication based on inbound client certificate. It's recommended to use basic authentication for allowing a client to authenticate itself against the CF server based on user credentials (`clientid` and `clientsecret`)
- You can use a maximum number of 10 JMS queues.
- There is no service level agreement with regards to the availability of the platform.

For more information, see [Trial Accounts and Free Tier](#).

Related Information

[Subscribing to Process Integration](#)

[Configuring User Access to Cloud Integration](#)

[Provisioning the Tenant](#)

[Creating Service Instances](#)

Subscribing to Process Integration

Subscribe to the Process Integration application from the Subscriptions page in the SAP BTP cockpit.

Prerequisites

i Note

You can't subscribe to Process Integration or Cloud Integration service independently anymore. To provide a comprehensive integration experience, Cloud Integration is only available as a capability of the SAP Integration Suite. For a new subscription of Cloud Integration, subscribe to SAP Integration Suite. See: [Initial Setup of SAP Integration Suite](#).

Configuring User Access to Cloud Integration

Create and modify application roles and assign users to these roles.

Prerequisites

You're subscribed to Process Integration service in the Cloud Foundry environment.

Context

As an administrator of SAP Cloud Integration in the Cloud Foundry environment, you can group application roles in role collections. Typically, these role collections provide authorizations for certain types of users. Once you've created a role collection, you pick the roles for that role collection. For more information on role collections, see [Roles and Role Collection](#).

Finally, you assign the role collection to the users provided by the SAP ID service.

Procedure

- 1. Go to your subaccount in SAP BTP cockpit, and choose ►Security ►Users. ►
- 2. Choose **Create** to add a new user.
- 3. Enter the **User Name** and **E-Mail**, and choose **Create**.
- 4. Choose the user and select ... under **Role Collections** section and choose **Assign Role Collection**.
- 5. In the resulting dialog box, select the role collections that you want to assign and choose **Assign Role Collection**.

i Note

Clear your web browser cache and cookies before navigating to **Go to Application**.

The following role collections are available:

Role Collections	
Role Collection	You Can...
PI_Read_Only	See Read-only Persona
PI_Business_Expert	See Business Expert Persona
PI_Integration_Developer	See Integration Developer Persona
PI_Administrator	See Tenant Administrator Persona

Provisioning the Tenant

Provision a Cloud Integration tenant and receive a consumer-specific URL to access the application.

Prerequisites

i Note

You can neither provision a new tenant using your existing Process Integration or Cloud Integration subscription nor subscribe to Cloud Integration anymore. For a new subscription of Cloud Integration, subscribe to SAP Integration Suite. See: [Initial Setup of SAP Integration Suite](#).

Creating Service Instances

Create Process Integration runtime service instances to access the endpoints after deploying the integration flows.

Context

i Note

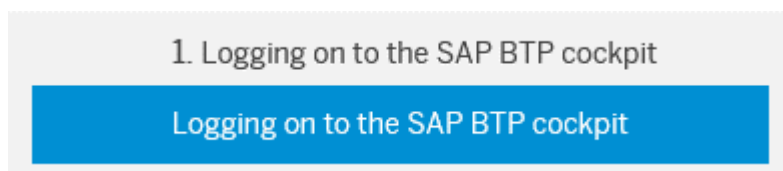
You can neither create a new service instance for your existing Process Integration or Cloud Integration subscription nor subscribe to Cloud Integration anymore. For a new subscription of Cloud Integration, subscribe to SAP Integration Suite. See: [Initial Setup of SAP Integration Suite](#).

Initial Setup of SAP Cloud Integration in the Neo Environment

This quick start guide provides all the information you need to quickly onboard after subscribing to the SAP Cloud Integration edition.

The following steps tell you how to complete the onboarding:

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

When you subscribe to any of the SAP Cloud Integration editions, you receive one or two e-mails from SAP, depending on the edition of SAP Cloud Integration that you have purchased. Log on to SAP BTP cockpit with your SAP S-user ID. For more information about the different commercial models, see <https://cloudplatform.sap.com/pricing.html>.

If you have chosen the subscription-based license model, you don't need to perform steps 2 (Creating a Subaccount) and 3 (Enabling and Configuring a Tenant). These actions are already done by SAP. You can go straight to step 4.

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.

2. Setting Up Your Account Model

Creating a Subaccount

Please note that image maps are not interactive in PDF outputs.

Create subaccounts in your global account. This allows you to divide your account model and structure it according to your business needs. You can find the Display Name of your subaccount in the welcome e-mail. For more information about accessing subaccounts, see [Navigate to Global Accounts and Subaccounts](#).

3. Configuring the SAP Cloud Integration Service

Enabling and configuring a Tenant

Disabling a Tenant

Enable the **Process Integration** service. Once the process integration service is enabled, you can configure a tenant.

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.

4. Managing Member Authorizations

Assigning Users and Roles

Adding new Administrators (Optional)

Verifying Access for Users

Please note that image maps are not interactive in PDF outputs.

Once you have obtained access to SAP Cloud Integration, you can add new users and assign the required roles to them. Then you can start creating and running your integration flows.

Related Information

[Getting Started with Integration Flow Development](#)

Logging on to SAP BTP Cockpit

Log on to your global account in the SAP BTP cockpit.

Prerequisites

You are assigned the Administrator role for the global account.

Context

When your organization has purchased the corresponding license, an e-mail is sent to the SAP S-user ID specified in the contract. The e-mail message contains the link for logging on to the system and the credentials (user and password) for the specified SAP S-user ID.

i Note

- If you have not received this e-mail, the most likely reason is that your user ID was not specified in the order form. Check with your internal team who was responsible for signing the contract, and check which e-mail ID or S-user ID was provided to the SAP Account Manager in the order form.
- Check with your SAP Account Manager which S-user ID was provided in the order form.
- Contact the SAP Customer Success Team [here](#) .
- If you are still facing issues, create a ticket using the component LOD-HCI-PI-OPS-PROV.

Procedure

1. Use the link in your welcome e-mail to log on to the SAP BTP cockpit. The global account **Overview** page opens, displaying global account information, including the number of subaccounts and regions in your global account, and service usage information. For more information, see [Log On to Your Global Account](#).
2. Choose the global account provided in your welcome e-mail. As a next step access your subaccounts already created by SAP or create new subaccounts.

Related Information

[Creating a Subaccount](#)

Creating a Subaccount

Use a subaccount to host applications and services in production or non-production environments.

Prerequisites

You are authorized to access services using the consumption-based and subscription-based models.

Context**i Note**

The following task is only required if you have purchased a consumption-based license model.

"Consumption-based" means that you pay for the time you use the service, and allows users to consume Cloud Integration services based on this strategy. The advantage of a consumption-based account is that you can configure and provision Cloud Integration tenants with minimal intervention from SAP. This method of enablement allows users to provision tenants either in production or non-production environments.

i Note

The global accounts provided with a consumption-based account are different from the global accounts offered in the existing subscription model.

Visit the [blog](#) , to understand the steps involved for the Tenant Administrator to activate and provision a Cloud Integration service. Read the [FAQ Document](#)  that gives you more insights on consumption-based commercial model for SAP BTP.

The tenant administrator must perform the procedure mentioned here to enable the SAP Cloud Integration service.

Procedure

1. Log on to the SAP BTP Cockpit.
2. Choose **New Subaccount** to create a new subaccount.
3. Enter the details in the new subaccount dialog box and save. For more information about creating subaccounts, see [Create Subaccounts Using the Cockpit](#).

Enabling and Configuring a Tenant

Enable the Cloud Integration service for provisioning a tenant in Cloud Integration and use this procedure to activate enterprise messaging for JMS capabilities.

Context

The tenant administrator performs these tasks before provisioning a tenant to host services in Cloud Integration.

i Note

The following tasks are only required if you have purchased a consumption-based license model.

Procedure

1. From the newly created subaccount navigate to **Services** > **Service Marketplace** and then choose the **Process Integration** tile.
2. On the service page, choose **Create** for creating your instance or subscription.

i Note

Once you have enabled the service, a new subscription is created for the provision application.

3. The green **Enabled** badge appears.
4. On the service page, choose **Configure Process Integration** to configure and provision a tenant.
5. Select the required environment for the tenant.
6. Choose **Provision** and wait for a few minutes until the provisioning task is complete.

i Note

Consumption-based account holders must always create a new account to provision a new tenant. The tenant administrator cannot provision a new tenant for an existing account.

7. (Optional) Use the **Enterprise Messaging** service to design and deploy integration flows configured with JMS capabilities such as JMS and AS2 adapters. Perform the steps below to activate the service for processing messages asynchronously:
 - a. Go to the **Enterprise Messaging** tab, and choose **Activate**.
 - b. **Activate** to start the service and wait for a few minutes until the task is complete.
8. (Optional) You can choose to edit an existing tenant environment based on your operational needs.

i Note

For Non-Production Cloud Integration tenants, as per contract, billing will include only Cloud Integration instance charges and additional connections will not be billed. For more information, see [2784487](#) .

Next Steps

Once the tenant is activated, you need to assign the roles on the tenant before choosing **Go to Service**. For information about how to assign users and grant them the necessary user roles, see [Assigning Users and Roles](#).

i Posting Instructions

After you have assigned the user roles, launch a new browser instance to access the service page of the subaccount. Then choose **Go to Service** to access the tenant.

Related Information

[Managing Users and Role Assignments, Neo Environment](#)

Disabling a Tenant

Tenant administrators can disable a tenant in Cloud Integration if the user wants to discontinue the service.

Context

You can use the Cloud Integration Cockpit to disable tenants assigned to your consumption-based account.

i Note

You cannot disable a tenant if the provisioning process is still running.

Procedure

1. On the service overview page, choose **Disable** to disable the subaccount.

i Note

Disabling a tenant deletes all the data related to the tenant permanently, so you must be sure before you disable a tenant since this process is irreversible.

2. Choose **OK** to confirm the action.

i Note

If the process **fails**, please retry or create an incident as described in SAP KBA [2589823](#) .

Authorizing Users or User Groups

You need to authorize selected users or user groups to access the tenant and to perform specific tasks.

Prerequisites

- You have the Administrator role for the subaccount. You have the user IDs of the members that you want to add.
- Only users with a valid S-user or P-user ID can be added as members of the tenant.

Context

You perform these steps to authorize selected people to work on the account as part of the integration team, you assign roles to the associated users. You might also need to authorize technical users of sender systems to process messages on the tenant.

Once you have verified that you have administrator access and have added any additional administrators required, you can assign users or groups of users who will work on SAP Cloud Integration scenarios and grant them the necessary authorizations.

Furthermore, you can give permissions on two different detail levels:

- You can give permissions for a typical set of tasks associated with a persona such like an integration developer. In that case, you assign an authorization group to the user.

More information: [Personas](#)

- You can give the permission for an individual task such like monitoring messages. In that case, you assign an individual role to the user.

More information: [Tasks and Permissions](#)

i Note

An authorization group is composed of a set of individual roles.

You can give permissions either for individual users or for user groups.

Related Information

[Assigning Users and Roles](#)

[Assigning User Groups and Roles](#)

Assigning Users and Roles

Associate individual users with authorization groups or roles.

Context

The following procedure shows how to give permissions for an individual user.

Procedure

1. In SAP BTP cockpit, select your subaccount and then choose ► **Security** ► **Authorizations** ►.
 2. In field **User** enter the user ID and click **Show Assignments**.
- As expected, no roles are shown.
3. Click **Assign**.
 4. In dialog **Assign roles to user <user ID>** specify the following settings:

Select the relevant **Subaccount** and in **Application**, select the one that ends with tmn.

Assign roles to user



Subaccount: ▼

Application: tmn ▼

Role: AuthGroup.IntegrationDe ▼

Note: Changes will affect new sessions only.

Save

Cancel

As **Role** select the desired authorization group (which start with AuthGroup) or role and click **Save**.

For example, if you like to authorize the user to perform typical integration developer tasks, assign the authorization group **AuthGroup.IntegrationDeveloper**.

If you like to assign a role (for example, `ESBMessaging.send`) to authorize a (technical) user to process messages on the tenant, as **Application** you need to select the one that ends with `iflmap`.

5. Repeat this step for all user-to-role assignments you like to specify.

Assigning User Groups and Roles

Associate user groups with authorization groups or roles

Context

The following procedure shows how to give permissions for a user group.

Procedure

1. In SAP BTP Cockpit, select your subaccount and then choose **Security > Authorizations**.

2. On page **Authorization Management** choose the tab **Groups** >> **New Group**.
3. Enter a Group Name (for example, myGroup) and click **Save**.
4. Select the newly created group and under **Individual Users** assign users to the group. To assign a user, click **Assign**, enter the user Id and click **Save**.
5. Under **Roles** assign the authorization groups or roles to the group. Follow the same procedure as described under [Assigning Users and Roles](#) for step 4.

Authorization Management

UsersGroupsToken

Assign Web Roles and Users to Individual Groups

Groups (All: 3)

New Group

Name	Actions
myGroup	

Roles

Assign

Unassign All

Subaccount	Application	Role	Actions
	tmn	AuthGroup.Integrat...	Unassign

Individual Users

Assign

Unassign All

User

Adding New Administrators (Optional)

You specify all users who should get assigned the administrator role for the subaccount.

Prerequisites

You have the Administrator role for the subaccount. You have the user IDs of the members that you want to add.

Context

SAP grants administrator rights to the S-user ID specified in the order form. This user can grant administrator rights to other users in this subaccount.

Procedure

1. In the Cockpit, choose **Members**.



Platform Roles

Roles

473

Add Members

Enter up to 100 SAP user IDs. To separate them, you can use commas, spaces, semicolons, or line breaks.

User IDs:

Assign roles:

- ☐ Administrator (predefined role)
- ☒ Developer (predefined role)
- ☐ Support User (predefined role)
- ☐ Application User Admin (predefined role)
- ☐ Cloud Connector Admin (predefined role)
- ☐ SCCTuneller

Comment (optional, visible to all members):

Add Members

Cancel

Next Steps

- The **Cloud Connector Admin** role is not mandatory for all users and depends on your requirements. Check question 16 in the [Security FAQs](#). Also, you may not need the **Cloud Connector Admin** role during onboarding.
- If you have more than one tenant, you must add members to each tenant separately.
- For the latest documentation and detailed instructions on how to add members to an account, see [Adding Members to an Account](#).

Verifying Access for Users

Verify whether all the users that you have added to the subaccount have access to the SAP Cloud Integration application.

Procedure

- Launch the URL <https://xxxxx-tmn.hci.sa1.hana.ondemand.com/itspaces> that you will find in the welcome e-mail in a browser (Internet Explorer or Google Chrome).
- Enter your S-user or P-user ID and password to log on to the application.

i Note

If you are unable to verify access, perform the following steps:

If you get an authentication error or any other issues, please check that you have assigned the right role to the S- or P-user that you are verifying access for.

If you get an **Access Denied** error even though you have correctly assigned the required user roles, please check the SSO certificates in your browser. The browser might be using another user for the SSO logon instead of the S-user that you defined in the roles and authorizations.

If you are still facing issues, create a ticket using the component LOD-HCI-PI-MPL. The SAP Cloud Operations team will look into the issue and provide a solution.

Security FAQs

How can new users and authorizations be added once a customer gets the SAP Cloud Integration tenant? Who is authorized to add new users?

When SAP provides a tenant, administrator permissions are given to the S-user ID provided by the customer in the order form during contract signing. This administrative user can go to the SAP BTP cockpit and add additional users, and assign them roles and authorizations. Since SAP Cloud Integration uses SAP Cloud Identity provider by default, all the users must have valid S-user or P-user IDs.

You can also configure Cloud Integration to use your own custom identity provider. For more information, see [Using Custom IDP with SAP Cloud Integration](#).

Where can I find a list of all roles and authorizations that can be assigned to users?

More information:

[Personas](#)

Which recommendations are given for assigning roles to users?

The customer has full control on giving permissions to users on a tenant.

A key part of an integration project is the development and deployment of integration content (for example, integration flows). The related permissions are defined by the authorization group `AuthGroup.IntegrationDeveloper` and `AuthGroup.Administrator`. Note that this authorization group provides extensive permissions. Therefore, take into account special considerations when assigning this authorization group to a user.

More information:

[Identity and Access Management](#)

How can I contact SAP Cloud Integration Operations support for information or issues related to tenant provisioning and security?

Create a ticket on component LOD-HCI-PI-OPS.

Are CA-signed certificates mandatory for transport-level authentication? Which scenarios require CA-signed certificates?

More information:

[Transport Level Security](#)

Where can I find a list of CAs approved by SAP?

[Load Balancer Root Certificates Supported by SAP](#)

I want to use the same signed certificate for multiple systems. Can I put * in the Common Name field (for example, *.xxxxx.com) while the certificate is being signed by the CA? Does SAP allow this?

SAP recommends using the full host name in the Common Name (CN) field for both inbound and outbound scenarios, but technically does support the wildcard character in the CN field (for certificate-based client authentication only). For HTTPS outbound scenarios (where SAP manages the CA-signed key pairs), SAP uses the full host name in the CN field.

Can I use self-signed certificates for HTTPS certificate-based client authentication (also referred to as dual authentication)?

No, self-signed certificates are not supported for inbound connections to SAP Cloud Integration. For outbound connections, we recommend using a CA-signed base certificate.

Which scenarios support self-signed certificates? Can I use them for message-level encryption and signing?

You can use self-signed certificates for message-level encryption and signing. However, we recommend using CA-signed certificates.

Who maintains and manages the keystore? Can control be given to the end customer?

SAP provides some keys by default, but keystore management is now a self-service, so you can manage your keystore yourself.

More information:

[Managing Keystore Entries](#)

What is the procedure for using certificates for message-level encryption and signing?

You can use the certificates that are in the keystore provided by SAP during tenant provisioning. If you want to use your own key pair, you can manage it yourself using the self-service. There are different ways in which you can sign and encrypt message content (for example, PGP, X.509).

More information:

[Message Level Security](#).

Do I need to make any special requests when connecting to the SFTP/SMTP server?

The following ports are opened by default:

- For SFTP/SSH: port 22
- For SMTP: ports 25, 465, and 587

Do I need to make any special requests for HTTP(S) for outbound connectivity?

By default, port 443 and all HTTP ports 1024 and higher are opened.

Which IP addresses for the SAP Cloud Integration landscape do I need to configure in my own firewall for inbound connections (IP allowlisting)?

See [Virtual System Landscapes](#).

Where can I find details on SAP Data Centers and security?

You can find this information on the SAP website under SAP Data Centers Information.

More information: <https://www.sap.com/about/cloud-trust-center/data-center.html> 

What is SAP Cloud Connector? Is it mandatory?

SAP Cloud Connector is a complementary offering. It needs to be installed on premise and is an integral component of SAP BTP. It acts as a reverse proxy and creates a secure tunnel with the customer's own SAP Cloud Integration account. SAP Cloud Integration can route calls via SAP Cloud Connector for HTTP-based protocols (for example, SOAP, OData IDoc XMLs). SAP Cloud Connector is the preferred mode of communication for SAP BTP customers. However, it is not mandatory and customers can use other reverse proxy software (for example, Web Dispatcher).

More information:

[Outbound/On-Premise: Reverse Proxy or SAP Cloud Connector](#)

Related Information

[Connecting a Customer System to Cloud Integration](#)

Transport Level Security

SAP Cloud Integration Inbound Connection

Protocol	Related Adapters	Authentication Method	Required Certificates	Where to Get Required Certificates	CERT Usage in Customer Sender or Receiver Systems	Customer-CA Signed CERT Required?	CERT Usage in Cloud Integration Keystore
HTTPS	HTTP, SOAP, IDoc, OData and other HTTP based sender adapters	Basic Authentication	Root CA of SAP Cloud Integration/Load Balancer	You can use the self-service provided by SAP	Need to import Root CA of SAP Cloud Integration/Load Balancer in the backend system's key store	No	Not required Note: Users requiring basic authentication must have the role ESBMessaging.send role in SAP Cloud Integration tenant. It needs to be assigned on the IFLMAP node.

Protocol	Related Adapters	Authentication Method	Required Certificates	Where to Get Required Certificates	CERT Usage in Customer Sender or Receiver Systems	Customer-CA Signed CERT Required?	CERT Usage in Cloud Integration Keystore
HTTPS	HTTP, SOAP, IDoc, OData and other HTTP based sender adapters	Certificate based client authentication	Root CA of SAP Cloud Integration/Load Balancer	You can use the self-service provided by SAP	Need to import Root CA of SAP Cloud Integration/Load Balancer in the backend system's key store	No	Not required
			Public key for certificate based client authentication	Customer must generate a key pair using any tool, generate CSR (certificate signing request) and get it signed by CA. List of allowed CAs are mentioned in the operations guide.	Customer needs to import the signed key pair along with Root CA in their sender's system keystore.	Yes	Not Required Note: Customer needs to provide the public key of the signed CA client certificate in the integration flow configuration on sender system after selecting authentication type as certificate based.

Cloud Integration Outbound Connection

Protocol	Related Adapters	Authentication Method	Required Certificates	Where to Get Required Certificates	CERT Usage in Customer Sender or Receiver Systems	Customer-CA Signed CERT Required?	CERT Usage in Cloud Integration Keystore
HTTPS	HTTP, SOAP, IDoc, OData and other HTTP based sender adapters	Basic Authentication	Root and intermediate CAs of the customer	Root and intermediate CAs should be provided by the customer	Not required	Yes	The root and intermediate certificates of the CA approved certificate needs to be added to the SAP Cloud Integration keystore. You can use the self-service to add it to the keystore.

Protocol	Related Adapters	Authentication Method	Required Certificates	Where to Get Required Certificates	CERT Usage in Customer Sender or Receiver Systems	Customer-CA Signed CERT Required?	CERT Usage in Cloud Integration Keystore
							Note: Users needing basic authentication must be deployed as user credentials on SAP Cloud Integration and name of this credential should be specified in the respective technical adapter settings
HTTPS	HTTP, SOAP, IDoc, OData and other HTTP based sender adapters	Certificate based client authentication	Root and intermediate CAs of the customer	Root and intermediate CAs should be provided by the customer	Not required	Yes	The root and intermediate certificates of the CA approved certificate needs to be added to the SAP Cloud Integration keystore. You can use the self-service to add it to the keystore.
			SAP Cloud Integration Public Key for certificate based client authentication	You can use the self service to manage keystore.	Public Key (or client certificate should be imported in customer server's keystore. Root and intermediate certificate should be imported in the customer server trust keystore.	No (yes only if customer wants to use own key pair for client authentication)	SAP will generate the signed certificate and will upload it in the keystore of SAP Cloud Integration tenant (or will store the certificates provided by customer). Customer would need to mention the alias name of the certificate

Protocol	Related Adapters	Authentication Method	Required Certificates	Where to Get Required Certificates	CERT Usage in Customer Sender or Receiver Systems	Customer-CA Signed CERT Required?	CERT Usage in Cloud Integration Keystore
							in adapter settings.
HTTP	HTTP	Basic Authentication	NA	NA	NA	NA	NA
LDAP	LDAP	Simple Authentication	NA	NA	NA	NA	NA

Direction	Protocol	Related Adapters	Authentication Method	Required Certificates	Where to Get Required Certificates	CERT Usage in Customer Sender or Receiver Systems	Customer-CA Signed CERT Required?	CERT Usage in Cloud Integration Keystore
SAP Cloud Integration inbound/outbound	SSH	SFTP (Poll from SAP Cloud Integration)	Certificate based client authentication	Public key for certificate based client authentication	SAP generates a key pair and shares the public key with the customer. If you want to use your own key pair, you can use the self service to generate it and add it to the keystore.	You have to import/add this public key in designated location at SFTP server	Optional	SA tea ger pai an rsa in k wil on Inte ter key key procus
				Public key fingerprint of SFTP server	Public key fingerprint of SFTP server will be provided by SFTP administrator or SAP cloud ops team.		Optional	Pul SF mu me "kr file deq Clc Inte Clc Inte Cu mu to thi: doi clo

Direction	Protocol	Related Adapters	Authentication Method	Required Certificates	Where to Get Required Certificates	CERT Usage in Customer Sender or Receiver Systems	Customer-CA Signed CERT Required?	CE Clc Int Ke
SAP Cloud Integration Outbound	SMTP	Mail	Basic Authentication/CEAM-MD5	Root and intermediate CAs from the mail server for TLS	Root and intermediate CAs from the mail server for TLS	Not required	Yes	You ma key the ser

Initial Setup of SAP Cloud Integration in the Kyma Environment

In the Kyma environment, connect SAP Cloud Integration to your cluster and manage it using SAP BTP Service Operator.

As a prerequisite, you must have subscribed to the service [Process Integration](#) and the service instance [Process Integration Runtime](#) in the same SAP BTP subaccount.

For detailed information about how to connect SAP Cloud Integration to your cluster, refer to [Using SAP BTP Services in the Kyma Environment](#).

Environment-Specific Aspects Integration Developers Should Know

Cloud Integration is available in the following Cloud environments: Cloud Foundry and Neo. In both environments, Cloud Integration comprises - with a few exceptions - the same features for integration developers. For the exceptions, see SAP Note [2752867](#). However, disregarding these exceptions, some additional aspects that concern the setup of your integration scenarios still depend on the environment. This section provides you, the integration developer and/or tenant administrator, with an overview of the most important aspects that are specific to the environment where you run Cloud Integration.

i Note

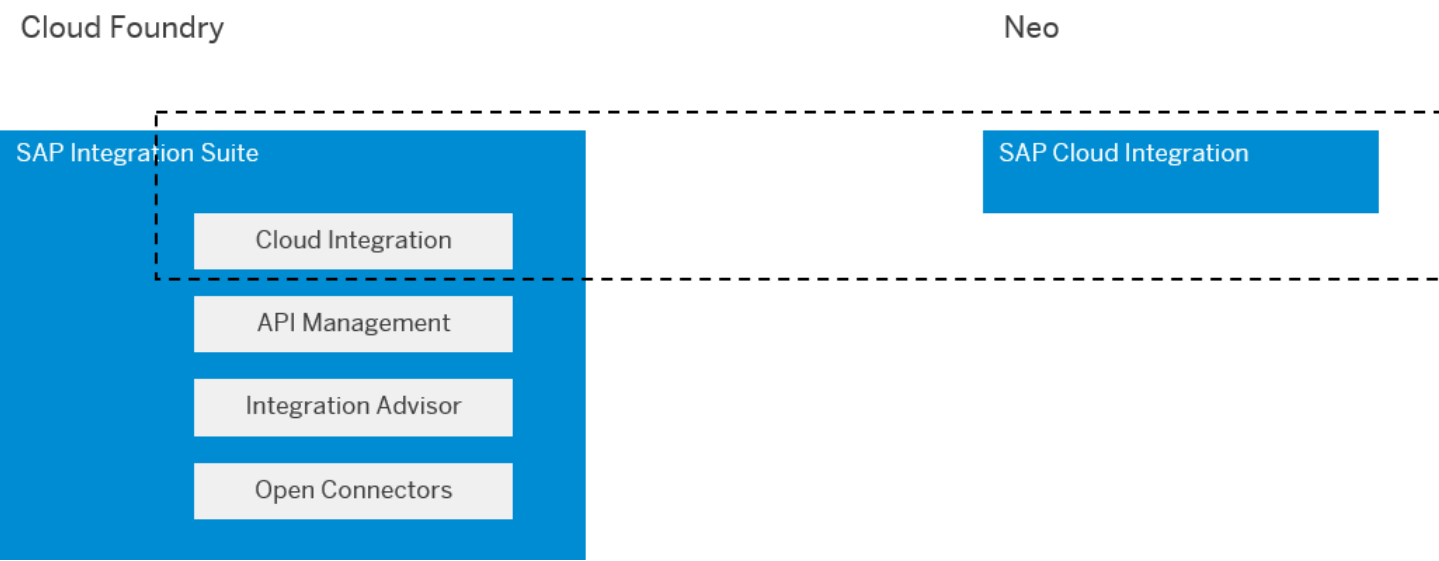
This information helps people who have operated Cloud Integration in the Neo environment and have migrated it to Cloud Foundry, for example.

Overview

Cloud Integration is operated in a cloud infrastructure: Physically, the software runs in data centers in different regions all over the world.

Environment	Product Availability	Infrastructure Provider
Cloud Foundry	Available as Cloud Integration capability of SAP Integration Suite. SAP Integration Suite comprises these capabilities: Cloud Integration, Integration Advisor, API Management, and Open Connectors. For more information on SAP Integration Suite, see SAP Integration Suite . SAP Integration Suite is also available as a trial version, see Welcome to SAP BTP Trial . To see how to set up SAP Integration Suite and how to use both capabilities Cloud Integration and API	Data centers of the cloud infrastructures of Amazon Web Services, Alibaba Cloud, and Microsoft Azure can be involved.

Environment	Product Availability	Infrastructure Provider
	Management together in a simple scenario, check out the scenario Request Product Details with an Integration Scenario👉. For Cloud Foundry, dedicated service plans are available. Depending on the service plan, some features used in Neo might not be available or only usable in a restricted way in Cloud Foundry. For more information, see SAP Note 2903776👉.	
Neo	Licensed as stand-alone service, SAP Cloud Integration.	The hardware that processes the messages is located exclusively in one, or multiple data centers owned by SAP.



i Note

In this section, the terms **SAP Cloud Integration** and **Cloud Integration** are used synonymously.

i Note

The component architecture differs depending on the cloud environment.

See:

- [Technical Landscape, Cloud Foundry Environment](#)
- [Technical Landscape, Neo Environment](#)

Connection Setup for Inbound Communication

Before starting with integration content design, you need to first set up inbound HTTP connections for the following use cases.

The ways how to configure inbound authentication differ in both environments because you need to enable sender applications to securely access certain resources on SAP BTP.

- Enabling sender systems to call integration flow endpoints.
- See: [Connection Setup for Inbound Communication - Integration Flow Endpoints](#)
- Enabling an API client to call the Cloud Integration OData API.

See: [Connection Setup for Inbound Communication - for API Clients](#)

In the Cloud Foundry environment, for the configuration of the most authentication options, you need to create an SAP BTP service instance. A service instance defines how external components can access a service (in this case, the Cloud Integration runtime or Cloud Integration resources exposed through the OData API) of SAP BTP. With a service instance, you define how to access a certain service or resource of SAP BTP, whereas the service key (generated by a service instance) contains the information required for a client to access the service (for example, credentials). Depending on whether you like to set up a connection to integration flow endpoints or to API resources, you need to specify **integration-flow** or **api** as **plan**. If you've operated Cloud Integration in the Neo environment, these concepts are likely new for you. The following SAP Community blog illustrates the configuration of the mentioned entities when addressing integration flow endpoints: [Integration Suite – Accessing Cloud Integration Runtime](#) .

User and Authorization Management

See: [User and Authorization Management](#)

API-Based Artifacts

In Neo, only the following API-based artifact type is available: OData API.

In Cloud Foundry, certain service plans are available that come with the following API-based artifact types: OData API, REST API, and SOAP API (see also: SAP Note [2903776](#) .

Setting Up Content Transport

See: [Setting Up Content Transport](#)

Developing Custom Adapters

See: [Adapter Development](#)

Audit Logging

See: [Audit Logging](#)

System Scope

Cloud Integration comes with various options to temporarily store data. The system limits depend on the environment.

See: [What Is SAP Cloud Integration?](#)

Connection Setup for Inbound Communication - Integration Flow Endpoints

When configuring secure inbound HTTP connections, different authentication options can be used for the sender authenticate itself against Cloud Integration.

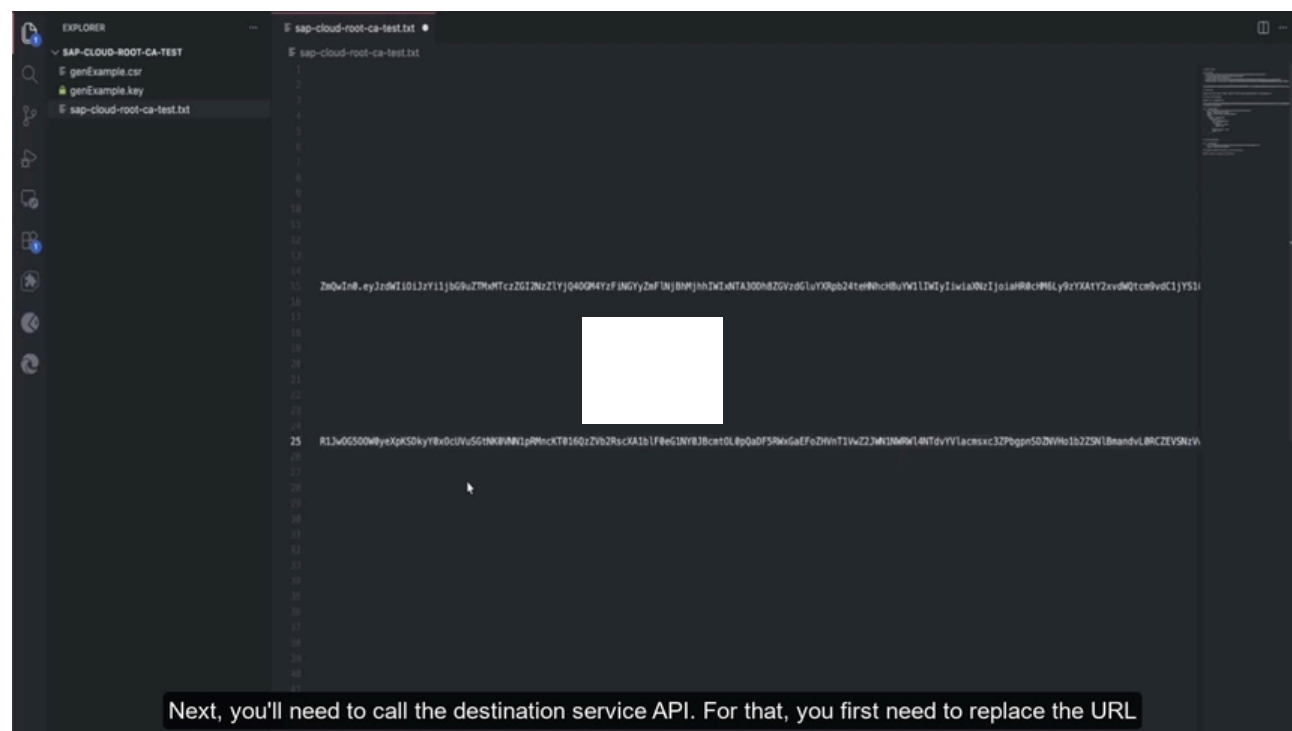
Cloud Foundry Environment

The following table provides an overview of the available authentication options and a summary of the configuration steps in the Cloud Foundry environment (for the connection to integration flow endpoints). The table provides a brief summary to indicate the key aspects. Note that the most secure/recommended options are listed on top in the table.

Note that the following description doesn't contain aspects that are common in both environments such like the role of the load balancer and the required security settings with regard to this component.

i Note

For **Client Certificate Authentication**, ensure that you have the required service keys and access configured. To generate SAP Cloud Root CA based certificates via the Destination Service REST API using a locally generated CSR (Certificate Signing Request), see [Manage Certificates Issued by the SAP Cloud Root CA](#) and the video below.



[Open this video in a new window](#)

Authentication Option	Configuration (Summary)	How it Works
Client certificate authentication	Go to SAP BTP cockpit and define a service key for the Process Integration service and integration-flow plan. When defining the service instance, specify the role that is to be used to grant access to the integration flow endpoint (you can either use the predefined role <code>ESBMessaging.send</code> or a custom role). When defining a service key for the service instance, enter the client certificate (public key) used by the sender to authenticate itself against Cloud Integration. i Note The client certificate is validated at the SAP BTP platform load balancer before the request reaches the Cloud Integration runtime. The certificate must	System checks if a service key is available that contains the client certificate provided by the sender. If a service key is available, the system then checks if the associated service instance has a role specified that grants permissions to call the integration flow endpoint.

Authentication Option	Configuration (Summary)	How it Works
	include the Client Authentication Extended Key Usage (EKU) with OID 1.3.6.1.5.5.7.3.2. Certificates without this EKU are rejected during the TLS handshake. Ensure that the required EKU is present while renewing certificates from public Certificate Authorities. Some public CAs no longer include this EKU in newly issued TLS certificates. Use certificates issued via SAP Cloud Root CA (for example via SAP BTP Destination Service) or configure a custom domain with a customer-managed certificate trust model. → Remember The maximum validity period for TLS Certificate 🖱️ is being reduced in phases: • Until March 15, 2026: Maximum validity is 398 days. • From March 15, 2026: Maximum validity is 200 days. • From March 15, 2027: Maximum validity is 100 days. • From March 15, 2029: Maximum validity is 47 days. As a result, server certificates are renewed more frequently. Configure your HTTP(S) clients to trust the appropriate Root CA instead of relying on individual server certificates. See TLS Certificate Lifetimes Will Officially Reduce to 47 Days DigiCert 🖱️ and Certificate Lifetime - Custom Domains 🖱️ i Note SAP Cloud Integration periodically renews load balancer server certificates as part of standard security maintenance. Configure your HTTP(S) clients to trust the appropriate Root CA instead of relying on a specific server certificate, intermediate certificate, or certificate chain. This allows server certificate renewals without requiring	

Authentication Option	Configuration (Summary)	How it Works
	customer-side trust configuration changes. You will be notified only if a change to the trusted Root CA requires updates to their trust configuration. See 3478796  Public CAs started excluding the Client Authentication EKU from newly issued certificates in September 2025. By May 2026 it may no longer be included by default. i Note Ensure that you have the required service keys and access configured. To generate SAP Cloud Root CA based certificates via the Destination Service REST API using a locally generated CSR (Certificate Signing Request), see Manage Certificates Issued by the SAP Cloud Root CA See: • Client Certificate Authentication for Integration Flow Processing	
OAuth	Go to SAP BTP cockpit and define a service key for the Process Integration service and integration-flow plan. Specify the role that is to be used to grant access to the integration flow endpoint. The generated service key contains the following properties: <code>clientid</code>, <code>clientsecret</code>, and <code>tokenurl</code>. See: OAuth with Client Credentials Grant for Integration Flow Processing	For the client credentials grant variant of OAuth, authentication at runtime comprises two HTTP requests: 1. In a first request (addressed to the token service addressed by the <code>tokenurl</code>), the sender provides <code>clientid</code> and <code>clientsecret</code> and gets back from the token service an access token. 2. In a second request (addressed to the integration flow endpoint), the sender provides the access token and gets access to the integration flow. Other grant types can be configured as well using the service key information).
Basic authentication (associated with an OAuth client)	Perform the same steps as for OAuth. See: Basic Authentication with clientid and clientsecret for Integration Flow Processing	In just one request, the sender uses <code>clientid</code> and <code>clientsecret</code> as user credentials to directly access the integration flow endpoint without the need to request an access token first.
Basic authentication of a user registered at an identity provider (IdP) (this option isn't	Register a user at an identity provider (for example, SAP's default identity provider SAP ID Service). Using SAP BTP cockpit,	With username and password (known to the identity provider), the sender can call the integration flow endpoint.

Authentication Option	Configuration (Summary)	How it Works
considered to be secure enough for productive scenarios)	you assign to the user a role that grants permission to call integration flow endpoints. See: Basic Authentication of IdP User for Integration Flow Processing	

When defining a service key, the required role needs to be added in JSON representation. You can get the JSON representation of the role from the Cloud Integration **Monitor** section (under **Manage Security** in the **User Roles** tile).

Neo Environment

The following table provides an overview of the available authentication options and a summary of the configuration steps in the Neo environment (for the connection to integration flow endpoints). The table provides a brief summary to indicate the key aspects. Note that the most secure/recommended options are listed on top in the table.

Note that the following description doesn't contain aspects that are common in both environments such like the role of the load balancer and the required security settings with regard to this component.

Authentication Option	Configuration (Summary)	How it Works
Client certificate authentication with certificate-to-user mapping	Create and deploy a Certificate-to-User Mapping artifact on the Cloud Integration tenant. This artifact relates a user with a client certificate (used by the sender to authenticate itself against Cloud Integration when calling an integration flow). When defining the integration flow (sender adapter), for Authorization select User Role and specify role to be used to grant access to the integration flow endpoint. i Note The client certificate is validated at the SAP BTP platform load balancer before the request reaches the Cloud Integration runtime. The certificate must include the Client Authentication Extended Key Usage (EKU) with OID 1.3.6.1.5.5.7.3.2. Certificates without this ECU are rejected during the TLS handshake. Ensure that the required ECU is present while renewing certificates from public Certificate Authorities. Some public CAs no longer include this ECU in newly issued TLS certificates. Use certificates issued via SAP Cloud Root CA (for example via SAP BTP Destination Service) or configure a custom domain with a customer-managed certificate trust model.	System checks if a Certificate-to-User Mapping artifact exists that fits to the client certificate provided by the sender. It checks if the associated user has the required permission to call the integration flow.

Authentication Option	Configuration (Summary)	How it Works
	Public CAs started excluding the Client Authentication EKU from newly issued certificates in September 2025. By May 2026 it may no longer be included by default. Go to SAP BTP cockpit and assign to the user this role. See: • Setting Up Inbound HTTP Connections (with Certificate-to-User Mapping), Neo Environment	
Client certificate authentication (no certificate-to-user mapping) This option is secure but, compared to the usage of certificate-to-user mapping, not recommended. The reason: As the certificate is specified as part of the integration flow, each certificate change requires a redeployment of the integration flow. A downtime of the integration flow is the consequence.	When defining the integration flow (sender adapter), for Authorization select Client Certificate and provide the client certificate that is to be used by the sender when calling the integration flow endpoint. See: • Setting Up Inbound HTTP Connections (with Client Certificate Authentication), Neo Environment	System checks if client certificate provided by the sender is associated with integration flow endpoint. Furthermore, system checks the permissions of the sender by evaluating the certificate's subject/issuer distinguished name.
OAuth	Go to SAP BTP cockpit and define an OAuth client. For Authorization Grant, select the option Client Credentials. On saving, a client ID and secret is generated. It's recommended to use the option to get a JSON Web Token (JWT) as access token. Assign to user <code>oauth_client_<client ID></code> a role that grants access to the integration flow (<code>ESBMessaging.send</code> or a custom role). See: • Setting Up Inbound HTTP Connections (with OAuth), Neo Environment	For the client credentials grant variant of OAuth, authentication at runtime comprises two HTTP requests: 1. In a first request (addressed to the token service), the sender provides ID and Secret from the OAuth client and gets back from the token service an access token. The URL of the token service can be found in the Branding tab of the OAuth client. 2. In a second request (addressed to the integration flow endpoint), the sender provides the access token and gets access to the integration flow. Other grant types can be configured as well.
Basic authentication of a user registered at an identity provider (IdP) (this option isn't considered to be secure enough for productive scenarios)	Register a user at an identity provider (for example, SAP's default identity provider SAP ID Service). Using SAP BTP cockpit, you assign to the user a role template that grants permission to call integration flow endpoints. See: Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment	With username and password (known to the identity provider), the sender can call the integration flow endpoint.

Connection Setup for Inbound Communication - for API Clients

The Cloud Integration OData API allows you to access Cloud Integration resources (for example, integration content, Partner Directory content, or message processing logs) through an API. Using API-based access, you can write programs that process Cloud Integration resources. When configuring secure inbound HTTP connections, different authentication options can be used to authenticate API clients against the Cloud Integration OData API.

For more information on the OData API, see [OData API](#).

Cloud Foundry Environment

The following table provides an overview of the available authentication options and a summary of the configuration steps in the Cloud Foundry environment (for the connection to API resources). The table provides a brief summary to indicate the key aspects. Note that the most secure/recommended options are on top, the less secure ones further below in the table.

Authentication Option	Configuration (Summary)	How it Works
OAuth	Go to SAP BTP cockpit and define a service key for the Process Integration service and api plan. Specify the role that is to be used to grant access to the Cloud Integration resource. For example, if you like to access message processing logs through the OData API, you need to specify role template <code>MonitoringDataRead</code>. The generated service key contains the following properties: <code>clientid</code>, <code>clientsecret</code>, and <code>tokenurl</code>. See: OAuth with Client Credentials Grant for API Clients	For the client credentials grant variant of OAuth, authentication at runtime comprises two HTTP requests: 1. In a first request (addressed to the token service reachable by the <code>tokenurl</code>), the API client provides <code>clientid</code> and <code>clientsecret</code> and gets back from the token service an access token. 2. In a second request (addressed to the Cloud Integration resource), the API client provides the access token and gets access to the Cloud Integration resource. For modifying calls, an CSRF-Token is required.
Basic authentication of a user registered at an identity provider (IdP) (this option isn't considered to be secure enough for productive scenarios)	Register a user at an identity provider (for example, SAP's default identity provider SAP ID Service). Using SAP BTP cockpit, you assign to the user a role template to be used to grant permission to access to the Cloud Integration resource (for example, <code>MonitoringDataRead</code> when you like to access message processing logs). See: Basic Authentication of an IdP User for API Clients	With username and password (known to the identity provider), the API client can access the Cloud Integration resource. For modifying calls, an CSRF-Token is required.

Neo Environment

The following table provides an overview of the available authentication options and a summary of the configuration steps in the Neo environment (for the connection to API resources). The table provides a brief summary to indicate the key aspects. Note that the most secure/recommended options are on top, the less secure ones further below in the table.

Authentication Option	Configuration (Summary)	How it Works
OAuth	Go to SAP BTP cockpit and define an OAuth client. For Authorization Grant select the option Client Credentials. On saving, a client ID and secret is generated. Assign to user <code>oauth_client_<client ID></code> a role that grants access to the Cloud Integration resource. See: Setting Up OAuth Inbound Authentication with Client Credentials Grant for API Clients	For the client credentials grant variant of OAuth, authentication at runtime comprises two HTTP requests: 1. In a first request (addressed to a service with an URL as described at Setting Up OAuth Inbound Authentication with Client Credentials Grant for API Clients), the API client provides ID and Secret from the OAuth client and gets back an access token. 2. In a second request (addressed to the Cloud Integration resource), the sender provides the access token and gets access to the Cloud Integration resource. For modifying calls, an CSRF-Token is required.
Basic authentication of a user registered at an identity provider (IdP) (this option isn't considered to be secure enough for productive scenarios)	Register a user at an identity provider (for example, SAP's default identity provider SAP ID Service). Using SAP BTP cockpit, you assign to the user a role template to be used to grant permission to access to the Cloud Integration resource. See: Setting Up Inbound Authentication of an IdP User for API Clients	With username and password (known to the identity provider), the API client can access the Cloud Integration resource. For modifying calls, an CSRF-Token is required.

IP Allowlisting for Inbound Requests

To protect Cloud Integration against unwanted inbound requests, you can configure IP allowlisting.

→ Tip

IP allowlisting is a security measure that restricts access to a system based on a predefined list of approved IP addresses. Only connections originating from these approved IP addresses are allowed, all others are blocked.

Cloud Foundry Environment

Currently, you need to configure IP allowlisting by integration flow design. The following SAP Community blog provides a solution using the header `x-forwarded-for` in a script that blocks requests from a particular IP address: [IP Allowlisting in SAP Cloud Integration](#) .

Neo Environment

Configure IP allowlisting on platform level by setting up custom domains as explained under [Configuring Custom Domains](#).

User and Authorization Management

When defining permissions for sending applications to access Cloud Integration (either integration flow endpoints or Cloud Integration resources through the OData API), you need to deal with roles.

A typical concept is to group permissions for individual activities along personas - fictitious persons that are associated with typical task areas. For example, the integration developer persona is in charge of tasks such like integration flow development and deployment and of monitoring message processing.

- In Cloud Foundry, there are **role collections** that contain all roles associated with a dedicated persona. An individual role can be defined based on predefined role templates available on SAP BTP.
- In Neo, there are **authorization groups** that contain all roles associated with a dedicated persona.

For more information and a comparison of Cloud Foundry / Neo entities, check out the topics:

- [Personas](#)
- [Tasks and Permissions](#)

Managing users and permissions is done differently in Cloud Foundry and in Neo.

- Cloud Foundry: [SAP Authorization and Trust Management Service in the Cloud Foundry Environment](#)
- Neo: [User Management for SAP BTP, Neo Environment](#)

Due to the difference in the user and permission management, also the configuration of access policies is different in both environments:

- Cloud Foundry: [Creating Custom Roles for Access Policies, Cloud Foundry Environment](#)
- Neo: [Creating Custom Roles for Access Policies, Neo Environment](#)

Related Information

[Identity and Access Management](#)

Setting Up Content Transport

For the transport of integration content across different tenants, different options are available:

- Manual export and import
- Usage of CTS+
- Usage of the cloud-based Transport Management

These options are identical independent of the environment (Cloud Foundry or Neo). However, setting up transport management using the cloud-based Transport Management service is different in both environments. Here's a brief summary of the differences.

- In the Cloud Foundry environment: In order to set up content transport with Transport Management Service, you need to activate the Content Agent service. Furthermore, you need to define various destinations between source and target tenant of the content transport and Content Agent. For more information, see:

[Enabling Content Transport, Cloud Foundry Environment](#)

[Introducing SAP Content Agent service: Enhanced Transport Capabilities for SAP Cloud Integration Content](#) 

- In the Neo environment: In order to set up content transport with Transport Management Service, you need to activate Lifecycle Management service. Furthermore, you need to define various destinations between source and target tenant of the content transport and Lifecycle Management. For more information, see:

[Enabling Content Transport, Neo Environment](#)

[Cloud Integration – Using Transport Management Service for a Simple Transport Landscape](#) 

Adapter Development

For the development of custom adapters, you need to use Eclipse (independent of the environment).

- In the Cloud Foundry environment, you can then deploy and manage the custom adapter as **Integration Adapter** artifact using the Cloud Integration **Design** and **Monitor** application.
- In the Neo environment, you deploy the adapter using Eclipse.

See: [Developing Custom Adapters](#)

Audit Logging

Audit log features are different in both environments:

- In the Cloud Foundry environment, you can use SAP Audit Log service as described under [Audit Logging in the Cloud Foundry Environment](#).
- In the Neo environment, you can use the audit log retrieval API as described under [Audit Logging in the Neo Environment](#).

Additionally, you can use the Cloud Integration **Monitor** application to find audit log information. Choose a tile in section **Access Logs** (see [Access Logs, Neo Environment](#)).

Environment Variables

You can use environment variables in integration flows to address technical details such like, for example, the region where Cloud Integration is deployed.

i Note

Example use case:

You can get the tenant name from an environment variable and use this information to specify a tenant-specific integration flow configuration. For example, you can specify different backend URLs for test and prod tenant, respectively.

There's a difference of these variables depending on whether you run Cloud Integration in the Neo or Cloud Foundry environment. Because of that, when you migrate from Neo to Cloud Foundry, you need to adapt all environment variables used in your integration content as a post-migration step.

The following table provides the mapping of environment variables in Neo and Cloud Foundry.

Variable Name (Neo)	Variable Name (Cloud Foundry)	Description
HC_APPLICATION Example value: abcd01iflmap	TENANT_NAME Example value: xyz001	Sub domain of worker application (associated with application identifier for worker node)
HC_APPLICATION_URL Example value: abcd01iflmap.uvwxy.eu1.hana.ondemand.com	TENANT_NAME + IT_SYSTEM_ID + IT_TENANT_UX_DOMAIN Example value: xyz001.it-cpi001.cfapps.eu10.hana.ondemand.com	URL of the worker application sub domain
HC_HOST Example value: hana.ondemand.com	IT_TENANT_UX_DOMAIN Example value: cfapps.eu10.hana.ondemand.com	Base URL of the SAP BTP region host where the application is deployed
HC_LOCAL_HTTP_PORT Example value: 9001	PORT Example value: 8080	HTTP port of the application bound to localhost
HC_OP_HTTP_PROXY_HOST	VCAP_SERVICES	Host of the HTTP Proxy for on-premise connectivity
HC_OP_HTTP_PROXY_PORT Example value: 20003	VCAP_SERVICES Example value: 20003	Port of the HTTP Proxy for on-premise connectivity
HC_REGION Example value: EU_1	IT_TENANT_UX_DOMAIN Example value: cfapps.eu10.hana.ondemand.com	Region where the application is deployed

For more information, refer to:

- [Using Cloud Environment Variables](#) (Neo)
- [Cloud Foundry Environment Variables](#) ➦

Connecting a Customer System to Cloud Integration

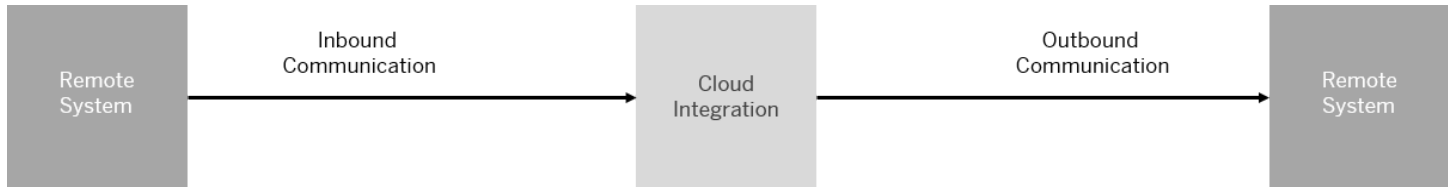
You can set up the technical connection between a tenant and different kinds of remote systems (in many cases located in the customer landscape).

Throughout this documentation we assume the following basic setup of technical components and communication paths: A remote system (which is not specified) is being connected to one of the tenants that are assigned to the customer. The remote system can act either as a sender or a receiver of messages. The setup and the detailed configuration procedure differ according to the communication direction that is being set up: whether a remote system is supposed to send a message to the integration platform or the other way round.

Throughout this documentation, the terms *inbound* and *outbound* reflect the perspective of the integration platform.

- **Inbound** refers to message processing from a remote system (in many cases, located in the customer landscape) to Cloud Integration. Here, the integration platform is the server.

- **Outbound** refers to message processing from the integration platform to a remote system (where the integration platform is the client).



Related Information

[Introduction](#)

[Configuring Inbound Communication](#)

[Configuring Outbound Communication](#)

[Setting up Message-Level Security Use Cases](#)

[Concepts of Secure Communication](#)

Introduction

You can connect various kinds of remote systems to the cloud-based integration platform using protocols such as HTTP/S, SSH and SMTP/S. Each communication protocol comes with certain options to protect the message exchange (security options).

Kind of Systems to Connect to Cloud Integration

To give you an idea of which **kinds of remote systems** can be connected to the integration platform, here are some typical examples (this is not a complete list):

- On-premise systems, for example, SAP systems based on SAP NetWeaver
- SFTP servers
- Cloud applications, for example, SAP SuccessFactors or SAP Cloud for Customer
- Other systems such as e-mail servers or SOAP clients

Depending on the kind of system to connect, a certain communication **protocol** is to be considered, as will be explained in the next section.

To support dedicated kinds of systems (through dedicated communication protocols), the integration platform provides certain **adapters**. An adapter allows you to configure the details of the technical communication channel between the remote system and the integration platform.

Supported Protocols

First task when setting up an integration scenario is to set up a secure transport channel between the remote system and Cloud Integration. The following protocols can be used: Hypertext Transfer Protocol Secure (HTTPS), SSH File Transfer Protocol (SFTP) and Simple Mail Transfer Protocol (SMTP), respectively SMTP secured with transport layer security (SMTPS).

i Note

Note that HTTPS is based on the Transport Layer Security (TLS) protocol.

The following table provides more information on the different aspects to consider for each protocol.

Protocols

Protocol	Call Direction	On Premise (Mandatory)	On Premise (Recommended)	Further Aspects to Consider
HTTP, HTTPS	Inbound	HTTP/S sender system (for example, SAP ERP Central Component)	HTTP/S proxy	Firewall to set up and configure
HTTP, HTTPS	Outbound	HTTP/S receiver system (for example, SAP ERP Central Component)	Web Dispatcher OR SAP Cloud Connector	Firewall to set up and configure
SSH	Outbound	SFTP server (to store files)	Tooling for ssh key management	Virus scanner on inbound directory
SMTP, SMTPS	Outbound	Mail server	SMTPS (SMTP over SSL/TLS) support of mail server	Virus scanner on inbound mail boxes

For each protocol, different authentication options are supported - ways how the connected systems prove their trustworthiness against each other during connection setup. Connection setup is performed differently, depending on whether inbound communication (when a remote system as a sender calls Cloud Integration) or outbound communication (when Cloud Integration calls a remote system which, in turn, is then considered as the receiver) is configured. The detailed procedure also depends on the chosen protocol and authentication option.

i Note

Basic authentication is recommended for test purposes only. For productive scenarios, we recommend that you use *certificate-based authentication*.

Adapters

The following figure illustrates some options for kinds of systems to connect to Cloud Integration. Both communication directions are considered: systems sending messages to Cloud Integration and systems that receive messages from Cloud Integration. The figure also shows which communication protocols and the Cloud Integration adapters that are to be configured in order to enable Cloud Integration to connect to the respective kind of system. Note that the figure only shows some typical use cases and is not complete.

i Note

Adapters exchange data with remote components that might be outside the scope of SAP. Make sure that the data exchange complies with your company's policies.

The following table lists the available adapters:

Adapter

Feature	Description
AmazonWebServices	Connects SAP Cloud Integration to Amazon Web Services.
Sender adapter	The adapter supports the following protocols: • S3: Simple Cloud Storage • SQS: Simple Queue Service

Feature	Description
	See: AmazonWebServices Sender Adapter
AmazonWebServices Receiver adapter	Connects SAP Cloud Integration to Amazon Web Services. The adapter supports the following protocols: • S3: Simple Cloud Storage • SQS: Simple Queue Service • SNS: Simple Notification Service • SWF: Simple Workflow Service See: AmazonWebServices Receiver Adapter
AMQP Sender adapter	Enables SAP Cloud Integration to consume messages from queues or topic subscriptions in an external messaging system. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocols: TCP, WebSocket See: Configure the AMQP Sender Adapter
AMQP Receiver adapter	Enables SAP Cloud Integration to send messages to queues or topics in an external messaging system. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocols: TCP, WebSocket See: Configure the AMQP Receiver Adapter
AMQP for SAP Event Mesh Sender adapter	Enables SAP Cloud Integration to consume messages from SAP Event Mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: WebSocket See: AMQP Sender for SAP Event Mesh
AMQP for SAP Event Mesh Receiver adapter	Enables SAP Cloud Integration to send messages to SAP Event Mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: WebSocket See: AMQP Receiver for SAP Event Mesh
AMQP for SAP Integration Suite, advanced event mesh Sender adapter	Enables SAP Cloud Integration to consume messages from SAP Integration Suite, advanced event mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: AMQ Sender for SAP Integration Suite, advanced event mesh
AMQP for SAP Integration Suite, advanced event mesh Receiver adapter	Enables SAP Cloud Integration to send messages to SAP Integration Suite, advanced event mesh. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: AMQP Receiver for SAP Integration Suite, advanced event mesh
AMQP for Microsoft Azure Service Bus	Enables SAP Cloud Integration to consume messages from Microsoft Azure Service Bus.

Feature	Description
Sender adapter	Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Sender Adapter
AMQP for Microsoft Azure Service Bus Receiver adapter	Enables SAP Cloud Integration to send messages to Microsoft Azure Service Bus. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter
AMQP for Solace PubSub+ Sender adapter	Enables SAP Cloud Integration to consume messages from Solace PubSub+. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Sender Adapter
AMQP for Solace PubSub+ Receiver adapter	Enables SAP Cloud Integration to send messages to Solace PubSub+. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter
AMQP for Apache Qpid Broker-J Sender adapter	Enables SAP Cloud Integration to consume messages from Apache Qpid Broker-J. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP, WebSocket See: Configure the AMQP Sender Adapter
AMQP for Apache Qpid Broker-J Receiver adapter	Enables SAP Cloud Integration to send messages to Apache Qpid Broker-J. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP, WebSocket See: Configure the AMQP Receiver Adapter
AMQP for Apache ActiveMQ 5 / Apache ActiveMQ Artemis Sender adapter	Enables SAP Cloud Integration to consume messages from Apache ActiveMQ 5 / Apache ActiveMQ Artemis. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Sender Adapter
AMQP for Apache ActiveMQ 5 / Apache ActiveMQ Artemis Receiver adapter	Enables SAP Cloud Integration to send messages to Apache ActiveMQ 5 / Apache ActiveMQ Artemis. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter
AMQP for IBM MQ Sender adapter	Enables SAP Cloud Integration to consume messages from IBM MQ. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP

Feature	Description
	See: Configure the AMQP Sender Adapter
AMQP for IBM MQ Receiver adapter	Enables SAP Cloud Integration to send messages to IBM MQ. Supported message protocol: AMQP (Advanced Message Queuing Protocol) 1.0 Supported transport protocol: TCP See: Configure the AMQP Receiver Adapter
Ariba Sender adapter	Connects SAP Cloud Integration to the Ariba Network. Using this adapter, SAP and non-SAP cloud applications can receive business-specific documents in commerce eXtensible Markup Language (cXML) format from the Ariba network. The sender adapter allows you to define a schedule for polling data from Ariba. See: Configure the Ariba Sender Adapter
Ariba Receiver adapter	Connects SAP Cloud Integration to the Ariba network. Using this adapter, SAP and non-SAP cloud applications can send business-specific documents in commerce eXtensible Markup Language (cXML) format to the Ariba network. Receiver adapter See: Configure the Ariba Receiver Adapter
AS2 Sender adapter	Enables SAP Cloud Integration to exchange business-specific documents with a partner through the Applicability Statement 2 (AS2) protocol. Sender adapter: Can return an electronic receipt to the sender of the AS2 message (in the form of a Message Disposition Notification (MDN)) See: Configure the AS2 Sender Adapter
AS2 Receiver adapter	Enables SAP Cloud Integration to exchange business-specific documents with a partner through the Applicability Statement 2 (AS2) protocol. See: Configure the AS2 Receiver Adapter
AS4 Sender adapter	Enables SAP Cloud Integration to securely process incoming AS4 messages using Web Services. The AS4 sender adapter is based on the ebMS 3.0 specification that supports the ebMS handler conformance profile. • Supports one-way/ebMS3 push message exchange pattern (MEP). • Support on-way/ebMS3 pull that allows the message party to pick the corresponding message from the partner. • Supports signature verification and decryption of the message. • Generates receipts after processing the incoming AS4 message. • Allows you to set a size limit for the body and attachment of an incoming message. See: AS4 Sender Adapter
AS4 Receiver adapter	Enables SAP Cloud Integration to establish a connection between any two message service handlers (MSHs) for exchanging business documents. The AS4 receiver adapter uses the Light Client conformance policy and supports only message pushing for the sending MSH and selective message pulling for the receiving MSH. Receiver adapter: • Supports one-way/push message exchange pattern (MEP) that involves the transfer of business documents from a sending MSH to a receiving MSH. • Supports one-way/selective-pull message exchange pattern (MEP) that involves the receiving MSH initiating a selective pull request to the sending MSH. The sending MSH responds by sending the specific user message.

Feature	Description
	Supports storing and verification of receipts. See: AS4 Receiver Adapter
BigCommerce GraphQL Receiver adapter	Connects SAP Integration Suite tenant to a BigCommerce GraphQL Admin APIs using the GraphQL message protocol. The GraphQL message protocol is used to connect to the GraphQL-based Admin APIs of a BigCommerce store. See
BigCommerce REST Receiver adapter	Connects SAP Integration Suite tenant to a BigCommerce REST Admin APIs, using the REST message protocol. The REST message protocol is used to connect to the REST-based Admin APIs of a BigCommerce store. See
Data Store Sender adapter	Enables SAP Cloud Integration to consume messages from a data store. See: Data Store Sender Adapter
ELSTER Receiver adapter	Enables SAP Cloud Integration to send a tax document to the ELSTER server. ELSTER (acronym for the German term <i>Elektronische Steuererklärung</i>) is used in German fiscal management to process tax declarations exchanged over the Internet. The adapter supports the following operations: Getting the version of the ERiC (ELSTER Rich Client) library, validating a tax document, and sending a tax document. See: ELSTER Receiver Adapter
Facebook Receiver adapter	Enables SAP Cloud Integration to access and extract information from Facebook based on certain criteria such as keywords or user data. Using OAuth, the SAP BTP tenant can access resources on Facebook on behalf of a Facebook user. See: Facebook Receiver Adapter
FTP Sender adapter	Enables SAP Cloud Integration to connect to a remote system using TCP (Transmission Control Protocol) to receive files from the system. FTP stands for File Transfer Protocol. The sender adapter allows you to define a schedule for polling data from the connected system. See: Configure the FTP Sender Adapter
FTP Receiver adapter	Enables SAP Cloud Integration to connect to a remote system using TCP (Transmission Control Protocol) to write files to the system. FTP stands for File Transfer Protocol. See: Configure the FTP Receiver Adapter
HTTPS Sender adapter	Establishes an HTTPS connection between SAP Cloud Integration and a sender system. See: HTTPS Sender Adapter
HTTP Receiver adapter	Establishes an HTTP connection between SAP Cloud Integration and a receiver system. Receiver adapter: - Supports HTTP 1.1 only (target system must support chunked transfer encoding and may not rely on the existence of the HTTP Content-Length header) - Supports the following methods: DELETE, GET, HEAD, POST, PUT, TRACE

Feature	Description
	Method can also be determined dynamically by reading a value from a message header or property during runtime. See: HTTP Receiver Adapter
IDoc Sender adapter	Allows SAP Cloud Integration to exchange Intermediate Document (IDoc) messages with a sender system that supports communication via SOAP Web services. A size limit for the inbound message can be configured for the sender adapter. See: Configure the IDoc Sender Adapter
IDoc Receiver adapter	Allows SAP Cloud Integration to exchange Intermediate Document (IDoc) messages with a receiver system that supports communication via SOAP Web services. See: Configure the IDoc Receiver Adapter
JDBC Receiver adapter	Allows SAP Cloud Integration to connect to a JDBC (Java Database Connectivity) database and to execute SQL commands on the database. See: JDBC Receiver Adapter
JDBC for DB2 (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to DB2 (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for DB2 (On-Premise)
JDBC for Microsoft SQL Server (Cloud) Receiver adapter	Allows SAP Cloud Integration to connect to Microsoft SQL Server (Cloud) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Microsoft SQL Server (Cloud)
JDBC for Microsoft SQL Server (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to Microsoft SQL Server (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Microsoft SQL Server (On-Premise)
JDBC for Oracle (Cloud) Receiver adapter	Allows SAP Cloud Integration to connect to Oracle (Cloud) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Oracle (Cloud)
JDBC for Oracle (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to Oracle (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for Oracle (On-Premise)
JDBC for PostgreSQL (Cloud) Receiver adapter	Allows SAP Cloud Integration to connect to PostgreSQL (Cloud) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for PostgreSQL (Cloud)
JDBC for SAP ASE Service (Neo) Receiver adapter	Allows SAP Cloud Integration to connect to SAP ASE Service (Neo) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP ASE Service (Neo)
JDBC for SAP HANA Cloud Receiver adapter	Allows SAP Cloud Integration to connect to SAP HANA Cloud using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP HANA (Cloud)

Feature	Description
JDBC for SAP HANA Platform (On-Premise) Receiver adapter	Allows SAP Cloud Integration to connect to SAP HANA Platform (On-Premise) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP HANA Platform (On-Premise).
JDBC for SAP HANA Service (Neo) Receiver adapter	Allows SAP Cloud Integration to connect to SAP HANA Service (Neo) using JDBC (Java Database Connectivity) and to execute SQL commands on the database. See: JDBC for SAP HANA Service (Neo).
JMS Sender adapter	Enables asynchronous messaging by using message queues. The sender adapter consumes messages from a queue. The messages are processed concurrently. To prevent situations where the JMS adapter tries again and again to process a failed (large) message, you can store messages (where the processing stopped unexpectedly) in a dead-letter queue after two retries. Certain constraints apply with regard to the number and capacity of involved queues, as well as for the headers and exchange properties defined in the integration flow before the message is saved to the queue (as described in the product documentation). See: Configure the JMS Sender Adapter
JMS Receiver adapter	Enables asynchronous messaging by using message queues. The receiver adapter stores messages and schedules them for processing in a queue. The messages are processed concurrently. See: Configure the JMS Receiver Adapter
Kafka Sender adapter	Allows SAP Cloud Integration to connect to an external Kafka broker via Kafka protocol and to fetch Kafka records (messages). See: Configure the Kafka Sender Adapter
Kafka Receiver adapter	Allows SAP Cloud Integration to connect to an external Kafka broker via Kafka protocol and to send Kafka records (messages). See: Configure the Kafka Receiver Adapter
Mail Sender for IMAP Sender adapter	Enables SAP Cloud Integration to read e-mails from an e-mail server using the Internet Message Access Protocol (IMAP) protocol. To authenticate against the e-mail server, you can send the user name and password in plain text or encrypted (the latter only if the e-mail server supports this option). You can protect inbound e-mails at the transport layer with IMAPS and STARTTLS. The sender adapter allows you to define a schedule for polling data from the connected system. For more information on possible threats when processing e-mail content with the Mail adapter, see the product documentation. See: Mail Sender for IMAP
Mail Sender for POP3 Sender adapter	Enables SAP Cloud Integration to read e-mails from an e-mail server using the Post Office Protocol (POP3) protocol. To authenticate against the e-mail server, you can send the user name and password in plain text or encrypted (the latter only if the e-mail server supports this option). You can protect inbound e-mails at the transport layer with POP3S and STARTTLS.

Feature	Description
	The sender adapter allows you to define a schedule for polling data from the connected system. For more information on possible threats when processing e-mail content with the Mail adapter, see the product documentation. See: Mail Sender for POP3
Mail Receiver adapter	Enables SAP Cloud Integration to send e-mails to an e-mail server. To authenticate against the e-mail server, you can send the user name and password in plain text or encrypted (the latter only if the e-mail server supports this option). - You can protect outbound e-mails at the transport layer with STARTTLS or SMTPS. - You can encrypt outbound e-mails using S/MIME (supported content encryption algorithms: AES/CBC/PKCS5Padding, DESede/CBC/PKCS5Padding). See: Configure the Mail Receiver Adapter
Microsoft Dynamics CRM Receiver adapter	Connects SAP Cloud Integration to Microsoft Dynamics Customer Relationship Management (CRM). See: Microsoft Dynamics CRM Receiver Adapter
OData Sender adapter	Connects SAP Cloud Integration to systems using the Open Data (OData) protocol in either ATOM or JSON format (only synchronous communication is supported). Supported versions: OData version 2.0 - The adapter receives incoming requests in either ATOM or JSON format. - Supported operations: Create (POST), Delete (DELETE), Query (GET), Read (GET), Update (PUT) Using the GET or POST method, the sender adapter can also invoke operations that are not covered by the standard CRUD (Create, Retrieve, Update, and Delete) methods (function import). See: Configure the OData Sender Adapter
OData Receiver adapter	Connects SAP Cloud Integration to systems using the Open Data (OData) protocol. Supported versions: - OData version 2.0 Supported operations: Create (POST), Delete (DELETE), Merge (MERGE), Query (GET), Read (GET), Update (PUT), Patch (PATCH) - OData version 4.0 Supported operations: Create (POST), Query (GET), Delete (DELETE), Update (PUT), and Patch (PATCH) - The outgoing request payload must be in XML format. See: Configure the OData V2 Receiver Adapter Configure the OData V4 Receiver Adapter
ODC Receiver adapter	Connects SAP Cloud Integration to SAP Gateway OData Channel (through transport protocol HTTPS). Supported operations: Create (POST), Delete (DELETE), Merge (MERGE), Query (GET), Read (GET), Update (PUT) See: ODC Receiver Adapter

Feature	Description
OpenConnectors Receiver adapter	Connects SAP Cloud Integration to more than 150 non-SAP Cloud applications that are supported by SAP Open Connectors. • Uses APIs to fetch data from specific third-party applications. • Is designed to handle large volumes of incoming data. • Supports messages in both JSON and XML format, for request and response calls. • Allows you to define specific values for variables. See: OpenConnectors Receiver Adapter
ProcessDirect Sender adapter	Connects an integration flow with another integration flow deployed on the same tenant. An integration flow with a ProcessDirect sender adapter (as consumer) consumes data from another integration flow. N:1 cardinality of producer and consumer integration flows is supported. See: Configure the ProcessDirect Sender Adapter
ProcessDirect Receiver adapter	Connects an integration flow with another integration flow deployed on the same tenant. An integration flow with a ProcessDirect receiver adapter (as producer) sends data to another integration flow. N:1 cardinality of producer and consumer integration flows is supported. See: Configure the ProcessDirect Receiver Adapter
RFC Receiver adapter	Connects SAP Cloud Integration to a remote receiver system using Remote Function Call (RFC). RFC is the standard interface used for integrating on-premise ABAP systems to the systems hosted on the cloud using SAP Cloud Connector. The adapter supports SAP NetWeaver, version 7.31 or higher. See: RFC Receiver Adapter
Salesforce Sender adapter	Connects SAP Cloud Integration to Salesforce. See: Salesforce Sender Adapter
Salesforce Receiver adapter	Connects SAP Cloud Integration to Salesforce. See: Salesforce Receiver Adapter
SFTP Sender adapter	Connects SAP Cloud Integration to a remote system using the SSH File Transfer protocol to read files from the system. SSH File Transfer protocol is also referred to as Secure File Transfer protocol (or SFTP). Supported versions: SSH version 2 (as specified at The Secure Shell (SSH) Protocol Architecture ) , SSH File Transfer Protocol (SFTP) version 3 or higher. The sender adapter allows you to define a schedule for polling data from the connected system. See: Configure the SFTP Sender Adapter
SFTP Receiver adapter	Connects SAP Cloud Integration to a remote system using the SSH File Transfer protocol to write files to the system. SSH File Transfer protocol is also referred to as Secure File Transfer protocol (or SFTP). Supported versions: SSH version 2 (as specified at The Secure Shell (SSH) Protocol Architecture ) , SSH File Transfer Protocol (SFTP) version 3 or higher.

Feature	Description
	See: Configure the SFTP Receiver Adapter
SOAP SOAP 1.x Sender adapter	Exchanges messages with a sender system that supports Simple Object Access Protocol (SOAP) 1.1 or SOAP 1.2. The message exchange patterns supported by the sender adapter are one-way messaging or request-reply. The adapter supports Web services Security (WS-Security). A size limit for the inbound message can be configured for the sender adapter. See: Configure the SOAP (SOAP 1.x) Sender Adapter
SOAP SOAP 1.x Receiver adapter	Exchanges messages with a receiver system that supports Simple Object Access Protocol (SOAP) 1.1 or SOAP 1.2. The adapter supports Web services Security (WS-Security). See: Configure the SOAP (SOAP 1.x) Receiver Adapter
SOAP SAP RM Sender adapter	Exchanges messages with a sender system based on the SOAP communication protocol and SAP Reliable Messaging (SAP RM) as the message protocol. SAP RM is a simplified communication protocol for asynchronous Web service communication that does not require the use of Web Service Reliable Messaging standards. A size limit for the inbound message can be configured for the sender adapter. See: Configure the SOAP (SAP RM) Sender Adapter
SOAP SAP RM Receiver adapter	Exchanges messages with a receiver system based on the SOAP communication protocol and SAP Reliable Messaging (SAP RM) as the message protocol. SAP RM is a simplified communication protocol for asynchronous Web service communication that does not require the use of Web Service Reliable Messaging standards. See: Configure the SOAP (SAP RM) Receiver Adapter
SuccessFactors REST Sender adapter	Connects SAP Cloud Integration to a SuccessFactors sender system using the REST message protocol. The adapter supports the following operations: GET See: Configure the SuccessFactors REST Sender Adapter
SuccessFactors REST Receiver adapter	Connects SAP Cloud Integration to a SuccessFactors receiver system using the REST message protocol. The adapter supports the following operations: GET, POST See: Configure the SuccessFactors REST Receiver Adapter
SuccessFactors SOAP Sender adapter	Connects SAP Cloud Integration to SOAP-based Web services of a SuccessFactors sender system (synchronous or asynchronous communication). The adapter supports the following operations: Query See: Configure the SuccessFactors (SOAP) Sender Adapter
SuccessFactors SOAP Receiver adapter	Connects SAP Cloud Integration to SOAP-based Web services of a SuccessFactors receiver system (synchronous or asynchronous communication). The adapter supports the following operations: Insert, Query, Update, Upsert See: Configure the SuccessFactors SOAP Receiver Adapter
SuccessFactors OData V2 Receiver adapter	Connects SAP Cloud Integration to a SuccessFactors system using OData V2. Features of OData version 2.0 supported by the adapter: Operations: GET (get single entity as an entry document), PUT (update existing entry with an entry document), POST (create new entry from an entry document), DELETE (Delete an entry from an entry document)

Feature	Description
	document), UPSERT (combination of Update OR Insert) • Query options: \$expand, \$skip, and \$top • Server-side pagination • Client-side pagination • Pagination enhancement: Data retrieved in chunks and sent to Cloud Integration • Deep insert: Creates a structure of related entities in one request • Authentication options: Basic authentication • Reference links: Link two entities using the <link> tag See: Configure the SuccessFactors OData V2 Receiver Adapter
SuccessFactors OData V4 Receiver adapter	Connects SAP Cloud Integration to a SuccessFactors system using OData V4. Features of OData version 4.0 supported by the adapter: • Operations: GET, POST, PUT, DELETE • Navigation • Primitive types supported according to OData V4 specification • Structural types supported for create/update operations: Edm.ComplexType, Edm.EnumType, Collection(Edm.PrimitiveType) and Collection(Edm.ComplexType) See: SuccessFactors OData V4 Receiver Adapter
SugarCRM Receiver adapter	Connects SAP Cloud Integration to SugarCRM. See: SugarCRM Receiver Adapter
Twitter Receiver adapter	Enables SAP Cloud Integration to access Twitter and read or post tweets. Using OAuth, SAP Cloud Integration can access resources on Twitter on behalf of a Twitter user. See: Twitter Receiver Adapter
XI Sender adapter	Connects SAP Cloud Integration to a remote sender system that can process the XI message protocol. See: Configure the XI Sender Adapter
XI Receiver adapter	Connects SAP Cloud Integration to a remote receiver system that can process the XI message protocol. See: Configure the XI Receiver Adapter

As well as the *transport-level security* options, you can also secure the communication at *message level*. This protects the content of the exchanged messages by means of digital encryption and signatures. Various security standards are available to do this: PKCS#7, XML Digital Signature, OpenPGP, and WS-Security.

Related Information

[Connectivity \(Adapters\)](#)

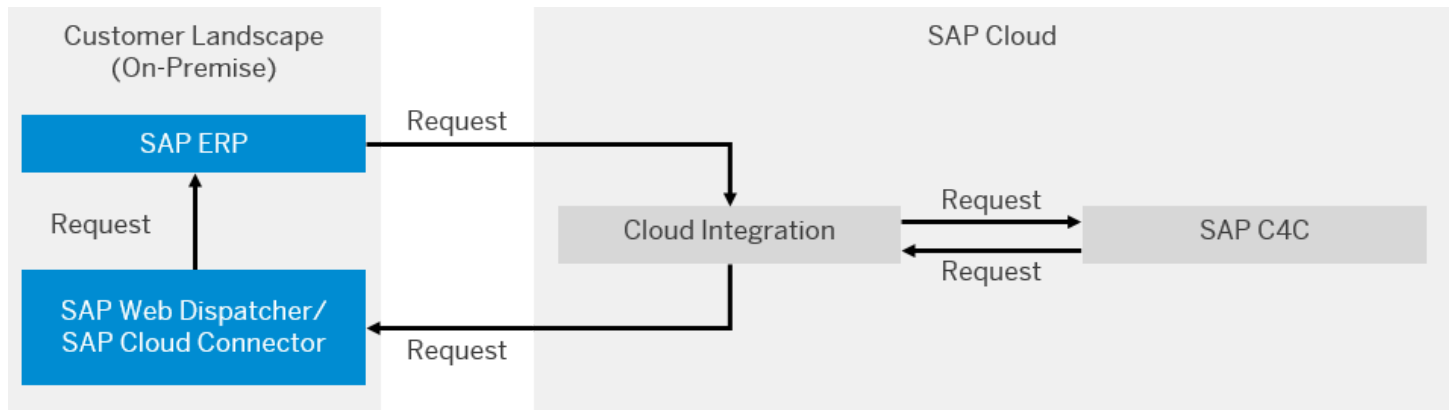
[Operating Model](#)

[Connecting to an On-Premise Landscape \(Example Setup\)](#)

Connecting to an On-Premise Landscape (Example Setup)

To give you an idea of what the technical landscape behind a real life integration scenario looks like, here's an example for the SAP Cloud for Customer (C4C)-to-SAP ERP integration scenario. In this scenario, SAP's own cloud solution SAP Cloud for Customer (C4C) is connected with an on-premise SAP Enterprise Resource Planning (ERP) system through Cloud Integration.

The following figure shows a typical setup of components:



The left side of the figure covers the communication of Cloud Integration with the on-premise system in the customer landscape.

The setup contains components that all are connected by HTTPS communication. Typical adapters are the IDoc adapter for the connection between the on-premise system and Cloud Integration, and the SOAP adapter for the connection between SAP Cloud for Customer and Cloud Integration (within the SAP Cloud).

The lower path shows the connection from Cloud Integration to the on-premise system, which is located in the customer landscape. This is the outbound communication from the perspective of the integration platform, but is an inbound connection from the perspective of the customer landscape. Therefore, to protect the components in the customer landscape from remote calls from the Internet, a load balancer component is required – which is either a Web Dispatcher component or the SAP Cloud Connector.

The upper path shows the connection from the on-premise system to Cloud Integration. From the *perspective* of Cloud Integration, this is an inbound connection and, therefore, again a load balancer is required to protect the tenant that actually processes the message against remote calls. This is the BIG-IP load balancer, which is involved in all HTTPS inbound requests by default, and isn't shown in the figure for the sake of simplicity. Also, this component is preconfigured by SAP and doesn't require any further configuration for such a scenario.

Configuring Inbound Communication

Configuring inbound communication means setting up the connection between a remote sender system and the integration platform. Inbound communication refers to message processing from a remote system, often located in the customer landscape, to Cloud Integration. Here, the integration platform is the server.

Sender Systems You Can Connect to the Integration Platform

You can connect different kinds of sender systems to the integration platform, for example:

- A cloud application, for example, SAP SuccessFactors
- An on-premise application, for example, SAP ERP

i Note

If an SAP system based on Application Server ABAP sends requests to Cloud Integration and there are 2 or more worker nodes enabled on Cloud Integration side, you can receive an HTTP/1.1 403 authentication error. The root cause is that the SAP kernel encodes the cookies' value by default, which breaks the load-balancing feature. To solve the issue, set profile parameter `ict/disable_cookie_urlencoding` to 1 or 2 depending on kernel level. For more information, see SAP note [2681175](#) .

- A SOAP client
- An e-mail server
- An SFTP server

In this case, the integration platform reads files from the SFTP server (polling).

To enable communication with such a variety of systems, Cloud Integration supports the following kinds of connections:

- HTTP connections that allow
 - Sender systems to call integration flow endpoints (through one of the adapters based on the HTTP protocol like, for example, the HTTPS adapter or the SOAP adapter)
 - API clients to call the OData API
- SFTP (SSH File Transfer Protocol) connections
- Connections to an FTP server using the Secure File Transfer Protocol (FTPS)
- Connections to an e-mail server using the mail sender adapter
- Connections to a Java Message System (JMS) message broker
- Connections to an external message broker using the Advanced Message Queuing Protocol (AMQP)

i Note

The procedure to set up HTTP connections depends on whether you use Cloud Integration in the Cloud Foundry or in the Neo environment.

Related Information

[Configuring Inbound HTTP Connections, Cloud Foundry Environment](#)

[Setting Up Inbound SFTP Connections](#)

[Setting Up Inbound Mail Connections](#)

[Tutorial: Set Up Inbound OAuth Client Credentials Grant Authentication for API Clients with SAP-Generated Certificate](#) 

[Tutorial: Set Up Inbound OAuth Client Credentials Grant Authentication for Senders Calling Integration Flows with SAP-Generated Certificate](#) 

Configuring Inbound HTTP Connections, Cloud Foundry Environment

Set up secure HTTP connections to enable:

- Sender systems to call integration flow endpoints (through one of the adapters based on the HTTP protocol like, for example, the HTTPS adapter or the SOAP adapter)

See: [Setting Up Inbound HTTP Connections \(Integration Flow Processing\)](#)

- API clients to call the Cloud Integration OData API

See: [Setting Up Inbound HTTP Connections \(for API Clients\)](#)

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

i Note

Systems acting as clients for SAP Cloud Integration endpoints are required to use SNI. See SAP Note [2752867](#) .

Related Information

- [Configuration Checklist for Inbound Authentication](#)
- [Creating Service Instance and Service Key for Inbound Authentication](#)
- [Setting Up Inbound HTTP Connections \(Integration Flow Processing\)](#)
- [Setting Up Inbound HTTP Connections \(for API Clients\)](#)

Configuration Checklist for Inbound Authentication

The following tables provide a summary of the configuration settings for the available inbound authentication options.

i Note

Most options are configured using SAP BTP service instances and service keys (see [Creating Service Instance and Service Key for Inbound Authentication](#)).

Those options based on users registered in an identity provider (IdP), don't require service instance/key configuration.

→ Tip

You can also use an application programming interface (API) to manage service instances and service keys. To learn more about how to use the API to retrieve information on the service instances and service keys configured for inbound communication, check out [Managing Service Instances and Service Keys Using an API](#).

Configuration Checklist (Sender Calling Integration Flow Endpoint)

Authentication Option	Service Instance			Service Key			
	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
Client certificate (for senders calling integration flow) Using SAP certificate See: Client Certificate Authentication	integration-flow	Keep standard role <code>ESBMessaging.send</code> or use one or more custom roles.	Client Credentials	Certificate	n.a.	Specify validity in days.	Specify key size.

	Service Instance			Service Key			
Authentication Option	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
for Integration Flow Processing							
Client certificate (for senders calling integration flow) Using own (external) certificate See: Client Certificate Authentication for Integration Flow Processing				External Certificate	Add PEM-encoded X.509 certificate.	n.a.	n.a.
OAuth client credentials grant (for senders calling integration flow) Using clientId and clientsecret to authenticate against token server See: OAuth with Client Credentials Grant for Integration Flow Processing				ClientId/Secret	n.a.	n.a.	n.a.
OAuth client credentials grant (for senders calling integration flow) Using SAP certificate to authenticate against token server				Certificate	n.a.	Specify validity in days.	Specify key size.

	Service Instance			Service Key			
Authentication Option	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
See: OAuth with Client Credentials Grant for Integration Flow Processing							
OAuth client credentials grant (for senders calling integration flow) Using own certificate to authenticate against token server See: OAuth with Client Credentials Grant for Integration Flow Processing				External Certificate	Add PEM-encoded X.509 certificate.	n.a.	n.a.
Basic authentication With clientId and clientsecret See: Basic Authentication with clientId and clientsecret for Integration Flow Processing				ClientId/Secret	n.a.	n.a.	n.a.
Basic authentication With IdP user See: Basic Authentication of IdP User for Integration Flow Processing	n.a.						

Configuration Checklist (API Client Calling OData API)

	Service Instance			Service Key			
Authentication Option	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
Client certificate (for API clients calling OData API) Using SAP certificate See: Client Certificate Authentication for API Clients	api	Role as described at Tasks and Permissions	Client Credentials	Certificate	n.a.	Specify validity in days.	Specify key size.
Client certificate (for API clients calling OData API) Using own (external) certificate See: Client Certificate Authentication for API Clients				External Certificate	Add PEM-encoded X.509 certificate.	n.a.	n.a.
OAuth client credentials grant (for API clients calling OData API) Using clientId and clientsecret to authenticate against token server See: OAuth with Client Credentials Grant for API Clients				ClientId/Secret	n.a.	n.a.	n.a.
OAuth client credentials grant (for API clients calling OData API)				Certificate	n.a.	Specify validity in days.	Specify key size.

	Service Instance			Service Key			
Authentication Option	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
Using SAP certificate to authenticate against token server See: OAuth with Client Credentials Grant for API Clients							
OAuth client credentials grant (for API clients calling OData API) Using own certificate to authenticate against token server See: OAuth with Client Credentials Grant for API Clients				External Certificate	Add PEM-encoded X.509 certificate.	n.a.	n.a.
Basic authentication (for API clients calling OData API) With IdP user See: Basic Authentication of an IdP User for API Clients	n.a.						

i Note
Note Related to Role Configuration

Depending on the chosen inbound authorization option, you define permissions for sender systems to call integration flow endpoints in different ways:

Option	Summary of Steps
Basic authentication of a user registered at an identity provider (IdP)	Use SAP BTP cockpit to define a role collection that contains the predefined role template MessagingSend and assign the role collection to the IdP user (under Security > Trust Configuration >). The role template MessagingSend is provided by default in your

Option	Summary of Steps
	subaccount to define permissions for sender systems to call integration flow endpoints for this use case. See: Basic Authentication of IdP User for Integration Flow Processing
Authentication with an OAuth client (service instance)	Use the Cloud Integration Monitor application and select the User Roles tile under Manage Security. When doing this, you can either use the predefined role <code>ESBMessaging.send</code> or create a custom role. Create service instance and service key using SAP BTP cockpit. During this step, you need the role specified with the User Roles tile. See: • Client Certificate Authentication for Integration Flow Processing • OAuth with Client Credentials Grant for Integration Flow Processing • Basic Authentication with clientId and clientsecret for Integration Flow Processing

Creating Service Instance and Service Key for Inbound Authentication

With a service instance, you define how to access a certain SAP BTP service. In the context of SAP Cloud Integration, a service instance is the definition of an OAuth client.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Create a service instance to implement inbound communication. A service instance is an OAuth client (with grant type `Client Credentials`).

i Note

How to specify the parameters depends on the plan and authentication option.

For more information on the inbound authentication options for senders calling integration flow endpoints (**integration-flow** plan):

- [Client Certificate Authentication for Integration Flow Processing](#)
- [OAuth with Client Credentials Grant for Integration Flow Processing](#)
- [Basic Authentication with clientId and clientsecret for Integration Flow Processing](#)

For more information on the inbound authentication options of API clients calling the SAP Cloud Integration OData API (**api** plan):

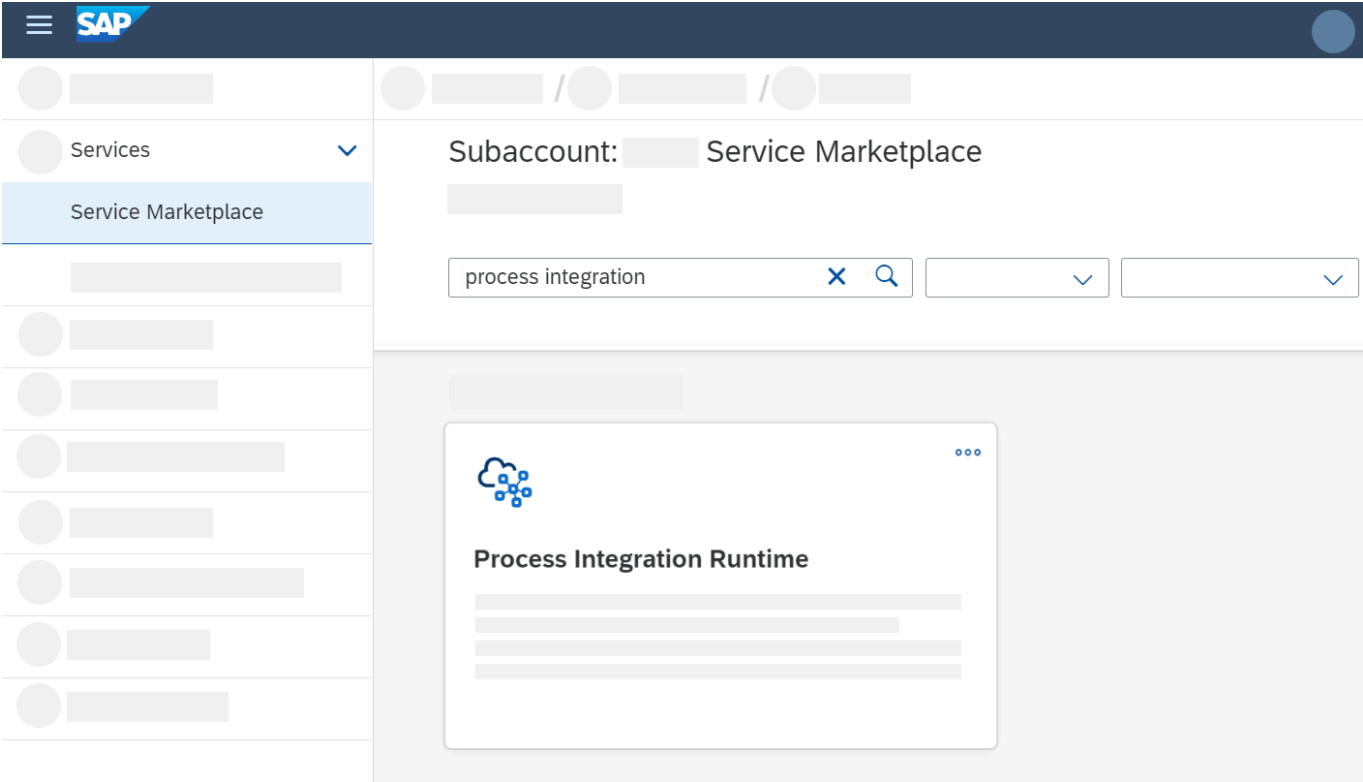
- [Client Certificate Authentication for API Clients](#)
- [OAuth with Client Credentials Grant for API Clients](#)

Creating Service Instance

- 1. Go to SAP BTP cockpit.
- 2. Select the subaccount that hosts your SAP Cloud Integration application.
- 3. Choose your subaccount, navigate to **Services** **Service Marketplace**, and select **Process Integration Runtime**.

i Note

The Process Integration Runtime tile is only displayed when you've created a runtime instance and a space.



- 4. Choose **Create**.
- 5. In the **New Instance or Subscription** dialog box, **Process Integration Runtime** is already preselected as **Service**.
- 6. Specify the following parameters:

Parameter	Value
Plan	Depending on the use case, select one of the following options: integration-flow To define inbound authentication of senders calling integration flow endpoint (on an SAP Cloud Integration worker node). See: Set Up SAP Integration Suite api To define inbound authentication of API clients calling the SAP Cloud Integration OData API.
Runtime Environment	Cloud Foundry
Space	Select a space (for example dev).

Parameter	Value
Instance Name	Enter a meaningful short name. i Note We recommend you use a CLI-friendly name to enable the managing of your instances with the SAP BTP command line interface as well. CLI-friendly name is a short string (up to 32 characters) that only contains alphanumeric characters (A-Z, a-z, 0–9), periods, underscores, and hyphens. Your instance name can't contain white spaces if you want your instance name to be CLI-friendly.

7. Choose **Next**.

8. Configure instance parameters. Choose how to enter your details, via **Form** or **JSON**.

We recommend choosing **Form** as the more convenient option.

i Note

Selecting **JSON**, you can also pass these parameters in a valid JSON object that contains service-specific configuration parameters, provided either in-line or in a file (see [Specifying Service Instance and Service Key Parameters in JSON Format](#)).

Specify the following parameters.

Parameter	Value
Roles	The selection of roles depend on the chosen option for Plan. - When as Plan you've chosen integration-flow, you can either keep the standard role <code>ESBMessaging.send</code> or enter a custom role. The default role <code>ESBMessaging.send</code> is already predefined. To define a custom role, go to the Monitor view of , and select the User Roles tile in the Manage Security section (for more information, see Managing User Roles, Cloud Foundry Environment). i Note Custom roles for this use case are not defined using the cockpit. You're able to add multiple roles by selecting enter after each role. The default is set to the standard role (<code>ESBMessaging.send</code>). → Tip When defining a service instance with integration-flow plan, you assign a role to it that enables the associated user to process the integration flow on the worker node. Simply spoken, this role defines permission for a sender to process an integration flow. - When as Plan you've chosen api, select one or more roles as provided in the dropdown list. These roles define permissions for API clients to access certain SAP Cloud Integration resources using the OData API. Choose the role depending on the resource you like to access using the OData API (see Tasks and Permissions).

Parameter	Value
Grant-types	Select Client Credentials. i Note The list of supported grant types is: - o Authorization Code - o Client Credentials - o Password - o Refresh Token - o SAML2 Bearer - o JWT Bearer
Redirect-uris (optional)	Enter the redirect URIs for authorization code grant type. Select Enter after filling out your uri and proceed with the next uri.
Access Token Validity (in seconds)	Define the period of time for which your access token is valid in seconds. Enter any value from 3600 to 86400 (1- 24 hours). The following values have been set as default: - o for plan api: 43200 (12 hours) - o for plan integration-flow: 3600 (1 hour)

9. Optional: Choose [Next](#) to review and verify the instance details.

10. Choose [Create](#).

Creating Service Key

With this step, you create a service key for the instance.

1. Using SAP BTP cockpit, enter your subaccount and go to [Instances and Subscriptions](#).

You can see your instances in a table.

2. Select the service instance.

3. Under ... *Actions*, choose [Create Service Key](#).

4. Enter a name for the service key under [Service Key Name](#). You can use up to 32 characters.

5. Configure instance parameters. Choose how to enter your details, via [Form](#) or [JSON](#).

We recommend choosing [Form](#) as the more convenient option.

Specify the following parameters.

Parameter	Value
Key Type	There are the following options. o ClientId/Secret: To define a service key that contains a clientId and clientsecret.

Parameter	Value
	◦ Certificate: Provides a x509 certificate issued by SAP. Select this option to have SAP BTP generate a client certificate for you. Client credentials grant is required to be able to use this key type. ◦ External Certificate: Allows to map an existing x509 certificate to a service key. Select this option to use a client certificate generated with another application than SAP BTP. Client credentials grant is required to be able to use this key type. See: Service Key Types
External Certificate (only applicable if for Key Type the option External Certificate has been chosen)	Add External Certificate: Enter the certificate that you exported from the certificate-generating application to your system. Enter the PEM-encoded X.509 certificate. → Tip <i>PEM</i> stands for <i>Privacy Enhanced Mail</i> and is a common format for X.509 certificates. It contains base64-encoded data with the string <code>-----BEGIN CERTIFICATE-----</code> at the beginning and the string <code>-----END CERTIFICATE-----</code> at the end of the character sequence. Example: <pre>-----BEGIN CERTIFICATE-----MIIHyDCCBrCgAwIB[. . .]CAq8Tn7kSFDmVnrXe6v8hcQ==-----END CERTIFICATE-----</pre> Don't enter the whole certificate chain. Make sure that the certificate is signed by a Certification Authority supported by the load balancer (see Load Balancer Certificates Supported by SAP). You can only use a single certificate once across all existing service instances. To assign multiple roles, don't create multiple service instances. Instead, maintain multiple roles within one service instance.
Pin Certificate (only available for plan integration-flow and only applicable if for Key Type the option External Certificate has been chosen)	Defines whether the exact client certificate is required for authentication. If this setting is enabled (=default), the incoming client certificate used during authentication must match the service key exactly. In case the client certificate is renewed, a new service key with the new certificate has to be created. See Key Types. If this setting is disabled, only the subjectDN and issuerDN of the incoming client certificate will be compared. This allows to renew client certificates for authentication without changing the service key thus reducing configuration effort. The following conditions need to be fulfilled for a renewed certificate: ◦ The renewed certificate has the same subjectDN and issuerDN as the certificate configured in the service key. ◦ The issueDate of the renewed certificate must be issued after the previously accepted certificate. i Note Once a new certificate was used for authentication purposes, the previously used certificate (with an older issue date) will not be accepted anymore. ⚠ Caution You must ensure that there are no conflicts between certificates uploaded in different service keys. A call with a service key will be rejected at runtime, if there are:

Parameter	Value
	- multiple certificates with the same subjectDN and issuerDN where pinning is disabled; - or one certificate with pinning enabled and another certificate with the same subjectDN and issuerDN disabled.
Validity in days (only applicable if for Key Type the option Certificate has been chosen)	Define the validity in days by selecting a number between 1 and 365.
Key Size (only applicable if for Key Type the option Certificate has been chosen)	The default for the key size is set to 2048.

i Note

Selecting **JSON**, you can also pass these parameters in a valid JSON object that contains service-specific configuration parameters, provided either in-line or in a file (see [Specifying Service Instance and Service Key Parameters in JSON Format](#)).

6. Choose **Create**.

7. Choose the newly created service key to display the details of the service key. You need the values of the service key for later reference.

Depending on the chosen **Key Type**, the service key contains certain parameters. The following table lists the parameters that are required to configure the client application:

Service Key Parameters

Key Type	Parameters Contained in Service Key
ClientId/Secret	clientid clientid to be used as credential when requesting the access token from the token server. clientsecret clientsecret to be used as password when requesting the access token from the token server. tokenurl URL of token server that issues the access token. url URL to address service.

Key Type	Parameters Contained in Service Key
Certificate	- clientid clientid of service key. - certificate PEM-encoded certificate chain (to be used by the sending application to authenticate itself against token server or application). The certificate chain contains a root certificate supported by SAP (see Load Balancer Root Certificates Supported by SAP). - key Private key of SAP-generated key pair. - tokenurl URL of token server that issues the access token. - url URL to address service. i Note To enable the related HTTP client to support this authentication option, you need to format the certificate (including the certificate chain) and the key accordingly. In particular, make sure to replace all \n in the SAP-generated certificate or key by line breaks. A suitable certificate, for example, would then look like the following: <pre>-----BEGIN CERTIFICATE----- MIIFtDCCA5ygAwIBAgIQCUFIj6cfjiSfZi/ZvVU6IDANBgkqhkiG9w0BAQsFADB5+ LvHPhNDM3rMsLu06agF4JTb08ANYtWQTx0PVrZKJu+8fcIaUp7MVBIVZ -----END CERTIFICATE-----</pre> i Note The generated certificate also contains additional parameters under certificatedetails. When for Key Type, you've chosen Certificate, the following applies for these parameters: - The values for the parameters issuerdn, serialnumber, and subjectdn are determined by SAP. - The value of parameter validuntil is calculated from the entry that you've selected for Validity in days when defining the service key. See: Service Key Types
External Certificate	- clientid clientid of service key. - certificate

Key Type	Parameters Contained in Service Key
	PEM-encoded certificate (to be used by the sending application to authenticate itself against token server or application). ◦ tokenurl URL of token server that issues the access token. ◦ url URL to address service. A service key with this Key Pair doesn't contain a private key because the corresponding key pair has been generated with another application than SAP BTP.

i Note

You have two options to display these parameters:

- **Form**

Displays content of service key in a user-friendly list easy to consume.

- **JSON**

Displays content of service key in JSON format.

You have the following options:

- Copy these values to your clipboard or to a text editor.
- Download the service key.

You need these values when specifying the required credentials or certificate values associated with the sending application.

Updating Service Instance

You can update an existing service instance. To do that, perform the following steps.

1. Using SAP BTP cockpit, enter your subaccount and go to **Instances and Subscriptions**.

You can see your instances in a table.

2. Select the service instance.
3. Under ... *Actions*, choose **Update**.
4. Check out the parameter settings and, if necessary, change them.
5. Choose **Update Instance**.

i Note

Because of caching mechanisms in SAP Cloud Integration, it can take up to 1 hour until role changes are considered.

→ Tip

You can also use an application programming interface (API) to manage service instances and service keys. To learn more about how to use the API to retrieve information on the service instances and service keys configured for inbound

Related Information

[Tutorial: Set Up Inbound OAuth Client Credentials Grant Authentication for API Clients with SAP-Generated Certificate](#) 

Specifying Service Instance and Service Key Parameters in JSON Format

Service Instance Parameters in JSON Format (Examples)

In this example (for [integration-flow](#) plan), role `ESBMessaging.send` and `Client Credentials` grant type is used.

Sample Code

```
{
  "grant-types": [
    "client_credentials"
  ],
  "redirect-uris": [

  ],
  "roles": [
    "ESBMessaging.send"
  ]
}
```

When you have defined custom roles to protect integration flow endpoints individually, you can also specify multiple roles separated by a comma.

Example:

Sample Code

```
{
  "grant-types": [
    "client_credentials"
  ],
  "redirect-uris": [

  ],
  "roles": [
    "ESBMessaging.send",
    "myRole1",
    "myRole2"
  ]
}
```

In this example (for [api](#) plan), role `MonitoringDataRead` and `Client Credentials` grant type is used.

With this role assignment, the API client can access message processing logs on the tenant using the `MessageProcessingLogs` entity.

Sample Code

```
{
  "roles": [
    "MonitoringDataRead"
  ],
  "grant-types": [
    "client_credentials"
  ]
}
```

You can also specify a list of multiple roles, for example:

Sample Code

```
{
  "roles": [
    "MonitoringDataRead",
    "WorkspaceArtifactsDeploy"
  ],
  "grant-types": [
    "client_credentials"
  ]
}
```

Service Key Parameters in JSON Format (Examples)

This example shows the JSON content for **Certificate** key type:

Sample Code

```
{
  "key-type": "certificate",
  "validity": 365,
  "key-length": 2048
}
```

This example shows the JSON content for **External Certificate** key type:

Sample Code

```
{
  "key-type": "certificate_external",
  "X.509": "-----BEGIN CERTIFICATE-----MIIHyDCCBrCgAwIB[... ]CAq8Tn7kSFDmVnrXe6v8hcQ==-----END",
  "validity": 365,
  "key-length": 2048
}
```

In this example, the value for the [X.509](#) parameter is the PEM-encoded certificate to be provided with this service key.

Service Key Types

You can select different service key types depending on the use case.

There are the following options.

Service Key Types

Key Type	Description	Supported Authentication Options
ClientId/Secret	Service key contains a clientId and clientsecret (client credentials).	For senders calling an integration flow endpoint: • OAuth with Client Credentials Grant for Integration Flow Processing (using client credentials to get OAuth access token) • Basic Authentication with clientId and clientsecret for Integration Flow Processing (using client credentials to directly call integration flow) For API clients: OAuth with Client Credentials Grant for API Clients (using client credentials to get OAuth access token)
Certificate	Service key contains a clientId and an x509 client certificate generated by SAP. Select this option to have SAP BTP generate a client certificate and key pair for you. When defining a service key with this Key Type, you can specify a Validity in days parameter (up to 365 days). SAP generates a client certificate and public/private key pair together with the service key. The certificate parameters of the service key (for example, the issuer DN and the serial number) are then determined by SAP. The validity of the certificate (validuntil parameter of the certificate) is calculated based on the entry that you've selected for Validity in days when defining the service key. The certificate needs to be signed by a certification authority that is supported by the load balancer. See: Load Balancer Root Certificates Supported by SAP	For senders calling an integration flow endpoint: • Client Certificate Authentication for Integration Flow Processing (using client certificate to directly call integration flow) • OAuth with Client Credentials Grant for Integration Flow Processing (using client certificate to get OAuth access token) For API clients: • Client Certificate Authentication for API Clients (using client

Key Type	Description	Supported Authentication Options
	Furthermore, the certificate is signed by a certificate authority (CA) that is supported by SAP BTP.	certificate to directly call OData API)
External Certificate	Service key contains a clientId and an x509 client certificate generated by a tool of your choice (other than SAP BTP). If you choose this option, you have to make sure that the certificate gets signed by a certificate authority supported by SAP BTP. See: Load Balancer Root Certificates Supported by SAP	• OAuth with Client Credentials Grant for API Clients (using client certificate to get OAuth access token)

The list of CAs that can sign the certificate depends on the scenario.

Scenario	Certificate to be signed by ...
Sender uses client certificate to directly call integration flow (Client Certificate Authentication for Integration Flow Processing).	The certificate needs to be signed by a certification authority that is supported by the load balancer. See: Load Balancer Root Certificates Supported by SAP
Sender uses client certificate to call token server, and in a second step uses the access token to call integration flow (OAuth with Client Credentials Grant for Integration Flow Processing).	The certificate must be signed by a CA from the list specified in the following topic: See: Trusted Authorities for X.509 Certificates

Managing Service Instances and Service Keys Using an API

Manage service instances and service keys using an application programming interface (API).

Using the Cloud Foundry V3 API, you can implement sequences of API calls to manage service instances and service keys automatically on your SAP Business Technology Platform subaccount.

The API supports write and read access. You can, for example, create and update service instances, as well as get the details of service instances and service keys.

→ Tip

For example, assume that you have configured a large number of service instances and service keys and you like to get an automatic notification for service keys that soon expire. To solve this challenge, you can implement a suitable sequence of API calls to get the service key's expiration dates as a final result.

The Cloud Foundry V3 API is described under <https://v3-apidocs.cloudfoundry.org/version/3.133.0/index.html> .

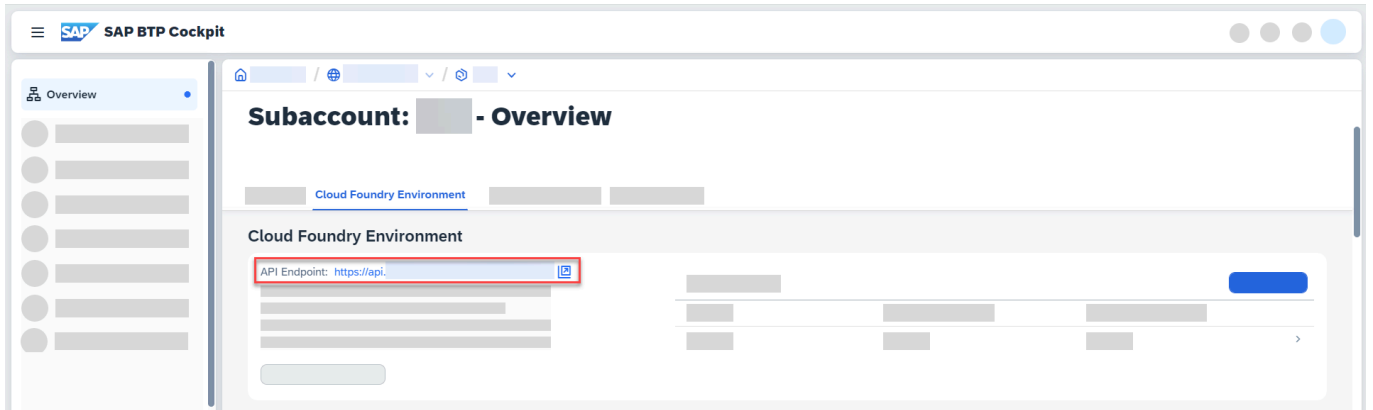
The API documentation provides you with the paths and the supported methods.

To use this API, you need a user with the Space Developer role for the Cloud Foundry space associated with the service instances.

API Endpoint Address

To get the base URL for the API calls, perform the following steps:

1. Go to SAP Business Technology Platform cockpit and select your subaccount (for which you like to manage the service instances and service keys).
2. On the **Overview** page, go to section **Cloud Foundry Environment**.



The **API Endpoint** entry is the base URL to use.

For example, assume that you like to get a list of all service instances defined on your subaccount.

In the API documentation, you find the following method and API path:

GET /v3/service_instances

That means, you have to send a GET request to the following address:

https://<API Endpoint address>/v3/service_instances

Authentication

You can authenticate your API client against the Cloud Foundry API with OAuth Password Credentials grant.

That means, you use the user credentials of the SAP Business Technology Platform space user to get an access token from a token server first. In a second call, you use the access token to call the API.

If you use Postman as your API client, there's a convenient way to configure the OAuth call sequence in one user interface.

Using Postman, set up the API call in the following way:

1. Open Postman.
2. In the address field, enter the address of the request (for example, https://<API Endpoint address>/v3/service_instances) and select the method (for example, **GET**).
3. For **Authorization** select **OAuth 2.0**.
4. Specify the following settings.

GET
https://api. /v3/service_instances
Send

Authorization
Cookies

Type
OAuth 2.0

Add authorization data to
Request H...

Current Token

Access Token

Header Prefix

Configure New Token

Configuration Options

Grant Type
Password Credentials

Access Token URL
https://uaa.

Client ID
cf

Client Secret
Client Secret

Username

Password

Client Authentication
Send client credentials in body

Clear cookies

Get New Access Token

Keep the default settings, except the following ones. Specify these parameters in the following way:

Parameter	Description
Grant Type	Select Password Credentials .
Access Token URL	Enter the address of the token server that issues the access token to request the API endpoint. Assume that your API endpoint address has the following form: https://api.<myAPI> In this case, for Access Token URL enter: https://uaa.<myAPI>/oauth/token
Client ID	cf
Username	Enter the user name of the user that has the corresponding permissions on the subaccount.
Password	Enter the password of the user.

Parameter	Description
	i Note For 2 factor authentication users, enter the password and second factor authentication, for example: <password>: <pin><rsa-code>. This may vary depending on the configured second factor.

5. select [Get New Access Token](#).

6. Select [Send](#).

You get a response that contains the list of service instances.

Call Sequence to Get Service Key Details

Let's assume that you are an integration developer and you've configured a set of service instances and service keys on your subaccount. Furthermore, you are interested in the expriy dates of the service keys associated with those service instances that control inbound communication to your integration flow endpoints. In other words, you are interested in the expiry dates of all service keys associated with service instances with service plan `integration-flow`.

To get the details of the service keys you are interested in, you can implement the following sequence of API calls.

API Call	API Call	Description
(1) Get all service instances with service plan <code>integration-flow</code>.	<code>https://<API Endpoint address>/v3/service_instances?service_plan_names=integration-flow</code>	As response, you get a list of all service in Let's assume you get 2 service instances, This is part of an example response: <pre> "resources": [{ "guid": "xyz-789-a "created_at": "202 "updated_at": "202 "name": "test2", "tags": [], "last_operation": "type": "creat "state": "succ "description": "updated_at": "created_at": }, ... { "guid": "uvw-567-d "created_at": "202 "updated_at": "202 "name": "test2", "tags": [], </pre>

API Call	API Call	Description
		<pre> "last_operation": "state": "succ "updated_at": "created_at": }, </pre>
(2) Get the service keys configured for the service instances retrieved from previous call.	<code>https://<API Endpoint address>/v3/service_credential_bindings?service_instance_guids=<guid1>,<guid2>...</code> To filter for the service instances retrieved from the previous call, you provide the service instance guids in a comma-separated list for the query parameter <code>service_instance_guids</code> (check out the Cloud Foundry API documentation for details about the query parameters). i Note Example: <code>https://<API Endpoint address>/v3/service_credential_bindings?service_instance_guids=xyz-789-abc,uvw-567-def</code>	As response, you get a list of service keys Let's assume you get 2 service keys, one service key is defined). This is part of an example response: <pre> "resources": [{ "guid": "abcd-1234 "created_at": "202 "updated_at": "202 "name": "testkey", "type": "key", "last_operation": "state": "succ "description": "type": "creat "updated_at": "created_at": { "guid": "efgh-5678 "created_at": "202 "updated_at": "202 "name": "testABC_1 "type": "key", "last_operation": "state": "succ "description": "type": "creat "updated_at": "created_at": }, </pre>
(3) Get details of individual service key	<code>https://<API Endpoint address>/v3/service_credential_bindings/<guid>/details</code> You can use the <code>guid</code> parameter retrieved from the previous call to get details of an individual service key.	For each service key retrieved from the pr As response, you get the service key deta If the service key is configured with an ex Part of an example response:

API Call	API Call	Description
retrieved from previous call.	i Note Example: https://<API Endpoint address>/v3/service_credential_bindings/abcd-1234/details You need to repeat call (3) for all service key guids retrieved from the previous call to get the details for all relevant service keys.	<pre>"certificatedetail "issuerdn": "C "subjectdn": " "validuntil": "serialnumber" }</pre>

Update Service Instance that Contains Deprecated client_x509 Grant Type

This information is relevant for you when you operate scenarios using Cloud Integration inbound authentication with client certificates and service instances that currently use the `client_x509` grant type.

The `client_x509` grant type is deprecated and should be replaced by the `client_credentials` grant type. Client certificate-based inbound authentication continues to be supported. Instructions for updating the service instance are provided next.

Caution

If the `client_x509` grant type is configured for affected service instances, updates fail, and new service keys cannot be created. To avoid disruption, update the grant type to `client_credentials`.

Updating Affected Service Instances

To find out if a service instance is affected, perform the following steps:

1. SAP BTP cockpit and select  **Services**  **Instances and Subscriptions**  
2. Select a service instance.

i Note

Only service instances with service **Process Integration Runtime** are relevant.

3. Select **View Parameters**.

The screenshot shows the SAP BTP Cockpit interface. On the left, the 'Instances and Subscriptions' section is active. The main panel displays the configuration for a 'Process Integration Runtime' instance. The instance is in a 'Created' state. A 'View Parameters' popup is open, showing the 'grant-types' field with the value 'client_x509'. The instance is currently 'Created'.

4. In the popup, choose the **JSON** tab.

5. If under **Grant-types** there is the entry **client_x509**, this service instance is affected.

```
{
  "roles": [
    "ESBMessaging.send"
  ],
  "grant-types": [
    "client_x509"
  ],
  "redirect-uris": [],
  "token-validity": 4200
}
```

6. Replace the entry **client_x509** with **client_credentials**.

7. Update instance.

See also: [Cloud Integration on CF - How to Setup Secure HTTP Inbound Connection with Client Certificates](#) (SAP Community blog)

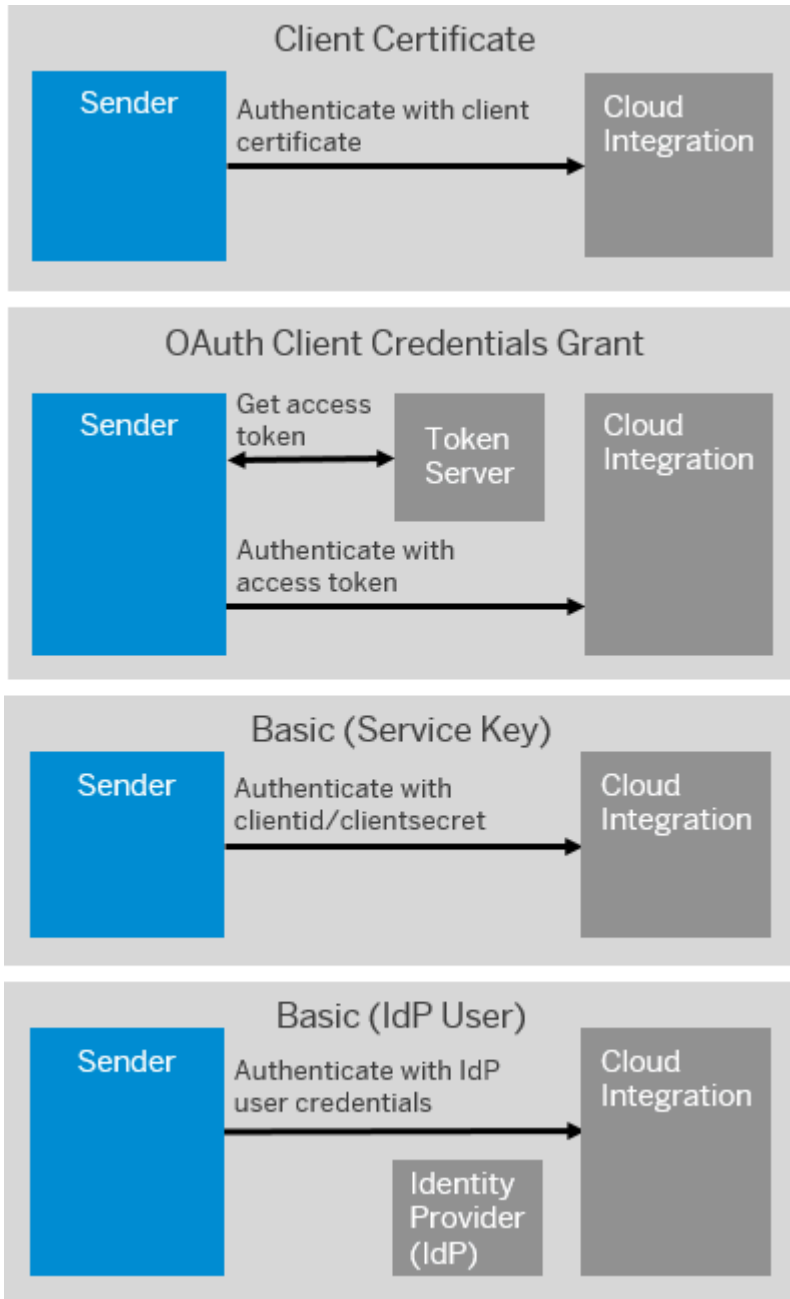
Setting Up Inbound HTTP Connections (Integration Flow Processing)

Enable a sender system to send messages to Cloud Integration over the HTTP protocol.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

The following figure shows the authentication options for **User Role** authorization.



Please note that image maps are not interactive in PDF outputs.

i Note

Note Related to Application Server ABAP

If an SAP system based on Application Server ABAP sends requests to Cloud Integration and there are 2 or more worker nodes enabled on Cloud Integration side, you can receive an HTTP/1.1 403 authentication error. The root cause is that the SAP kernel encodes the cookies' value by default, which breaks the load-balancing feature. To solve the issue, set profile parameter `ict/disable_cookie_urlencoding` to 1 or 2 depending on kernel level. For more information, see SAP note [2681175](#).

i Note

Note Related to Role Configuration

Depending on the chosen inbound authorization option, you define permissions for sender systems to call integration flow endpoints in different ways:

Option	Summary of Steps
Basic authentication of a user registered at an identity provider (IdP)	Use SAP BTP cockpit to define a role collection that contains the predefined role template <code>MessagingSend</code> and assign the role collection to the IdP user (under Security > Trust Configuration >). The role template <code>MessagingSend</code> is provided by default in your subaccount to define permissions for sender systems to call integration flow endpoints for this use case. See: Basic Authentication of IdP User for Integration Flow Processing
Authentication with an OAuth client (service instance)	Use the Cloud Integration Monitor application and select the User Roles tile under Manage Security. When doing this, you can either use the predefined role <code>ESBMessaging.send</code> or create a custom role. Create service instance and service key using SAP BTP cockpit. During this step, you need the role specified with the User Roles tile. See: • Client Certificate Authentication for Integration Flow Processing • OAuth with Client Credentials Grant for Integration Flow Processing • Basic Authentication with clientId and clientsecret for Integration Flow Processing

Related Information

[Creating Service Instance and Service Key for Inbound Authentication](#)

[Configuration Checklist for Inbound Authentication](#)

[Tutorial: Set Up Inbound OAuth Client Credentials Grant Authentication for Senders Calling Integration Flows with SAP-Generated Certificate](#) 

Client Certificate Authentication for Integration Flow Processing

The sender authenticates itself with a client certificate when calling the integration flow deployed on the worker node. At runtime, the system checks if a service key is available that contains the client certificate provided by the sender. If a service key is available, the system then checks if the associated service instance has a role specified that grants permissions to call the integration flow endpoint.

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

→ Tip

This authentication method is considered to be the recommended and secure option for HTTP inbound connections. Another recommended and secure option is [OAuth with Client Credentials Grant for Integration Flow Processing](#).

As client certificate, you can either use an own (*external*) certificate or one generated by SAP (see [Service Key Types](#)).

For more information, check out:

- [Client Certificate Authentication \(Inbound\), Cloud Foundry Environment](#) (explains the concepts and how this authentication option works)
- [Cloud Integration on CF – How to Setup Secure HTTP Inbound Connection with Client Certificates](#)  (SAP Community blog)

Procedure

1. Look up the role to be used to authorize the sender to call integration flow endpoint.

This role is to be specified as **User Role** parameter for the corresponding sender adapter of the integration flow to be called.

This can be either the standard role `ESBMessaging.send` or a custom role (see [Managing User Roles, Cloud Foundry Environment](#)).

To check out the roles defined for your tenant, go to the SAP Cloud Integration **Monitor** section, and under **Manage Security**, select the **User Roles** tile.

2. Get the sender client certificate from the administrator of the sender system.
3. In SAP BTP cockpit, select the subaccount that hosts your SAP Cloud Integration virtual environment and create a service instance and service key.

Proceed as described under [Creating Service Instance and Service Key for Inbound Authentication](#).

For this use case, specify the service instance and service key parameters as follows:

	Service Instance			Service Key			
Option (Certificate Type)	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
SAP certificate	integration-flow	Keep standard role <code>ESBMessaging.send</code> or use one or more custom roles.	Client Credentials	Certificate	n.a.	Specify validity in days.	Specify key size.
External certificate	integration-flow	Keep standard role <code>ESBMessaging.send</code> or use one or more custom roles.	Client Credentials	External Certificate	Add PEM-encoded X.509 certificate.	n.a.	n.a.

If you use an external certificate, the certificate needs to be signed by a certification authority that is supported by the load balancer.

See: [Load Balancer Root Certificates Supported by SAP](#)

4. Configure the sender system.
 - a. Make sure that the sender keystore contains the root certificate of the load balancer server certificate.

Get this certificate using the Cloud Integration **Connectivity Test** (pointing to the integration flow endpoint address). From downloaded `.zip` file, select the `*.cer` file of the root certificate and import this into the sender system keystore.

More information: [Using the Connectivity Test to Get the Load Balancer Server Root Certificate](#)
 - b. Make sure that the sender keystore contains a client certificate that is signed by one of the CAs supported by the load balancer.

More information: [Load Balancer Root Certificates Supported by SAP](#)

5. Configure the inbound communication for the related integration flow.

- a. Go to the SAP Cloud Integration **Design** section and edit the relevant integration flow.
- b. Create a sender channel with the adapter type that supports this authentication option, and select the connection for the associated sender adapter.
- c. For **Authorization** choose **User Role** and specify the role. You can keep the default role name **ESBmessaging.send**. You can also select a custom role if you want to use a dedicated role to control authorization to the process the integration flow.

i Note

If for **Authorization** you alternatively select **Client Certificate**, you can set up a specific variant of client certificate authentication. Using this variant, sender authorization is checked on the tenant by evaluating the subject/issuer distinguished name (DN) of the certificate (sent together with the inbound request). However, we don't recommend this option anymore because it has the following disadvantages:

- When the client certificate is renewed, the integration flow needs to be redeployed.
- Because only the DN's are checked, and not the whole certificate, the security level is decreased.

- d. After you have finished configuring the integration flow, including the processing steps for your scenario, deploy the integration flow on the tenant.

To do this, save the integration flow and select **Deploy**.

Next Steps

Configure the request from the sender to the integration flow endpoint.

With the request, the sender has to pass on a certificate chain that contains a root certificate supported by the load balancer (see [Load Balancer Root Certificates Supported by SAP](#)). Otherwise, the load balancer doesn't pass on the client certificate to SAP Cloud Integration.

- When you use an SAP-generated client certificate (with **Key Type** set to **Certificate**), the service key contains a certificate chain and a private key (see [Creating Service Instance and Service Key for Inbound Authentication](#)). The certificate chain contains already a root certificate supported by the load balancer.

You can use these values to configure the request.

i Note

To enable the related HTTP client to support this authentication option, you need to format the certificate (including the certificate chain) and the key accordingly. In particular, make sure to replace all \n in the SAP-generated certificate or key by line breaks.

A suitable certificate, for example, would then look like the following:

```
-----BEGIN CERTIFICATE-----
MIIFtDCCA5ygAwIBAgIQCUFIj6cfjiSfZi/ZvVU6IDANBgqhkiG9w0BAQsFADB5
.....
.....+
LvHPHNDM3rMsLu06agF4JTb08ANYtWQTx0PVRZKJu+8fcIaUp7MVBIVZ
-----END CERTIFICATE-----
```

- When you use an external certificate (with **Key Type** set to **External Certificate**), the service key displays only the public key certificate provided by you (see [Creating Service Instance and Service Key for Inbound Authentication](#)). To configure the

request, use the key pair exported from the application used to generate the client certificate.

OAuth with Client Credentials Grant for Integration Flow Processing

You can configure OAuth authentication, in particular the Client Credentials Grant variant, for inbound calls from sender systems to the integration platform. That way, the sender (client) application is granted access to the associated worker node through OAuth authentication.

Prerequisites

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

i Note

This option is a recommended and secure way to set up HTTP inbound connections. Another recommended and secure option is [Client Certificate Authentication for Integration Flow Processing](#).

Simply spoken, this authentication is established using the following sequent steps:

1. The sender authenticates itself at the SAP BTP token server.

There are 2 options to authenticate against the token server:

- Using clientId and clientsecret from the service key
- Using a client certificate from the service key

If you use a client certificate, you can either use an own (*external*) client certificate or a client certificate generated by SAP (see [Service Key Types](#)).

→ Tip

For a step-by-step description of how to set up this use case, check out the following tutorial:

[Set Up Inbound OAuth Client Credentials Grant Authentication for Senders Calling Integration Flows with SAP-Generated Certificate](#) 📄

⚠ Caution

If you use an *external* client certificate (**Key Type** parameter set to **External Certificate** when creating the corresponding service key, see [Creating Service Instance and Service Key for Inbound Authentication](#)), the certificate must be signed by a CA from the list specified in the following topic:

See: [Trusted Authorities for X.509 Certificates](#)

2. Token server issues access token.
3. Sender authenticates itself with access token when calling the integration flow deployed on the worker node.

For more information, check out: [OAuth Authentication with Client Credentials Grant \(Inbound\), Cloud Foundry Environment](#) (explains the concepts and how this authentication option works).

To set up this authorization option, perform the following steps.

Procedure

1. Look up the role to be used to authorize the sender to call integration flow endpoint.

This role is to be specified as **User Role** parameter for the corresponding sender adapter of the integration flow to be called.

This can be either the standard role `ESBMessaging.send` or a custom role (see [Managing User Roles, Cloud Foundry Environment](#)).

To check out the roles defined for your tenant, go to the SAP Cloud Integration **Monitor** section, and under **Manage Security**, select the **User Roles** tile.

2. In SAP BTP cockpit, select the subaccount that hosts your SAP Cloud Integration virtual environment and create a service instance and service key.

Proceed as described under [Creating Service Instance and Service Key for Inbound Authentication](#).

For this use case, specify the service instance and service key parameters as follows:

	Service Instance			Service Key			
Option (Authentication At Token Server)	Plan	Roles	Grant- types	Key Type	External Certificate	Validity	Key Size
ClientId and clientsecret	integration-flow	Keep standard role <code>ESBMessaging.send</code> or use one or more custom roles.	Client Credentials	ClientId/Secret	n.a.	n.a.	n.a.
SAP certificate	integration-flow	Keep standard role <code>ESBMessaging.send</code> or use one or more custom roles.	Client Credentials	Certificate	n.a.	Specify validity in days.	Specify key size.
External certificate	integration-flow	Keep standard role <code>ESBMessaging.send</code> or use one or more custom roles.	Client Credentials	External Certificate	Add PEM-encoded X.509 certificate.	n.a.	n.a.

3. Configure the sender system.

Make sure that the sender keystore contains the root certificate of the load balancer server certificate.

Get this certificate using the SAP Cloud Integration **Connectivity Test** (pointing to the integration flow endpoint address). From downloaded .zip file, select the *.cer file of the root certificate and import this into the sender system keystore.

More information: [Using the Connectivity Test to Get the Load Balancer Server Root Certificate](#)

4. Configure inbound communication for the related integration flow.

- a. Go to the SAP Cloud Integration **Design** section and edit the relevant integration flow.

- b. Create a sender channel with the adapter type that supports this authentication option, and select the connection for the associated sender adapter.
- c. For **Authorization**, choose **User Role** and specify the role. Keep the role name **ESBmessaging.send** pre-entered by default in the **User Role**. You can also select a custom role if you want to use a dedicated role to control authorization to the process the integration flow.
- d. After you have finished configuring the integration flow, including the processing steps for your scenario, deploy the integration flow on the tenant.

To do this, save the integration flow and choose **Deploy**.

Next Steps

When you've accomplished the configuration steps below, you've generated a service key that contains the following information:

- When using `clientId` and `clientsecret` to call token server:

Service key contains OAuth client credentials (`clientId` and `clientsecret`) and the URL of the OAuth authorization service (`tokenurl`).

- When using a client certificate to call token server:

Service key contains a client certificate and the URL of the OAuth authorization service (`tokenurl`).

To set up an OAuth workflow with the client credentials grant, you need to do the following:

We assume that you're using an HTTP client (for example, Postman) to call the integration flow endpoint.

1. Call the authorization service to get the access token for the integration flow endpoint:

In your HTTP client (calling the integration flow), set up a POST request with the following parameters:

As server address, use the following URL:

`<tokenurl from service key>?grant_type=client_credentials`

→ Tip

The `<tokenurl from service key>` part of the URL is given by value of the `tokenurl` field of the service key.

- When using `clientId` and `clientsecret` to call token server:

Choose the appropriate authentication option and make sure to pass on with the request the values of `clientId` and `clientsecret` from the service key.

- When using a client certificate to call token server:

Choose the appropriate authentication option and make sure to pass on the client certificate with the request.

With the request, the sender has to pass on a certificate chain that contains a root certificate supported by the load balancer (see [Load Balancer Root Certificates Supported by SAP](#)). Otherwise, the load balancer doesn't pass on the client certificate to SAP Cloud Integration.

- When you use an SAP-generated client certificate (with **Key Type** set to **Certificate**), the service key contains a certificate chain and a private key (see [Creating Service Instance and Service Key for Inbound Authentication](#)). The certificate chain contains already a root certificate supported by the load balancer.

You can use these values to configure the request.

i Note

To enable the related HTTP client to support this authentication option, you need to format the certificate (including the certificate chain) and the key accordingly. In particular, make sure to replace all \n in the SAP-generated certificate or key by line breaks.

A suitable certificate, for example, would then look like the following:

```
-----BEGIN CERTIFICATE-----
MIIFtDCCA5ygAwIBAgIQCUFIj6cfjiSfZi/ZvVU6IDANBgkqhkiG9w0BAQsFADB5
.....+
LvHPhNDM3rMsLu06agF4JTb08ANYtwQTx0PVrZKJu+8fcIaUp7MVBIVZ
-----END CERTIFICATE-----
```

- When you use an external certificate (with **Key Type** set to **External Certificate**), the service key displays only the public key certificate provided by you (see [Creating Service Instance and Service Key for Inbound Authentication](#)). To configure the request, use the key pair exported from the application used to generate the client certificate.

The response contains the access token.

2. Call the integration flow endpoint:

For the address of the call, enter the endpoint address of the integration flow.

Choose the appropriate authentication option and make sure to pass on with the request the access token that you retrieved as a response from the first HTTP call.

i Note

Example

When using Postman, for **Authorization**, select **OAuth 2.0** and in the **Access Token** field enter the access token that you retrieved as a response from the first HTTP call.

Basic Authentication with clientId and clientsecret for Integration Flow Processing

Use this procedure to connect a sender system to SAP Cloud Integration.

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

When you select this option, the user associated with the sender system's request is authenticated based on the user credentials (using basic authentication. `clientId` and `clientsecret`) that are generated with a service key.

i Note

This option is **not** recommended for productive scenarios.

For more information, see: [Basic Authentication](#) (explains the concepts and how this authentication option works).

In detail, perform the following steps:

→ Tip

The configuration steps to create service instance and service key are the same as for the option [OAuth with Client Credentials Grant for Integration Flow Processing](#) (when using `clientId` and `clientsecret` to call token server).

`clientId` and `clientsecret` from the service key are directly used as credentials to authenticate the sender to call the integration flow.

Procedure

1. Look up the role to be used to authorize the sender to call integration flow endpoint.

This role is to be specified as **User Role** parameter for the corresponding sender adapter of the integration flow to be called.

This can be either the standard role `ESBMessaging.send` or a custom role (see [Managing User Roles, Cloud Foundry Environment](#)).

To check out the roles defined for your tenant, go to the SAP Cloud Integration **Monitor** section, and under **Manage Security**, select the **User Roles** tile.

2. In SAP BTP cockpit, select the subaccount that hosts your SAP Cloud Integration virtual environment and create a service instance and service key. However, during runtime, no access token is retrieved from the token server.

Instead of an access token, the values of `clientId` and `clientsecret` from the service key are used as user credentials to access the integration flow endpoint.

Proceed as described under [Creating Service Instance and Service Key for Inbound Authentication](#).

For this use case, specify the service instance and service key parameters as follows:

Service Instance			Service Key			
	Roles	Grant-types		External Certificate	Validity	Key Size
integration-flow	Keep standard role <code>ESBMessaging.send</code> or use one or more custom roles.	Client Credentials	ClientId/Secret	n.a.	n.a.	n.a.

3. Configure the sender system.

Make sure that the sender keystore contains the root certificate of the load balancer server certificate.

Get this certificate using the SAP Cloud Integration **Connectivity Test** (pointing to the integration flow endpoint address). From downloaded .zip file, select the *.cer file of the root certificate and import this into the sender system keystore.

More information: [Using the Connectivity Test to Get the Load Balancer Server Root Certificate](#)

4. Configure the inbound communication for the related integration flow.

- a. Go to the SAP Cloud Integration **Design** section and edit the relevant integration flow.
- b. Create a sender channel with the adapter type that supports this authentication option, and select the connection for the associated sender adapter.
- c. For **Authorization**, choose **User Role** and specify the role. Keep the role name `ESBmessaging.send` pre-entered by default in the **User Role**. You can also select a custom role if you want to use a dedicated role to control

authorization to the process the integration flow.

- d. After you have finished configuring the integration flow, including the processing steps for your scenario, deploy the integration flow on the tenant.

To do this, save the integration flow and choose **Deploy**.

Related Information

[Basic Authentication](#)

[Defining Permissions for Senders to Process Messages on a Runtime Node](#)

Basic Authentication of IdP User for Integration Flow Processing

Use this procedure to connect a sender system to SAP Cloud Integration using basic authentication of a user registered by an identity provider (IdP).

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

⚠ Caution

This authentication option can't be used when operating SAP Cloud Integration on Alibaba Cloud.

On Alibaba Cloud, SAP ID Service isn't used as default IdP. Therefore, also basic authentication with SAP ID Service can't be used on Alibaba Cloud.

i Note

When setting up trust relationships in SAP BTP cockpit, in most cases *SAP ID service* is used as default identity provider. For more information about adding users to SAP ID Service, see [SAP ID Service](#). In the BTP cockpit, the role name to be used in the Role Collection is `MessagingSend`, which corresponds to the user role `ESBmessaging.send` on Cloud Platform Integration side.

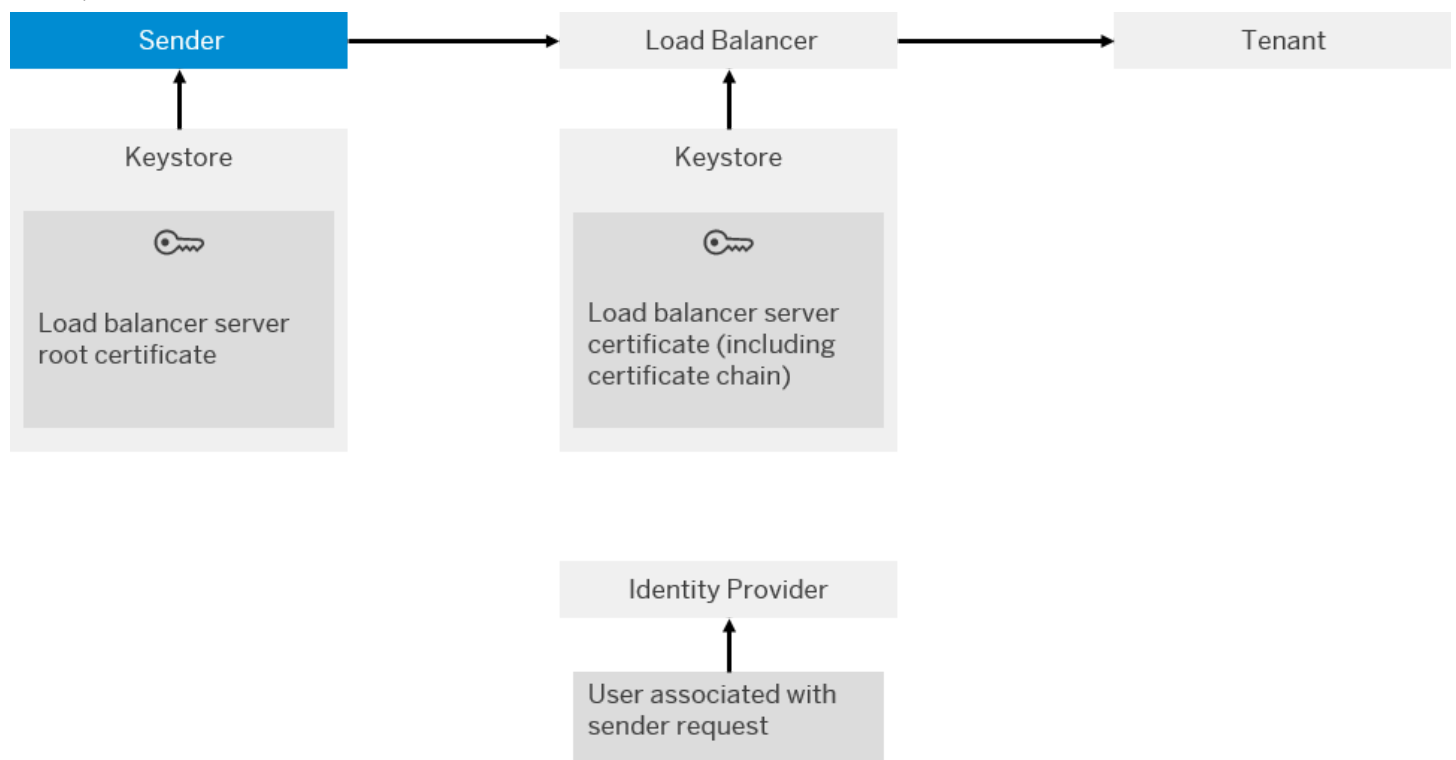
However, you've the option to define a custom IdP as your default IdP.

i Note

This option is **not** recommended for productive scenarios.

If you like to use SAP Identity Authentication Service as custom IdP, you need to make this IdP as your default IdP. To do that, perform the steps described at [Setting Up SAP Identity Authentication Service as Custom IdP for Basic Authentication, Cloud Foundry Environment](#).

The following figure shows the components and the involved security artifacts:



The table provides an overview of the required digital keys and their purpose in the authentication process, and summarizes the required configuration steps. Note that when setting up secure communication of different systems, typically administrators associated with the different systems need to accomplish configuration tasks in a coordinated way and to exchange public keys.

Security Artifact	Used to ...	Configuration Steps
Load balancer server root certificate	Make the sender trust the load balancer.	Sender administrator: Get certificate using the Cloud Integration Connectivity Test (pointing to endpoint address of integration flow).
Load balancer server certificate (including certificate chain)	Qualify load balancer as trusted component (for senders that like to connect to it).	No action required as this artifact is maintained by the operator of the cloud infrastructure.

Using SAP BTP cockpit, assign to the user a role that is to be used to authorize the sender to call the integration flow endpoint. You can either assign the predefined role `ESBMessaging.send` or a custom role.

In detail, perform the following steps.

Procedure

1. In SAP BTP cockpit, select the subaccount.
2. Select **Security** > **Role Collections**.
3. Click the **+** icon to create a new role collection.
4. In the **Create Role Collection** dialog, enter a name and (optional) a description of the role collection and choose **Create**.
5. Select the newly created role collection.
6. Click **Edit**.
7. Open the dropdown list of field **Role Name** and select the desired combination of role (left entry) and application identifier (right entry).

Select the role that grants permission for a sender to call the integration flow. By default, the role template **MessagingSend** is available (which corresponds to the predefined role **ESBMessaging . send**).

→ Tip

This predefined role grants permissions to call an integration flow endpoint.

You can also define a custom role for that purpose. To do that, go to the SAP Cloud Integration **Monitor** section and under **Manage Security** select the **User Roles** tile. On the next screen, you can create a new role.

More information: [Managing User Roles, Cloud Foundry Environment](#)

After you've performed that step, you find the newly defined role in SAP BTP cockpit next to the application starting with **it - rt** (when selecting the subaccount under **► Security ► Roles ►**). Using this role, you can define a role collection in the same way as described for the previous steps. When configuring the sender adapter of the integration flow, you can select the newly defined role. That way, you control access to individual integration flows using separate custom roles.

→ Tip

The **Application Identifier** parameter identifies the SAP BTP application. The **it - rt** application represents Cloud Integration when accessed at runtime.

Note that remote components can connect to Cloud Integration at different levels, where the level is expressed by different application identifiers.

- To configure access to Cloud Integration resources as a dialog user (designing integration flows, for example) or an API client, you connect the remote system to an application with an **Application Identifier** starting with **it !**.
- To configure access to Cloud Integration runtime resources (integration flows) from a sender, you need to connect the sender to an **it - rt** application.

8. Choose **Add**.

9. Choose **Save**.

10. In the navigation area, select **► Security ► Trust Configuration ►**.

11. Select **Default identity provider**.

12. In the navigation area, select **► Security ► Users ►**. Enter the email address of the IdP user and click **Show Assignments**.

If the user isn't known yet to the tenant, a message is displayed. Confirm the message by choosing **Add User**.

13. Choose **Assign Role Collection**.

14. Select the role collection that you defined and choose **Assign Role Collection**.

i Note

Due to caching mechanisms in SAP Cloud Integration, it can take up to 1 hour until role changes are considered.

15. In the sender adapter of the integration flow that you want to call for **Authorization**, select **User Role**.

Results

A sender can now call the integration flow endpoints using the IdP user with the configured settings.

Setting Up Inbound HTTP Connections (for API Clients)

An application programming interface (API) allows you to access Cloud Integration data, for example, monitoring data.

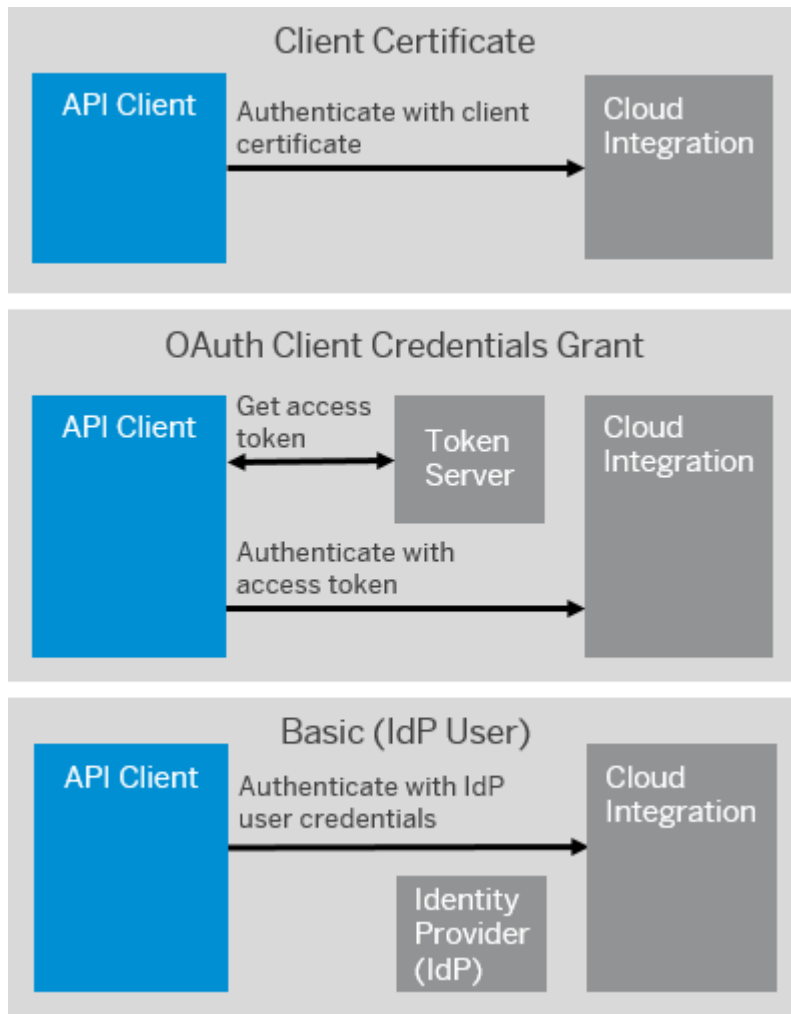
i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

When configured, an API client sends an HTTP request to the OData API of Cloud Integration to access certain resources. For example, you can access message processing logs stored on the Cloud Integration tenant.

For more information on the available Cloud Integration API resources, see [API Details](#).

There are different options for the API client to authenticate itself against Cloud Integration.



Please note that image maps are not interactive in PDF outputs.

Related Information

[Configuration Checklist for Inbound Authentication](#)

[Creating Service Instance and Service Key for Inbound Authentication](#)

[OData API](#)

[Tutorial: Set Up Inbound OAuth Client Credentials Grant Authentication for API Clients with SAP-Generated Certificate](#)

Client Certificate Authentication for API Clients

The API client authenticates itself with a client certificate when calling the Cloud Integration OData API. At runtime, the system checks if a service key is available that contains the client certificate provided by the sender. If a service key is available, the system then checks if the associated service instance has a role specified that grants permissions to call the API resource.

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

→ Tip

This authentication method is considered a secure option for HTTP inbound connections.

As client certificate, you can either use an own (*external*) certificate or one generated by SAP, see [Service Key Types](#).

! Restriction

External certificates are not supported for Edge Integration Cell local API.

For more information, check out [Client Certificate Authentication \(Inbound\), Cloud Foundry Environment](#) (explains the concepts and how this authentication option works).

Procedure

1. Look up the role to be used to authorize the API client to access the related Cloud Integration resource using the API.

See: [Tasks and Permissions](#)

2. In SAP BTP cockpit, select the subaccount that hosts your SAP Cloud Integration virtual environment and create a service instance and service key.

Proceed as described under [Creating Service Instance and Service Key for Inbound Authentication](#).

For this use case, specify the service instance and service key parameters as follows:

Option (Certificate Type)	Service Instance			Service Key			
	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
SAP certificate	api	Select role according to the API resource to access. See: Tasks and Permissions	Client Credentials	Certificate	n.a.	Specify validity in days.	Specify key size.
External certificate	api	Select role according to the API resource to access. See: Tasks and Permissions	Client Credentials	External Certificate	Add PEM- encoded X.509 certificate.	n.a.	n.a.

Next Steps

Configure the request from the API client to the Cloud Integration OData API, see [HTTP Calls and URI Components](#).

With the request, the API client has to pass on a certificate chain that contains a root certificate supported by the load balancer, see [Load Balancer Root Certificates Supported by SAP](#). Otherwise, the load balancer doesn't pass on the client certificate to SAP Cloud Integration.

- When you use an SAP-generated client certificate (with **Key Type** set to **Certificate**), the service key contains a certificate chain and a private key, see [Creating Service Instance and Service Key for Inbound Authentication](#). The certificate chain contains already a root certificate supported by the load balancer.

You can use these values to configure the request.

i Note

To enable the related HTTP client to support this authentication option, you need to format the certificate (including the certificate chain) and the key accordingly. In particular, make sure to replace all \n in the SAP-generated certificate or key by line breaks.

A suitable certificate, for example, would then look like the following:

```
-----BEGIN CERTIFICATE-----
MIIFtDCCA5ygAwIBAgIQCUFIj6cfjiSfZi/ZvVU6IDANBgkqhkiG9w0BAQsFADB5
.....
.....+
LvHPHNDM3rMsLu06agF4JTb08ANYtWQTx0PVrZKJu+8fcIaUp7MVBIVZ
-----END CERTIFICATE-----
```

- When you use an external certificate (with **Key Type** set to **External Certificate**), the service key displays only the public key certificate provided by you, see [Creating Service Instance and Service Key for Inbound Authentication](#). To configure the request, use the key pair exported from the application used to generate the client certificate.

OAuth with Client Credentials Grant for API Clients

The API is protected by OAuth and role-based authorization.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

i Note

This option is a recommended and secure way to set up HTTP inbound connections.

Simply spoken, this authentication is established using the following sequent steps:

1. The API client authenticates itself at token server.

There are 2 options to authenticate against the token server:

- Using clientId and clientsecret from the service key
- Using a client certificate from the service key

If you use a client certificate, you can either use an own ("external") certificate or one generated by SAP.

2. Token server issues access token.

3. API client authenticates itself with access token when calling the OData API.

For more information, check out: [OAuth Authentication with Client Credentials Grant \(Inbound\), Cloud Foundry Environment](#) (explains the concepts and how this authentication option works).

To set up this authorization option, perform the following steps.

1. Look up the role to be used to authorize the API client to access the related Cloud Integration resource using the API.

See: [Tasks and Permissions](#)

2. In SAP BTP cockpit, select the subaccount that hosts your SAP Cloud Integration virtual environment and create a service instance and service key.

Proceed as described under [Creating Service Instance and Service Key for Inbound Authentication](#).

For this use case, specify the service instance and service key parameters as follows:

Option (Authentication At Token Server)	Service Instance			Service Key			
	Plan	Roles	Grant-types	Key Type	External Certificate	Validity	Key Size
ClientId and clientsecret	api	Select role according to the API resource to access. See: Tasks and Permissions	Client Credentials	ClientId/Secret	n.a.	n.a.	n.a.
SAP certificate	api	Select role according to the API resource to access. See: Tasks and Permissions	Client Credentials	Certificate	n.a.	Specify validity in days.	Specify key size.
External certificate	api	Select role according to the API resource to access. See: Tasks and Permissions	Client Credentials	External Certificate	Add PEM- encoded X.509 certificate.	n.a.	n.a.

When you've accomplished the configuration steps, you've generated a service key that contains the following information:

- When using `clientId` and `clientsecret` to call token server:

Service key contains OAuth client credentials (`clientId` and `clientsecret`) and the URL of the OAuth authorization service (`tokenurl`).

- When using a client certificate to call token server:

Service key contains a client certificate (PEM-encoded) and the URL of the OAuth authorization service (`tokenurl`).

To set up a sequence of requests for an OAuth workflow with the client credentials grant, you need to do the following.

i Note

For the following instructions, we assume that you're using an HTTP client (for example, Postman) to call the integration flow endpoint.

1. Call the authorization service to get the access token for the integration flow endpoint:

In your HTTP client (calling the integration flow), set up a POST request with the following parameters:

As server address, use the following URL:

`<tokenurl from service key>?grant_type=client_credentials`

- When using `clientId` and `clientsecret` to call token server:

Choose the appropriate authentication option and make sure to pass on with the request the values of `clientId` and `clientsecret` (from the service key).

- When using a client certificate to call token server:

Choose the appropriate authentication option and make sure to pass on with the request the client certificate.

With the request, the API client has to pass on a certificate chain that contains a root certificate supported by the load balancer (see [Load Balancer Root Certificates Supported by SAP](#)). Otherwise, the load balancer doesn't pass on the client certificate to SAP Cloud Integration.

- When you use an SAP-generated client certificate (with **Key Type** set to **Certificate**), the service key contains a certificate chain and a private key (see [Creating Service Instance and Service Key for Inbound Authentication](#)). The certificate chain contains already a root certificate supported by the load balancer.

You can use these values to configure the request.

i Note

To enable the related HTTP client to support this authentication option, you need to format the certificate (including the certificate chain) and the key accordingly. In particular, make sure to replace all `\n` in the SAP-generated certificate or key by line breaks.

A suitable certificate, for example, would then look like the following:

```
-----BEGIN CERTIFICATE-----
MIIFtDCCA5ygAwIBAgIQCUFIj6cfjiSfZi/ZvVU6IDANBgkqhkiG9w0BAQsFADB5
.....
.....+
LvHPhNDM3rMsLu06agF4JTb08ANYtWQTx0PVRZKJu+8fcIaUp7MVBIVZ
-----END CERTIFICATE-----
```

- When you use an external certificate (with **Key Type** set to **External Certificate**), the service key displays only the public key certificate provided by you (see [Creating Service Instance and Service Key for Inbound Authentication](#)).

[Authentication](#)). To configure the request, use the key pair exported from the application used to generate the client certificate.

The response contains the access token.

2. Call the OData API:

For the address of the call, enter the address of the OData API resource and the query options (see [HTTP Calls and URI Components](#)).

See: [OData API](#)

Choose the appropriate authentication option and make sure to pass on with the request the access token that you retrieved as a response from the first HTTP call.

Basic Authentication of an IdP User for API Clients

Using this option, the API client is authenticated based on user credentials associated with a user registered at an identity provider (IdP).

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

⚠ Caution

This authentication option is supported for compatibility reasons.

However, it is not recommended to use this option for API calls in productive environments.

Therefore, if possible, use one of the other supported authentication options:

- [Client Certificate Authentication for API Clients](#)
- [OAuth with Client Credentials Grant for API Clients](#)

⚠ Caution

This authentication option can't be used when operating SAP Cloud Integration on Alibaba Cloud.

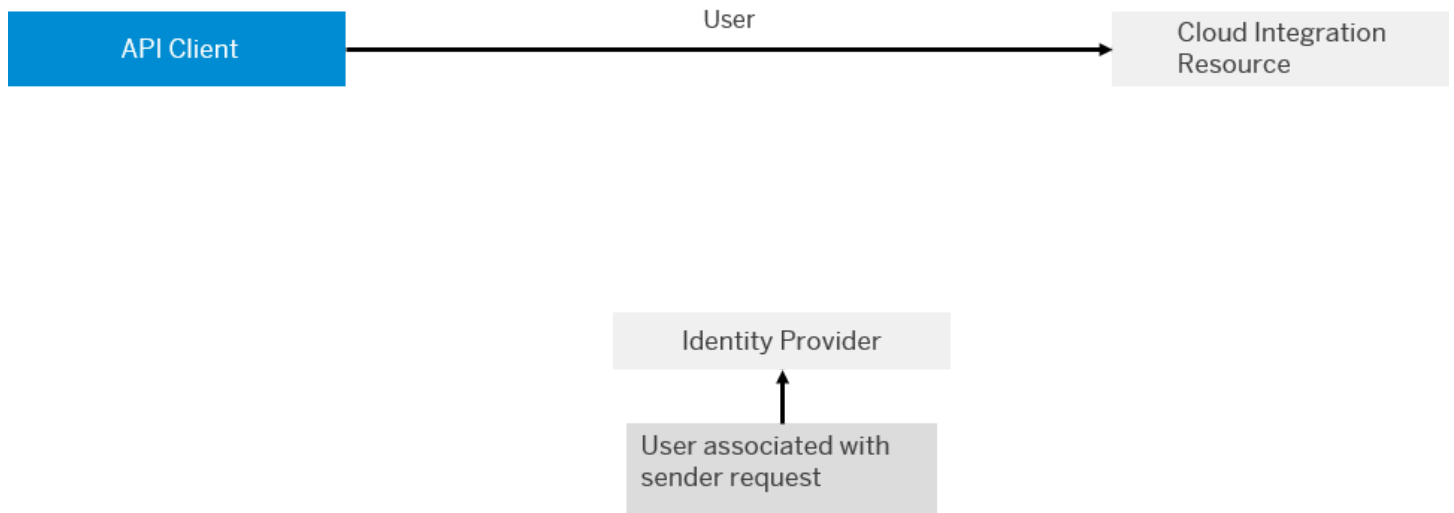
On Alibaba Cloud, SAP ID service isn't used as default IdP. Therefore, also basic authentication with SAP ID Service can't be used on Alibaba Cloud.

i Note

When setting up trust relationships in SAP BTP cockpit, in most cases *SAP ID service* is used as default identity provider. However, you've the option to define a custom IdP as your default IdP.

If you like to use SAP Identity Authentication Service as custom IdP, you've to make this IdP as your default IdP. To do that, perform the steps described at [Setting Up SAP Identity Authentication Service as Custom IdP for Basic Authentication, Cloud Foundry Environment](#).

The following figure shows the components and the involved security artifacts:



Using SAP BTP cockpit, assign to the user a role that is to be used to authorize the API client to call the OData API. Which role you assign, depends on the Cloud Integration resource you like to access through the API. For more information, see [API Details](#).

In detail, perform the following steps.

1. Using SAP BTP cockpit, select your subaccount.
2. Go to ► **Security** ► **Role Collections** ►
3. Choose **Create (+)** to create a new role collection.
4. Specify a role collection name (for example, **MonitoringAPI**).

Let's assume that you want to access monitoring information with the OData API (using the **MessageProcessingLogs** resource).

5. Choose **Create**.
6. Select the newly created role collection.
7. Select **Edit**.
8. In the **Role Name** drop down list, select the role you like to assign.

Assign a role that grants permission to access certain data through the API. In our example, we want to access monitoring data through the API, so we select the predefined **MonitoringDataRead** role.

For more information about the available predefined roles, see [Tasks and Permissions](#).

→ Tip

Make sure you select a role with an application identifier that starts with **it!**.

The **Application Identifier** parameter identifies the SAP BTP application.

Note that remote components can connect to Cloud Integration at different levels, where the level is expressed by different application identifiers.

- To configure access to Cloud Integration resources as a dialog user (designing integration flows, for example) or an API client, you connect the remote system to an application with an **Application Identifier** starting with **it!**.
- To configure access to Cloud Integration runtime resources (integration flows) from a sender, you need to connect the sender to an **it-rt** application.

9. Choose **Add**.

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Repeat to add more roles to the role collection, if desired.

10. Choose **Save**.

11. Go to **Security > Users**.
12. Select the user for which you like to grant API access.
13. Under **Role Collections**, select **Assign Role Collection**.
14. Select the role collection that you've created in the previous steps.
15. Choose **Assign Role Collection**.

You can now call the resource of the OData API from an API client using the credentials of the IdP user. For more information on the address of the API call, see [HTTP Calls and URI Components](#).

Basic Authentication with clientId and clientsecret for API Clients

You can use basic authentication based on the user credentials (clientId and clientsecret) that are generated with the service key.

Context

When you select this option, the user associated with the sender system's request is authenticated based on the user credentials (using basic authentication. clientId and clientsecret) that are generated with a service key.

i Note

This option is **not** recommended for productive scenarios.

For more information, see: [Basic Authentication](#) (explains the concepts and how this authentication option works).

In detail, perform the following steps:

Procedure

1. Look up the role that's used to authorize the API client to access the related Cloud Integration resource using the API. This can be either a standard role such as AuthGroup_ReadOnly, or a custom role. For more information, see [Tasks and Permissions](#).
2. In SAP BTP cockpit, select the subaccount that hosts your virtual environment and create a service instance and service key.

Proceed as described under [Creating Service Instance and Service Key for Inbound Authentication](#).

For this use case, specify the service instance and service key parameters as follows:

Service Instance			Service Key			
	Roles	Grant-types		External Certificate	Validity	Key Size

Service Instance			Service Key			
	Roles	Grant-types		External Certificate	Validity	Key Size
api	Keep a standard role such as AuthGroup_ReadOnly, or use one or more custom roles.	Client Credentials	ClientId/Secret	n.a.	n.a.	n.a.

3. Call the OData API.

- For the address of the call, enter the address of the OData API resource and the query options (see [HTTP Calls and URI Components](#)).
- Choose **Basic Authentication** as your authentication option, and make sure to use the clientid / clientsecret that you generated with the service key.

For more information, see [OData API](#).

Configuring Inbound HTTP Connections, Neo Environment

HTTP connections enable:

- Sender systems to call integration flow endpoints (through one of the adapters based on the HTTP protocol like, for example, the HTTPS adapter or the SOAP adapter)
- API clients to call the OData API

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Related Information

[Setting Up Inbound HTTP Connections \(Integration Flow Processing\), Neo Environment](#)

[Setting Up Inbound HTTP Connections \(for API Clients\), Neo Environment](#)

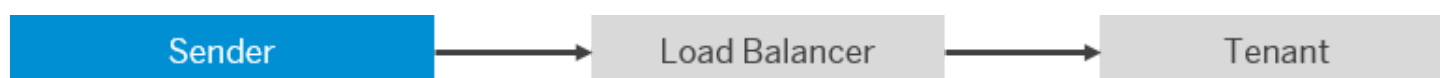
Setting Up Inbound HTTP Connections (Integration Flow Processing), Neo Environment

You can use various sender adapters (for example, the SOAP adapters, the IDoc adapter, and the HTTP adapter) to connect the tenant to a sender system so that the sender can send messages to Cloud Integration over the HTTP protocol.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

The following figure illustrates the basic setup for HTTP inbound communication:



Authorization	Authentication	Description	How to configure (checklist) ...
User role	Client-Certificate with certificate-to-user mapping	Load balancer authenticates sender based on a client certificate and, if the check is successful, forwards the certificate's issuer and subject DNs to the tenant in the message header. Tenant evaluates if a certificate-to-user mapping is defined (for the certificate) and, if so, checks whether the user (derived from the certificate-to-user mapping) is authorized to process the integration flow on the tenant. This step is performed based on user-to-role assignments (defined for the subaccount for the runtime node) and by checking the user role specified in the sender adapter. i Note The client certificate is validated at the SAP BTP platform load balancer before the request reaches the Cloud Integration runtime. The certificate must include the Client Authentication Extended Key Usage (EKU) with OID 1.3.6.1.5.5.7.3.2. Certificates without this ECU are rejected during the TLS handshake. Ensure that the required ECU is present while renewing certificates from public Certificate Authorities. Some public CAs no longer include this ECU in newly issued TLS certificates. Use certificates issued via SAP Cloud Root CA (for example via SAP BTP Destination Service) or configure a custom domain with a customer-managed certificate trust model. → Remember The maximum validity period for TLS Certificate 🗝 is being reduced in phases: Until March 15, 2026: Maximum validity is 398 days.	Sender administrator: Configure sender keystore (generate sender key pair; import CA root certificate supported by load balancer). Tenant administrator: - Define Certificate-to-User Mappings artifact (to map sender client certificate to user). - In the integration flow / sender adapter, choose the User Role authorization option and specify the role (either keep the role <code>ESBmessaging.send</code> or enter a custom role defined for the runtime node). - In SAP BTP cockpit, assign the <code>ESBMessaging.send</code> role to the user (or define own role for runtime node and assign to user).

Authorization	Authentication	Description	How to configure (checklist) ...
		• From March 15, 2026: Maximum validity is 200 days. • From March 15, 2027: Maximum validity is 100 days. • From March 15, 2029: Maximum validity is 47 days. As a result, server certificates are renewed more frequently. Configure your HTTP(S) clients to trust the appropriate Root CA instead of relying on individual server certificates. See TLS Certificate Lifetimes Will Officially Reduce to 47 Days DigiCert and Certificate Lifetime - Custom Domains i Note SAP Cloud Integration periodically renews load balancer server certificates as part of standard security maintenance. Configure your HTTP(S) clients to trust the appropriate Root CA instead of relying on a specific server certificate, intermediate certificate, or certificate chain. This allows server certificate renewals without requiring customer-side trust configuration changes. You will be notified only if a change to the trusted Root CA requires updates to their trust configuration. See 3478796 Public CAs started excluding the Client Authentication EKU from newly issued certificates in September 2025. By May 2026 it may no longer be included by default. i Note We recommend using this option for HTTP inbound connections. See also: Supported by the following sender adapters: HTTPS, IDoc, SOAP (SOAP	

Authorization	Authentication	Description	How to configure (checklist) ...
		RM, SOAP 1.x), AS2, OData More information: Client Certificate Authentication and Certificate-to-User Mapping (Inbound) , Neo Environment	
Client certificate	Client-Certificate without certificate-to-user mapping	Load balancer authenticates sender based on a client certificate and, if the check is successful, forwards the certificate's issuer and subject DNs to the tenant in the message header. Tenant checks the permissions of the sender by evaluating the certificate's subject/issuer distinguished name (DN). This is a secure option, but as the client certificate is defined for each integration flow, it has the disadvantage that each time you change (and redeploy) the integration flow, there is a brief downtime. Secondly, when the client certificate is renewed, you need to redeploy the integration flow. Supported by the following sender adapters: HTTPS, IDoc, SOAP (SOAP RM, SOAP 1.x), AS2, OData More information: Client Certificate Authentication (Inbound), Neo Environment	Sender administrator: Configure sender keystore (generate sender key pair; import CA root certificate supported by load balancer). Tenant administrator: In integration flow / sender adapter, choose Client Certificate authorization and specify the certificate.
User role	Basic	Sender sends user credentials (user name and password) in the message header. This option is not recommended for productive usage. Supported by the following adapters: HTTPS, IDoc, SOAP (SOAP RM, SOAP 1.x), AS2, OData More information: Basic Authentication	Sender administrator: Enable sender to provide user credentials with the request. Tenant administrator: - In integration flow / sender adapter, choose the User Role authorization option and specify the role (either keep the role <code>ESBmessaging.send</code> or enter a custom role defined for the runtime node). - In BTP cockpit, assign the <code>ESBMessaging.send</code> role to the user (or define own role for runtime node and assign to user).
User role	OAuth Client Credentials Grant	Grants access to resources of SAP Cloud Integration without the need to share passwords with the client. Supported by the following sender adapters: SOAP (SOAP 1.x), SOAP (SAP RM), HTTPS i Note	More information: OAuth Client Credentials Grant

Authorization	Authentication	Description	How to configure (checklist) ...
		Usage of JSON Web Token (JWT) is also supported for authentication. The advantage of using JWT is that at runtime no additional steps are required to have an identity provider validate the token. Therefore, this feature results in a better performance under high load when a token is used for multiple calls within the limit of its validity period.	
User role	OAuth SAML Bearer Destination	Grants access to resources of SAP Cloud Integration without the need to share passwords with the client. You need to set up principal propagation to forward the identity of the user associated with the sender from the sender account to the receiver account. An SAML 2.0 Bearer Assertion is used to authenticate the sender as well as to request the OAuth 2.0 access token from an OAuth 2.0 authorization server (hosted in the SAP cloud). Supported by the following sender adapters: SOAP (SOAP 1.x), SOAP (SAP RM), HTTPS	More information: OAuth SAML Bearer Destination

Related Information

[Authentication and Authorization Options \(Inbound\)](#)

[Setting Up Inbound HTTP Connections \(with Certificate-to-User Mapping\), Neo Environment](#)

[Setting Up Inbound HTTP Connections \(with Client Certificate Authentication\), Neo Environment](#)

[Setting Up Inbound HTTP Connections \(with Basic Authentication\), Neo Environment](#)

[Setting Up Inbound HTTP Connections \(with OAuth\), Neo Environment](#)

Setting Up Inbound HTTP Connections (with Certificate-to-User Mapping), Neo Environment

Using this option, authentication of a sender is performed based on a client certificate. With a certificate-to-user mapping, the certificate is mapped to a user, whose authorizations are checked on the tenant.

Prerequisites

SAP has provided you or your organization with an account and tenant. Your tenant administrator has assigned you the integration developer permissions.

Context

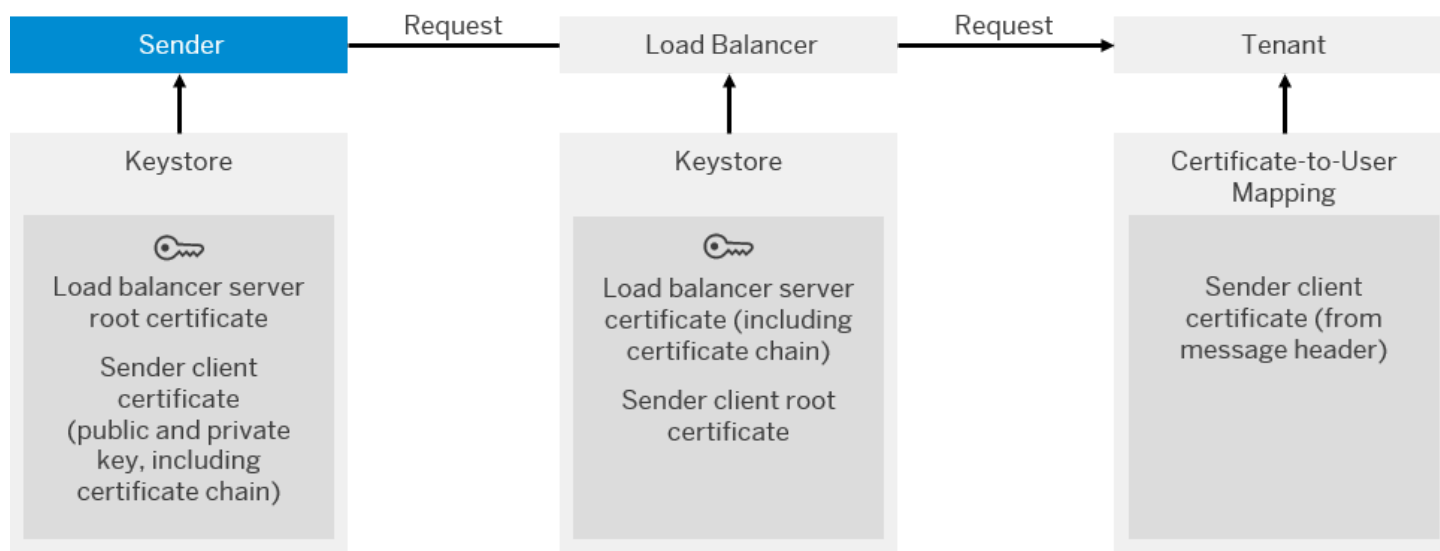
i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

i Note

This is the recommended and secure option for HTTP inbound connections.

The following figure shows the involved components and digital keys.



The table summarizes the required security artifacts required to set up this inbound authentication scenario and the configuration steps to be accomplished by the integration developer/tenant administrator and the administrator of the involved sender system.

- For an overview of the procedure how to set up this authentication option, check out the numbered list below the following table.
- For more information on how this authentication option works at runtime, check out: [Client Certificate Authentication and Certificate-to-User Mapping \(Inbound\), Neo Environment](#)
- For an end-to-end description of the procedure, check out the following blog: [Cloud Integration – How to Setup Secure HTTP Inbound Connection with Client Certificates](#)

Certificates for Inbound Message Processing

Security Artifact	Used at runtime to ...	Configuration Steps
Load balancer server root certificate	Make the sender trust the load balancer.	Sender administrator: Get certificate using the Cloud Integration Connectivity Test (pointing to endpoint address of integration flow).
Load balancer server certificate (including certificate chain)	Qualify load balancer as trusted component (for senders that like to connect to it).	No action required as this artifact is maintained by the operator of the cloud infrastructure.
Sender client certificate (public and private key, including certificate chain)	Authorize sender to call integration flow. At runtime, system checks if a Certificate-to-User Mapping artifact exists that fits to the client certificate provided by the	Tenant administrator: Creates and deploys a Certificate-to-User Mapping artifact and adds sender

Security Artifact	Used at runtime to ...	Configuration Steps
	sender. It checks if the associated user has the required permission to call the integration flow.	client certificate to it. This key pair is to be signed by a CA supported by the load balancer. Only root certificates are being imported into the load balancer keystore. Therefore, the whole certificate chain must be assigned to the certificate to enable the connected component to evaluate the chain of trust.
Sender client root certificate	Sign sender client certificate.	Sender administrator: Get sender client certificate signed by a certificate authority (CA) supported by the load balancer. The root certificates supported by the load balancer are listed at Load Balancer Root Certificates Supported by SAP.

The tenant administrator also needs to deploy a **Certificate-to-User Mapping** artifact on the tenant.

This artifact is required to map the client certificate transferred with the inbound request to a user (for which the permission to process messages have been specified).

Furthermore, the tenant administrator goes to SAP BTP cockpit and assigns a role to be used to authorize the sender to call integration flow endpoint. You can either specify the predefined role `ESBMessaging.send` or a custom role.

Procedure

1. Configure the sender system.

- a. Make sure that the sender keystore contains the root certificate of the load balancer server certificate.

Get this certificate using the Cloud Integration **Connectivity Test** (pointing to the integration flow endpoint address). From downloaded .zip file, select the *.cer file of the root certificate and import this into the sender system keystore.

More information: [Using the Connectivity Test to Get the Load Balancer Server Root Certificate](#)

- b. Make sure that the sender keystore contains a client certificate that is signed by one of the CAs supported by the load balancer.

More information: [Load Balancer Root Certificates Supported by SAP](#)

2. Authorize the user (to be related to the client certificate in the certificate-to-user mapping below) to process messages on the runtime node.

You perform user and authorization management using SAP BTP Cockpit. You have the following options:

- o Assign the user (for example, user myUser) the role `ESBMessaging.send` (predefined by SAP to define permission to process messages on the runtime node).
- o Define a custom role for the runtime node.

3. Configure role-based authorization for the related integration flow.

To open the design tool for integration flows, open a browser and enter the **Web UI URL** that you received from SAP in the mail informing you that your tenant has been provided.

To create and design integration flows, go to the **Design** tab.

- a. Open the integration flow with the integration designer and click the connection for the associated sender adapter.
- b. Choose **User Role** as the **Authorization** and specify the role against which to check inbound authorization.
- c. After you have finished configuring the integration flow (including the processing steps for your scenario), deploy the integration flow on the tenant.

4. Define the certificate-to-user mapping.

- a. Export the sender client certificate from the sender keystore to your local computer.
- b. Create a **Certificate-to-User Mappings** artifact.

You perform this step using the Web-based Monitoring application.

Use the same URL as for the integration flow design tool and go to the **Monitoring** tab.

To create a new artifact or edit an existing one for the tenant, click the **Certificate-to-User Mappings** tile under **Manage Security**.

When specifying the properties of the **Certificate-to-User Mappings** artifact, select the sender client certificate from your hard disk and enter the user that is authorized to process messages on the tenant (user myUser from above).

More information:

[Managing Certificate-to-User Mappings, Neo Environment](#)

Related Information

<https://blogs.sap.com/2017/06/05/cloud-integration-how-to-setup-secure-http-inbound-connection-with-client-certificates/> 

[Client Certificate Authentication and Certificate-to-User Mapping \(Inbound\), Neo Environment](#)

[Managing Certificate-to-User Mappings, Neo Environment](#)

[Load Balancer Root Certificates Supported by SAP](#)

[Defining Permissions for Senders to Process Messages on a Runtime Node](#)

Setting Up Inbound HTTP Connections (with Client Certificate Authentication), Neo Environment

Using this option, authentication of a sender is performed based on a client certificate (which is specified in the sender channel of the integration flow).

Context

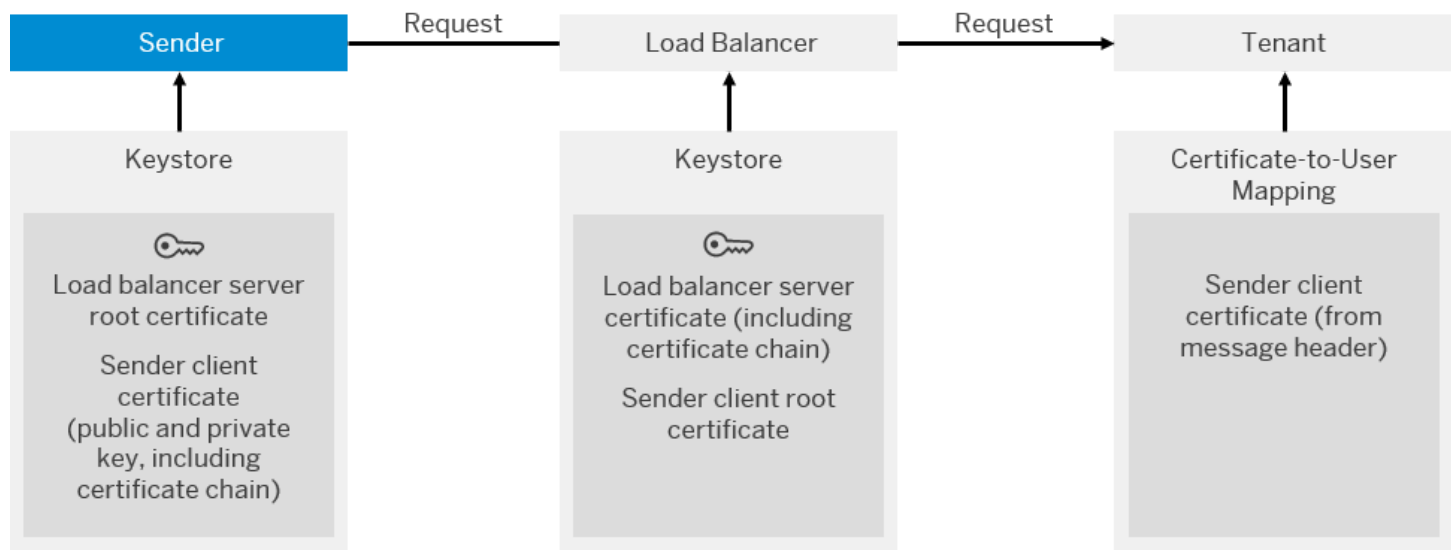
i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

i Note

This option is secure but not recommended when compared to the usage of certificate-to-user mapping (see [Setting Up Inbound HTTP Connections \(with Certificate-to-User Mapping\), Neo Environment](#)). The reason: Each certificate change requires downtime because certificate is specified as part of the integration flow.

The following figure shows the involved components and digital keys.



The table summarizes the required security artifacts required to set up this inbound authentication scenario and the configuration steps to be accomplished by the integration developer/tenant administrator and the administrator of the involved sender system.

Certificates for Inbound Message Processing

Security Artifact	Used at runtime to ...	Configuration Steps
Load balancer server root certificate	Make the sender trust the load balancer.	Sender administrator: Get certificate using the Cloud Integration Connectivity Test (pointing to endpoint address of integration flow).
Load balancer server certificate (including certificate chain)	Qualify load balancer as trusted component (for senders that like to connect to it).	No action required as this artifact is maintained by the operator of the cloud infrastructure.
Sender client certificate (public and private key, including certificate chain)	Authorize sender to call integration flow. At runtime, system checks if client certificate provided by the sender is associated with integration flow endpoint. Furthermore, system checks the permissions of the sender by evaluating the certificate's subject/issuer distinguished name.	Tenant administrator: Specifies the sender client certificate in the sender channel of the integration flow. This key pair is to be signed by a CA supported by the load balancer. Only root certificates are being imported into the load balancer keystore. Therefore, the whole certificate chain must be assigned to the certificate to enable the connected component to evaluate the chain of trust.
Sender client root certificate	Sign sender client certificate.	Sender administrator: Get sender client certificate signed by a certificate authority (CA) supported by the load balancer. The root certificates supported by the load balancer are listed at Load Balancer Root Certificates Supported by SAP .

Procedure

1. Configure the sender system.

- a. Make sure that the sender keystore contains the root certificate of the load balancer server certificate.

Get this certificate using the Cloud Integration **Connectivity Test** (pointing to the integration flow endpoint address). From downloaded .zip file, select the *.cer file of the root certificate and import this into the sender system keystore.

More information: [Using the Connectivity Test to Get the Load Balancer Server Root Certificate](#)

- b. Make sure that the sender keystore contains a client certificate that is signed by one of the CAs supported by the load balancer.

More information: [Load Balancer Root Certificates Supported by SAP](#)

2. Configure client certificate authorization for the related integration flow endpoint.

- a. Create a new integration flow or open an existing integration flow.
- b. Open the integration flow and click the connection for the associated sender adapter.
- c. Select **Client Certificate** as the **Authorization**.
- d. Choose **Add**.
- e. Choose **Select** and browse to the sender client certificate (for example, <UserID>.crt) from your local file system (or enter the **Subject DN** (information used to authorize the sender) and **Issuer DN** (information about the Certificate Authority that issues the certificate) manually).

Related Information

[Client Certificate Authentication \(Inbound\), Neo Environment](#)

[Load Balancer Root Certificates Supported by SAP](#)

[Blog: Cloud Integration – How to Setup Secure HTTP Inbound Connection with Client Certificates](#) ➦

Setting Up Inbound HTTP Connections (with OAuth), Neo Environment

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

The following use cases can be implemented for inbound communication:

- Using the OAuth Client Credentials Grant scenario to support system-to-system communication
- Using an OAuth SAML bearer destination to implement principal propagation between accounts

More information about these concepts:

[Protecting Applications with OAuth 2.0](#)

[OAuth 2.0 Specification](#) ➦

i Note

This option is supported for the following sender adapter types: SOAP (SOAP 1.x), SOAP (SAP RM), HTTPS, and OData.

Related Information

[OAuth Client Credentials Grant](#)

[OAuth SAML Bearer Destination](#)

OAuth Client Credentials Grant

You can configure OAuth authentication, in particular the Client Credentials Grant variant, for inbound calls from sender systems to the integration platform. This gives the sender (client) application access to the associated runtime node through OAuth authentication.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

It works as follows: A client requests access to a protected a virtual environment, for example, a runtime node that is to be used to process messages received by the client application. The initial request is sent to an OAuth authorization server, which is part of SAP Cloud.

After the client has been authenticated successfully by the OAuth authorization server, it's provided with the access tokens that are required to process messages on the associated runtime node. In terms of OAuth, the client uses the access token to get access to the protected resources that are represented by the virtual environment of a runtime node.

This process is executed without any manual interaction, and is therefore best suited to system-to-system communication.

For a step-by-step description of how to set up such an authentication scenario, check out the following SAP Community blog:

[Cloud Integration – Inbound HTTP Connections using OAuth Client Credentials Grant](#)👉.

→ Tip

It's highly **recommended** to use JSON Web Token (JWT) for authentication.

A JWT contains the signed information required for the authentication of the caller (for example, issuer of the token and expiry date). Therefore, the runtime node can validate the token without the need to call the authorization service. If instead of a JWT, you use an access token retrieved from the authorization service by a call without the option `&token_format=jwt` in step 3a, the situation is different: In this case, each time the related integration flow endpoint is called, the runtime node has to communicate with the authorization service to validate the identity and the authorizations of the caller. Therefore, using JWT allows you to implement more robust integration scenarios with a higher performance.

Therefore, this feature results in a better performance under high load when a token is used for multiple calls within the limit of its validity period.

The JWT provided by the OAuth authorization server contains the calling user and is digitally signed by the identity provider. Therefore, SAP Cloud Integration can validate the user information without contacting the identity provider.

More information about these concepts:

[Protecting Applications with OAuth 2.0](#)

[OAuth 2.0 Specification](#)👉

i Note

This option is supported for the following sender adapter types: SOAP (SOAP 1.x), SOAP (SAP RM), HTTPS, and OData.

Configuring OAuth with a Client Credentials Grant

In the SAP BTP cockpit, perform the following steps:

1. Register the client application as the OAuth client in the consumer account. In the **Security** >> **OAuth** section, open the **Clients** tab.

Also specify a subscription to restrict the authorizations associated with the access token on the particular runtime node.

i Note

You can only subscribe to runtime nodes with node type `iflmap` or `hcioem`.

Perform this step as described in [Register an OAuth Client](#).

To enable this security setting for the abovementioned scenario (client application sending messages to the cloud-based integration platform), specify the following information when registering the OAuth client:

- o In **Subscription**, select the VM name of the runtime node that ends with the node type, for example, `... iflmap`.

You can only register applications for node type `iflmap` or `hcioem`.

- o Enter a client ID.

You can either get a client ID from the client or you can choose one. You then have to forward this ID to the client.

- o In **Authorization Grant**, choose **Client Credentials**.
- o Enter a secret (as assigned to the client application).
- o Specify a **Token Lifetime** to increase the security level.

2. In the **Security Authorizations** section, assign the user with the name `oauth_client_<client ID>` to the `ESBMessaging.send` role in the subscription of the consumer account (for the `iflmap/hcioem` node).

Perform this step as described in [Defining Authorizations for Integration Team Members](#).

3. On the client side, perform the following steps:

- a. To get an access token in JSON Web Token (JWT) format, perform a POST HTTPS call to `https://<Token Endpoint address>?grant_type=client_credentials&token_format=jwt`.

Example:

```
https://oauthservices-<consumer-account>.<landscape host name>/oauth2/api/v1/token?grant_type=client_credentials&token_format=jwt
```

i Note

You can also perform a POST HTTPS call to the following address (without `&token_format=jwt`):

```
https://<Token Endpoint address>?grant_type=client_credentials
```

However, it is recommended that you use JSON Web Token to get a more robust scenario.

To find the Token Endpoint address, go to **Security** > **OAuth**. In the **Branding** tab of the OAuth client created in step 1, in the **OAuth URLs** section, the URL is displayed under **Token Endpoint**.

OAuth Settings

Branding
Clients
Platform API

Branding

Logo Image: [Browse...](#)

☒ Default Theme
☐ Custom Theme

Background Color:

Button Color:

Button Text Color:

[Save](#)

OAuth URLs

Authorization Endpoint: [https://\[redacted\]](#)

Token Endpoint: [https://\[redacted\]/token](#)

End User UI: [https://\[redacted\]](#)

Use basic authentication where the client ID is the user and the secret is the password. This call returns the access token.

Example:

Sample Code

```
{
  "access_token": "8271a067 .... 07c6880",
  "token_type": "Bearer",
  "expires_in": 0,
  "scopes": []
}
```

- b. Perform an HTTPS call to the endpoint URI with the HTTP header with the name “Authorization” and value “Bearer <access token>”.

You can repeat the call several times before the access token becomes invalid. Then execute step **a.** again.

OAuth SAML Bearer Destination

You can enable principal propagation between different accounts.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

If you have chosen this option, the identity of the user associated with the sender (client) application is forwarded from the sender account to the receiver account. It is a prerequisite for this scenario that the authentication method *OAuth 2.0* is used, in particular, the OAuth 2.0 SAML bearer assertion flow.

A Security Assertion Markup Language (SAML) 2.0 Bearer Assertion is used to authenticate the client as well as to request the OAuth 2.0 access token from an OAuth 2.0 authorization server (hosted in the SAP cloud).

To configure the scenario, an OAuth2SAMLBearerAssertion destination has to be specified on the sender account.

More information:

- [SAML Assertion Authentication](#)
- [Principal Propagation to OAuth-Protected Applications](#)

i Note

This option is supported for the following sender adapter types: SOAP (SOAP 1.x), SOAP (SAP RM), HTTPS, and OData.

Create Connection of Sender and Receiver Account with Trusted Identity Providers

Make sure that the settings for SAML communication between SAP BTP and a trusted identity provider are specified. This communication has to be established for both the sender and receiver account.

In this way you establish a trust relationship between the sender and receiver account.

i Note

Note the following remarks related to the identity providers of the sender and receiver account:

- You can assign different identity providers to sender and receiver accounts.
- Sender account: You must **not** assign the default SAP ID Service as the identity provider.
- Receiver account: You can assign the default SAP ID Service for testing purposes. This identity provider is configured by default and has a landscape-dependent **Local Service Provider** name.

Perform the following steps for both the sender and receiver account:

To configure the settings, go to SAP BTP cockpit and choose ► **Security** ► **Trust** ►.

Proceed as described under ID Federation with the [Application Identity Provider](#).

Configure OAuth in Receiver Account

Configure the OAuth settings for the receiver account. In this way, you register the client application as the OAuth client.

Go to SAP BTP cockpit and choose ► **Security** ► **OAuth** ► (Clients tab).

Proceed as described under [OAuth 2.0 Configuration](#).

Note the following specific settings:

- As **Subscription**, select the VM name of the runtime node (that ends with the node type, for example, ... iflmap).
You can only register applications for node type iflmap or hcioem.
- Enter a client ID.

You can either get a client ID from the client or you can choose one (you then have to forward this ID to the client).

- As **Authorization Grant**, choose **Client Credentials**.
- Enter a secret (as assigned to the client application).
- Specify a **Token Lifetime** to increase the security level.

Configure Trust to Sender Local Service Provider in the Receiver Account

In the receiver account, configure a trust relationship to the sender's local service provider.

Note that here the local service provider of the sender account takes the role of an additional trusted entity provider for the receiver account.

To configure the settings, go to the SAP BTP cockpit and choose ► **Security** ► **Trust** ► (Trusted Entity Provider tab).

Proceed as described under

Proceed as described under [Application Identity Provider](#).

(subsection **Configure Trust to the SAML Identity Provider**).

As **Name**, enter the **Local Service Provider** name from the sender account.

Enter the **Signing Certificate** as specified for the sender's local service provider.

Specify User Group in Receiver Account and Enable User Group to Process Message on Runtime Node

In the receiver account, perform the following tasks:

1. Create a user group.

To configure the settings, go to the SAP BTP cockpit and choose ► **Security** ► **Authorizations** ►. On the **Groups** tab, create a new group.

2. Create a mapping of the user group to the local sender service provider.

You have the following options:

- Specify a default group, which means that all users logged in via the sender's local service provider are assigned to this user group.

To configure the settings for the default group, go to the SAP BTP cockpit and choose ► **Security** ► **Trust** ►. On the **Trusted Identity Provider** tab, go to the identity provider specified previously. On the **Groups** tab, choose **Add Default Group** and enter the name of the newly created user group.

- Define mapping rules based on the user attributes (such as e-mail address).

3. Assign the user group to the **ESBMessaging.send** role.

You perform this step to enable all users that are assigned to the user group created to execute integration flows on the runtime node application.

To configure the settings, go to the SAP BTP cockpit and choose ► **Security** ► **Authorizations** ►. On the **Groups** tab, select the group defined previously and choose **Assign**. Select the role **ESBMessaging.send**.

Create OAuth2SAMLBearerAssertion Destination in the Sender Account

To configure the settings, go to the SAP BTP cockpit and choose **Destinations**.

- As **Type** select **HTTP**.
- As **Proxy Type** select **Internet**.
- As **Authentication** select **OAuth2SAMLBearerAssertion**.
- As **Audience** enter the **Local Provider Name** of the receiver account.
- As **Client Key** specify the key that identifies the consumer to the authorization server. This key must contain the ID of the client created above.
- As **Token Service URL** enter the **OAuth token URL** for the receiver account. You can find the value to be entered in the receiver account, SAP BTP cockpit, under ► **Security** ► **OAuth** ►. On the **Branding** tab in section **OAuth URLs**, the URL is displayed under **Token Endpoint**.
- As **Token Service User** specify the user for basic authentication for the OAuth server (if required). This entry must contain the ID of the client created above.
- As **Token Service Password** specify the **Password for Token Service User** (if required). This entry must contain the secret of the confidential client.
- As **Additional Property** add the property `authnContextClassRef` with the following value:
`urn:oasis:names:tc:SAML:2.0:ac:classes:X509`.

Related Information

[Setting Up Principal Propagation \(Example Scenario\)](#)

Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment

Context

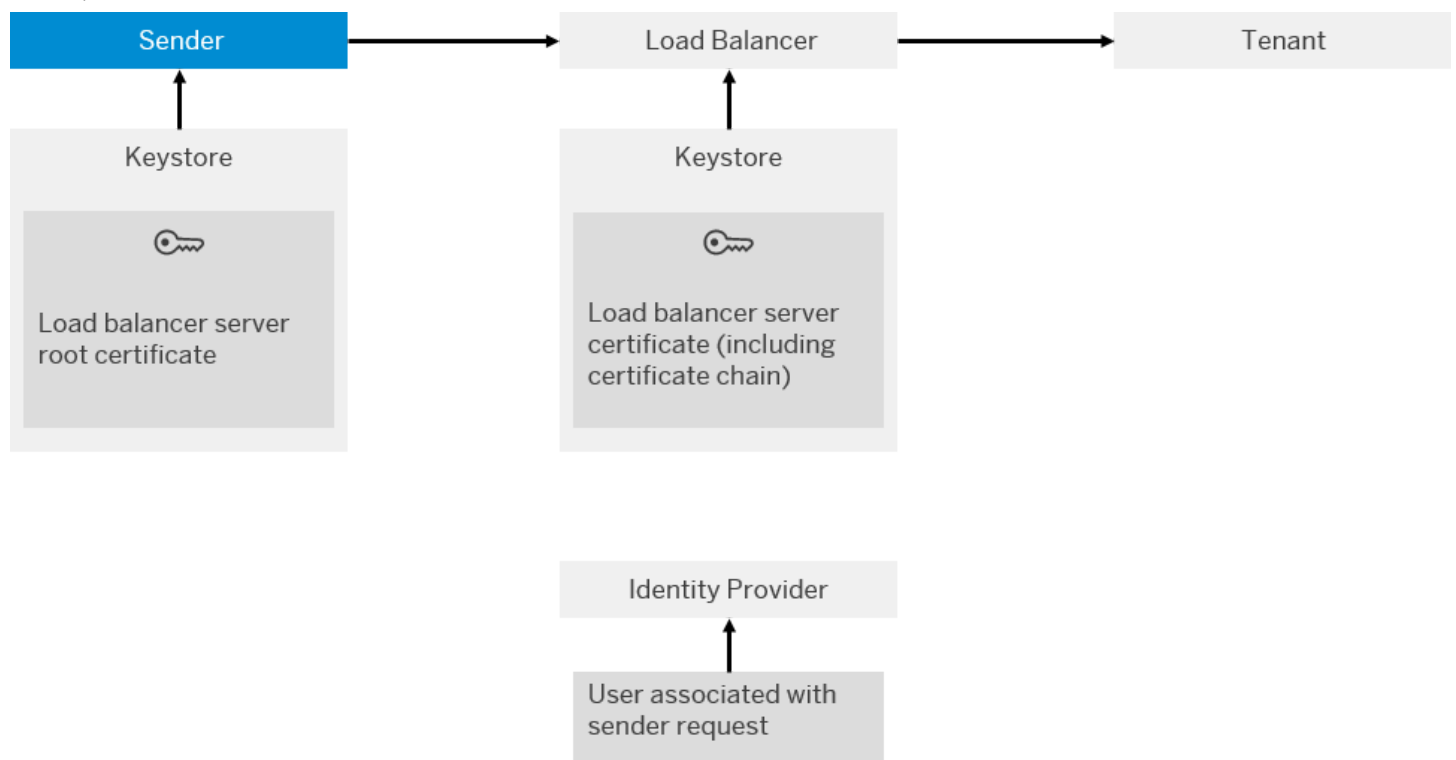
i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

i Note

This option is **not** recommended for productive scenarios.

The following figure shows the components and the involved security artifacts:



The table provides an overview of the required digital keys and their purpose in the authentication process, and summarizes the required configuration steps. Note that when setting up secure communication of different systems, typically administrators associated with the different systems need to accomplish configuration tasks in a coordinated way and to exchange public keys.

Security Artifact	Used to ...	Configuration Steps
Load balancer server root certificate	Make the sender trust the load balancer.	Sender administrator: Get certificate using the Cloud Integration Connectivity Test (pointing to endpoint address of integration flow).
Load balancer server certificate (including certificate chain)	Qualify load balancer as trusted component (for senders that like to connect to it).	No action required as this artifact is maintained by the operator of the cloud infrastructure.

Procedure

1. Configure the **sender system**.

This detailed procedure depends on the type of sender system and will not be covered here.

However, make sure to take care of the following:

- To enable the sender for this authentication option, a communication user has to be created for the sender system.
- The keystore of the sender needs to contain the load balancer server root certificate (which identifies the certification authority that has signed the load balancer server certificate).

Only with such a setup, the sender (client) can trust the load balancer's server certificate and, that way, establish a trust relationship to the load balancer (as required for HTTPS communication).

- Make sure that the message sent from the sender to the tenant contains this user in the message header.

2. Authorize the communication user of the sender system user to process messages on the runtime node.

You perform user and authorization management using SAP BTP Cockpit. You have the following options:

- o Assign to the user the role `ESBMessaging.send` (predefined by SAP to define permission to process messages on the runtime node)
- o Define a custom role for the runtime node.

3. Configure the related integration flow.

To open the design tool for integration flows, open a browser and enter the **Web UI URL** you have received from SAP in the mail that informs you that your tenant has been provided.

To create and design integration flows, go to the **Design** tab.

- a. Create a sender channel (with adapter type that supports this authentication option) and click the connection for the associated sender adapter.
- b. As **Authorization** choose **User Role** and specify the role (either keep the role name `ESBmessaging.send` pre-entered by default in the **User Role** field, or enter a custom role (in case it is available).
- c. After you have finished configuring the integration flow (including the processing steps for your scenario), deploy the integration flow on the tenant.

Related Information

[Developing Integration Content with SAP Cloud Integration](#)

[Defining Permissions for Senders to Process Messages on a Runtime Node](#)

Setting Up Inbound HTTP Connections (for API Clients), Neo Environment

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Related Information

[Setting Up OAuth Inbound Authentication with Client Credentials Grant for API Clients](#)

[Setting Up Inbound Authentication of an IdP User for API Clients](#)

Setting Up OAuth Inbound Authentication with Client Credentials Grant for API Clients

The API is protected by basic authentication and OAuth.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

1. Register the client application as the OAuth client in the consumer account using the SAP Cloud Integration SAP BTP cockpit (in the **Security** > **OAuth** section, go to the **Clients** tab).

Also specify a subscription in order to restrict the authorizations associated with the access token on the particular runtime node.

Perform this step as described under [Register an OAuth Client](#).

To enable this security setting for the above-mentioned scenario (client application sending messages to the cloud-based integration platform), specify the following information when registering the OAuth client:

- o As **Subscription**, select the application of the tenant management node (that ends with the node typetmn).

You can only register applications for node type tmn.

- o Enter a client ID.

You can either get a client ID from the client or you can choose one (you then have to forward this ID to the client).

- o As **Authorization Grant**, choose **Client Credentials**.
- o Enter a secret (as assigned to the client application).
- o Specify a **Token Lifetime** to increase the security level.

2. Assign the user with name `oauth_client_<client ID>` to the respective role in the subscription of the consumer account (for the tmn node).

To do this, select the **Security Authorizations** section.

Perform this step as described under [Defining Authorizations for Integration Team Members](#).

3. On the client side, perform the following steps:

- a. Perform a POST HTTPS call to `https://oauthservices-<consumer-account>.<landscape host name>/oauth2/api/v1/token?grant_type=client_credentials`.

For the URL part `https://oauthservices-<consumer-account>.<landscape host name>/oauth2/api/v1/token`, you can find the value that you need to enter in the receiver account, SAP BTP cockpit, under ► **Security** ► **OAuth** ► On the **Branding** tab in section **OAuth URLs**, the URL is displayed under **Token Endpoint**.

Use basic authentication where the client ID is the user and the secret is the password. This call returns the access token.

Example: **Fetch OAuth-token request:**

⌵ Sample Code

```
POST /oauth2/api/v1/token?grant_type=client_credentials HTTP/1.1
Host: oauthservices-<consumer account>.<landscape host name>
Authorization: Basic xxxxxxxxxxxxx
Accept: */*
```

Response:

⌵ Sample Code

```
{
  "access_token": "8271a067126f0aa93b46c2fe07c6880",
  "token_type": "Bearer",
  "expires_in": 0,
  "scopes": []
}
```

- b. **GET request for MPL OData API** (use the token from above):

Sample Code

```
GET /api/v1/MessageProcessingLogs HTTP/1.1
Host: <subaccount>-tmn.<landscape host name>
Authorization: Bearer 8271a067126f0aa93b46c2fe07c6880
Accept: text/xml
```

Perform an HTTPS call to the endpoint URI with the HTTP header with name “Authorization” and value “Bearer <access token>”.

i Note

For modifying calls in the Neo environment, a CSRF-Token is required in the same way as for basic authentication.

Related Information

[HTTP Calls and URI Components](#)

Setting Up Inbound Authentication of an IdP User for API Clients

Using this option, the API client is authenticated based on user credentials associated with a user registered at an identity provider (IdP).

In order to protect the API against CSRF (cross-site request forgery) attacks, modifying operations (for example, POST, DELETE) should be used in conjunction with session-based authentication and client-side CSRF handling.

It is a prerequisite that the client has HTTP cookies enabled, so that the session cookie set by the server is sent back by the client. If the client does not support HTTP cookies, the `Cookie` header can also be set manually. Before being able to execute modifying operations, the client needs to obtain a valid CSRF token from the server. This token has to be requested with a GET request by setting the `Using SAP BTP cockpit`, perform the following steps

X-CSRF-Token HTTP header with value `Fetch`. The server will then pass a CSRF token in the HTTP response header `X-CSRF-Token`. The token is only valid for the current HTTP session (identified by the session cookie) and needs to be passed in a special HTTP header (`X-CSRF-Token`) in subsequent requests.

i Note

If you use a custom IDP, refer to [Using Custom IDP with SAP Cloud Integration](#)

Example fetch request:

Sample Code

```
Using SAP BTP cockpit, perform the following
GET /api/v1/ HTTP/1.1
Host: <tmn host>.hci.int.sap.hana.ondemand.com
Authorization: Basic xxxxxxxxxxxxx
Accept: */*
X-CSRF-Token: Fetch
```

Example fetch response (headers only):

Sample Code

```

HTTP/1.1 200 OK
Cache-Control: private
Expires: Thu, 01 Jan 1970 00:00:00 UTC
Set-Cookie: JSESSIONID=3E4F4FD64EE7244E0A137292E4C758CA5B938883A5EE059472ECAB545D26E2DA; Path=/a
Set-Cookie: JTENANTSESSIONID_w1fc74894=a%2BgZSQv3aqD01H1ph6cgQTKj%2B%2FwWBH6iUvhLmed3%2Bcg%3D; D
X-CSRF-Token: 662592DAB9491668BB9844B7C3284BE7
DataServiceVersion: 1.0
Date: Tue, 06 Dec 2016 11:58:52 GMT
Content-Type: application/atomsvc+xml; charset=utf-8
Content-Length: 2730
Server: SAP

```

Example POST request (headers only):

Sample Code

```

POST /api/v1/IntegrationRuntimeArtifacts HTTP/1.1
Host: <tmn host>.hci.int.sap.hana.ondemand.com
Cookie: JSESSIONID=3E4F4FD64EE7244E0A137292E4C758CA5B938883A5EE059472ECAB545D26E2DA; BIGipServer
Accept: application/atom+xml
X-CSRF-Token: 662592DAB9491668BB9844B7C3284BE7
Content-Length: 9184
Expect: 100-continue
Content-Type: multipart/form-data; boundary=-----2fb3a93483eb87fd

```

Example POST response:

Sample Code

```

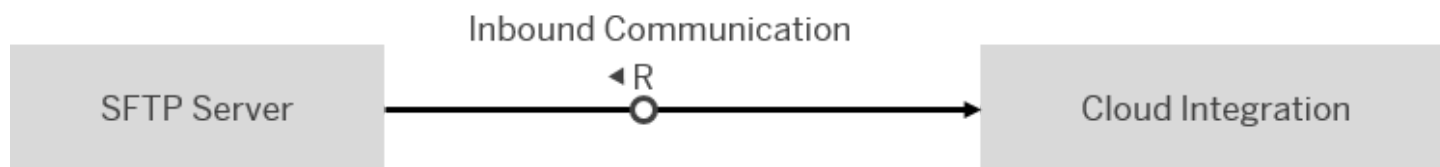
HTTP/1.1 202 Accepted
DataServiceVersion: 2.0
Date: Tue, 06 Dec 2016 11:59:22 GMT
Content-Type: application/atom+xml; charset=utf-8
Content-Length: 0
Server: SAP

```

Setting Up Inbound SFTP Connections

Using the SFTP sender adapter, you connect an SAP Cloud Integration tenant with an SFTP server so that the tenant can read data from the SFTP server (in a process referred to as polling).

In other words, the tenant sends a request to the SFTP server, but the data flow is in the opposite direction, from the SFTP server to the tenant, as illustrated in the figure. The direction of the request is indicated by the arrow next to the **R** notation in the following figure, the direction of the data flow by the direction of the connection arrow.



The following table lists the options for setting up secure connections for the different protocols. Consider the following table as a connection setup checklist. For a detailed description of the available properties for integration flow design, see the documentation of the individual adapter.

Authentication	Description	How to configure (checklist) ...
Public key	Tenant sends request to SFTP server to read files from the SFTP server (referred to as polling). SFTP server authenticates the tenant based on a public key. Using this authentication option, the user (performing the file transfer) is authenticated by the public key associated with the user. Recommended configuration option for secure communication is public key authentication. More information: How SFTP Works	Administrator of SFTP server: • Create user account. • Import public key (alias <code>id_rsa</code> or <code>id_dsa</code>, as provided by tenant administrator, see below) and import to SFTP server. Tenant administrator: • In the integration flow for the SFTP sender adapter, choose Public Key authentication and specify the User (which is defined on the SFTP server). • Add host key of the SFTP server to known hosts file and deploy (as SSH Known Hosts artifact) on the tenant. • Maintain private key pair (alias <code>id_rsa</code> or <code>id_dsa</code>) in the tenant Keystore. • Export public key (alias <code>id_rsa</code> or <code>id_dsa</code>) from Keystore and hand over to SFTP server administrator. More information: Setting Up Inbound SFTP Connections (Details)
User name/password	Tenant sends request to SFTP server to read files from the SFTP server (referred to as polling). SFTP server authenticates the tenant based on a public key. Using this authentication option, the user (performing the file transfer) is authenticated based on credentials (user name/password). More information: How SFTP Works	Administrator of SFTP server: • Create user account. Tenant administrator: • Specify user name/password in a User Credentials artifact and deploy artifact on tenant. • In the integration flow for the SFTP sender adapter, choose User Name/Password authentication and specify the User Credentials artifact (and enter the credentials there). • Add host key of the SFTP server to known hosts file and deploy (as SSH Known Hosts artifact) on the tenant.

Authentication	Description	How to configure (checklist) ...
		More information: Setting Up Inbound SFTP Connections (Details)

Related Information

[Cloud Integration - How to Setup Secure Connections to SFTP Server](#) 

[Setting Up Inbound SFTP Connections \(Details\)](#)

Setting Up Inbound SFTP Connections (Details)

Configuring the SFTP Client (Tenant)

In certain cases, you have the option to choose between the following authentication options for SFTP connectivity in the SFTP adapter:

- User Name / Password
- Public Key

The configuration depends on the chosen option.

If you have selected **Public Key**, perform the following steps:

1. Create a known hosts file and enter the required data (SFTP server host name, public key algorithm, and public key).
2. Generate an SFTP key pair and import it into the tenant keystore.
3. Deploy the keystore and the known hosts file as artifacts on the tenant.

If you have selected **User Name / Password**, perform the following steps:

1. Create a User Credentials artifact containing the credentials to be used when the SFTP client connects to the SFTP server.
2. Deploy the artifact on the tenant using the Web UI (Monitoring application).

Configuring the SFTP Server (from which data is to be read)

Configure the authorized keys file on the SFTP server. It has to contain the public key of the SFTP client (tenant).

Who performs this task depends on whether the SFTP server is hosted by the customer or by SAP.

Configuring the Integration Flow

Configure the **SFTP sender adapter** to specify the technical details of how the data is to be read from the SFTP server.

Related Information

[Creating SFTP Keys](#)

[Inbound SFTP With Public Key Authentication](#)

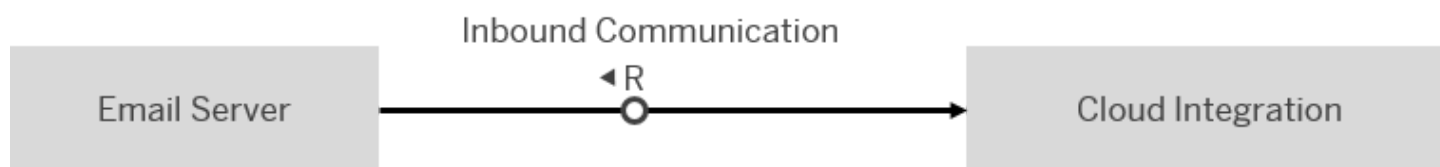
[Configure the SFTP Sender Adapter](#)

[Blog: Dynamically Configure the SFTP Receiver Adapter](#) 

Setting Up Inbound Mail Connections

Using the mail sender adapter, you connect the tenant with an e-mail server so that the tenant can read data from the e-mail server (in a process referred to as polling).

In other words, the tenant sends a request to the e-mail server, but the data flow is in the opposite direction, from the e-mail server to the tenant, as illustrated in the figure. The direction of the request is indicated by the arrow next to the **R** notation in the following figure, the direction of the data flow by the direction of the connection arrow.



Using the sender mail adapter, you can connect to mail servers through the following protocols: IMAP, POP3.

The following table lists the options for setting up secure connections for the different protocols. Consider the following table as a connection setup checklist. For a detailed description of the available properties for integration flow design, see the documentation of the individual adapter.

Authentication	Description	How to configure (checklist) ...
Encrypted user/password	The mail sender adapter can download e-mails from an e-mail server and access the e-mail body content as well as attachments. User name and password are hashed before being sent to the server.	- Create and deploy a User Credentials artifact that contains the credentials (user name and password) of the e-mail account owner. - In the integration flow / mail receiver adapter, specify the mail adapter settings. In particular, as Credential Name specify the name of the User Credentials artifact to use for this connection.
Plain user/password	The mail sender adapter can download e-mails from an e-mail server and access the e-mail body content as well as attachments. User name and password are sent in plain text (only use together with SSL or TLS).	

Related Information

[Mail Adapter](#)

Configuring Outbound Communication

Outbound communication refers to message processing from the integration platform to a remote system (where the integration platform is the client).

Configuring outbound communication means setting up the connection of a remote receiver system with the integration platform.

Receiver Systems You Can Connect to the Integration Platform

You can connect the following kinds of receiver systems to the integration platform (examples):

- A cloud application, for example, an SAP cloud application like SuccessFactors or SAP Cloud for Customer

- An on-premise application, for example SAP ERP

You can connect on-premise systems (located in the customer system landscape) such as SAP systems. Typical use cases for this are hybrid integration scenarios, where an on-premise SAP application (for example, SAP ERP) is integrated with an SAP cloud application (for example, SAP Cloud for Customer or SAP SuccessFactors).

- An e-mail server

In this case, the integration platform sends e-mails to the e-mail server (for an e-mail address specified in the related adapter).

- An SFTP server

In this case, the integration platform writes files to the SFTP server.

SAP Cloud Integration supports the following kinds of connections: HTTP connections, SFTP (SSH File Transfer Protocol) connections, and connections to an e-mail server using the mail sender adapter.

Related Information

[Setting Up Outbound HTTP Connections](#)

[Setting Up Outbound SFTP Connections](#)

[Setting Up Outbound Mail Connections](#)

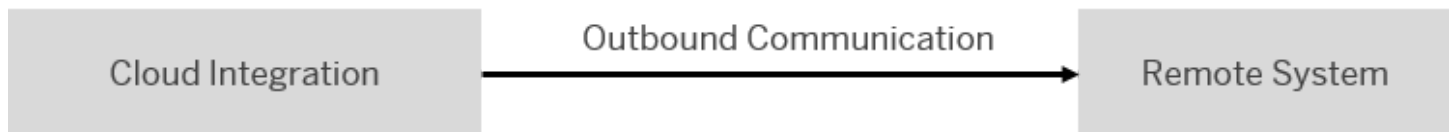
Setting Up Outbound HTTP Connections

You can use various receiver adapters (for example, the SOAP adapters, the IDoc adapter, and the HTTP adapter) to connect the tenant to a receiver system through the HTTP protocol.

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

The following figure illustrates the basic setup for HTTP outbound communication:



Outbound Connections to On-Premise Systems in the Customer Landscape

For connections like this (when SAP Cloud Integration sends a message to the on-premise system) you have to make sure that the on-premise business systems connected to the cloud are not directly exposed to the Internet.

Therefore, a further component is interconnected between the on-premise system and the integration platform in the SAP Cloud that protects the on-premise system against external calls (from the Internet).

There are two different options for this component:

- SAP Cloud Connector
- Reverse proxy (for example, SAP Web Dispatcher)

The following table lists the options for setting up secure connections for the different protocols. Consider the following table as a connection setup checklist. For a detailed description of the available properties for integration flow design, see the documentation of the individual adapters and integration flow steps.

Authentication	Description	How to configure (checklist) ...
Basic	Cloud Integration authenticates itself against receiver system is based on user credentials (username and password). Supported by the following receiver adapters: AS2 (only for Enterprise license), OData, HTTP, IDoc, ODC, SOAP (SOAP 1.x), SOAP (SAP RM), SuccessFactors More information: Basic Authentication	Receiver administrator: Configure keystore so that it contains certificate that is signed by a certification authority (CA) which is also part of the tenant keystore. Tenant administrator: • Make sure that the tenant Keystore contains receiver server root certificate. • In integration flow / sender adapter, as Authentication option choose Basic and specify name of User Credentials artifact. • Create and deploy a User Credentials artifact that contains username and password. More information: Setting Up Outbound HTTP Connections (with Basic Authentication)
Client certificate	Cloud Integration authenticates itself against receiver system is based on a client certificate. Supported by the following receiver adapters: Ariba, AS2 (only for Enterprise license), OData, HTTP, IDoc, SOAP (SOAP 1.x), SOAP (SAP RM) More information: Client Certificate Authentication (Outbound)	Receiver administrator: Configure keystore. This keystore must contain a certificate that is signed by a certification authority (CA) which is also part of the tenant keystore. Furthermore, it must contain the tenant client root certificate (that identifies CA that has signed the tenant client certificate), and a receiver server certificate (signed by CA with which the tenant has a trust relationship). Tenant administrator: • Make sure that the tenant Keystore contains receiver server root certificate (which is accepted by the receiver). • Make sure that the tenant Keystore contains a client certificate (including public and private key) and receiver server root certificate which is accepted by the receiver. • In integration flow / sender adapter, as Authentication option choose Client Certificate. You have the option to specify the User Private Key Alias (to refer to a specific key pair in the tenant keystore). If you don't specify this attribute, any valid key pair from the tenant Keystore will be used for authentication. More information: Setting Up Outbound HTTP Connections (with Client Certificate Authentication)
Principal propagation	Cloud Integration authenticates itself against the receiver system by forwarding the identity (principal) of the user (associated with the	More information: Setting Up Principal Propagation (Example Scenario)

Authentication	Description	How to configure (checklist) ...
	inbound request) to SAP Cloud Connector and from there to the receiver system (for example, an on premise SAP system). Supported by the following receiver adapters: OData, HTTP, IDoc, ODC, SOAP (SOAP 1.x), SOAP (SAP RM)	
OAuth	Cloud Integration authenticates itself against the receiver system by using OAuth.	More information: Setting Up Outbound HTTP Connections (with OAuth)
OAuth with Twitter and Facebook adapter	Using the Twitter or Facebook receiver adapter, you can connect Cloud Integration to Twitter or Facebook using OAuth. Supported by the following receiver adapters: Twitter, Facebook More information: OAuth 2.0	• Configure the Twitter or Facebook API to accept calls from Cloud Integration (during this step, you generate the OAuth credentials). • Configure the Twitter or Facebook adapter and specify the name of the Secure Parameter artifacts (for the required OAuth credentials). • Create and deploy a Secure Parameter for each required OAuth credential.

i Note

When you connect an on-premise (receiver) system to the integration platform, you need to interconnect either a reverse proxy or an SAP Cloud Connector between the on-premise system and the integration platform.

More information: [Outbound/On-Premise: Reverse Proxy or SAP Cloud Connector](#)

You can access the following link to see the list of available landscapes and respective IP addresses used by : [Landscape Hosts](#).

Related Information

[Outbound/On-Premise: Reverse Proxy or SAP Cloud Connector](#)

[Setting Up Outbound HTTP Connections \(with Basic Authentication\)](#)

[Setting Up Outbound HTTP Connections \(with Client Certificate Authentication\)](#)

[Twitter Receiver Adapter](#)

[Facebook Receiver Adapter](#)

Outbound/On-Premise: Reverse Proxy or SAP Cloud Connector

When you connect an on-premise (receiver) system to the integration platform, you need to interconnect either a reverse proxy or an SAP Cloud Connector between the on-premise system and the integration platform in the SAP Cloud.

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

Overview

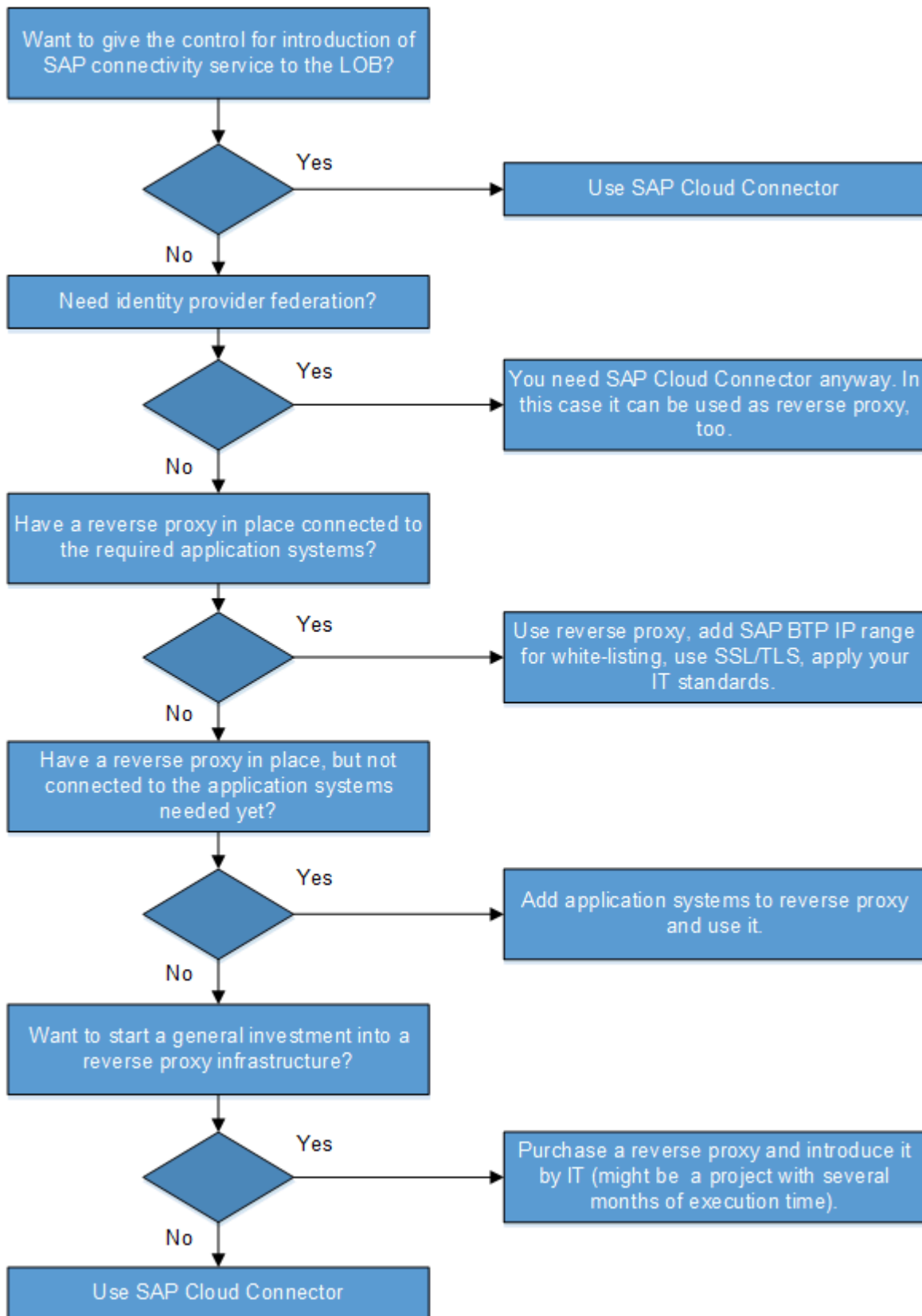
To decide which option is the best one for your use case, refer to the following table.

Decision Matrix: Reverse Proxy versus SAP Cloud Connector

Advantages When Using Reverse Proxy	Advantages When Using SAP Cloud Connector
Existing reverse proxy and demilitarized zone (DMZ) infrastructure can be reused for cloud scenarios: No additional components need to be operated on-premise.	- No inbound ports pointing to the on-premise network need to be opened. - Firewall and DMZ remain unchanged (for example, no additional IP white-listing). - Direct attacks (for example, DDOS) from the Internet are not possible.
IT-based, centralized approach with more re-use potential, independent of SAP BTP cockpit or Cloud Integration infrastructure.	De-central, simple solution that can be administered by LOBs and subsidiaries.
Usage for other cloud scenarios besides SAP BTP cockpit/Cloud Integration-connectivity of backends.	Usage for other SAP BTP cockpit-related scenarios, for example, extension apps, possible.
Additional capabilities might be provided by the reverse proxy (load balancing, application gateway, rules, and so forth, depending on the used product).	Synchronous native RFC-client call from SAP BTP cockpit supported in addition (that means, outside Cloud Integration).
Several reverse proxy instances per target landscape in one Cloud Integration tenant.	Propagation of cloud user identity to on-premise system is supported.
Monitoring and control included in the IT processes, tools and concepts.	Monitoring and control native on SAP BTP cockpit (for example, SAP BTP cockpit, User, Security).
Re-use of existing license, but separate license needed of a reverse proxy is not used yet.	License comes with Cloud Integration Standard/Professional Edition.
Third party support needed, except if the SAP Netweaver Web Dispatcher is used as reverse proxy.	SAP support in case of issues or feature requests.

Decision Graph

To decide whether to use a reverse proxy or SAP Cloud Connector, you can follow the decision graph as outlined in the following figure and described further next.



1. Independent of your current IT infrastructure setup, you want to give the control for introduction of SAP connectivity service to the line of business (LOB) - for example, because of timing, or special solution in a subsidiary or segment.

- Yes: Use SAP Cloud Connector.
- No: Go to next question.

2. Need identity provider federation?

For example, users are managed via on-premise Idap servers like Microsoft Active Directory.

- You need SAP Cloud Connector anyway. In this case it can be used as reverse proxy, too.

- No: Go to next question.

3. Have a reverse proxy in place connected to the required application systems?

- Use reverse proxy, add SAP BTP cockpit IP range for white-listing, use SSL/TLS, apply your IT standards.
- No: Go to next question.

4. Have a reverse proxy in place, but not connected to the application systems needed yet?

- Add application systems to reverse proxy and use it.
- No: Go to next question.

5. Want to start a general investment into a reverse proxy infrastructure?

- Purchase a reverse proxy and introduce it by IT.

Caution

This might be a project with several months of execution time.

- No: Use SAP Cloud Connector.

More Information

SAP Cloud Connector

[Outbound: SAP Cloud Connector](#)

[SAP Cloud Connector](#) (product documentation)

[Technical connectivity between cloud and on-premise systems via the SAP Cloud Connector](#)  (SAP Community article)

SAP Web Dispatcher

[SAP Web Dispatcher](#) (product documentation)

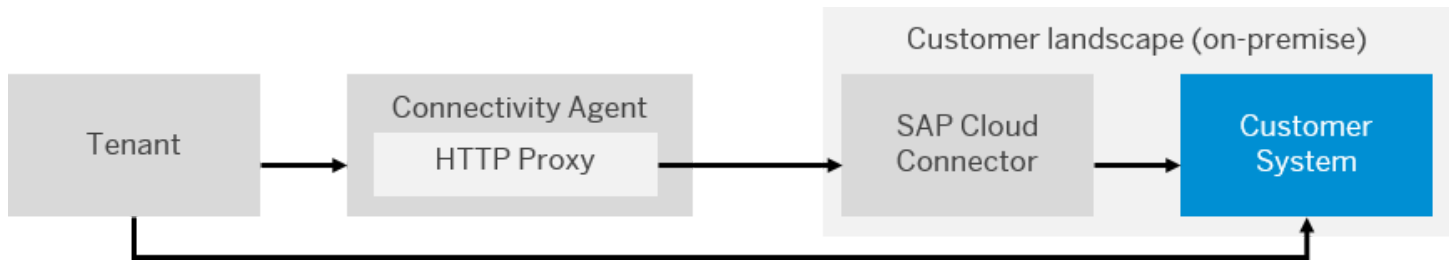
Outbound: SAP Cloud Connector

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

SAP Cloud Connector (SCC) runs as on premise agent in a secured network and acts as a reverse invoke proxy between the on premise network and SAP Cloud Integration. Due to its reverse invoke support, you don't need to configure the on premise firewall to allow external access from the cloud to internal systems.

You can configure an outbound connection from the tenant via SAP Cloud Connector (SCC). The following figure illustrates how the connection is set up and the basic components of the scenario.



You need to install and configure the SAP Cloud Connector on your on premise systems for this mode of outbound communication. For more information on installing and configuring SCC, you can refer to [Using SAP Cloud Connector with Cloud Integration Adapters](#).

See also:

[Installation and Configuration of SAP Cloud Connector](#) (SAP Community blog)

Using SAP Cloud Connector with Cloud Integration Adapters

You can use the SAP Cloud Connector (SCC) with Cloud Integration adapters to communicate with SAP on premise systems. You have to install cloud connector on the on premise system/s that you want to communicate with, and configure it. This is a one-time activity after which you can use cloud connector with Cloud Integration adapters.

Perform the following steps to use cloud connector with Cloud Integration adapters.

1. Install SAP Cloud Connector on your on premise system. For more information, see [Installing the Cloud Connector](#).
2. For SAP BTP, **Cloud Foundry** environment, you need to create a role collection for your subaccount. Follow the mentioned steps next:
 - a. From your subaccount, navigate to **Security > Role Collections**.
 - b. Choose Add role collection⁺ and enter a value for **Name**(Example: CloudConnector) and **Description** and choose **Create**.
 - c. Open the newly created role collection and choose **Edit**.
 - d. Under the **Roles** section, choose Add role collection⁺ and assign the role **Cloud Connector Administrator** from the drop-down list.
 - e. Under the **Users** section, choose Add role collection⁺ and enter the **ID** and **E-Mail** of the user to whom you need to provide access to connect to the cloud connector. Let the value of the **Identity Provider** be **Idap**.
 - f. Choose **Save**.
3. Set up mutual authentication between the cloud connector and a backend system. For more information, see [Initial Configuration](#) and [Initial Configuration \(HTTP\)](#).
4. Enable the web application to connect to access backend system on the intranet. For more information, see [Configuring Access Control \(HTTP\)](#).

You can now connect to on premise systems using Cloud Integration adapters by selecting **on-premise** value in **Proxy Type** field dropdown list.

→ Remember

Here are some important considerations while using SAP Cloud Connector with Cloud Integration adapters:

- Ensure that the receiver URL starts with `http://` while configuring the integration flow.
- Always use **Basic Authentication** as the *<Authentication Type>*.
- Ensure that you deploy the credentials that enables access to the backend system that you are trying to connect to.
- Ensure that you use the correct **Location ID** of the cloud connector that you want to establish a connection to. You can find this in the configuration of the cloud connector in the target system. For more information, you can also see this blog: [Connecting multiple Cloud Connectors to an account in SAP BTP](#)
- Ensure that the hostname has a period within it if **Session Reuse** is enabled for the integration flow. See SAP Note [0003502167](#)

Setting Up Outbound HTTP Connections (with Client Certificate Authentication)

Using this option, authentication of Cloud Integration calling a receiver is performed based on a client certificate.

Context

The following figure shows the involved components, digital keys, and storage locations. For more information on the tenant keystore that comes with Cloud Integration, see [Keystore](#).



The table summarizes the required security artifacts required to set up this inbound authentication scenario and the configuration steps to be accomplished by the integration developer/tenant administrator and the administrator of the involved sender system.

- For an overview of the procedure how to set up this authentication option, check out the numbered list below the following table.
- For more information on how this authentication option works at runtime, check out: [Client Certificate Authentication \(Outbound\)](#)

Security Artifact	Used to ...	Configuration Steps
Tenant client certificate (private/public key pair including certificate chain)	Authorize Cloud Integration to call receiver. At runtime, the identity of the Cloud Integration tenant is checked by the receiver by evaluating the client certificate chain of the tenant. i Note In many cases, there is a multilevel setup of Certification Authorities (CAs) so that a certificate is signed by an intermediate CA. The trustability of the intermediate CA is guaranteed by another intermediate CA one level higher, and so on, up to the root CA at the top of the certificate chain. In this case, it is necessary to assign the certificate chain to the certificate, to enable the connected component (which has imported only the root CA into its keystore) to evaluate the chain of trust. i Note Outbound client certificate authentication uses a certificate chain based on SAP Cloud Root CA to SAP Cloud Platform Client CA. Receiver systems that validate the certificate chain must ensure that the SAP Cloud Root CA is trusted otherwise outbound connectivity may fail after certificate renewal. Public CAs started excluding the Client Authentication EKU from newly issued certificates in September 2025. By May 2026 it may no longer be included by default.	Tenant administrator: - Generate key pair or use preinstalled one. You can use the preinstalled key pair with alias sap_cloudintegrationcertificate. This key pair is already part of the tenant keystore (provided by SAP together with the tenant). i Note This key pair is not preinstalled when you operate a Cloud Integration trial tenant. - Hand over the public key (tenant client certificate) to the receiver administrator. In the tenant keystore, check out the Key Pair entry used for this connection and can download the public part from there. More information: Keystore Managing Keystore Entries

Security Artifact	Used to ...	Configuration Steps
Tenant client root certificate (identifies CA that has signed the tenant client certificate)	Sign tenant client certificate. This certificate is required to identify the root CA that is at the top of the certificate chain that ultimately guarantees the trustability of the tenant client certificate.	Receiver administrator: • Get tenant client root certificate from tenant administrator. • Add certificate to receiver keystore.
Tenant client certificate (public key)	Check trustworthiness of the Cloud Integration tenant at the receiver side based on this certificate.	Receiver administrator: • Get tenant client certificate from tenant administrator. • Add certificate to receiver keystore.
Receiver server certificate (signed by CA with which the tenant has a trust relationship)	Qualify receiver as trusted component (for Cloud Integration tenants that like to connect to it). This certificate is required to identify the receiver (to which the tenant connects as the client) as a trusted server.	Receiver administrator: • Create server certificate (key pair) and import it into the receiver keystore. This certificate can be a certificate chain where the top-level certificate is a root certificate issued by a dedicated CA. When you configure an integration with an SAP system as receiver, check out the corresponding integration guide (which you typically find as part of the related integration content package). In such a guide, you find more information on the required certificates. • Download root certificate from the receiver keystore and make it available to the tenant administrator.
Receiver server root certificate	Make Cloud Integration trust the receiver. This certificate is required to identify the root CA that is at the top of the certificate chain that ultimately guarantees the trustability of the receiver server certificate.	Tenant administrator: Import this certificate into the tenant keystore.

In the related receiver adapter, as **Authentication** choose **Client Certificate**. Optionally, you can enter a **Private Key Alias** to specify a dedicated key to be used for this step.

Procedure

1. Maintain the tenant keystore.

To enable the tenant to authenticate itself as client against the receiver, a keystore with a valid client certificate has to be deployed on the tenant.

Note that the tenant provided initially by SAP has already a keystore deployed that contains an initial set of security artifacts. The already available key pair might be suitable to set up the outbound connection.

The keystore also has to contain a certificate of the certification authority (CA) that has signed the server certificate of the receiver system.

Import the receiver server root certificate into the tenant keystore. To get this certificate, you've the following options:

- Get certificate handed over by receiver administrator.
- Apply the outbound **Connectivity Test** against the receiver system.

More information: [Performing Connectivity Tests](#)

For an example how to get such a certificate for an email server (as receiver system), check out: [Update the Tenant Keystore with the Certificates Required by the Mail Server](#)

2. Configure the receiver keystore.

In the same way as for the tenant keystore, generate a public/private key pair, create a certificate signing request and get the certificate signed by a CA. Note that this must be the CA which root certificate is also obtained in the tenant keystore.

More information:

[Creating X.509 Keys](#)

3. Configure the security-specific settings in the related integration flow.

- a. Open the SAP Cloud Integration design section for integration flows.
- b. To create and design integration flows, go to the **Design** tab.
- c. Open the related receiver adapter (that is used to specify the connection of the tenant with the receiver system) and as **Authentication** choose **Client Certificate**.

Optionally, you can enter a **Private Key Alias** to specify a dedicated private key from the tenant keystore (tenant client certificate) to be used for this step.

Related Information

[Client Certificate Authentication \(Outbound\)](#)

Setting Up Outbound HTTP Connections (with OAuth)

Using this option, authentication of Cloud Integration calling a receiver is performed using OAuth.

Context

You can configure different OAuth grant types.

For detailed information on the supported grant types and the involved components, check out [OAuth 2.0](#).

The set of supported OAuth grant types depends on the receiver adapter type. Therefore, a step-by-step description of the required configuration steps can only be provided for a particular use case with a dedicated receiver adapter type involved.

Nevertheless, the general sequence of steps to configure this authentication option is:

Procedure

1. Get the details for OAuth connection from the receiver system to be connected. This includes, for example, the address of the token service that issues the OAuth access token on behalf of the receiver.
2. Depending on the receiver adapter type and the desired OAuth grant type to implement, create one of the following artifacts. To do that, go to the **Monitor** section and select the **Security Material** tile under **Manage Security**.
 - **OAuth2 Client Credentials** (see [Deploying an OAuth2 Client Credentials Artifact](#))
 - **OAuth2 SAML Bearer Assertion** (see [Deploying an OAuth2 SAML Bearer Assertion](#))
 - **OAuth2 Authorization Code** (see [Deploying an OAuth2 Authorization Code](#))
3. In the receiver adapter of the related integration flow, choose the corresponding **Authentication** option and specify the **Credential Name** (to point to the artifact from step 2).

Example

Check out the following SAP Community blogs to find detailed instructions how to set up scenarios with a given receiver adapter and OAuth grant type:

- [SAP Cloud Integration – OAuth2 Client Credentials Support in OData V2 Adapter](#) 
- [SAP Cloud Integration – OAuth2 SAML Bearer/X.509 Certificate Authentication Support in SuccessFactors Connector](#) 
- [Cloud Integration – Call Microsoft Graph API with OAuth 2.0 Authorization Code](#) 

Setting Up Outbound HTTP Connections (with Basic Authentication)

Context

This option was referred to as *basic authentication* in former releases. It is based on user credentials.

Procedure

1. Create and deploy a tenant keystore that contains the receiver server root certificate.
This certificate is required to identify (authenticate) the receiver system as trusted server.
2. Create and deploy the credentials on the tenant.
These are user name and password that are used to authenticate the tenant calling the receiver system.
 - a. Use the same URL like for the integration flow design tool and choose the **Monitor** section.
 - b. Click the **Security Material** tile under **Manage Security**.
 - c. To create a new **User Credentials** artifact or edit an existing one for the tenant, choose **Add**.
 - d. On the **Add User Credentials** page, enter the attributes (**Credential Name**, **User** and **Password**) and choose **OK**.
3. Configure the security-specific settings in the related integration flow.
 - a. Open the SAP Cloud Integration design section for integration flows.
 - b. To create and design integration flows, go to the **Design** section.
 - c. Open the related receiver adapter (that is used to specify the connection of the tenant with the receiver system) and as Authentication choose **Basic**; then enter the credential name.

This is the name of the **User Credentials** artifact that you have deployed on the tenant in a previous step.

4. Configure the receiver keystore.

The keystore needs to contain a certificate that is signed by a certification authority (CA) which is also contained in the tenant keystore.

More information:

[Creating X.509 Keys](#)

Related Information

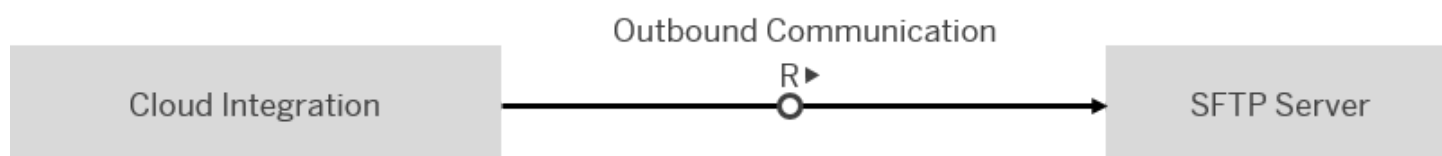
[Basic Authentication](#)

[Deploying a User Credentials Artifact](#)

Setting Up Outbound SFTP Connections

Using the SFTP receiver adapter, you connect the tenant with an SFTP server so that the tenant can write data to the SFTP server.

In other words, the tenant sends a request to the SFTP server, and the data flow is in the same direction, from the tenant to the SFTP server, as illustrated in the figure. The direction of the request is indicated by the arrow next to the **R** notation in the following figure, the direction of the data flow by the direction of the connection arrow.



The following table lists the options for setting up secure connections. Consider the following table as a connection setup checklist. For a detailed description of the available properties for integration flow design, see the documentation of the individual adapter.

Authentication	Description	How to configure (checklist) ...
Public Key	Public Key authentication (recommended): Tenant sends request to SFTP server to write files to the SFTP server. SFTP server authenticates the tenant based on a public key. With this is authentication option, the user (performing the file transfer) is authenticated by the public key associated with the user. More information: How SFTP Works	Administrator of SFTP server: • Create user account and provide user to the tenant admin. • Import public key (as provided by tenant administrator, see below) and import to SFTP server. Tenant administrator: • Maintain private key pair (Create or Add) in the tenant Keystore. Provide an alias and reuse this alias in the subsequent steps. • In the integration flow for the SFTP receiver adapter, choose Public Key, provide the alias of the key in the key store and specify the User (which is defined on the SFTP server and provided by the server admin).

Authentication	Description	How to configure (checklist) ...
		• Add host key of the SFTP server to known hosts file and deploy (as SSH Known Hosts artifact) on the tenant. • Export public key (the very same which you've just created or added) from Keystore and hand over to SFTP server administrator. More information: Setting Up Inbound SFTP Connections (Details).
User name/password	Tenant sends request to SFTP server to write files to the SFTP server. SFTP server authenticates the tenant based on a public key. Using this authentication option, the user (performing the file transfer) is authenticated based on credentials (user name/password). Supported by SFTP sender adapter. More information: How SFTP Works	Administrator of SFTP server: • Create user account. Tenant administrator: • Specify user name/password in a User Credentials artifact and deploy artifact on tenant. • In the integration flow for the SFTP receiver adapter, choose User Name/Password authentication and specify the User Credentials artifact (and enter the credentials there). • Add host key of the SFTP server to known hosts file and deploy (as SSH Known Hosts artifact) on the tenant. More information: Setting Up Inbound SFTP Connections (Details).

Related Information

[Setting Up Outbound SFTP Connections \(Details\)](#)

Setting Up Outbound SFTP Connections (Details)

Configuring the SFTP Client (Tenant)

In certain cases, you have the option to choose between the following authentication options for SFTP connectivity in the SFTP adapter:

- User Name/Password
- Public Key

The configuration depends on the chosen option.

When you've selected **Public Key**, perform the following steps:

1. Create a known hosts file and enter the required data (SFTP server host name, public key algorithm, and public key).

2. Generate an SFTP key pair and import it into the tenant keystore.
3. Deploy the keystore and the known hosts file as artifact on the tenant.

When you've selected **User Name/Password**, perform the following steps:

1. Create a User Credentials artifact that contains the credentials based on which the SFTP client connects to the SFTP server.
2. Deploy the artifact on the tenant using the Web UI (Monitoring application).

i Note

You can also load a known_hosts file from the Partner Directory. To point to the Partner Directory content, you need to set the following property in the integration flow before calling the SFTP receiver adapter:

`SAP_FtpPdUri`

You can use, for example, a Content Modifier for this purpose.

The value of the property needs to apply to the following format:

`pd:partnerId:parameterId:Binary`

Configuring the SFTP Server (To Which Data Is to Be Written)

Configure the authorized keys file on the SFTP server. It has to contain the public key of the SFTP client (tenant).

Who performs this task depends on whether the SFTP server is hosted by the customer or by SAP.

Configuring the Integration Flow

Open the related integration flow and configure the **SFTP receiver adapter** to specify the technical details of how the data is to be written to the SFTP server.

Related Information

[Blog: Dynamically Configure the SFTP Receiver Adapter](#) 

[How SFTP Works](#)

[Creating SFTP Keys](#)

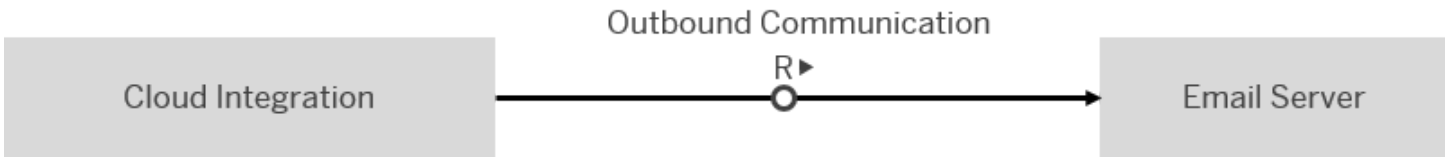
[Configure the SFTP Receiver Adapter](#)

[Outbound SFTP With Public Key Authentication](#)

Setting Up Outbound Mail Connections

Using the mail receiver adapter, you connect the tenant with an e-mail server so that the tenant can send emails to the e-mail server.

In other words, the tenant sends a request to the e-mail server, and the data flow is in the same direction, from the tenant to the e-mail server, as illustrated in the figure. The direction of the request is indicated by the arrow next to the **R** notation in the following figure, the direction of the data flow by the direction of the connection arrow.



Using the mail receiver adapter, you can connect to mail servers through the SMTP protocol.

The following table lists the options for setting up secure connections. Consider the following table as a connection setup checklist. For a detailed description of the available properties for integration flow design, see the documentation of the individual adapter.

Authentication	Description	How to configure (checklist) ...
Encrypted user/password	User name and password are hashed before being sent to the server.	• Create and deploy a User Credentials artifact that contains the credentials (user name and password) of the e-mail account owner. • In the integration flow / mail receiver adapter, specify the mail adapter settings. In particular, as Credential Name specify the name of the User Credentials artifact to use for this connection.
Plain user/password	User name and password are sent in plain text (only use together with SSL or TLS).	

Related Information

[Mail Adapter](#)

Detailed Steps

Related Information

- [Creating X.509 Keys](#)
- [Creating SFTP Keys](#)
- [Creating Keys for Message Level Security](#)
- [Securely Exchanging Key Material](#)
- [Using the Connectivity Test to Get the Load Balancer Server Root Certificate](#)

Setting Up SAP Identity Authentication Service as Custom IdP for Basic Authentication, Cloud Foundry Environment

You can set up SAP's custom identity provider (IdP) to configure basic inbound authentication for sender systems to call an integration flow or for API clients to access the OData API.

When setting up trust relationships in SAP BTP cockpit, in most cases SAP ID service is used as default identity provider. However, you've the option to define a custom IdP as your default IdP.

This procedure only works for SAP Identity Authentication Service and isn't supported for non-SAP IdPs.

You can use only one identity provider for basic authentication. You can either use the SAP default identity provider (SAP ID service) or SAP Identity Authentication Service as custom IdP.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Perform the following steps:

1. Configure SAP Custom IdP with OpenID Connect.

More information: [Establish Trust and Federation Between UAA and Identity Authentication](#)

2. Create a service instance for XS user authentication and authorization service (XSUAA) under the `apiaccess` plan.

Furthermore, create a service key for the service instance, and request an access token from the authorization service associated with the custom IdP (using the content of the service key when sending the request).

More information: [Get API Access](#)

3. Using the token retrieved from the previous step, perform another HTTP call to perform a patch request at:

```
https://api.authentication.<landscape
domain>.hana.ondemand.com/sap/rest/authorization/v2/securitySettings
```

i Note

You can find the landscape domain in the address of your SAP Cloud Integration application:

```
https://<tenant name>.it-<Cloud Integration system name>.cfapps.<landscape
domain>.hana.ondemand.com/itspaces
```

Example for landscape domain: eu10

Send the following body along with the request:

```
{ "defaultIdp": "sap.custom" }
```

You can now set up basic authentication for users registered by the custom IdP for the following use cases:

- Sender component calls integration flow endpoint, see: [Basic Authentication of IdP User for Integration Flow Processing](#).
- API client calls Cloud Integration OData API, see: [Basic Authentication of an IdP User for API Clients](#).

Switching Back to SAP ID Service

If you like to switch back to SAP default identity provider (SAP ID service), perform the following steps.

Using a REST API client, perform a PATCH request as described for the default IdP in the following SAP Community blog: [Cloud Integration: Enable SAP IAS \(Identity Authentication Service\) as Custom IdP for Basic Inbound Authentication in Cloud Foundry Environment](#) 

To make sure to switch back to SAP ID service, use the following settings for the request:

Operation	PATCH
URL	Value from <code>apiurl</code> of service key, suffixed with <code>/sap/rest/authorization/v2/securitySettings</code> Example

	https://api.authentication.sap.hana.ondemand.com/sap/rest/authorization/v2/securitySetti
Authorization	Bearer Token
Token	Value retrieved from previous step as described in SAP Community blog: Cloud Integration: Enable SAP IAS (Identity Authentication Service) as Custom IdP for Basic Inbound Authentication in Cloud Foundry Environment 👉
Header	Content-Type with value as application/json
Body	<pre>{ "defaultIdp": "sap.default" }</pre>

Creating X.509 Keys

You need X.509 keys to configure communication with certificate-based authentication over HTTPS and if you want to configure digital encryption and signing of messages with security standards PKCS#7 and XML Digital Signature.

1. Generate a key pair.

See: [Generating a Key Pair](#)

2. Upload the signing response that you receive from the CA to the keystore.

See:

3. Request a signed certificate from a certification authority

To enable the tenant to communicate as client with the customer system, you have to import a client certificate to the tenant client keystore. This certificate has to be signed by a certification authority (CA). The procedure depends on the CA and is not documented here.

4. Upload the signing response that you received from the CA to the keystore.

See: [Updating a Key Pair with a Signing Response](#)

Generating a Key Pair

Context

This section covers all steps to generate a new private key pair using the SAP Cloud Integration **Monitor** application.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, choose **Create**.
3. Choose **Key Pair** from the popup Menu.
4. Fill out the required data.
 - Enter an **Alias**
 - Define the key specific values

- Define a validity period

i Note

The recommended key size is 4096 bit.

5. Choose **Deploy** to create the key pair.

Next Steps

You can download the signing request.

Downloading a Certificate Signing Request

Download a certificate signing request to send to a certification authority.

Context

When a certificate is originally created, it is self-signed. It has to be signed by a certification authority (CA) before it can be used for productive scenarios. To get a certificate signed by a CA, you first need to download a certificate signing request (CSR) in the Keystore Monitor.

i Note

This option is not available for key pairs with the alias `id_dsa` or `id_rsa`, or SAP key pairs.

Procedure

1. Open the SAP Cloud Integration **Monitor** application.
2. Choose the **Keystore** tile in the **Manage Security** section.
3. On the **Current** tab, select a key pair.
4. Choose the  (**Actions**) icon, then select **Download Signing Request**. Alternatively, you can click the key pair alias to open the key pair details, and then choose **Download** > **Signing Request**.
5. A file with the name `<alias>.csr` is downloaded.

Results

You have downloaded a CSR to your computer.

Next Steps

You send the CSR to a certification authority, who will provide a signing response.

Related Information

[Updating a Key Pair with a Signing Response](#)

Creating SFTP Keys

You can set up reliable file transfer based on SSH File Transfer Protocol (SFTP). SFTP is an enhancement of the Secure Shell (SSH) network protocol.

Context

This section covers all steps to generate the required security artifacts for a tenant to be connected as SFTP client to an SFTP server.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. In the **Current** tab, choose **Create**.
3. Choose **SSH Key** from the popup Menu.
4. Fill out the required data.

Note

The recommended Key Size is 4096 bit.

5. Choose **Create** to add the SSH key pair to the keystore.

Next Steps

You can download the public key in OpenSSH format and configure it as an authorized key for the required user on the SFTP server.

Related Information

[Generating a Key Pair](#)

Maintaining SSH Known Hosts for SFTP Connectivity

Context

The known hosts configuration is used to set up secure connectivity between Cloud Integration and an SFTP server using Secure Shell (SSH). It contains the addresses and public keys of trusted hosts (SFTP servers) that Cloud Integration can connect to.

SFTP stands for *Secure File Transfer Protocol*.

→ Tip

To get the host key for a trusted server, you can use the SSH connectivity test. See [SSH Connectivity Tests](#).

❖ Example

When your integration flow contains an SFTP receiver adapter with **Proxy Type** set to **On Premise**, the known host entry must match the virtual host name configured in the adapter.

For instance,

- SFTP server name: `ftp123.myorg.com`
- SFTP receiver adapter address: `orgftp:22`

In this case, the known host entry must contain the virtual host name (orgftp) instead of the server host name (ftp123.myorg.com).

Perform the following steps to create or update the content of the known hosts file and to redeploy the updated file on the tenant.

Working with Known Hosts Entries

You can create known host entries either by importing them from a file or by adding them manually using the Known Hosts dialog.

1. Navigate to the **Monitor** section in the SAP Cloud Integration, and go to **Manage Security** > **Security Material** tile.
2. Do one of the following:
 - To create a new known hosts configuration, choose **Create** and select **Known Hosts (SSH)**.
 - To update an existing configuration, select an existing Known Hosts (SSH) artifact and choose **Edit**.

i Note

The dialog opens with an empty table for new configurations. For existing configurations, the table is automatically populated with deployed entries.

3. In the **Create SSH Known Hosts** dialog, you can:
 - Add entries manually using **Add**.
 - Edit entries directly in the table.
 - Remove entries using the  *delete* icon.
 - Import entries from a file using **Browse**.

⚠ Caution

Using **Browse** replaces all existing known host entries in the table with the entries from the imported file.

4. After reviewing the entries and confirming the changes, choose **Deploy**.

The known hosts configuration is deployed and becomes effective for SFTP connectivity.

Updating a Known Hosts File

Procedure

1. If you want to update an already deployed known hosts artifact on the **Manage Security Material** page, locate the **Known Hosts (SSH)** artifact that you want to update.
2. Choose the  *Edit* button under the **Actions** tab.
3. In the **Edit Known Hosts (SSH)** dialog, update the required entries by adding, editing, or deleting them directly in the table.
4. Choose **Deploy** to apply the updated configuration.

⚠ Caution

SFTP scenarios might encounter failures during the period where the latest known host is not deployed in SAP Cloud Integration, so it's recommended to perform these changes during a designated maintenance window when no critical

SFTP integration flows are running and the business load is reduced. Once the known host is properly updated, these failures will automatically recover during the next schedule or execution.

Known Host File Format

Procedure

- To create a new known hosts file, create a text file on your local system and add the entries using the structure shown below:

Code Syntax

```
[<host name>] :<IP address> <public key algorithm> <base64 encoded public key>
```

```
[<host name>] :<port> <public key algorithm> <base64 encoded public key>
```

- Each line must contain the host name or IP address, the public key algorithm, and the public key. This information is typically provided by the participant system administrator during onboarding.

Sample Code

```
[mysite.myorg.corp]:29418 ssh-rsa
AAAAB3NzaC1yc2EAAAADjkzfkjxc7awsL4bRmfvLQF+l/c8ldB1onyFSH19plyfxdliuglzigutzutzuftzftTvAXXS21NYkGlxQ==
```

Tip

Add each trusted host in a separate row.

Troubleshooting the Known Hosts File

Context

Issue	Description	Solution
The connectivity test fails with a reject HostKey error	This issue usually occurs when the knownhost file in SAP Cloud Integration lacks a hostkey for the same host and algorithm.	Verify that the knownhost entries in SAP Cloud Integration are correctly updated.
The connectivity test fails with a HostKey has been changed error	This issue typically arises when the knownhost file in SAP Cloud Integration contains a hostkey for the same host and algorithm, but it does not match the host key sent by the SFTP server during the key exchange.	Ensure the knownhost entries in SAP Cloud Integration are properly updated.
The integration flows still fails with the error messages reject HostKey or HostKey has been changed during the execution, despite asuccessful connectivity test		Restart the integration flow to resolve the issue. i Note Restarting the integration flow is not always necessary after deploying the

Issue	Description	Solution
		correct known host file, as the error typically auto-recovers after the next poll or execution. This step is included as a fallback mechanism to clear stale knownhost entries if auto-recovery does not occur by the next poll or execution.

Related Information

[Deploying an SSH Known Hosts Artifact](#)

Creating Keys for Message Level Security

Related Information

- [Creating OpenPGP Keys](#)
- [Creating Keys for the Usage of PKCS#7, XML Digital Signature and WS-Security](#)

Creating OpenPGP Keys

You use the tool gpg4win to create the required keys for the usage of OpenPGP.

This section covers the creation of OpenPGP keys for tenants managed by SAP.

This description does not apply to tenants managed by customers. Customers might have their own OpenPGP key management processes.

The OpenPGP keys are maintained on the Windows VM on which the keys of the X.509 certificates are also maintained.

The kind of keys required depends on the use case and the role of the tenant for which the keys are created.

The following table lists the possible use cases and the required kinds of keys.

i Note

As soon as you start gpg4win, files are created for the PGP Public Keyring and PGP Secret Keyring.

OpenPGP Keys for the Tenant

Role of Tenant	Chosen Kind of Message Protection	Required Keys
Sender (outbound communication)	Encrypts outbound payload	PGP Public Keyring (contains receiver's public key to encrypt payload)
	Encrypts and signs outbound payload	PGP Public Keyring (contains receiver's public key to encrypt payload) PGP Secret Keyring (contains tenant's secret key to sign payload)

Role of Tenant	Chosen Kind of Message Protection	Required Keys
Receiver (inbound communication)	Decrypts inbound payload	PGP Secret Keyring (contains tenant's secret key to decrypt payload)
	Decrypts and verifies inbound payload	PGP Secret Keyring (contains tenant's secret key to decrypt payload) PGP Public Keyring (contains the sender's public key to verify payload) for verifying

Related Information

[How OpenPGP Works](#)

[Creating PGP Keys for Encryption \(Tenant Is Sender\)](#)

[Creating PGP Keys for Encryption and Signing \(Tenant Is Sender\)](#)

[Creating PGP Keys for Decryption \(Tenant Is Receiver\)](#)

[Creating PGP Keys for Decryption and Verifying \(Tenant Is Receiver\)](#)

[Securely Exchanging Key Material](#)

Creating PGP Keys for Encryption (Tenant Is Sender)

Prerequisites

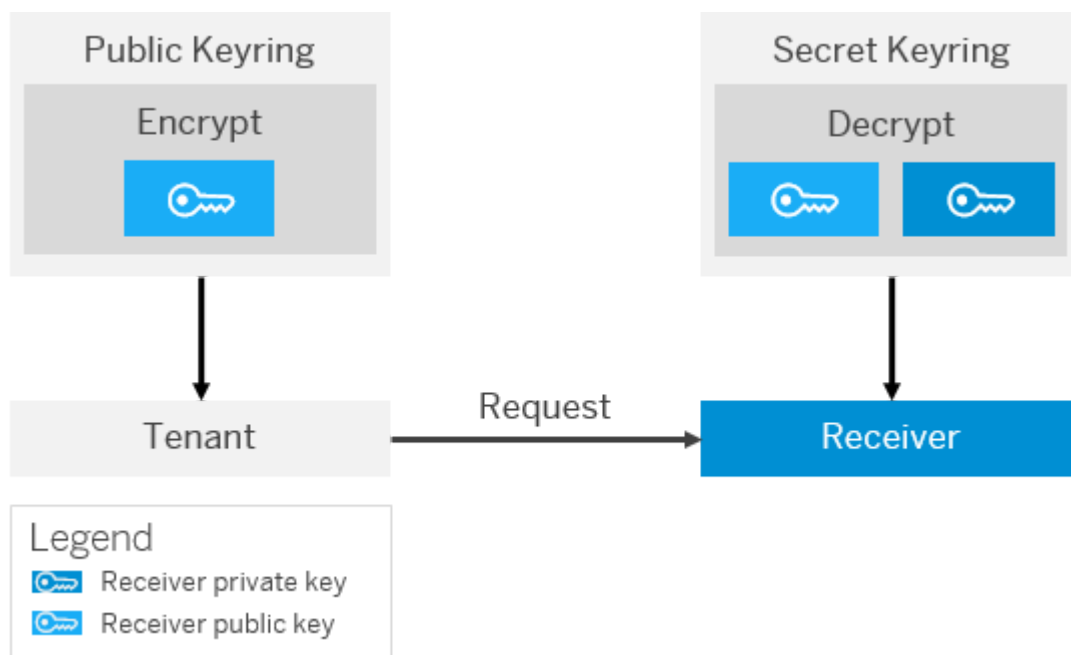
You have installed gpg4win, created the tenant-specific directory, and created a key pair.

Context

For this use case, the following key artifact has to be deployed on the tenant:

A **PGP Public Keyring** that contains the **receiver's** public key (required by the tenant to encrypt the payload)

The following figure shows the required entities to be configured for the tenant (on the left).



Procedure

1. Obtain the public key from the receiver.

We recommend using a secure channel to ensure that the information originates from the correct source and that it has not been changed on its way. A signed email would be suitable, for example.

2. Import the receiver's public key into the PGP Public Keyring.
3. If a secure channel has not been used to obtain the public key from the receiver, verify the fingerprint of the public key.

One option is to phone the owner of the public key and compare the fingerprint.

Next Steps

Deploy the **PGP Public Keyring** on the tenant.

Related Information

[Installing gpg4win](#)

[Creating Tenant-Specific File Directories](#)

[Starting the GPA Tool](#)

[Creating a Key Pair](#)

[Importing a Public Key](#)

[Securely Exchanging Key Material](#)

Creating PGP Keys for Encryption and Signing (Tenant Is Sender)

Prerequisites

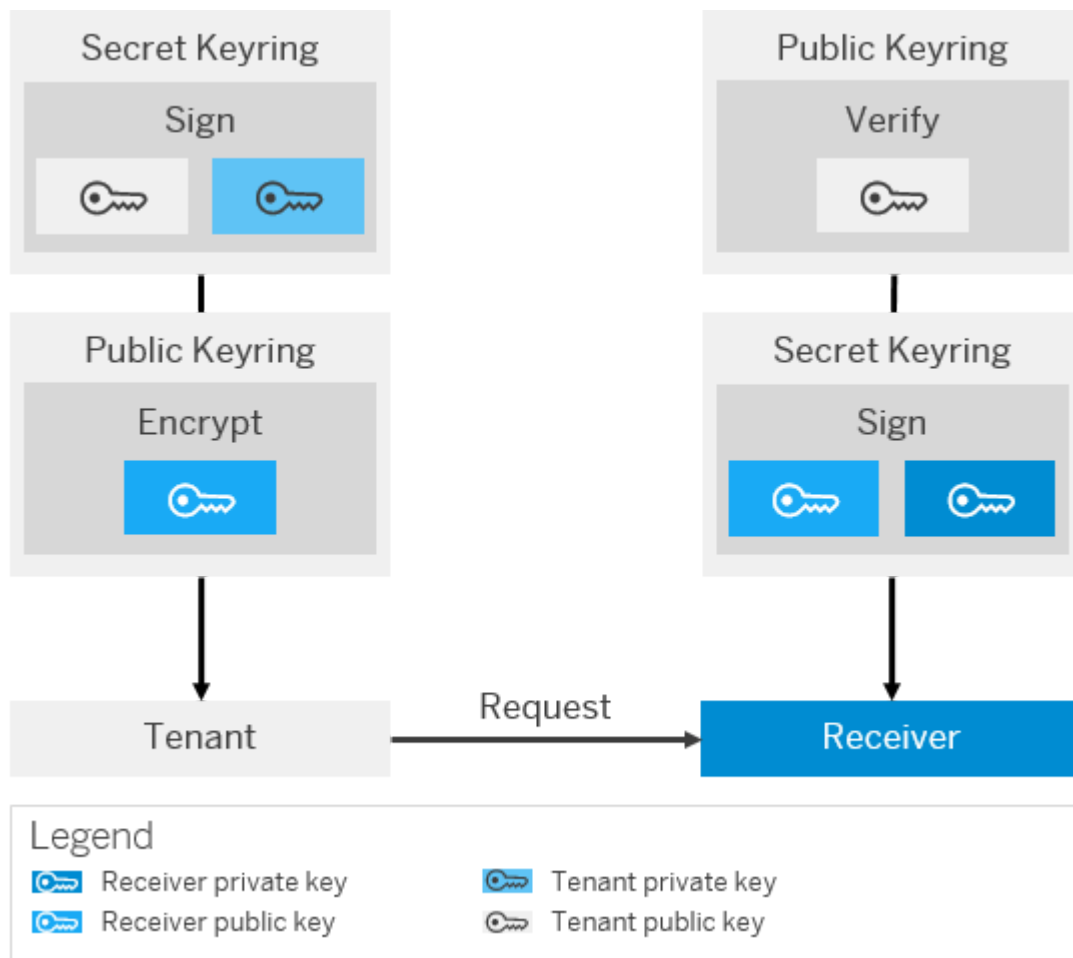
You have installed gpg4win, created the tenant-specific directory, and created a key pair.

Context

For this use case, the following key artifacts have to be deployed on the tenant:

- A **PGP Secret Keyring** that contains the tenant's private key (required by the tenant to sign the payload)
- A **PGP Public Keyring** that contains the **receiver's** public key (required by the tenant to encrypt the payload)

The following figure shows the required entities to be configured for the tenant (on the left).



Procedure

1. Start the GPA tool and create a new key.

This action creates a PGP Secret Keyring containing a private/public key pair.

2. Obtain the public key from the receiver.

We recommend using a secure channel (for example, encrypted email) for this information exchange.

3. Import the receiver's public key into the PGP Public Keyring.

4. If a secure channel was not used to obtain the public key from the receiver, verify the fingerprint of the public key.

5. Export the public key from the tenant's PGP Public Keyring.

6. Provide the receiver with the public key (ideally through a secure channel).

The receiver has to import the tenant's public key into its PGP Public Keyring.

Next Steps

Deploy the **PGP Public Keyring** and the **PGP Secret Keyring** on the tenant.

Related Information

[Installing.gpg4win](#)

[Creating Tenant-Specific File Directories](#)

[Starting the GPA Tool](#)

[Creating a Key Pair](#)

[Importing a Public Key](#)

[Exporting the Public Key](#)

Creating PGP Keys for Decryption (Tenant Is Receiver)

Prerequisites

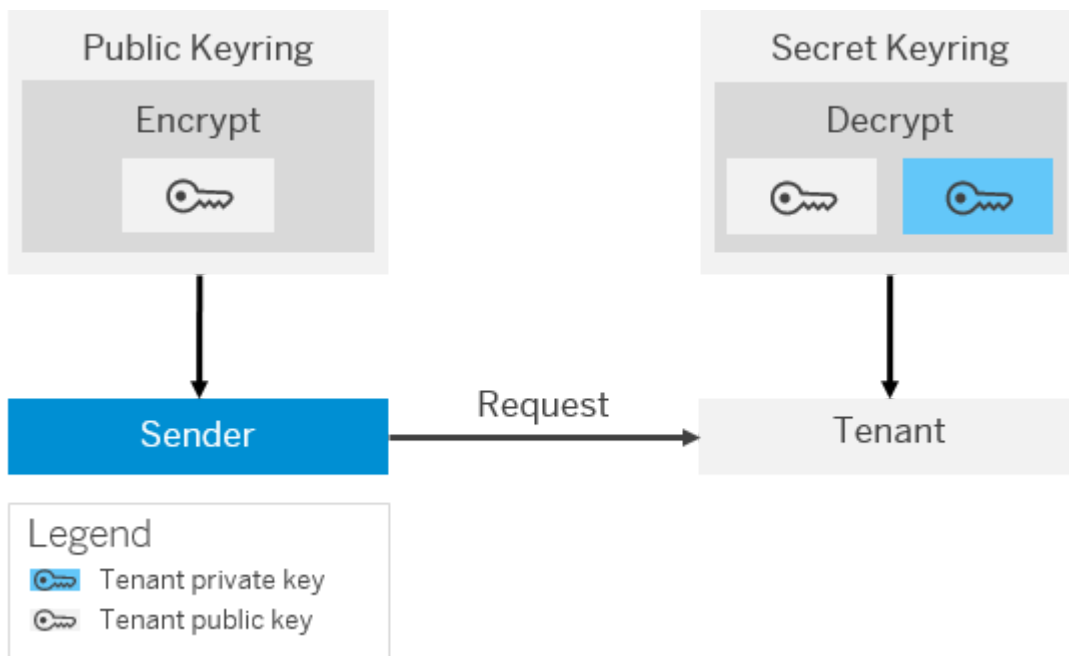
You have installed gpg4win, created the tenant-specific directory, and created a key pair.

Context

For this use case, the following key artifact has to be deployed on the tenant:

A **PGP Secret Keyring** that contains the **tenant's** private key (required by the tenant to decrypt the payload)

The following figure shows the required entities to be configured for the tenant (on the right).



Procedure

Start the GPA tool and create a new key.

This action creates a PGP Secret Keyring containing a private/public key pair.

Next Steps

Deploy the **PGP Secret Keyring** on the tenant.

Related Information

[Installing gpg4win](#)

[Creating Tenant-Specific File Directories](#)

[Starting the GPA Tool](#)

[Creating a Key Pair](#)

[Securely Exchanging Key Material](#)

Creating PGP Keys for Decryption and Verifying (Tenant Is Receiver)

Prerequisites

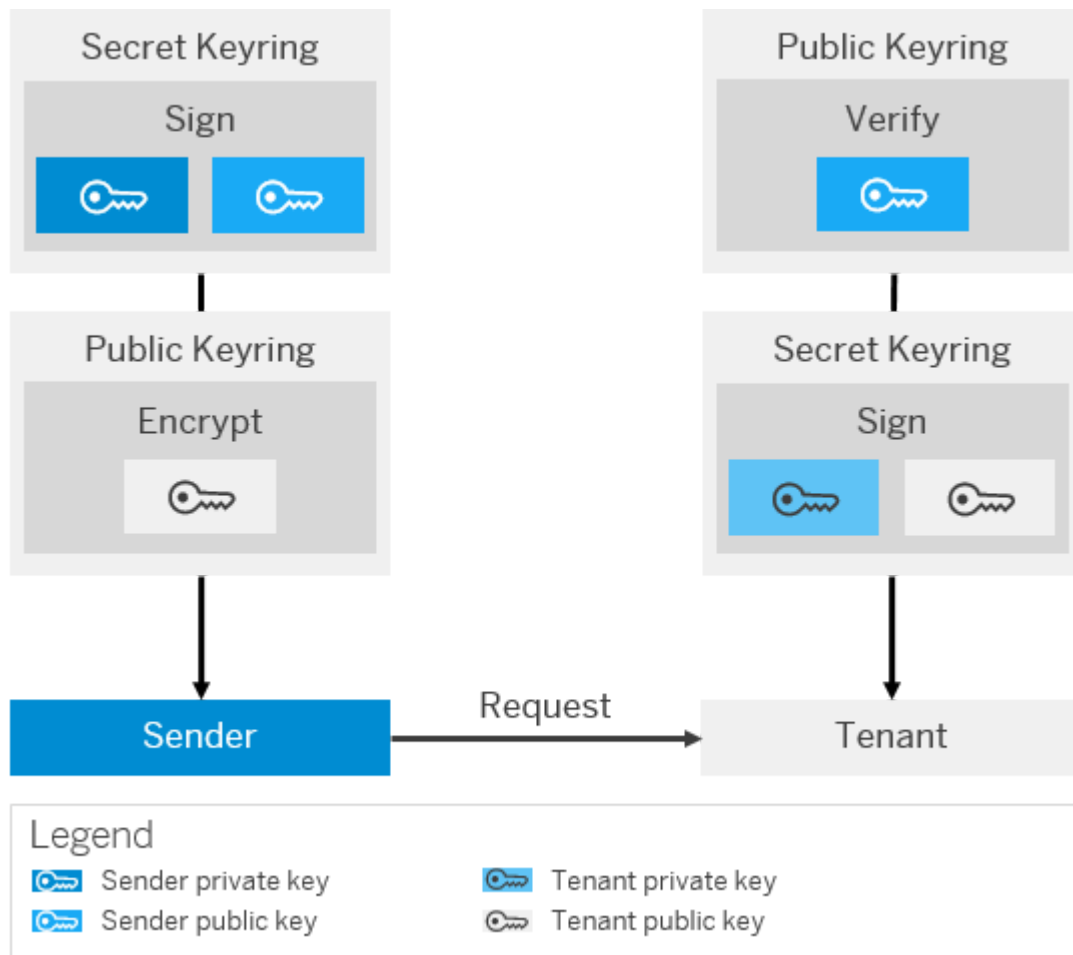
You have installed gpg4win, created the tenant-specific directory, and created a key pair.

Context

For this use case, the following key artifacts have to be deployed on the tenant:

- A **PGP Public Keyring** that contains the **sender's** public key (required by the tenant to verify the payload obtained from the sender)
- A **PGP Secret Keyring** that contains the tenant's private key (required by the tenant to decrypt the payload obtained from the sender)

The following figure shows the required entities to be configured for the tenant (on the right).



Procedure

1. Start the GPA tool and create a new key.

This action creates a PGP Secret Keyring containing a private/public key pair.

2. Obtain the public key from the sender.

We recommend using a secure channel (for example, encrypted email) for this information exchange.

3. Import the sender's public key into the PGP Public Keyring.
4. If a secure channel was not used to obtain the public key from the sender, verify the fingerprint of the public key.
5. Export the public key from the tenant's PGP Public Keyring.
6. Provide the sender with the public key (ideally through a secure channel).

The sender has to import the tenant's public key into its PGP Public Keyring.

Next Steps

Deploy the **PGP Public Keyring** and the **PGP Secret Keyring** on the tenant.

Related Information

[Installing.gpg4win](#)

[Creating Tenant-Specific File Directories](#)

[Starting the GPA Tool](#)

[Creating a Key Pair](#)

[Exporting the Public Key](#)

[Importing a Public Key](#)

[Securely Exchanging Key Material](#)

Using gpg4win to Create PGP Keys

Related Information

[Installing.gpg4win](#)

[Creating Tenant-Specific File Directories](#)

[Starting the GPA Tool](#)

[Creating a Key Pair](#)

[Exporting the Public Key](#)

[Importing a Public Key](#)

[Using the GNU Privacy Guard Command Line Tool](#)

Installing gpg4win

We recommend that you use gpg4win to create OpenPGP key material.

Context

gpg4win is a free software and can be downloaded from the Internet.

Procedure

1. Download version 2.3.4 gpg4win from: <https://files.gpg4win.org/gpg4win-2.3.4.exe> ➦ .
2. When being asked to check the components to install, make sure that:

GPA is **selected**.

Kleopatra is **deselected**.

Creating Tenant-Specific File Directories

A PGP Secret Keyring and a PGP Public Keyring have to be maintained for each tenant that uses OpenPGP. The GPA tool cannot maintain several PGP Secret or Public Keyrings at the same time. Therefore, you have to create a separate directory for each tenant, where you have to configure GPA and the launching of GPA separately (otherwise, keys from different tenants will be stored in the same keyring).

Context

The following procedure shows how you can achieve the described setup using Gnu Privacy Assistant. To facilitate the usage of the software, we provide a set of simple configuration files to download.

Caution

The following description, together with the configuration files, show a possible way how to use Gnu Privacy Assistant. We cannot give any guarantee that the software (in combination with the configuration files) works in the desired way.

Start Gnu Privacy Assistant (separately for each tenant).

Procedure

1. For each tenant (using OpenPGP), create a separate file directory for maintaining the keyrings.
2. Copy the following three files into this file directory:
 - `gpa.conf`
 - `gpg.conf`
 - `run_gpa.bat`

You can download the files at:

[Files for OpenPGP Key Management](#)

Download the `.zip` file and extract the content on your computer.

3. Adapt the file `run_gpa.bat` by entering the path to the tenant-specific directory.

These files are required to configure the usage of the GPA tool.

The file `run_gpa.bat` sets the shell variable `GNUPGHOME` to the tenant-specific directory.

The files `gpa.conf` and `gpg.conf` contain configurations for GPA and GPG. The file `gpg.conf`, for example, determines the strength of the applied encryption. Read the comments in the configuration files for further details.

Next Steps

You can now start creating keys.

Starting the GPA Tool

Start the GPA tool to manage keys.

Procedure

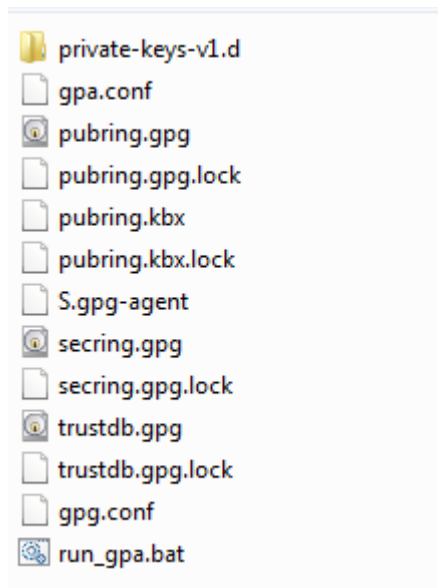
Double-click the `run_gpa.bat` file in the relevant tenant-specific directory.

If you start GPA without executing `run_gpa.bat`, gpa will use the default GNUPGHOME directory.

Next Steps

As soon as you have started the GPA tool, the following files are created for the PGP Public Keyring and PGP Secret Keyring: `pubring.gpg` and `secring.gpg` (see the following screenshot of the tenant-specific directory after tool launch).

These files have to be deployed later on the tenant as PGP Public Keyring and PGP Secret Keyring.



Creating a Key Pair

Context

OpenPGP provides the option of defining two kinds of keys: primary keys and subkeys. There is no general recommendation for when to use which type.

Usually, a primary key is created for certification and signing, and a subkey is created for encryption for each tenant that uses OpenPGP, but this is just a recommendation.

Procedure

1. Start the GPA tool (by double-clicking `run_gpa.bat` in the tenant-specific directory).
2. In the main menu, select **Window > Key Maintenance**.
3. In the menu of the following window, select **New > New Keys**.
4. In the **Generate Key** dialog, keep the Algorithm and Key Size (RSA, 2048), and specify the following attributes.

For **Name**, enter a string according to the following naming convention:

`<speaking tenant name> <tenant alias>.hci.sap.com`

For `<speaking tenant name>`, you can use the name of the company, for example (like Citi).

Leave the **Email** and **Comment** fields empty.

Select **Expires** and chose a period of 2 years.

5. Choose **OK**.

6. Enter a password (passphrase).

Note that all private keys in the secret keyring must have the same password.

There's also the option to have multiple secret keys in a PGP secret keyring (each with a passphrase). When using PGP secret keys for Cloud Integration, all secret keys must have the same passphrase.

7. The key is generated.

If you select the key entry, more details are displayed.

On the **Subkeys** tab, the usage of the related subkeys is displayed.

Related Information

[Deploying a PGP Secret Keyring](#)

Exporting the Public Key

Export a public key in order to make it available for your communication partner (sender or receiver).

Context

Your communication partner needs the public key for the related activities such as signing the message (when this is a sender) or verifying the message (when this is a receiver).

Procedure

1. Start the GPA tool and select the key that is to be exported.
2. Choose **Export**.
3. Select a location on your local disk and specify a file name (extension **.pub**).
4. Choose **Save**.

Results

When you open the public key file with a text editor, it looks like this (example):

```
-----BEGIN PGP PUBLIC KEY BLOCK-----
Version: GnuPG v2.0.22 (MingW32)
mQENBFN01NoBCACn3GcPFGBJlpLcVanYq2/oWiuwmd1c2ZWos+rQnyEClZtr3DN/
dKJlBpMI4JwsE8ztsTRnk900DPaDdHWL0euVm61Utoa5tNdoaA0tpdtldX3ZdJ7S
fIN2IMkd++trTwhNGWzY1uQfZP30eaKmJVedMCz/v55JH0THFHLKbdfkseGkprKm
+d3TIwD9V2aX2Iikf6FmlQyWSTbI96q078V2N0iDqPSRA2V8Qa+B0qDAQCIHnb1n
9S1xkj94mk3SlhkhayuM0MPPKI7H/7al5ug0BijjqAczEGJzpzj6sVYUfFoQREGB
AK5+Weekhgpa2n39IHUYTdhlTgMTMXpsgzrABEBAAG0JVBldGVyIEEdldHNjaGUg
PHBldGVyLmdldHNjaGVAc2FwLmNvbT6JATkEEwECACMFALN01NoCGw8HCwkIBwMC
AQYVCAIJCgsEFgIDAQIeAQIXgAAKCRczSBnHmxBNSeaVB/0TWgHEE74SEZLg1IBW
vqGXzCZvPGXrh8AN+2TQGBvXTZ564/uc4U3RU0CvdAAth76rj9eesaAsVnd1Zw8k
7C107mvg0omX4oPtJy94KbR831HHwiD+yfnm18Eq0STQwUBcHnqTFjiKX6a0g6UX
CscwFHC1utlfoK4NI8KAJxFB037ld7d2moJRJljqcD6bHeCB8Hv16QzA3cpFTBW/
ns4abVj88SdVN5igm7R64mkTMK0iaJ6NL958rfJ1Q2lEns8Z1WtcBdYLSs5JxSqB
9XgT
=jEwk
-----END PGP PUBLIC KEY BLOCK-----
```

Importing a Public Key

You can import public keys provided by your communication partner.

Context

The administrators of the sender or receiver system provide the public keys that need to be imported into the tenant's PGP Public Keyring.

Procedure

1. You obtain the public key from the sender or receiver administrator (either by e-mail or by download from a key server).
2. Start the GPA tool and select the key.
3. Choose **Import**.
4. Browse for the key on your local disk and add it to your keyring.
5. After the import, verify the fingerprint of the imported key.

This is important because the key could have been tampered with during its transfer from the sender or receiver.

The fingerprint is displayed in the GPA tool on the **Details** tab.

One option to verify the correctness of the fingerprint is to contact the sender/receiver administrator by phone or signed e-mail and ask whether the fingerprint is correct.

Using the GNU Privacy Guard Command Line Tool

The GNU Privacy Guard command line tool provides additional functions for working with OpenPGP keys.

The GPA graphical tool only contains a subset of functions that might be relevant when configuring scenarios using OpenPGP. Some use cases might require you to remove a subkey or add a new subkey. This can only be done with the command line tool.

When using the command line tool, make sure that you always specify the tenant home directory in the commands, in order to make changes for a specific tenant.

Example:

```
gpg --homedir=C:/tenantMyCompany --edit-key MyCompany
```

This command edits the key **MyCompany** in the tenant directory **C:/tenantMyCompany**.

To consult the manual for further details, run the command: `gpg --help`.

i Note

For more information, check out the following SAP Community blog:

[Cloud Integration – Import and Export PGP Secret Key – Change PGP Secret Key Password](#) 

This blog shows you how to import and export keys, and how to change key passwords.

Creating Keys for the Usage of PKCS#7, XML Digital Signature and WS-Security

To set up message level security scenarios based on PKCS#7, XML Digital Signature or WS-Security, the required keys are created in the same way as for transport level security HTTPS.

Setting up message level security based on PKCS#7, XML Digital Signature or WS-Security requires the generation of public-private key pairs of type **X.509** – the same standard as is used for transport level security SSL.

Therefore, technically, you can use the same public key pairs for message level and transport level security (HTTPS).

Depending on the scenario, however, separate key pairs might be required.

Keep in mind that you can set up message level security on top of another transport security (like, for example SFTP). In that case, you in any case have to generate key pairs based on X.509 standard.

To generate a new public-private key pair, proceed as described for transport level security SSL. In particular, proceed in the same way as described for the configuration of certificate-based outbound authentication (HTTPS).

Note the following in addition:

- If you have already generated a keystore file and a separate key pair should be used for message level security, you can use the same keystore file, import the certificates required for message level security, and re-deploy the keystore file on the relevant tenant.
- To implement digital signature, a certificate from the sender is needed (the public key of the sender is required to verify the signature – in other words, to decrypt the digest).
- To implement digital encryption, a certificate from the receiver is needed (the public key of the receiver is required to encrypt the symmetric encryption key).

Related Information

[Message-Level Security](#)

[Creating X.509 Keys](#)

Securely Exchanging Key Material

In many cases, communication partners need to exchange public keys in order to establish a secure connection.

To establish a secure communication between software systems, communication partners use asymmetric (or public) key technology and work with private/public key pairs. In some cases, public keys have to be exchanged between the partners at certain points of the configuration process.

You need to apply certain measures when exchanging key material to ensure that you do not compromise the security of your scenario.

Public Keys

When exchanging public keys (for example, X.509 certificates), make sure that the keys cannot be manipulated by a third party during the transfer.

You have the following options:

- Use a secure communication channel for the key exchange.

For example, you can use PGP-encrypted and -signed e-mail or a secure collaboration platform like SAP Jam.

- Verify the sender (for example, using a signature) and check whether the sender is authorized to provide this key material.
- Verify that the content was not manipulated (usually using a signature).

If you can't use a secure communication channel, check the integrity of the keys by other means, such as the following:

- In the case of X.509 certificates, check that the certificate is valid and that it has been issued by a trusted certification authority (CA).
- Use a separate communication channel (for example, phone) to verify the fingerprint of the key with the sender.

Private Keys

Private keys are even more sensitive than public keys. Sharing your private key with others will allow them to read your encrypted messages and sign messages with your signature.

Whenever possible, avoid exchanging private key material.

In exceptional cases where you have to exchange private keys, apply one of the following measures:

- Use an encrypted container with a password (like PKCS#12 or Java Keystore).
- Transfer the password through a separate communication channel (for example, phone).
- Use secure communication channels. **Never** use plain e-mail or plain HTTP.
- SAP can provide you with a process for exchanging keys in a secure manner.

Example: Exchanging a Public Key

To configure a scenario that includes digital encryption of message content, the following main steps are required, including the exchange of a public key:

1. A potential receiver R of the message generates a public/private key pair (that contains the receiver's public key **PubKey_R** and the associated private key **PrivKey_R**).
2. R provides a potential sender S of messages with the public key **PubKey_R**. To do this, R communicates with S using a private SAP Jam group that is only accessible for dedicated people associated with R and S.

However, R does not disclose the private key.
3. S imports **PubKey_R** into the keystore of the software system that is involved in the scenario on the sender side.
4. S encrypts the message with public key **PubKey_R** and sends the encrypted message to the receiver.
5. R decrypts the message with the private key **PrivKey_R**.

Using the Connectivity Test to Get the Load Balancer Server Root Certificate

You can use the outbound connectivity test to get the load balancer server root certificate (which maybe required to set up inbound HTTP communication).

1. Open the SAP Cloud Integration **Monitor** section.
2. Under **Manage Security** select the **Connectivity Tests** tile.
3. Choose **TLS**.
4. As **Host** enter the address of the worker node.

You need to enter the worker node URL, as the sender system is supposed to connect to the worker node (through the load balancer component).

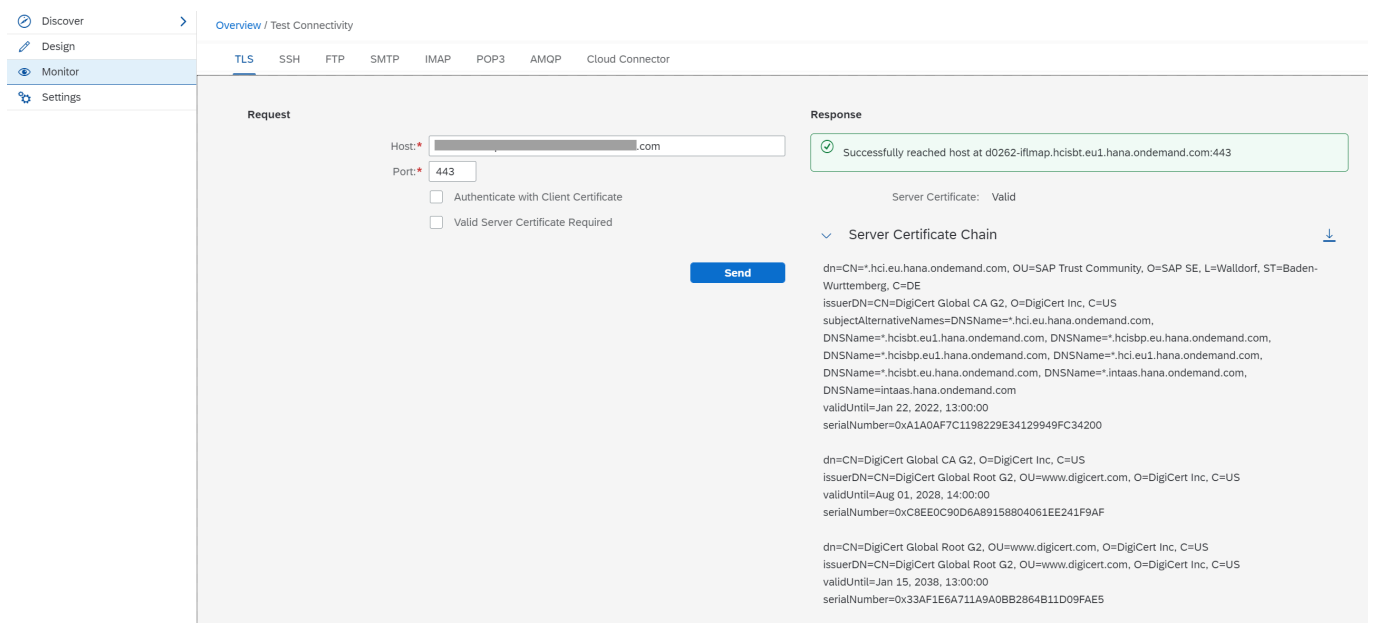
i Note

To get this address, open the **Monitor** section and click a tile under **Manage Integration Content**. Select a deployed integration flow that has an HTTP-based sender adapter (for example, an HTTPS sender adapter) and copy the URL displayed under **Endpoints**.

Delete the part after the backslash (/). The worker node URL has the following form (example):

`mytenant-iflmap.hcisbt.eu1.hana.ondemand.com`

5. Deselect the option **Validate Server Certificate** and run the test.



6. Choose the **Download server certificates** icon.
- A .zip file is created and stored on your computer.
7. Extract the .zip file.

From the extracted files (with file extension .cer) you need to use the root certificate.

Concepts of Secure Communication

There are several options to protect the message exchange. You can secure the communication on transport level by selecting the HTTPS or SFTP protocol and installing specific authentication methods. In addition to that, you can set up methods to encrypt and decrypt the content of the message and to digitally sign and verify the message.

Related Information

[Basics](#)

Basics

Familiarize yourself with the basics of secure communication.

Related Information

[HTTPS-Based Communication](#)
[SFTP-Based Communication](#)
[Message-Level Security](#)
[Certificate Management](#)

HTTPS-Based Communication

Certain adapters support communication based on Hypertext Transfer Protocol Secure (HTTPS).

HTTPS uses the Transport Layer Security (TLS) protocol.

i Note

Mutual TLS (mTLS) is equivalent to client certificate authentication. While setting up the TLS connection, client and server exchange certificates. With mTLS, not only server certificates, but also client certificates are validated based on the signatures provided by certification authorities. For more information, see [Client Certificate Authentication \(Outbound\)](#) and [Keystore](#).

Timeout Behavior for HTTP communication

In HTTP communication spanning multiple components (for example, from a sender, through the load balancer, to Cloud Integration, and from there to a receiver), the actual timeout period is influenced by each of the timeout settings of the individual components that are interconnected between sender and receiver. To be more precise, the timeout is influenced by those components in your landscape that can control the Transmission Control Protocol (TCP) session. The component or device with the lowest number set for the idle session timeout defines the timer.

- When considering inbound communication through HTTP-based sender adapters, note that in particular the load balancer has its own timeout setting that influences the overall timeout. For inbound scenarios, SAP recommends that you design them so no communication is expected to wait longer than 10 minutes. In general, it is recommended to keep waiting times as short as possible to avoid unnecessary load on the network components.

In inbound scenarios, Cloud Integration uses a Java web server to receive and process inbound HTTP requests. This web server has an important timeout parameter that controls how long it is allowed to wait while reading the request body. Cloud Integration sets this parameter to 20 seconds (to mitigate the risk of DoS attacks with fragmented HTTP requests). If the delay between the URI and the HTTP request body, the delay between body chunks (in case of chunked transfer encoding), or the delay between individual network packets delivering the HTTP request body exceeds this timeout, the request will be cancelled and an error message returned to the client.

- When considering outbound communication, note that you can configure a dedicated timeout in the involved HTTP-based receiver channel. However, the timeout setting has no influence on the TCP timeout when the receiver or any additional component interconnected between Cloud Integration and the receiver has a lower timeout. For example, consider that you have configured a receiver channel timeout of 10 minutes and there's another component involved with a timeout of 5 minutes. If nothing has been transferred for a certain amount of time, the connection is closed after 5 minutes have passed.

Related Information

[Authentication and Authorization Options \(Inbound\)](#)

[Authentication Options \(Outbound\)](#)

[Load Balancer Root Certificates Supported by SAP](#)

Authentication and Authorization Options (Inbound)

When a client calls a server using a secure communication channel, two different kinds of checks are performed subsequently.

- Authentication

Verifies the identity of the calling entity.

- Authorization

Checks what a user or other entity is authorized to do (for example, as defined by roles assigned to it). In other words, the authorization check evaluates the access rights of a user or other entity.

When a client calls a server, it is first authenticated and, in a subsequent step, the authorization check is performed.

We use **inbound** to refer to the communication direction when a sender system sends a message to the integration platform.

Combinations of Authentication and Authorization (Inbound)

For inbound communication based on HTTPS, the authentication and authorization options can be combined in a specific way.

Combination of Authentication/Authorization Options

Authentication Option ...	Can Be Used with the Following Authorization Option ...
Basic authentication The sender (client) authenticates itself against the server based on user credentials (user name and password). The HTTP header of the inbound message (from the sender) contains the user name and password.	Role-based authorization For this user, the authorizations are checked based on user-to-role assignments defined on the tenant. i Note When you use Cloud Integration in the Cloud Foundry environment, as user credentials you can also use clientid and clientsecret from a Process Integration service instance with plan integration-flow and client_credentials grant type.
Client-certificate authentication and certificate-to-user mapping (only in the Neo environment) The sender (client) authenticates itself against the server based on a digital client certificate. Furthermore, this certificate is mapped to a user (based on the information contained in a Certificate-to-User Mapping artifact deployed on the tenant). i Note You can map multiple certificates to the same user (n:1 certificate-to-user mappings possible).	Role-based authorization For the user derived from the certificate-to-user mapping, the authorizations are checked based on user-to-role assignments defined on the tenant.
Client-certificate authentication	Subject/Issuer DN authorization check of a certificate

Authentication Option ...	Can Be Used with the Following Authorization Option ...
(without certificate-to-user mapping) The sender (client) authenticates itself against the server based on a digital client certificate.	In a subsequent authorization check, the permissions of the sender are checked on the tenant by evaluating the distinguished name (DN) of the client certificate of the sender.
OAuth Grants access to resources of SAP Cloud Integration without the need to share passwords with the client. i Note This option is supported for the following sender adapter types: SOAP (SOAP 1.x), SOAP (SAP RM), HTTPS, and OData.	Role-based authorization

i Note

It is not recommended to use client certificate authentication (without certificate-to-user mapping). Instead of this, it is recommended to use client certificate authentication with certificate-to-user mapping (which is a more secure way of authentication).

More information: [OAuth 2.0 Specification](#) ➦

Note that there are major differences for the setup of inbound connections depending on whether you use Cloud Integration in the Cloud Foundry or Neo environment, see: .

For detailed instructions on how to set up the different authentication options, see: [Environment-Specific Aspects Integration Developers Should Know](#).

- [Configuring Inbound HTTP Connections, Cloud Foundry Environment](#)
- [Configuring Inbound HTTP Connections, Neo Environment](#)

Related Information

[Protecting Applications with OAuth 2.0](#)

[Authentication Options \(Inbound\)](#)

[Authorization Options \(Inbound\)](#)

Authentication Options (Inbound)

For inbound communication, different ways are supported how the sender can authenticate itself against Cloud Integration.

We use **inbound** to refer to the communication direction when a sender system sends a message to the integration platform.

- **Basic authentication**

The calling entity is authenticated based on credentials (user name and password).

- **Client-certificate authentication and certificate-to-user mapping**

The calling entity is authenticated based on a certificate, and the certificate is mapped to a user (for which the authorization check is executed in a subsequent step).

- Client-certificate authentication
- OAuth 2.0

OAuth allows you to set up authentication scenarios without the need to share credentials.

See:

- [Protecting Applications with OAuth 2.0](#)
- [OAuth 2.0 Specification](#) ➦

Related Information

[Basic Authentication](#)

[Client Certificate Authentication and Certificate-to-User Mapping \(Inbound\), Neo Environment](#)

[Client Certificate Authentication \(Inbound\), Neo Environment](#)

[Setting Up Inbound HTTP Connections \(with OAuth\), Neo Environment](#)

Client Certificate Authentication (Inbound), Cloud Foundry Environment

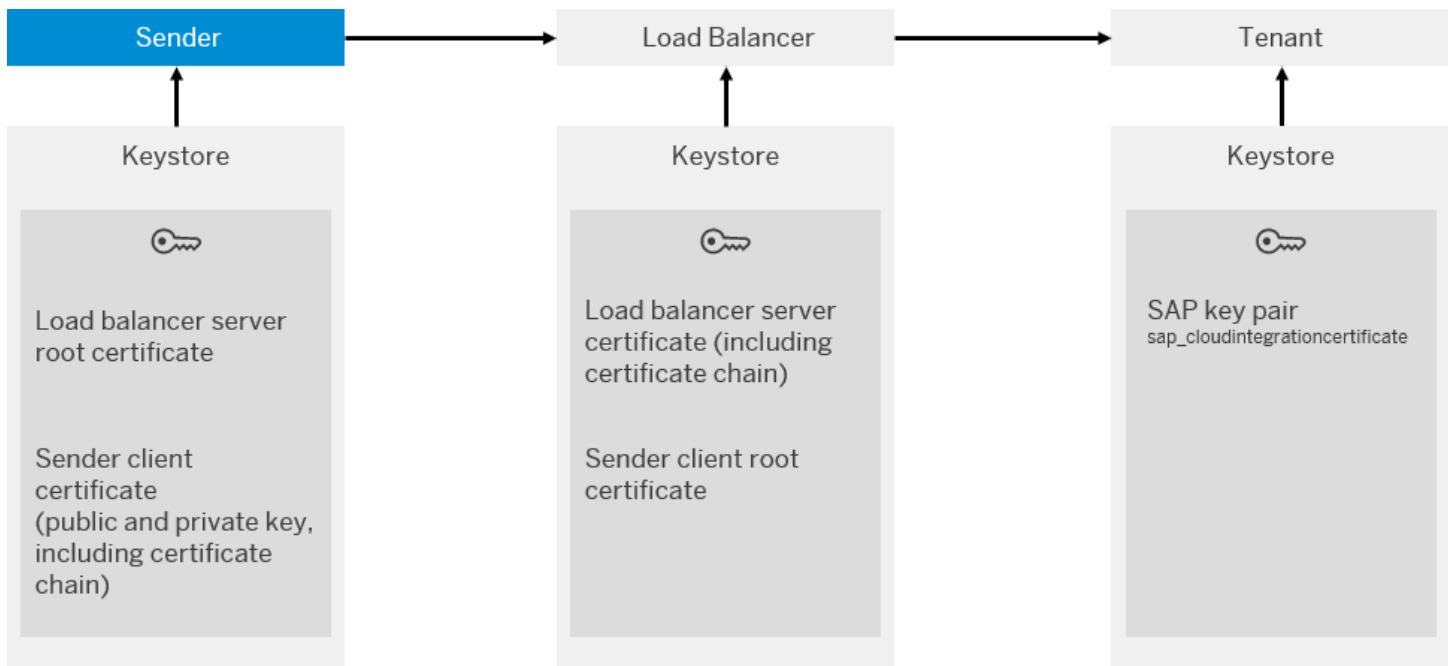
Authentication of a sender is performed based on a client certificate. At runtime, the system checks if a service key is available that contains the client certificate provided by the sender. If a service key is available, the system then checks if the associated service instance has a role specified that grants permissions to call the integration flow endpoint.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

How it Works

The following figure shows the involved components and digital keys. For more information on the tenant keystore that comes with Cloud Integration, see [Keystore](#).



The table provides an overview of the required digital keys and their purpose in the authentication process, and summarizes the required configuration steps. Note that when setting up secure communication of different systems, typically administrators associated with the different systems need to accomplish configuration tasks in a coordinated way and to exchange public keys.

i Note

For inbound HTTP connections, a load balancer component connects the remote sender system and the Cloud Integration tenant.

The load balancer terminates each inbound Transport Layer Security (TLS) request and establishes a new one for the connection to the tenant where the message is processed.

Security Artifact	Used to ...	Configuration Steps
Load balancer server root certificate	Make the sender trust the load balancer.	Sender administrator: Get certificate using the SAP Cloud Integration Connectivity Test (pointing to endpoint address of integration flow).
Load balancer server certificate (including certificate chain)	Qualify load balancer as trusted component (for senders that like to connect to it).	No action required as this artifact is maintained by the operator of the cloud infrastructure.
Sender client certificate (public and private key, including certificate chain)	Authorize sender to call integration flow. At runtime, system checks if there's a service key that contains a client certificate that matches client certificate provided with the sender's request.	Tenant administrator: Create service instance (using SAP BTP cockpit) and generate service key. Add sender client certificate (provided by sender administrator) to service key. This key pair is to be signed by a CA supported by the load balancer. Only root certificates are being imported into the load balancer keystore. Therefore, the whole certificate chain must be assigned to the certificate to enable the connected component to evaluate the chain of trust.
Sender client root certificate	Sign sender client certificate.	Sender administrator: Get sender client certificate signed by a certificate authority (CA) supported by the load balancer. The root certificates supported by the load balancer are listed at Load Balancer Root Certificates Supported by SAP .
SAP key pair (alias: sap_cloudintegrationcertificate) (not available on trial tenant)	Enable internal communication between involved SAP BTP microservices.	No action required - this key pair is preinstalled and maintained by SAP. i Note This key pair is not preinstalled when you operate a Cloud Integration trial tenant. Therefore, client certificate inbound authentication is not supported for trial.

When defining the service key, the tenant administrator also specifies the role to be used to authorize the sender to call integration flow endpoint. You can either specify the predefined role `ESBMessaging.send` or a custom role.

→ Tip

Based on this setup of keys, the communication is established at runtime in the following way:

The sender connects to the load balancer and verifies the load balancer certificate. On the other way round, the load balancer verifies if the certificate sent by the sender system is valid. It's important that the client certificate installed on the sender system is signed by a certificate authority that is supported by the load balancer.

If the check is successful, the system checks if a service key is available that contains the sender's client certificate. If that is the case, the role specified for the associated service instance is checked. If this role is identical to the one specified in the sender adapter of the integration flow endpoint (addressed by the request), the message can be processed.

For more information, check out this SAP Community blog: [Cloud Integration on CF – How to Setup Secure HTTP Inbound Connection with Client Certificates](#) .

For more information on how to set up this option, see [Client Certificate Authentication for Integration Flow Processing](#).

OAuth Authentication with Client Credentials Grant (Inbound), Cloud Foundry Environment

You can configure OAuth Client Credentials Grant authentication for inbound calls from sender systems or API clients to the integration platform.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

How It Works - Inbound Authentication for a Sender Calling an Integration Flow

The sender (client) application is granted access to the associated worker node through OAuth authentication, Client Credentials Grant.

Using this variant, the authentication workflow is established at runtime in the following way:

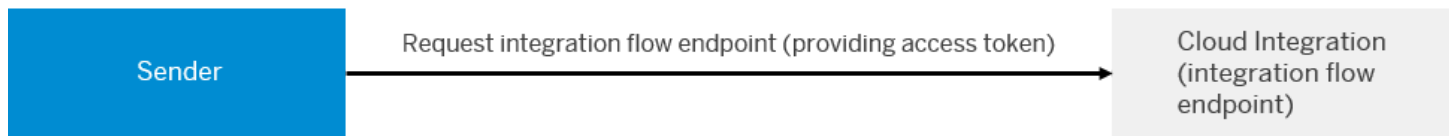
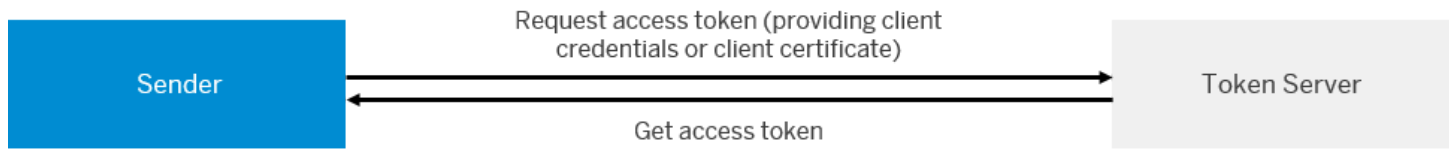
1. In a first call, the sender requests an access token from the SAP BTP token server.

There are the following options for the sender to authenticate itself against the token server:

- o Using client credentials
- o Using a client certificate (either generated by SAP or by a custom tool)

2. In a second call, the sender uses the access token to call the integration flow endpoint.

If the access token is accepted and the authorization check is successful, the integration flow can be processed.



→ Tip

For detailed information on how to set up this option, see [OAuth with Client Credentials Grant for Integration Flow Processing](#).

Let's summarize the required steps and settings:

- To define the way how the sender can call the integration flow endpoint, you create a service instance (service plan [integration-flow](#)) and a service key using SAP BTP cockpit.

When defining the service instance, you specify the role that is to be used to authorize the sender to call integration flow endpoint. You can either specify the predefined role `ESBMessaging.send` or a custom role. Furthermore, the role has to correspond to the one specified in the sender adapter of the integration flow that is addressed by the call.

- For the service instance, you furthermore create a service key. In the service key, you define how the sender is to be authenticated against the token server (either using client credentials or a client certificate).

Depending on the chosen option, the service key generated for the service instance contains values for various properties. To establish the call from the sender to the token server, the values for the following properties are required:

- If you've chosen the option to use client credentials: `clientid`, `clientsecret`, `tokenurl`.
- If you've chosen the option to use a client certificate: `clientid`, `certificate`, `key`, and `tokenurl`.

The `tokenurl` value is the address of the token server.

These values are required in to set up the call to get the access token from the token server.

When you've configured service instance and service key accordingly, the authorization workflow mentioned earlier uses the relevant information at runtime in the following way:

1. The sender uses the service key data to call the token server and get the access token.
2. The sender uses the access token to call the integration flow endpoint.

If the access token is accepted, the system checks the role specified for the associated service instance. If this role is identical to the one specified in the sender adapter of the integration flow endpoint (addressed by the request), the integration flow can be processed.

How It Works - Inbound Authentication for an API Client Calling the OData API

The API client is granted access to the Cloud Integration API resource through OAuth authentication, Client Credentials Grant.

Using this variant, the authentication workflow is established at runtime in the following way:

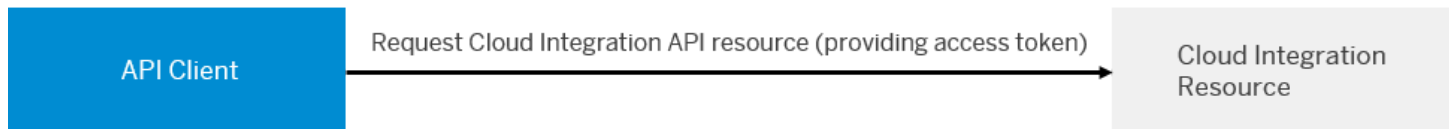
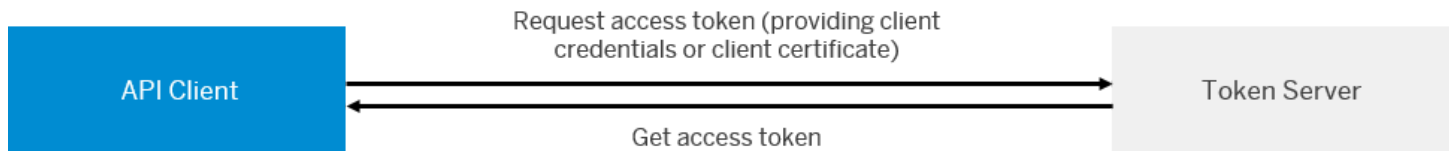
1. In a first call, the API client requests an access token from the SAP BTP token server.

There are the following options for the API client to authenticate itself against the token server:

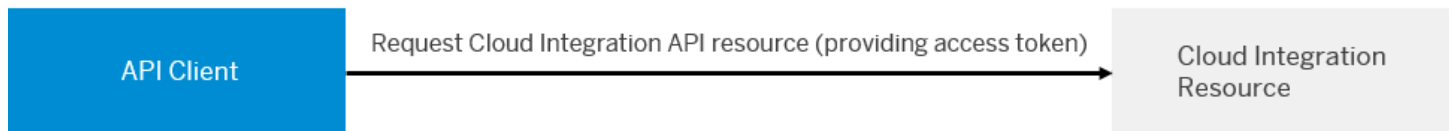
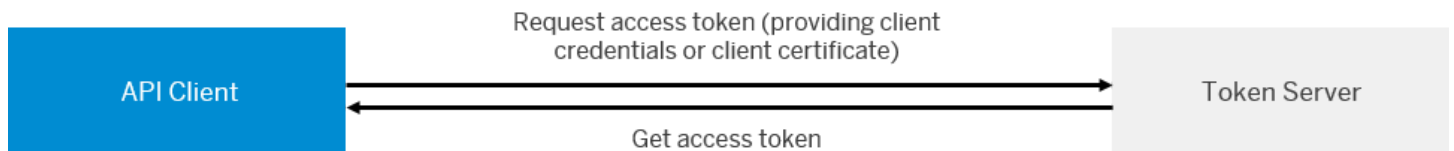
- Using client credentials
- Using a client certificate (either generated by SAP or by a custom tool)

2. In a second call, the API client uses the access token to call the Cloud Integration API resource.

If the access token is accepted and the authorization check is successful, the API client can access the Cloud Integration API resource.



By providing client credentials or a client certificate, the API client requests an access token from the SAP BTP token server. Consequently, the token server will provide the access. Then, the API client uses the access token to call the Cloud Integration API resource. The API client can access the Cloud Integration API resource, if the access token is accepted and the authorization check was successful.



→ Tip

For detailed information on how to set up this option, see [OAuth with Client Credentials Grant for API Clients](#).

Let's summarize the required steps and settings:

- To define the way how the API client can call the Cloud Integration API resource, using SAP BTP cockpit, you create a service instance (service plan [api](#)) and associate it with a role that is to be used to authorize the API client to call the OData API. Which role you assign, depends on the Cloud Integration resource you like to access through the API. For more information, see [API Details](#). Furthermore, you generate a service key for the service instance.

- For the service instance, you furthermore create a service key. In the service key, you define how the API client is to be authenticated against the token server (either using client credentials or a client certificate).

Depending on the chosen option, the service key generated for the service instance contains values for various properties. To establish the call from the API client to the token server, the values for the following properties are required:

- If you've chosen the option to use client credentials: `clientid`, `clientsecret`, `tokenurl`.
- If you've chosen the option to use a client certificate: `clientid`, `certificate`, `key`, and `tokenurl`.

The `tokenurl` value is the address of the token server.

These values are required in to set up the call to get the access token from the token server.

When you've configured service instance and service key accordingly, the authorization workflow from above uses the relevant information at runtime in the following way:

1. The API client uses the service key data to call the token server and get the access token.
2. The API client uses the access token to call the Cloud Integration API.

If the access token is accepted, the system checks the role specified for the associated service instance. If this role complies with the set of roles required to access the addressed Cloud Integration API resource, the call is accomplished successfully.

Client Certificate Authentication and Certificate-to-User Mapping (Inbound), Neo Environment

This option includes an authentication step based on a digital client certificate and the mapping of the certificate to a user.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

With a certificate-to-user mapping, a certificate is mapped to a user, and that way the user can be authenticated based on a certificate.

i Note

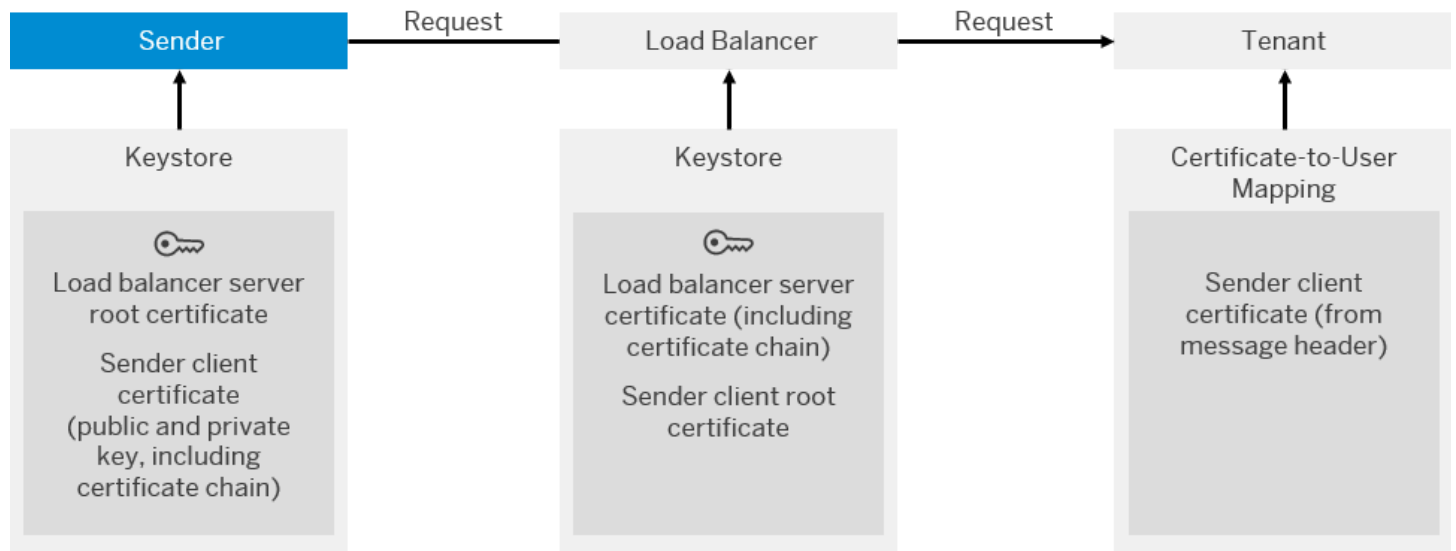
Note that multiple certificates can be mapped to one user (n:1 certificate-to-user mappings possible).

Certificate-to-user mappings make sure that a user is always associated with the certificate as a whole, not only with one attribute of it (for example the common name (CN)). As different certificates can have the same CN, mapping only the CN to a user name bears the risk that different certificates can be mapped accidentally to the same user. Using certificate-to-user mappings circumvents this risk.

For the user defined that way, in a subsequent step, an authorization step is being executed.

How it Works

The following figure shows the complete setup of components and security artifacts required for this option.



When you have configured this authentication option, the authentication of the user is performed in the following way at runtime:

The TLS connection of the sender system and the integration platform is terminated and newly established by the load balancer. This means, that first the load balancer authenticates itself against (as server) the sender based on the load balancer server certificate. Vice versa, the sender authenticates itself against the load balancer as client using the sender client certificate.

To enable the sender to communicate that way with the load balancer, the sender administrator has to make sure that the sender client certificate is signed by one of the certification authorities that are supported by the load balancer.

The load balancer sets the following message header fields:

- **SSL_CLIENT_CERT**
Contains the Base64-encoded sender client certificate.
- **SSL_CLIENT_USER**

When the authentication is been executed successfully, the load balancer writes the sender client certificate (base 64-encoded) into the message header (field `SSL_CLIENT_CERT`). The tenant then maps the sender client certificate to a user based on the certificate-to-user mapping which is deployed on the tenant.

i Note

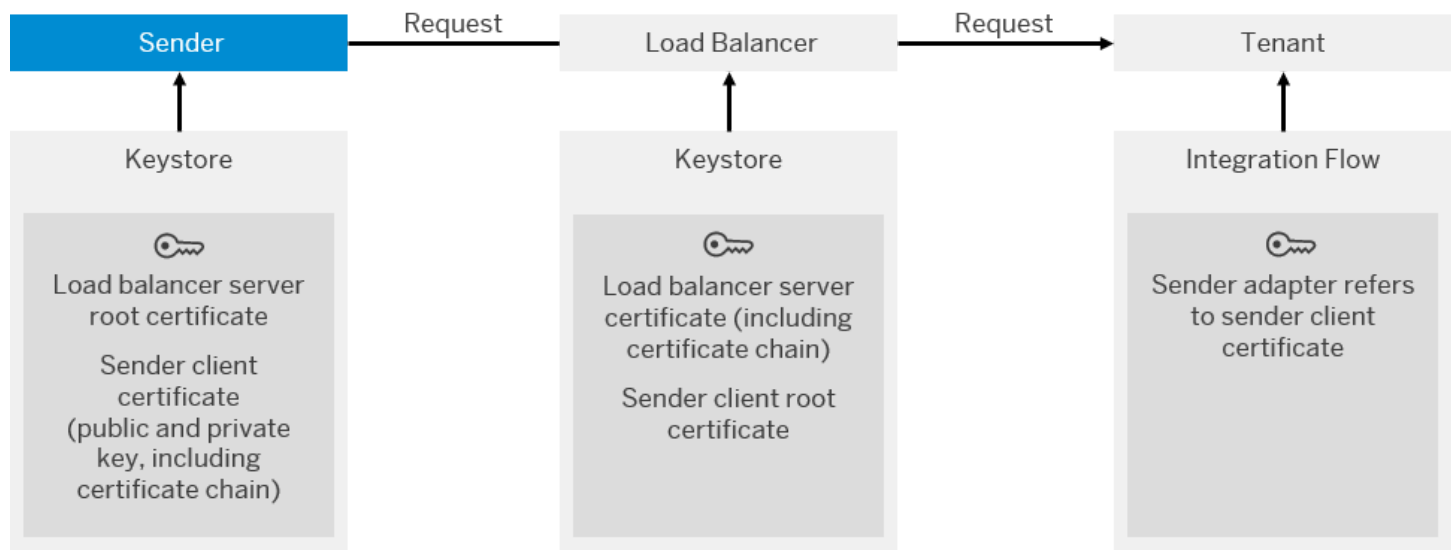
In a subsequent step, the authorization check is executed for the default role (`ESBMessaging.send`) provided by SAP or a custom role configured in an adapter. You can define a custom role. For more information, read the blog on [How to Setup Secure HTTP Inbound Connection with Client Certificates](#).

Client Certificate Authentication (Inbound), Neo Environment

This option includes an authentication step based on a digital client certificate.

How it Works

The following figure shows the complete setup of components and security artifacts required for this option.



When you have configured this authentication option, the authentication of the user is performed in the following way at runtime:

The TLS connection of the sender system and the integration platform is terminated and newly established by the load balancer. This means, that first the load balancer authenticates itself against (as server) the sender based on the load balancer server certificate. Vice versa, the sender authenticates itself against the load balancer as client using the sender client certificate.

To enable the sender to communicate that way with the load balancer, the sender administrator has to make sure that the sender client certificate is signed by one of the certification authorities that are supported by the load balancer.

The load balancer sets the following message header fields:

- **SSL_CLIENT_CERT**
Contains the Base64-encoded sender client certificate.
- **SSL_CLIENT_USER**

i Note

Mutual TLS (mTLS) is equivalent to client certificate authentication. While setting up the TLS connection, client and server exchange certificates. With mTLS, not only server certificates, but also client certificates are validated based on the signatures provided by certification authorities. For more information, see [Client Certificate Authentication \(Outbound\)](#) and [Keystore](#).

Required Security Material

Certificates for Inbound Message Processing

Keystore	Certificate	Description
Sender keystore	Load balancer server root certificate (identifies CA that has signed the load balancer server certificate)	This certificate is required to identify the root CA at the top of the certificate chain that ultimately guarantees the trustability of the load balancer server certificate. In many cases, there is a multilevel setup of Certification Authorities (CAs) so that a certificate is signed by an intermediate CA. The trustability of the intermediate CA is guaranteed by another intermediate CA one level higher, and so on, up to the root CA at the top of the certificate chain. In this case,

Keystore	Certificate	Description
		it is necessary to assign the certificate chain to the certificate, to enable the connected component (which has imported only the root CA into its keystore) to evaluate the chain of trust.
	Sender client certificate	This certificate is required to authenticate the sender (client) when calling Cloud Integration. On the Cloud Integration tenant side, this certificate is required to configure the authorization check.
Load balancer keystore	Load balancer server certificate	This certificate is required to identify the load balancer as a trusted server (to which clients like the sender system can connect).
	Sender client root certificate	This certificate is required to identify the root CA at the top of the certificate chain that ultimately guarantees the trustability of the sender client certificate. There is a list of CAs that are supported by the load balancer. More information: Load Balancer Root Certificates Supported by SAP

For sakes of completeness, note that always a tenant keystore (not depicted in the figure) needs to be available to enable the system to do an additional outbound communication step that is required for technical purposes: The basic technical connectivity of a cluster is checked on a regular basis, as soon as the cluster is active. For this purpose, every 30 seconds the tenant management node sends an HTTPS request to an assigned runtime node via the load balancer. This simulates an external call to the runtime node. To enable this communication, a keystore needs to be deployed on the tenant, containing a valid client certificate that is accepted by the load balancer as well as the root certificate of the same. If this keystore is not available or contains an invalid certificate, the cluster will raise an error. The keystore and required certificate are provisioned by SAP together with the tenant.

i Note

In a subsequent authorization check, the permissions of the sender are checked on the tenant by evaluating the distinguished name (DN) of the client certificate of the sender. The client certificate of the sender is being passed through to the tenant by the load balancer (in the message header). To provide the tenant with the information on the correct client certificate to be expected from the sender, a corresponding setting has to be made in the related integration flow.

Basic Authentication

Basic authentication allows a client to authenticate itself against the server based on user credentials.

⚠ Caution

Consider that we do **not** recommend to use basic authentication in productive scenarios because of the following security aspects:

Basic authentication has the risk that authentication credentials, for example, passwords, are sent in clear text. Using TLS (transport-layer security, also referred to as Secure Sockets Layer) as transport-level encryption method (when using HTTPS as protocol) makes sure that this information is nevertheless encrypted on the transport path. However, the authentication

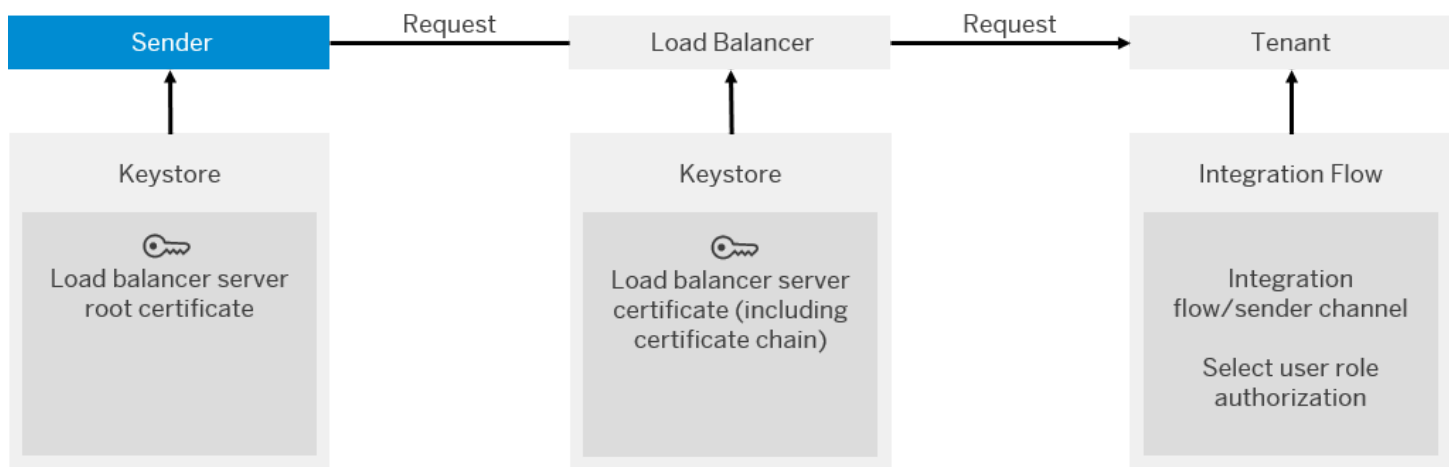
credentials might become visible to SAP-internal administrators at points in the network where the TLS connection is terminated, for example, load balancers. If logging is not done properly at such devices, the authentication credentials might become part of log files. Also network monitoring tools used at such devices might expose the authentication information to administrators. Furthermore, the person to whom the authentication credentials belong (in the previous example, the password owner) needs to maintain the password in a secure place.

i Note

When you use Cloud Integration in the Cloud Foundry environment, as user credentials you can also use `clientid` and `clientsecret` from a [Process Integration Runtime](#) service instance with plan [integration-flow](#).

How it Works

The following figure shows the setup of components required for inbound basic authentication.



i Note

For inbound HTTP connections, a load balancer component connects the remote sender system and the SAP Cloud Integration tenant.

The load balancer terminates each inbound Transport Layer Security (TLS) request and establishes a new one for the connection to the tenant where the message is processed.

These are the steps at runtime:

The HTTP header of the inbound message (from the sender) contains user name and password. To protect these credentials during the communication step, the connection is secured using TLS (SSL).

This includes a step where the load balancer authenticates itself as server against the sender based on a certificate. To enable this security measure, the keystore of the load balancer contains a server certificate signed by a certification authority. To be more precise, the keystore of the load balancer contains a complete certificate chain from (including all intermediate certificates). On the other side of the communication, the keystore of the connected sender system must contain the load balancer server root certificate. That is the certificate that identifies the certification authority (CA) that signed the load balancer's server certificate (on top of the certificate chain).

The other way round, the identity of the sender is checked by SAP evaluating the credentials (user and password) against the user.

It is also depicted in the figure that the authentication option needs to be activated for the corresponding integration flow.

Required Security Material

To enable the sender system to authenticate itself against the integration platform with basic authentication, a communication user has to be created for the sender.

The following figure provides an overview of the involved security artifacts and storage locations.

Certificates for Inbound Message Processing

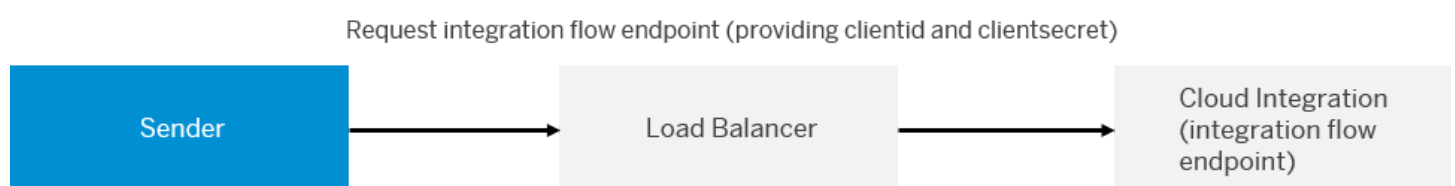
Keystore	Certificate	Description
Sender keystore	Load balancer server root certificate (identifies CA that has signed the load balancer server certificate)	This certificate is required to identify the root CA at the top of the certificate chain that ultimately guarantees the trustability of the load balancer server certificate. In many cases, there is a multilevel setup of Certification Authorities (CAs) so that a certificate is signed by an intermediate CA. The trustability of the intermediate CA is guaranteed by another intermediate CA one level higher, and so on, up to the root CA at the top of the certificate chain. In this case, it is necessary to assign the certificate chain to the certificate, to enable the connected component (which has imported only the root CA into its keystore) to evaluate the chain of trust.
Load balancer keystore	Load balancer server certificate	This certificate is required to identify the load balancer as a trusted server (to which clients like the sender system can connect).

For sakes of completeness, note that always a tenant keystore (not depicted in the figure) needs to be available to enable the system to do an additional outbound communication step that is required for technical purposes: The basic technical connectivity of a cluster is checked on a regular basis, as soon as the cluster is active. For this purpose, every 30 seconds the tenant management node sends an HTTPS request to an assigned runtime node via the load balancer. This simulates an external call to the runtime node. To enable this communication, a keystore needs to be deployed on the tenant, containing a valid client certificate that is accepted by the load balancer as well as the root certificate of the same. If this keystore is not available or contains an invalid certificate, the cluster will raise an error. The keystore and required certificate are provisioned by SAP together with the tenant.

When defining the service instance, the tenant administrator also specifies the role to be used to authorize the sender to call integration flow endpoint. You can either specify the predefined role `ESBMessaging.send` or a custom role.

The service key generated for the service instance contains values for the properties `clientid` and `clientsecret`. These values are required to call the integration flow endpoint.

Based on this setup of keys and OAuth credentials, the communication is established at runtime in the following way:



Related Information

[Using Custom IDP with SAP Cloud Integration](#)

Authorization Options (Inbound)

For inbound HTTPS requests, two different ways to check the authorization of the caller can be configured.

We use **inbound** to refer to the communication direction when a sender system sends a message to the integration platform.

- **Role-based authorization**

The permissions of the calling entity (user) are checked based on a user-to-role assignments configured in the associated identity provider.

In the related sender adapter, you can assign the role based on which the inbound authorization is to be checked for the integration flow.

- **Subject/Issuer DN authorization check**

The distinguished name (DN) of a certificate (associated with the calling entity) is checked.

Subject/Issuer DN authorization check can be defined for individual integration flows.

Related Information

[Role-Based Authorization](#)

[Subject/Issuer DN authorization check](#)

Role-Based Authorization

This option allows you to define permissions for users in the connected identity provider (by default, SAP Identity Service) and to perform an authorization check based on these settings.

For HTTPS requests sent to Cloud Integration, it is checked if the role **ESBMessaging.send** is assigned to the user.

The permissions of the sending client are checked according to roles assigned to the user in the associated identity provider

User management (which includes the assignment of permissions to users) is performed by the tenant administrator using SAP BTP cockpit.

Subject/Issuer DN authorization check

It is checked (for a specific integration flow) if the subject/issuer distinguished name (DN) of the assigned certificate matches the incoming certificate.

If yes, this specific integration flow can be processed. The authorization check is performed based on the distinguished name (DN) of the client certificate. The DN has to be specified when configuring the relevant integration flow.

Authentication Options (Outbound)

For outbound communication through HTTPS (when the tenant sends a message to a receiver), the following authentication options are supported.

- **Basic authentication**

The calling entity (tenant) is authenticated based on credentials (user name and password)

- **Client-certificate authentication**

The calling entity (tenant) is authenticated based on a certificate.

- **OAuth**

Related Information

[Basic Authentication](#)

[Client Certificate Authentication \(Outbound\)](#)

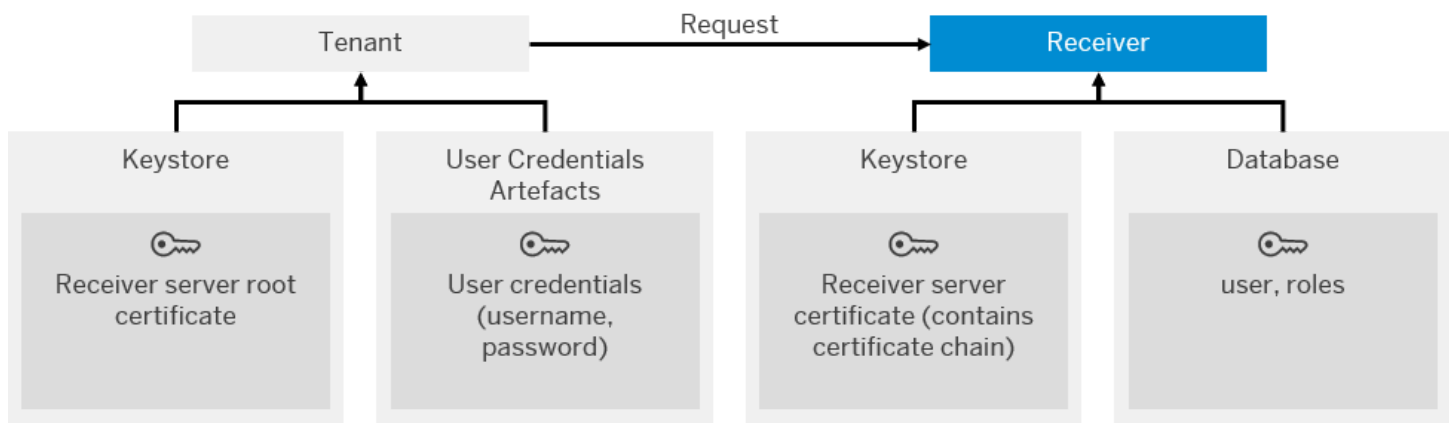
[OAuth 2.0](#)

Basic Authentication

Basic authentication allows a the tenant to authenticate itself against the receiver through credentials (user name and password).

How it Works

The following figure shows the setup of components required for this authentication option.



Basic authentication for HTTPS-based outbound calls works the following way:

1. The tenant (client) sends a message to the customer back-end system.

The HTTP header of the message contains user credentials (name and password).

To protect the user credentials during the communication step, the connection is secured using SSL.

2. The customer back-end authenticates itself as server against the tenant using a certificate (the customer back-end identifies itself as trusted server).

To support this, the keystore of the customer back-end system must contain a server certificate signed by a certification authority. To be more precise, the keystore must contain the complete certificate chain. On the other side of the communication, the keystore of the connected tenant must contain the customer back-end server root certificate.

3. The tenant is authenticated by the customer back-end by evaluating the credentials against the user stored in a related data base connected to the customer back-end.

Required Security Material

Keystore	Security Element	Description
Keystore (tenant-specific) More information: Keystore	Receiver server root certificate	This certificate is required to identify the root CA that is at the top of the certificate chain that ultimately guarantees the trustability of the receiver server certificate.
Receiver keystore	Receiver server certificate (signed by CA with which the tenant has a trust relationship)	This certificate is required to identify the receiver (to which the tenant connects as the client) as a trusted server.
User credentials artifact	User and password	With these credentials the tenant authenticates itself as client at the receiver system.

Client Certificate Authentication (Outbound)

The following figure shows the setup of components required for this authentication option.

How it Works



The tenant authenticates itself against the receiver based on a certificate.

This authentication option works the following way:

1. The tenant sends a message to the receiver.
2. The receiver authenticates itself (as trusted server) against the tenant when the connection is being set up.

In this case, the receiver acts as server and the authentication is based on certificates.
3. Authentication of the tenant: The identity of the tenant is checked by the receiver by evaluating the client certificate chain of the tenant.

As prerequisite for this authentication process, the client root certificate of the tenant has to be imported into the receiver keystore (prior to the connection set up).

As CA who provides the root certificate, *Cyber trust Public Sure Server SV CA* is used.

Steps 2 and 3 are referred to as mutual *SSL handshake*.

4. Authorization check: The permissions of the client (tenant) are checked in a subsequent step by the receiver.

→ Remember

If your tenant contains many key pairs in the Keystore, the tenant takes significantly more time to authenticate itself against the target endpoint.

OAuth 2.0

OAuth 2.0 allows a user to grant a client access to a protected resource (hosted by a resource server). The user typically restricts the access of the client and doesn't allow full access.

OAuth 2.0 (Open Authorization) is an open standard for authorization. It enables users, for example the owners of a protected source, to grant clients restricted access (scope) to their data, that is, the protected source without sharing their authorization details. This data is hosted by a resource server (in terms of Cloud Integration outbound communication, the receiver system). This means, users restrict access and keep credentials private. In Cloud Integration, the Twitter, Facebook, HTTP, Mail, OData, SFSF, and AMQP adapters support the OAuth 2.0 authorization standard.

OAuth 2.0 uses the following components:

Component/Term	Description	In the context of Cloud Integration outbound communication, this is ...
Resource owner	Owns the data and allows access to it.	Administrator of the receiver system.
Resource server	Hosts data and accounts of the resource owner.	Receiver system (functions of the receiver system called by Cloud Integration) The protected resources hosted on the resource server are the capabilities or functions of a receiver system called by an integration flow in the context of an integration scenario.
OAuth client	Party that wants to access the data of a resource owner that needs to be authorized.	The Cloud Integration runtime component that calls the receiver.
Token service	Service that issues OAuth access tokens. A token service is implemented on a system that is referred to in terms of OAuth as an authorization server .	Is in general provided or hosted by the receiver system's organization.
Token service URL	Address of the token service that issues the access token.	
Authorization URL	Address required to ask for for an authorization code from a resource owner.	

Component/Term	Description	In the context of Cloud Integration outbound communication, this is ...
	With the authorization code, an access token can be fetched from the token service. This component is only required for the OAuth 2.0 Authorization Code grant type.	
Client credentials	Comprises client ID and client secret and is used to get an access token (from the token service) when using the OAuth 2.0 Client Credentials grant type.	
Access token	Is used to get access to the protected resource.	

When talking about Cloud Integration outbound communication towards a receiver system, protected resources are the capabilities of the receiver system addressed by the integration flow (outbound communication). The client in this scenario is the Cloud Integration worker where the integration flow is deployed.

There are three grant types for OAuth 2.0. Grant types refer to the possible ways in which an application can get an access token.

- OAuth 2.0 Client Credentials
- OAuth 2.0 SAML Bearer Assertion
- OAuth 2.0 Authorization Code

A specific OAuth variant is used with the Cloud Integration Twitter and Facebook adapter.

OAuth 2.0 Client Credentials Grant

When OAuth 2.0 client credentials grant is implemented, the client gets access to the protected resources in two steps: After presenting a set of client credentials, the client fetches an access token from the token service. In a subsequent step, the client uses the access token to get access to the protected resources.

In detail, OAuth 2.0 client credentials grant is implemented in the following way for Cloud Integration outbound communication:

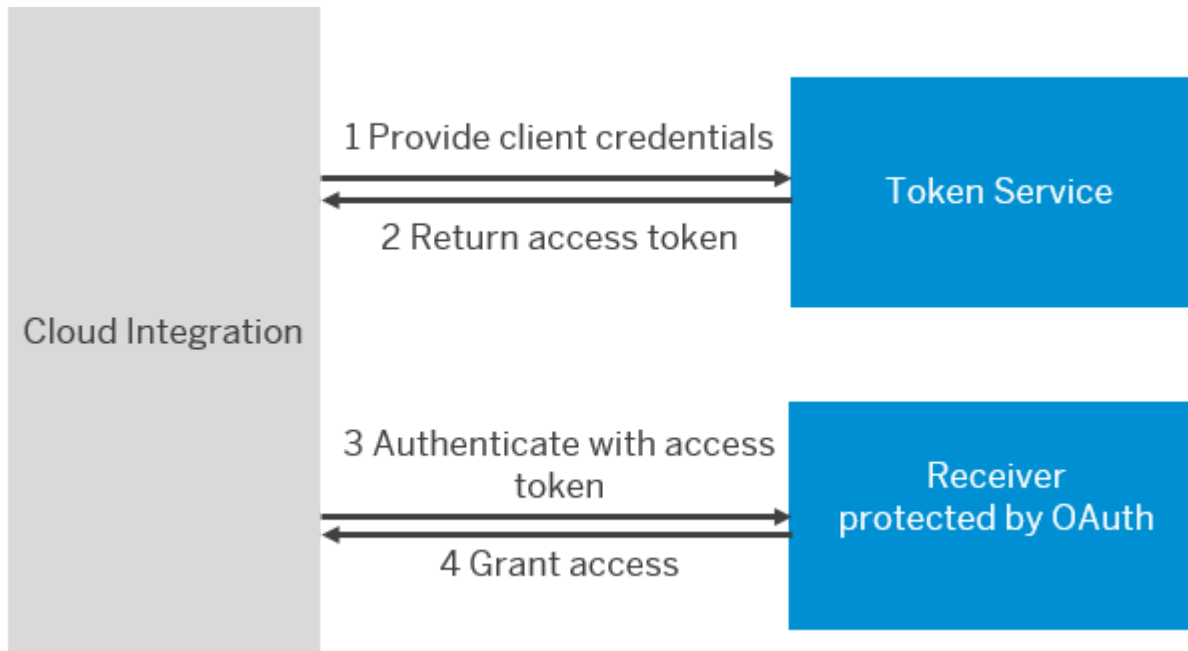
At **design time**, the integration developer performs the following tasks:

1. The integration developer requests the client credentials such as Client ID, Client Secret and token service URL from the administrator of the receiver system
2. Once the receiver system administrator has shared the credentials and the URL, the integration developer specifies an **OAuth2 Client Credentials** artifact using information requested before and deploys the artifact on the Cloud Integration tenant. Likewise, the integration developer specifies the name of the **OAuth2 Client Credentials** artifact when configuring the receiver adapter of the integration flow involved (that is used to connect to the receiver system).

i Note

See: [Deploying an OAuth2 Client Credentials Artifact](#)

After these steps have been executed successfully, at **runtime**, the authorization workflow is processed as outlined in the following chart:



1. Cloud Integration connects to the token service and presents the credentials.
2. The token service authenticates the client credentials and (if they are valid) provides an access token in return.
3. Cloud Integration authenticates itself against the receiver system (that hosts the protected resources in OAuth terms) with the help of the access token and requests access to the protected source.
4. The receiver system validates the access token and (if it's valid) grants access to the protected resource.

OAuth 2.0 SAML Bearer Assertion Grant

When OAuth 2.0 SAML Bearer Assertion grant is implemented, the client (Cloud Integration) gets a SAML Bearer Assertion from a (SAML) custom identity provider. Using the SAML Bearer Assertion, the client requests an access token from a token service in a next step (using the SAML bearer assertion to prove its identity). Finally, the token service validates the SAML Bearer Assertion and passes back the OAuth access token that can be used to access the protected resources finally.

A SAML Bearer Assertion defines a user context that can be propagated between different systems in a communication scenario – a scenario known as *Principal Propagation*. A SAML Bearer Assertion contains a user and a public certificate that identifies the user at a custom identity provider. The SAML Bearer Assertion enables a component to request an access token from a resource server for the given user context.

The following receiver system types are currently supported:

- An SAP SuccessFactors system
- An SAP BTP system (either Neo or Cloud Foundry)

→ Tip

For an example of how this grant type is used with an SAP SuccessFactors system, see:

[OAuth SAML Bearer Assertion Flow Example](#) (maps the explanation of the grant type to a concrete system landscape and use case).

[SAP Cloud Integration – Principal Propagation with SuccessFactors OData V2](#) (SAP Community blog describing step by step how to set up this example)

In detail, OAuth 2.0 SAML Bearer Assertion grant is implemented in the following way for Cloud Integration outbound communication:

At **design time**, the integration developer performs the following tasks:

1. The integration developer creates a trust relationship between the sender system and Cloud Integration.
2. The integration developer creates an OAuth client for Cloud Integration (required to define the connection from the sender to Cloud Integration).
3. The integration developer creates an OAuth client for the receiver system (required to define the connection from Cloud Integration to the receiver).

You specify the signing certificate of the certificate defined when setting up the trust relationship (that way, you exchange the public certificate for the custom identity provider associated with Cloud Integration with the receiver system).

During this step, a client key is created (that is needed to specify the security artifact for Cloud Integration). This is a key to access the API of the receiver system (API key).

4. The integration developer gets information such like token service URL, the type of receiver system, and additional information specific for the receiver system type. In case the **SuccessFactors** system type is chosen, a company ID is given that indicates the client instance used to connect to the SuccessFactors system. In case the **SAP Cloud BTP** (Neo or Cloud Foundry) system type is chosen, a token service user and password is given that indicates the user to access the token service

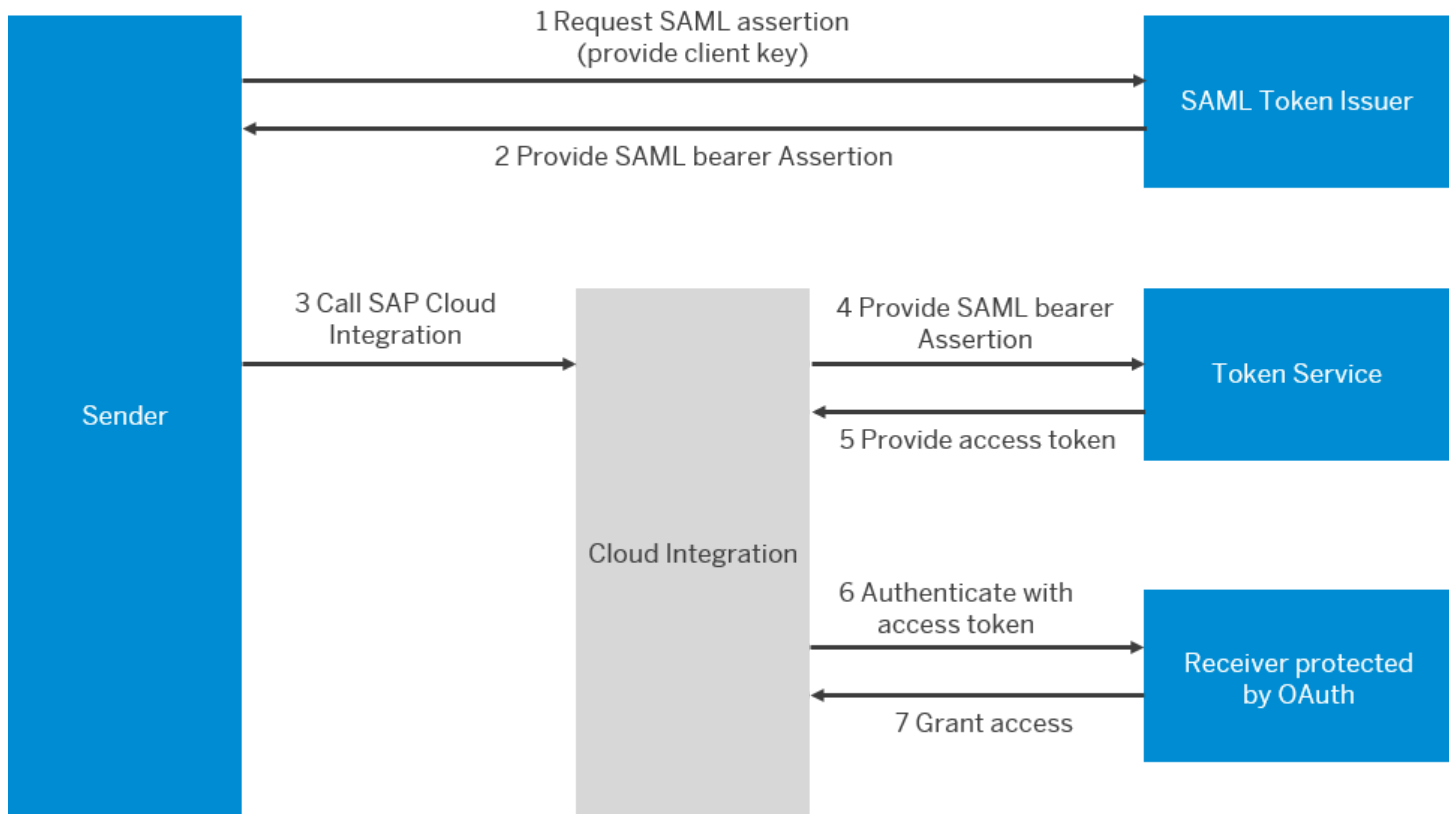
This information is needed to define the **OAuth2 SAML Bearer Assertion** artifact to be referred to in the related receiver adapter

5. Once the information is known, the integration developer specifies an **OAuth2 SAML Bearer Assertion** artifact using information requested before and deploys the artifact on Cloud Integration. Likewise, the integration developer specifies the name of the **OAuth2 SAML Bearer Assertion** artifact when configuring the receiver adapter of the integration flow involved (that is used to connect to the receiver system).

More information: [Deploying an OAuth2 SAML Bearer Assertion](#)

6. Set up the connection of the sender to Cloud Integration (for example, by defining a destination in the sender system). During this step, you need to specify token service URL, token service user, and password of the subaccount that hosts the SAP Cloud Integration tenant.

When these steps have been executed successfully, at **runtime**, the authorization workflow is processed as outlined in the following chart:



1. The sender sends the client key to the SAML token issuer.
2. The SAML token issuer authenticates the client key and, if valid, provides the sender with the SAML Bearer Assertion (for the given user context).
3. The sender requests processing of the related integration flow on the Cloud Integration tenant (and provides the SAML Bearer Assertion with the request).

This step is executed via an SAP BTP destination with `OAuth2SAMLBearer` authentication.
4. Cloud Integration connects to the token service and presents the SAML Bearer Assertion.
5. The token service validates the SAML Bearer Assertion and (if it is valid) provides an access token in return.
6. Cloud Integration authenticates itself against the receiver system with the help of access token and requests access to the protected resource. Note that the receiver system contains the protected resources in terms of OAuth.
7. The receiver system validates the access token and (if it's valid) grants access to the protected resource.

Related Information

[OAuth SAML Bearer Assertion Flow Example](#)

OAuth SAML Bearer Assertion Flow Example

In the example, OAuth SAML Bearer Assertion grant type is used with the SuccessFactors OData V2 receiver adapter.

You find a detailed description how to configure and set up this example step by step in the following SAP Community blog: [SAP Cloud Integration – Principal Propagation with SuccessFactors OData V2](#) .

We summarize how the components involved interact with each other, the steps how to configure the scenario, and the OAuth authorization workflow.

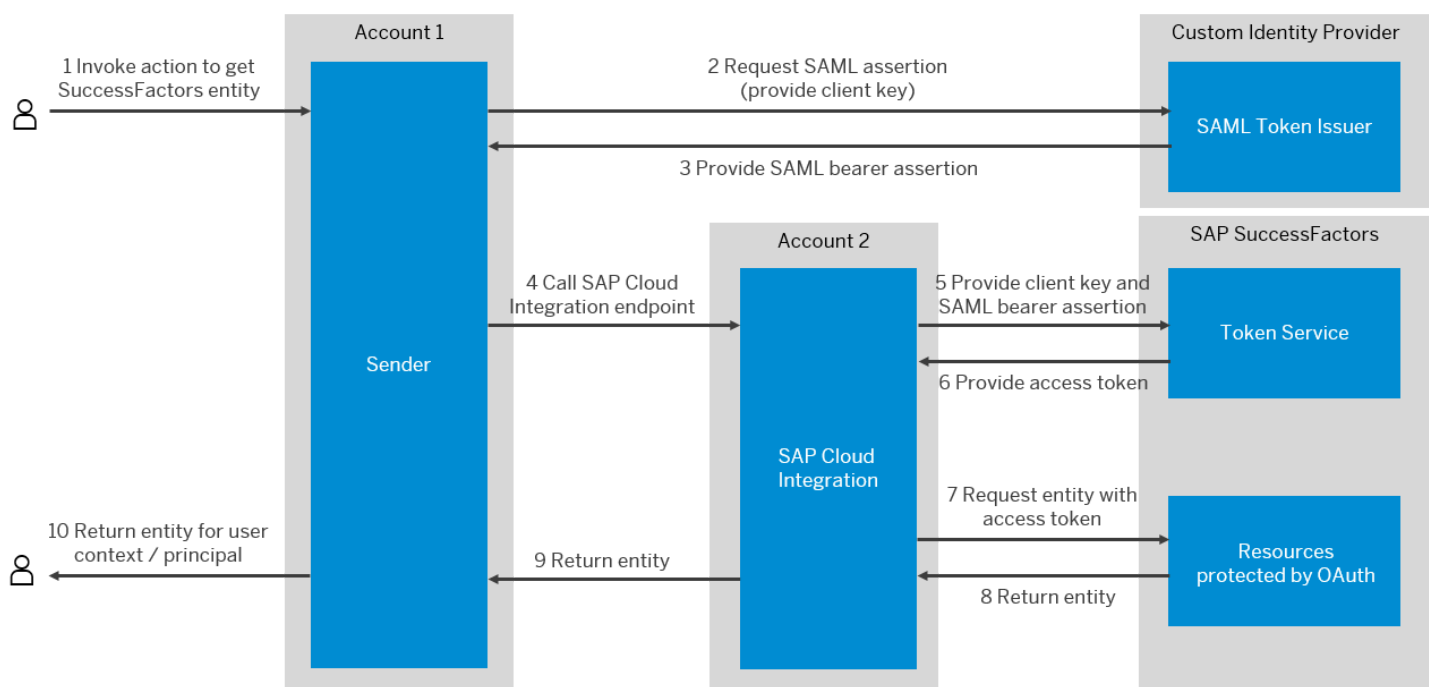
In this example, the user logs in to a sender app to fetch tasks retrieved from an SAP SuccessFactors system (which, in OAuth terms, contains the protected resources).

Cloud Integration is interconnected with the sender and SAP SuccessFactors. The connection to the SuccessFactors system is configured using the SuccessFactors OData V2 receiver adapter.

In this scenario, Cloud Integration fetches the tasks of the user logged in to the sender app. Therefore, the user context (principal) needs to be propagated from the sender app to Cloud Integration, and, finally, from Cloud Integration to SAP SuccessFactors. Principal propagation is achieved through the OAuth2 SAML Bearer assertion flow.

The sender app uses a custom identity provider that also acts as SAML token issuer.

The following figure maps the entities described generally in [OAuth 2.0 SAML Bearer Assertion Grant](#) to the concrete use case and system landscape given in the example.



When configured as described in the blog [SAP Cloud Integration – Principal Propagation with SuccessFactors OData V2](#), the authorization flow works in the following way.

1. The user (logged in to the sender app deployed on account 1) invokes an action to get an SAP SuccessFactors entity (task).
2. The sender app requests the SAML assertion from the custom identity provider (providing the client key).

The sender app communicates with the integration flow via an SAP BTP destination configured in account 1. When connecting to the custom identity provider to get the SAML assertion, account 1 communicates with the custom identity provider based on this destination.

3. The custom identity provider sends back the SAML assertion.
4. The sender calls the Cloud Integration endpoint (of the related integration flow).

This step is executed via an SAP BTP destination with `OAuth2SAMLBearer` authentication.

5. Cloud Integration connects to the token service (part of SAP SuccessFactors) providing the information stored in the [OAuth2 SAML Bearer Assertion](#) credentials artifact (deployed on the Cloud Integration tenant and referred to in the SuccessFactors receiver adapter).
6. The token service returns the OAuth access token.
7. Cloud Integration uses the access token to request the SAP SuccessFactors entity.

8. SAP SuccessFactors checks if the token is valid and, if that's the case, returns the entity.
9. Cloud Integration returns the SAP SuccessFactors entity to the sender app.
10. The sender app returns the entity for the given user context.

OAuth 2.0 Authorization Code Grant

The grant type Authorization Code is the most complex grant type and offers an additional level of security. The user password is never visible to the OAuth client.

In detail, OAuth 2.0 Authorization Code grant is implemented in the following way for Cloud Integration outbound communication:

As a prerequisite to initiate the Authorization Code grant workflow, the integration developer performs the following tasks:

i Note

The integration developer and the account user are typically the same person just with different roles.

1. The integration developer registers an application, so an OAuth 2.0 client (with client Id, client secret, authorization URL, and token service URL) is created. For the Mail adapter, the integration developer creates the OAuth 2.0 client in the Microsoft Active Directory tenant.

i Note

For the registration of the application, you need to specify a Redirect URI which is used by the Token Service to return the authorization code to the SAP Cloud Integration tenant. Determine the Redirect URI in the following way:

- a. Log into SAP Cloud Integration and check your host name in the browser address field:

`https://<host name>/itspaces`

- b. Use the <host name> to construct the following redirect URI:

`https://<host name>/itspaces/odata/api/v1/OAuthTokenFromCode`

2. The integration developer uses SAP Cloud Integration and creates an **OAuth2 Authorization Code** credentials artifact and deploys it on the SAP Cloud Integration or Edge Integration Cell tenant. During this step, the integration developer specifies the parameters **Client ID**, **Client Secret**, **Authorization URL**, and **Token Service URL** based on the values generated when creating the OAuth client in the previous step. After this step, the credential is in status **Unauthorized**.

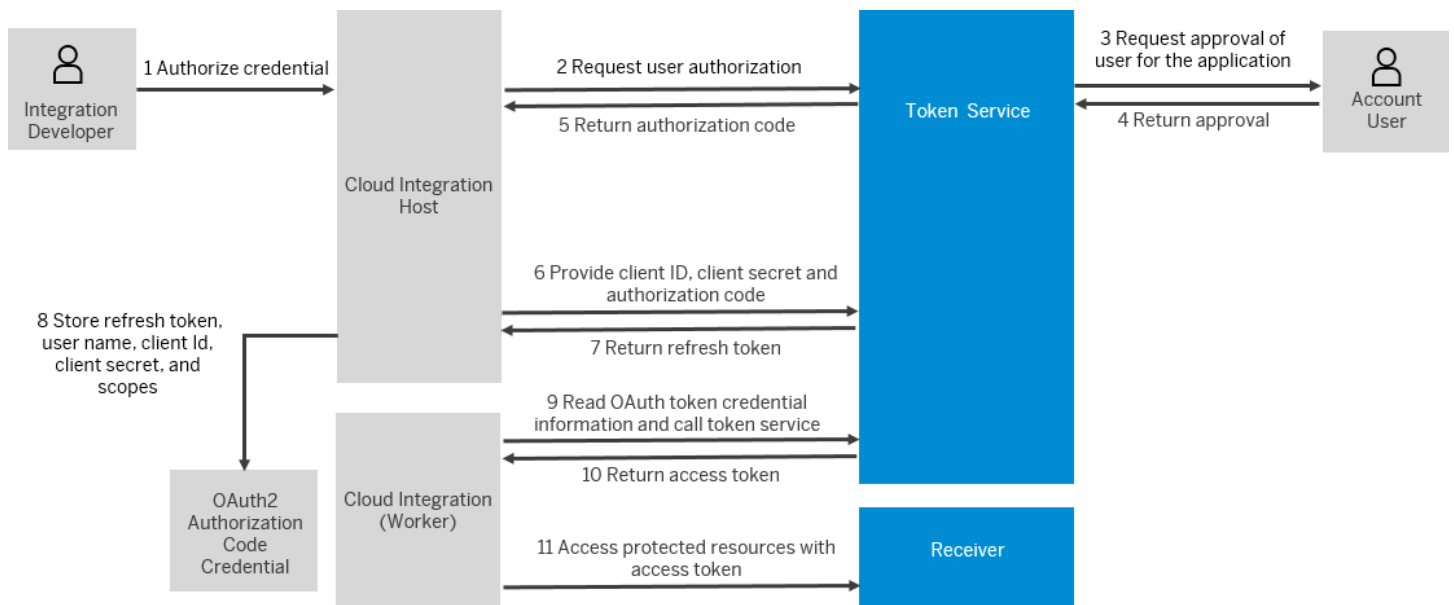
See: [Deploying an OAuth2 Authorization Code](#).

i Note

Note that Edge Integration Cell (EIC) maintains only one active refresh token at a time. Therefore, each credential must be assigned to a single runtime only.

When these steps have been executed successfully, at the authorization workflow is processed as outlined in the following chart. Note that the workflow depicted in the figure comprises user actions (of the integration developer) and system steps (executed by the Cloud Integration worker).

The following chart explains the flow for OAuth 2.0 Authorization Code:



1. The integration developer authorizes the **OAuth2 Authorization Code** credentials artifact using the Cloud Integration Web UI. TEST
2. Triggered by the **Authorize** action, Cloud Integration requests user authorization for certain scopes of the application from the token server (Authorization URL is used).
3. The token service prompts a user login screen and requests the approval of the user for the app.
4. The account user grants approval.
5. After the user gave their approval, the token service returns the authorization code of the user to the SAP Cloud Integration tenant (Redirect URL).
6. Cloud Integration calls the OAuth 2.0 token endpoint of the token service with the client ID, client secret, and the authorization code (Token Service URL with "authorization_code" grant type is used).
7. The token service checks the request and sends a refresh token as a response. It also sends an access token that is ignored by the SAP Cloud Integration tenant (Token Service URL with "authorization_code" grant type is used).
8. Cloud Integration stores the refresh token and the user name together with the client Id, client secret, and the scopes in the OAuth 2.0 Authorization Code.

i Note

For EIC runtime, at this point, the credentials are stored on Cloud Integration are synced to EIC.

The status of the **OAuth2 Authorization Code** credentials artifact changes to **Deployed**. Or the status will change to **Stored** to signal that the credentials are not synced to Edge Integration Cell (EIC).

9. The Cloud Integration worker reads OAuth 2.0 Authorization Code information and calls the token service with client Id, client secret, and refresh token (Token Service URL with "refresh_token" grant type is used).

i Note

If the refresh token on EIC is expired, the credential state will remain as **Deployed** in the UI. This is a rare occurrence, as there's a scheduled job that automatically updates a refresh token before it expires.

10. The token service sends back an access token (Token Service URL with "refresh_token" grant type is used).
11. The Cloud Integration worker connects to the receiver system protected by OAuth 2.0 and requests access with the access token and user name, if necessary.

i Note

- **Refresh Token:** A refresh token must be valid at least for 3 days, although Microsoft allows shorter validity periods (up to 10 minutes).
- The refresh token is automatically updated before it expires by a scheduled job.
- The maximum number of OAuth2 Authorization Code credentials is 500 in Cloud Foundry and 60 in Neo (including the Microsoft 365 credentials).

i Note

Behavior for credentials deployed with Edge Integration Cell:

- The credentials are stored on Cloud Integration and synced to Edge Integration Cell. Then, the credential token refresh operations are handled only on EIC.

Therefore, if any issues occur and a new token generated on Cloud Integration expires before reaching EIC, scenarios related to this expired refresh token will fail on EIC. In such cases, re-authorize the credential to obtain a new refresh token. This new valid refresh token will then be successfully synchronized to EIC.

- Refresh token updates performed by SAP (system user) are not displayed in the Web UI as modified by SYSTEM_USER for credentials assigned to EIC.
- If the token data synchronized to EIC is lost, the status shown in the UI remains unchanged until the next periodic synchronization. Because the updated refresh token will only exist on EIC, any data loss requires a re-authorization through the UI.
- Refresh tokens that expire and are invalidated aren't supported by EIC when deploying an OAuth2 Client Authorization Code with Generic Provider. For more information, see [Deploying an OAuth2 Client Authorization Code with Generic Provider](#).

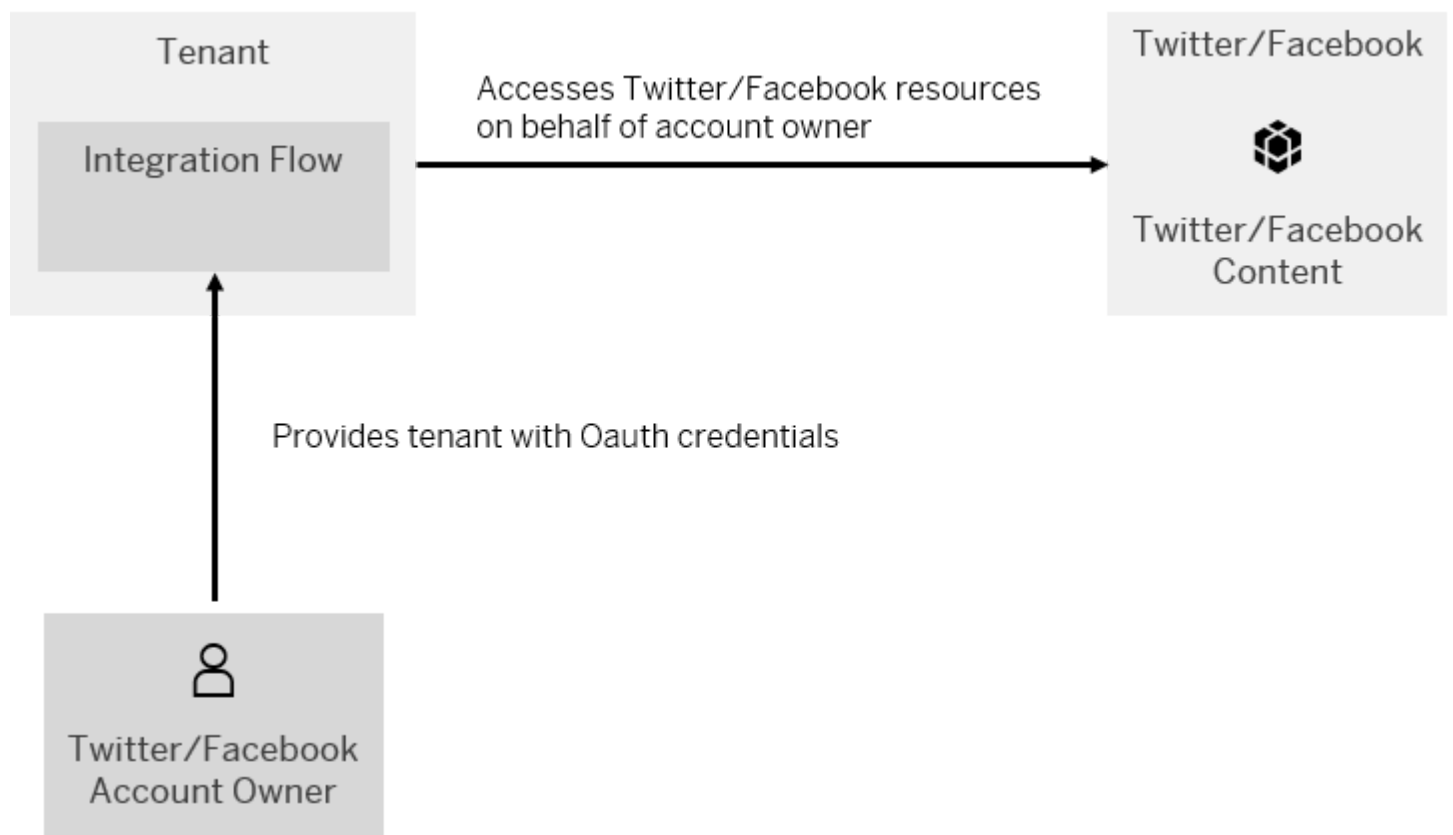
OAuth 2.0 Used With Twitter and Facebook Adapter

OAuth can be used for outbound communication with the Twitter or Facebook receiver adapter. In this case, the OAuth 2.0 roles are used in the following way:

- The tenant is the client that accesses Twitter or Facebook (as resource server).
- The Twitter or Facebook account owner is the user (that owns the protected resources which is Twitter or Facebook content).
- Using an API (for Twitter or Facebook), the user generates the OAuth 2.0 credentials (client credentials as well as token credentials) required in order to access the protected resources.
- The user provides the client (tenant) with the OAuth 2.0 credentials in the following way:

For each OAuth 2.0 credential, a separate **Secure Parameter** artifact is created and deployed on the tenant. In the Twitter or Facebook adapter, the credential names are specified.

The following figure illustrates the OAuth 2.0 communication flow for this use case.



Adapters Supporting OAuth 2.0 (Outbound Communication)

The following receiver adapters support OAuth 2.0.

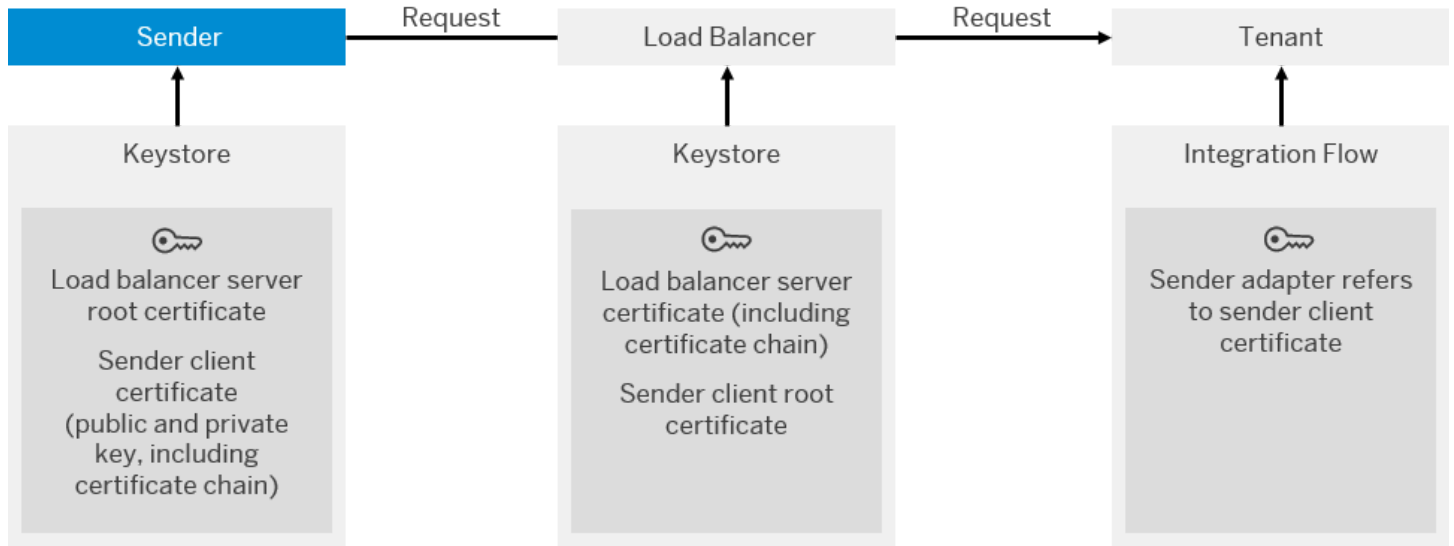
Receiver Adapter	OAuth 2.0 Client Credentials Grant	OAuth 2.0 SAML Bearer Assertion Grant	OAuth 2.0 Authorization Code Grant	OAuth 2.0 for Twitter/Facebook Adapter
AMQP/Websocket	✓			
Facebook				✓
HTTP	✓	✓		
Mail			✓	
OData/V2	✓	✓		
OData/V4	✓			
SuccessFactors/OData V2		✓		
SuccessFactors/ SOAP		✓		
Twitter				✓

Load Balancer Root Certificates Supported by SAP

The load balancer supports a certain list of root certificates.

A system sending a message to the Cloud-based integration platform using HTTPS as secure transport channel is not directly connected to the tenant. Instead of this, a load balancer component is interconnected that terminates all inbound HTTPS requests, and re-establishes a new secure connection.

To set up a secure connection between a sender system and the integration platform, you therefore need to make sure that the sender system's keystore contains a client certificate that is signed by one of those certification authorities (CAs) that are trusted by the load balancer component of SAP.



For more information on the **root certificates** that are supported by the load balancer, check out SAP Note [2801396](#).

i Note

A specific certificate that identifies a certification authority (CA) is referred to as *root certificate*. Such a certificate is typically not signed by any other authority, as it is *at the root* of a certificate chain.

The load balancer component is owned by SAP, and you, the customer, don't need to care how it is configured. However, you need to make sure that the client certificate in your sender keystore is signed by one CA that is listed at SAP Note [2801396](#).

Be aware that only **root certificates** are being imported into the *Keystore* of the *SAP Load Balancer*. Therefore you as a customer must always assign the whole certificate chain to the certificate to enable the connected component to evaluate the chain of trust.

SFTP-Based Communication

A tenant can connect **as SFTP client** to an SFTP server (the latter either hosted at SAP or in the customer landscape).

Depending on the direction of data flow (whether the tenant reads data from the SFTP server or writes data to it), either an SFTP sender adapter or SFTP receiver adapter is involved.

Files are stored on the SFTP server in specific directories referred to as *mailboxes*. For each mailbox, a user is specified in order to control access to the data.

In certain cases, you have the option to choose between the following authentication options for SFTP connectivity in the SFTP (sender or receiver) adapter:

- User Name/Password
- Public Key

User Name/Password Authentication

The tenant connects to the server with a user and authenticates itself against the SFTP server with a password.

The user credentials (user name and password) are stored in a User Credentials artifact which has been deployed on the tenant prior to connection set up.

Public Key Authentication

In order to set up secure connection between the SFTP client and SFTP server, a combination of symmetric and asymmetric keys is applied.

- Symmetric (session) keys are used in order to encrypt and decrypt data within a data transfer session.
- Asymmetric key pairs (on client and server side) are used in order to encrypt and decrypt the session keys.

Symmetric and asymmetric keys are used by a client and a server exchanging data via SFTP in the following way:

1. The client connects to the server.
2. The server sends his public key to the client.
3. The client checks if the server is a trusted participant by evaluating a known_hosts file at client's side: if the server's public key is listed there-in, the identity of the server is confirmed.
4. The client generates a session key (to be used for one data transfer session).
5. The client encrypts the session key with the public key of the server.
6. The client sends the encrypted session key to the server. As public and private key of one party are mathematical correlated with each other, the server can decrypt the session key using its private key.
7. The session can now be continued in an encrypted way.
8. As part of the secure data transfer (using the session key exchanged by the step before), the client sends its public key to the server.
9. The server checks if the public key of the client is known to him (evaluating an authorized_keys file on the server side).
10. The server encrypts a random number with the client's public key and sends it to the client.
11. The client decrypts the random number with its private key and sends the unencrypted random number back to the server. That way, the client authenticates itself on server side.

Related Information

[Inbound SFTP With Public Key Authentication](#)

[Outbound SFTP With Public Key Authentication](#)

How SFTP Works

A tenant can connect as **SFTP client** to an SFTP server (the latter either hosted at SAP or in the customer landscape).

Depending on the direction of data flow (whether the tenant reads data from the SFTP server or writes data to it), either an SFTP sender adapter or SFTP receiver adapter is involved.

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4. The client generates a session key (to be used for one data transfer session).
5. The client encrypts the session key with the public key of the server.
6. The client sends the encrypted session key to the server. As public and private key of one party are mathematical correlated with each other, the server can decrypt the session key using its private key.
7. The session can now be continued in an encrypted way.
8. As part of the secure data transfer (using the session key exchanged by the step before), the client sends its public key to the server.
9. The server checks if the public key of the client is known to him (evaluating an authorized_keys file on the server side).
10. The server encrypts a random number with the client's public key and sends it to the client.
11. The client decrypts the random number with its private key and sends the unencrypted random number back to the server. That way, the client authenticates itself on server side.

Related Information

[Inbound SFTP With Public Key Authentication](#)

[Outbound SFTP With Public Key Authentication](#)

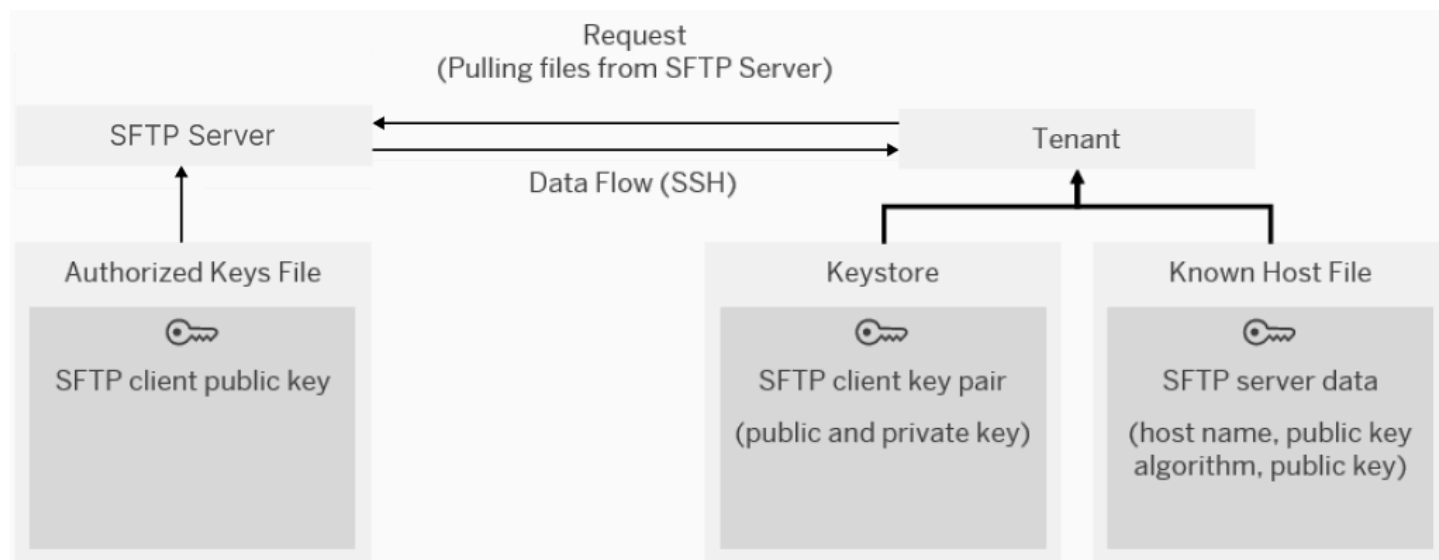
Inbound SFTP With Public Key Authentication

For an SFTP client connected to an SFTP server using the Public Key authentication option, the following artifacts have to be generated and stored at the locations summarized in the following table. The table also shows which artifacts need to be exchanged between the client and the server (during the onboarding process):

SFTP Client Side	SFTP Server Side
Public keys of all connected SFTP servers A public key is used in order to authenticate the SFTP server (as known host) on the SFTP client side. Public keys of all connected SFTP servers are stored in a <i><known_hosts></i> file on the client side. i Note The <i><known_hosts></i> file contains the public keys and addresses of the trusted SFTP servers. The client checks if the server is a trusted participant by evaluating a <i><known_hosts></i> file on the client side: If the server's public key is listed there, the identity of the server is confirmed. i Note Generating the public key of the SFTP server is the task of the expert that hosts the SFTP server.	Public keys of all connected SFTP clients (used in order to authenticate the SFTP clients on the SFTP server side) This file has to be stored in an <i><authorized_keys></i> file on the SFTP server. i Note Generating this public key is the task of the expert that hosts the SFTP client.
Private key of SFTP client (stored on client) i Note The private key of the SFTP client can be either an RSA private key file or a DSA private key file. The private key (together with its associated public key) has to be stored in a keystore. i Note Generating this private key is the task of the expert that hosts the SFTP client.	Private key of SFTP server (stored on server) i Note Generating this public key is the task of the expert that hosts the SFTP server.

A tenant can connect as an SFTP client to an SFTP server (the latter either hosted at SAP or in the customer landscape).

The following figure shows the basic setup of components used for SFTP for inbound communication (when the data flow is directed from an SFTP server to the tenant).



To specify the technical details of the message flow from the SFTP sender to the tenant (SFTP client), a **sender SFTP adapter** has to be configured for the related integration flow.

Outbound SFTP With Public Key Authentication

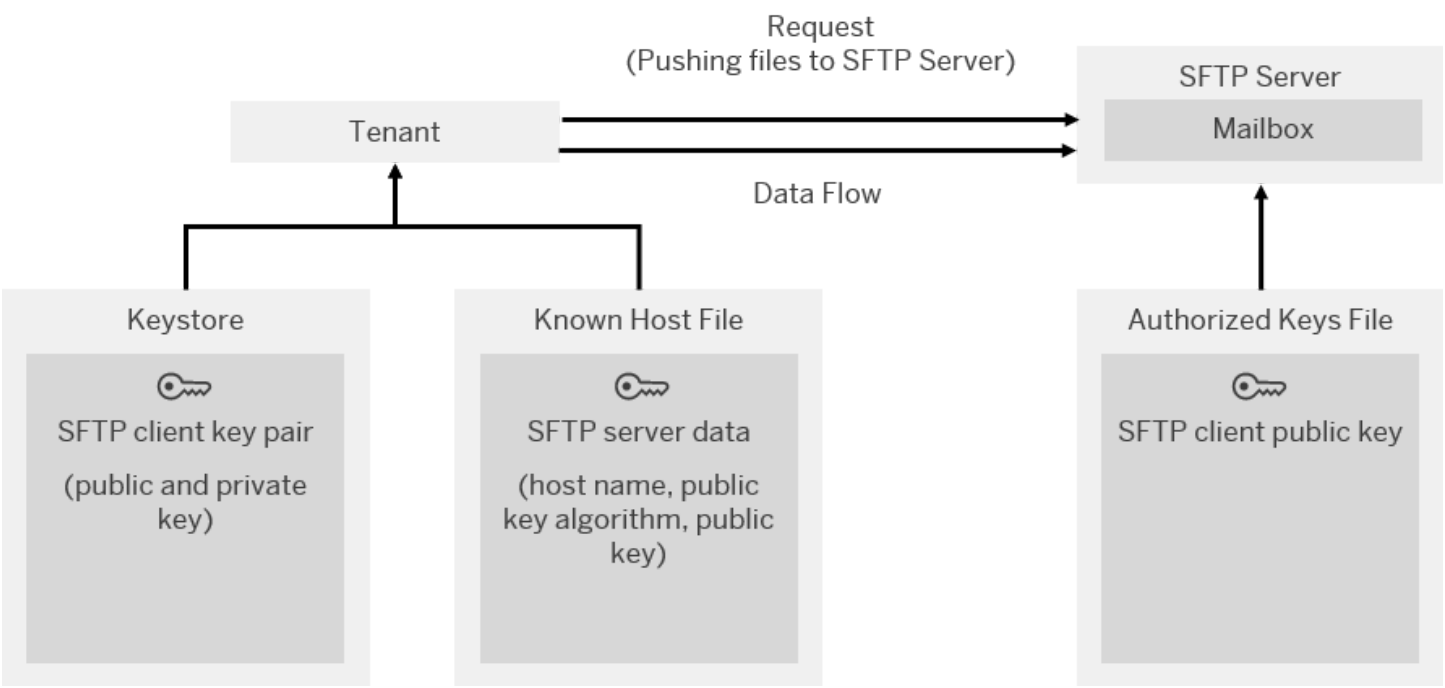
For an SFTP client connected to an SFTP server using the Public Key authentication option, the following artifacts have to be generated and stored at the locations summarized in the following table. The table also shows which artifacts need to be exchanged between the client and the server (during the onboarding process):

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Private key of SFTP client (stored on client) i Note The private key of the SFTP client can be either an RSA private key file or a DSA private key file. The private key (together with its associated public key) has to be stored in a keystore. i Note	Private key of SFTP server (stored on server) i Note Generating this public key is the task of the expert that hosts the SFTP server.

SFTP Client Side	SFTP Server Side
Generating this private key is the task of the expert that hosts the SFTP client.	

A tenant can connect as an SFTP client to an SFTP server (the latter either hosted at SAP or in the customer landscape).

The following figure shows the basic setup of components used for SFTP for outbound communication (when the data flow is directed from the tenant to an SFTP server).



To specify the technical details of the message flow from the tenant (SFTP client) to the SFTP server, an **SFTP receiver adapter** has to be configured for the related integration flow.

Message-Level Security

Several standards are supported to protect the message content (message-level security).

Message-level security features allow you to digitally encrypt/decrypt or sign/verify a message (or both). The following standards and algorithms are supported.

Message-Level Security Options		
Security Standard	Security Feature	Supported Algorithms
PKCS#7/CMS Enveloped Data and Signed Data PKCS#7/CMS provides a syntax for data that has cryptography applied to it, such as digital signatures or digital encryption. The CMS specification can be found at: http://tools.ietf.org/html/rfc5652	Encryption/decryption of message content	Supported algorithms (by the symmetric key) for content encryption (format/Mode/Padding Scheme): AES/GCM/NoPadding, AES/CCM/NoPadding, AES/ARCFOUR/ECB/NoPadding, Camellia/CBC/PKCS5Padding, CAST5/CBC/PKCS5Padding, DES/CBC/PKCS5Padding, DESede/CBC/PKCS5Padding, RC2/CBC/PKCS5
	Signing/verification of payload	Supported algorithms for content signing (digest and encryption algorithm): SHA3-256/RSA, SHA3-384/RSA, SHA3-512/RSA, SHA512/RSA, SHA384/RSA, SHA224/RSA, SHA/RSA, RIPEMD128/RSA, RIPEMD160/RSA, RIPEMD256/RSA, RIPEMD160andMGF1/RSA-ISO9796-2-2-3, SHAandMGF1/RSA-ISO9796-2-2-3, SHA512/DSA, SHA3-384/DSA, SHA3-256/DSA, SHA3-224/DSA, SHA512/DSA, SHA256withDSA, SHA224withDSA, SHA/DSA, SHA3-224/ECDsa, SHA3-256/ECDsa, SHA3-384/ECDsa, SHA512/ECDsa

Security Standard	Security Feature	Supported Algorithms
Digitally signing a message is based on the CMS type Signed Data. Digitally encrypting or decrypting the content of a message is based on the CMS type Enveloped Data.		384/ECDSA, SHA3-512/ECDSA, SHA512/ECDSA, SHA384/ECDSA, SHA256/ECDSA, SHA1/ECDSA. The generated signature conforms to the CAdES-BES (CMS Advanced Electronic Signature) signature standard according to the ETSI TS 101 733 V1.7.4, 1.8.1, 1.8.3, 2.1.1. published at: https://www.etsi.org/deliver/etsi_ts/101700_101799/101733/02.02.01_60/t
PKCS#7/CMS Enveloped Data and Signed Data	Encryption/decryption and signing/verification of payload	Signature algorithms: AES/GCM/NoPadding, AES/CCM/NoPadding, MD5/RSA, RIPEMD160/RSA, RIPEMD256/RSA, SHA/RSA, SHA224/RSA, SHA256/RSA, SHA512/RSA. This is a subset of the algorithms that are supported for PKCS#7/CMS Enveloped Data. The generated signature does not conform to the CAdES-BES (CMS Advanced Electronic Signatures) signature standard.
Basic Digital Signature Option (<i>Simple Signer</i>)	Signing payload	Supported algorithms for content signing (digest and encryption algorithm): RIPEMD160andMGF1/RSA-ISO9796-2-2-3, RIPEMD128/RSA, RIPEMD160/RSA, SHA/RSA, SHA/DSA, SHA224/RSA, SHA256/RSA, SHA384/RSA, SHA512/RSAandMGF1/RSA-ISO9796-2-2-3, SHA256withDSA, SHA224withDSA, SHA256/RSA, SHA3-384/RSA, SHA3-512/RSA, SHA3-512/DSA, SHA3-384/DSA, SHA3-224/DSA, SHA512/DSA, SHA384/DSA, SHA3-224/ECDSA, SHA3-256/ECDSA, SHA3-512/ECDSA, SHA512/ECDSA, SHA384/ECDSA, SHA256/ECDSA, SHA224/ECDSA, SHA1/ECDSA.
Open Pretty Good Privacy (PGP)	Encryption/decryption of message content	Supported symmetric key algorithms for content encryption (symmetric key 128, 192, and 256-bit key, Blowfish (128 bit key, 16 rounds), CAST5 (128 bit key), DESede with 168-bit key, Twofish with 256-bit key. DES is not supported.
	Encryption/decryption and signing/verification of the message	Supported signature algorithms for PGP signing: MD5, RIPE-MD/160, SHA-1, SHA-384, SHA-512.
XML Signature	Signing/verification of payload	Supported signature algorithms: SHA1/DSA, SHA1/RSA, SHA256/RSA, SHA512/RSA, SHA224/ECDSA, SHA256/ECDSA, SHA384/ECDSA, SHA512/ECDSA.
XML Advanced Electronic Signature (XAdES) Supported XAdES forms: XAdES Basic Electronic Signature and XAdES Explicit Policy based Electronic Signature	Signing payload	The same signature algorithms as for XML Signature are supported.
WS-Security	Signing/verification of SOAP body	The default signature algorithm is set by the data in the certificate, that is, or http://www.w3.org/2000/09/xmldsig#rsa-sha1 or http://www.w3.org/2000/09/xmldsig#rsa-sha1 .
	Encryption/decryption of message content	The default signature digest algorithm is: http://www.w3.org/2000/09/xmldsig#sha1 .

Strong encryption is supported for the following algorithms:

- AES/CBC/PKCS5Padding
- Camellia/CBC/PKCS5Padding

For these algorithms, the key lengths 192 and 256 are possible.

Caution

Algorithms starting with SHA1, MD2, or MD5 are still supported for compatibility reasons, but they no longer meet today's security requirements. Therefore, we recommend using stronger algorithms where possible. Check with your security experts or authorities like NIST for more detailed security recommendations.

Recommendations

Some algorithms (like MD2, MD5, DES or RC4) are still supported for legacy reasons, but they are not considered secure any more. We recommend that you check the official recommendations from National Institute of Standards and Technology (NIST) or European Union Agency for Network and Information Security (ENISA) for advice on algorithms and key strengths (for example, at: <https://www.enisa.europa.eu/activities/identity-and-trust/library/deliverables/algorithms-key-sizes-and-parameters-report> ).

Related Information

[How PKCS#7 Works](#)

[How XML Signature Works](#)

[How WS-Security Works](#)

[How OpenPGP Works](#)

How PKCS#7 Works

You have the option to sign and encrypt message payloads based on PKCS#7/CMS Enveloped Data and Signed Data (PKCS stands for Public Key Cryptography Standards).

Related Information

[Signing and Verifying a Message Using PKCS#7/CMS](#)

[Encrypting and Decrypting the Content of a Message Using PKCS#7/CMS](#)

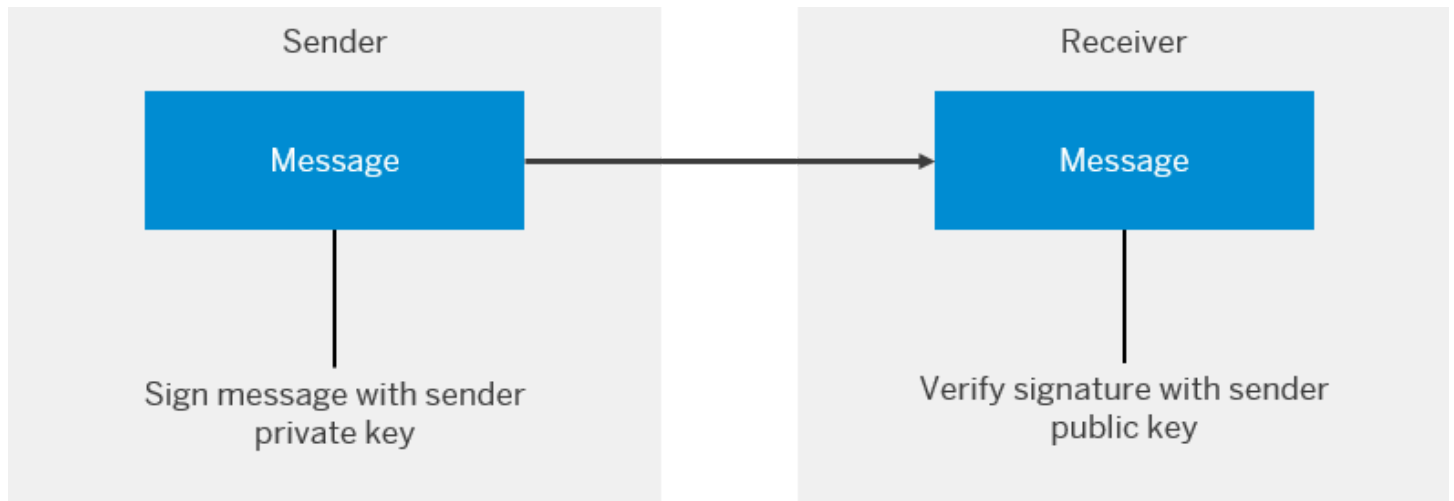
Signing and Verifying a Message Using PKCS#7/CMS

A digital signature ensures the authenticity of a message that way that it guarantees the identity of the signer and that the message wasn't altered after signing.

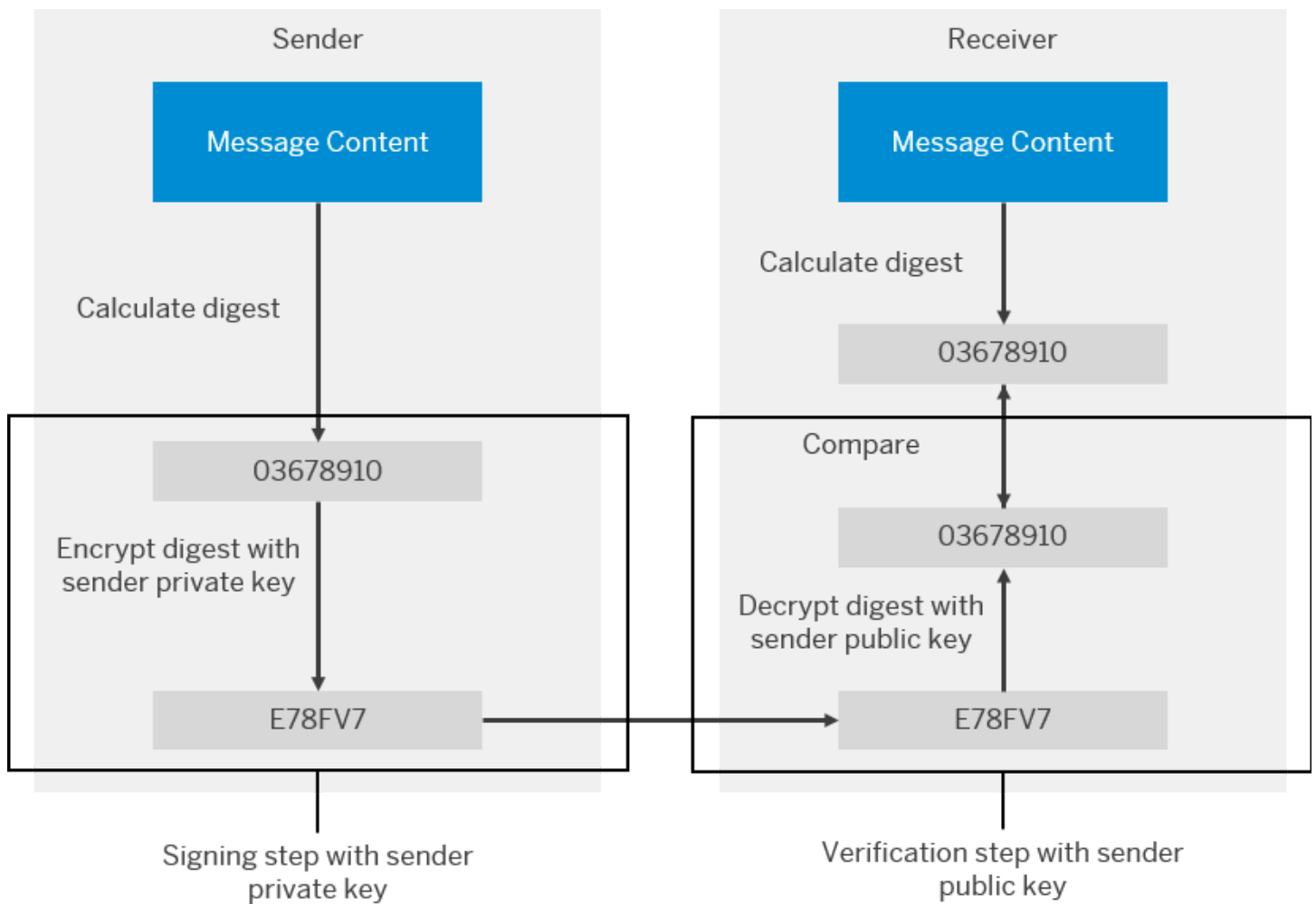
Process Overview

In simple terms, digitally signing and verifying a message works in the following way:

1. The sender signs the message using its own private key.
2. The receiver verifies the signature by using the public key associated with the sender's private key.



On a more technical level, the signing and verifying process based on the PKCS#7/CMS standard works in the following way.



i Note

To support this use case, Cloud Integration comes with the following integration flow components: **PKCS7/CMS Signer** and **PKCS7 Signature Verifier**.

When referring to *sender* and *receiver* in the context of Cloud Integration and the PKCS#7/CMS-based signing and verifying process, we therefore mean more specifically:

- The **PKCS7/CMS Signer** step (processed by the sender component in the figure above)
- The **PKCS7 Signature Verifier** step (processed by the receiver component in the figure above)

Steps Performed by the Sender

At runtime, the sender performs the following steps:

1. Calculates from the message content a digest (also referred to as *hash value*) using a digest algorithm.

→ Tip

A digest (also referred to as *hash value*) is an expression with a fixed size calculated from an input value (which can be of any size). Performing the same hash algorithm on the same data results in the same digest. However, the inverse operation isn't possible: The input value can't be reproduced out of the hash value.

2. Encrypts the digest using a private key (type RSA or DSA). This is actually the signing step.

Supported algorithms for content signing (digest and encryption algorithm): SHA3-224/RSA, SHA3-256/RSA, SHA3-384/RSA, SHA3-512/RSA, SHA512/RSA, SHA384/RSA, SHA256/RSA, SHA224/RSA, SHA/RSA, RIPEMD128/RSA, RIPEMD160/RSA, RIPEMD256/RSA, MD5/RSA, MD2/RSA, RIPEMD160andMGF1/RSA-ISO9796-2-2-3, SHAandMGF1/RSA-ISO9796-2-2-3, SHA3-512/DSA, SHA3-384/DSA, SHA3-256/DSA, SHA3-224/DSA, SHA512/DSA, SHA384/DSA, SHA256withDSA, SHA224withDSA, SHA/DSA, SHA3-224/ECDSA, SHA3-256/ECDSA, SHA3-384/ECDSA, SHA3-512/ECDSA, SHA512/ECDSA, SHA384/ECDSA, SHA256/ECDSA, SHA224/ECDSA, SHA1/ECDSA.

3. Sends the encrypted digest (which corresponds to the signature) together with the message content to the receiver.

Steps Performed by the Receiver

At runtime, the receiver performs the following steps:

1. Retrieves the information required to verify the signature.

In particular, the receiver retrieves the message content, the signature (that means, the encrypted digest of the message content), the digest algorithm, and the public key (associated with the private key that was used to sign the data). In more detail, the **PKCS7 Signature Verifier** performs the following steps:

i Note

There are different options how the sender transfers this information to the receiver (in a format called *Signed Data*).

See also: [Options to Handle Signed Data](#)

- a. Checks if the property `SapCmsSignedData` exists.

If this property exists, the **PKCS7 Signature Verifier** takes the property value as Signed Data object and assumes that the message body contains the content to be verified.

- b. If the property `SapCmsSignedData` does not exist, the **PKCS7 Signature Verifier** checks if the header `SapCmsSignedData` exists.

If the header exists, the **PKCS7 Signature Verifier** takes the header value as Signed Data object and assumes that the message body contains the content to be verified.

- c. If the header does not exist, the **PKCS7 Signature Verifier** assumes that the message body contains a Signed Data object which includes the content.

2. Decrypts the digest with the public key (which is related to the senders' private key). The public key has the type RSA or DSA.
3. Calculates the digest out of the content of the message (which has been sent to it by the sender).

The receiver uses the same digest algorithm that the sender had used.

i Note

PKCS#7 ensures that the digest algorithm is transferred together with the signature of the message and therefore available for the receiver.

This calculation is based on the message content. In case the message content has been transferred encrypted, a decryption step is needed before this step.

4. Compares the decrypted digest (from the sender) with the one calculated at receiver side. In case both values (digests) are identical, the signature is verified.

Options to Handle Signed Data

To enable the receiver (using the PKCS7 Signature Verifier) to verify the signed message, there needs to be a standard way to pass on the information required to verify the message signature from the sender to the receiver. This is where PKCS#7 Signed Data comes into play. Signed Data is a specific format that can be used to transfer this information from the sender to the receiver.

The **PKCS7/CMS Signer** step supports different ways to handle Signed Data.

- Signed Data can be transferred together with the message content as part of the message body. To configure this option, in the **PKCS7/CMS Signer** step select the option **Include Content in Signed Data**.
- Signed Data can be transferred in an Exchange property (**SapCmsSignedData**). To enable this option, in the **PKCS7/CMS Signer** step deselect the option **Include Content in Signed Data**.

Note that the signer provides no option any more to automatically store the Signed Data object in a header – for reasons of security: If stored in a header, the Signed Data object is by default sent together with the message content to a receiver, if no other measures are taken.

Up to version 1.2 of the **PKCS#7/CMS Signer** you can choose to include the signed data in the **SapCmsSignedData** header. From version 1.3 of the **PKCS#7/CMS Signer** onwards, you can include the signed data in the **SapCmsSignedData** property.

However, if you still like to use a header on purpose to transfer the Signed Data object to another component, you can define this header in a dedicated step, for example, a content modifier.

As an additional option, you can base64-encode the Signed Data object (independent of the option you have chosen for **Include Content in Signed Data**). To do this, select the option **Encode Signed Data with Base64**.

Likewise, the **PKCS7 Signature Verifier** step can handle the Signed Data object (received from the sender) in different ways. Depending on how the verifier expects to receive the Signed Data object, you can select one of the following options when configuring the **PKCS7 Signature Verifier** step:

Option	Description
Header/Property is Base64-Encoded	Select this option if you expect that the Signed Data object is transferred to the receiver that verifies the message either in a header or a property (with the name SapCmsSignedData) and, additionally, the value is base64-encoded.
Body is Base64-Encoded	Select this option if you expect that the message body contains either the Signed Data object including the content or the content alone base64-encoded.

Encrypting and Decrypting the Content of a Message Using PKCS#7/CMS

Encode the content of a message in such a way that only authorized parties can read it.

Digital encryption using PKCS#7/CMS works two-stage based on symmetric and asymmetric key technology:

1. The sender encrypts the content of the message using a symmetric key.

i Note

The following algorithms for content encryption (by the symmetric key) are supported (format Cipher/Operation Mode/Padding Scheme): DESede/CBC/PKCS5Padding, DES/CBC/PKCS5Padding, AES/CBC/PKCS5Padding, ARCFOUR/ECB/NoPadding, Camellia/CBC/PKCS5Padding, RC2/CBC/PKCS5Padding, CAST5/CBC/PKCS5Padding.

2. The sender encrypts the symmetric key using a public key.

i Note

To encrypt the symmetric key, a public key of type RSA (with the cipher – or algorithm – RSA/ECB/PKCS1Padding) is used for each recipient.

3. The sender sends the encrypted message and the encrypted symmetric key to the receiver.
4. The receiver decrypts the symmetric key using a private key (which is related to the public key used by the sender).

i Note

For this decryption step, you need a private key of type RSA.

5. The receiver decrypts the content of the message using the decrypted symmetric key.

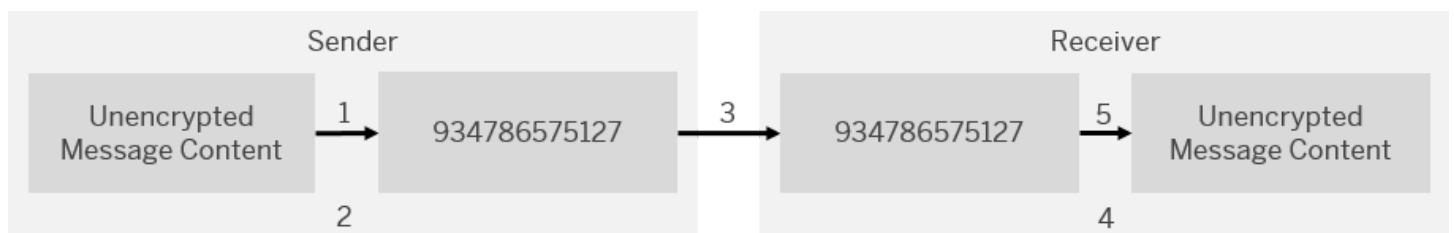
i Note

Strong encryption is supported for the following algorithms:

- AES/CBC/PKCS5Padding
- Camellia/CBC/PKCS5Padding

For these algorithms also the key lengths 192 and 256 are possible.

The following figure illustrates the process of digitally encrypting and decrypting the content of a message.



How XML Signature Works

A digital signature ensures the authenticity of a message that way that it guarantees the identity of the signer and that the message was not altered after signing. You have the option to digitally sign and validate a message based on the XML Signature

standard (issued by the W3C consortium). Applying this standard means that the digital signature of a document itself is stored as an XML element.

XML Signature can be applied to any XML document.

The following options for XML Signature are supported:

Options to Apply XML Signature

Option	Description
Enveloped Signature	Digital signature/validation is applied to XML element that contains the signature as an element (the Signature element). Using this option, the digital signature is part of the XML document to be signed/validated.
Enveloping Signature	Digital signature/validation is applied to content within an Object element which is part of the Signature element. That way, the Signature elements acts as an envelope for the signed content. Using this option, the digital signature is part of the XML document to be signed/validated.

i Note

You configure the usage of XML Signature in the related integration flow.

For more information on the supported signature algorithms and canonicalization methods, see: [Sign the Message Content with XML Digital Signature](#).

Background Information

In a simplified view, when configured correctly, digitally signing a message based on XML Signature implies the following main steps:

1. The sender of the message canonicalizes the XML message content that is to be signed.

Canonicalization transforms the XML document to a standardized (reference) format. This step is required because an XML document can have more than one valid representations. Calculating a digest out of two different representations of the same document (according to step 2) results in different digests (or hash values). This would make the whole signing/validating process invalid.

2. Out of the canonicalized XML document, a digest is calculated using a digest algorithm.

3. The sender builds up a **SignedInfo** element that contains the digest.

4. The sender canonicalizes the **SignedInfo** element.

5. The sender builds a second digest for the **SignedInfo** element which contains the first digest.

6. The sender encrypts the digest using its private key.

7. The sender builds up the **SignatureValue** element which contains the encrypted digest from step 5 (the signature).

8. The message is sent to the receiver.

Digitally verifying (validating) a message based on XML Signature works the following way:

1. The receiver decrypts the encrypted digest (which is part of the **SignatureValue** element of the received message) using the sender's public key.
2. The receiver calculates the digest out of the **SignedInfo** element of the message.
3. The receiver compares the two digests that result out of steps 1 and 2.

That way it is the authenticity of the sender is checked.

4. The receiver canonicalizes the XML message content.
5. The receiver calculates the digest out of the XML message content.
6. The receiver compares the digest that results from the canonicalized message content with that one contained in the **SignedInfo** element of the message.

That way, it is made sure that the content of the message has not been altered during message processing.

How WS-Security Works

Messages can be protected according to the WS-Security standard.

There are the following options:

- Digitally sign a message (and the other way round to verify a signed message)
- Digitally sign a message and to encrypt the message content (and the other way round to verify a message and to decrypt the message content)

i Note

For more information on the WS-Security standard, see https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=wss .

Related Information

[WS-Security Configuration for the Sender SOAP 1.x Adapter](#)

[WS-Security Configuration for the Receiver SOAP 1.x Adapter](#)

How OpenPGP Works

You can use Open Pretty Good Privacy (Open PGP) to digitally sign and encrypt messages.

OpenPGP gives you the following options to protect communication at message level:

- You can encrypt a payload.
- You can sign and encrypt a payload.

OpenPGP does not support signing without encryption or just verifying without decryption. The tenant expects either an encrypted payload or a signed and encrypted payload.

During runtime, the encryptor/signer processor signs and encrypts the body of the inbound message and returns the resulting OpenPGP message in the body of the outbound message.

The required keys are stored in OpenPGP keyrings. The following types of keyrings exist:

PGP Keyrings

Type of Keyring	Description
PGP secret keyring	Contains the public/private key pairs of the sender. It can contain multiple key pairs, each identified by a user ID. The private key is protected with a passphrase. For PGP secret keyrings deployed on tenants, the same passphrase has to be used to access all private keys of the PGP secret keyring.
PGP public keyring	Contains the public keys (related to the private keys that are stored in the PGP secret keyring of the communication partner).

OpenPGP Signing/Verifying

A digital signature ensures the authenticity of a message by guaranteeing the identity of the signer and that the message has not been altered since signing.

A message is digitally signed and verified as follows:

1. The sender calculates a digest (or hash value) from the message content using a digest algorithm.

The following hash algorithms are supported for PGP signing:

- o For DSA key: SHA-1, SHA224, SHA256, SHA384, SHA512
- o For RSA key: MD5, SHA-1, RIPE-MD/160, SH256, SHA384, SHA512, SHA224

2. The sender encrypts the digest using a private key (type RSA or DSA). This is the actual signing step.

The private key is looked up in the sender's PGP secret keyring.

3. The encrypted hash value, together with the hash algorithm that has been used, is written to the signature element that is sent to the receiver together with the payload (as PGP signature format). The key ID of the signer of the private key is also written to the PGP signature format.

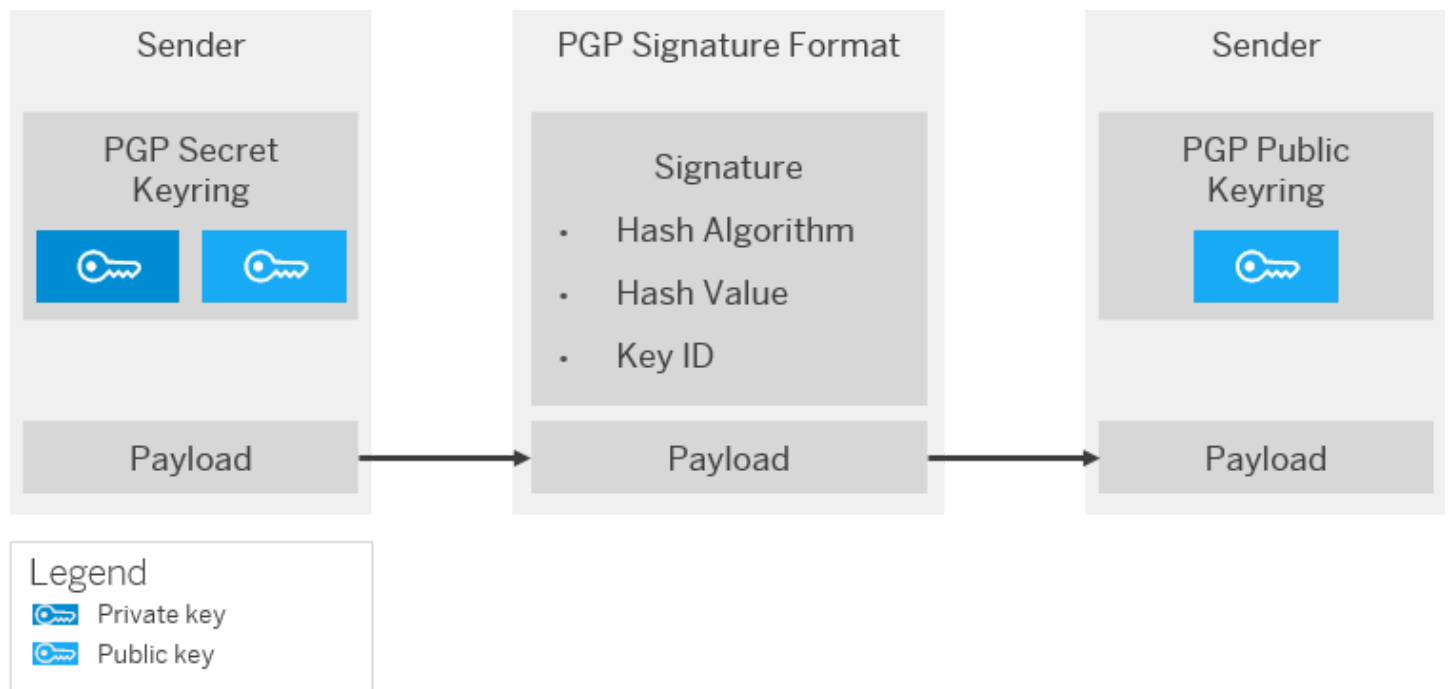
4. The receiver obtains the PGP signature format.

5. The receiver selects the key ID from the signature and uses the key ID to look up the right public key in the receiver's PGP public keyring. This is the public key that corresponds to the private key used to sign the payload.

In addition, the receiver checks whether the user ID (associated with the key ID) corresponds to an allowed user.

6. The receiver decrypts the hash value (and verifies the payload) using the public key.

The following figure illustrates the concept.



OpenPGP Encrypting/Decrypting

Digital encryption allows you to encode the content of a message in such a way that only authorized parties can read it.

A message is digitally encrypted and decrypted as follows:

1. The sender generates a symmetric key.
2. The sender encrypts the payload with the symmetric key.

The following symmetric key algorithms for content encryption (symmetric key algorithms) are supported:

TripleDES (168bit key derived from 192), CAST5 (128 bit key, as per [RFC2144]), Blowfish (128 bit key, 16 rounds), AES with 128, 192, and 256-bit key, Twofish with 256-bit key

DES is not supported.

3. The sender looks up a public PGP key in the PGP public keyring.
4. The sender encrypts the symmetric key using the public PGP key (from the PGP public keyring).

You can use the following key types to encrypt the symmetric key: RSA and Elgamal (DAS is not supported).

5. The sender writes the encrypted symmetric key and the key ID into the Encryption Info element of the message.

The key ID is used to identify the public key used for encryption (as the PGP public keyring can contain more than one public key).

The Encryption Info element is sent to the receiver, together with the encrypted payload.

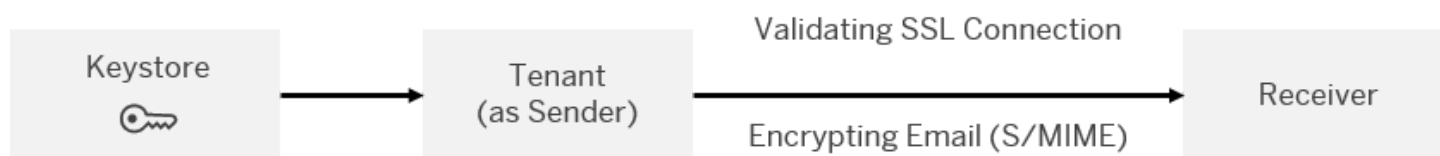
6. The receiver obtains the message and, based on the key ID (in the Encryption Info element), looks up the correct private key (associated with the public key used to encrypt the payload) in the PGP secret keyring.

A passphrase is required to access the private key.

7. The receiver decrypts the symmetric key with the private key from the PGP secret keyring.
8. The receiver decrypts the payload with the symmetric key.

There is an option to compress data before the encryption step. The following compression algorithms are supported: ZIP [RFC1951], ZLIB [RFC1950], BZip2.

The following figure illustrates the concept.



The runtime supports the following features:

- Signing with several private keys (the resulting OpenPGP message then contains several signatures).
- Encryption with several public keys.

More precisely, the symmetric encryption key can be encrypted by several public keys (the resulting OpenPGP message then contains several *Public Key Encrypted Session Key* packets).

- Optional: OpenPGP compression and base 64 output or input.
- OpenPGP allows you to apply two different kinds of keys: primary keys and subkeys. (For simplicity, these are not differentiated in the figures above.)

When you generate OpenPGP keys, a primary key with at least one subkey is created. Only the primary key can be used for certification, that is, to certify the trustworthiness of other keys. In addition, the primary key is also typically used to sign payloads. The subkey is used to encrypt payloads.

OpenPGP Message Format Specification

The OpenPGP message format is specified at <http://tools.ietf.org/html/rfc4880> . An OpenPGP message is composed of a sequence of packets. The following table contains the most important packet types.

OpenPGP Message Format - Packets

Packet Type	Description
Public Key Encrypted Session Key	Session key encrypted with a public key, key ID of the public key, and public-key algorithm
Signature	Binding between a public key and some data. There are several types of signature packets: The certification, direct key, and subkey binding signature can be self-signed. The version 4 signature packet may also contain meta-information about the signature such as creation time, issuer, or key expiration time. The version 3 signature is deprecated.
Symmetric Key Encrypted Session Key	A symmetric key (also called session key) encrypted with a symmetric key; a symmetric algorithm is used. This packet is not supported.
One-Pass Signature	Placed at the beginning of the message before the data. It contains sufficient information to allow the system to start calculating the signature before the actual signature packet (which is after the data) is reached. There can be several such packets. One packet contains the public key algorithm, the hashing algorithm, the key ID

Packet Type	Description
	of the signing key, and an indicator whether the signatures should be nested or not. A zero value indicates that the next packet is another One-Pass Signature packet that describes another signature to be applied to the same message data. i Note Nested signatures are not supported. However, several signatures over the same data in one PGP message are supported.
Public Key	
Public Subkey	Contains similar information to a public key package, but it denotes a subkey.
Secret Key	Contains all the information that is found in a public key packet, but also includes the secret key (encrypted private key).
Secret Subkey	Contains similar information to a secret key package, but it denotes a subkey.
Compressed Data	Typically, this packet contains the contents of an encrypted packet, or follows a Signature or One-Pass Signature packet, and it contains a literal data packet.
Symmetrically Encrypted Data	Data encrypted with a symmetric key (using a symmetric key algorithm). The symmetric cipher used may be specified in a Public-Key or Symmetric-Key Encrypted Session Key packet that precedes the Symmetrically Encrypted Data packet. This packet uses a variant of the cipher feedback mode (CFB) (as defined at http://tools.ietf.org/html/rfc4880).
Literal Data	Contains plain data (binary or text).
User ID	Indicates the holder of a key. The package contains the user name, e-mail address, and comment of the keyholder.
User Attribute	Variant of the User ID packet, which can contain more information about the user. It is only used together with key material. This packet is not supported.
Sym. Encrypted and Integrity Protected Data	Variant of the Symmetrically Encrypted Data packet. It contains data that is encrypted with a symmetric key algorithm (using a symmetric key algorithm) and is protected against modification by the SHA-1 hash algorithm (less strong than a signature, but stronger than bare CFB encryption). It does not use Open PGP CFB mode but pure CFB mode.

Restrictions for the Input Message Structure (for Decryptor/Verifier)

The input payload must have the following packet sequence:

Public Key Encrypted Session Key ..., Sym. Encrypted and Integrity Protected Data | Sym. Encrypted Data, (Compressed Data,) (One Pass Signature ...), Literal Data, (Signature ...)

Entries in brackets are optional, ellipses indicate repetition, commas represent sequential composition, and '|' separates alternatives.

For example, the Compressed Data packet is optional.

Restrictions for the Output Message Structure (for Encryptor/Signer)

The output PGP message is restricted to the following packet sequence:

Public Key Encrypted Session Key ..., Sym. Encrypted and Integrity Protected Data | Sym. Encrypted Data, Compressed Data, (One Pass Signature ...), Literal Data, (Signature ...)

Entries in brackets are optional, ellipses indicate repetition, commas represent sequential composition, and '|' separates alternatives.

This does mean the following:

- A symmetric key cannot be encrypted with another symmetric key.

The symmetric key that encrypts the payload cannot be encrypted by another symmetric key (which is, for example, generated from a password). OpenPGP allows this (see Symmetric Key Encrypted Session Key packet).
- Compression cannot be switched off. The Compressed Data packet is always mandatory.
- Encryption is always mandatory. It is not possible to only sign data.
- Only one password for all private keys in the keyring can be used. This simplifies password maintenance.
- Nested signatures are not supported: If there are multiple signatures in the PGP message, they all contain the same hash value built over the original payload. OpenPGP does allow nested signatures where the enclosing signature is a signature of the enclosed PGP message including the enclosed signatures.
- DSA keys can only be combined with certain hash algorithms.

Certificate Management

Depending on the applied transport- and message-level security option, different types of security artifacts need to be managed and deployed on the tenant.

- X.509 certificates

Used for transport-level security TLS and for message-level security using PKCS#7, WS-Security, and XML Digital Signature.

They are stored in a Java keystore.
- PGP keys

Used for message-level security using Open PGP.
- Known hosts files

Required for transport-level security SFTP.

SFTP keys are also stored in a Java keystore.

Related Information

X.509 Certificates

X.509 certificates (that comply with the X.509 standard) are used for transport-level security TLS and for message-level security using PKCS#7, WS-Security, and XML Digital Signature.

Elements of X.509 Certificates

This topic does not explain the standard in detail, but points out the following important elements of an X.509 certificate.

A digital certificate provides a public key that is signed by a certification authority (CA).

Elements of X.509 Certificates

Element	Description
Issuer	Specifies the CA (that issued and signed the certificate).
Subject	Specifies the entity associated with the public key of the certificate.
Distinguished Name (DN)	Comprises the issuer, the subject, and other attributes. A DN is a unique identifier of the certificate.

When you specify a certificate, you have to define additional attributes such as a company name, a country or region identification, and so on.

Related Information

- [Keystore](#)
- [Requirements for Keystore Passwords](#)
- [Certificate Chains](#)

Keystore

Certificates and key pairs are stored in one keystore per tenant, referred to also as tenant keystore.

Keystore Usage

A keystore is used to secure message exchange both at transport level and at message level.

Transport-level security (HTTPS outbound connections from the SAP Cloud Integration tenant to a remote system)

- Supporting client certificate authentication

You can protect HTTP outbound connections by specifying client certificate authentication when configuring the related receiver adapter. If you do that, the receiver system authenticates the tenant (the client) based on a client certificate.

To make this authentication option work, the tenant keystore needs to contain a *client certificate* which is a signed key pair containing a private and a public key.

During the TLS handshake, one of the key pairs whose certificate chain is trusted by the server is selected for the TLS communication. If the server does not have a certificate of an appropriate certification authority (CA) in its trust store, the communication fails because the server cannot authenticate the client. If the server trusts several key pairs, one key pair is chosen at random for the connection.

If you want to avoid random selection, you can specify an alias of a key pair entry in the related receiver adapter, so that only this specific key pair can be used in the TLS communication (use the [Private Key Alias](#) parameter for this purpose). If the keystore contains only one key pair or the server only trusts one key pair, this measure is not necessary. In some cases it is necessary to adapt the chain of the key pair. For example, if the chain of the key pair contains only the public certificate and the server contains only the root CA certificate, then you need to add the intermediate certificate to the chain of the key pair.

More information: [Client Certificate Authentication \(Outbound\)](#)

- Enabling the tenant to establish a trust relationship to the receiver system

The SAP Cloud Integration tenant also needs to establish a trust relationship to the receiver in such a way that the receiver can authenticate itself against SAP Cloud Integration. In this case, authentication is accomplished based on a *server certificate* (as the receiver plays the role of a server). As prerequisite for this security measure, the tenant keystore needs to contain a (server) *root certificate* that is also trusted by the receiver.

Even in case you specify basic authentication when configuring the related receiver adapter, you need to make sure that the tenant keystore contains a valid root certificate that is also trusted by the receiver.

Message-level security

The keystore also contains the public and private keys used for message-level security (signing and encryption). Public keys are used in the signature verification steps (XML Signature, PKCS#7/CMS Signature Verification, WebService Security) and in the encryption steps (PKCS#7/CMS, WebService Security) of integration flows. Private keys are used in the signature creation steps (XML Signature, PKCS#7/CMS Signature, WebService Security) and decryption steps (PKCS#7/CMS, WebService Security) of integration flows. In these steps, the relevant keystore entries are referenced by their aliases. We recommend using different keys for message- and transport-level security. Keep in mind that the expiration date of the certificates is not checked in the encryption/decryption steps and in the signing steps.

Note that certain adapters (like the SOAP 1.x and the AS2 adapter) support options to sign/verify and encrypt/decrypt message content based on the Web Services Security (WS-Security) standard. To support such scenarios, the tenant keystore also needs to contain certain X.509 keys.

Keystore Content

There are the following entry types:

- [Key Pair](#) entry

Consists of a private key and its X.509 certificate chain.

All private keys of a keystore are encrypted with the same password. This password is also used as the keystore password (for checking the integrity of the keystore). The keystore is never stored in the same database as the encrypted/signed application data. The password is stored in a separate database.

The certificate chain typically consists of the public key certificate and the intermediate certification authority (CA) certificate with which the signature of the public key certificate can be verified.

- [Certificate](#) entry

In many cases this is an X.509 root certificate.

Keystore Management

A tenant keystore contains both entries owned by the tenant administrator (tenant owner) and entries owned by SAP. SAP-owned entries cannot be changed or deleted by the tenant administrator and entries owned by the tenant administrator cannot be changed or deleted by SAP.

More information: [Managing Keystore Entries](#)

i Note

There is a dedicated naming convention for keystore aliases to indicate the owner of the keystore entry:

Alias names of SAP-owned entries start with `sap_` or are `hcicertificate`, `hcicertificate1`, `hcimsgcertificate`.

SAP Cloud Integration does not verify the signatures of the certificates during the upload. Therefore, the user who uploads the certificates is responsible for ensuring that the signatures of the certificates are verified before the upload. Note that root certificates in particular must always be verified manually in any case.

Keystore Entries Preinstalled by SAP

When a customer starts using Cloud Integration, certain keystore entries have already been made available by SAP.

Keystore Entry	Purpose
One signed Key Pair entry with the alias <code>sap_cloudintegrationcertificate</code> .	- For outbound client certificate authentication - In the Cloud Foundry environment, for inbound client certificate authentication (required to enable internal communication between the involved BTP microservices)
Certain Certificate entries which are also owned by SAP.	These are root certificates that the customer can use to set up connections with other SAP cloud systems such like SAP SuccessFactors, for example.

Certificate Chains

The trust relationship between a client and a server using TLS authentication is usually based on chain certificates.

When using the X.509 standard, a key pair used for the TLS handshake is usually signed by a certification authority (CA). This means that the server can assume that the public key (included in the certificate) provided by the client originates from a trusted source.

The X.509 standard allows you to build up hierarchical trust models. In such a model (also referred to as a *certificate chain*), many certification authorities (CAs) are involved on different hierarchy levels. This means that the certificate that identifies the CA as a trusted participant can itself be signed by a CA at a higher level in the hierarchy. This means that a number of (intermediate) CAs can be arranged above the actual client certificate. The highest level CA is called the root CA.

The following figure shows a certificate chain with two intermediate CAs:



We assume that the tenant is connected as a client to an external component (which can be referred to as the server or receiver system).

To establish SSL connectivity, the server is provided with the root CA certificate and nothing else. To make sure that a trust relationship between client and server can be established nevertheless, the client certificate (of the tenant) used for the SSL handshake has to contain the whole certificate chain. In other words, the client certificate has to include all intermediate CAs (excluding the root CA). This enables the server to evaluate and calculate the whole chain of trust.

Therefore, during connection setup (onboarding), the tenant key pair (client certificate) has to be assigned the whole certificate chain.

→ Tip

To find out the certificate chain of the server, you can use the TLS Outbound Connection Test (accessible in the Monitoring application). This test also helps you to find out whether you have the correct CA certificate in the keystore to validate the server certificate chain (see option [Validate Server Certificate](#) of the Outbound Connection Test).

Related Information

[TLS Connectivity Tests](#)

Requirements for Keystore Passwords

To protect a keystore, you have to specify a password when creating the keystore.

You have to apply the following rules when specifying passwords for keystores:

- The password must have a minimum length of 8 characters.
- The password must contain characters of **at least three** of the following groups:
 - Lower-case Latin characters (a-z)
 - Upper-case Latin characters (A-Z)
 - Base 10 digits (0-9)
 - Non-alphabetic characters (!@#\$%...)
- The password must **not** contain any characters from outside the standard ASCII table like, for example, German umlaut characters (<ü>).

i Note

Example for password compliant with the above rule:

<xB+gku!kjh>

PGP Keys

PGP public and secret keys (the latter containing a private key) can be uploaded to the tenant via separate keyrings. The PGP Public Keyring contains Transferable Public Keys as defined in section 11.1 of the Open PGP specification (<https://tools.ietf.org/html/rfc4880> 🗑) and the secret keyring contains Transferable Secret Keys as defined in section 11.2.

PGP keys are used in the PGP Encryptor and Decryptor step. You should only add PGP Public keys to the PGP public Keyring if you trust this key. Typically you check the fingerprint of the public key. The same security measures must be taken for the secret keys that you use in the secret keyring. The encryption and signing steps do also work with expired certificates.

For the PGP Secret Keyring, the same precautions as for the X.509 keystore must be taken because it contains private keys.

Known Hosts File

Known hosts files are relevant for SFTP communication. The known hosts file contains the host names and the public keys of the trusted SFTP servers. You should only have entries for those servers in the file which are used by the integration flows of the tenant and which you trust.

Security Elements

To set up the secure communication between a tenant and a sender/receiver system, certain security elements have to be created and - in some cases - exchanged between the involved components (the tenant on the one side and the sender/receiver system on the other side of the communication).

For example, to set up SSL communication using certificate-based authentication between a tenant and a receiver system, X.509 certificates are required. Those private keys owned by the tenant are to be part of a Java keystore that is to be deployed on the tenant, whereas the private keys owned by the receiver are to be part of the receiver system keystore. To complete the security setup, each keystore also has to contain the public key of the connected partner. In our example, the Java keystore of the tenant has to contain the receiver public key, and the receiver keystore has to contain the tenant public key.

This section provides a summary for each security option of how the required security elements have to be distributed among the involved components (tenant and sender/receiver systems).

Related Information

[Security Elements \(Transport-Level Security\)](#)

[Security Elements \(Message-Level Security\)](#)

Security Elements (Transport-Level Security)

Each transport-level security option requires a specific set of security elements.

The following tables provide a summary of how the required security elements (in **bold letters**) have to be distributed among the involved components (tenant and sender/receiver systems).

Transport-Level Security

Security Option	Direction	Required by tenant administrator ...	... to do the following	Required by sender/receiver administrator ...	... to do the following
HTTPS – basic authentication	Inbound (sender calls tenant)	User name (to be provided by sender administrator). This is the user under which the customer system is	Grant the required authorizations to enable this user to call the tenant.	Load balancer root certificate (to be provided by tenant administrator) Is required for the SSL communication	Import into the keystore of the sender system.

Security Option	Direction	Required by tenant administrator ...	... to do the following	Required by sender/receiver administrator ...	... to do the following
		to call SAP Cloud Integration.		step (can be obtained via the URL of the runtime node provided in the tenant mail by SAP).	
				User name and password (to be provided by tenant administrator)	Enable the sender to support basic authentication.
	Outbound (tenant calls receiver)	Receiver server root certificate (to be provided by receiver administrator) Is required to enable HTTPS communication with the receiver system (server).	Import into the tenant keystore (and deploy the keystore on the tenant).		
		User credentials (to be provided by receiver administrator) These are the user credentials under which the tenant is to call the receiver system.	Define the User Credentials artifact (to be deployed on the tenant).		
HTTPS – certificate-based	Inbound (sender calls tenant)	Sender client root certificate (to be provided by sender administrator)	Check whether the CA the customer system used to get its client certificate signed is already part of the load balancer (server) keystore.	Load balancer server root certificate (to be provided by tenant administrator)	Import into client PSE of the sender system.
		Sender client certificate (to be provided by sender administrator)	Configure the authorization check in the integration flow.		
				List of trusted root certificates supported by load balancer (to be provided by tenant administrator)	Select a certification authority from the list for the certificate signing request for the client certificate.

Security Option	Direction	Required by tenant administrator ...	... to do the following	Required by sender/receiver administrator ...	... to do the following
	Outbound (tenant calls receiver)	Receiver server root certificate (to be provided by receiver administrator)	Import into tenant keystore (if not already there).	Tenant client root certificate (to be provided by tenant administrator)	Import into the server PSE of the receiver system.
				Tenant client certificate (to be provided by tenant administrator)	Define the client certificate-to-user mapping for the configuration of authorization checks.
SFTP	Outbound (tenant as SFTP client sends a request to an SFTP server)	SFTP server (receiver) public key (to be provided by SFTP server (receiver) administrator) Is required by tenant to check whether SFTP server can be trusted.	Add to known_hosts file (to be deployed as Known Hosts artifact on tenant).	Tenant public key (to be provided by tenant administrator) Is used to authenticate tenant as a trusted SFTP client on the SFTP server side.	Add to authorized_keys file on the SFTP server side.

Security Elements (Message-Level Security)

The configuration of secure message exchange requires the exchange of public keys (or other security-related information) between the involved parties. Each message-level security option requires a specific set of keys to be exchanged.

The following tables provide a summary of how the required security elements (in **bold letters**) have to be distributed among the involved components (tenant and sender/receiver systems).

Message-Level Security

Security Option/Standard	Direction	Protection Method on Tenant	Required by tenant administrator ...	... to do the following	Required by sender/receiver administrator ...	... to do the following
PKCS#7, WS-Security, XML Digital Signature (uses X.509 certificates) XML Digital Signature: only sign/encrypt	Inbound (sender calls tenant)	Decrypt			Tenant public key certificate (to be provided by tenant administrator) Is used to encrypt the message from the sender (that is to be encrypted by the tenant).	Import into sender keystore

Security Option/Standard	Direction	Protection Method on Tenant	Required by tenant administrator ...	... to do the following	Required by sender/receiver administrator ...	... to do the following
		Verify	Sender public key certificate (to be provided by sender administrator) Is used by the tenant to verify the signature of the message sent from the sender system.	Import into tenant keystore.		
	Outbound (tenant calls receiver)	Encrypt	Receiver public key certificate (to be provided by receiver administrator) Is used by the tenant to encrypt the message (sent to the receiver).	Import into tenant keystore.		
		Sign			Tenant public key certificate (to be provided by tenant administrator) Is used by the receiver to verify the message sent from the tenant.	Import into receiver keystore
OpenPGP (uses PGP keys)	Inbound (sender calls tenant)	Decrypt			Tenant public key (to be provided by tenant administrator) Is used to encrypt the message from the sender (that is to be encrypted by the tenant). To make sure that the public key originates from the correct source and that it has not been	Import into sender PGP public keyring

Security Option/Standard	Direction	Protection Method on Tenant	Required by tenant administrator ...	... to do the following	Required by sender/receiver administrator ...	... to do the following
					changed on its way, consider the note below this table.	
		Verify	Sender public key (to be provided by sender administrator) Is used by the tenant to verify the signature of the message sent from the sender system. To make sure that the public key originates from the correct source and that it has not been changed on its way, consider the note below this table.	Import into tenant PGP public keyring.		
	Outbound (tenant calls receiver)	Encrypt	Receiver public key (to be provided by receiver administrator) Is used by the tenant to encrypt the message (sent to the receiver). To make sure that the public key originates from the correct source and that it has not been changed on its way, consider the note below this table.	Import into tenant PGP public keyring.		
		Sign			Tenant public key (to be provided by tenant administrator)	Import into receiver PGP public keyring

Security Option/Standard	Direction	Protection Method on Tenant	Required by tenant administrator ...	... to do the following	Required by sender/receiver administrator ...	... to do the following
					Is used by the receiver to verify the message sent from the tenant. To make sure that the public key originates from the correct source and that it has not been changed on its way, consider the note below this table.	

i Note

Relevant for the SAP-managed operating model: When **exchanging public PGP keys**, note the following:

To ensure that the information originates from the correct source and that it has not been changed on its way, the key should be exchanged using a secure channel (for example, encrypted e-mail).

If a secure channel is not available, the person who receives the public key from the key owner has to **verify the fingerprint** of the public key. One option is to phone the owner of the public key and compare the fingerprint.

Setting up Message-Level Security Use Cases

On top of the secure transport channel (that is based either on HTTPS or SFTP), you can additionally protect the message exchange by digital encrypting and signing the message.

To do that, you can use different security standards.

Related Information

[Inbound: Message-Level Security With PKCS#7, XML DigitalSignature](#)

[Inbound: Message-Level Security with OpenPGP](#)

[Outbound: Message-Level Security With PKCS#7, XML DigitalSignature](#)

[Outbound: Message-Level Security with OpenPGP](#)

Inbound: Message-Level Security With PKCS#7, XML DigitalSignature

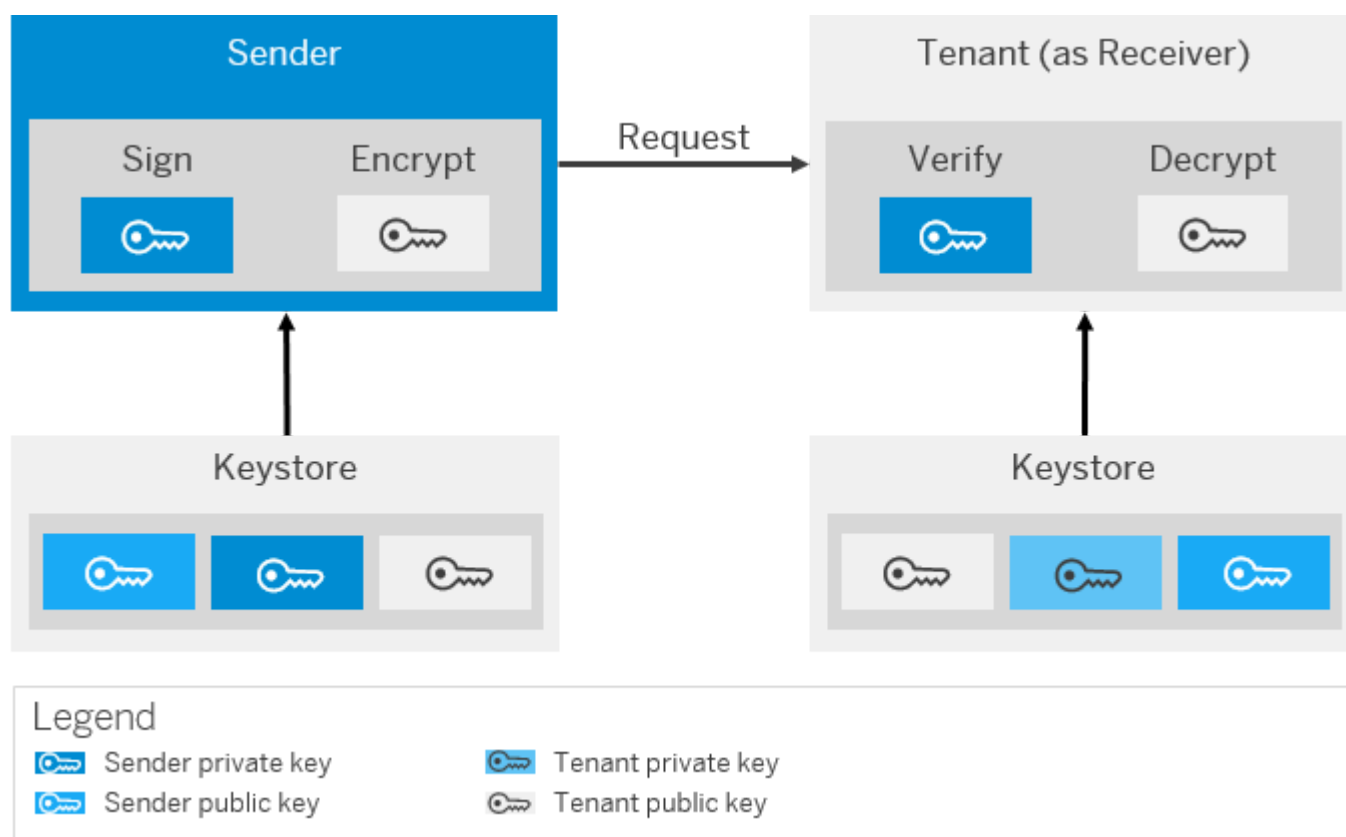
On top of a secure transport channel (for example, based on HTTPS), you have the option to implement message-level security capabilities. That way, you can protect the message by applying digital signing or encryption. Asymmetric key technology is used in the following way to implement these features:

Keys for Message-Level Security

Key Type	Usage
Private key	Used by a sender to sign a message
	Used by a receiver to decrypt a message (that has been encrypted by a sender)
Public key	Used by a receiver to verify a message (signed by a sender)
	Used by a sender to encrypt a message

In the inbound case, the tenant acts as receiver that either decrypts or verifies a message.

To implement message-level security for the standards PKCS#7, WS-Security, and XML Digital Signature, you use X.509 certificates (the same type of certificates as used for HTTPS-based transport-level security). However, note that different keys are usually used for message-level security and SSL transport-level security. XML Digital Signature supports only the use cases of signing/verifying messages.



Configuring the Sender

Configure the sender keystore in the following way:

- Generate a key pair (and get it signed by a CA).
- Import the tenant public key into the sender keystore.

Provide the tenant administrator with the public key (is used to verify messages sent to the tenant).

Configuring the Integration Flow Steps for Message-Level Security

Depending on the desired option, configure the security-related integration flow steps.

- Configure the **Verifier** (PKCS7 or XML Signature Verifier) step.

Specify the **Public Key Aliases** in order to select the relevant keys from the tenant keystore.

- Configure the **Decryptor** (PKCS7) step.

Make sure that you specify the **Public Key Aliases** for all expected senders (only if you have specified **Enveloped or Signed and Enveloped Data** or **Signed and Enveloped Data** for **Signatures in PKCS7 Message**).

These are the public key aliases corresponding to the private keys (of the expected senders) that are used to sign the payload. The public key aliases specified in this step restrict the list of expected senders and, in this way, act as an authorization check.

In general, an alias is a reference to an entry in a keystore. A keystore can contain multiple public keys. You can use a public key alias to refer to and select a specific public key from a keystore.

Related Information

[How PKCS#7 Works](#)

[How XML Signature Works](#)

[How WS-Security Works](#)

[Creating Keys for the Usage of PKCS#7, XML Digital Signature and WS-Security](#)

[Define PKCS#7/CMS Decryptor](#)

Inbound: Message-Level Security with OpenPGP

On top of a secure transport channel (for example, based on HTTPS), you have the option to implement message-level security capabilities. That way, you can protect the message by applying digital signing or encryption. Asymmetric key technology is used in the following way to implement these features:

Keys for Message-Level Security

Key Type	Usage
Private key	Used by a sender to sign a message
	Used by a receiver to decrypt a message (that has been encrypted by a sender)
Public key	Used by a receiver to verify a message (signed by a sender)
	Used by a sender to encrypt a message

In the inbound case, the tenant acts as receiver that either decrypts or verifies a message.

To implement message-level security for OpenPGP, you use PGP keys.



Configuring the Sender

1. Generate and configure the PGP keys and the storage locations (PGP secret and public keyrings) for the sender system.
2. Import the related public keys from the tenant into the public PGP keyring of the sender and finish the configuration of the sender system.

Provide the tenant administrator with the public key (is used to verify messages sent to the tenant).

Configuring the Integration Flow Steps for Message-Level Security

Configure the security-related integration flow steps.

Configure the **Decryptor** (PGP) and **Verifier** (PGP) step.

When signatures are expected, make sure that you specify the **Signer User ID of Key(s) from Public Keyring** for all expected senders.

Based on the signer user ID of key(s) parts, the public key (for message verification) is looked up in the PGP public keyring. The signer user ID of key(s) key parts specified in this step restrict the list of expected senders and, in this way, act as an authorization check.

Related Information

[How OpenPGP Works](#)

[Creating OpenPGP Keys](#)

[Define PGP Decryptor](#)

Outbound: Message-Level Security With PKCS#7, XML DigitalSignature

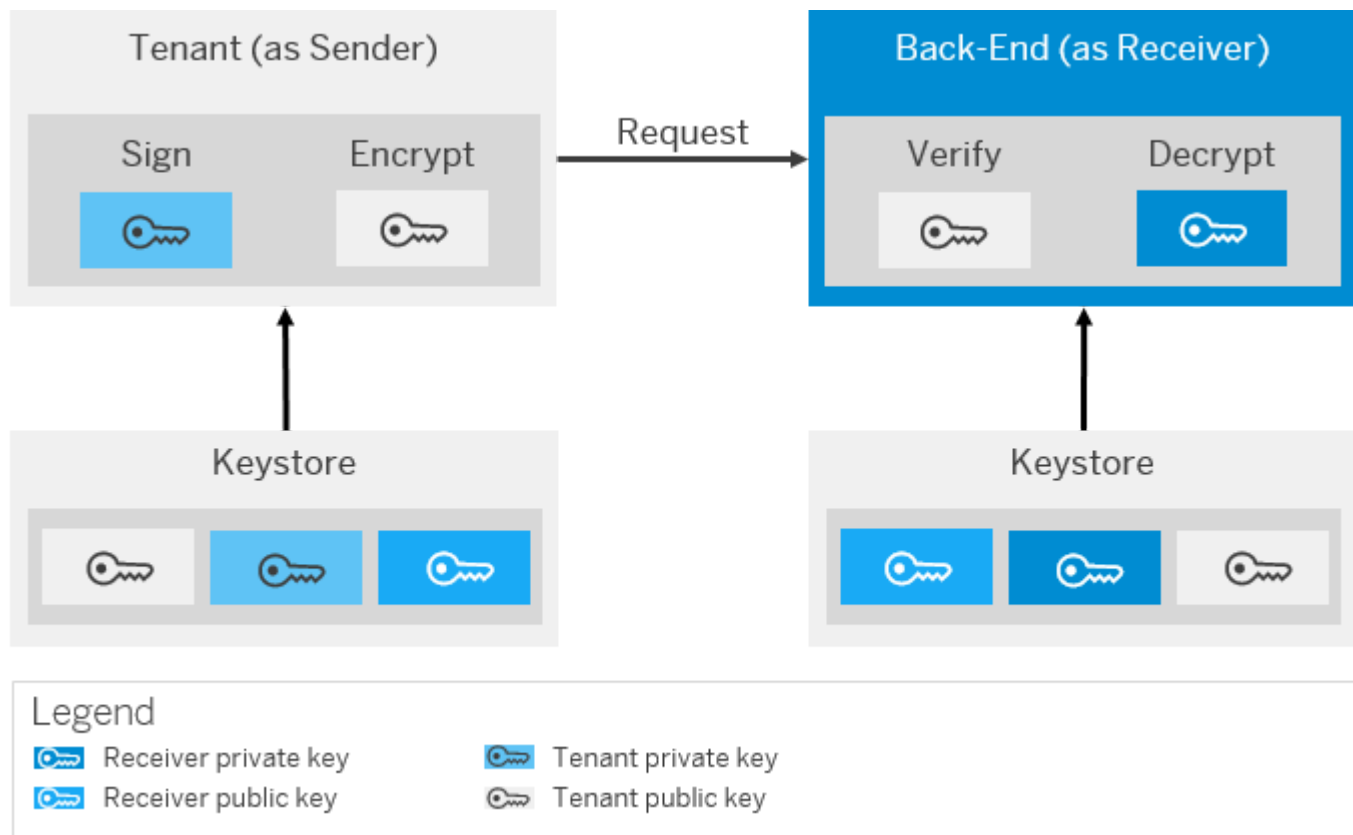
On top of a secure transport channel (for example, based on HTTPS), you have the option to implement message-level security capabilities. That way, you can protect the message by applying digital signing or encryption. Asymmetric key technology is used in the following way to implement these features:

Keys for Message-Level Security

Key Type	Usage
Private key	Used by a sender to sign a message
	Used by a receiver to decrypt a message (that has been encrypted by a sender)
Public key	Used by a receiver to verify a message (signed by a sender)
	Used by a sender to encrypt a message

In the outbound case, the tenant acts as sender that either encrypts or signs a message.

To implement message-level security for standards PKCS#7, WS-Security, and XML Digital Signature, you use X.509 certificates (the same type of certificates as used for HTTPS-based transport-level security). However, note that different keys are usually used for message-level security and SSL transport-level security. XML Digital Signature supports only use cases for signing and verifying messages.



Configuring the Receiver

Configure the receiver keystore in the following way:

- Generate a key pair (and get it signed by a CA).
- Import the tenant public key into the receiver keystore.

Provide the tenant administrator with the public key (is used to encrypt messages sent to the receiver).

Configuring the Integration Flow Steps for Message-Level Security

Depending on the desired option, configure the security-related integration flow steps.

- Configure the **Encryptor** (PKCS7) step.
Specify the **Public Key Aliases** in order to select the relevant key from the tenant keystore. In case you have selected **Signed and Enveloped Data** (as **Signatures**), you also need to specify the **Private Key Alias** to select the relevant private key for signing.
- Configure the **Signer** (PKCS7 or XML Digital Signer) step.
Make sure that you specify the **Private Key Aliases** to select the desired keys from the keystore.

In general, an alias is a reference to an entry in a keystore. A keystore can contain multiple public keys. You can use an alias to refer to and select a specific key from a keystore (as shown for the Signer step in the figure below).

PKCS7/CMS Encryptor

*Name:

PKCS7Encryptor1

Block Size(in bytes):

2048

☐ Encode Body with Base64

Signatures:

Enveloped Data Only

Content Encryption Algorithm:

AES/CBC/PKCS5Padding

Secret Key Length:

256

*Receiver Public Key Alias:

☐ Public Key Alias

☐ SOAPReceiver1

Related Information

[How PKCS#7 Works](#)

[How XML Signature Works](#)

[How WS-Security Works](#)

[Creating Keys for the Usage of PKCS#7, XML Digital Signature and WS-Security](#)

[Sign the Message Content with PKCS#7/CMS Signer](#)

[Encrypt and Sign the Message Content with PKCS#7/CMS Encryptor](#)

Outbound: Message-Level Security with OpenPGP

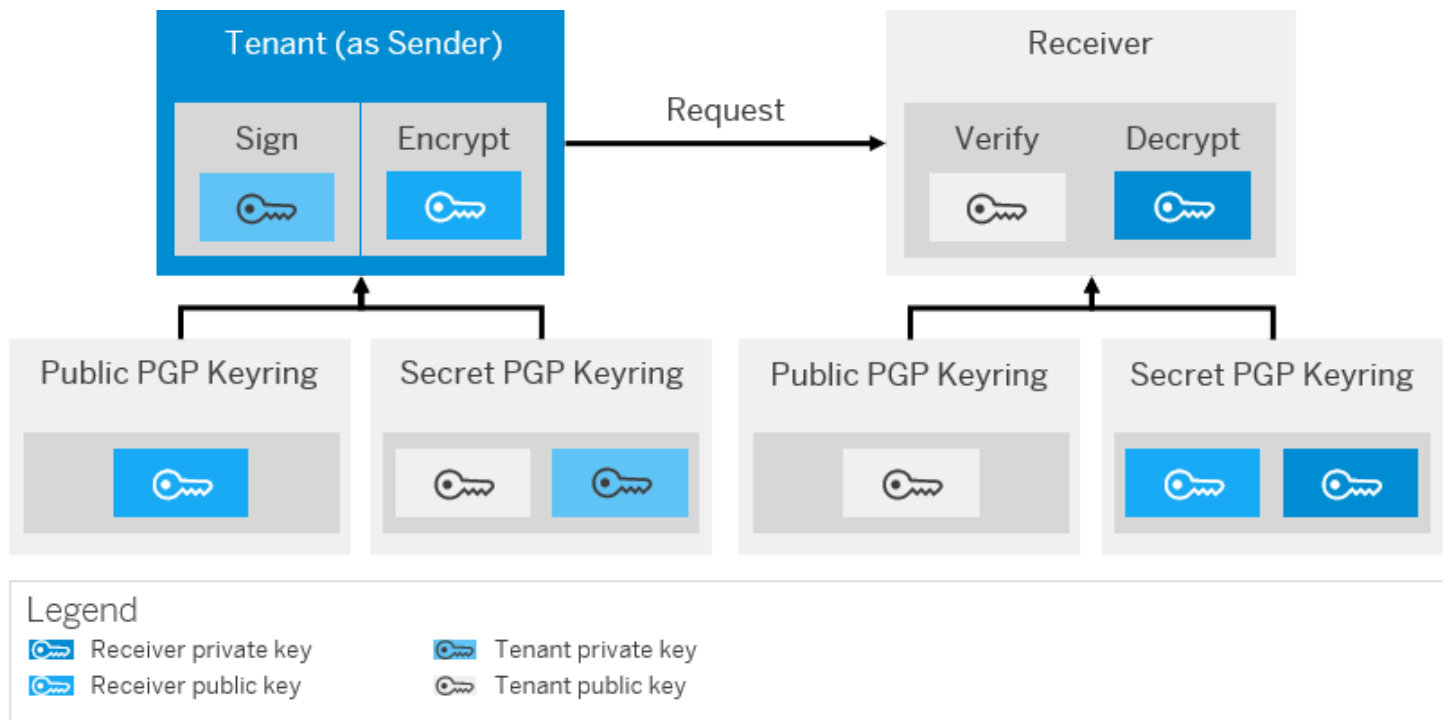
On top of a secure transport channel (for example, based on HTTPS), you have the option to implement message-level security capabilities. That way, you can protect the message by applying digital signing or encryption. Asymmetric key technology is used in the following way to implement these features:

Keys for Message-Level Security

Key Type	Usage
Private key	Used by a sender to sign a message
	Used by a receiver to decrypt a message (that has been encrypted by a sender)
Public key	Used by a receiver to verify a message (signed by a sender)
	Used by a sender to encrypt a message

In the outbound case, the tenant acts as sender that either encrypts or signs a message.

To implement message-level security for OpenPGP, you use PGP keys.



Configuring the Receiver

1. Generate the PGP keys and the storage locations (PGP secret and public keyrings) for the receiver system.
2. Import the related public keys from the tenant into the public PGP keyring of the receiver and finish the configuration of the receiver system.

Provide tenant administrator with the public key (used to encrypt messages sent to the receiver).

Configuring the Integration Flow Steps for Message-Level Security

Depending on the desired option, configure the security-related integration flow steps.

Configure the **Encryptor** (PGP) step.

- Specify the **User ID of Key(s) from Public Keyring** in order to select the relevant public receiver keys from the PGP public keyring.
- If you want to sign the payload, specify the **Signer User ID of Key(s) from Secret Keyring** in order to select the relevant private key from the PGP secret keyring. The private key is used to sign the message.

Related Information

[How OpenPGP Works](#)

[Creating OpenPGP Keys](#)

[Define PGP Encryptor](#)

Specific Use Cases

Related Information

[Setting Up Principal Propagation \(Example Scenario\)](#)

Setting Up Principal Propagation (Example Scenario)

Use principal propagation to forward the principal (identity of a user) across several connections in a complex system landscape.

Cloud Foundry Environment

For more information on how to set up a scenario with principal propagation end-to-end, check out:

[Principal Propagation in SAP Integration Suite](#) (SAP Community blog series)

In the following scenarios described there the component is involved:

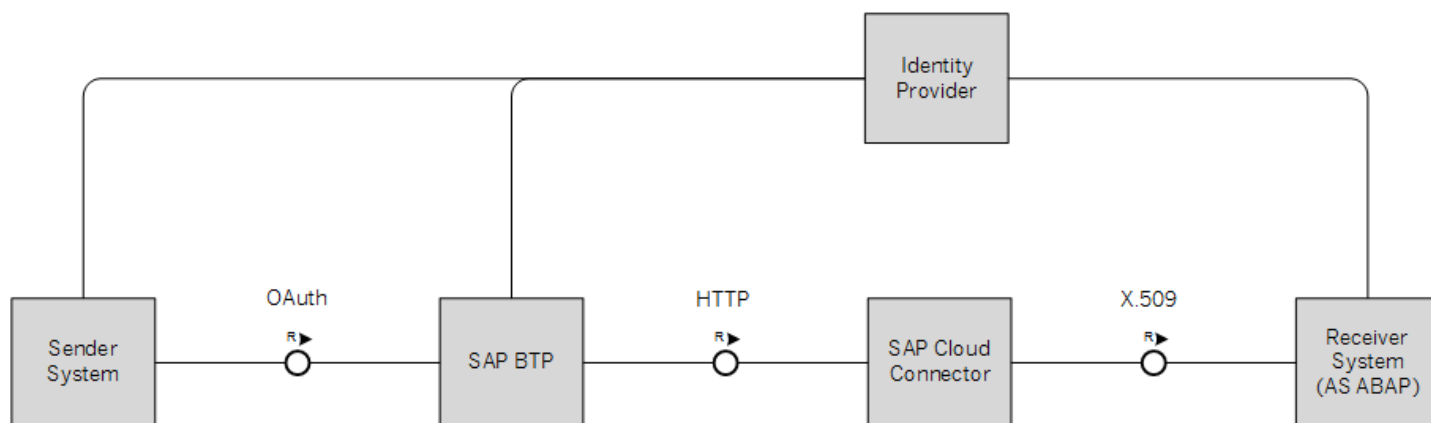
- Propagate identity from an external system to an On-Premise SAP Backend with mediation
- Propagate identity from a BTP application to an On-Premise SAP Backend with mediation

i Note

The adapters that allow you to set the authorization method **Principal Propagation** support such a scenario. However, when setting up such a scenario that spans multiple components and systems, you are responsible for configuring each component in the right way.

Neo Environment

In the following example setup, the principal of the inbound user is forwarded to SAP Cloud Connector, and from there to the back-end receiver system.



We assume the following:

- In this example, the authentication option OAuth is used (using OAuth SAML Bearer Destination) for inbound communication (from the sender to SAP BTP).
- An on-premise SAP system based on Application Server ABAP is used as the receiver system,.

The on-premise receiver system is connected to SAP BTP through SAP Cloud Connector.

⚠ Caution

Using SAP Cloud Connector is a mandatory when configuring principal propagation.

- The receiver system is associated with an identity provider, which mediates a trust relationship between the sender, SAP BTP, and the receiver.
- To establish an outbound connection (from SAP BTP to SAP Cloud Connector), an adapter that supports principal propagation is used (for example, the HTTP receiver adapter).
- All systems that communicate with each other have to provide the same user. This can be achieved by using an identity provider, as indicated in the figure above as an example setup.

To configure principal propagation for this setup, perform the following steps.

1. Enable OAuth (with SAML Bearer Destination) for the inbound connection from the sender to SAP BTP.

More information: [OAuth SAML Bearer Destination](#)

i Note

Note that currently only the following (sender) adapter types can be used on the inbound side: HTTPS, SOAP (SOAP 1.x), SOAP (SAP RM), and IDoc.

2. In the receiver channel of the integration flow, as **Authorization** option, enable **Principal Propagation**.

More information: [HTTP Receiver Adapter](#)

→ Remember

When you want to use Principal Propagation as the authentication method to connect with an on-premise system, don't pass any authorization headers. Follow the approach recommended by SAP BTP Connectivity. See: [Authentication to the On-Premise System](#).

3. Prepare SAP Cloud Connector to support principal propagation with X.509 certificates (for the communication with the receiver system).

You need a certificate chain with at least one intermediate certification authority. The intermediate certification authority signs a short-lived certificate, which is used for principal propagation. Use the user name (associated with the identity to be propagated) as the subject common name (subject CN) of this certificate.

SAP Cloud Connector forwards the identity (to be propagated) in a short-living X.509 certificate in HTTP header SSL_CLIENT_CERT.

4. In SAP Cloud Connector, configure the trust relationship with the SAP BTP application.

More information:

[Connectivity for the Neo Environment](#)

5. Configure the receiver system. You need to do the following:

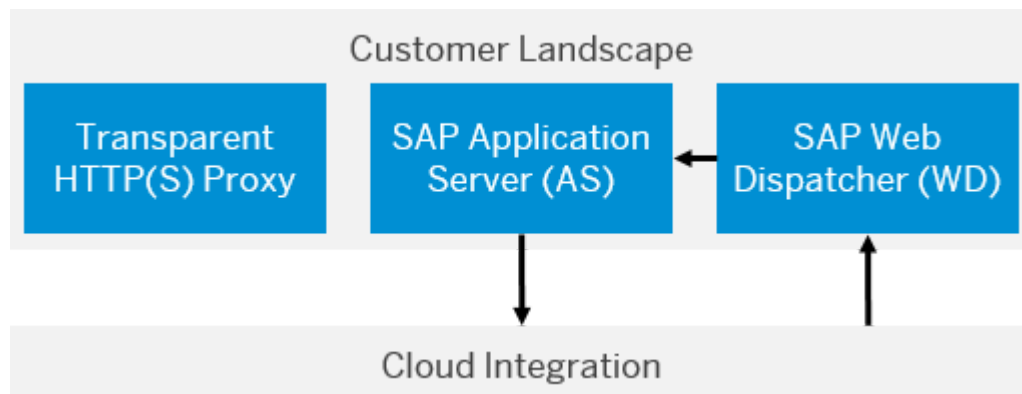
- Configure the receiver system to trust the certificate of the SAP Cloud Connector.
- Configure the Internet Communication Manager (ICM) to trust the system certificate for principal propagation.
- Map the short-living certificate (from SAP Cloud Connector) to the user (whose identity is being propagated).

More information: [Configuring Principal Propagation to an ABAP System for HTTPS](#) 

Technical Landscape for On Premise-On Demand Integration

As one example for certificate-based connectivity, customer intends to connect a customer-based SAP on-premise system (based on SAP Application Server ABAP with Cloud Integration).

The following figure illustrates the system landscape.



i Note

We use the following abbreviations in this documentation:

- **AS** for SAP Application Server
- **WD** for SAP Web Dispatcher

In the proposed system landscape, SAP Web Dispatcher is used in the on premise customer landscape to receive incoming calls from Cloud Integration. SAP Web Dispatcher (as reverse proxy) is the entry point for HTTPS requests into the customer system landscape.

Communication Cloud Integration to SAP Application Server

In the proposed landscape, two SSL connections have to be implemented on the way in between Cloud Integration and AS, because WD - interconnected in between - terminates all SSL calls from Cloud Integration. Therefore, the following trust relationships have to be implemented:

- Trust relationship between WD and Cloud Integration.

As this connection spans the Internet, it is strongly recommended to use certificates that are signed by a certification authority (CA) that both parties (WD and Cloud Integration) trust.

- Trust relationship between WD and AS.

As this connection resides within the customer landscape, it might be an option to use self-signed certificates for this connection.

i Note

For reasons of simplicity, within this guide we assume that self-signed certificates are used for this connection.

The following table summarizes the required certificates and the related keystores.

Keystores		
Keystore	Certificate/Key	Description

Keystore	Certificate/Key	Description
Cloud Integration client keystore	Cloud Integration client certificate (private and public key)	Required to authenticate Cloud Integration as sender of messages. This security artifact has to be generated at SAP side and contains the public and private key of Cloud Integration. The certificate has to be signed by a certification authority (CA) that both SAP (Cloud Integration) and the customer (WD) trust.
	WD server root certificate (of the CA that has signed the server certificate)	Required to authenticate WD as receiver of messages. This certificate identifies the CA that has signed the WD server certificate.
WD server keystore (SSL server PSE)	Cloud Integration client root certificate	Required to identify Cloud Integration as trusted communication partner. This certificate identifies the CA that has signed the Cloud Integration client certificate.
	WD server certificate	Required to authenticate WD as trusted communication partner to receive calls. This certificate is signed by the CA to which both WD and Cloud Integration have established a trust relationship.
WD client keystore (SSL client PSE)	WD client certificate (private and public key)	Required to authenticate WD as sender of messages. This security artifact has to be generated at customer side and contains the public and private key of the WD. As the related communication path resides within the customer landscape, it might be sufficient to use a self-signed certificate. i Note Customers can extend the use case in a way that also this certificate is signed by a CA. This is not covered in this guide.
AS server keystore (SSL server PSE)	WD client certificate (public key)	Required to authenticate WD as sender of messages. This public key has to be imported into the AS server keystore.

Communication SAP Application Server to Cloud Integration

In the proposed landscape, the SSL connection is **not** terminated on the way in between AS and Cloud Integration (transparent proxy). Therefore, a trust relationship has to be set up between AS and Cloud Integration.

As this connection spans the Internet, it is strongly recommended to use certificates that are signed by a certification authority (CA) that both parties (AS and Cloud Integration) trust.

The following table summarizes the required certificates and the related keystores.

Keystores		
Keystore	Certificate/Key	Description
AS client keystore	AS client certificate (private and public key)	Required to authenticate AS as sender of messages. This security artifact has to be generated at customer side and contains the public and private key of AS. The certificate has to be signed by a certification authority (CA) that both SAP (Cloud Integration) and the customer (AS) trust.
	Cloud Integration server root certificate	Required to authenticate Cloud Integration as trusted receiver of messages. This certificate identifies the CA that has signed the Cloud Integration server certificate.
Cloud Integration server keystore	AS client root certificate	Required to authenticate AS as sender of messages. This certificate identifies the CA that has signed the AS client certificate. This artifact has to be provided by the customer for SAP during the connection setup process, and the expert at SAP side has to import it into the Cloud Integration server keystore.
	Cloud Integration server certificate	Required to authenticate Cloud Integration as trusted communication partner to receive calls. This certificate is signed by the CA to which both AS and Cloud Integration have established a trust relationship.

Using Custom IDP with SAP Cloud Integration

SAP Cloud Integration supports the usage of custom identity providers. By default, SAP Cloud Integration uses the global standard [account.sap.com](#) as the identify provider. However, you can use a tenant-specific Identity Authentication Service or your own SAML based identity provider.

For more information on configuring custom IDP, see [Authentication](#).

Change Account Configuration to Basic Authentication While Using a Tenant-Specific Identity Authentication Service

i Note

These instructions are relevant only when you use SAP Cloud Integration in the Neo environment.

Let us consider a scenario where you are using a custom IDP and you are able to log in to SAP Cloud Integration web application. However, when you try to connect to the operations server using **Basic Authentication**, you will be unable to log in using your custom IDP credentials

In SAP Cloud Integration this is due to the usage of the Eclipse UA and the OData API. If you consider using the same credentials for these two use cases, you need to change your account configuration to basic authentication.

You can change this configuration using the SAP BTP cockpit. For more information, see [Basic Authentication](#).

→ Remember

Only [account.sap.com](#) and [Identity Authentication Service](#) can be configured for basic authentication. You are not allowed to use any arbitrary custom IDP for this use case.

Set up Custom IDP within Cloud Foundry Environments

i Note

These instructions are relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

For more information see: [Setting Up SAP Identity Authentication Service as Custom IdP for Basic Authentication, Cloud Foundry Environment](#)

If you open a support ticket on the component **BC-CP-CF-SEC-IAM** to get help from SAP.

Setting Up OAuth Configurations in Customer Account, Neo Environment

With a REST API you can manage roles and their assignments to users.

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

You can set up OAuth configurations in the customer account for enabling the API using the following steps. These steps are one time manual steps per customer account when provisioning the Cloud Integration tenant.

Procedure

1. Get **OAuth Client Credentials** from the **Customer Account**.
 - a. Select **OAuth** tab.
 - b. Click on **Create API Client** button to create Client credentials from **Platform API** tab.
 - c. Select the check-box **Authorization Management** and click **Save**.

d. Make note of the credentials from the popup as these credentials are used to create HTTP destination named **OAuthTokenDestination** in the consumer account as shown below.

2. Create a destination named **OAuthTokenDestination** from the account's cockpit as shown below.

Destination Configuration

* Name:

Type:

HTTP

Description:

* URL:

Proxy Type:

Internet

Authentication:

BasicAuthentication

User:

Password:

Save

Cancel

Destination Property, Value	Description
Name, OAuthTokenDestination	
Type, HTTP	
Url, https://api.{landscapeHost}/oauth2/apitoken/v1/?grant_type=client_credentials	OAuth Token Endpoint Url. Landscape host: PROD: neo.ondemand.com FACTORY: hana.ondemand.com STAGING: staging.hanavlab.ondemand.com CANARY: sap.hana.ondemand.com For data center Rot (Europe): https://hana.ondemand.com For data center Ashburn (USA): https://us1.hana.ondemand.com For data center Sydney (Asia-Pacific): https://ap1.hana.ondemand.com For data center Phoenix (USA - West) https://us2.hana.ondemand.com
Proxy Type, Internet	
Authentication, Basic Authentication	
User	Client Id of the customer account as obtained in Step 1-E
Password	Client secret of the customer account as obtained in Step 1-E